

Essentials of Production & Operations Management



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Unit - 1.1: Brief History of Production and Operation Management, the Service Revolution, Definition

Objectives:

At the end of this unit, you will be able to:

- Learn what is production management
- What is operations management
- Learn about production and operation management
- Understand brief history of production and operation management
- Understand its service revolution

Introduction



In the competitive world of today, productivity is the main concern of any business to sustain in the industry. The relationship between the output i.e., Products and services and input i.e., resources consumed in providing them is productivity of a business. The ratio of Aggregate output to the aggregate input is called productivity.

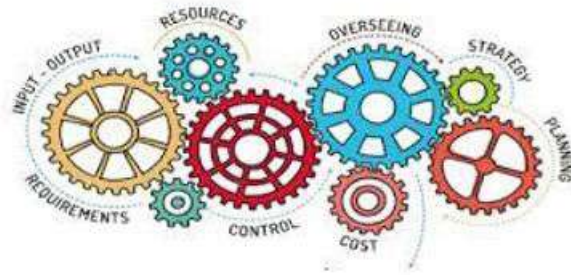
$$\text{Productivity} = \text{Output/Input}$$

where, Output is the goods and services and, Input is the Labour, material, Capital, Energy and Management. Therefore, to improve productivity wastes should be minimized and Ineffective time should be reduced. (Ineffective Time is wherein the worker or machine or both are idle due to the shortcomings of the management or the worker).

Operations Management helps concentrating on the core businesses, squeezes out the waste and focuses on differentiating between competitors in a meaningful way. The significance of Operations management lies in examining the process by which the goods and services are created and utilized, and to use the available knowledge and techniques to resolve problems.

An Operations Management is only said to be successful only if it is able to make world class companies by its ability. For example: Bajaj Auto has focused on operations Management to emerge as lowest cost manufacturer of two-wheelers in the world. TATA steel is the lowest cost steel producer, internationally.

What is Production and Operations Management?



The very essence of any business is to understand needs of customer and provide the services and goods to fulfil their needs, and in process create value for customers and solve their problems. Production and operations management helps in creating goods and services by applying business organization and management concepts.

1.1.1 History of Production and Operations Management

- Pre-industrial Revolution

A Scottish philosopher – and the father of modern economics - Adam Smith was the one of the first people to address the issues of operations management. In 1776 Smith wrote “The Wealth of Nations,” in which he described the division of labour. According to Smith, by dividing their tasks workers can produce their products more efficiently, than if the same number of workers each built product from start to finish. In 1913, this concept was later be used by Henry Ford with the introduction of the assembly line. An assembly line is a production process that breaks up the manufacturing process of a product into a pre-defined sequence that further divides the labour and their tasks.

- Post-industrial Revolution

The Industrial revolution was transition to new manufacturing process, which included going from hand production methods to machines. Machinery allowed factories to grow in capacity and greatly increased their output, during Industrial revolution. Despite this growth, there was considerable inefficiency in production. Frederick Winslow Taylor and Ford helped to overcome these inefficiencies in the early 20th century. Taylor developed a scientific approach for operations management, collecting all the data about production, then analysing this data and using the same to make improvements to operations. Ford introduced assembly line production which resulted in increased efficiency in production by and also improved the supply chain through just-in-time delivery.

- Post World War II

During the second world war, technological developments created new possibilities for the managers looking to improve their operations. The development of computational technology in specific, allowed for a greater degree of data to be analysed by firms leading to efficiencies. The abilities of computers have continued to increase aggressively, allowing for a high degree of communication and data analysis. Modern producers are now able to track their inventory from raw materials, through production and delivery.

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- Modern Day

In today's operations management, Quality management systems are very popular as it provides guidance for mapping, improving and monitoring operations processes. A variety of quality management systems, like the ISO systems and Six Sigma are in use among top firms, with an aim to increase the efficiency of business processes. In spite of the fact that operations management has typically dealt with the manufacturing process, the growth of the service industry has created a field of service operations management.

Further, in 1960's and 1970's scholars started to note the similarity of problems being faced by operation so they started treating operation system as a proper system. Here is the sequence in which some important concepts related to operation were developed is as follows:

JIT (Just in time)

A revolution was brought about in how the production should be carried out by Japanese in 1980's. JIT is designed to achieve high volume of production using minimum inventory cost wherein the parts or raw materials required arrives from the suppliers at the work station exactly when needed along with the total Quality Control; aggressively trying to remove the causes of production defects. JIT is the core stone of many production practices. For example, famous for its inventory system, Toyota motor orders parts of the car only when it receives new car orders.

Manufacturing Strategy Paradigm

Throughout the history of manufacturing process many paradigms were proposed to ease the processes of manufacturing management. In the beginning of 20th Century, Mass Manufacturing was created by Henry Ford. This paradigm, now called as Current Mass Manufacturing (CMM) is still used by companies worldwide.

Lean Manufacturing (LM) which started to emerge in mid-fifties in Japan, was consolidated in the 1970s. And, at the end of the 80s and early 90s, three other important paradigms were developed:

- Time-based competition (or competing against time) also called Responsive Manufacturing, by Kritchanchai and MacCarthy (1998) and Fernandes and MacCarthy (1999), initially proposed by Stalk and Hout (1990);
- Mass Customisation, which started in 1987 with Stanley Davis in his famous book: "Future perfect" (Davis, 1987);
- Agile Manufacturing, became known and was popularised in 1991 by a group of researchers (see Goldman et al (1991)) from the Iacocca Institute, at Lehigh University, in the United States, proposing Agile Manufacturing as a new paradigm for the environment of this period and which predicted how competitiveness would develop in the following 20 years.

The Paradigms of manufacturing focuses on the concept of trade-offs, which focuses that management should try to have focus of their operation which they can work extremely well. For example, low cost can be a factory focus but it cannot be coupled with high quality, as for low-cost company will have to let go high quality.

Services and Productivity

Service industry is a very diverse industry from airlines to zoos, hence it is very difficult to make a single system as bench mark. Many companies have successfully standardized their operations, which has made sure that same quality of food or services are being provided by one branch in a city is same as that of its branch in another city. This standardization helps to keep a check on the quality and attain a goodwill among customers as well.

TQM (Total Quality Management)

Another development is the focus on TQM which basically ensure the quality of organization's output, including goods and services at every step of operation rather at the end. It is a customer focus process and ensures that all the employees work towards the common goal of improving products and services. Many awards in this category are important: for example, for outstanding quality management system Aldridge National Quality Award Which is given to the manufacturer. The ISO 9000 Certification given by international organization for standardization. Basically, it provides global quality standards for manufacturers.

BPR (Business Process Re-engineering)

In 1990's, during the economic recession, the Idea of business process re-engineering was popularized, which focused on analysis and design of workflows and business processes within an organization. The basic idea is to look at all the processes of an organization and eliminate all the steps that are non-valued added and automate the remaining steps to achieve the desired outcome.

Supply Chain Management

Today's is the time of competition of one supply chain with the other supply chain enhancing the importance of establishing long-term relationships with suppliers.

Internet Commerce

Internet has brought a revolutionary change in the world. Nowadays every information is there on web pages and search engines. The use of Web pages, search engines has changed the way people collect information and communicates. It has also made a remarkable impact on the operation system, as we can combine various systems to attain a smooth flow of operational systems.

- Historical Milestones in Operation Management

Managing the transformation of inputs into goods and services- this is what we can think of what operations management does and we can see that as a function which is as old as time. We will be able to find Operations management at work everywhere. Organizing the first Olympic games, building the Great Wall of China, or erecting the Egyptian pyramids, operations management was at work in all these great organizational efforts.

Operations management did not emerge as a formal field of study, many events helped shape operations management however, until the late 1950s and early 1960s, when scholars began to recognize that all production systems face a common set of

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problems and to stress the systems approach to viewing operations processes. Here, we will describe some of the most important of these historical milestones and explain their influence on the development of operations management. Later we will take a glance at some current trends in operations management.

The Industrial Revolution

- The Industrial Revolution brought a significant impact on the way goods are produced today. Before this revolution, products were made by hand by skilled craftspeople in their shops or homes. Each product had a unique feature, and was done with great care and thoroughness, made by one person.
- The Industrial Revolution changed all of this. In 1770s, it started with the development of a number of inventions that relied on machine power instead of human power.
- The most important invention was the steam engine, which was invented by James Watt in 1764. The steam engine provided a new source of power that was used to replace human labour in textile mills, machine-making plants, and other facilities. The Here emerged the concept of the factories.
- In addition, railroads allowed a wider distribution of goods which was one of the advances in transportation led by steam engine. The concept of division of labour was introduced at the same time. This concept would become one of the important ideas behind the development of the assembly line, as first described by Adam Smith in 1776.
- When each task of production of goods is broken into series of small, element tasks and is performed by a different worker is called as Division of labour. Due to repetition of the task, worker becomes highly specialized in that task. Division of labour allows higher volumes to be production, which when coupled with the advances in transportation of steam-powered boats and railroads, opened up even faraway and unapproachable markets.
- A few years later, in 1790 the concept of interchangeable parts was introduced by Eli Whitney. Prior to that, every part used in a production process was unique. Interchangeable parts are those that can be substitute for each other, and are standardized so that every item in a batch of items fits equally. This concept meant that we could move from one-at-a-time production to mass production, for example, in the manufacture of watches, cars, furniture, clocks, computers etc.

Scientific Management

- Near the end of 20th Century, Scientific management was an approach to management promoted by Frederick W. Taylor, is also known as Taylorism. Through scientific management he sought to improve economic efficiency, increase worker productivity and organizational output.
- His concept had two key features. First, Taylor assumed that all the workers are motivated only by money, so he introduced the idea of “a fair day’s pay for a fair day’s work”, which means if a worker did not achieve his target work in a day, then he is not eligible to get paid. Taylor believed that worker productivity is governed by scientific laws and these laws can be discovered through measurement,

analysis, and observation. Workers are to be paid in direct proportion to how much they produce and not more than that. The second feature of this approach was the separation of the planning and doing functions in a company.

- Many people disliked the scientific management approach, especially workers, who thought that management is being unfair to them and used these methods to unfairly increase output without paying them accordingly. Still, many companies adopted the scientific management approach in operations management. For example, piece-rate incentives, in which workers are paid in direct proportion to their output, came out of this movement.
- Also, Taylor introduced, stopwatch time studies, a widely used method of work measurement. In stopwatch time studies, observations are made and recorded of a worker performing a task over many cycles, this information is further used to set a time standard for performing the particular task. This method is still used today to set a time standard for short and repetitive tasks.
- Henry Ford used these techniques in his factories making Scientific management approach popular. Ford introduced moving assembly line to produce cars, along with scientific management with interchangeable parts to develop the concept of mass production. These innovations helped him increase production and efficiency at his factories.

Total Quality Management

- Total Quality Management is a term that originated in 1950's. In order to remain competitive, companies have been forced to focus on improving quality, as customers demand ever higher quality in their products and services.
- Total quality management (TQM) is a concept declared by "quality gurus"- W. Edwards Deming, which eliminates causes of product defects, aggressively seeks to improve product quality. With TQM, everyone in the company is responsible for quality. Practiced by some companies in the 1980s, TQM became prevalent in 1990s and is an area of operations management that no competitive company has been able to ignore.
- Total quality management is very crucial and its importance is demonstrated by the number of companies achieving ISO 9000 certification. ISO 9000 is a family of quality management systems developed for global manufacturers by the International Organization for Standardization (ISO) to control trade into the then-emerging European Economic Community (EEC). Today many companies require their suppliers to meet the standards as a condition for obtaining contracts, with ISO 9000 as a global set of standards.

The Computer Age

- In the 1970s the use of computers in business became extensively popular. Many quantitative models developed by management science could be employed on a larger scale with computers. With important effects in areas such as forecasting, scheduling, and inventory management, data processing became easier.
- Material requirements planning (MRP), was developed for inventory control and scheduling. In order to compute huge Inventory requirements and develop

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schedules for production of thousands of items, Material requirements planning was able to process huge amounts of data processing which was impossible before the age of computers.

- The aggressive growth in computing capabilities continues to impact operations management. Just-in-Time (JIT), developed in Japan in the 1980s, that is designed to achieve high-volume production using minimal amounts of wastes. This is achieved through coordination of the flow of materials and production of goods to meet customer demand exact in time, quantity and quality, which is a management philosophy, termed as just-in-time.
- However, JIT is much more than the coordinated movement of goods. It helps companies employs teams of workers to achieve continuous improvement in processes and organizational efficiency by minimizing all organizational waste. Though, JIT was first used in manufacturing, it has been implemented in the service sector as well. For example, in the food service industry. It is also credited with helping turn many companies around and is used by companies such as Honda, Toyota, and General Motors. JIT improves businesses and will continue to transform businesses in the future.

The Human Relations Movement

- Human Relations Movement was founded by sociologist George Elton Mayo. In early twentieth century, the scientific management movement and its philosophy dominated. Mayo did a series of experiments known as Hawthorne studies, which were conducted at a Western Electric plant in Hawthorne, Illinois, in the 1930. It focused on exploring link between employee satisfaction and workplace productivity. Its aim was to study the effects of environmental changes, such as changes in lighting and room temperature, on the productivity of assembly-line workers.
- The findings were unexpected. the productivity of the workers continued to increase irrespective of the environmental changes made. Elton Mayo, a sociologist from Harvard, came to the conclusion that the workers were actually motivated by the attention they were given. Hawthorne effect is all about workers responding to the attention they are given. The study of these findings by many sociologists and psychologists led to the human relations movement, an entirely new philosophy based on the recognition that factors other than money can contribute to worker productivity.
- This new philosophy had a tremendous impact on the development of operations management, as more and more concepts were implemented that motivate workers by making their jobs more interesting and meaningful.
- Hawthorne studies showed that scientific management had made jobs too repetitive and boring, hence, it adopted the approach of Job enlargement in which workers are given a larger portion of the total task to do.
- Another approach was adopted wherein workers are given a greater role in planning, hence giving more meaning to jobs is job enrichment. Recent studies have shown that environmental factors in the workplace, such as adequate lighting and ventilation, can have a major impact on productivity. However, this does not contradict the principle that attention from management is a positive factor in

motivation along with wide variety of factors including pay, social relationships, interests and attitudes.

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Business Process Reengineering

- Re-engineer is to redesign or restructure. Business process reengineering means redesigning a company's processes, practices or the way work is done, to increase efficiency, improve quality, and reduce costs.
- In many companies' things are done in a certain way for years, which has been passed down over the years. Often managers say, "Well, we've always done it this way." Reengineering requires asking why things are done in a certain way, questioning assumptions, and then redesigning the processes. Operations management is a key player in a company's reengineering efforts.
- To implement BPR a five-step approach is suggested to radically change the workflow:
 1. Develop business Objectives.
 2. Identify the processes to be re-designed.
 3. Understand and measure the existing processes.
 4. Identify IT levels.
 5. Develop and design a prototype of new process.
- Historical Evolution of Production and Operations Management

Operations and production management has been recognised as an important factor in a country's economic growth, for over two centuries.

The traditional view of manufacturing management came into existence eighteenth century when Adam Smith recognised the economic benefits of specialisation of labour. He recommended breaking of jobs down into subtasks and recognises workers to specialised tasks in which they would become highly skilled and efficient. In the early twentieth century, F.W. Taylor implemented Smith's theories and developed scientific management. From then till 1930, many techniques were developed prevailing the traditional view. Brief information about the contributions to manufacturing management is shown in the Table below:

Historical Outline of Operations Management

Date	Contribution	Contributor
1776	Labour Specialization in manufacturing	Adam Smith
1799	Interchangeable parts and cost accounting	Eli Whitney and others
1832	Division of labour by skills; assigning jobs by skills; basics of time study	Charles Babbage
1900	Scientific management time study and work study developed; dividing planning and doing of work	Frederick W. Taylor
1900	Motion of study of jobs	Frank B. Gilbreth
1901	Scheduling techniques for employees, machines jobs in manufacturing	Henry L. Gantt

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1915	Economic lot sizes for inventory control	F.W. Harris
1927	Human relations; the Hawthorne studies	Elton Mayo
1931	Statistical inference applied to product quality: quality control charts	W.A. Shewart
1935	Statistical sampling applied to quality control: inspection sampling plans	H.F. Dodge & H.G. Roming
1940	Operations research applications in World War II	P.M. Blacker and others
1946	Digital computer	John Mauchily and J.P. Eckert
1947	Linear programming	G.B. Dantzig, Williams & others
1950	Mathematical programming, on-linear and stochastic	A. Charnes, W.W. Cooper processes & others
1951	Commercial digital computer: large-scale computations available.	Sperry Univac
1960	Organizational behaviour: continued study of people at work	L. Cummings, L. Porter
1970	Integrating operations into overall strategy and policy, Computer applications to manufacturing, Scheduling and control, Material requirement planning (MRP)	W. Skinner J. Orlicky and G. Wright
1980	Quality and productivity applications from Japan: robotics, CAD-CAM	W.E. Deming and J. Juran

- It was when F.W. Taylor's works become popular and managers developed techniques that focussed on economic efficiency in manufacturing and Production.
- Workers were studied in great detail to eliminate wasteful efforts and achieve greater efficiency. Psychologists, socialists and other social scientists began to study people and human behaviour in the working environment.
- In addition, more sophisticated analytical approaches were contributed by economists, mathematicians, and computer socialists. With the 1970s emerges two distinct changes in our views. The most obvious of these, reflected in the new name operations management was a shift in the service and manufacturing sectors of the economy.
- As service sector became more prominent, the change from 'production' to 'operations' emphasized the broadening of our field to service organizations. The second, more suitable change was the beginning of an emphasis on synthesis, rather than just analysis, in management practices.

1.1.2 The Service Revolution

Services

Services are intangible parts of our economy, where no physical goods are transferred from seller to buyer, Services are non-physical, as opposite to goods

which we can touch or handle. Example of services are: Banking Services, medical treatments, education services.

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Services can be divided into – Modern services and traditional services.

1. Modern services are linked to information communication technology (ICT). They can be broken in a value chain just like goods. The reason is that they can be electronically transported internationally through satellite and telecom networks. There has been a constant increase in the number of services that can be transported digitally--processing insurance claims; call centres, desktop publishing; the remote management and maintenance of IT networks; completing tax returns; transcribing medical records; compiling audits; health records, and financial research and analysis.
2. On the other hand, the traditional services and require face to face interaction and are less ICT-intensive. These include transport, trade, hotel, restaurant, beauty shops, barbers, education, health and other government and community services. It is modern services that are developing rapidly thanks to the 3Ts: growing tradability, more sophisticated technology and reduced transport costs.

Can Service Sector Contribute to Growth?

For a long time, services have been the main source of growth in rich countries, and are now also a rich source in poor countries. Services have become a larger share of GDP in poor countries. There has been rapid development of modern, commercial services - business processing, finance, insurance, and communications. Modern service productivity growth, in turn, is driven by the 3T's: tradability, technology and transportability.

For many decades, this has been argued that economic development was achieved through developing a labour-intensive manufacturing sector, and Service sector has given rise to stagnant growth. Services were considered as low skilled, low wage and low productivity sector and today services are among the most dynamic sectors in the economy, even in poorer counties.

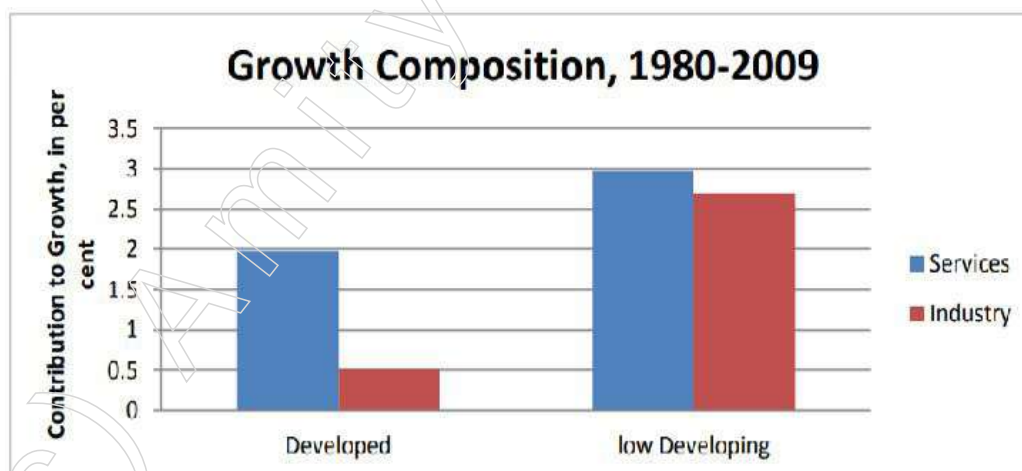


Fig: Source of Growth in Rich and Poor countries

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In the figure above, it shows the composition of services and industry to GDP growth from the year 1980 to 2009 for rich and poor countries. In both cases, the contribution of services to total growth is higher than industry's contribution. In poor countries, service have contributed more to growth than in rich countries.

Now Service industry is the fastest growing sector in global trade and more than 75 percent of global economy is now accounted by services, out of which 45 percent is developing countries. The average growth of services exports has increased from poor countries than that of rich countries. Services export is faster than goods exports. Hence, Globalization of services has enabled developing countries to look towards services as a source of growth.

Let's compare two largest growing economies:

India and China are two fastest growing economies, having different patterns of growth. In the figure below compares the share of service value added in GDP in India, China and OCED (Organization for economic co-operation and development) Countries. India has much bigger share of service sector than that of China. China has much bigger manufacturing sector and has a reputation for exporting manufactured goods, whereas, India has large services sector and has acquired its global image for exporting services.

Now, here comes in mind. Can service sector be as dynamic as manufacturing sector? Can service-led-growth contribute to job creation and poverty reduction?

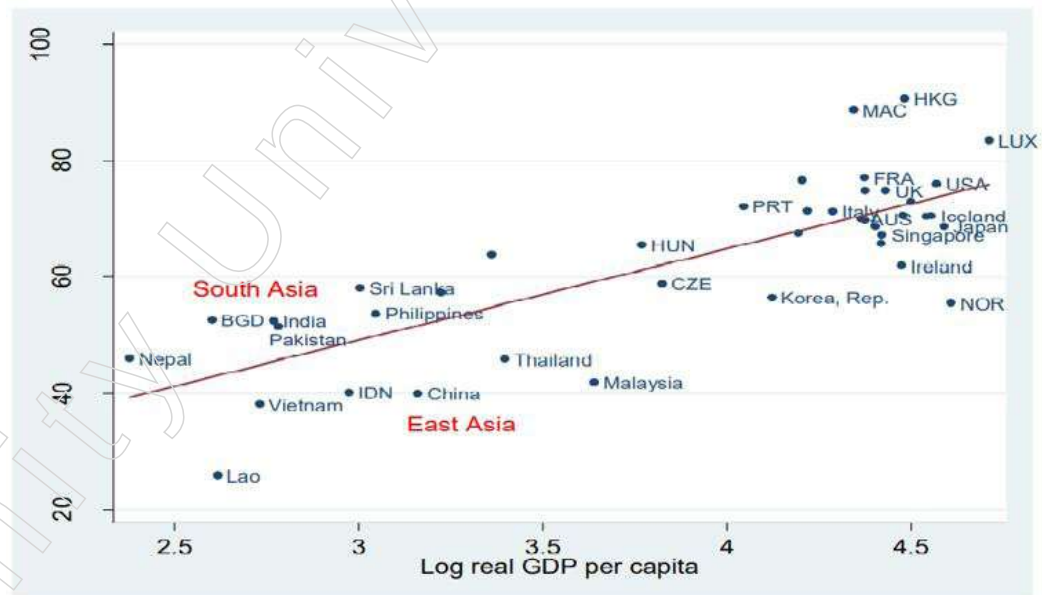


Fig: Comparing the contribution of manufacturing and services sectors to GDP in China and India

Can Service Sector be Dynamic and Contribute to Growth?

- Services has been main source of growth in rich countries and poor countries. Larger share of GDP is of services in poor countries and productivity growth in services have exceeded in most of the poor countries. This is largely explained by the rapid development of modern, commercial services—finance, insurance, business processing and communications. Modern service

productivity growth, in turn, is driven by the 3T's: tradability, technology and transportability.

- In developing countries, the share of world service exports has increased from 14 percent in 1990 to 21 per in 2008. The average growth of service exports from poor countries has exceeded that of rich countries during the last two decades. There has been an accelerated growth in service exports than goods exports. In brief, the globalization of service has enabled developing countries to tap into services as a source of growth.
- India's growth pattern in the 21st century is remarkable as it contradicts a seemingly iron law of development. This law—which is now conventional wisdom— states that industrialization is the only route to rapid economic development for developing countries. It goes further to say that as a result of globalization the pace of development can be explosive. But the potential for explosive growth is distinctive to manufacturing only (UNIDO 2009). This is no longer the case.

Role of Service-led Growth in Poverty Reduction and Job Creation

Growth in service sector is directly correlated with poverty reduction than growth in agriculture. Service sector growth has led to reduction in both urban and rural poverty rates. With an increase in their service sector share, some states like Andhra Pradesh, Kerala and Tamil Nadu, have experienced a significant decrease in urban poverty.

Services contribute to poverty reduction via two channels. Directly, they provide the largest source of new job growth. Indirectly, they provide the income that, when spent, drives further demand for goods and services and jobs to produce these.

Some estimates suggest that the indirect effect of a growing service sector can be larger than the direct effect, in addition to direct job creation. For instance, India's IT industry association NASSCOM estimates that if one job is created in the IT sector, four additional jobs are created in the rest of the economy due to high levels of consumption spending by professionals employed in this sector.

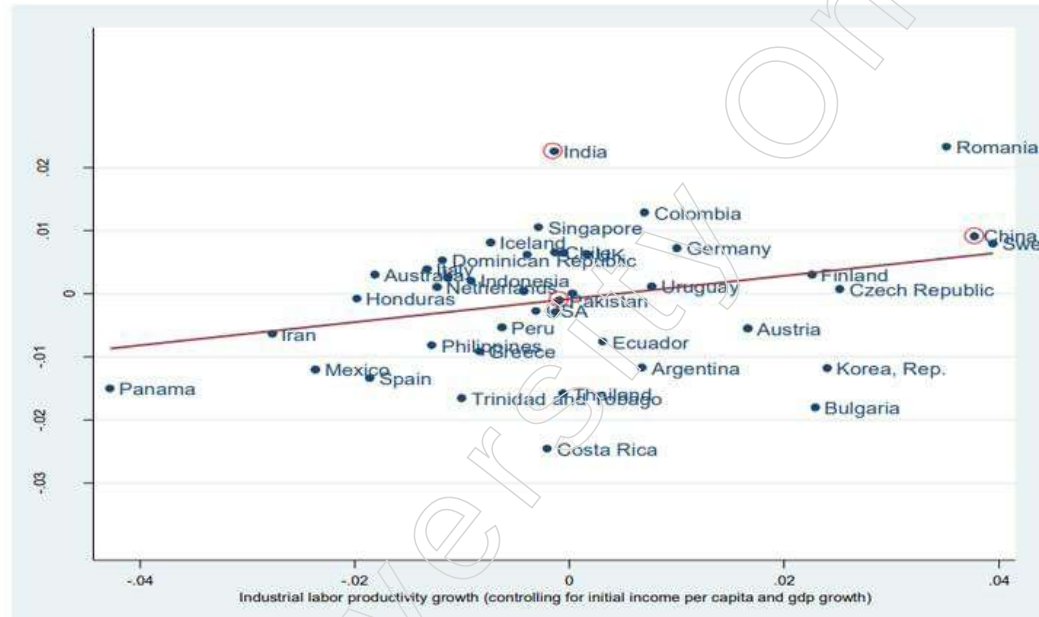
In service sector, wage growth has been considerably higher than in manufacturing and agriculture in recent years in India. Service sector wages in utilities, trade, transport, and even rural finance improved, while manufacturing wages fell in the early 2000s in both rural and urban areas despite rapid economic growth. In fact, in many sectors, rural wages may have increased faster than in urban areas, possibly reflecting the rising rural-urban migration over time that is taking place in India.

The internal rural-urban migration and links between rural and urban labour markets, allows the modern impersonal service sector in India to contribute to overall poverty reduction, even though modern services are concentrated in urban areas.

The question here is that: Is the service sector more productive in India than in China, and manufacturing sector more productive in China compared to India? The Figure given below compares the service labour productivity and industry labour productivity for a large sample of countries, after controlling for initial GDP per capita and GDP growth. India is way above the line. It shows much higher labour productivity in the service sector compared to manufacturing industry. On the other hand, China shows much higher labour productivity in manufacturing industry compared to services.

Notes

There are two tipping points that are driving higher service labour productivity growth rates. First, as low-income countries have the benefit of low wage rates, scale economies, and knowledge spill overs, like India, are coming up with the global production possibility frontier in services. In essence, the service sector in India has behaved like the manufacturing sector in China. Second, high income countries like Singapore, Sweden, United Kingdom, and United States are pushing the global production possibility frontier in services through innovation.



Lowest female labour participation rates are what currently South Asia suffers from. In countries like in India, Pakistan and Sri Lanka, only around one-third of all women of working age are actually working or looking for work. Internationally, countries with high employment in services tend to have the highest participation of women in the labour market. The development of service industries, therefore, brings new workers into the labour force, making the contribution to aggregate growth even larger.

The employment of women plays an important role in poverty reduction. Households where women have jobs their incomes are significantly higher. This enhanced economic status for women in turn reduce the number of children per household and also drive higher levels of education and health care for those children, resulting in an increase in household savings and the ability to accumulate assets that generate additional income, hence increasing the standard of living.

Can Service-led Growth be Sustained?

- Yes, it can. Service is the largest sector in the world, accounting for more than 70 percent of global output. The Service Revolution has altered the characteristics of services. Services can now be produced and exported at low cost. The idea of services being non-transportable, non-tradable, and non-scalable no longer holds and is old for a host of modern impersonal services that are moved across borders over the internet, digitized and stored electronically, and scaled into giant global businesses.

- Developing countries can sustain a service led growth as there is a huge room for catch up and convergence. Technological change in ICT has progressed at a dizzying pace. The range of business processes that can be digitized and traded globally is constantly expanding: processing insurance claims; desktop publishing; the remote management and maintenance of IT networks; compiling audits; completing tax returns; transcribing medical records; financial research and analysis. The list of possible activities is almost endless.
- And, poor countries take advantage of this trend. India has been the first one, but many other poor countries are also finding it easier to generate productivity growth in services than in industry. A strong telecommunications backbone is what Modern services need. In service exports use of the internet, personal computers and telephone lines are all independently significant in-service exports.
- Now an active component of industrial policy in many developing countries is promoting FDI selectively in telecom sectors (Alfaro and Charlton, 2007). Globalization of services provides many opportunities for late-developing countries to find niches, beyond manufacturing, where they can be successful. Taking advantage of these opportunities requires a government that energetically takes steps to accelerate services growth, through a variety of policies. Services may provide the easiest and fastest route out of poverty for many poor countries. India's development experience offers hope for the late comers to development.

1.1.3 Production and Operation Management Definition

- Production Management

Definition

Production management may be defined as, the processes of effectively planning, coordinating and controlling the production, that is the operations of that part of an enterprise, and the performance of the management activities with regards to selecting, designing, operating, Controlling and updating production system. It means to say that production and operations Management is responsible for the actual transformation of raw materials into finished products. (iv) Production management is defined as management function which plans, organizes, coordinates, directs and controls the material supply and Processing activities of an enterprise, so that specified products are produced through specified methods to meet an approved sales programme. These activities are being carried out in such a manner that Labour, Plant and Capital available are used to the best advantage of the organization.

- Operations Management

Definition

There are 2 ways in which operation management is viewed – Traditional View and Modern View:

Notes

1) Traditional View – Transformation Approach

Operation management is transformation system, this is what the traditional definition considers. According to this view - Operations management involves the systematic direction and control of all the processes that transform resources, which includes inputs into finished goods or services, which involves output, for customers or clients. A transformation process is any group of activities that takes one or more inputs, transforms and adds value to them, and provides outputs for customers or clients. Where the inputs are raw materials, it is relatively easy to identify the transformation involved. For example: when milk is transformed into cheese and butter. And, where the inputs are information or people, the nature of the transformation may be less obvious. For example, a hospital transforms ill patients (the input) into healthy patients (the output).

2) Value Driven Approach

The second approach is a value drive approach to operations management. The value driven approach starts by recognizing businesses as a set of processes each of which has inputs outputs, and structure. Each process has a number of jobs to do and each process should be measured on the basis of how effective it is in achieving the desired outcomes, with least wastage.

This approach is based on the four core business processes describing the functioning of a business organization. There can be many more core processes depending upon the business and how it is structured.

- a) Understanding Customer needs
- b) Develop strategy to develop product
- c) Secure materials to satisfy demand
- d) Managing Strategic planning processes
- Production and Operations Management

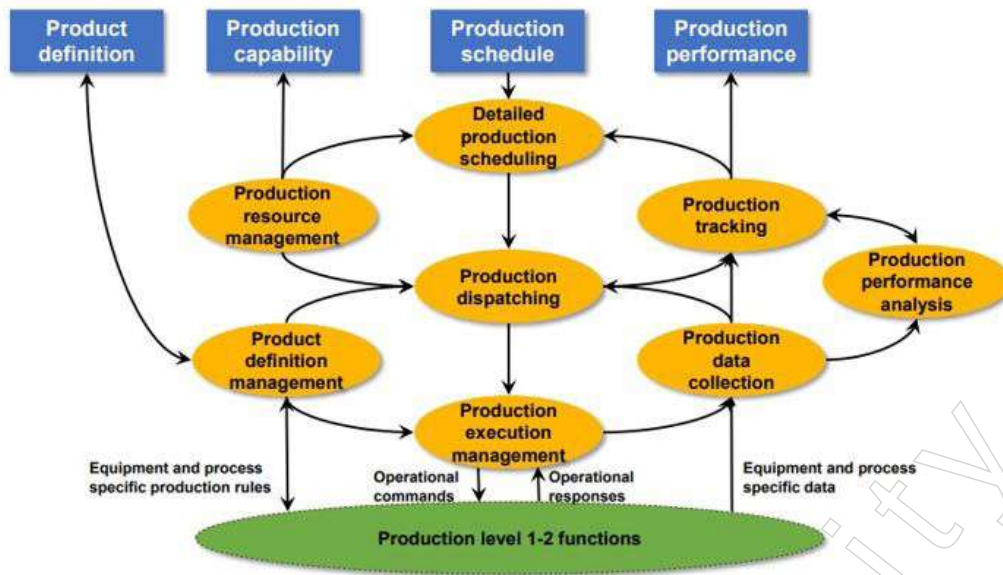
Definition

It is defined as the design, operation, and improvement of transformation process, which converts the various inputs into the desired outputs of product and services according to the specifications in the amount and by scheduled demand at a minimum cost and as per the policies of the organization.

The Production function directly affects customer experience and there by future of organization itself. With an aim to add value to product or service which will create a strong and long-lasting customer relationship or association, production function provides insights to real product needs of customers.

Operations management works for an organization, in order to deliver value to customers it is essential to Identify the customer needs and convert that into a specific product or service (numbers of products required for specific period of time). And, based on product requirement do back-ward working to identify raw material requirements. And, engage internal and external vendors to create supply chain for raw material and finished goods between vendor → production facility → customers.

Production Operations Management



Notes

Check Your Understanding

1. The father of modern economics - Adam Smith wrote:
 - a) The wealth of nations
 - b) Poor Economics
 - c) Why Nations Fail
 - d) Good Economics for hard Times
2. Which process did Henry Ford used to break his manufacturing processes?
 - a) Consumer Needs
 - b) Quality Control
 - c) Planning
 - d) Assembly Line
3. Manufacturing Strategy Paradigm is also known as:
 - a) Mass Manufacturing
 - b) Current Mass Manufacturing
 - c) Lean Manufacturing
 - d) Agile Manufacturing
4. Responsive Manufacturing was proposed by:
 - a) Kritchanchai and MacCarthy
 - b) Fernandes and MacCarthy
 - c) Stalk and Hout
 - d) Lehigh University
5. Steam Engine was invented by:
 - a) Frederick W. Taylor

Notes

- b) James Watt
 - c) Henry Ford
 - d) Edwards Deming
6. Redesigning a company's processes, practices or the way work is done, to increase efficiency, improve quality, and reduce costs is known as:
- a) Operations Management
 - b) Business Process Reengineering
 - c) Production Management
 - d) All the above
7. Operations Management has two views. These are:
- a) Strategic and Planning View
 - b) Traditional and Modern View
 - c) Capability and Scheduling
 - d) Production and Performance

Summary

- Operations Management helps concentrating on the core businesses, squeezes out the waste and focuses on differentiating between competitors in a meaningful way.
- The very essence of any business is to understand needs of customer and provide the services and goods to fulfil their needs, and in process create value for customers and solve their problems.
- According to Smith, by dividing their tasks workers can produce their products more efficiently, than if the same number of workers each built product from start to finish.
- Technological developments created new possibilities for the managers looking to improve their operations. The development of computational technology in specific, allowed for a greater degree of data to be analysed by firms leading to efficiencies.
- Quality management systems are very popular as it provides guidance for mapping, improving and monitoring operations processes. A variety of quality management systems, like the ISO systems and Six Sigma are in use among top firms, with an aim to increase the efficiency of business processes.
- Service industry is a very diverse industry from airlines to zoos, hence it is very difficult to make a single system as bench mark.
- The Industrial Revolution brought a significant impact on the way goods are produced today. Before this revolution, products were made by hand by skilled craftspeople in their shops or homes
- Railroads allowed a wider distribution of goods which was one of the advances in transportation led by steam engine.
- Material requirements planning (MRP), was developed for inventory control and scheduling. In order to compute huge Inventory requirements and develop schedules for production of thousands of items.

- Services are intangible parts of our economy, where no physical goods are transferred from seller to buyer, Services are non-physical, as opposite to goods which we can touch or handle.
- Services has been main source of growth in rich countries and poor countries. Larger share of GDP is of services in poor countries and productivity growth in services have exceeded in most of the poor countries.
- India's growth pattern in the 21st century is remarkable as it contradicts a seemingly iron law of development.
- In service sector, wage growth has been considerably higher than in manufacturing and agriculture in recent years in India.
- The Production function directly affects customer experience and there by future of organization itself. With an aim to add value to product or service which will create a strong and long-lasting customer relationship or association,
- Operations management for an organization, in order to deliver value to customers it is essential to Identify the customer needs and convert that into a specific product or service (numbers of products required for specific period of time).

Activity

1. Ques: Design an organizational structure with necessary structure and design for product manufacture and services. Visit Mother Dairy or any reputed organization and understand their Structural and operational processes and activities they pursue to deliver their products and services.

Questions

1. What is difference between India and China's Economy? Can services contribute to growth and leads to Poverty reduction and job creation? And, how this can be sustained? Compare the two largest growing economies and state that how growing economy of India contradicts the "iron law of development".
2. What is the history of Operations Management? Describe the historical milestones in operations management? Who is the Father of Operations management and what was his contribution in bringing revolution in the industry?

Glossary

- Operations Management: It helps concentrating on the core businesses, squeezes out the waste and focuses on differentiating between competitors in a meaningful way.
- Just-In-Time: It is designed to achieve high volume of production using minimum inventory cost wherein the parts or raw materials required arrives from the suppliers at the work station exactly when needed along with the total Quality Control; aggressively trying to remove the causes of production defects.
- Total quality management: TQM is a concept declared by "quality gurus"- W. Edwards Deming, which eliminates causes of product defects, aggressively seeks to improve product quality.
- Business Process Reengineering: The Idea of business process re-engineering was popularized, which focused on analysis and design of workflows and business processes within an organization.
- Material requirements planning: (MRP), was developed for inventory control

Notes

and scheduling. In order to compute huge Inventory requirements and develop schedules for production of thousands of items, Material requirements planning was able to process huge amounts of data processing which was impossible before the age of computers.

- **Services:** These are intangible parts of our economy, where no physical goods are transferred from seller to buyer, Services are non-physical, as opposite to goods which we can touch or handle.
- **Production management:** It may be defined as, the processes of effectively planning, coordinating and controlling the production, that is the operations of that part of an enterprise, and the performance of the management activities with regards to selecting, designing, operating, Controlling and updating production system.
- **Traditional View:** According to this view - Operations management involves the systematic direction and control of all the processes that transform resources, which includes inputs into finished goods or services, which involves output, for customers or clients.
- **Value Driven Approach:** The value driven approach starts by recognizing businesses as a set of processes each of which has inputs outputs, and structure. Each process has a number of jobs to do and each process should be measured on the basis of how effective it is in achieving the desired outcomes, with least wastage.
- **Production and Operations Management:** The Production function directly affects customer experience and there by future of organization itself. Operations management works for an organization, in order to deliver value to customers it is essential to Identify the customer needs and convert that into a specific product or service (numbers of products required for specific period of time).

Further Readings

1. Production and Operation Management - S. Anil Kumar, N.Suresh
2. Production and Operation Management – Gowtham Kumar
3. Satyabhama – School of Management Studies
4. Production and Operation Management - Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra

Check your Understanding

1. a)
2. d)
3. b)
4. c)
5. b)
6. b)
7. b)

Unit - 1.2: Factors Affecting Operations Management

Notes

Objectives

In this unit, you will

- Learn Operations Management
- Learn about Production
- Understand Classification by Production System
- Identify Ways of production
- Explain Importance of higher productivity
- Learn Models of Production system
- Understand Factors Affecting Production
- Learn Factors Affecting Operations Management
- Explain Competitive Priorities in operations management

Introduction

Operations management (OM) is the administration of business practices to create the highest level of efficiency possible within an organization. It is concerned with converting materials and labour into goods and services as efficiently as possible in order to maximize the profit of an organization. Operations management teams attempt to balance costs with revenue to achieve the highest net operating profit possible.

It involves utilizing resources from staff, materials, equipment, and technology. Goods are acquired, developed, and delivered to clients based on client needs and the abilities of the company.

various strategic issues, including determining the size of manufacturing plants and project management methods and implementing the structure of information technology networks are also handled by operations management. Some of the other operational issues include the management of inventory levels, which includes work-in-process levels and raw materials acquisition, materials handling, maintenance policies and quality control.

Production

A process which involves transformation of raw material (input) into desired product or service (output) by means of process and of resources, to meet the consumer's demand is Production.

According to Elwood Butta, "Production is a process by which goods or services are created." And, to deliver value for customers in products and services, it is essential for the company to do the following:

1. Identification of the customer's needs and convert them into a specific product or service (i.e., numbers of products required for specific period of time).
2. Identifying the need of raw material required to meet the product requirement.

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3. Engage various vendors, internally and externally to create supply chain for raw material and finished goods between vendor → production facility → customers.

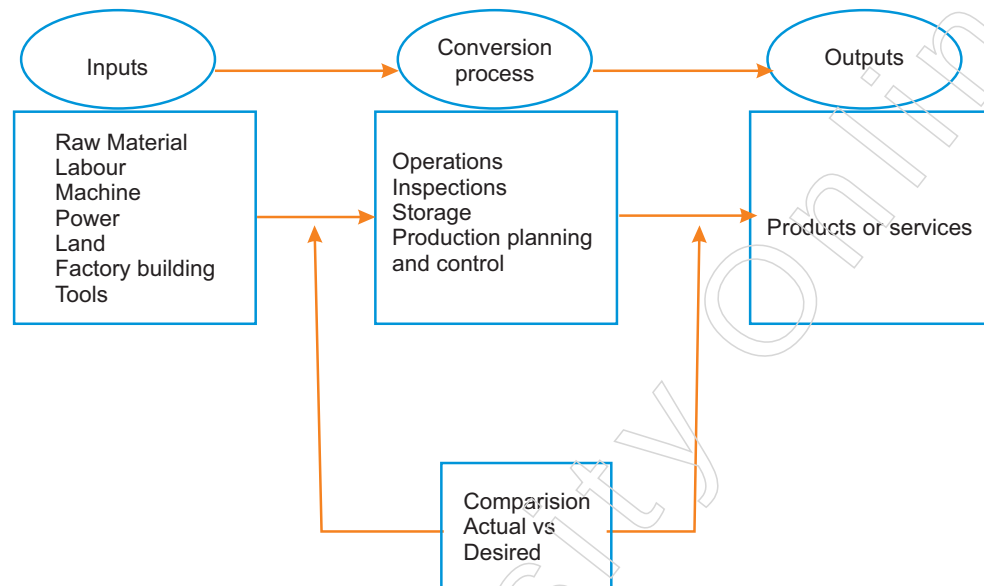


Fig: Production System

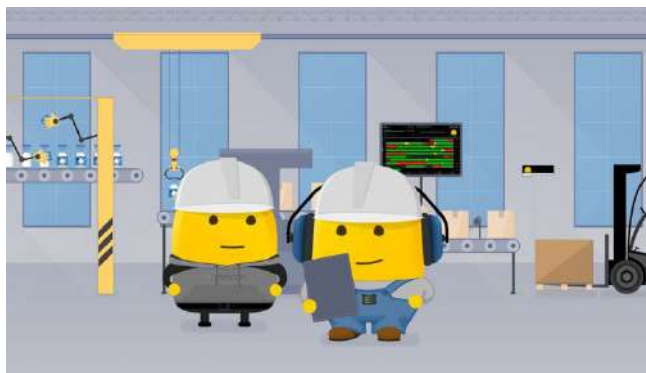
Production System

The Production System of an organization which involves activities that combines flowing resources within a defined system and transform them in a controlled manner to add value in accordance with the policies communicated by the management. A simplified Production system is shown in the above figure.

Production system has following characteristics:

- Production is an organized activity, so every production system has an objective.
- It provides useful outputs from various inputs.
- This system works with all other organisation system as it cannot operate in isolation.
- There is always a feedback system, which is essential to improve and control the system performance.

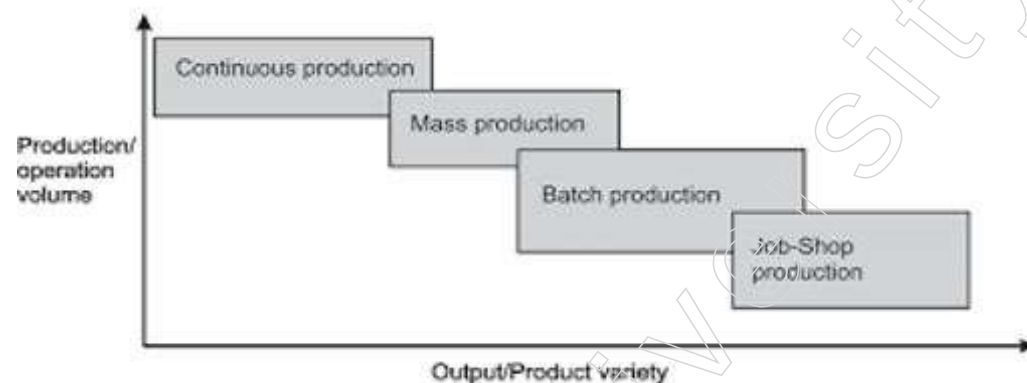
Ways of Production



- **Production by Disintegration:** In this the desired output is achieved by separating the contents from raw materials. Example: fuel oils, crude oils etc.
- **Production by Integration:** It involves production by assembling various components or parts of the products to get desired product. Example: two-wheelers, four-wheelers etc., in which various parts required are assembled and final product is manufactured.
- **Production by Modification or Improvement:** This involves change in chemical and mechanical parameters of the raw materials of input without making changes in the physical attributes of the raw material.

For Example: Annealing process (heating at high temperatures and then cooling), is example of production by modification or improvement.

Classification by Production System



- **Job Shop Production** – It refers to the manufacturing facility in which a variety of custom products are made in small batches or few quantities of products designed and produced, as per the specification of the customers within prefixed time and cost. Each job demands unique technological requirements, demands processing on machines in a certain sequence. This system may be further classified as.
 - (i) **The Job produced only once:** In this the customer visits the firm, places his order and takes delivery of the product and leaves. He may not come again and book the order for the same product. The firm has to plan for material, process and manpower only after receiving the order from the customer and there is no scope of pre-planning of the product.
 - (ii) **The job produced at irregular intervals:** Here, the customer visits the firm to place orders for the same type of the product at irregular intervals and the firm will not have any idea of customer's visit. Here also planning for materials, process and manpower will start only after taking the order from the customer.
 - (iii) **The Jobs Produced periodically at regular intervals:** In this system, the customer arrives at regular intervals at the firm to place orders for the same type of product and the firm knows very well that the customer visits at regular intervals. Therefore, it can plan for materials, and process and manpower and have them in a master file.

Notes

Advantages

Following are the advantages of Job-shop Production:

1. Varieties of products can be produced, because of general purpose machines and facilities.
2. As each job gives learning opportunities, operators become more skilled and competent.
3. The potential of the operators can be utilized to its fullest.
4. Due to challenging environment, opportunity exists for Creative methods and innovative ideas.

Limitations

Following are the limitations of Job-shop Production:

1. Cost is very high due to frequent set up changes.
 2. Higher inventory cost due to higher level of inventory at all levels.
 3. Production planning is complicated.
 4. Space requirement is large.
- Batch Production – In this manufacturing process, the components or goods are produced in groups or batches and not in continuous stream. A limited number of products are produced at regular intervals. In this, a setup is used for the production of item in a batch and change of set up is done as per the requirement for processing the next batch. As compared to job order production, manufacturing lead-time and cost are also lower. This system also can be classified under three categories.
 - (i) A batch produced only once: In this the customer places the order of his specification and the size of the order is greater than that of job production order. The firm will have to plan for the resources or acquire resources after taking the order from the customer.
 - (ii) A Batch produced at irregular intervals: The customer places the order whenever needed or need arises. As the frequency of order is irregular, the firm can maintain a file of its detailed plans and it can refer to its previous files and start production.
 - (iii) A Batch Produced periodically at known Intervals: In this, the firm either receives order from the customer at regular intervals or it may produce the product to satisfy the demand. Hence, it will have a well-designed file of its plans, material requirement and instructions for the ready reference. As the frequency of regular The Batch Production System becomes Mass Production System as the frequency of regular orders goes on increasing.

Advantages

Following are the advantages of Batch Production:

1. Better utilization of Plant and machinery.

2. Functional specialization is promoted.
3. Unlike job order production, cost per unit is lower.
4. Investment in plant and machinery is less.
5. Flexibility to accommodate and process number of products.

Limitations

Following are the limitations of Batch Production:

1. Due to irregular and longer flows material handling is complex.
 2. Production planning and control is complex.
 3. Work in process inventory is higher compared to continuous production.
 4. Due to frequent changes in set up, the cost of set up is high.
- Continuous Production: Continuous Production system is the specialized manufacture of identical products on which the machinery and equipment is fully engaged. Where there are large quantities produced due to high rate of demand is normally associated with continuous production. Hence the advantage of automatic production is taken. This system is classified as:
 - (i) Mass Production: In this, same type of product is produced to meet the demand of an assembly line or the market. This production system requires good planning for material, process, maintenance of machines and instruction to operators. Flow of materials, components and parts is continuous and without any back tracking, hence, materials are produced in bulk, therefore, Production planning and control is easy.
 - (ii) Flow Production: The type of product and its relation to the plant are the differences between Mass and Flow Production. In Mass Production large number of products are produced which are identical in nature. If the demand falls or ceases then the machinery and equipment, after slight modifications or alterations be used for manufacturing products of similar nature. In flow production, the plant and equipment are designed for a specified product. Therefore, if the demand falls for the product or ceases, the plant cannot be used for manufacturing other products, it will be a scrap.

The examples for the above discussed production system are:

- (i) Job Production Shop: Tailor shop; cycle and other vehicles repair shops etc.
- (ii) Batch Production Shop: Readymade dresses companies, Cosmetics manufacturing companies etc.
- (iii) Mass Production Shops: Components of industrial products.
- (iv) Flow Production: Car assembly plant, Sugar factory, Oil refineries etc.

Advantages

Following are the advantages of Continuous Production:

1. Standard product and process sequences.
2. Rate of production is high with reduced cycle time.

Notes

3. Due to Line balancing, capacity utilization is high.
4. Material handling is completely automatic; hence, manpower is not required.
5. Due to high volume of production, unit cost is low.

Limitations

Following are the limitations of Continuous Production:

1. There is no flexibility to accommodate and process number of products.
 2. Requires very high investment for setting flow lines.
 3. Limited Product differentiation.
- Mass Production: Mass Production Manufacture of discrete parts or assemblies using a continuous process are called Mass Production. A system having very large volume of production comes under this category. The machines are arranged in a line or product layout. Product and process standardization exists and all outputs follow the same path.

Advantages:

1. Reduced cycle time and higher rate of production.
2. Higher capacity utilisation due to line balancing.
3. Less skilled operators are required.
4. Low process inventory.
5. Low manufacturing cost per unit.

Limitations:

1. If one machine breakdown then entire production line will stop.
2. There will be major change in Line layout, with the changes in the product design.
3. High investment is required in production facilities.
4. The cycle time is determined by the slowest operation.

Importance of Higher Productivity:



The Benefits derived from higher productivity are as follows:

- A higher productivity helps to improve the profits by cutting cost per unit.
- As a result of higher productivity, profits are high. These profits can be transferred to the consumers in form of lower priced Products or better-quality products.
- The profits earned from higher productivity can also be shared with workers or employees by paying them at higher rate.
- Higher profits to entrepreneur, due to higher productivity can be re-invested in meeting future financial needs of the concern.
- More employment opportunities would be generated.
- Overall productivity reflects the efficiency of production system.
- It leads to higher per capita income.
- The same output is produced with lesser input.
- More output is produced with more input.
- The proportional increase in output is more than the proportional increase in input.

Models of Production System

A model is an informative representation of reality that captures the essential features of an object/system/process. There are 3 types of models such as physical, schematic and mathematical.

- I. Physical Model: Physical model is replica of a physical object with a change of scale.
 - a. For big/huge structure of physical object: small scale (Ex. solar system)
 - b. For microscopic objects: magnified scale (Ex. Atomic model)
- II. Schematic Model: These are 2-D models which represents
 - Price fluctuations with year.
 - Symbolic chart of activities in sequence for a job.
 - Maps of routings
 - Networks of timed events. The pictorial aspects are useful for good demonstration purposes.
- III. Mathematical Model: Formulas and equations have long been the servants of physical sciences. The important aspects of a system/problem can be represented in mathematical form by using variables, parameters and functions. This is called mathematical model. We can learn how the real system will behave under various conditions, by analysing and manipulating the mathematical model.

Factors Affecting Production

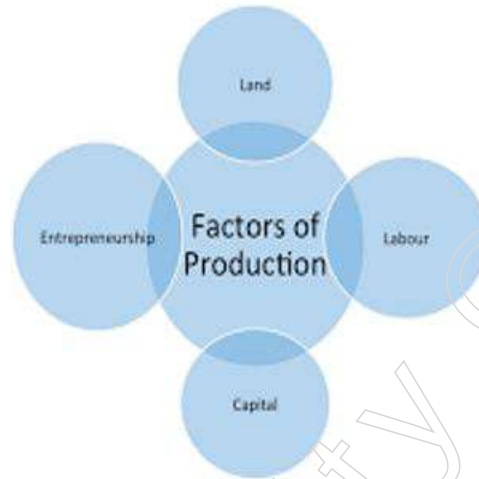
Objectives:

After this lesson, students will be able to:

- Tell what are factors of Production?

Notes

- Describe the resources for Production.
- Understanding of the key concepts of production.



Factors of Production

A very important economic concept which outline the elements needed to produce a good or service for sale is factors of production. They are commonly broken down into four elements: land, labour, capital, and entrepreneurship. However, labour and capital are sometimes referred as the two primary factors of production, depending on specific circumstances, at times one or more factors of production might be more important than the others.

Economic growth comes with increasing the quality and quantity of factors of production. Improved economic growth raises the standard of living, by lowering the cost of production and increasing wages. Factors of Production are the building blocks of an economy.

As we can understand how the least cost combinations of factors are obtained by a firm. The theory of cost of production depends the combination of factors employed. From the point of view of cost of production, factors of production are divided into Fixed and Variable Factors. Fixed factors are those whose cost do not change with change in output and variable factors are those whose costs changes with change in output. In short run a firm can cover the cost of variable factors at least and will continue to produce but in long-run it must cover the cost of production of both fixed and variable factor.

Factors of production are also divided into divisible and indivisible factors. When their inputs can be adjusted to the output, then the factors are said to be divisible. Labour is divisible as the number of labourers may be reduced in keeping with the output of the firm. Divisible factors lead to the “economies of scale” for a firm by adjusting the number of factors to the output of the firm.

Indivisible factors are those which are available in minimum sizes, and are lumpy, such as machines, entrepreneur, etc. They also lead to economies of scale, but at a faster pace.

Following are the four Factors of Production: Land, Labour, Capital and Entrepreneurship.

Notes**Land**

- “Land” as a factor of production refers to all natural resources and is quite a broad category. These resources are gifts which are given by nature. Some typical examples of natural resources are water, oil, copper, natural gas, coal, and forests.
- The use of land can range from land used for agriculture to that used for commercial real estate, as well as the natural resources derived from land. These resources are renewable, such as forests, or non-renewable such as oil or natural gas.
- The income which is earned from land or other such natural resources is called rent. In spite of the fact that, land is undeniably crucial to most forms of production, just how essential it really differs and depends on the industry. For instance, land is a very crucial and of central focus to all agriculture but is very much less important to a tech company that is quite literally operating in the virtual sphere.
- “Land was designated as the origin of economic value by the physiocrats, a collection of French economists who came before the better-known classical political economists (Smith, Ricardo, Marx, and others).”

Labour

- Labour, is also an important factor of production and involves any human input. Any contribution or work done by people contributing to production is Labour. The workforce's skills, education, and motivation level describe the quality of labour. In general terms, the higher the quality of labour, there will be a more productive the workforce. According to early, influential political economists, labour was considered to be the main source of economic value.
- Examples of labour range from the very physical to primarily mental work that goes into production. On the mental side, are the laborers like artists producing art, or programmers creating software. On the physical side of labour can be food service workers, construction workers, or factory workers.
- What you earn or get paid, when you contribute as a labour resource to the production of goods and services is wages. This is the income earned by the Labour resource. It is the largest source of income for most people.
- In production, wages are paid on the basis of time invested as well the level of skills they have. Laborers with a good training and education are considered to be “highly skilled” and are paid higher wages than less trained workers. These skilled or highly-trained workers are described as “human capital”.

Capital

- Here capital refers not to money but to the capital goods or man-made resources such as tools, infrastructures, used in production of other goods and services. They are used in commercial production, is what separates them from more widely used consumer goods. In contrast to the economists of the past, who described land or labour as the main source of value. The modern, mainstream/neoclassical economists typically consider capital to be the main source of value.
- Some other examples of capital include hammers, forklifts, conveyor belts, computers, doctor's MRI machines and delivery vans. Other than this kind of

Notes

machinery; office furniture like conference tables and desk chairs also fall under the umbrella of capital. An increase in capital goods means an increase in the productive capacity of the economy and the income earned by owners of capital resources is called interest.

- Through this interest more goods can produce and the standard of living rises. Ever-improving capital is what is needed, and is important because of what follows its production. Personal and private capital is distinct from the capital we are describing here. For instance, your personal vehicle is not a capital good in this sense; however, a taxi or other vehicle used in some form of business is considered to be a capital good.
- It is important that when companies need to prevent falling profits, they may spend less money on capital to compensate. On a contrary, when companies and, more broadly, economies are growing, they are likely to spend on capital goods so as to be able to raise levels of production.

Entrepreneurship

- An entrepreneur is someone who combines all the other factors of production, who takes on the economic risk involved in bringing the other factors of production together in order to produce a good or service for consumers. For an economic growth at all scales entrepreneurs plays an important role, helping to build many of the largest firms in the world as well as some of the small businesses in your neighbourhood.
- Governments always do their best to encourage entrepreneurs, using the right combination of policies to make starting a business accessible, as they help to contribute to economic growth. The payment an entrepreneur receives is referred to as profit, and functions as a reward for the risk they take.
- Most entrepreneurs start small businesses, not the big companies that we all know the names of. Out of six million companies in the United States, a total of 5.8 million are small businesses and 65 percent of new jobs are created by these small businesses. "Entrepreneurs can acquire the money for their new small businesses by issuing what's known as an initial public offering on the stock market, with shares of small-cap stocks."

Operations Management

Operations management (OM) is the administration of business practices to create the highest level of efficiency possible within an organization. It is concerned with converting materials and labour into goods and services as efficiently as possible in order to maximize the profit of an organization. Operations management teams attempt to balance costs with revenue to achieve the highest net operating profit possible. It involves utilizing resources from staff, materials, equipment, and technology. Goods are acquired, developed, and delivered to clients based on client needs and the abilities of the company.

various strategic issues, including determining the size of manufacturing plants and project management methods and implementing the structure of information technology networks are also handled by operations management. Some of the other operational issues include the management of inventory levels, which includes work-in-process

levels and raw materials acquisition, materials handling, maintenance policies and quality control.

Skills Required of an Operations Manager

The skills required for effective and efficient operations management and those needed in both logistics and supply chain management runs parallelly. Excellent organizational ability is very important in successfully enhancing efficiency and driving productivity as an operations manager.

In order to get the flow seamlessly, one must be able to understand the series of processes within a company, and in this sense the role is directly related to supply chain management. Meanwhile, in setting up these processes in practice represents logistics, the coordination is involved; The combination of understanding and coordinating the work of a company are central to becoming a successful operations manager.

1.2.1 Factors Affecting Operations Management

- Global Competition

This refers to products or services offered by competing companies that serve international customers. Global competition also affects the operation management and decision making of the businesses. For example: Marks and Spencer Ltd. - a multinational retailer, they have variety of clothing options offered for their respective customers but other competitions worldwide affecting the customer base of the organization (Barracough, 2016). Hence, other than online, they must focus on the Asian market and other markets to grab more customers and attract them to compete globally.

- Material and Capital Security

Capital and raw materials are also the most crucial ingredients of operation management. The organization must keep a close observation to secure them and ensure the proper utilization of the materials.

- Quality, Customer Care and Expenditure Challenge

Customers want a quality product with better service and affordable cost. But maintaining three of them can maximize the cost of production. Hence, providing customers with better product and service quality is a big challenge for an organization.

- Social Responsibilities

Each company needs to perform social welfare for the local community and maintain good relations with the society. Because any social issue can lead to damage the goodwill of the company's business condition. By contributing towards the society and by helping society to grow in certain manner and showcasing that as well, can improve its relation with the society.

- Expansion of Technology Advancement

Technology is an important factor to make various processes of an organization like manufacturing and processing of operations easier and convenient for the employees of the organization. The technology advancement also helps an to target new markets,

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customer base also resulting in its expansion. Any organization must make their technological advancement decision very carefully and to use them in a very specific way to make the best use of it.

- **Legal Considerations**

An organization needs to be concerned about their legal rules and regulations in various sections like tax, export-import, vat, tariffs, and government policies. Any disobedience legally, can affect the business in a negative way and hamper the business conditions. So, the decisions should be taken with the proper observation of the current legal situations whenever and wherever necessary.

- **Business Ethics**

Business ethics should be followed. These are the moral principles that act as guidelines for the way a business conducts itself. A business must follow each and every primary business ethics of the business environment and must practice their business operations by maintaining the code of ethics so the business policies are not hampered with any adverse condition.

Competitive Priorities in Operations Management

In operations management, competitive priorities are an important variable for operations managers. Strategic focus is provided on building specific manufacturing capabilities, by Competitive priorities that can improve a plant's position in the market. Decisions with regards to the capacity, technology, production process, planning, control, etc. are guided by strategic focus. The level of consistency between emphasised competitive priorities and matching decisions concerning operational structure and infrastructure, are dependent on the strength of an operations strategy.

According to Porter, cost leadership or differentiation, are two primary routes for competitiveness. In contrast, according to Leong, there are actually five dimensions of competitive priorities within manufacturing: cost, flexibility, quality, delivery performance and innovativeness.

We have to make trade-offs within competitive priorities. No business can normally be on top in all aspects, in a competitive market. The purpose of the priorities is to assist in the mission of positioning an organization with an advantage vs its rivals.

Five Competitive Priorities

Competitive Priority	Manufacturing Implications
Cost	Monitoring costs, and distribute it effectively on the different products.
Quality	Making sure that the product and its process perform and comply with specification.
Flexibility	Capability to manage volume and product mix changes, equipment and workforce-related
Delivery performance	Reliable and quick delivery of goods, stock availability.
Innovativeness	Time-to-market, how quickly new products or processes can be implemented.

On a “high level” a manufacturing strategy can represent the competitive priorities. The priorities must be split up into decision categories, in order to develop a manageable strategy. The idea of order winners and order qualifiers may be utilized, as a way to prioritize among the list of competitive priorities. “Order-winners for a product are the qualities which make an organization receive a new order. Order-qualifiers, in contrast, are required to be able to compete in a marketplace.” To establish a basis for the manufacturing strategy, is the main objective of analysing a company's order-winners and order-qualifiers. Identifying order-winners and order-qualifiers needs to be done regularly, as it is not a onetime activity, as demand of markets changes as time passes, along with that the order-winners and order-qualifiers.

Check Your Understanding

- 1) Economic growth comes with increasing _____ and _____ of factors of production.
 - a) Quality and Quantity
 - b) Labour and Capital
 - c) Wages and Cost
 - d) Fixed Factor and Variable Factor
- 2) _____ are the moral principles that act as guidelines for the way a business conducts itself.
 - a) Social Responsibilities
 - b) Business Ethics
 - c) Legal Considerations
 - d) Corporate strategies
- 3) Divisible factors lead to the _____ for a firm by adjusting the number of factors to the output of the firm.
 - a) Public Offering
 - b) Economies of scale
 - c) Small -Cap Stocks
 - d) Comparative Advantage
- 4) Important elements of Factors of production are:
 - a) Entrepreneurship
 - b) Land
 - c) Labour
 - d) All the above
- 5) _____ are those whose cost do not change with change in output.
 - a) Divisible Factors
 - b) Indivisible Factors

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- c) Fixed factors
 - d) Variable Factors
- 6) _____ takes on the economic risk involved in bringing the other factors of production together in order to produce a good or service for consumers.
- a) Industry
 - b) Economy
 - c) Government
 - d) Entrepreneur

Summary

- Operations management (OM) is the administration of business practices to create the highest level of efficiency possible within an organization. It is concerned with converting materials and labour into goods and services as efficiently as possible in order to maximize the profit of an organization.
- A very important economic concept which outline the elements needed to produce a good or service for sale is factors of production.
- Global Competition refers to products or services offered by competing companies that serve international customers.
- Business ethics should be followed. These are the moral principles that act as guidelines for the way a business conducts itself.
- Order-winners for a product are the qualities which make an organization receive a new order. Order-qualifiers, in contrast, are required to be able to compete in a marketplace.
- Economic growth comes with increasing the quality and quantity of factors of production. Improved economic growth raises the standard of living, by lowering the cost of production and increasing wages.
- The theory of cost of production depends the combination of factors employed. From the point of view of cost of production, factors of production are divided into Fixed and Variable Factors.
- Land as a factor of production refers to all natural resources and is quite a broad category. These resources are gifts which are given by nature. Examples: are water, oil, copper, natural gas, coal, and forests.
- Here capital refers not to money but to the capital goods or man-made resources such as tools, infrastructures, used in production of other goods and services.
- Laborers with a good training and education are considered to be "highly skilled" and are paid higher wages than less trained workers.
- Entrepreneurs can acquire the money for their new small businesses by issuing what's known as an initial public offering on the stock market, with shares of small-cap stocks.
- According to Porter, cost leadership or differentiation, are two primary routes for competitiveness. In contrast, according to Leong, there are actually five

dimensions of competitive priorities within manufacturing: cost, flexibility, quality, delivery performance and innovativeness.

- Each company needs to perform social welfare for the local community and maintain good relations with the society. Because any social issue can lead to damage the goodwill of the company's business condition.

Activity

1. Visit any organization and see how the factors of production and Operations Management are affecting and contributing in its growth in economy.

Question and Exercises

1. What are the factors of Operations Management? And, how an organization can improve its plant's position in the market through Competitive priorities.
2. What are Factors of Production. Describe the various factors of production. And, how they contribute in Economic Growth.
3. Who is an entrepreneur? How does he contribute to Economic growth using Factors of production?

Glossary

- Operations management (OM) is the administration of business practices to create
- Operations management (OM) is the administration of business practices to create the highest level of efficiency possible within an organization.
- Operations Management is concerned with converting materials and labour into goods and services as efficiently as possible in order to maximize the profit of an organization.
- Global Competition refers to products or services offered by competing companies that serve international customers.
- Business ethics should be followed. These are the moral principles that act as guidelines for the way a business conducts itself.
- A very important economic concept which outline the elements needed to produce a good or service for sale is factors of production. They are commonly broken down into four elements: land, labour, capital, and entrepreneurship.
- "Land" as a factor of production refers to all natural resources and is quite a broad category. These resources are gifts which are given by nature. Some typical examples of natural resources are water, oil, copper, natural gas, coal, and forests.
- Labour, is also an important factor of production and involves any human input. Any contribution or work done by people contributing to production is Labour.
- capital refers not to money but to the capital goods or man-made resources such as tools, infrastructures, used in production of other goods and services.
- An entrepreneur is someone who combines all the other factors of production, who takes on the economic risk involved in bringing the other factors of production together in order to produce a good or service for consumers.

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- Fixed factors are those whose cost do not change with change in output and variable factors are those whose costs changes with change in output.

Further Readings

1. Production and Operations Management; Prepared and Compiled By:
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Check you Understanding - Answers

- 1) a)
- 2) b)
- 3) b)
- 4) d)
- 5) c)
- 6) d)

Unit - 1.3: Role – Scope and Functions of Production and Operations Management

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Objectives:

In this unit, you will be able to

- Learn about production management
- Ps of production management
- Understand operations management
- Framework for managing operations
- Role of production and operation management
- Scope of production and operation management
- Function of production and operation management

Introduction

Production Management

It may be defined as:

Production management is defined as the performance of the management activities with regards to selecting, designing, operating, Controlling and updating production system and, actual transformation of raw materials into finished products by effectively planning, coordinating and controlling the production process, and this is what production and operations management is responsible for, and, that is the operations of that part of an enterprise.

The management function which plans, organizes, coordinates, directs and controls the material supply and Processing activities of an enterprise, so that specified products are produced by specified methods to meet the set sales is defined as Production management. In this process Labour, Plant and Capital available are used to the best advantage of the organization.

Objectives of Production Management

In order to produce goods services of right quality and quantity at the right time and right manufacturing cost, is the objective of the production management.

1. **Right Quality** - The best quality is not necessarily the right quality. The quality is determined by the cost of the product and the technical characteristics as suited to the specific requirements and customer's needs.
2. **Right Quantity** – The products should be manufactured in right number by manufacturing firm. If the products are produced exceeding the demand the capital will be blocked in the form of inventory and if the quantity is produced in short of demand, then it will lead to shortage of products.

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3. **Right Time** - One of the important parameters to judge the effectiveness of production department is "timeliness of delivery". Hence, the production department should make optimal utilization of input resources to achieve its objective.
4. **Right Manufacturing Cost** - Manufacturing costs are pre-defined before the product is actually manufactured. Hence, all attempts should be made to produce the products within the pre-defined cost, so as to reduce the variation between actual and the pre-defined cost.

Five P's of Production Management

The division of production management functions in to 5 P's (product, plant, programme, processes and people) will provide useful conceptual framework for the various activities performed by production or operations manager.

The Five P's includes:

- The Product: The link between production and marketing is a Product. This is not enough if only the customer requires a product but the organization should also be capable of producing the product. As per the product policy of the organization, an agreement is reached between the various functions on the following aspects of the product. These are:
 1. Performance
 2. Reliability and Quality.
 3. Ergonomics and Aesthetics.
 4. Quantity and selling price.
 5. Delivery schedule.

To arrive at the above, the external and the internal factors should be given due consideration, which affects various aspects such as market needs, existing culture and legal constraints and the environmental demands. Thus, the major policy decisions regarding variety of product mix are going to affect the producing system.

- The Plant: The plant accounts for major investment (fixed assets). The plant should match the needs of the product market, the worker and the organization. The plant is concerned with:
 1. Design and layout of building and offices.
 2. Maintenance of equipment, Reliability, perfection.
 3. Safety of operations.
 4. The financial constraint.

Plant layout should be such that it should allow for smooth movement of men and materials with minimum back tracking which deals with physical arrangement of plants and machineries within the selected site. The type of the layout is dependent on production type, volume of demand, etc.

- The Process: Process is a method with which a product is manufactured. There are number of alternative methods of creating a product. It is essential

to select the one best method, which attains the objectives. In deciding about the process, it is necessary to examine the following factors:

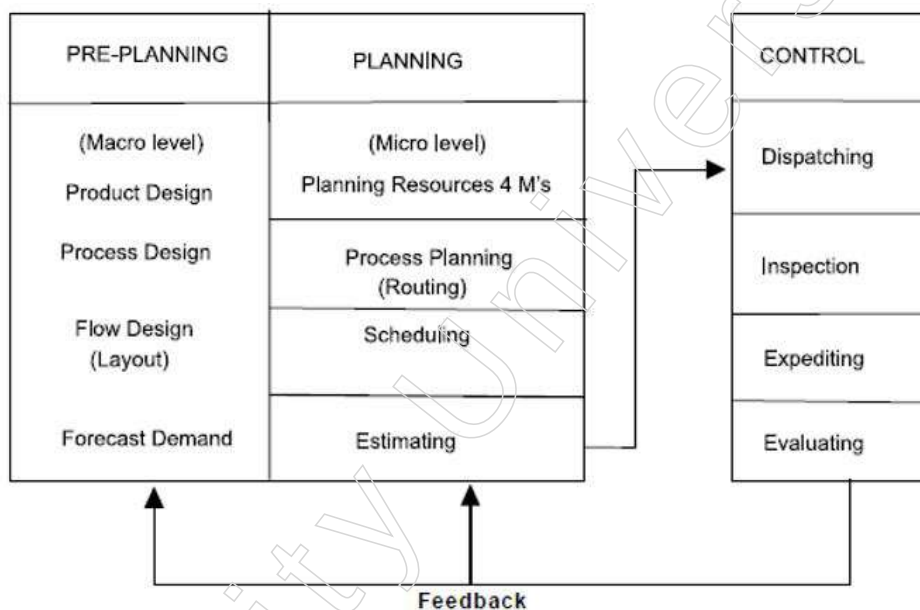
1. Available capacity
 2. Available Manpower skills
 3. Production type
 4. Plant Layout
 5. Safety
 6. Maintenance required
 7. Manufacturing costs
- The Programme: The timetable of production is what is referred to as programme. Thus, the programme prepares schedules for:
 1. Purchasing
 2. Transforming
 3. Maintenance
 4. Cash
 5. Transport and Storage
 - The People: Production depends upon people. The people vary in their skills, attitudes, and expectations from the work. Thus, to make the best use of available human resource, there should be a good match between people and jobs which may lead to job satisfaction. The production manager should be involved in issues like:
 1. Wages/salary administration.
 2. Conditions of work/safety.
 3. Motivation.
 4. Training of employees.
 - Functions of Production Management Department
- The size of the firm is what functions of Production Management depend upon. There is a difference in the functioning of the firms on the basis of their size.
- In Small Firms - The production Manager may have to look after production planning and control along with Personnel, Production and Operations Management Marketing, Finance and Purchase functions.
 - In Medium Sized Firms - They may have separate managers for Personnel, marketing and Finance functions. But the production planning and control and Purchase and stores may be under the control of Production management department.
 - In Large Sized Firms - The Production Management activities are confined to the management of production activities only. Here are some of the functions, which are looked after by Production management department. They are as follows:

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- (i) **Materials:** It is necessary that the production manager has complete and sound knowledge of the materials and the properties of the product. To appropriately select materials for the product. In order to satisfy the changing needs of the design in the product and availability of material resumes, research on materials is necessary to find alternatives.
- (ii) **Methods:** Identifying the sequence of processes, and finding the best method for the process, to search for the methods to suit the available resources are some of the activities of Production Management.
- (iii) **Machines and Equipment:** Production management department takes care of selection of suitable machinery for the process desired, designing the maintenance policy and design of layout of machines.
- (iv) **Estimating:** Production management department does a thorough estimation of Production times and production costs, to fix up the Production targets and delivery dates and to keep the production costs at minimum. In competitive environment this will help the management to decide what should be done in arresting the costs at desired level.
- (v) **Loading and Scheduling:** The Production Management department has to schedule a time table for various production activities, specifying when to start and when to finish the process. The scheduling is to be done keeping in mind the loads on hand and availability of the capacities of facility.
- (vi) **Routing:** The production department fixes the flow lines for various raw materials, components etc., from the stores to the packaging of finished product, so that all the concerned knows what exactly is happening on the shop floor.
- (vii) **Despatching:** There are various documents such as Job Cards, Route sheets, Move Cards, Inspection Cards for each and every component of the product, which the Production Management department has to prepare in set of five copies. These documents are to be released from Production Management department to give green signal to start the production. The shop floor activities will follow the instructions given in these documents. Activity of releasing the document is known as dispatching.
- (viii) **Follow Up:** After dispatching the document, the management wants to know whether the activities are being carried out as per the plans or not. The engineers go round the production floor along with the plans, compare the actual with the plan and give the feedback of the progress work to the management. This will help the management to evaluate the plans.
- (ix) **Inspection:** Here inspection is concerned with, the inspection activities during production, but there is a separate quality control department which does the quality inspection, and is not under the control of Production Management. The reason of separate quality department is, if the quality inspection is given to production Management, then there are chances of qualifying the defective products also. For example, Teaching and examining of students is given to the same person, then there is a possibility of passing all the students in the first grade. To avoid such situations an external person is employed to keep a check and does corrections, so that judgement is done correctly.

- (x) Evaluation: Evaluation is a very necessary step and the Production department must evaluate itself and its contribution in fulfilling the corporate and departmental objectives, which is necessary for setting up the standards for future. Production management department alone must do Routing, Scheduling, Loading, Dispatching and expediting, irrespective of the size of the firm. The reason is that this department knows very well regarding materials, Methods, and available resources etc.
- If the firms are small, all the above-mentioned functions (I to x) are to be carried out by Production Management Department.
 - In medium sized firms in addition to Routing, Scheduling and Loading, Dispatching and expediting, some other functions like Methods, Machines may be under the control of Production Management Department.
 - In large firms, there is a Separate department for Methods, Machines, Materials and others but routing, loading and scheduling are the sole functions of Production Management.

And, all the above ten functions are categorized in three stages, that is Preplanning, Planning and control stages as shown in figure below:



Benefits Derived from Efficient Production Management

An efficient Production Management will give benefits to the various sections of the society. They are:

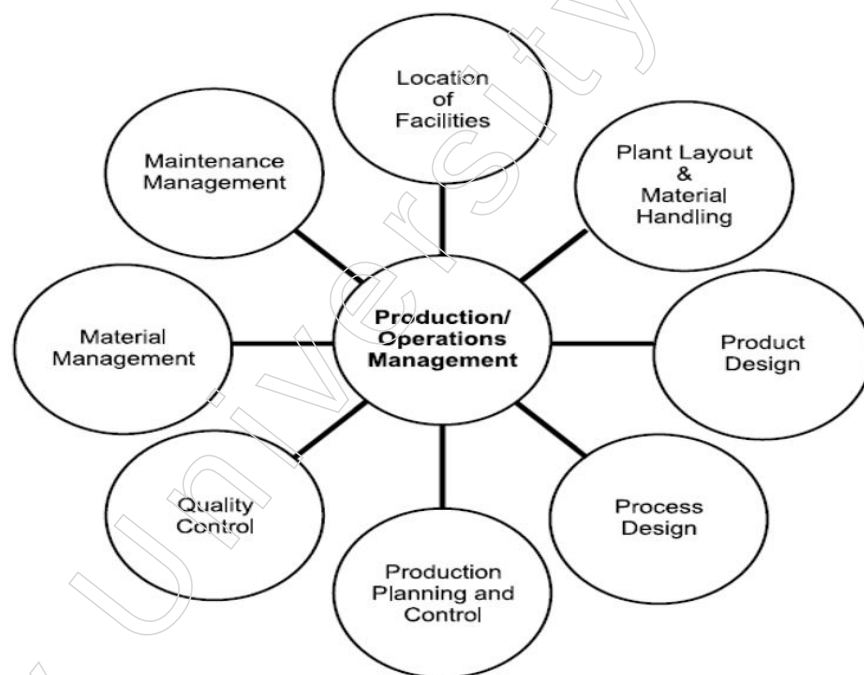
- Consumer is benefited as due to improved industrial Productivity, increased use value in the product. Products are made available to the consumers him at right place, at right price, at right time, in desired quantity and of desired quality.
- Investors get increased security for their investments, adequate market returns, and creditability and good image in the society.
- Employee gets appropriate and adequate Wages, Job security, improved working conditions and increased Personal and Job satisfaction.

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- Suppliers: Will get confidence in management and their bills can be realized without any delay and, community also enjoys Benefits from economic and social stability.
- The Nation will achieve prospects and security because of increased Productivity and healthy industrial atmosphere.

Scope of Production Management

Production management is defined as the performance of the management activities with regards to selecting, designing, operating, Controlling and updating production system and, actual transformation of raw materials into finished products by effectively planning, coordinating and controlling the production is what production and operations management is responsible for, and, that is the operations of that part of an enterprise.



Following are the activities which are listed under production and operations management functions:

1. Location of facilities
2. Plant layouts and material handling
3. Product design
4. Process design
5. Production and planning control
6. Quality control
7. Materials management

1. Location of Facilities:

- This is a long-term decision for operations of a business, which involves long term commitment about the geographically static factors that affect a

business organization. It is an important strategic level decision-making for an organization.

- The selection of location is a very important decision as large investment is made in building plant and machinery. If the location is not good or not proper the it may lead to the waste of all the investments made in the plant and machinery equipment.
- Hence, company's expansion plan and policy, diversification plan for the products, changing sources of raw materials and many other factors is what the location of plant depends upon. The purpose of the location study is to find the optimal location that will result in the greatest advantage to the organization.

2. Plant Layout and Material Handling

- The physical arrangement of facilities refers to as Plant layout. It is the configuration of departments, work centres and equipment in the conversion process. The overall objective of the plant layout is to design a physical arrangement that will meet the required output with quality and quantity most economically.
- According to James Moore, "Plant layout is a plan of an optimum arrangement of facilities including personnel, operating equipment, storage space, material handling equipment and all other supporting services along with the design of best structure to contain all these facilities".
- The process of moving of materials from the store room to the machine and from one machine to the next during the process of manufacture is known as Material Handling. It is a specialized activity for a modern manufacturing concern, with 50 to 75% of the cost of production, this cost can be reduced by proper section, operation and maintenance of material handling devices. Material handling devices results in increases in output, improves quality, speeds up the deliveries and decreases the cost of production. Hence, material handling is very important and of prime consideration in the designing new plant and several existing plants.

3. Product Design

- Designing is converting ideas into reality, and product design deals with the same. A business organization designs, develop and introduce new products as a survival and growth strategy. The biggest challenge faced by an organization is developing new products and launching them in the market.
- The entire process of product starts from need identification to physical manufactures of product which involves the following functions: marketing, product development, and manufacturing. Product development means translating the customers' needs given by marketing department into technical specifications and designing the various features into the product to these specifications.
- Selecting the processes by which the product can be manufactured is the responsibility of Manufacturing Department. Product design and development bridges the link between marketing, customer needs and expectations and the activities required to manufacture the product.

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4. Process Design

- A macroscopic decision-making of an overall process route for converting the raw material into finished goods is process design. These decisions surround the selection of a process, choice of technology, process flow analysis and layout of the facilities.
- Hence, the analysis of the workflow for converting raw material into finished product and to select the workstation for each included in the workflow and is an important decision in process design.

5. Production Planning

- The process of planning process in advance and setting the exact route of items, fixing the starting and finishing dates for each item, to give production orders to businesses and to follow up the progress of products according to orders is called as Production Planning and Control.
- The principle of production planning and control is – “First Plan Your Work and then Work on Your Plan”. Planning is deciding in advance what to do, how to do it, when to do it and who has to do it, as it includes planning, routing, scheduling, dispatching and follow-up.
- Through planning we can make a map road of where we are and where we have to go. Routing is the selection of path which each part of the product will follow, while being transformed from raw material to finished products, as routing determines the most advantageous path to be followed from department to department. Scheduling determines the fixation time and date of each operation, and also determines the sequence of operations to be followed, whereas, Dispatching is concerned with the starting the processes.

6. Quality Control

- A system which maintains desired level of output of quality in a product or service by systematically controlling various factors that affect the quality of the product. Controlling quality aims at prevention of defects at the source, relying on effective feedback system and corrective action procedure.
- Quality control is an entire collection of activities, which ensures that the operation will produce the optimum quality products at minimum cost. The main objectives of quality control are:
 - ◆ To increase company's income by making the production more acceptable to the customers i.e., by working towards customer satisfaction by providing products as needed or desired by the customer.
 - ◆ Reduction in losses. Hence reducing companies cost.
 - ◆ To produce optimal quality at less price.
 - ◆ To ensure customer satisfaction with productions or services or high-quality level, to build customer goodwill.
 - ◆ Ensuring quality control through inspections.

7. Materials Management

- Materials management is that aspect of management function which is primarily concerned with the acquisition, control and use of materials needed

and flow of goods and services connected with the production process having some predetermined objectives in view. The main objectives of materials management are:

- ◆ Minimises material cost.
 - ◆ To purchase, receive, transport and store materials efficiently and to reduce the related cost.
 - ◆ To cut down costs through simplification, standardisation, value analysis, import substitution, etc.
 - ◆ To trace new sources of supply and to develop cordial relations with them in order to ensure continuous supply at reasonable rates.
 - ◆ To reduce investment tied in the inventories for use in other productive purposes and to develop high inventory turnover ratios.
- Operations Management

Definition

There are 2 ways in which operation management is viewed:

- Traditional View and,
- Modern View.

Traditional View is also known as the Transformation Approach. The traditional definition view operation management to be a transformation system. According to the traditional view, operations management is the systematic direction and control of the processes that transform resources known as inputs, into finished goods or services for customers or clients known as outputs.

A transformation process is any activity or group of activities which takes one or more inputs, then transforms and adds value to them, and provides outputs for customers or clients. Some transformation processes are simple. Like the transformation process of milk into cheese and butter is relatively easy to identify as the inputs are the raw materials. Other than this, the nature of the transformation may be less obvious, where the inputs are information or people. For example, a hospital transforms ill patients (the input) into healthy patients (the output).

Transformation processes include:

- physical characteristics of materials or customer changes.
- Location of materials, information or customers changes.
- Ownership of materials or information changes.
- storage or accommodation of materials, information or customers
- Purpose or form of information changes.
- Physiological or psychological state of customers changes.

Often all three types of input – materials, information and customers, are transformed by the same organisation. For example, the process of withdrawing money from a bank account involves information about the customer's account, materials such as cheques and currency, and the customer himself. Another example is, treatment

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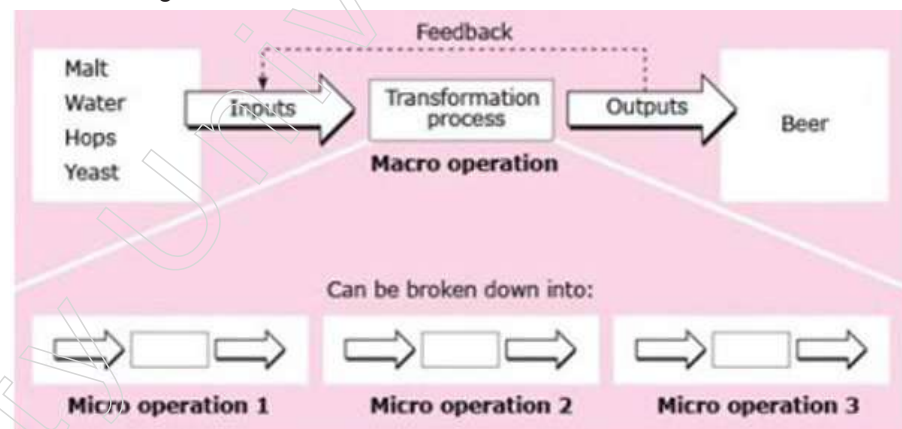
of a patient in hospital involves not only the customer's state of health, but also the information about the patient any materials used in treatment.

One useful way of categorizing different types of transformation is into:

- manufacture – the physical creation of products (for example cars)
- transport – the movement of materials or customers (for example a taxi service)
- supply – change in ownership of goods (for example in retailing)
- service – the treatment of customers or the storage of materials (for example hospital wards, warehouses).

Several different transformations are usually required to produce a good or service. The overall transformation process is described as the macro-operation, and the more detailed transformations or steps in between this macro-operation is known as micro-operations. For example, the macro-operation in a brewery is making beer, shown below in Figure. The micro-operations include:

- milling the malted barley into grist.
- wort is formed by mixing the grist with hot water
- cooling down the wort and transferring it to the fermentation vessel
- adding yeast to the wort and fermenting the liquid into beer
- filtering the beer to remove the spent yeast
- decanting the beer into casks or bottles.



Value Driven Approach or Modern Approach

The second approach to operations Management is a value drive approach. The value driven approach recognizes that a business is a set of processes each of which has inputs, outputs, and structure. Each process has a job or work to do and each process should be evaluated or measured in terms of how effective it is in achieving the desired outcomes. There are four core business processes describing the functioning of a business organization, mentioned below:

- 1) **Determine Customer Needs:** This is the very crucial part for any organization to determine customer's needs in order to support the firm demand, its forecasting needs its product design and development activities. And, in order to do the same, it is necessary to monitor the competitive environment. The supporting business processes are engaged in marketing the products and providing after

sales service. Customer satisfaction should be measured through feedbacks. There is also a need to understand the specific needs of different market segments and the nature of the competitive environment.

- 2) **Develop Product Strategy:** Creation of products that customer desire involves marketing, operations and engineering activities. To support the design of new products or introduce product improvements, ability to evaluate product concepts should be there. At slow pace, one can focus more on delighting customers by finding better ways to incrementally improve products that already exist. But as the pace of business increases, the greater is the need to be aware of the competitive challenges that new technologies and competitors introduce into the marketplace.
- 3) **Secure Processes and Materials to Satisfy Demand:** Starting from the selection of raw materials from vendors and the ultimate delivering and servicing of the product for the customer are the management activities which include operations planning and control processes and managing the product transformation processes. In addition, the business logistics and the supply chain process play a critical role, hence, should be managed effectively.
- 4) **Manage Strategic Planning Processes:** Support business processes are essential to all organizations. The strategic planning process defines the firm as well as its own operations resource management function and creates an organization design that is suited to the competitive environment, and also provides or enhances the human capital needed by other functions to effectively carry out their tasks.

Operations management activities are mostly involved in the second and third core processes. Operations management as a value creating activity contributes to the customer satisfaction processes by helping in designing and developing products that have the capability to satisfy the customer's functional need with the desired level of design quality and cost.

Characteristics of Operations Management

The Characteristics of operations management of a core business are:

- To Understand the customer needs.
- To Understand the organisation's financial performances.
- Great Planner (Either on time Delivery or Business Development or New product development).
- Effective communicator.
- Perform under pressure.
- Tracks & Measures Staff Performance.

Scope of Operations Management

1. Product Design and Selection:

- A selection of product makes a system efficient or inefficient. Hence, all the objectives should be kept in mind while selecting the right product, which is very important.

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- Designing is converting ideas into reality, product design deals with the same. A business organization designs, develop and introduce new products as a survival and growth strategy. The biggest challenge faced by an organization is developing new products and launching them in the market.
- Selecting the processes by which the product can be manufactured is the responsibility of Manufacturing Department. Product design and development bridges the link between marketing, customer needs and expectations and the activities required to manufacture the product.

2. Planning and Selection:

- Taking decisions about technology, machines and equipment is involved in selection process. A macroscopic decision-making of an overall process route for converting the raw material into finished goods is process design.
- Hence, the analysis of the workflow for converting raw material into finished product and to select the workstation for each included in the workflow and is an important decision in process design.

3. Location Facilities:

- It is the most important facility as it looks for long term decisions; a wrong decision can make it pay a lot. The operation management helps to select that particular location where distribution, production cost and location cost is less.
- The selection of location is a very important decision as large investment is made in building plant and machinery. If the location is not good or not proper the it may lead to the waste of all the investments made in the plant and machinery equipment.

4. Material Handling Facilities and Plant Layout:

- According to James Moore, "Plant layout is a plan of an optimum arrangement of facilities including personnel, operating equipment, storage space, material handling equipment and all other supporting services along with the design of best structure to contain all these facilities".
- The process of moving of materials from the store room to the machine and from one machine to the next during the process of manufacture is known as Material Handling. It is a specialized activity for a modern manufacturing concern, with 50 to 75% of the cost of production, this cost can be reduced by proper section, operation and maintenance of material handling devices.

5. Capacity Planning

The level of output of the conversion process over a period of time refers to as Capacity. Industry creates challenging problems in capacity planning, requiring in the long run, expansion and contraction of the major facilitates in the conversion process. Marginal costing, linear programming are some tools which help in capacity planning. There are 3 types of capacity planning:

- Product Capacity Planning: This ensures enough products or ingredients available for your deliverables.
- Workforce capacity planning: This ensure having enough team members and work hours available to complete the job.

- Tool Capacity Planning: This ensure having enough tools to complete the jobs.

Objectives of Operation Management

- To Improve productivity by better utilization of resources like men, material, method etc.,
- To have very less or no defects.
- To effectively and effectively utilize plants and machinery.
- To Improve material handling equipment.
- To have proper control on raw materials, semi-finished goods and finished goods.
- To achieve proper cash management.
- To enhance production staff with respect to their salary, wages etc.

Functions of Operation Management

- Finance - The main function of operations management is Finance, which should not be wasted in unproductive tasks. It should ensure that all finance of the organization is utilized for the manufacturing of useful goods or services to satisfy consumer wants.
- Operation – The operation management is basically concerned with planning, organizing, directing and controlling of daily routine operations of an organization. The operation activities help operation manager to ensures that all activities are going effectively and efficiently.
- Strategy – Another main function of operation management is strategy formulation. The operation manager should have pre-planned tasks. The organization can optimize their resources and develop a competitive edge over competitors by formulation of plans and tactics.
- Product Design – The operations manager is responsible for designing the product according to the market trends and demands, and also must ensure that innovative techniques are incorporated within the product and its quality is maintained.
- Maintaining Quality – It is the duty of operations manager to ensure a better quality of products. Quality should not be compromised and the manager should also not compromise with the quality of Products. They should work on efficient quality management and should supervise all tasks, if any defects are found they should take steps to rectify such defects.

A Framework for Managing Operations

“Managing operations can be enclosed in a frame of general management function as shown in Figure below. Operation managers are concerned with planning, organizing, and controlling the activities which affect human behaviour through models.”

- Planning – Planning is thinking before you do. Activities that set a course of action and guide future decision-making is planning. The operations manager defines the objectives for the operations subsystem of the organization, and the policies, and procedures for achieving the objectives. The role and focus of operations in the

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- organization's overall strategy is clarified in this stage, which also involves product planning, facility designing and using the conversion process.
- **Organizing** – It is a set of activities that identify and establishes a structure of tasks and authority. A structure of roles and the flow of information within the operations subsystem is established by Operation managers, which determine the activities required to achieve the organizational goals and assign authority and responsibility for carrying them out.
 - **Controlling** – It is a set of activities which compares the actual performance with the set ones, that assure the actual performance in accordance with planned performance. It is very important to ensure that the plans for the operations subsystems are accomplished, the operations manager must keep a check and exercise control by measuring actual outputs and comparing them to planned operations management. Controlling costs, quality, and schedules are the important functions here.
 - **Behaviour** - Operation managers are concerned with how their efforts to plan, organize, and control affect human behaviour. They also want to know how the behaviour of subordinates can affect management's planning, organizing, and controlling actions. Their interest lies in decision-making behaviour.
 - **Models** – While in the process of planning, organising and controlling the conversion process, the operation managers face many problems and must make many decisions. They can use models like aggregate planning models for examining how best to use existing capacity in short-term, break-even analysis to identify break even volumes, linear programming and computer simulation for capacity utilisation, decision tree analysis for long-term capacity problem of facility expansion, simple median model for determining best locations of facilities etc. and can simplify their difficulties.

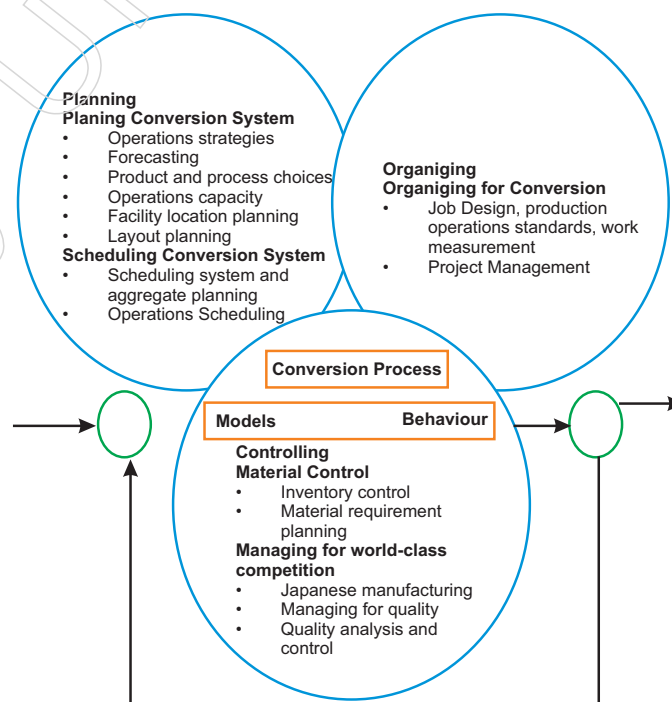


Fig: General Model for Managing Operation

Managing Global Operations

- “Globalisation means deployment of business facilities and operations around the world. Globalisation can be defined as a process in which geographic distance becomes a factor of diminishing importance in the establishment and maintenance of cross border economic, political and socio-cultural relations. It can also be defined as worldwide drive toward a globalised economic system dominated by supranational corporate trade and banking institutions that are not accountable to democratic processes or national governments.”
- Mentioned below are the four developments, which has propelled the trend toward globalisation. These are:
 1. Improved communication technologies and transportation;
 2. Opened financial systems;
 3. Increase in imports demand; and
 4. Reduced import quotas and other trade barriers.
- Setting up a firm abroad is not easy; it involves added complexities in its operations. Global Markets for sure impose new standards and quality for the products. Managers should not give priorities to domestic markets first and then global markets later, rather they must think globally and act locally. Also, keeping a good understanding of their competitors is very important.
- Some other important challenges of managing multinational operations include different management style, unfamiliar laws and regulations, different languages and customs, and different costs. Managing global operations would also focus on the following key issues:
 - To properly utilise and acquire the following concepts and those related to global operations, supply chain, logistics, etc.
 - Associating global historical events to key drivers in global operations from different perspectives.
 - Developing criteria for conceptualisation and evaluation of different global operations.
 - Associate success and failure cases of global operations to political, social, economic and technological environments.
 - To visualize trends in global operations.
 - To develop an understanding of the world vision regardless of their country of origin, residence or studies in a respectful way of perspectives of people from different races, studies, preferences, religion, politic affiliation, place of origin, etc.

Objectives of Operation Management

- Customer Service

The primary objective of an operating systems is to satisfy the customer wants. Hence, customer service is a key objective of operations management. The operating system must provide something to a specification which can satisfy the customer in terms of cost and timing. Thus, the primary objective can be satisfied by providing the

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'right thing at a right price at the right time'. Generally, an organization aims consistently to achieve certain standards and operations manager will always try attempting to achieve these standards. Hence, the operations manager's decisions will be influenced by the objectives to achieve the required customer service.

- **Resource Utilisation**

Another main objective of operating systems is to utilise resources for the satisfaction of customer wants effectively, i.e., customer service must be provided with the achievement of effective operations through efficient use of resources. As inadequate customer service or inefficient use of resources leads to commercial failure of an operating system. Operations management is essentially concerned with the utilisation of resources, i.e., obtaining maximum out of the resources or minimising loss of resources or wastage. Operations management is also concerned with the achievement of both satisfactory customer service and resource utilisation. Both the objectives cannot be maximized at the same time, any improvement in one will often give rise to deterioration in the other. Hence, a satisfactory performance must be achieved on both objectives. The balance established between both and within these basic objectives will be influenced by market considerations, competitions, the strengths and weaknesses of the organization, etc. Hence, when the objectives are set, operations managers should make a contribution towards the same.

- **Production and Operations Management**

It is defined as the design, operation, and improvement of transformation process, which converts the various inputs into the desired outputs of product and services according to the specifications in the amount and by scheduled demand at a minimized cost.

Difference between Production and Operations Management

Production and operations management are more similar than different: In Production management manufacturing products is of prime concern, and the same is termed as production management, whereas management of services is somewhat broader in scope are called operations management. The line between products-based and services-based organizations is blurring rapidly as well— car manufactures need to service their cars and the retailers manufacture their own brand labels.

Production and operations management involve three main types of decisions, typically made at three different stages:

1. **Production Planning.** The first decision which an operations manager faces comes at the planning stage. At this stage, managers decide where, when, and how production will occur and determine site locations and obtain the necessary resources.
2. **Production Control.** This stage involves, the decision-making process and focuses on controlling quality and costs, scheduling, and the actual day-to-day operations of running a factory or service facility.
3. **Improving Production and Operations.** This is the final stage of operations management which focuses on developing more efficient methods of producing the firm's goods or services.

All three decisions are ongoing and may occur simultaneously. In the following unit, we will take a closer look at the decisions and considerations firms face in each stage of production and operations management.

Need for Production and Operation Management

- i) **Produce Right Quality of Product:** The best quality is not necessarily the right quality. The quality is determined by the cost of the product and the technical characteristics as suited to the specific requirements and customer's needs.
- ii) **Right Quantity:** The products should be manufactured in right number by manufacturing firm. If the products are produced exceeding the demand the capital will be blocked in the form of inventory and if the quantity is produced in short of demand, then it will lead to shortage of products.
- iii) **Right Time:** One of the important parameters to judge the effectiveness of production department is "timeliness of delivery". Hence, the production department should make optimal utilization of input resources to achieve its objective.
- iv) **Right Manufacturing Cost:** Manufacturing costs are pre-defined before the product is actually manufactured. Hence, all attempts should be made to produce the products within the pre-defined cost, so as to reduce the variation between actual and the pre-defined cost.

Objectives of Production and Operation Management

- Maximum output should be attained at lowest cost.
- To control wastage and pollution.
- To ensure resource utilization and optimum capacity.
- To ensure that quality of products should be attained.
- To make necessary changes in machinery and equipment.
- To ensure timely delivery of output.
- To maintain inventory.

Elements of Production and Operation Management

- i) **Planning:** Planning is thinking before you do. Activities that set a course of action and guide future decision-making is planning. The operations manager defines the objectives for the operations subsystem of the organization, and the policies, and procedures for achieving the objectives. The role and focus of operations in the organization's overall strategy is clarified in this stage, which also involves product planning, facility designing and using the conversion process.
- ii) **Organizing:** It is a set of activities that identify and establishes a structure of tasks and authority. A structure of roles and the flow of information within the operations subsystem is established by Operation managers, which determine the activities required to achieve the organizational goals and assign authority and responsibility for carrying them out.
- iii) **Controlling:** It is a set of activities which compares the actual performance with the set ones, that assure the actual performance in accordance with planned performance.

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It is very important to ensure that the plans for the operations subsystems are accomplished, the operations manager must keep a check and exercise control by measuring actual outputs and comparing them to planned operations management. Controlling costs, quality, and schedules are the important functions here.

- iv) Behaviour: Operation managers are concerned with how their efforts to plan, organize, and control affect human behaviour. They also want to know how the behaviour of subordinates can affect management's planning, organizing, and controlling actions. Their interest lies in decision-making behaviour.
- v) Model: While in the process of planning, organising and controlling the conversion process, the operation managers face many problems and must make many decisions. They can use models like aggregate planning models for examining how best to use existing capacity in short-term, break-even analysis to identify break even volumes, linear programming and computer simulation for capacity utilisation, decision tree analysis for long-term capacity problem of facility expansion, simple median model for determining best locations of facilities etc. and can simplify their difficulties.

1.3.1 Role of Production Management

1. Accomplishment of Firm's Objectives: A business firm to achieve all its objectives with the help of Production management. It produces products, which satisfy the customers' needs and wants. So, the firm will increase its sales. This will help it to achieve its objectives.
2. Enhances Reputation, Goodwill and Image: Satisfying customers is very important for any organization as it increases the firm's reputation, goodwill and image and a good image helps the firm to growth and expansion.
3. Helps to Introduce New Products: Production management also helps in introducing new products to the market through its Research and Development (R&D). This helps the firm to develop successful newer and better-quality products as they give full satisfaction to the customers.
4. Supports other Functional Areas: Production management also supports the other functional areas in an organization, such as marketing, finance, and personnel. Due to increase in sale, the marketing department will find it easier to sell good quality products, and the finance department will get more funds to invest. Due to better performance of production department, the personnel department will be able to manage the human resources effectively.
5. Helps to Face Competition: As production management helps produces products of right quantity, right quality, right price and at the right time, it helps the firm to face competition in the market, as the products are delivered to the customers as per their requirements.
6. Optimum Utilization of Resources: Production management facilitates optimum utilization of resources such as manpower, machines, etc. So, the firm can meet its capacity utilization objective. This will bring higher returns to the organization.
7. Minimizes Cost of Production: Production management helps to minimize the cost of production. It tries to maximize the output and 0 minimize the inputs. This helps the firm to achieve its cost reduction and efficiency objective.

8. Expansion of the Firm: Production management helps the firm to expand and grow. This is because it tries to improve quality and reduce costs. This helps the firm to earn higher profits. These profits help the firm to expand and grow.

- Role of Production Management to Customers and Society

1. Higher Standard of Living: With continuous research and development, production management will produce new and better varieties of products. People use these products and enjoy a higher standard of living.

2. Generates Employment: Production activities also creates many different job opportunities in the country, either direct or indirect. In production area direct employment is generated and, in supporting areas such as marketing, finance, customer support, etc., indirect employment is generated.

3. Spread Effect: Other sectors also expand because of Production. Companies making spare parts will expand and service sector such as banking, transport, communication, insurance, BPO, etc. will also expand. This expansion will lead to more job opportunities and boosts economy.

4. Creates Utility: Production creates Form Utility. Consumers can get form utility in the shape, size and designs of the product. Production also creates time utility, because goods are available whenever consumers need it.

5. Boosts Economy: Optimum utilization of resources and effective production of goods and services lead by Production management, ensures speedy economic growth and well-being of the nation.

- Role of Operation Management

1) Towards Customers: The objectives of the operation management totally depend upon customer's preferences and their requirements. Hence, Customers are the most affected by any business.

2) Towards Suppliers: Suppliers also get affected by the suppliers, on both the grounds including how they prosper themselves, and on how effective they are at supplying the operation.

3) Towards Shareholders: The better the operations are at producing goods and services, the more likely the whole business is to prosper and shareholders will be one of the major beneficiaries of this.

4) Towards Employees: If the company is prosperous, the employees will be generally better off, which includes the general working conditions which are determined by the way the operation is designed.

5) Towards Society: Although often having no direct economic connection with the company individuals and groups in society at large can be impacted by the way its operation managers behave.

1.3.2 Scope of Production and Operation Management

Production and operations management is concerned with the conversion of inputs into outputs, using physical resources(inputs), so as to provide the desired utilities to the customer(outputs), while achieving the other organizational objectives of effectiveness, efficiency and adoptability. By its primary concern for 'conversion by

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using physical resources, it distinguishes itself from other functions such as personnel, marketing, finance, etc.

Following are the activities which are listed under production and operations management functions:

1. Location of facilities
2. Plant layouts and Job design
3. Material handling
4. Product design
5. Process design
6. Production and planning control
7. Quality control
8. Materials management
9. Maintenance management
10. Automation

1. Location of Facilities:

- Selection of appropriate location must be done to ensure the availability of power supply, water supply, road conditions, nearness of raw materials, skilled labours etc. This is a long-term decision for operations of a business, which involves long term commitment about the geographically static factors that affect a business organization. It is an important strategic level decision-making for an organization.
- The selection of location is a very important decision as large investment is made in building plant and machinery. If the location is not good or not proper the it may lead to the waste of all the investments made in the plant and machinery equipment.
- Hence, company's expansion plan and policy, diversification plan for the products, changing sources of raw materials and many other factors is what the location of plant depends upon. The purpose of the location study is to find the optimal location that will result in the greatest advantage to the organization.

2. Plant Layout and Job Design:

- Preparation of an efficient plant layout for the establishment and accommodation of machines in the required sequence. In order to organize machines, tasks into a unit of work to achieve certain objectives, a job design must be prepared.
- The physical arrangement of facilities refers to as Plant layout. It is the configuration of departments, work centres and equipment in the conversion process. The overall objective of the plant layout is to design a physical arrangement that will meet the required output with quality and quantity most economically.

- Job design can be defined as the process of establishing employee's roles and responsibilities and the system and procedures that should use or follow. It is the process of creating job which enables an organization to achieve its goals through employee motivation.

3. Materials Handling:

- It is the process of ensuring the movement of raw materials and semi-finished goods inside the factory.
- The process of moving of materials from the store room to the machine and from one machine to the next during the process of manufacture is known as Material Handling. It is a specialized activity for a modern manufacturing concern, with 50 to 75% of the cost of production, this cost can be reduced by proper selection, operation and maintenance of material handling devices. Material handling devices results in increases in output, improves quality, speeds up the deliveries and decreases the cost of production. Hence, material handling is very important and of prime consideration in the designing new plant and several existing plants.

4. Product Design:

- Designing the product and conceiving the idea about its production. Product design considers the product size, weight, colour etc.
- Designing is converting ideas into reality, and product design deals with the same. A business organization designs, develop and introduce new products as a survival and growth strategy. The biggest challenge faced by an organization is developing new products and launching them in the market.
- The entire process of product starts from need identification to physical manufactures of product which involves the following functions: marketing, product development, and manufacturing. Product development means translating the customer's needs given by marketing department into technical specifications and designing the various features into the product to these specifications.

5. Process Design:

- It is the complete description of specific steps in the production process. This determines the production process which is most relevant.
- A macroscopic decision-making of an overall process route for converting the raw material into finished goods is process design. These decisions surround the selection of a process, choice of technology, process flow analysis and layout of the facilities.
- Hence, the analysis of the workflow for converting raw material into finished product and to select the workstation for each included in the workflow and is an important decision in process design.

6. Production and Planning Control:

- It means coordination of series of functions according to a plan which will economically utilizes the plant facilities and regulates orderly movement of

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goods. The process of planning process in advance and setting the exact route of items, fixing the starting and finishing dates for each item, to give production orders to businesses and to follow up the progress of products according to orders is called as Production Planning and Control.

- The principle of production planning and control is – “First Plan Your Work and then Work on Your Plan”. Planning is deciding in advance what to do, how to do it, when to do it and who has to do it, as it includes planning, routing, scheduling, dispatching and follow-up.

7. Quality Control:

- It is a staff function concerned with the prevention of defects in manufacturing so that, items may be made right way and ensure the quality standard.
- A system which maintains desired level of output of quality in a product or service by systematically controlling various factors that affect the quality of the product. Controlling quality aims at prevention of defects at the source, relying on effective feedback system and corrective action procedure.
- Quality control is an entire collection of activities, which ensures that the operation will produce the optimum quality products at minimum cost. The main objectives of quality control are:
 - o To increase company's income by making the production more acceptable to the customers i.e., by working towards customer satisfaction by providing products as needed or desired by the customer.
 - o Reduction in losses. Hence reducing companies cost.
 - o To produce optimal quality at less price.
 - o To ensure customer satisfaction with productions or services or high-quality level, to build customer goodwill.
 - o Ensuring quality control through inspections.

8. Inventory Management:

- It is the process of maintaining proper records of raw materials semi-finished goods and finished goods and, refers to the process of ordering, storing, selling, using company's inventory. There are four inventory methods:
- Just-in-time: Through this method companies can save significant amounts of money and reduce waste by keeping only the inventory they need to produce and sell products.
- Materials Requirement Planning: This inventory management is dependent on sales-forecast method, which means that manufacturers need to have the accurate sales records to ensure accurate planning of inventory needs and to communicate those needs with materials suppliers in a timely manner.
- Economic Order Quantity (EOQ) — This model is used for inventory management which includes calculating the number of units a company should add to its inventory with each batch order to reduce the total costs of its inventory while assuming constant consumer demand.
- Day's Sales of Inventory (DSI) — This method is a financial ratio which

indicates the average time in days that a company takes to turn its inventory, including goods that are a work in progress, into sales. Day Sales of Inventory is also known as the average age of inventory, days inventory outstanding (DIO), days in inventory (DII), and is interpreted in multiple ways.

9. Maintenance Management:

- This means maintain company's assets and resources, while controlling and cost, thereby ensuring maximum efficiency. An organization should continuously work towards improving its maintenance management. It is the process of formulating the corrective measures to stay in track with planned quality, time schedule and predetermined cost schedules.
- Maintenance management is very important as, when the resources are maintained properly, they keep the production stable and greatly minimize the chances of unplanned downtime. And, an unplanned down time leads to spike in unexpected costs, associated with things like repairs, overtime labour, spare parts etc.

10. Automation:

- This refers to a range of technologies and techniques of operating or controlling a productive process by electronic device and reducing human intervention to the minimum.
- In an enterprise, popular approaches to automation include business process automation (BPA) and robotic process automation (RPA). In general, the term BPA is used when it is to decide about how to apply the concept of automation to business processes, while RPA is used when discussing how to automate a specific, repetitive task.

1.3.3 Functions of Production and Operation Management

- 1) **Creation of Goods and Services:** Every production and operation department starts with creation of goods or services. In order to satisfy the needs of the customer, production and operations department include the physical assembling of goods and many customers care services.
- 2) **Profit:** Another main function of production and operations department is to produce a product or service and create profit and revenue for the company.
- 3) **Evaluation:** The production and operation department must work as a self-evaluating entity that monitors the quality, quantity, and cost of goods produced.
- 4) **Tasks:** This involves various tasks important for production, which includes forecasting, scheduling, purchasing, design, maintenance, people management, flow analysis, reporting, assembly and testing.
- 5) **Fulfilment:** This involves fulfilments of demand of the customers and ensures timely delivery of the output from production to customers.
- 6) **Analysis:** Critical path analysis, stock control analysis, utilization analysis, capacity analysis and just-in-time analysis of inputs, break- even analysis and metric analysis, all are Standard analysis function included in a production and operation department.

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Reasons for Production and Operation Management

We need Production and Operations management as it helps:

- Understanding the role played by the people in producing goods & services.
- Getting a clear picture about the factory.
- It helps in selecting a career.
- It has strategic use to the executives.
- Understands how important it is to Nation.

Check Your Understanding

1. Which of the following is not a function of production and operations management?
 - a) Location of facilities
 - b) Plant layouts and Job Design
 - c) Material handling
 - d) Making changes in machinery
2. Which of these is an objective of Production and Operations management?
 - a) control wastage and pollution
 - b) Right manufacturing cost
 - c) Right Quality
 - d) Right Time
3. _____ means deployment of business facilities and operations around the world.
 - a) Cross Border Economy
 - b) Globalisation
 - c) Planning
 - d) Customer Service
4. A framework of managing operations include:
 - a) Planning
 - b) Organizing
 - c) Models
 - d) All the above
5. Functions of Operation Management
 - a) Product Design
 - b) Maintaining Quality
 - c) Function
 - d) All the above
6. Which one is one of the developments, which has propelled the trend toward globalisation?
 - a) Democratic Processes

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- b) Opened Financial Systems
 - c) Managing Operations
 - d) Customer Service
7. These are the moral principles that act as guidelines for the way a business conducts itself is known as:
- a) Legal Consideration
 - b) Social Responsibilities
 - c) Business Ethics
 - d) Expenditure Challenge
8. _____ for a product are the qualities which make an organization receive a new order.
- a) Order-winners
 - b) Order-qualifiers
 - c) Winner-qualifiers
 - d) Qualifier-Order
9. _____ can be defined as the process of establishing employee's roles and responsibilities and the system and procedures that should use or follow.
- a) Plant Layout
 - b) Job design
 - c) Material Handling
 - d) Product Design
10. Which model is used for inventory management which includes calculating the number of units a company should add to its inventory with each batch order to reduce the total costs of its inventory?
- a) Economic Order Quantity
 - b) Material Requirement Quantity
 - c) Day sales of inventory
 - d) Just-in-Time
- Ans a)
11. This refers to a range of technologies and techniques of operating or controlling a productive process by electronic device and reducing human intervention.
- a) Process design
 - b) Quality Management
 - c) Inventory Management
 - d) Automation
12. _____ costs are pre-defined before the product is actually manufactured.

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- a) Designing
- b) Production
- c) Manufacturing
- d) Process

Summary

- Production management is defined as the performance of the management activities with regards to selecting, designing, operating, Controlling and updating production system and, actual transformation of raw materials into finished products by effectively planning, coordinating and controlling the production process, and this is what production and operations management is responsible for, and, that is the operations of that part of an enterprise.
- In order to produce goods services of right quality and quantity at the right time and right manufacturing cost, is the objective of the production management.
- Process is a method with which a product is manufactured. There are number of alternative methods of creating a product.
- Consumer is benefited as due to improved industrial Productivity, increased use value in the product. Products are made available to the consumers him at right place, at right price, at right time, in desired quantity and of desired quality.
- Activities which are listed under production and operations management functions - Location of facilities, Plant layouts and material handling, Product design, Process design, Production and planning control, Quality control, Materials management.
- The process of moving of materials from the store room to the machine and from one machine to the next during the process of manufacture is known as Material Handling.
- Plant layout is a plan of an optimum arrangement of facilities including personnel, operating equipment, storage space, material handling equipment and all other supporting services along with the design of best structure to contain all these facilities.
- Designing is converting ideas into reality, and product design deals with the same. A business organization designs, develop and introduce new products as a survival and growth strategy.
- There are 2 ways in which operation management is viewed: Traditional View and, Modern View.
- Managing operations can be enclosed in a frame of general management function as shown in Figure below. Operation managers are concerned with planning, organizing, and controlling the activities which affect human behaviour through models.
- Planning is thinking before you do. Activities that set a course of action and guide future decision-making is planning. The operations manager defines the objectives for the operations subsystem of the organization, and the policies, and procedures for achieving the objectives.

- Organizing It is a set of activities that identify and establishes a structure of tasks and authority.
- Controlling It is a set of activities which compares the actual performance with the set ones, that assure the actual performance in accordance with planned performance.
- Globalisation means deployment of business facilities and operations around the world. Globalisation can be defined as a process in which geographic distance becomes a factor of diminishing importance in the establishment and maintenance of cross border economic, political and socio-cultural relations.

Activity

1. Go through case study of any organization or visit any to find out how it manages its Production department and Operations department, and what measures does it takes to increase productivity with reduced cost and maintaining good quality of products.

Questions and Exercises

1. What is Production and Operations Management. Describe its role scope and functions.
2. Describe 2 ways in which operations management is viewed. Elaborate the framework of managing operations, and why we need operations management?
3. What are the factors affecting operations management? What are the competitive priorities in operations management? Also, describe its objectives.

Glossary

- Designing is converting ideas into reality, product design deals with the same. A business organization designs, develop and introduce new products as a survival and growth strategy.
- The traditional definition view operation management to be a transformation system. According to the traditional view, operations management is the systematic direction and control of the processes that transform resources known as inputs, into finished goods or services for customers or clients known as outputs.
- Materials management is that aspect of management function which is primarily concerned with the acquisition, control and use of materials needed and flow of goods and services.
- Quality control is an entire collection of activities, which ensures that the operation will produce the optimum quality products at minimum cost.
- The process of moving of materials from the store room to the machine and from one machine to the next during the process of manufacture is known as Material Handling.
- The physical arrangement of facilities refers to as Plant layout. It is the configuration of departments, work centres and equipment in the conversion process.

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- Production management is defined as the performance of the management activities with regards to selecting, designing, operating, Controlling and updating production system and, actual transformation of raw materials into finished products by effectively planning, coordinating and controlling the production.
- Order-winners for a product are the qualities which make an organization receive a new order. Order-qualifiers, in contrast, are required to be able to compete in a marketplace.”
- Planning is thinking before you do. Activities that set a course of action and guide future decision-making is planning.
- Organizing is a set of activities that identify and establishes a structure of tasks and authority.
- Controlling is a set of activities which compares the actual performance with the set ones, that assure the actual performance in accordance with planned performance.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
3. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding

1. d)
2. a)
3. b)
4. d)
5. d)
6. b)
7. c)
8. a)
9. b)
10. a)
11. d)
12. c)

Unit - 1.4: Criteria of Performance for the Production and Operation Management

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Objectives

In this unit, you will be able to

- Understand criteria of performances for operations management
- Learn how aims of performances affects production and operation management

Introduction

Operational management aims that a company must improve at various stages, to meet corporate strategy. For this, a company will identify the relevant operational performance objectives to measure and configure the environment, to enable the objectives to be accomplished. It is important to understand the objectives of operational performance within a manufacturing operation in order to boost production efficiency within your overall operation.

1.4.1 Criteria of Performance for the Production and Operation Management

Each of the production and operations part must be managed correctly, from the strategic planning stage, the execution stage, production monitoring, and even the final assessment of future innovation outcomes, so that the company remains competitive and profitable in its sector.

Three aims of performance of the Production and Operations Management system may be identified. They are,

- a) Efficiency
 - b) Customer satisfaction,
 - c) Effectiveness,
- i) The case of Efficiency is productive utilization of resources is clear. The optimal utilization of resource inputs is always a desired objective, irrespective of the type of organization, whether the organization is in 'private sector' or in the 'public sector', is a 'manufacturing or 'non-manufacturing' organization or a 'profit' or a 'non-profit' organization. The effectiveness has more dimensions to it, as it involves possible prioritization within the objectives, and optimality in the fulfilment of multiple objectives.
 - ii) Modern production and operations management has a set of target customers to serve to, the people working within, the region, country or society at large. Thus Production / Operations Management system, must satisfy many more customers along with being profitable and / or efficient. Depending upon the operations system, this effectiveness has to be again viewed in terms of the short and long-term horizons.
 - iii) Increase in effectiveness of operation system to do things right and broader concept, can be improved by Optimum, Good, Better operations management.

Notes

Effectiveness of operation system refers to doing right things, which is seven rights, they are: Right operation, Right Quantity, Right Quality, Right Supplier or Right Vendor, Right Time, Right Place and Right Price.

Basically, we can measure the efficiency and effectiveness of the operations system by the four dimensions, they are: (i) Cost, (ii) Quality, (iii) Dependability and (iv) Reliability. And, these directly relate to the competitiveness of the organization, both nationally and internationally. "Modern developments in better tools and techniques, methods and systems like Automation, Flexible manufacturing, Computer Aided Design, Production and Operations Management CAM (Computer-aided Manufacturing), CIM at management, Use of Robotics, Total Quality Management, OR Techniques etc, are taking place to achieve improvements in Cost, Quality, Dependability, Reliability and Flexibility and thus to help for better management."

Check Your Understanding

- 1) The aims of performance of the Production and Operations Management system may be identified. They are:
 - a) Efficiency
 - b) Customer satisfaction
 - c) Effectiveness
 - d) All the above
 - e) None of the above
- 2) Modern _____ and _____ management has a set of target customers to serve to, the people working within, the region, country or society at large.
 - a) Production and Operation
 - b) Customer Service and Customer Satisfaction
 - c) Corporate Level and Business level
 - d) Corporate Strategy and Business Strategy
 - e) None of the above
- 3) An organization can measure the efficiency and effectiveness of the operations system by the four dimensions, they are:
 - a) Cost
 - b) Quality
 - c) Dependability
 - d) All the above
 - e) None of the above
- 4) Modern developments in better tools and techniques, methods and systems like:
 - a) Automation,
 - b) Flexible manufacturing
 - c) Computer Aided Design

- d) Computer-aided Manufacturing
 - e) All the above
- 5) Effectiveness of operation system refers to doing right things, which include which right?
- a) Right operation
 - b) Right Customer
 - c) Right Strategy
 - d) Right Cost
 - e) None of Above

Summary

- We can measure the efficiency and effectiveness of the operations system by the four dimensions, they are: (i) Cost, (ii) Quality, (iii) Dependability and (iv) Reliability.
- Modern developments in better tools and techniques, methods and systems like Automation, Flexible manufacturing, Computer Aided Design, Production and Operations Management CAM (Computer-aided Manufacturing), CIM at management, Use of Robotics, Total Quality Management, OR Techniques etc.
- Effectiveness of operation system refers to doing right things, which is seven rights, they are: Right operation, Right Quantity, Right Quality, Right Supplier or Right Vendor, Right Time, Right Place and Right Price.
- The optimal utilization of resource inputs is always a desired objective, irrespective of the type of organization, whether the organization is in 'private sector' or in the 'public sector', is a 'manufacturing or 'non-manufacturing' organization or a 'profit' or a 'non-profit' organization.
- Operations Management system, must satisfy many more customers along with being profitable and / or efficient.

Activity

1. Mention or briefly study aims and objectives of any organizations working for a better Production and Operations management.

Questions and Exercise

1. What is Production and Operations Management. Describe aims of performance of the Production and Operations Management system may be identified.

Glossary

- Operational management aims that a company must improve at various stages, to meet corporate strategy by identifying relevant operational performance objectives to measure and configure the environment, to enable the objectives to be accomplished.
- Effectiveness of operation system refers to doing right things, which is seven rights, they are: Right operation, Right Quantity, Right Quality, Right Supplier or Right Vendor, Right Time, Right Place and Right Price.

Notes

- Efficiency is productive utilization of resources is clear. The optimal utilization of resource inputs is always a desired objective, irrespective of the type of organization, whether the organization is in 'private sector' or in the 'public sector', is a 'manufacturing or 'non-manufacturing' organization or a 'profit' or a 'non-profit' organization

Further Readings

1. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
2. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding

- 1) d)
- 2) a)
- 3) d)
- 4) e)
- 5) a)

Unit - 1.5: Operation Strategies

Notes

Objectives

In this unit, you will

- Learn strategic management process for production and operation
- Explain Hill framework for operations strategy formulation
- Understand core operational strategy areas
- Learn types of operations strategies
- Explain process of operations strategy
- Identify steps to write a strategic operations plan
- Understand operations strategy decision areas
- Learn the six elements of operations strategy are
- Understand importance of operations strategy
- Identify elements of operations strategy
- Learn components of corporate strategy
- Understand Levels Of Strategy
- Identify Linkage between Operations Management with Operations Strategy

Introduction

- An important aspect of operations management which is concerned with long term planning for a company's customer service and business strategies and focuses on the goals and aspirations of the company, as well as the actual plans for getting the business to achieve their goals.
- These strategies support the business in reaching its goals with proper processes and technologies in place. This means that strategic plans should not only include the identification of existing processes and technologies, but also considerations around the supply chain, their customers, their competitors, their own strengths and weaknesses, and the business facilities. An analysis of all the mentioned aspects should be conducted to identify how they support the business strategy, and what needs to be done for the optimization of their usefulness and verify operational capabilities.
- Proper strategic plans should be utilized for every product or service, new and old and during product development, companies can use these plans to make certain that there is an appropriate understanding of the manufacturing strategy and how it ties in to the sales strategy. This helps to make sure that costs are being optimized throughout the process and helps minimize the chances of operations functioning in unexpected ways.
- Updating plans allows for companies to make sure that their competitive priorities have not changed and that they are continually improving their products and services as new information arises. Tracking competitors and their offerings can

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help organizations identify not only the weaknesses of the competitor, but also their own. Organizations can then use this information to refine their strategies.

The set of decisions that are warranted in the operational processes to support the competitive of the business, to develop organizational strategies at three level of operation:

- Corporate level
- Business level
- Functional level



1.5.1 Operations Strategy

Organizations depends upon various strategies to achieve their goals in order to survive in today's competitive business environment. These strategies are carried for long run and carry a broad range of activities.

- Strategy can be seen to exist at 3 main levels:

3 LEVELS OF STRATEGY

Mastering Your Enterprise Strategy



1. Corporate Level Strategy

- Corporate level strategy is about selection of an optimal set of businesses and determining how they should be integrated to make a corporate as a whole.
- At the highest level these strategies provide long-span guidance for the whole organization, and tells, “what a business should do”? It also helps the firm to earn above average profits and create value for shareholders.

2. Business Level Strategy

- Business level strategy is about “How do we compete?” and how to gain competitive advantage over other rivals?
- What products and services that should be offered in the market defined at the corporate level – How do we compete in this business?
- This strategy aims at gaining competitive advantage by offering true value for customers while being unique and hard to imitate player in the market.

3. Functional Level Strategy

- This level strategy is concerned with “How to support business level strategy with functional level?” by improving the effectiveness of company’s operations within the organizational departments.
- These strategies involve setting up short term objectives, realizing which will lead to realization of business level strategy.

Strategic Management Process for Production and Operation

Strategic planning has to trickle down to various function areas of the business, for success of organizational strategic objective. In order to build strategy management, process a sequential process as below is followed:

- 1) Competition Analysis: In this step, current competition in the market and practices that are followed in the industry for operations and production vis-à-vis company policies is evaluated by the company.
- 2) Goal Setting: Next step involves to focus on the objective towards which the organization wants to move towards.
- 3) Strategy Formulation: The next step is breaking down of organizational goals into operations and production strategies.
- 4) Implementation: The final step is to convert operations and production strategies into day-to-day or everyday activities like production schedule, product design, quality management etc.

As organizations are always customer-centric, production and operation strategy for organization are built around them.

- Hill framework for Operations Strategy Formulation
- Operations strategy along with being a part of overall business or corporate strategy, is also crucial for competitiveness and success. Companies fail to keep up with changing markets and lose out to more strategic competitors, without a strong operations strategy.

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- Many companies, big and small, have struggled with operations strategy, often lacking in comparison with technologically savvy competitors. For example, Amazon, while constantly advancing technology such as drones for delivery, has pushed aside myriad brick-and-mortar retailers.
- In order to be effective and competitive, all parts of a company must work together, to be. All departments should contribute to the company mission and have strategies underlying the overall corporate/business strategy. In addition to having an operations strategy, they should also have functional area strategies in finance, IT, sales, marketing, human resources, and possibly other departments, depending on the type of business. "An operations strategy should guide the structural decisions and the evolution of operational capabilities needed to achieve the desired competitive position of the company as a whole," says Tim Laseter in his article "An Essential Step for Corporate Strategy."

Operation Strategy Framework

Corporate Objectives	Marketing Strategy	Order Winning Criteria	Manufacturing Strategy	
			Process Choice	Infrastructure
- Growth - Survival - Profit - Return on investment - Other financial measures	- Product markets and segments - Range - Mix - Volumes - Standardisation vs. customisation - Level of innovation - Leader vs. follower alternatives	- Price - Conformance quality - Delivery speed - Delivery reliability - Volume flexibility - Color range - Product range - Design - Brand image - Technical support - After sales support	- Choice of alternative processes - Tradeoffs embodied in process choice - Role of inventory in process configuration - Make of buy decisions and supply chain management - Capacity size - Capacity timing - Capacity location	- Functional support for operations - Manufacturing planning and control systems - Quality assurance and control - Manufacturing systems engineering - Clerical procedures - Compensation agreements - Work structuring - Organisational structure

Hill (2005) provides an iterative framework that links together the corporate objectives; which provide the organisational direction, the marketing strategy; which defines how the organisation will compete in its chosen markets, and the operations strategy; which provides capability to compete in those markets. The framework consists of five steps:

1. Define corporate objectives
2. Determine marketing strategies to meet these objectives
3. Assess how different products win orders against competitors
4. Establish the most appropriate mode to deliver these sets of products
5. Provide the infrastructure required to support operations

Step 1 - Corporate Objectives

A corporate objective establishes a direction for the organisation and performance indicators that allow progress in achieving those objectives to be measured. The objectives depend on: needs of external and internal stakeholders, and so will include

financial measures such as growth and profit rates, as well as employee practices such as skills development and appropriate environmental policies.

Step 2 - Marketing Strategy

This involves identifying how to compete in the target market. For example: Youngster sport persons are the target market for Adidas, hence their strategy evolves a product range pricing comfortless and value for money which attracts the target segment. Their advertisement and promotional strategies included sport personalities to attract the target age group.

Step 3 - How Do Products Win Orders in the Market Place

In Hill's methodology this is a crucial stage, where mismatches between the requirements of the organization's strategy and the operations' capability are revealed, if there are any. This step imparts the link between corporate marketing proposals and the operations processes and infrastructure necessary to support them.

This is achieved by translating the marketing strategy into a range of competitive factors (e.g., price, quality, delivery speed) on which the product or service wins orders. These external competitive factors provide the most important indicator as to the relative importance of the internal operations performance objectives.

An organization can measure its operation's performance in achieving its strategic goals with the five basic internal operation's performance objectives. These objectives are Quality, Speed, Dependability, Flexibility and Cost. At this stage, the nature of the markets should be clear that operations will serve, by identifying the relative importance of the range of competitive factors on which the product or service wins orders. Hill distinguishes between the following types of competitive factors which relate to securing customer orders in the marketplace.

- Order-winning factors – They are key reasons for customers purchasing the goods or services and raising the performance of the order-winning factor may secure more business.
- Qualifying factors – At certain level performance of qualifying factors must be to gain business from customers, but performance above this level will not necessarily gain further competitive advantage.

Step 4 - Delivery System Choice (Structural Decisions)

Process Types are ways of describing the general approach taken to designing and managing processes. The volume and variety of the products and services that an organisation process are the two important factors in process design, they are based on.

Step 5 - Provide the Infrastructure Required to Support Operations

Planning and Control in Operation is about reconciling market requirements i.e., demand with the operation's resources i.e., supply. Planning is determining a course of steps that should take place in the future. Control is when as the operation is ongoing, determining what action to take, if there is a significant deviation from the planned

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course of action. In reality planning and control activities are closely linked in an ongoing organization with customer always as a priority.

- **Core Operational Strategy Areas**

Different sources use different terms to describe strategy areas. Here's one way to categorize core strategies:

- 1) **Corporate Strategy:** This involves seeing the business as one. Each department of a company depends on other to achieve its organizational goals and desired outcomes. Overall company strategy, drives the company mission and interconnected departments.
- 2) **Customer-driven:** Operational strategies to meet the needs of a targeted customer segment and should include customer-driven approach to meet the needs of the target market.
- 3) **Core Competencies:** Strategies which develop the company's key strengths and resources which include well-trained staff, optimal business locations and marketing and financial expertise.
- 4) **Competitive Priorities:** To better provide a desired product or service, make strategies that differentiate the company in the market. An organization evaluates operational costs, the quality of product or service, time it takes to develop and deliver a good or service and the flexibility of a good or service with regard to variety, volume and customization.
- 5) **Product or Service Development:** Product design, value, and innovation strategies should be considered by the strategies for development of new products or services. In the due course of developing new customer products, a company can wait for the introduction of innovations on the market to improve and wait to see if a company's innovation is successful before moving forward.

- **Operations Strategy Examples**

With the rapidly changing marketplace, some companies have excelled in part due to their strong operations strategies, in recent years. Here a few examples:

Amazon: There was a time Amazon was known for books, but today it is known as the go-to platform for online shoppers of any product. Its distribution network is widely spread and they have even experimented with drone deliveries.

Apple Computers: Apple has been long recognized in operations circles for its operational excellence and supply chain management.

Walmart: This retailing giant has managed to undercut many competitors on the price and variety of a wide range of products.

FedEx: With excellent operations, FedEx made speed of delivery its calling card.

IKEA: The world's largest furniture retailer undercut many homes good.

- **Types of Operations Strategies**

- Operations decisions can be broadly categorized as structure or infrastructure. Structure means the physical attributes of operations, while

infrastructure refers to the people, systems, and software.

- Facilities, capacity to produce, process technology, and supply network are included in Structural decisions. Example: Many companies have to decide that how much they should outsource and how much they should handle in-house. Also, the structural decision on whether to build or expand a facility is an expensive one, which could affect the company for years ahead.
- Planning and control systems, quality management, work organization, human resources, new product development, and performance management of employees, all are included Infrastructure decisions.
- Operations strategy shares a vertical relationship with overall business/corporate strategy, and horizontal relationship with other functional strategies, such as strategies for marketing, sales, finance, IT, and HR.
- Another way to categorize operations strategies is top-down or bottom-up. That is, operations strategies might come down from business strategy, supporting it. Or, strategies might arise over time as a pattern of decisions within operations.
- Also, operations strategy can be market-led or operations-led. Operations strategy derives from a response to the market conditions, when it's market-led. Excellence in operations in a particularly savvy company drives the strategy, when it's operations-led.

Operations Strategy Framework

The ability to improve products, services, and processes is Operations strategy. Business/corporate strategy and a market/needs analysis, are considered to develop a strategy. Then, consider the competing priorities of cost, quality, time, and flexibility — and how you'll handle them.

You will need to have acceptable quality, price, reputation/years in business, and reliability, to exist in the market. To actually win more orders in the market, the factors change a bit. You need winning quality, price, speed of delivery, consistency of delivery, and reliability.

"These factors combine like this to provide an operations strategy framework, as outlined by lean transformation consultant Anand Subramanian:

- Firstly, we start with business/corporate strategy, laying out objectives, such as return on investment (ROI), profit, and growth.
- Next, we move to marketing strategy, wherein we consider factors such as customer segmentation, standardization vs. customization, innovation level, and leader-vs.-follower alternatives.
- Further, comes order-winning criteria such as quality, price, delivery speed, design, and after-sales support.
- Lastly, comes the operations/manufacturing strategy, which includes choices of structure - facilities and process, and infrastructure - planning/control systems and work organization. Feeding into that strategy are the elements of product/process design, inventory, quality management, human resources and job design, and maintenance."

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In a similar line, Slack and his co-authors outlined five performance objectives in their 2004 book, Operations Management:

- Cost: Which includes ability to compete on low price.
- Quality: This includes ability to compete on high quality.
- Speed: This includes ability to compete on fast delivery.
- Dependability: This explains ability to compete on reliable delivery.
- Flexibility: This is the ability to compete with new products/services, wide selection, and timings.

Process of Operations Strategy

- Formulation of operations strategy is the practical process of phrasing various objectives and decisions that make up the strategy.
 - a) It is majorly about different ways of aligning plans, activities and objectives.
 - b) It will be a relatively occasional activity, although operations strategy consideration may form part of the annual planning cycle. Many detailed formulation models have been developed.
- Monitoring and control involves tracking performance, scanning the environment, Interpreting the information that it detects, and respond appropriately.
- Controlling Risks can be controlled through
 - a) Prevention strategies, in which an operation seeks to prevent an event occurring beforehand.
 - b) Mitigating strategies is where an operation seeks eliminate any possible negative consequences from the event.
 - c) Recovery strategies in this an operation not only analyses but also accepts the consequences from an event and further undertakes to minimize, alleviate or compensate for them.
- Steps to Write a Strategic Operations Plan

Below are 15 steps for writing an effective strategic operations plan:

1. Choose the Right People: To compile the operations strategy plan or an operations plan, those with the right knowledge are required. Some businesses provide more strategy than others in their operations strategy plan.
2. Study the Overall Business Strategy Plan: Sometimes the operations strategy plan is included as a section of the overall business plan. In any case, the operations strategy plan should align with the business plan.
3. Develop Measurable Operations Goals: These should match up with the business plan. Don't do KPIs in a vacuum. Ensure that stakeholders have a say and agree to the numbers.
4. Gather Key People to Brainstorm Strategies: Work on strategies and approaches to reach goals and underlying tactics which includes specific steps and tasks to implement the strategy.

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5. Outline Your Major Points to Maintain Your Plan's Focus: To make the process or agenda clear we need to use headings, subheadings, and bulleted lists for clear organization of steps. This should carry over to your fully written plans, providing clear structure and easy scanning. Your plan might also include elements of a SWOT analysis: strengths, weaknesses, opportunities, and threats.
6. Keep Your Audience in Mind: Keeping audience in mind is very important. Write so that the audience should understand it, after all, plan is all about communication.
7. Include an Index: Index should be used for easy scanning of the plan and its sections, as index helps us to easily locate the requisite.
8. Use an Appendix: Use this for supplementary material or for items too detailed for the whole audience.
9. Include the Operations Budget: It is important to include or cross-reference or cross-link budget in the operations strategy plan. Show the rationale for key budget items, especially large expenses.
10. Include a "Stage of Development" Section: Understanding the current state of operations is very necessary along with what you're trying to accomplish and improve. Provide a high-level view of how you make your product, your supply chain, and quality control, further Identify risks and how you'll monitor them.
11. Include a Production Process Section: This goes into detail on the daily production process, and demonstrates that you've worked out the necessary specifics. For manufacturing, you would list plant details, equipment, assets, materials, special requirements, inventory, and quality control steps. You might include prototype and testing details, for a start-up.
12. If Necessary, Divide Other Sections by Product Family: You can also divide them by product, service, or different areas of operations. You might include overall strategies and tactics or consider them by section wise.
13. Use Flowcharts: Use of flowcharts and images and other graphics to make it more easily understandable.
14. Build in Flexibility: Explain how based on a changing market, you might adjust operations.
15. Regularly Monitor Your Goals: You need to regularly monitor your goals in order to see how your strategies and tactics are working. Adjust as necessary to keep ahead of the curve, a strong operations strategy plan is key to your success.

Table: Operations Strategy of McDonald's

Competitive Priorities			
Cost Reduction	Less Serving Time	High Quality	Flexibility
Use of efficient equipments	Fast, accurate, friendly services	5 Ps policy (product, price, promotion, place, people)	Mix flexibility

Notes

Direct Purchase from Farmer	Ready to serve products/ services	Quality centres	Product flexibility
Low-cost supply chain			Volume flexibility
Just-in-time strategy			

- Operations Strategy Decision Areas

1. Capacity Strategy

- This strategy concerns with how to configure capacity and facilities in general.
- 'What should be the overall level of capacity?', 'How many sites should the capacity be distributed across, and what size should they be?' are certain questions it includes.
- 'Should each site specialize in one or two activities or should it engage in a broad mixture of activities?'
- 'Exactly what should be the location of each site?', 'When should changes be done to overall capacity levels?'
- 'How big the change should be to each capacity?'
- 'Capacity expansion or reduction should be pursued at what pace?'; and many more.

2. Supply Network Strategy

- This concerns, how the operations are related to its interconnected network of other operations, including customers, customers' customers, suppliers, suppliers' suppliers, and so on.
- Should we attempt to manage the network in different ways depending on the types of market we are serving?; 'What is the number of suppliers that we must have?'
- 'Whether the nature of our relationship with our suppliers, purely be market-based or long-term partnerships?', 'Describe appropriate ways of managing different types of supplier relationships?'

3. Process Technology Strategy

- This concerns the choice and development of the systems, machines and processes that act directly or indirectly on transformed resources to convert them into finished products and services
- Importance of Operations Strategy
 - Designing of the production system
 - Facilities for production and services
 - Product or service design and development
 - Technology selection, development, and process development
 - Allocation of resources

6. Focus on facilities planning
1. Designing of the Production System: - Selection of the type of product design, processing system, inventory plan for finished goods, etc. are involved in designing of the production system. The product design has two varieties. • Customized product design • Standard product design. There are two types of production systems. They are product focused and process focused. Product-focused system is adopted where their mass production is done by using a group of machines. For example, products like automobiles, computers, etc. Process based system is an approach which views a business as a collection of processes.
2. Facilities for Production and Services: - Having certain specialization in production allows the firm to provide the customers with products of lower cost, faster delivery, on-time delivery, high product quality, and flexibility.
3. Product or Service Design and Development: - • Generating ideas • Creating the feasibility reports • Designing the prototype and testing • Preparing a production model • Evaluating economies of scale for production • Product Testing in the market • Getting feedback • Creating final product design and starting the production
4. Technology Selection and Process Development: - Once a product is selected an appropriate process and applicable technology is selected for an optimal production. As there are many alternatives, there are many challenges faced by an operations manager in this decision. The techno-economic analysis is required for each alternative which will help to decide the required technology.
5. Allocation of Resources: - The production units face continuous problems of allocating the scarce resources like capital, machines, equipment, materials, manpower, services, etc. Allocation at the right place and the time of production indicates the efficiency of the production planners. Optimum utilization of resources will enable economical production with minimized waste, optimum utilization of resources, and the best quality product demand - a sound operations strategy.
6. Facility, Capacity and Layout Planning: - The key decision areas for the operations manager are the location, layout, and facilities creation for the production, and are critical for achieving the competitiveness. The decision also influences the future expansion of the plant. While evaluating the alternatives, the operations manager will consider the raw materials available, access to market, etc. Enormous capital requirement is will be there and the planning is always long range. Here, the production process adopted and the technology pursued depicts the volume, quality, and cost of production.

Notes

The four elements of operations strategy (From Nigel Slack (2017) The Operations Advantage, Kogan Page.)



- Elements of Operations Strategy
 - Operations strategy is a top-down reflection of what the whole group or business wants to do. An operations strategy must reflect the decisions taken by the top management of an organization and which set the overall strategic direction in which an organization moves. This is called a 'top-down' approach to operations strategy. So, if the organization is a large, diversified corporation, its corporate strategy will consist of decisions about what types of business the group wants to be in, in which parts of the world they want to operate in, what businesses to acquire and what to divest, how to allocate its cash and other resources between its various businesses, and so on. Within the corporate group, each business unit will also need to put together its own business strategy, which reflects its individual mission and objectives, and how it intends to compete in its markets.
 - Operations strategy is a bottom-up activity where operations improvements cumulatively build strategy. In this the whole organization participates in the process of leading the organization. Employees are invited to participate in the goal setting – sometimes by feedback, or by stake in decision. This can lead to better engagement, performance and productivity as employees will be more aligned to the company's culture, mission and values. If an unforeseen event comes up during projects, targets are shifted through an open line communication between business executives and low-ranking employees. Hence employees can set their own targets and work towards achievement of organizational goals.
 - Operations Strategy is an inside-out activity which focuses on strengthening and building up internal strengths and abilities. In this, the company will take into account its own resources and make the best possible use of them efficiently. It actually focuses on what company is good at or what they love to do, what they stand for and who they are. For Example: Apple, they have clearly defined who they are and what they are good at, that they can get mobiles with such innovations that people wait for their latest launches time and time again.
 - Outside-in strategy focuses on the external, in which the company directs

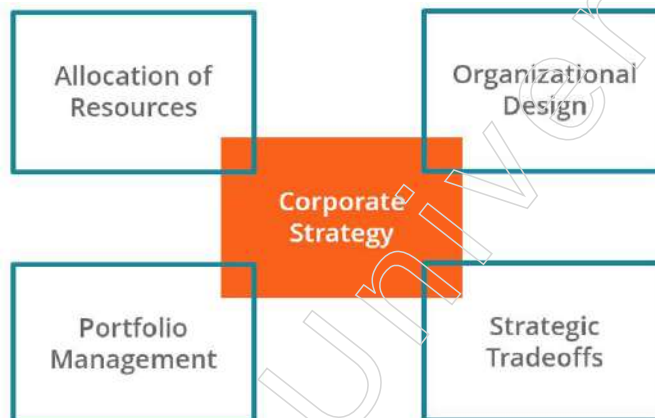
its attention towards the customers, concentrating towards high customer value. They put themselves in place of the customers and view themselves from their perspective. It related to how they can expand the demand of their products and services by tuning into customer's needs, by keeping up to their changing needs. For Example: Amazon, they started with online book store, but they went beyond what they do and asked "what customers really want?" "And, now they are big providers of cloud computing and web services and opened a broader array of opportunities.

- Components of Corporate Strategy

There are several important components of corporate strategy that leaders of organizations focus on. The main tasks of corporate strategy are:

1. Allocation of resources
2. Organizational design
3. Portfolio management
4. Strategic trade-offs

In the following sections, let's go through them.



1. Allocation of Resources

A firm usually focuses on allocation of two resources: people and capital. In an effort to maximize the value of the entire firm, leaders must determine how to allocate these resources in order to grow the complete rather than a part of it.

Key factors related to the allocation of resources are:

- People
 - o Identify the core competencies, ensuring they are well distributed across the firm.
 - o Moving leaders to the places they are needed most and add the most value which changes over a period of time, based on priorities.
 - o To ensure that an appropriate supply of talent is available to all businesses.
- Capital
 - o Allocating capital across businesses so it earns the highest risk-adjusted return

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- o Analysing external opportunities, mergers and acquisitions and allocating capital between internal (projects) and external opportunities.



2. Organizational Design

In order to create the maximum amount of value, organizational design needs to have a necessary corporate structure and related systems in place. Factors that leaders must consider are the corporate head office role and the reporting structure of individuals and business units – vertical hierarchy, matrix reporting, etc.

Key factors related to organizational design are:

- Head Office (Centralized vs Decentralized)
 - o Determining how much power to give business units.
 - o Decide whether to make decisions top-down or bottom-up.
 - o Influence on the strategy of business units.
- Organizational Structure (Reporting)
 - o Determine how to divide large initiatives and commitments into smaller projects.
 - o Integrating business units and business functions such that there are no redundancies.
 - o Allowing for the balance between risk and return to exist by separating responsibilities.
 - o To develop centres of excellence.
 - o Appropriate delegation of authority.
 - o Setting governance structures.
 - o Setting reporting structures (military / top-down, matrix reporting)

3. Portfolio Management

Portfolio management looks at the way business units complement each other, their correlations, and decides where the firm will “play” (i.e., what businesses it will or won’t enter).

Corporate Strategy related to portfolio management includes:

- Deciding what business to be in or to be out of.
- Determining the extent of vertical integration, the firm should have.
- Managing risk through diversification and reducing the correlation of results across businesses.
- To Create new strategic options by seeding new opportunities that could be heavily invested in, if appropriate.
- Monitoring the competitive landscape and ensuring the portfolio is well balanced relative to trends in the market

4. Strategic Trade-offs

Balancing the trade-offs between risk and return across the firm is one of the most challenging aspects of corporate strategy. It's important to have a holistic approach of all the businesses combined and ensure that the desired levels of risk management and return generation are being pursued.

Below are the main factors to consider for strategic trade-offs:

- Managing Risk
 - o Firm-wide risk is largely depending on the strategies it chooses to pursue
 - o True product differentiation, for example, is a very high-risk strategy that could result in a market leadership position or total ruin
 - o Many companies adopt a copycat strategy, by looking and learning from what other risk-takers have done and modify it slightly to benefit the organization.
 - o It's important to be fully aware of strategies to be adopted and associated risks across the firm.
 - o Some areas might require true differentiation (or cost leadership) but other areas might be better suited to copycat strategies that rely on incremental improvements.
 - o The degree of autonomy business units have is also very important in managing this risk.
- Generating Returns
 - o The possibility of higher rates of return are created by higher risk strategies. The examples above of true product differentiation or cost leadership could provide the most return in the long run if they are well executed.
 - o Swinging for the fences will lead to more home runs and more strikeouts, so it's important to have the appropriate number of options in the portfolio. These options can later turn into big bets as the strategy develops.
- Incentives
 - o Incentive structures plays a big role in how much risk and how much return managers seeks.
 - o The responsibilities of risk management and return generation, are necessary to be separated so that each can be pursued to the desired level.

Notes

- o This further helps to manage multiple overlapping timelines, ranging from short-term risk, to return to long-term risk, return and ensuring there is appropriate dispersion.
- Linkage between Operations Management with Operations Strategy
 - The various functions of the business which includes, operations, marketing, and finance etc., make long-range plans in order to support the competitive advantage being pursued by the business strategy - How does the function contribute to the business strategy? This means how the functions are aligned to attain the business level strategy.
 - The relationship of operations management with other functional areas in the overall organizational strategy is important to consider. The cost of goods sold is not the only area where costs controls take place. The business leaders need to consider all levels of costs from operations, sales, administrative and distribution. Strategies must match the overall mission and vision of the company.
 - For example, a wallet's manufacture has a mission to provide higher quality handmade wallets, then bulk automation with machines would not fit into this strategy. The Cost of Goods Sold may be more expensive because cost of labour in operations increases. In that case the company may have to increase the price of the products or to reduce costs elsewhere in the company, may be in the administrative areas.
 - Another example is that, if the company wants to expand and for that needs to increase the production. In this instance, costs will increase, but with proper operations management, double production won't mean that the COGS pricing will not double. The organizational strategy here is to scale up the production and sales in a way that enables a greater profit per-unit ratio, because production has achieved greater efficiency. It is very important to consider the effect of that strategy to the Cost of Goods Sold; it must also consider what the effect of that strategy is on the overall bottom line of the company. Every company has a goal to be, as profitable as possible and, having good operations management provides the efficiency to achieve the same.

Linkage Corporate & Operations Strategy



Check Your Understanding

1. Which of the following steps is not included in the strategic management process for production and operations?
 - a) Competitive Analysis
 - b) Goal Settings
 - c) Infrastructure
 - d) Implementation
 - e) None of the above
2. The core operational strategies include:
 - a) Corporate Strategy
 - b) Customer Driven
 - c) Competitive Priorities
 - d) Core Competencies
 - e) All the above
3. Which of the following are the elements of Operations Strategy?
 - a) Top-down
 - b) Bottom-up
 - c) Inside-out
 - d) Outside-in
 - e) All the above
4. Key factors of Allocation of resources include:
 - a) Organizational Structure
 - b) Managing Risks
 - c) People
 - d) Returns
 - e) None if the above
5. The levels of Strategy include:
 - a) Corporate level Strategy
 - b) Business Level Strategy
 - c) Functional Level Strategy
 - d) All the above
 - e) None of the above
6. If the company wants to expand and for that needs to increase the _____.
 - a) Cost
 - b) Production

Notes

Notes

- c) Supply
d) Profit
e) All the above
7. _____ strategy focuses on the external, in which the company directs its attention towards the customers, concentrating towards high customer value.
- a) Outside-In
b) Inside-out
c) Business level
d) Operational level
e) None of the above
8. In which strategy, an operation seeks to prevent an event occurring beforehand.
- a) Mitigation Strategy
b) Prevention Strategy
c) Recovery Strategy
d) Functional Strategy
e) Business strategy
9. _____ involves to focus on the objective towards which the organization wants to move towards.
- a) Manufacturing Strategy
b) Marketing strategy
c) Competitive Analysis
d) Goal Setting
e) None of the above
10. _____ involves identifying how to compete in the target market.
- a) Win Orders
b) Marketing Strategy
c) Qualifying Factors
d) Structural Decision
e) None of the above

Summary

- An important aspect of operations management which is concerned with long term planning for a company's customer service and business strategies.
- A corporate objective establishes a direction for the organisation and performance indicators that allow progress in achieving those objectives to be measured.
- Operations decisions can be broadly categorized as structure or infrastructure.

Structure means the physical attributes of operations, while infrastructure refers to the people, systems, and software.

- Planning and control systems, quality management, work organization, human resources, new product development, and performance management of employees, all are included Infrastructure decisions.
- Understanding the current state of operations is very necessary along with what you're trying to accomplish and improve.
- The product design has two varieties. • Customized product design • Standard product design.
- Having certain specialization in production allows the firm to provide the customers with products of lower cost, faster delivery, on-time delivery, high product quality, and flexibility.
- The key decision areas for the operations manager are the location, layout, and facilities creation for the production, and are critical for achieving the competitiveness.
- Balancing the trade-offs between risk and return across the firm is one of the most challenging aspects of corporate strategy.
- Many companies adopt a copycat strategy, by looking and learning from what other risk-takers have done and modify it slightly to benefit it slightly.

Activity

1. Visit any organization and write its operation strategy. Also describe its complete framework it works in for managing operations.

Question and Exercises

1. Describe the steps to write a strategic Operation plan.
2. What is Strategic Management? What are the elements of Operations strategy and what is its importance?
3. What is Hill's framework for Operations Strategy Formulation?

Glossary

- Inside-out: It is an activity which focuses on strengthening and building up internal strengths and abilities. In this, the company will take into account its own resources and make the best possible use of them efficiently.
- bottom-up: It is an activity where operations improvements cumulatively build strategy. In this the whole organization participates in the process of leading the organization.
- Inside-out: It is an activity which focuses on strengthening and building up internal strengths and abilities. In this, the company will take into account its own resources and make the best possible use of them efficiently.
- Outside-In: It is a strategy focuses on the external, in which the company directs its attention towards the customers, concentrating towards high customer value.

Notes

They put themselves in place of the customers and view themselves from their perspective.

- Corporate level strategy: It is about selection of an optimal set of businesses and determining how they should be integrated to make a corporate as a whole.
- Business level strategy: It is about “How do we compete?” and how to gain competitive advantage over other rivals?
- Functional Level Strategy: This level strategy is concerned with “How to support business level strategy with functional level?” by improving the effectiveness of company’s operations within the organizational departments.
- Prevention strategies: These are the strategies in which an operation seeks to prevent an event occurring beforehand.
- Mitigating strategies: These are the strategies where an operation seeks eliminate any possible negative consequences from the event.
- Recovery strategies: In these strategies an operation not only analyses but also accepts the consequences from an event and further undertakes to minimize, alleviate or compensate for them.

Further Readings

1. A Study of Factors Affecting the India’s Service Sector: An Econometric Analysis - Priya lashmi, Dr. Sandeep Kumar.

Check Your Understanding

1. c)
2. e)
3. e)
4. c)
5. d)
6. b)
7. a)
8. b)
9. d)
10. b)

Unit - 1.6: Effects of Growth of Service Sector on Operations Management

Notes

Objectives

In this unit, you will

- Learn what is service economy
- Understand benefits of operations management in service industry
- Explain factors responsible for growth of service sector
- Identify challenges and issues in service marketing
- Understand service life cycle
- Learn strategies for influencing customer perception

Introduction

Meaning of Service Economy

Service economy is an economy where economic activity is provision of services rather than production of goods. The size of the service sector is increasing in virtually all countries around the world. In developing economies, the service output is growing rapidly and often represents at least half of the GDP. Thus, Service economy is growing. The service economy in developing countries like India is mostly concentrated in financial services, health, and education.

Examples of services:

A shoe salesman sells shoes to the customer, and does not design or manufacture them.

A doctor does not manufacture medicines, but only provides diagnosis for diseases or injuries.

1.6.1 Effects of Growth of Service Sector on Operations Management

A smooth operations management process has many benefits for an organisation, including:

1) Product/Service Quality

- To check that the products are of best quality, we need to have a checklist that meets the objectives and goals of the company as well as meeting the customer's needs. There will be a set list of processes and a checklist to determine that everything is in order during the production process, which the Operations manager will do. This is necessary to ensure that everyone is aware of customer's needs and organizational objectives.
- Once the final product has been created, the operations manager will assess to ensure it meets the organisation goals and that the customer's needs are met.

Notes

Thereafter, the operations manager will review the pre-production process to ensure efficiency for the next creation.

2) Customer Satisfaction

- A customer's feedback can make or break a business. If the feedback is negative or customer's experience is negative then it could be a challenge to retain clients. That is why it is necessary to ensure that the customers' needs are at the forefront of your product or service.
- Quality management process will be deployed by the operations manager, which ensures that the products and service will meet the customers' needs. If the organisation is a service provider, the customer is the lifeblood. Then the operations manager will have processes in place to make sure that the service quality is not at all compromised and all are at the best. A returning customer means more for the bottom line.
- The elements used to help gain a satisfactory customer includes:
 - Quality Management: This ensures that the organisation maintains a consistent and good service.
 - Employee Capacity: This means that right people are placed in the right roles, which helps produce a good product/service.
 - Planning: This is done to ensure that there is no lapse in production and that goods/services are on time.
 - Enough Inventory: This should be maintained, to keep up with customer demand.

With these four elements in mind, the operations manager can meet the customers' needs.

3) Revenue Increase

An organisation having great product/service quality and customer satisfaction will have a goodwill or good image. This leads to increased revenue from a new customer base. As the revenue grows, it will have funds to launch new and innovative products/services or an increase in resources and technology.

4) Competitive Advantage

An effective operations management leads the business could be ahead of its competitors. If internal and external factors are managed well within an organisation, it could mean a good standing within the market.

5) Compliance

Every organization need to adhere to certain rules and regulations an organisation needs to adhere to. The operations manager will have certain controls in place to avoid fines and to make sure the organisation is running within a lawful manner.

6) Motivated Employees

Overall, the operations manager ensures that employees are aware of their roles within a company, which is very important as the employees may feel left out and

demotivated if they feel they're not contributing in a meaningful way. An operations manager helps define these roles to ensure that production is maximised and efficient.

Notes

Hence, Operations Management in Service:

- Operation management is a process that involves planning, organizing, managing, controlling and supervising the production and manufacturing processes. The major aim of an operation manager is to ensure timely delivery of the products and to successfully turn the raw materials into the finished products (input to output). Operations management plays a vital role to run any project successfully. Its benefits include:
- Operation management relates to similar management for every industry or business irrespective of their nature of the operation. Planning, organizing, staffing, monitoring controlling, directing and motivating are its significant elements. Operation management is obligatory for organizations to manage the daily activities smoothly.
- Operation management is the management of the various business activities that take place within an organization and contributes in making the products to align with customer's requirements. Operation management is the heart of an organization as it controls the entire operation. If the services are catering to the needs of the customers, then, they'll be sold at a rapid rate.
- Factors Contributing to the Growth of Service Sector
 1. High Demand for Services as Final Product: India was a new market for service sector. As a result of business outsourcing from developed countries to India, service sector started booming and there was very high demand for these services especially for banking, computer service, advertisement and communication. This high demand in turn led to a high growth rate of service sector.
 2. Liberalization and Economic Reforms: Growth of Indian service sector is also attributable to the liberalization and various economic reforms that were initiated in 1991. As a result of these reforms, various restrictions on the movement of international finance were reduced. Hence, various MNC's entered in India. This led to huge inflow of foreign capital, foreign direct investments and outsourcing to India. This also started making many changes in Indian market and encouraged the service sector growth.
 3. Structural Transformation: Indian economy experienced structural transformation that implies shift of economic dependence from primary to tertiary sector. Due to this transformation, there was increase in demand of services by other sectors which boosted the service sector.
 4. Advanced Technology and Growth of IT: IT sector new innovations and advancements enabled the use of internet, telecommunication, mobile phone and electronic transactions across different countries. They helped perform vital service businesses of the country. Highly skilled software resources are also found in India. They also contributed to the growth of the service sector in India.
 5. Increased Volume of Trade: Low tariff and non-tariff barriers on imports by

Notes

India are also responsible for high growth rate of service sector. The foreign trade reforms enabled the domestic products to interact and compete in the international markets.

6. Cheap Labour and Reasonable Degree of Skill in India: As the labour available in India was cheap and reasonable degree of skilled man power, outsourcing for developed countries to India became feasible and profitable. The business outsourcing in itself provides substantial encouragements (like development of human capital that requires services like good coaching centres and reputed institutions, etc.) for the growth of service sector.

- Challenges and Issues in Service Marketing

(a) Tangibility

A product is tangible, means the customer can touch and see the product before deciding to make a purchase. Packaging and presentation may attract the customer and compel them to purchase a product. Services, on the other hand, are not tangible, which can make them more difficult to promote and sell than a product.

(b) Relationship and Value

Customer's needs are fulfilled by products, so companies can use this to sell a product. A service is not about selling tangible products but is about selling a relationship and the value of the relationship between the buyer and seller of the service. For example, a car is something a buyer can touch and see as well as use it. A service, such as lifestyle coaching, for example, is not tangible. A lifestyle coach may be able to assist clients in creating a life plan and implementing steps to transform his life into one that the client wants to live, but it is not tangible that the client can touch or look at. Therefore, the client needs to perceive the value of the service, which can be harder to get across.

(c) One Versus Many

Multiple products are involved in marketing products, that makes up the line. For example, cleaning product manufacturers has just not one cleaning product but has a line of cleaning products to market, to serve the various needs of their customers. Services, on the other hand, typically have a single option. It can be harder to promote and sell the reputation of one single service over the benefits of many different products.

(d) Comparing Quality

Measuring the quality of service is as not easy as measuring quality of Products. If a customer buys a cleaning product to clean the kitchen sink and it doesn't do product, then the product is zero. On the other hand, it is harder to measure the quality of a service, as we may have to wait longer to see the effects of services done or it may also depend up our satisfaction levels which is again not measurable.

(e) Return Factor

If a customer purchases a product, and is not satisfied or the product is damaged, then it can be returned and customer can get a money back or at least to receive a

store credit. On the other hand, once a service is consumed as it is offered, so it lacks the return factor that a product has. Some service providers still overcome this by offering money-back guarantees.

- **Service Life Cycle**

The service life cycle has the same four stages as that of the product life cycle: introduction, growth, maturity and decline. The characteristics of each stage are the same. The only difference lies in the strategies that can be used.

I) Introductory Stage

A new service is said to be at the introductory stage, when it is first offered. As the case with goods, if the new services do not obtain acceptance by customers, then they never get past the first stage of the service life cycle. An advantage that services have over goods is that many new services can be introduced on a small scale and expanded if being appreciated by the market. This small-scale introduction reduces the financial risk associated with the introduction, making failure less costly. At this stage, strategies should be made

II) Growth Stage

During the growth stage when the industry is growing rapidly, most of the firms offering new services gets a positive cash flow. The price of services keeps increasing. Thus, results in increase in competition. The service providers must develop combination of features that allows it to exceed its competitors.

For e.g.: patient can learn about the incubation period for chicken pox by either talking to a nurse or dialling into a vast library of pre-recorded tapes. Second, patients can seek advice about routine illnesses such as congestion or abdominal pain.

III) Maturity Stage

During the maturity stage, industry sales level off. Competition becomes very high that the only way to survive is to gain the market share or increase sales is to take them away from a competitor. The competition is so fierce that only the strong survives. The result of this increased competition is a decrease in overall industry profits. Weaker firms will be shaken out of the industry. At this stage in the service life cycle, consumers can see very few distinguishable characteristics among the various firms in a service industry.

IV) Decline Stage

During the decline stage, industry sales totally decline and the revenue generation stops. This sales drop is often due to the current one has been abandoned and a new technology has been developed. For e.g., typewriter sale and repair services declined because typewriters were largely replaced by computers, which resulted in a need for a computer service technicians and computer programmers.

- **Strategies For Influencing Customer Perception**

1. **Measure and Manage Customer Satisfaction Service Quality:**

Notes

We need to track the trends, diagnose problems, and link to other customer focused strategies.

2. Aim for Customer Quality and Satisfaction in every Service Encounter.

Every service encounter is critical to customer retention. Therefore, many firms aim for zero defects or 100% satisfaction.

Clear documentation of all the points of contact between the organisation and its customers. And, development of understanding of customer expectation by developing appropriate strategy.

3. Plan for Effective Recovery

When customers have been disappointed with the services on the first try, then making it right the second time is very essential to maintain customer loyalty. This implies there is a need for service process and system analysis to determine the root cause of failure should be done and redesign the service system.

4. Visibility

Being visible is very important to stay in people's mind. Strategies should be made to make cost-effective methods to get the right visibility. Free available information for download, t-shirts with company logo, e-greetings and inspirational quotes are a few ways which can get you the right visibility.

5. Encourage Spontaneity:

Recruitment and selection procedures can be used to hire right employees, which means employees with strong service orientation, whose natural tendency is to be service minded. Strong service culture, employee empowerment, effective supervision and monitoring should be done.

6. Help Employees Cope with Problem Customers:

Employees need appropriate training and tools to deal with problem customers. Training the customers is essential so that they know what to expect and know the appropriate behaviour in the situations.

7. Manage the Dimensions of Quality at the Encounter Level:

Whenever the customer is encountering the service, the service firm should take appropriate measures to maintain the dimensions of service quality. As if the quality is bad it will be very difficult to keep up with customer perception about the company. Hence, strategies should be made to improve the same.

- In reality, the role of operations management is crucial to any business.

Imagine this: You are a company which manufactures plastic and supply various companies with their packaging. And there is a new client and they are looking forward to launch their products in a month or so. Where will you start? The operations manager will ensure that the correct budget is being allocated for the same, the right people are on the job, and also everyone involved is aware of the roles they play, which will further ensure that the deadline are met, within budget.

Employee engagement activity also improves with effective operations management, and defines the roles and responsibilities of the employees within an organisation. A strategic operations management plan in place ensures that employees' workflow is smooth and company's production remain unaffected, no matter whatever obstacles may come for an organization to face.

Notes

Check Your Understanding

- 1) Which of the following are the issues in service marketing?
 - a) Relationship and Value
 - b) One versus Many
 - c) Tangibility
 - d) Comparing Quality
 - e) All the above
- 2) _____ is an economy where economic activity being provision of services rather than production of goods.
 - a) Free Market Economy
 - b) Service economy
 - c) Capitalist Economy
 - d) Socialist Economy
 - e) None of the above
- 3) _____ ensures that the organization maintains a consistent and good service.
 - a) Employee Capacity
 - b) Quality Management
 - c) Planning
 - d) Inventory
 - e) All the above
- 4) Every organization need to adhere to certain rules and regulations an organisation needs to adhere to, known as _____.
 - a) Operations
 - b) Inventory
 - c) Competitive Advantage
 - d) Compliance
 - e) None of the above
- 5) _____ is done to ensure that there is no lapse in Production and that goods and services will be delivered on time.
 - a) Planning

Notes

- b) Organizing
- c) Controlling
- d) Quality Control

Summary

- In developing economies, the service output is growing rapidly and often represents at least half of the GDP.
- Low tariff and non-tariff barriers on imports by India are also responsible for high growth rate of service sector.
- Growth of Indian service sector is also attributable to the liberalization and various economic reforms that were initiated in 1991. As a result of these reforms, various restrictions on the movement of international finance were reduced.
- The service life cycle has the same four stages as that of the product life cycle: introduction, growth, maturity and decline.
- Recruitment and selection procedures can be used to hire right employees, which means employees with strong service orientation, whose natural tendency is to be service minded.
- Operation management relates to similar management for every industry or business irrespective of their nature of the operation.

Glossary

- Service Economy: Service economy is an economy where economic activity is provision of services rather than production of goods.
- Tangible: A product is tangible, means the customer can touch and see the product before deciding to make a purchase.
- Employee capacity: This means that right people are placed in the right roles, which helps produce a good product/service.
- Operation management: It is a process that involves planning, organizing, managing, controlling and supervising the production and manufacturing processes.

Further Readings

1. Effect Of the Growth of The Service Sector on Operations Management In a Developing Economy - Peter Kamoni and Dr. David Kiarie

Check Your Understanding

- 1) e)
- 2) b)
- 3) b)
- 4) d)
- 5) a)

Module 2 – Demand Forecasting

Notes

Structure:

- 2.1 Qualitative Forecasting Methods
 - 2.1.1 Educated Guess
 - 2.1.2 Consensus
 - 2.1.3 Delphi Method
 - 2.1.4 Historical Analogy
 - 2.1.5 Market Research
- 2.2 Quantitative Forecasting
 - 2.2.1 Linear Regression
 - 2.2.2 Moving Average
 - 2.2.3 Weighted Moving Average
 - 2.2.4 Exponential Smoothing
- 2.3 Analytical Tools For DSS For Operations Management
 - 2.3.1 Analytical Tools For DSS For Operations Management

Notes**Unit - 2.1: Qualitative Forecasting Methods****Objectives**

At the end of this unit, you will be able to:

- Understand what is forecasting
- Understand forecasting horizons
- Learn types of forecasts
- Understand applications of forecasting
- Identify general steps in the forecasting process
- See the role of forecasting in operations management
- Learn why is forecasting important
- Determine what to forecast
- Learn the basic steps in forecasting task
- Evaluate the forecast accuracy
- Learn the forecasting models
- Understand the concept of educated guess
- Learn qualitative forecasting techniques
- Understand quantitative methods

Introduction**Forecasting**

- Forecasting is a process which estimates the future event by casting forward past data. There is a predetermined way to obtain the estimate of the future, in which the past data is systematically combined. Now, let's understand what is prediction. Prediction is a process in which future events are estimated, based on subjective considerations other than just past data; and these subjective considerations need not be combined in a pre-planned way. Therefore, forecasting is estimating the future values of certain specified indicators relating to a decisional/planning situation. In certain situations, forecasting related to single indicator is sufficient, whereas, in some other situations forecast regarding several other indicators is necessary. The intended use of the forecast decides the number of indicators and the degree of detail required in the forecast.

- Forecasting is the process of making predictions on the basis of past and present data for the future based activities. This is most commonly by analysis of trends. A commonplace example might be estimating a variable of interest at some specified future date. Prediction is a similar, but more general term. Both might refer to formal statistical methods making use of - time series, cross-sectional or longitudinal data, or alternatively to less formal judgmental methods. Usage of forecast and prediction can differ between areas of application: for example, in hydrology, the terms “forecast” and “forecasting” is referred to as estimates of values at certain specific future times, while the term “prediction” is used for more general estimates, such as how many times or the number of times floods will occur over a long period.
- Risk and uncertainty are central to forecasting and prediction; and it is very important and a good practice indicate the degree of uncertainty attached to specific forecasts. In any case it is very much required that, the data should be up to date in order for the forecast to be as accurate as possible. In some cases, the data used to predict the variable of interest is itself forecasted.
- In this module, we explore forecasting in more detail, as there are many choices that can be made in developing a forecast. Accuracy is very much required and is very important when it comes to forecasts. If the demand of the product is overestimated, then the company could end up spending money on manufacturing, distribution, and servicing activities it won't need. For example: Data Impact, a software company, recently overestimated the demand for one of its new products. As the sales of the product didn't meet projections. The company lacked the cash available to pay to its vendors, utility providers, and others. As a result, Employees had to be terminated in many areas of the firm to trim costs.
- As you saw in case of overestimating, underestimating demand can also be just as devastating. A company launches marketing and sales campaigns to create demand for the product, whenever a company introduces a new product. But if the company can't deliver the amount of the product the market demands, then other competitors may steal sales, the firm might otherwise have captured. For Example: Inability of Sony into deliver the e-Reader in sufficient numbers made Amazon's Kindle more readily accepted in the market and their sales increased.
- The firm has to do more than just forecast the company's sales, which can be a complex process, because how much the company can sell will depend on as many factors such as – what will be the product cost, how competitors will react, and so forth. Hence, all these factors need to be taken into account while determining how much the company is likely to sell. As factors change, the forecast has to change as well. Thus, a sales forecast is actually a composite of a number of estimates and has to be dynamic as those other estimates change.
- The very first step is to determine market potential, or total industry-wide sales expected in a particular product category for the time period of interest, in this time period of interest might be the coming year, quarter, month, or any other time period. Nielsen, Gartner are some marketing research companies, which first estimate the market potential for various products and then sell that research to companies who produces those products.

Notes

- Once the firm has an idea of the market potential, the company's sales potential can be estimated. A firm's sales potential is the maximum total revenue it hopes to generate from a product or the number of units of it the company can hope to sell. The sales potential for the product is typically represented as a percentage of its market potential and equivalent to the company's estimated maximum market share for the time period. In your budget, you'll want to forecast the revenues earned from the product against the market potential, as well as against the product's costs.

Forecasting Horizons

The forecast horizon is the length of time or time period into the future for which forecasts are to be prepared. They vary from short-term forecasting horizons, which can be less than three months to long-term horizons, which can be more than two years.

- 1) Long term forecasting tends to be completed at high levels in the organization. The time frame for long term forecasting is generally considered longer than 2 years into the future. It requires a detailed knowledge about the products and markets due to the high degree of uncertainty. This is usually the case with new products entering into the market, emerging new technologies and opening new facilities. It is often that no historical data is available.
- 2) Medium term forecasting is one in which the time frame generally is for several months, up to 2 years into the future and is referred to as intermediate term and, both quantitative and qualitative forecasting methods may be used in this time frame.
- 3) Short term forecasting is almost done daily up to months in the future. These forecasts are used for operational decision making such as ordering and scheduling, inventory planning of the workforce. Usually, quantitative methods such as time series analysis are used in this short time frame.

Types of Forecasts

- 1) Economic Forecasts: This is a process in which future condition of an economy is predicted. They help form the basis of planning by describing the expected future behaviour of all parts or a part of an economy. It Predicts a variety of economic indicators, like money supply, inflation rates, interest rates, etc.
- 2) Technological Forecasts: It is an attempt to understand the potential direction, rate, characteristics, and effects of technological change, especially invention, innovation, adoption, their use. It is an important tool to anticipate and understand the pace of technological progress, innovation and effects of technological change on business, or whether this change will reach or exceed the key parameters of performance by future date.
- 3) Demand Forecasts: It is very important to predict the future demand for a company's products or services, so that production can be done accordingly, without any wastage. Proper demand forecasting gives businesses valuable information about their potential in their current market and other markets. This helps managers to make decisions about pricing, strategies related to business growth, and market potential. Without demand forecasting or poor forecasting will risk business by making poor decisions about their products and target markets, and this will further

lead to negative effects on inventory holding costs, customer satisfaction, supply chain management, and profitability.

Some Applications of Forecasting:

Forecasts are important to every business organization and for every significant management decision. Now, we will put light on some areas in which forecasting is widely used.

1) Sales Forecasting

A company needs to forecast the demand of the goods it is producing to sell, as manufacturers need to know how much to produce. Wholesalers and retailers need to know how much to stock. Understanding certain part of demand may lead to loss in sales, not happy customers, and perhaps allowing the competition to gain the upper hand in the marketplace. On the other hand, significantly overestimating demand also becomes very costly, as other support systems will be then required for rest of the produce. Hence, cost will increase due to:

- (1) Excessive inventory costs,
- (2) Forced price reductions,
- (3) Unnecessary production or storage capacity, and
- (4) Losing opportunities to make market more profitable goods.

Successful marketing and production managers understand the importance of obtaining good sales forecasts, as it is very essential to help trigger the forecast for production which in turn triggers the forecast of the raw materials needed for production.

2) Supply Chain Management

Forecasting helps in ensuring that the right product is at the place at the right time, through the supply chain. Accurate forecasting will result in reduced excess inventory and thus increase profit margin for the retailers. Hence, forecasting in supply chain management includes and the movement and storage of raw materials, work-in-process inventory, and finished goods from point of production to point of consumption.

3) Forecasting the Need for Raw Materials and Spare Parts

Although effective sales forecasting is a key for virtually any company, some organizations must rely on other types of forecasts as well. A prime example involves forecasts of the need for raw materials and spare parts. Many companies need to maintain an inventory of spare parts to enable them to quickly repair either own equipment or their products sold or leased to customers.

4) Forecasting Economic Trends

This is a process in which future condition of an economy is predicted. They help form the basis of planning by describing the expected future behaviour of all parts or a part of an economy. It Predicts a variety of economic indicators, like money supply, inflation rates, interest rates, etc. Government officials use economic forecasts to

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determine fiscal policies and business managers use the same to make monetary policies and plan future operating activities.

5) Forecasting Staffing Needs

In economies of the developed countries, there is a shifting emphasis from manufacturing to services. Now goods are being produced outside the country where labour is cheap, and are then imported. At the same time, there is an increasing number of business firms which specializes in providing some kind of service. For example: Travel, legal aid, financial, educational, health services, design, maintenance, tourism, entertainment etc. For such a company's sales forecasting means forecasting the demand for services, which then translates into forecasting staffing, i.e., human resource required to provide those services.

6) Forecasting in Education Environment

A good education institute typically plans its activities based on the demand of its different activities and services. The institute may forecast the number of students who graduate is more in which sector, and can accordingly plan the future requirements of its students. This may require a fresh orientation of the syllabus and faculty, development of required teaching materials/cases, recruiting new faculty with specific sector-oriented background, requisite experience and teaching skills. As another option, the management may decide that the future is more secure with the standard areas of operations. then it may continue with the original syllabus, etc.

7) Forecasting in a Rural Setting

Here is how we can understand this. Cooperative milk producers' union operates in a certain district. The products, which cooperative milk producers produces, the production capacities it has for the same, the manpower it recruits, and many such decisions are closely linked with the forecasts of the quantity of milk it may procure and the different milk products it may see. Milk is a product, which has a ready market, and is not difficult to sell. Thus, demand forecasting for products may not be a very dominant issue for the organization. However, milk forecast procurement forecast is a crucial issue for production planning process which includes making decision on what to produce, how to produce and when to produce. As raw milk is a highly perishable commodity and building up of adequate processing capacity is important for the dairy.

8) Ministry of Petroleum

The decisions related to the quantity of purchases to be made for various types of crude oils and petroleum products from different sources across the oil-exporting nations for the next few years needs to be made by the officials of this crucial ministry. Also, how much money need to be spent on the development of indigenous sources needs to be decided. These decisions involve/need Therefore, information on the future demand of different types of petroleum products and the change in the prices and the availability of crude oil and petroleum products in the country and the oil-exporting nations is required to make the required decision. This takes us back to the field of forecasting.

9) Department of Technology

It is an attempt to understand the potential direction, rate, characteristics, and effects of technological change, especially invention, innovation, adoption, their use. It is an important tool to anticipate and understand the pace of technological progress, innovation and effects of technological change, or whether this change will reach or exceed the key parameters of performance by future date. The concerned person or official decides to forecast the changing technology in this area with the help of a team of information technology experts. This is again a forecasting problem although of a much different type. This field of forecasting is known as technological forecasting.

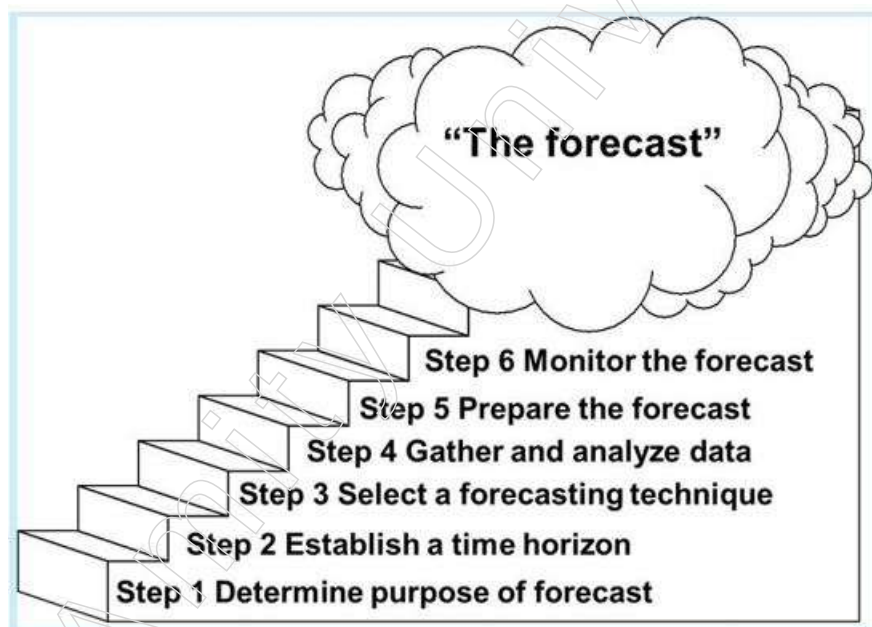
Therefore, Forecasting is the basis of corporate long-run planning. Forecasting provides the basis for budgetary planning and cost control, in the functional areas of finance and accounting.

Sales forecasting to plan new products, compensate sales personnel, and make other key decisions in marketing.

And, to make periodic decisions involving selection of process, capacity planning, and facility layout, as well as for continual decisions about production planning, scheduling, and inventory in Productions and operations.

As we have observed in the aforementioned examples, forecasting forms an important input in many business and social science-related situations.

General Steps in the Forecasting Process



The general steps in the forecasting process are as follows:

1) Identifying the General Need

We need to first identify the need or requirement or purpose as to why forecast needs to be done. Do we need forecast for sales or for demand or for anything else in an organization?

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2) Selecting the Period (Time Horizon) of Forecast

Then we need to decide the time period for which it is to be done, whether to forecast for a long period or short span of time period. The forecast horizon is the length of time or time period into the future for which forecasts are to be prepared. They vary from short-term forecasting horizons, which can be less than three months to long-term horizons, which can be more than two years.

3) Select Forecast Model to be Used

To select a model, we need to have knowledge of various forecasting models. Then only we will be able to understand in which situations which one is applicable, and how reliable each one of them is, also what type of data is required. On these considerations; one or more models can be selected.

4) Data Collection:

We must collect data from various appropriate sources. Data should be compatible with the requirement of the model. Data should not go that much in the past that it becomes outdated. The only point is that only relevant data should be collected and should meet the requirements of the model, which will help forecasting.

5) Prepare Forecast:

Using the data collected now we need to apply the model and calculate the value of the forecast.

6) Evaluate:

The forecast obtained through any of the model should not be used blindly. It should be first evaluated in terms of 'confidence interval' – usually through all forecast models we have methods of calculating the upper value and the lower value within which the given forecast is expected to remain, with certain specified level of probability. Evaluation can also be done from logical point of view whether the value obtained is logically feasible or not? It can also be evaluated against some related variable or phenomenon. Thus, based on evaluation it is possible or even sometimes advisable to modify the statistically forecasted value.

Operations Management

Operations management is chiefly concerned with planning, organizing and supervising in the contexts of production, manufacturing or the provision of services. As such, it is delivery-focused, ensuring that an organization successfully turns inputs to outputs in an efficient manner. The inputs themselves could represent anything from materials, equipment and technology to human resources such as staff or workers.

The Role of Forecasting in Operations Management

- In supply chain management, For the overall success of supply chain forecasting is very critical and is responsible for providing accurate information to their suppliers. I am not sure that the panic buying of the year 2020 could have been predicted or not, or else a better information would have prevented the empty

shelves. But I am certain that the Bull Whip Effect created by the panic buying must have attempted to ramp up the production to meet the increased demand, which would have resulted in excess stocks within a few months.

- In the short term, the forecasting is critical to the production of products, which includes the forecasting of the raw materials, components, assemblies and sub-assemblies. Forecasting of where the goods should be stored according to demand forecasts and also, the personnel required to make the supply chain operate effectively.
- In the long term, forecasts are required to predict the demand and requirement for new product. Also, how much of the new product to be stocked and where they should be stocked. All this is a part of long-term supply chain forecast.
- Operations management chain aims towards adding value by satisfying customer demand and it is necessary to meet the production, distribution and quality to meet and satisfy customers' demands for which forecasting is required. The forecast must be strong enough to ensure an uninterrupted flow of products or services for the customers.
- The strategic plan of the company must include some sort of forecasting in order to plan where the company wants to see itself down the line and what capacity the company must work at in order to meet these forecasts.
- Wall street keeps a close watch on ability to meet the forecasts in the strategic plan for publicly held companies. In order to meet their forecasts, sometimes companies play games with their forecasts and production. Krispy Kreme Doughnuts tried to do this.
- Several years ago, as Krispy Kreme tried to pump up their production and shipment numbers artificially and the resulting forecasts for future production. Seems Krispy Kreme reported the number of doughnuts that they were actually not selling and shipping. The company was actually shipping the doughnuts to their retail customers i.e., grocery stores, at the end of reporting periods with the understanding that the retailers would not be charged for the extra number of doughnuts shipped.
- Forecasting after the reporting period was closed out, Krispy Kreme had the retailers ship the excess doughnuts back. The quantity shipped and the not actual sales quantity was reported thus skewing the forecasts for future sales and drove the stock price through the roof while also creating a huge waste of doughnuts. Now what happened when this deceptive practice was discovered, that the price went from approximately \$40 per share to \$4 per share almost overnight. The darling of Wall Street crashed and seven years later the stock price is still depressed.

Why is Forecasting Important?

Forecasting has become valuable to businesses as it gives the ability to make informed business decisions and develop data-driven strategies. On the basis of current market conditions and predictions on how the future looks, the financial and operational decisions are being made. Past data is assembled and analysed to find patterns and is used to predict future trends and changes. Forecasting allows your company to

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be proactive instead of being reactive. There are 3 ways forecasting can help your organization excel:

1. Helps Set Goals and Plan

Based on current and historical data, forecasting allows businesses to set reasonable and measurable goals. In order to help businesses to decide what amount of change or growth will be determined as success, we need to have accurate data and statistics. Having these goals also helps to evaluate progress, and even adapt business processes wherever needed to continue on the desired path. There are certain tools such as CRM, that helps to visual forecasting and give insight into things like the sales pipeline, opportunities, and more.

2. Helps Budgeting

It would have been great if businesses would get endless supplies of money. Well, this is not the case. There is never enough money and time. Instead of spending money widely, a business needs to understand the best place where it can spend its money and dedicate its time to. Having visibility into potential trends and changes help businesses to know where to allocate their budget and time spent on certain offerings, whether it is upcoming product launch, investing in new staff or any other investment for a better future. "Budgeting quantifies the expectation of revenues that a business wants to achieve for a future period, whereas financial forecasting estimates the amount of revenue or income that will be achieved in a future period".

3. Helps Anticipate Change within the Market

In order to change or improve the outcome, necessary changes or adjustments can be made to the business strategies and current operations can also be altered by having insight into not only current data, but also projections of what could happen in the future. Forecasting helps position businesses which puts you in a position to become active instead of being reactive, which means you can adjust your sales forecast in accordance to the market trends, if the data is showing changes in consumer behaviour it is important to readjust to the market overall and optimize resources to stand out from the competition.

Determining what to Forecast

- In the early stages of a forecasting project, we need to first understand and decide what should be forecasted. For example, if forecasts are required for items in a manufacturing environment, it is necessary to find out, whether forecasts are needed for:
 - Every product line, or for groups of products?
 - Every sales outlet, or for outlets grouped by region, or only for total sales?
 - Weekly data, monthly data or annual data?
- It is also required to consider the forecasting horizon, which tells that whether the forecasts be required for one month in advance, for few months, or for ten years? Depending on what forecast horizon is most important, different types of models will be required.

- Now, here question comes that how frequently are forecasts needed or required? Forecasts that needed frequently are better done using an automated system rather than with methods that require careful manual work. It is definitely worth talking to people who will use the forecasts and ensure that you understand their needs.
- Once it has been determined what forecasts are needed or required, it is then necessary to find or collect the data on which the forecasts will be based. The data required for forecasting may already exist, as these days, a lot of data is recorded, and the forecaster's task is often to identify where and how the required data are stored.
- The data may include previous sales records of a company, previous years demand of the product, or an unemployment rate of a geographic region. A forecaster can spend large part of his/her time in locating collating the available data prior to developing suitable forecasting methods. This is further explained in five steps below:

The Basic Steps in Forecasting Task

A forecasting task usually involves five basic steps.

1) Problem Definition.

This is often the most difficult part of forecasting. In order to define the problem, we need to first understand the way the forecasts will be used, who will require the forecasts, and how the forecasting function fits within the organisation requiring the forecasts. A forecaster needs to spend time and discuss the issues with the people, who will be involved in collecting data, maintaining databases, and using the forecasts for planning the future.

2) Gathering Information.

Here at least two kinds of information are required: (a) statistical data, and (b) the accumulated expertise of the people who collect the data and use the forecasts. Often, if it is difficult to obtain enough historical data to be able to fit a good statistical model or if data is not available. In that case, we can go for judgmental forecasting methods. Sometimes, old data may be less useful due to structural changes in the system being forecast; then we may look for and choose to use only the most recent data. However, we must keep in mind that evolutionary changes will be handled by good statistical models. Hence, good data should be kept and not thrown unnecessarily.

3) Preliminary (Exploratory) Analysis.

You need to start by graphing the data. Check for any consistent patterns? Check for any important trend? Is seasonality important? Is there evidence of the presence of business cycles? Are there any outliers in the data, which would need explanation from experts? How strong are the relationships among the variables available for analysis? Various tools to forecast can be developed to help with this analysis.

4) Choosing and Fitting Models.

Based on the availability of historical data, the strength of relationships between

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the forecast variable and any explanatory variables, and the way in which the forecasts are to be used, the best model is selected. It is obvious to compare two or three potential models. Every model will not work in each situation. We need to see and understand which model sets the best with the given situation. Each model is itself an artificial construct that is based on a set of assumptions and usually involves one or more parameters which must be estimated using the known historical data.

5) Using and Evaluating a Forecasting Model.

Once a model has been selected and its parameters estimated, the model is used to make forecasts. The performance of the model can only be properly evaluated after the data for the forecast period have become available.

Evaluating the Forecast Accuracy:

- There are many ways to measure forecast accuracy. Some of these measures are the mean absolute forecast error, called the MAD (Mean Absolute Deviation), the mean absolute percentage error (MAPE) and the mean square error (MSE)

Error = Actual Observed value – Forecasted value

Absolute Percentage Error = (Error / Actual Observed Value) × 100

MAD = The average of the absolute errors

MAPE = The average of the Absolute Percentage Errors

MSE = The average of the squared errors

- It is very common that two forecasting models are ranked differently depending on the accuracy measure used. For example, unlike model B, model A may give a smaller MAD but a larger MSE. Why? Reason being that the MAD gives equal weight to each error and MSE gives more weight to large errors because they are squared. It is up to the manager, not the management scientist to decide whether MAD is more appropriate for his application or MSE is more appropriate for his application. The MSE is most often used in practice. Given a preferred accuracy measure, how do we know when our forecasts are good, bad, or indifferent? We can do it by comparing the accuracy of a given model with that of a benchmark model. A handy benchmark is the naïve model, which assumes that the value of the series next period will be as same as it is this period; i.e., say $F_{t+1} = X_t$. Where, F = forecast and X = the observed value. The subscript t is an index for the time period and the current period is t + 1.
- The initial step in any forecasting problem should be to use the naïve model to compute the benchmark accuracy. A model which is unable to beat the naïve model should be discarded. It may seem a waste of time to check model accuracy against that of the naïve model, but unless we do so, it becomes easy to choose an inappropriate forecasting model.
- The mean error measures are computed only for the last half of the data. We can evaluate the forecasting models by dividing the data in to two parts. Fitting consists in the first part of the model. In order to get the data “warmed

up”, Fitting leads to running the model through the first part of the data. Fitting data is also known as warm-up sample. The second part of the data is used to test the model and is called the forecasting sample. In the warm up sample expecting accuracy is really irrelevant. Accuracy in the forecasting sample is more important because the pattern of the data often changes over time. The forecasting sample evaluates how well the model tracks such changes.

- There are no statistical rules on where to divide the data into warm-up samples and forecasting samples. There may not be enough data to have two samples. A good rule of thumb is to put two complete seasons of seasonal data or at least six non seasonal data points in the warm-up sample. If the data is less, then there is no need to bother with two samples. In long time series, it is common practice to simply divide the data in half.

Forecasting Models

Depending on the data you have available and time horizon you are using, you will want to apply different forecasting models. While there are a lot of models one could use, these fall into three broad types:

- 1) Judgmental (Qualitative): This type of forecast is made subjectively by the forecaster. The Delphi method, scenario building, and statistical surveys which are based on intuition and a collection of opinions from managers and experts are included in Judgemental forecasting.
- 2) Time-series (Quantitative): Time series forecasts is solely based on use of historical data to make predictions about the future and strategic decision making. Data corresponds to a specific period, such as monthly inputs over a span of ten years. These forecasts rely on the assumptions that past patterns will repeat in the future, so these data inputs are used to create long term forecasts.
- 3) Associative Models (Quantitative): Associative models are also known as casual models, which uses data corresponding to various underlying factors to predict the desired variable or another variable of interest. In this, the relationship between the factors and the variable are assumed to predict more complex patterns. Some methods include regression analysis and autoregressive moving averages.

2.1 Qualitative Forecasting Methods

Objectives

- Learn about Educated Guess
- Understand Consensus
- Explain Delphi Method
- Identify Historical Analogy
- Learn about Market Research

Introduction

Qualitative forecasting is a method of making predictions about a company's

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finances that uses judgement from experts, rather than numerical analysis. Qualitative forecasting is being performed by expert employees by identifying and analysing the relationship between existing knowledge of past operations and potential future operations.

Estimations are being made by the experts about how a company might perform in the future based on the opinions they offer and the information they collect from other sources, like staff polls or market research.

These types of forecasting methods are subjective in nature and are based on judgments, opinions, intuition, emotions, or personal experiences. They do not rely on any rigorous mathematical computations.

2.1.1 Educated Guess

Estimating an unknown quantity produces the best-educated guess possible based on the available data. We all want or often need to know estimates of things that are just plain impossible to know exactly. Here are some examples of unknowns to be estimated:

- Next quarter's sales.
- What the government will do next to our tax rates.
- How the population of Chicago will react to a new product.
- How your portfolio of investments will fare next year.
- The productivity gains of a change in strategy.
- The defect rate in a manufacturing process.
- The winners in the next election.
- The long-term health effects of tablet computer screens.

When reliable data is available statistics can shed light on some of these situations by producing a good, educated guess. Keep in mind that all statistical estimates are just guesses and thus, are often wrong. However, they will serve their purpose when they are close enough to the unknown truth to be useful. One could decide how much attention to give them as a manager, if one knew how accurate these estimates were (even approximately).

An indication of the amount of uncertainty or error involved in the guess, is provided by Statistical estimation, accounting for the consequences of random selection of a sample from a large population. The confidence interval gives probable upper and lower bounds on the unknown quantity being estimated, which means that you can say that – “I am not sure exactly what the answer is, but I am quite confident it's between these two numbers”.

The confident intervals show how reliable an estimated value actually is, Hence, you must routinely expect to see confidence intervals and if you do not then you should ask for them.

For example, in the forecasting statement, there is certainly some information that sales next quarter are expected to be \$11.3 million.

However, some additional and deeper understanding comes that you are 95% confident that next quarter's sales will be somewhere in between \$5.9 million and \$16.7 million.

The confidence interval puts the guess in perspective and helps you avoid the tendency to treat the expected single number as very precise when, as a matter of fact, it might not be precise at all.

It is crucial in every company to predict about future of the business. There are possibilities that some forecasts might fail and others succeed in providing realistic and correct predictions, which contributes to the success of the company. There are several techniques for making good forecasts, but the bottom line is the same: Forecast basically is an educated guess and information increases the accuracy.

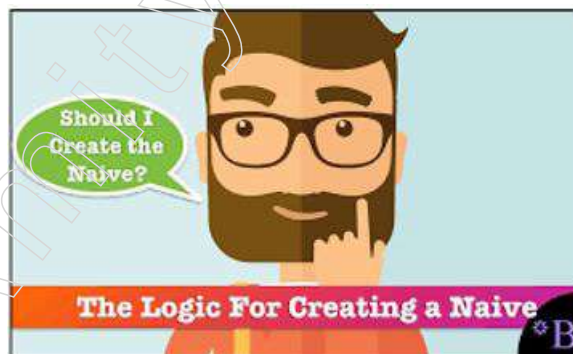
Michael J. Burry made one of the best examples of an educated guess, regarding the subprime mortgage crisis, which he saw it coming years before anyone else could see it and here's how he did it: He actually used the existing information for analysis.

There are five general types of forecasts, however, let's first drive through these three basic questions.

- What is the purpose of the forecast and exactly how will it be used? This helps the forecaster to estimate and understand how much and what resources will be used in the process.
- What are the parts and components of the system or event about which the forecast is going to be made? This helps those involved in the forecasting to have a better understanding of the interacting variables.
- How big is the role of the past in forecasting the future? To answer this question, take into account big changes in the system, not the recent ones, but the critical events of the past.

Now that we pointed out the three main questions, let's now discuss about the five general methods for forecasting.

1. Naive Forecasting



This is the base of forecasting, and simple data of the past is looked at without considering causal factors, seasonal variations or cyclical trends. Forecasters using this method might just take the data of the same period from the past or even may use the average from that period.

Naive forecasting is the technique in which the next period's forecast is done

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without predictions or adjusting the factors, by using the last period's sales. Forecasts produced using a naïve approach are very much equal to the final observed value. It is very easy to use naïve forecasting method, and it is suitable for finance and sales departments, and ensures that these departments work to improve the company. Other, more advanced, forecasting methods include the moving averages method (MA), linear trend forecasting, and exponential smoothing.

In naïve forecasts there are no calculations or formulas, there only is an assertion of the actual sales numbers. In certain cases, naïve forecasting can accurately predict situations, while others can be problematic, as it only considers only the previous period to forecast the next period. Thus, historical sales data is the first and foremost requirement for naïve forecasting and factors such as seasonality remains untouched. Oftentimes, salespeople will use naïve forecasting to make goals as a way to always ensure that they are either improving or maintaining their contribution to the company.

How to Calculate a Naïve Forecast?

The naïve forecasting method in Microsoft Excel provides information about the "system" and does not generate a final forecast. The "system" refers to the type of market or industry in which you are forecasting.

As previously mentioned, "calculating" a naïve forecast is a simple technique. We just need to consider the previous month's sales and use it to forecast the sales for the next period.

Using the Naïve Method

The naïve method of forecasting commands that we must use the previous period to forecast for the next period. To demonstrate the pros and cons of this method we've created a % difference column, to show you the % of variance between the Actual Sales column and the forecast. This will show you how accurate the forecast actually is. You can see the equation we have used, = IF(Actual Sales=0,0,(Naïve Forecast/Actual Sales)-1) , in the image below the same has been shown:

	A	B	C	D
1	Period	Actual Sales	Naïve Forecast	% Difference
2	1/1/2012	\$11,367.91		0.00%
3	2/1/2012	\$12,367.47		C3)-1)
4	3/1/2012	\$15,433.65		0.00%
5	4/1/2012	\$16,235.66		0.00%

What this equation it means that the % will be positive, if the forecast is less than the actual sales within that time period. A positive percentage means that what we actually sold is greater than what we actually forecast and, a negative percentage means that we sold less than the forecast indicated we would sell off.

To calculate a naïve forecast you need to simply take the sales of previous month of sales and plug it in next to the adjacent period. The equation for this method, =(Previous months actual sales), is shown below:

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B2				=B2
	A	B	C	D
1	Period	Actual Sales	Naïve Forecast	% Difference
2	1/1/2012	\$11,367.91		0.00%
3	2/1/2012	\$12,367.47	=B2	0.00%
4	3/1/2012	\$15,433.65		0.00%
5	4/1/2012	\$16,235.66		0.00%

D3				=IF(C3=0,0,(B3/C3)-1)
	A	B	C	D
1	Period	Actual Sales	Naïve Forecast	% Difference
2	1/1/2012	\$11,367.91		0.00%
3	2/1/2012	\$12,367.47	\$11,367.91	8.79%
4	3/1/2012	\$15,433.65		0.00%
5	4/1/2012	\$16,235.66		0.00%

Only when you will apply this equation, you'll notice that the equation has projected a positive percentage within 10%. That is a pretty good forecast relatively speaking, moving in the right direction. It is usually nicer to surpass your forecast, which means promising less and selling more, but it can also lead to some inventory planning problems, which may require proper arrangement for inventory as well.

For now, let's apply the equation to the entire 3 years of sales periods we have to see what we get.

	Period	Actual Sales	Naïve Forecast	% Difference
1	1/1/2012	\$11,367.91		0.00%
2	2/1/2012	\$12,367.47	\$11,367.91	8.79%
3	3/1/2012	\$15,433.65	\$12,367.47	24.79%
4	4/1/2012	\$16,235.66	\$15,433.65	5.20%
5	5/1/2012	\$23,041.25	\$16,235.66	41.92%
6	6/1/2012	\$26,572.81	\$23,041.25	15.33%
7	7/1/2012	\$27,823.28	\$26,572.81	4.71%
8	8/1/2012	\$23,889.31	\$27,823.28	-14.14%
9	9/1/2012	\$14,062.15	\$23,889.31	-41.14%
10	10/1/2012	\$9,458.80	\$14,062.15	-32.74%
11	11/1/2012	\$10,577.96	\$9,458.80	11.83%
12	12/1/2012	\$12,743.64	\$10,577.96	20.47%
13	1/1/2013	\$21,279.31	\$12,743.64	66.98%
14	2/1/2013	\$24,573.62	\$21,279.31	15.48%
15	3/1/2013	\$31,447.50	\$24,573.62	27.97%
16	4/1/2013	\$34,911.29	\$31,447.50	11.01%
17	5/1/2013	\$46,999.74	\$34,911.29	34.63%
18	6/1/2013	\$52,449.48	\$46,999.74	11.60%
19	7/1/2013	\$53,702.30	\$52,449.48	2.39%
20	8/1/2013	\$48,717.87	\$53,702.30	-9.28%
21	9/1/2013	\$26,902.49	\$48,717.87	-44.78%
22	10/1/2013	\$14,957.61	\$26,902.49	-44.40%
23	11/1/2013	\$13,527.05	\$14,957.61	-9.56%

One of the very first things we can discern from this forecast is that we have periods of positive (sold more than we forecast) and negative (sold less than we forecasted) periods.

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25	12/1/2013	\$16,712.18	\$13,527.05	23.55%
26	1/1/2014	\$28,985.00	\$16,712.18	73.44%
27	2/1/2014	\$35,456.77	\$28,985.00	22.33%
28	3/1/2014	\$56,333.33	\$35,456.77	58.88%
29	4/1/2014	\$52,072.64	\$56,333.33	-7.56%
30	5/1/2014	\$68,181.37	\$52,072.64	30.94%
31	6/1/2014	\$73,961.71	\$68,181.37	8.48%
32	7/1/2014	\$75,256.88	\$73,961.71	1.75%
33	8/1/2014	\$65,884.46	\$75,256.88	-12.45%
34	9/1/2014	\$40,489.61	\$65,884.46	-38.54%
35	10/1/2014	\$26,491.45	\$40,489.61	-34.57%
36	11/1/2014	\$22,249.60	\$26,491.45	-16.01%
37	12/1/2014	\$22,201.59	\$22,249.60	-0.22%
38	1/1/2015	\$41,613.86	\$22,201.59	87.44%
39	2/1/2015	\$51,226.35	\$41,613.86	23.10%
40	3/1/2015	\$64,219.78	\$51,226.35	25.36%
41	4/1/2015	\$60,594.81	\$64,219.78	-5.64%
42	5/1/2015	\$80,052.91	\$60,594.81	32.11%
43	6/1/2015	\$86,682.38	\$80,052.91	8.28%
44	7/1/2015	\$97,384.42	\$86,682.38	12.35%

These positive and negative periods are important to analyse because they potentially provide us with information about seasonality trends. These trends can help us to understand the performance of a product during particular times of the year.

2. Qualitative and Quantitative



• Qualitative Forecasting

Personal opinions, is what Qualitative forecasting us based upon. Some examples are:

- The Delphi method (The Delphi method is a forecasting method based on the results of questionnaires sent to a panel of experts. Several rounds of questionnaires made and circulated, and the anonymous responses are aggregated and shared with the group after each round. Since questions are asked in multiple rounds and the panel is told what the group thinks as a whole. The experts adjust their answers according to the group responses provided to them in the subsequent rounds. The Delphi method is the one which seeks to reach the correct response through consensus.)
- Informed opinions and the historical life cycle analogy (This is a judgmental forecasting technique based on identifying history that is comparable to a present situation, such as the history of a similar object or event, and using that past pattern to predict future).

Qualitative Forecasting Techniques

Qualitative forecasting techniques come from the experiences and instincts of the business experts. These forecasting techniques includes interpretation of data

combined with the professional expertise you've developed over time on the job, and cannot be just guessed. For example, if you want qualitative information for projecting sales for the year, you need to estimate the impact of a new ad campaign or promotion your company is planning, also keep a check at the effects that new technologies might have on consumer purchasing and take into account recent social vogue and trends. By holding focus groups of customers to discuss and gauge their reactions to several new product features your company is considering, demand can be forecasted.

Advantages of Qualitative Forecasting

Flexibility: Business owners have the flexibility they need to explore the expert opinion, judgment, and intuition of their industry's leaders without being held back by rigid numerical data, by utilizing qualitative methods.

Intuition: When sales data is not adequate, qualitative demand forecasting methods are often much more accurate and desirable among business owners. For example, if any firm is launching a new product that is very different from any other item which is currently available, they would not have the past data on hand to forecast for its demand. Qualitative forecasting can, thus, fill these gaps of knowledge to complete an accurate forecast.

Disadvantages of Qualitative Forecasting

Errors in Judgment: Despite a hunch about what to expect within the business can be accurate, there are many a times when experts in this field are dealing with the unknown. As a result, the qualitative approach can be susceptible to human errors as it is so heavily dependent on executive opinion.

Unexpected Changes: Sudden environmental changes such as harsh weather, as well as governmental and economic activity shifts, can derail the accuracy of qualitative techniques.

As this forecasting technique may or may not take into account unexpected occurrences.

Bias: Although the experts and consumers involved in market research aim to remain unbiased, their responses can be heavily affected by personal biases. Individuals who are too optimistic, will likely to develop a forecast which is overly optimistic.

- **Quantitative Method**

Analysing past numerical data is quantitative forecasting. Anything with data in form of numbers could be a source for the quantitative method.

Quantitative Forecasting Techniques

Quantitative forecasts often use previous years data, such as previous sales and revenue figures, yearly or previous year's production and financial reports and website traffic statistics to forecast future data. For Example: looking at seasonal sales data, can help you plan next year's production and labour needs based on last year's monthly or quarterly figures. Projections based on statistical modelling, trend

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analyses or other information from expert sources such as government agencies, trade associations and academic institutions, are used by Quantitative forecasting.

Advantages of Quantitative Forecasting

Addresses Historic Data: When conducting quantitative methods, businesses are able to address the company's history in a way which is not influenced by personal feelings or opinions. From revenue and sales to expenses, businesses have the unbiased past data which they need to make informed decisions about the company's future.

Exposes Patterns: Patterns of spending, sales, and scheduling within the business, can clearly be exposed by numerical data. This type of forecasting shows trends over a specific time period and whether these patterns are consistent from year to year. As a result, in order to meet expected annual averages, owners can make changes to the workforce or inventory control processes.

Attracts Stakeholders: When businesses have solid data to back up their need for investors or loans, the quantitative technique works in their favour. Before making any commitments stakeholders want to see the company's bottom line and cash flows. Thus, showcasing historical trends gives businesses the fuel they need to appeal for more funding.

Disadvantages of Quantitative Forecasting

Lacks Detail: Quantitative forecasting methods produce clear numbers needed to make important decisions but it still lacks intuition and experience. This type of forecasting does not allow businesses to account for external factors only years of experience within the industry can be revealed.

Cost: In many cases, qualitative demand forecasting methods are less expensive to employ. More staff and analyst are required to Compile, analyse, and organize quantitative data to uncover trends and patterns. However, by using affordable business forecasting tools and software to automate the process, this concern can be easily addressed.

3. Causal Forecasting



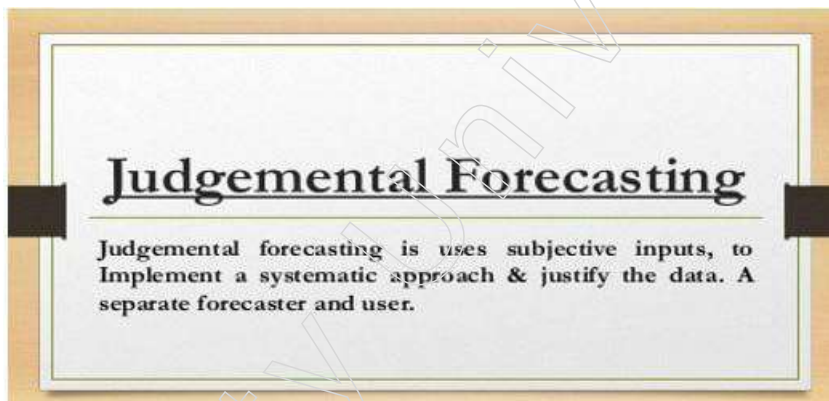
This method is based on an assumption that mathematical function which uses current variables that can be also used to forecast the future value of variables. For example, information about how many tickets sold of a movie could be of use to predict the number of movie-related T-shirt sales as well. This prediction is done determine what type of impact those anticipated variables will have on the consumer demand,

the type of pricing that the market will be able to support in the future, and what those changes would mean or affect the future of the company. This type of forecasting is helpful to companies in the development of sales and advertising for the upcoming period, and is also helpful in several other ways.

There are several elements that go into a causal forecasting model. Typically, the process will begin with an assessment of the market as it currently stands, which also includes the current position of the company within that market. From there, a need to identify both dependent and independent variables come up, that are likely to put some influence on the direction that the market will take over a specified period of time. Once there is a reasonable projection of what will happen in the future to the market as a whole, it is possible to apply those same variables and their cumulative effect to the business operation itself.

The ability to prepare for what is most likely to occur in the future, is one of the benefits of causal forecasting. Depending on the outcome of forecast of the future situations, the company may find it advantageous to begin increasing production now in anticipation of an increased demand for its products at a later date. At the same time, the results of the causal forecasting may indicate upcoming economic circumstances that would make wise decision by reducing the production now in order to prevent being left with huge inventories during some sort of recession or other crisis in the market and the general economy.

4. Judgmental Forecasting



Swift polling, is the perfect tool to get started, which provides various poll and survey-related services. Examples of some of the services are: real-time SMS and web polling, PowerPoint integration, Facebook and Twitter integration, as well as open-ended questions. All these tools can be used to conduct high efficiency forecasting. For example, many betting websites started using polls for the upcoming games, where users can vote the outcomes of the available matches. This is useful where you cannot individually interview all the people with knowledge in a given field, however, you can connect with them through polls. Just keep in mind the targeting, you wouldn't want non-professionals to take part in your survey or poll, in that case you would get an imprecise forecast.

Various forecasting models of the judgmental kind utilize subjective and intuitive information to make predictions. For instance, there are times when there is no data available for reference. Launching a new product or facing unpredictable market

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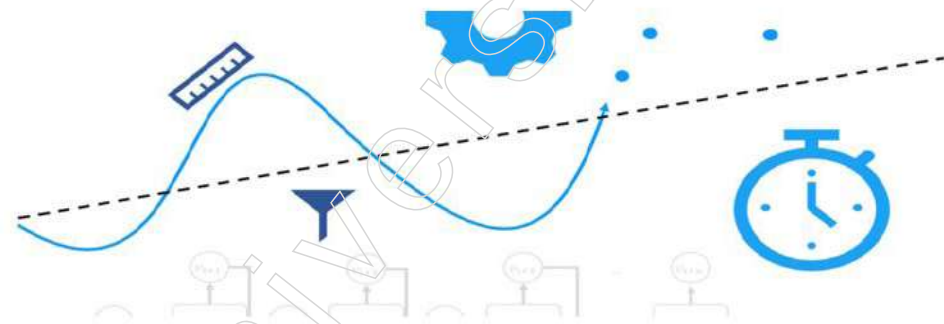
conditions also creates situations in which judgmental forecasting models prove beneficial.

Here are some characteristics of judgmental models:

- Takes a subjective, opinionated approach
- Assumes specific variables
- Comes with limitations
- Accuracy improves with the addition of new information

This type of forecasting model is especially helpful in the field of research and development. Focus groups and expert panels can provide understanding that no computerized model would have. For instance, companies can better assess their direction when developing specific product features, by surveying a group of people about what they look for in a product.

5. Time Series



In order to predict future outcomes, this type of forecasting employs historical data. Data is recorded over a specified period of time, such as a company's sales by quarter, because past patterns have a tendency to repeat themselves. Time series forecasts are mostly for long-term projects. The most successful people are ones who can see what's coming before anybody else does, so use this knowledge to make informed predictions. You and your company can make profits from the success.

As the key to reliable forecasting, this type of model uses historical data. You'll be able to visualize patterns of data better when you know how the variables interact, whether in terms of hours, weeks, months or years.

While there are many other methods of completing a time series model, you can follow these general steps in a spreadsheet to estimate outcomes using information obtained from recent analytical data:

1. Is your time-based data available for use (time series and values series)?
2. Input the compiled data involving time or duration in the first column of the sheet.
3. Put the remaining values you want to forecast in the next column.
4. Select the relevant data.
5. Click Data tab, then select Forecast Group, then choose Forecast Sheet.

6. Access the sheet, then select the line or bar graph option you want to use.
7. In the Forecast End box, determine your end date and hit Create.

Once the forecasting model is set, the next step will be then to interpret it to formulate your best estimation of the future.

2.1.2 Consensus

Strategic consensus occurs when many or numerous people within an organization have similar views regarding what is most important for the organization to succeed.

Strategic consensus is the level of agreement within an organization regarding the relative importance of cost, quality, delivery and flexibility to the organization's operational goals, as well as the relationships between these competitive priorities and operational policies.

Objectives

Characteristics of consensus decision-making include:

- Collaboration: Participants contribute to a shared proposal and shape it into a decision that meets the concerns of all group members as much as possible.
- Cooperation: Participants in an effective consensus process should aim to reach the best possible decision for the group and all of its members, rather than competing for personal preferences.
- Egalitarianism: All group of members of a consensus decision-making body should be afforded, as much as possible, equal input into the process. All members participating have the opportunity to present and amend proposals.
- Inclusion: As many stakeholders, as possible who can be involved, should be involved in the consensus decision-making process.
- Participation: The consensus process should actively ask for the input and participation of all decision-makers.
- Alternative to Common Decision-making Practices

Consensus decision-making can be a good alternative to commonly practiced group decision-making processes. Robert's Rules of Order, for instance, is a guide book used by number of organizations. This book allows the structuring of debate and passage of proposals that can be approved through majority voting. It does not emphasize on the goal of full agreement. Critics of such a process believe that it can involve debate full of conflicts and the formation of competing factions. These dynamics may harm relationship among the group members and undermine the ability of a group to cooperatively implement a contentious decision. Consensus decision-making attempts to address the beliefs of such problems. Supporters claim that outcomes of the consensus process include:

- Better decisions: By including the input of all stakeholders the resulting proposals may better address all potential concerns.
- Better implementation: It is a process that includes and respects all parties, and generates as much agreement as possible. Hence, it sets the stage for greater cooperation in implementing the resulting decisions.

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- Better group relationships: A cooperative, collaborative group atmosphere can foster greater group togetherness and interpersonal connection.
- Roles

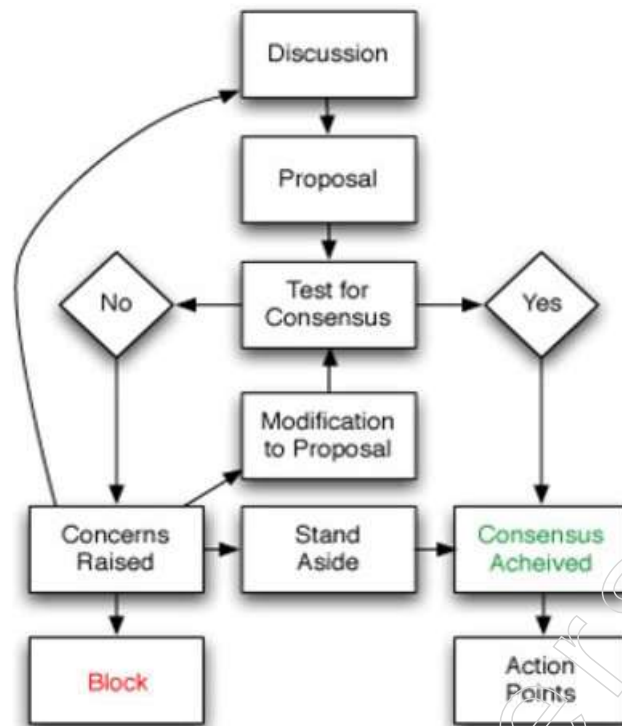
The consensus decision-making process often has several designed roles to make the process run more effectively and efficiently. Although the name and nature and other factors of these roles varies from group to group, the most common are the facilitator, a timekeeper, an empath and a secretary or notes taker. All decision-making bodies does not use all of these roles, although the facilitator position is almost always filled, and some groups use supplementary roles, such as a Devil's advocate or greeter. In order to build the experience and skills of the participants, and prevent any perceived concentration of power, some decision-making bodies rotate these roles through the group members.

The common roles in a consensus meeting are:

- **Facilitator:** As the name itself implies, the role of the facilitator is to help in making the process of reaching a consensus decision easier. Facilitators takes up the responsibility for moving through the agenda on time; ensuring the group adheres to the mutually agreed-upon mechanics of the consensus process; and, if necessary, also suggests alternate or additional discussion or decision-making techniques, such as go-arounds, break-out groups or role-playing. Some consensus groups use two co-facilitators. Shared facilitation is often adopted to disperse the perceived power of the facilitator and create a system whereby a co-facilitator can pass off facilitation duties if he or she becomes more personally engaged in a debate.
- **Consensor:** The team of consensors accepts those relevant proposals which conform with the UN Charter on Human Rights; for displaying an initial list of these options; to draw up a balanced list of options to represent the entire debate; to analyse the preferences cast in any subsequent MBC ballot; and, if needed, to determine the composite decision from the two most popular options.
- **Timekeeper:** The purpose of the timekeeper is to ensure the decision-making body keeps to the schedule set in the agenda. A variety of techniques is used by effective timekeepers to ensure the meeting runs on time including: giving time updates frequently, ample warning of short time, and keeping individual speakers from taking an excessive or unnecessary amount of time.
- **Empath or Vibe Watch:** This is called and is responsible for monitoring the 'emotional climate' of the meeting, taking note of the body language and other non-verbal cues of the participants. Defusing potential emotional conflicts, maintaining a climate free of threatening effect and also being aware of potentially destructive power dynamics, such as sexism or racism within the decision-making body, are the primary responsibilities of the empath.
- **Note Taker:** The role of the notes taker or secretary is to document or take a note of all the decisions, discussion and action points of the decision-making body.

Consensus means that each person must agree to support the implementation of the decision, regardless of whether or not he/she agrees with the decision.

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There two critical factors necessary for obtaining consensus.

Factor #1 – Membership in the Decision Team

The above definition has a key corollary: membership is defined and known. This is necessary so that each member of the group knows who is in and who is out of the implementation group. For larger groups, you need to first define and develop the core team, extended team, and the network of relationships in an organization.

Factor #2 – A Clear Signal that the Individual will Support the Decision

There should to be a way to recognize the 100% agreement of project participants. All the members of the group must visibly signal their agreement to support the implementation. The signals can range from verbal statements to physical signals, like “thumbs up or thumbs down” technique.

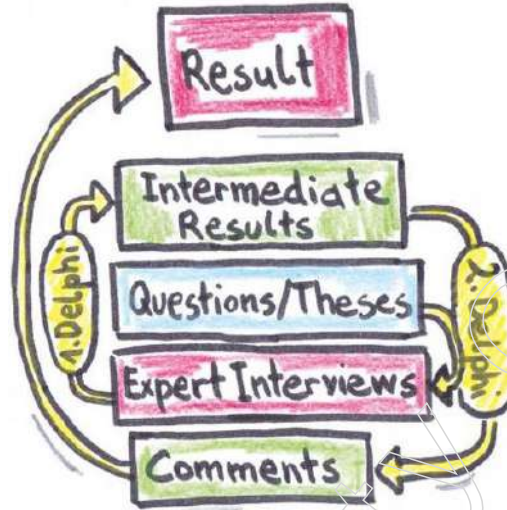
Benefits of Consensus

Consensus is an important tool as:

- It assures that people will work according to the decisions that had been taken. It is a basic discipline that surfaces and removes obstacles from full implementation.
- It also supports commitment, and is defined as the willingness to invest resources in the face of uncertain outcomes.

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2.1.3 Delphi Method



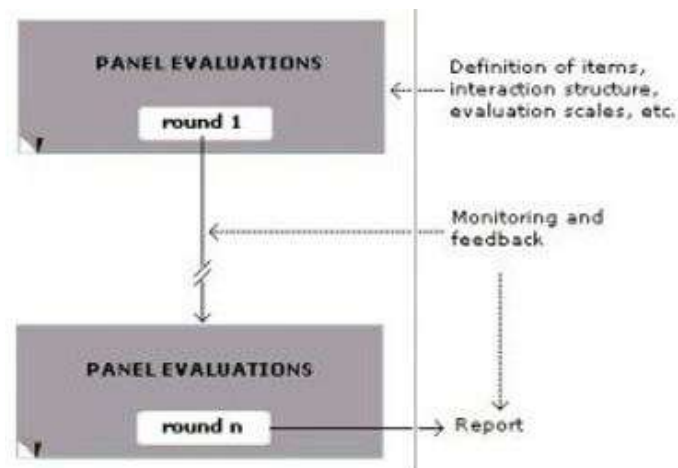
The Delphi method is a forecasting process framework which is based on the results of multiple rounds of questionnaires sent to a panel of experts. After every round of questionnaires, the experts are presented with an accumulated summary of the last round, allowing each expert personnel to adjust their answers according to the group response. This process provides the benefits of expert analysis with elements of the wisdom of crowds.

Originally, Delphi method has been developed as a systematic, interactive forecasting method which relies on a panel of experts, and is a structured communication technique, in which questionnaires in two or more rounds are there. After each round, an anonymous summary of the experts forecasts from the previous round as well as the reasons they provided for their judgments, is provided by the facilitator.

The anonymous aggregated responses are shared after each round to allow them to adjust their answers in subsequent rounds until they are able to reach their final response through consensus. As in most judgmental forecasting, the Delphi method also has its shortcomings or difficulties. The time required to complete this method is long and finding the right mix of industry experts and keeping them involved throughout the product life-cycle is quite difficult.

Forecasts (or decisions) from a structured group of individuals are more accurate than those from unstructured groups, is what principle of Delphi is based on. The technique can also be adapted for use in face-to-face meetings, and is then called mini-Delphi or Estimate-Talk-Estimate (ETE). Delphi has certain advantages over another structured forecasting approach, prediction markets and has been widely used for business forecasting as well.

The Delphi Method Communication Structure



Delphi Method Process

First, based on the topic to be examined, the group facilitator selects a group of experts.

Once all participants are confirmed, each member of the group is sent a questionnaire with instructions to comment on each topic based on their personal opinion, experience, or previous research.

The questionnaires are returned to the facilitator who groups the comments and prepares copies of the information. A copy of the compiled comments is sent to each participant, along with the opportunity to comment further.

At the end of each comment session, all questionnaires are returned to the facilitator who decides if another round is necessary or if the results are ready for publishing.

The questionnaire rounds can be repeated as many times as necessary to achieve a general sense of consensus.

Advantages of the Delphi Method

The Delphi method seeks to aggregate opinions from a diverse set of experts, and it can be done without bringing everyone together for a physical meeting. Since the responses of the participants are anonymous, individual panellists do not worry much about repercussions of the opinions they provide. Consensus can be reached over time as opinions are swayed, to make the method very effective.

Disadvantages of the Delphi Method

Delphi technique does not produce a right or wrong answer and also does not lead to interactions such as live discussion. A live discussion can sometimes produce a better example of consensus, as ideas and perceptions are introduced and is further broken down to be reassessed. Response times with the Delphi method can be long, which slows down the rate of discussion, as it relies on expert response which takes minimum of 2 weeks.

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Pros/Cons of Delphi Technique

Pros	Cons
Avoids biases and obstacles associated with interacting groups	Time consuming. May take as little as several weeks or as much as several months
Generates fairly useful information and high-quality solutions	Highly structured format doesn't offer conditions change

- Characteristics of the Delphi
 - The Delphi process has two distinct forms. The common among them is the “paper-and-pencil” version, which is commonly referred to as a “Delphi Exercise.” There is a small monitor team which designs a questionnaire, which is then sent to a larger group of respondents. After the questionnaire is returned with the responses, the monitor team summarizes the results and, based upon the results, they again develop a new questionnaire for the respondent group. The respondent group is usually given at least one opportunity to do the re-evaluation of its original answers based upon examination of the group response.
 - To certain extent, this form of Delphi is a consolidation of a polling procedure and a conference procedure which attempts to shift a significant amount of an effort needed for individuals to communicate from the larger group of respondents to the smaller monitor team. We shall denote this form as conventional Delphi.
 - A newer form is sometimes called as “Delphi Conference,” which replaces the monitor team to a large degree by a computer in which programming has been done to carry out the compilation of the group results. This latter approach has the advantage of turning the process into a real-time communications system by eliminating the delay caused in summarizing each round of Delphi. However, in this it does require that the characteristics of the communication be well defined, before Delphi is undertaken.
 - Whereas, monitor team can adjust these characteristics, through paper-and-pencil Delphi exercise as a function of the group responses. This latter form shall be labelled real-time Delphi. Usually Delphi, undergoes four distinct phases, whether it be conventional or real-time:
 - ◆ The first phase is characterized by exploration of the subject which is under discussion, wherein each individual contributes additional information he feels is relevant to the issue.
 - ◆ The second phase involves reaching an understanding of how the group views the issue, i.e., points where the members agree or disagree with each other and what they mean by relative terms such as importance, feasibility or desirability.
 - ◆ In the third phase, disagreement is explored, if there is any disagreement, to bring out the underlying reasons for the differences and possibly to evaluate them.

- ◆ The last phase, which is the final evaluation phase. This occurs when all previously gathered information has been initially examined and evaluations are sent consideration.

Select the Above Characteristic or the Below One

Key characteristics – The following key characteristics of the Delphi method help the participants to focus on the issues at hand and separate Delphi from other methodologies

- Anonymity of the Participants – The identity of the participants is not revealed, they remain anonymous, even after the completion of the final report. This prevents the authority, personality, or reputation of some participants from dominating others in the process.
- Structuring of Information Flow – The initial step is taken by collecting contributions from the experts in the form of answers to questionnaires and their comments to these answers. The interactions among the participants is controlled by the panel director, by processing the information and remove any irrelevant content. The negative effects of face-to-face panel discussions is avoided and usual problems of group dynamics is solved.
- Regular Feedback – Participants needs to comment on their own forecasts, the responses that others have given and on the progress of the panel as a whole as well. They can also revise their earlier statements at any moment. While in regular group meetings participants tend to stay with their previously stated point of views and sometimes conform too much to the group leader, which the Delphi method prevents.
- Role of the Facilitator – The facilitator or leader is the person coordinating the Delphi method, and facilitates the responses of their panel of experts, whosoever are selected for a reason, usually that they hold knowledge on an opinion or view.
- Delphi Method Process

1. Identify the Issue and Objective

It is important to understand the issue you are trying to solve and know what you want to achieve by using the Delphi method. You must know or should be fully aware of what you are trying to forecast.

2. Choose a Group of Experts and a Facilitator

The experts can be people who are internal or external to an organization. The facilitator should be someone with knowledge and experience with research and data collection and should take a neutral position.

3. Round One Questionnaire

The facilitator provides the first set of questionnaires for the experts. Usually there are open-ended questions in this round, as it let the experts to brainstorm their ideas. Then, the facilitator gathers all the answers from the questionnaire and provides summary report of answers to the experts. The expert's identities remain anonymous in the summary report, in order to encourage them to state their opinions freely.

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4. Round Two Questionnaire

The second questionnaire has to be created by analysing the answers that you get from the first round. Similarities between answers should be identified and eliminate irrelevant content should be eliminated, so that the second questionnaire can go in the way where there is consensus within the experts.

When the experts answer the second questionnaire, their opinions may or may not change after reading the summary report from the first round and understanding the other expert's opinions. After completing the second round of questionnaire, the facilitator hands out a second summary report of answers to the experts, like did in the first round.

5. Round Three Questionnaire

Now proceeding to the third round, the idea is the same as that of the second round. After analysing the answers from the second questionnaire, the third questionnaire should be created. The experts will answer the third questionnaire based on their opinions from the summary report of answers in the second round.

You may choose to stop, if you feel that you've reached enough consensus between the experts, and all their forecasts agree with each other or you may continue to go through other additional rounds of questionnaires if the case is not so.

Limitations and Variations

Applying the Delphi method can really be time consuming method. In a Delphi method setting, it is impossible to reach the final forecasts in hours or minutes, which can be done through group meetings, in which final forecasts can possibly be reached in hours or even minutes. The panel may lose interest and cohesiveness, if it takes a long time to reach a consensus in a Delphi setting.

"In a group setting, personal interactions can lead to quicker and better clarifications of qualitative justifications. "Estimate-talk-estimate" is a variation which is applied to the Delphi method, wherein, the experts can interact between iterations, though the forecast submissions can still remain unnamed. A drawback of this variation is the possibility of the loudest person exerting undue influence."

2.1.4 Historical Analogy

Historical analogy – This method primarily forecasts the demand for a new product, which may be accurate and cheap. It develops a best fit forecast, based on forecasts and past data of any similar or relevant existing products.

Let's take an example: By forecasting the demand of phone cover of iPhone 6, we can base on the sales of iPhone 6, now, forecasting the demand of iPhone 6 will provide a base for the sales of iPhone 5; or forecasting the demand of a new type of camera film can provide a base to the sales of the company's latest camera. The weak point of this is that it relies on analogy being correct, and there is no guarantee that the new product demand will match with that of the forecasted or not.

A historical analogy covers the approaches applied in reaching techniques to direct a judgmental forecast, useful in creating an understanding of history. Analogous

aspects are helpful in creating situations to predict the longer term. The cross comparison between a product and its substitutes helps in arising with a pattern to forecast the longer-term sales of a given product.

It calls for a comparison between the historical performance of a product and using the knowledge to assess the longer-term performance of an in-depth product. It covers situations and events along with comparison between products but also covers situation and events.

The structured analogy process for new product forecasting has six main steps:

Step 1 – Query: You need to find a set of candidate products that have same attributes to the new product.

Step 2 – Filter: Then remove inappropriate or outlier products from the set of candidate products manually.

Step 3 – Cluster: Candidate products must be clustered according to their sales pattern and make use of judgment to manually select the most suitable cluster to serve as the surrogate products.

Step 4 – Model: Then, Select the most suitable statistical model for the cluster of surrogate products.

Step 5 – Forecast: Forecast the new product, Using the statistical model.

Step 6 – Override: Make manual adjustments wherever necessary to the statistical model's forecast.

2.1.5 Market Research

The process of determining the feasibility of a new service or product through research conducted directly with potential customers is market research. A company to discover the target market and get opinions and other feedback from consumers about their interest in the product or service, through market research.

In-house research can be conducted in this type of research or can be conducted by a third-party company that specializes in market research. It can be done by conducting surveys, product testing, and focus groups. Compensation is provided to test samples in form of product samples or paid a small stipend for their time. Market research is a critical element in the research and development (R&D) of a new product or service.

Market research is a critical tool in can be used by companies to determine or understand the need of the consumers, developing products that those consumers will use, and maintaining a competitive advantage over other companies in their industry.

Understanding Market Research

The purpose of market research is to look at the market associated with a particular good or service to find out how the audience will receive it. For this information is gathered for the purpose of market segmentation and product differentiation, which can be used to alter advertising efforts or determine which features are seen as a priority to the consumer.

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A business must take part in a variety of tasks to complete the market research process. It needs to gather information which based on the market sector being examined. The resulting data should be analysed and interpreted to determine the presence of any patterns or relevant data points, that it can use in the decision-making process.

- **How Market Research Gathers Information**

Market research consists of a combination of information which is primary and which has been gathered by the company or by a person hired by the company, and secondary information, or what has been gathered by an outside source.

1) Primary Information

Primary information is the data directly collected by the company or that has been collected by a person or business hired to conduct the research. This type of information is generally divided into two categories: exploratory and specific research.

Exploratory research functions via more open-ended questions and is a less structured option, and it results in questions or issues being presented that the company may need to address. Specific research also finds answers to issues that were identified previously and are often brought to attention through exploratory research.

2) Secondary Information

Secondary information is data that an outside entity has already accumulated. This can include population information from government census data, trade association research reports, or presented research from some other business operating within the same market sector.

- **The Development of Market Research**

In Germany during the 1920s, formal market research began. At the same time, in the United States market research took off during the advertising boom of the Golden Age of Radio. Companies that advertised on the radio began to realize the demographics that were revealed by how different radio shows were sponsored.

1) Face-to-face Interviews

From there, companies were developed for interviewing people on the street about publications that they read, to find out whether they recognise any of the ads or brands within the ads that were published in the magazines or newspapers the interviewer showed them. To see how effective those ads were, data collected from these interviews were compared to the circulation of the publication. From these early techniques, market research and surveys were altered to make them suitable for new use.

2) Phone Research

Data collection then shifted to the telephone, making face-to-face contact unnecessary. Further, this method improved the market research model greatly by collecting the information very quickly and in an orderly fashion through telephone operator who could collect information or organize focus groups.

3) Online Market Research

Many market researchers have shifted their activities online as well, as more people are spending more time online. Even if the platform may have changed, data collection is still mainly done in a survey-style form. This leads to availability of more options now, as now people can themselves choose to sign up and take surveys and offer opinions when they have time, instead of companies only actively seeking participants by finding them on the street or by cold calling them on the phone. This makes the process far less obstructive and less rushed since people can do so on their own time and by their own will.

Steps Business Operators should Keep in Mind during the Research Process

Because of the constantly changing business environment, it can be difficult to keep up with the pace of all the changes. Due to technological advancements and the emergence of social media, the entire concept of marketing has changed, when it comes to market research. So, as a business leader, you must ensure that your marketing teams are using all of their resources efficiently.

Here are six tips business operators should keep in mind during the research process:

1. **Remember Research is an Ongoing Process.** The motive of a research is usually to gain knowledge on consumers and what they want and need. Consumers' needs are also going to change continuously, with the ever-changing society we live in. If you want to keep up with the changing trends, you need to understand those trends and patterns, you need to watch them regularly, for which you need to utilize research to find and analyse trends and patterns. A Continuous monitoring of both the behaviour of your consumers and your competitors will allow you to manage your business in a positive way.
2. **Do not Ignore Opportunity Costs.** There are also opportunity costs in research, just as there are opportunity costs in business decisions. By doing no research at all the benefits your company might have received will be less, than the benefits your company could have received by taking an alternative action or conducting new research which might increase the potential.
3. **Steer Clear of Research Method Short-sightedness.** Research cannot work with one method; you need to have many different research method options because one method cannot work for all research projects. When you set out to answer your own research objectives, you must not limit yourself to only one method. Sometimes, multiple courses of action are required such as a survey, focus group and reviewing the secondary data.
4. **Outline Your Objectives Before Starting Research.** The very first step is defining your objectives in guiding successful market research. Along with that, list the questions you want to answer through the research you plan on, conducting this will help you and your team to stay on task. Keeping your main objectives and questions in the back of your head, while you conduct your research, will ensure that the project is not a waste of your time and money.
5. **Study Your Customers.** By simply tracking their purchases, you cannot understand your customers. Humans are both cognitive and emotional beings, therefore, you

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need to measure their actions and emotions to fully understand their behaviour. When you do research for your company, you need to study customer's emotional drivers and their rational thoughts as well. You can create a better marketing strategy, if you can understand your customer in all aspects.

6. **Share Research Results with Employees.** Once you complete your market research and gain useful knowledge, you need to share it. Information which is beneficial for your company needs to be communicated with all your employees, especially if the new data is going to require strategy changes. Publicizing your research within your company can also show your teams the benefits of conducting market research.

Three Key Objectives of Market Research

A market research project may usually have 3 different types of objectives.

1. **Administrative:** Through proper planning, organization, and both human and material resources control, helps a company or business development, thus satisfying all specific needs within the market, at the right time.
2. **Social:** Satisfying consumer's specific needs through a required product or service. The product or service should comply with the requirements and preferences of a customer when it's consumed.
3. **Economical:** While being new to the market, we need to determine company's economical degree of success or failure they can have, or otherwise introducing new products or services, and thus providing certainty to all actions to be implemented.

Why is Market Research Important?

One of the best ways of achieving customer satisfaction, reducing customer churn and elevating business, is conducting research. Here are the reasons why market research is important and should be considered in any business:

- **Valuable Information:** It helps making business plan and strategize by providing information and opportunities about the value of existing and new products.
- **Customer-centric:** This helps in determining what the customers need and want. Marketing is customer-centric and understanding the customers and their needs will help businesses to design products or services that best suit them.
- **Forecasts:** Businesses can also forecast their production and sales, by understanding the needs of customers. Market research also helps in determining optimum inventory stock.
- **Competitive Advantage:** Market research is a vital tool to carry out comparative studies to stay ahead of competitors. Businesses can devise business strategies that can help them stay ahead of their competitors.
- **Types of Market Research: Market Research Methods and Examples**

Whether an organization or business wishes to know purchase behaviour of consumers or the likelihood of consumers paying a certain cost for a product, market research helps in drawing meaningful conclusions.

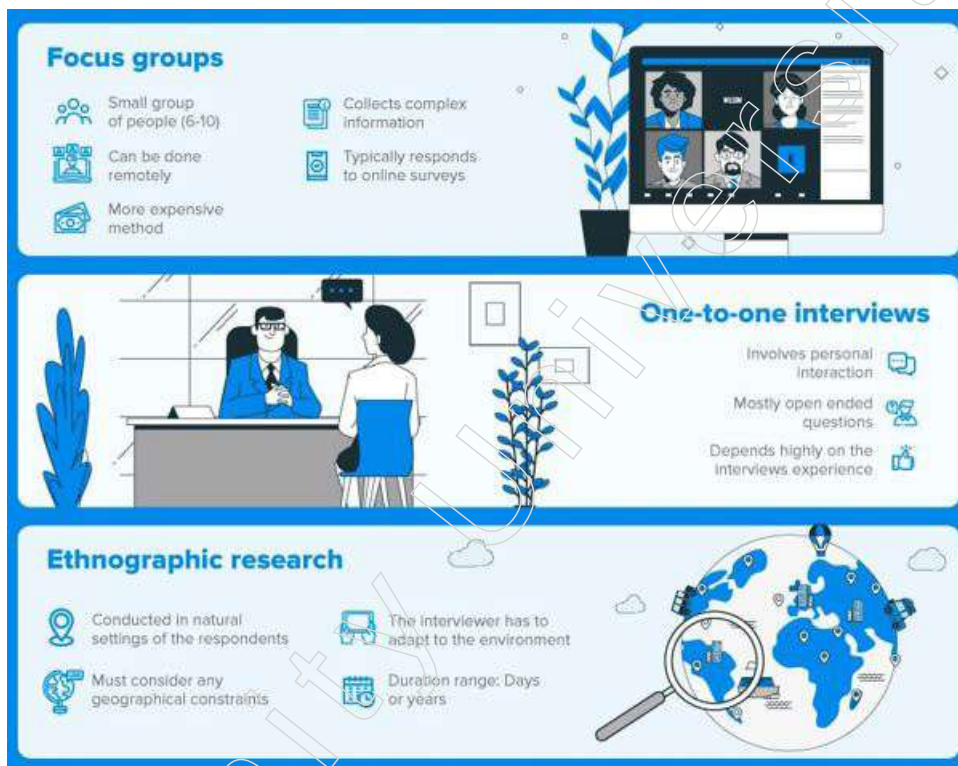
Depending on the methods and tools required, following are the types:

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1. **Primary Market Research:** This is a combination of both Qualitative and Quantitative Research: Primary market research involves a process, where organizations or businesses get in touch with the end users or employ a third party to carry out relevant studies to collect data. The data collected can be qualitative data, which is non-numerical data or quantitative data, which is numerical or statistical data.

One can gather two types of information, while conducting primary market research: Exploratory and Specific. Exploratory research is open ended, where a problem is explored by asking open ended questions in a detailed interview format usually with a small group of people also known as sample. In this, the sample size is small and is restricted to 6-10 members. Specific research, on the other hand, is more pinpointed and is used to solve the problems that are identified by exploratory research.

As earlier mentioned, primary market research is a combination of both qualitative market research and quantitative market research. Qualitative market research study involves semi-structured or unstructured data collected through some of the commonly used qualitative research methods like:



- **Focus Groups:** The most commonly used Focus group is one of the commonly used qualitative research methods. It involves a small group of people (6-10) who typically respond to online surveys sent to them. Through focus group the information can be collected remotely and can be done without any personal interacting with the group members, and is a more expensive method as it is used to collect complex information.
- **One-to-one Interview:** This method involves personal interaction in the form of an interview, where a series of questions is being asked by the researcher to collect information or data from the respondents. These questions are mostly open-ended questions and asked in a way to facilitate responses. This

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method is heavily dependent on the ability and experience of the interviewer to ask questions that calls up responses.

- **Ethnographic Research:** This type of in-depth research is done in the natural settings of the respondents. This method requires the interviewer to adapt himself/herself to the natural environment of the person being interviewed which could be a city or a remote village. There could be a Geographical constraint, which can be a hindering factor in conducting this kind of research. The time span of Ethnographic research can vary from a few days to a few years.

Qualitative research methods are used by organizations to conduct structured market research which uses online surveys, questionnaires and polls to gain statistical insights to make informed decisions.

This method was used to conduct through using pen and paper. This has now evolved to sending structured online surveys to the respondents to gain actionable insights. Researchers tend to use modern and technology-oriented survey platforms to structure and design their survey to evoke maximum response from respondents.

Through a well-structured mechanism, data can be easily collected and reported and necessary action can be taken with all the information that is made available first hand.

2. **Secondary Market Research:** Secondary research uses information that is assembled by outside source like government agencies, media, chambers of commerce etc. This information is issued in newspaper, magazines, books, company website, free government and nongovernment agencies and so on. Secondary source makes use of the following:

- **Public Sources:** Public sources such as library are an awesome way of assembling free information. Government libraries usually offer services free of cost and a researcher can document available data and facts.
- **Commercial Sources:** Although reliable commercial sources are expensive. Local newspapers, magazines, journal, television media are good commercial sources to gather data and facts.
- **Educational Institutions:** Although this is not a very popular source of collecting information, many universities and educational institutions are considered to be a very rich source of information and more research projects are carried out there than any business sector.

- **Steps for Conducting Market Research**

If a researcher already knows what to do in various situations that arises during the investigation will actually save researcher's time and reduce problems. Today successful enterprises use powerful survey software for market research which helps them to conduct thorough research under a unified platform and therefore, provide actionable insights much faster with very less problems.

Following are the steps to conduct an effective market research.

Step #1: Define the Problem

To conduct a research researchers must have a well-defined knowledge of the subject of research which will help researchers when they ask questions. These questions should be such that they are directed towards solving the problems and they have to be adapted to the project. Questions must be written clearly so that the respondents easily understand them. Researchers can conduct a small test with a group of people to know if the questions asked are understandable or not and will they be enough to gain insightful results.

Objective of the research should be written in a precise way and should include a brief description of the information that is needed and the way in which it will obtain it. The researchers should know and should be able to answer this question “why are we doing the research?”

Step #2: Define the Sample

In order to carry out market research, researchers need a representative sample that can be collected using one of the many sampling techniques. A representative or exemplar sample is a small number of people which reflect, as accurately as possible, a larger number of groups.

- Collecting information from a wrong population will lead to waste of resources which an organization cannot do. It is important that the population represents characteristics that matter to the researchers and that they need to investigate, are in the chosen sample.
- Take into account that marketers will always be prone to fall into a bias in the sample because there will always be people who do not answer the survey as they are busy, or answer the survey incompletely, in that case researchers may not get the required data.
- In terms of sample size, the larger the sample size, the more likely it is to be representative of the entire population. A bigger representative sample allows the researcher to be more confident than the persons included are the ones they require, as well as eliminate bias. As a result, they should have representative and balanced samples if they wish to avoid inaccuracy in our surveys.
- Scientific sampling is what all serious surveys are practically based on, which is based on statistical and probability theories.

There are two ways to obtain a representative sample:

- **Probability Sampling:** The sample will be chosen at random in probability sampling, ensuring that every person of the population has the same chance of being selected and included in the sample group. Researchers should make sure they have up-to-date information on the population from which they will draw the sample and conduct a survey of the majority to guarantee that the sample is representative.
- **Non-probability Sampling:** In non-probability sample, the aim is to obtain a more balanced representative sample. Being aware of our group's demographic features will surely help in narrowing the profile of the targeted

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sample and defining the variables that researchers are interested in, such as gender, age, and place of residence. Researchers can create an efficient representative sample for us by knowing these criteria before getting the information. In case of sample is not representative, there is a margin of error. The researchers should select equal number of men and women, If the representative sample is of 100 employees. Though the sample size is critical, it does not ensure accuracy. More than just quantity, representativeness is linked to the sampling frame, or the list from which persons are picked up, as in a survey.

Step #3: Carry Out Data Collection

An important step is to create a data collection instrument. There will be research errors if they do not respond to a survey or answer it incorrectly. If data is collected correctly, then this can be avoided.

Step #4: Analyse the Results

Points of market research process are totally linked to one another. Even if all the above is executed well, but no accurate analysis of the results is there, then the decisions made in consequence will not be appropriate. If an in-depth analysis is being done without leaving loose ends will be effective in gaining effective solutions. Data analysis captured should be written clearly in a report, so that effective decisions can be made on that basis.

Analyse and interpret the results is to look for a wider meaning to the obtained data. All the previous phases have been developed to arrive at this moment.

What criteria can be used to evaluate the outcomes? Age, sex, profession, and number of interviewees will be the only quantifiable data collected; the rest will be feelings and experiences relayed to us by the interlocutors. For this, we use a tool called empathy map, which compels us to put ourselves in the shoes of our customers in order to find the traits that would enable us to better match our products and services to their requirements and interests.

The interpretation is normally carried out after the research has been well planned, the hypotheses have been adequately specified, and the indicated collection method has been followed.

Step #5: Make the Research Report

When presenting the findings, researchers should consider what they want to achieve with this research report, and they should not assume that the survey framework is the best way to do the analysis. Many researchers make the error of presenting their reports in the same sequence as their questions and fail to recognise the value of storytelling.



The greatest analysts advise the following when it comes to writing good reports: To convey the findings, use an inverted pyramid format, beginning with the most important questions about the firm that prompted the research. Instead of gathering evidence, start with the conclusions and give them fundamentals. After this, the researcher can provide details to the readers who have the time and interest.

Step #6: Make Decisions

A company or researcher should never ask, "Why do market research?" Instead, they should just do it!

Market research allows researchers to learn an extensive range of facts and information, such as, intentions of consumer purchase, or also provides comments about the growth of the target market. They can also obtain useful information that will assist them in predicting the the prices of their product or service and find a point of equilibrium that will beneficial to them and the consumers.

Take decisions! Act and implement.

- Benefits of an Efficient Market Research
 - Make well-informed Decisions: The manner in which management makes decisions affects an organization's growth. Management can make company decisions based on gathered data that back up their knowledge and experience using market research tools. Market research is useful for determining market trends; thus, it should be conducted on a regular basis to gain a full understanding of your customers.
 - Gain Accurate Information: Market research delivers true and precise information that can help the firm plan for any future errors. By thoroughly

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researching the market, a company will certainly advance and, as a result, will get an advantage over its existing competitors.

- **Determine the Market Size:** The size of the market can be evaluated by the researcher that must be covered in case of selling a product or service in order to make profits.
- **Choose an Appropriate Sales System:** Select a precise sales system based on what the market demands, and the product/service may then be positioned in the market accordingly.
- **Learn about Customer Preferences:** It's important to understand how customers' preferences (and tastes) evolve over time so that the business can cater to their choices, shopping patterns, and financial level. Based on the individual needs of consumers, researchers can decide the type of product that must be made or sold.
- **Gather Details about Customer Perception about the Brand:** Market research, in addition to producing data, aids a researcher in understanding how buyers perceive a company or brand.
- **Examine Customer Communication Methods:** Market research can help you communicate with present and potential customers.
- **Profitable Company Investment:** It is a wonderful investment for any business since it provides essential information and tells researchers how to go on the correct track and make the necessary sales.
- **5 Market Research Tips for Businesses**

Tip #1: Define the Objective of your Research.

Consider what you're attempting to accomplish with your business next before embarking on your research mission. Do you want to increase foot traffic to your business? Or do you want to boost your sales? Or do you want to turn one-time buyers into regulars? Identifying your goal can aid you in tailoring the rest of your study and future marketing materials. Having a research objective will help you figure out what kind of data you'll need.

Tip #2: Learn about Your Target Customers.

The most crucial thing to keep in mind is that your company caters to a specific type of client. Defining your unique consumer has a number of benefits, including helping you figure out what language to use in your marketing materials and how to approach creating relationships with them. You can locate the greatest products and services to market to your target consumer if you take the time to define them.

Whatever you want to learn everything you can about your target market. This knowledge can be gathered by observation and research about the kind of customers who visit your type of business. For starters, knowing their age and salary is beneficial. What do they do for a living? What's their marital status and education level?

Tip #3: Recognize that Knowing Who You Serve Helps You Define Who You Do Not.

Let's look at a great example from Dan Kennedy, the master of copywriting. He

claims that if you're building a fine dining steakhouse focusing on indulgent cuisine, you know you're not going to appeal to vegetarians or dieters right away. You can build better marketing messages that speak to your target clients now that you have this knowledge.

It's fine to exclude certain customers from your target market. In fact, recognising who you don't cater to can be crucial to your small business's growth. Why? Simply said, being small has the advantage of allowing you to connect deeply with a specific target group. You want to concentrate your efforts on the correct customer who has already expressed an interest, and is compelled to spend money on your offer.

Spreading yourself too thin by attempting to be all things to all people will only dilute your primary message. Keep your attention on your target buyer instead. Define them, go deep, and you'll be able to figure out how your products and services can best serve them.

Tip #4: Learn From Your Competition.

This works for both brick and mortar and online businesses since it helps you to put yourself in the shoes of your customer and gain a fresh perspective on your company. Take a look around the web and in your neighbourhood. If possible, pay a visit to one of your competitors' stores. If you own an Italian restaurant, for example, eat at another Italian restaurant in your area or in the next township.

Look for what's being done well and wrong as you experience the business through the eyes of a customer.

Do you notice any areas that need to be addressed or improved? In comparison, how are you handling things? What is the product's and customer service's quality? Are the customers satisfied? Take a good look at their target market as well. Who else is coming to their store? Are these the same types of folks that give you money? You may get a lot of information by asking these questions and conducting in-person research to help you identify your unique selling position and generate even better offerings for your customers.

Tip #5: Get Your Target Customers to Open Up and Tell You Everything.

Because it allows you to go inside your customer's thoughts, a good customer survey is one of the most valuable market research tools. Take criticism as a learning tool to steer you in the correct direction, but keep in mind that some input may be harsh.

It's easy to make a survey. Inquire about what your customer feels you're doing well and what you can do better. You may also ask them what sorts of items and services they'd want to see you offer, which can provide you with invaluable information into how to better monetise your business. Many clients will be more than happy to provide feedback. Customers those complete surveys can even receive a prize, such as a special voucher for their next purchase.

An insight & research repository is a centralised research management platform that allows you to gain insights from past and current market research. You may use a tool like this to leverage earlier research to get to insights faster, build on previous market research and develop trendlines, and apply research methodologies that have succeeded in the past, among other things.

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- **Need of Business for Market Research**

One of the most effective ways to obtain insight into your client base, rivals, and the entire market is to conduct market research. The purpose of market research is to provide your firm with the data it requires to make informed decisions.

It's especially crucial for small firms seeking to figure out whether a new business idea is viable, expanding into new markets, or introducing a new product or service. For a more in-depth look at how market research might benefit small firms, continue reading.

Competition: According to a Business Insider report, 72 percent of small firms are focused on boosting income. Businesses can acquire insight into rival behaviour by doing research. You can learn how to position your product or offering by studying the strengths and shortcomings of your competitors. Small businesses must understand what products and services competitors offer, as well as their pricing point, in order to be successful.

Customers: Many small firms believe they have a good grasp of their customers, only to find out later that their assumptions were incorrect. You may construct a profile of your average consumer by performing research and learning about their buying habits, how much they're prepared to spend, and which characteristics appeal to them. Furthermore, and probably more crucially, you can discover what motivates someone to choose your product or service over one offered by a competitor.

Opportunities: Market research can be used to identify potential opportunities, whether they be products or services. You can gain insights into complementing products and services by understanding more about your customers. Consumer needs evolve throughout time, impacted by new technologies and changing circumstances, and you may discover new needs that aren't being satisfied, which can lead to new business chances.

Forecast: Customers and a small business are both affected by the success of the local and national economies. When customers are concerned, they are less likely to spend money, which has an impact on the firm. Businesses can learn if consumers are enthusiastic or pessimistic about the economy's direction by doing consumer research and making adjustments as needed. If the economy appears to be deteriorating, a small business owner can decide to postpone the launch of a new product.

Check Your Understanding

- 1) Risk and uncertainty are central to _____ and _____.
 - a) Forecasting and Prediction
 - b) Demand and Forecasting
 - c) Predictions and Supply
 - d) Economy and Technology
 - e) None of the above
- 2) _____ is a process in which future condition of an economy is predicted.
 - a) Technological Forecasts

- b) Economic forecasts
 - c) Demand Forecasts
 - d) Sales Forecasting
 - e) None of the above
- 3) The Delphi method, scenario building, and statistical surveys which are based on intuition and a collection of opinions from managers and experts are included in _____.
- a) Time-Series Forecasting
 - b) Associative Forecasting
 - c) Data Forecasting
 - d) Judgemental Forecasting
 - e) None of the above
- 4) _____ is a method of making predictions about a company's finances that uses judgement from experts, rather than numerical analysis.
- a) Quantitative forecasting
 - b) Naive Forecasting
 - c) Qualitative forecasting
 - d) Casual forecasting
 - e) None Of the above
- 5) _____ is the technique in which the next period's forecast is done without predictions or adjusting the factors, by using the last period's sales.
- a) Naive forecasting
 - b) Casual Forecasting
 - c) Judgemental Forecasting
 - d) Time-Series Forecasting
 - e) None of the above
- 6) _____ is mainly concerned with planning, organizing and supervising in the contexts of production, manufacturing or the provision of services.
- a) Swift Polling
 - b) Judgemental Forecasting
 - c) Time Series Forecasting
 - d) Operations management
 - e) None of the above
- 7) What is the full form of MAPE?
- a) Mean absolute percentage error
 - b) Model of absolute percentage error

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- c) Method of absolute percentage error
 - d) Mean absolute periodic error
 - e) None of the above
- 8) What is the full form of MAD?
- a) Mean Analysis Deviation
 - b) Mean Absolute Deviation
 - c) Mean Associative deviation
 - d) Mean Assumed Deviation
 - e) None of the above
- 9) What is the full form of MSE?
- a) Model Square Error
 - b) Mean square error
 - c) Method Square Error
 - d) Model Sequence Error
 - e) None of the above
- 10) _____ is solely based on use of historical data to make predictions about the future and strategic decision making.
- a) Time series forecasts
 - b) Naïve Forecasts
 - c) Qualitative Forecasts
 - d) Demand Forecasts
 - e) None of the above
- 11) _____ is a method of making predictions about a company's finances that uses judgement from experts, rather than numerical analysis.
- a) Quantitative Forecasting
 - b) Judgemental Forecasting
 - c) Qualitative forecasting
 - d) Accurate Forecasting
 - e) None of the above
- 12) _____ is the technique in which the next period's forecast is done without predictions or adjusting the factors, by using the last period's sales.
- a) Naive forecasting
 - b) Swift Polling
 - c) Egalitarianism
 - d) Time Keeping
 - e) None of the above

- 13) _____ is the method which is based on an assumption that mathematical function which uses current variables that can be also used to forecast the future value of variables.
- a) Judgemental Forecasting
 - b) Casual Forecasting
 - c) Time Series Forecasting
 - d) Consensus
 - e) None of the above
- 14) The _____ is a forecasting process framework which is based on the results of multiple rounds of questionnaires sent to a panel of experts.
- a) Quantitative Method
 - b) Historical Analogy Method
 - c) Delphi method
 - d) Market Research Method
 - e) None of the above
- 15) Delphi Method Process involves which of the following steps:
- a) Identify the issue and objective
 - b) Round Two Questionnaire
 - c) Choose a group of experts and a facilitator
 - d) Round three questionnaire
 - e) All the above

Summary

- forecasting is estimating the future values of certain specified indicators relating to a decisional/planning situation.
- Accuracy is very much required and is very important when it comes to forecasts. If the demand of the product is overestimated, then the company could end up spending money on manufacturing, distribution, and servicing activities it won't need.
- Proper demand forecasting gives businesses valuable information about their potential in their current market and other markets.
- In the long term, forecasts are required to predict the demand and requirement for new product.
- Forecasting has become valuable to businesses as it gives the ability to make informed business decisions and develop data-driven strategies.
- The Delphi method, scenario building, and statistical surveys which are based on intuition and a collection of opinions from managers and experts are included in Judgemental forecasting.

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- In naive forecasts there are no calculations or formulas, there only is an assertion of the actual sales numbers.
- Qualitative forecasting techniques come from the experiences and instincts of the business experts.
- Swift polling, is the perfect tool to get started, which provides various poll and survey-related services.
- Delphi method has been developed as a systematic, interactive forecasting method which relies on a panel of experts, and is a structured communication technique, in which questionnaires in two or more rounds are there.

Activity

1. Visit an organization, meet an operations manager and understand when and how forecasting has been done, and for what all activities has it been done yet. Also understand if forecasting is a regular practice in the organization or is it not required, and what all forecasting method has been used for the same?

Questions and Exercises

1. Briefly explain what is Forecasting? Explain the steps and applications of Forecasting.
2. What is Qualitative Forecasting? Explain its methods as well.
3. What is Market Research? Explain its types and why businesses need market research and describe the steps for conducting market research and its benefits?

Glossary

- Forecasting is the process of making predictions on the basis of past and present data for the future based activities.
- The forecast horizon is the length of time or time period into the future for which forecasts are to be prepared.
- Long term forecasting tends to be completed at high levels in the organization. The time frame for long term forecasting is generally considered longer than 2 years into the future.
- Medium term forecasting is one in which the time frame generally is for several months, up to 2 years into the future
- Short term forecasting is almost done daily up to months in the future.
- Economic forecasts: This is a process in which future condition of an economy is predicted.
- Demand forecasts: It is very important to predict the future demand for a company's products or services, so that production can be done accordingly, without any wastage.
- Associative models are also known as casual models, which uses data corresponding to various underlying factors to predict the desired variable or another variable of interest.
- The Delphi method is a forecasting method based on the results of questionnaires

sent to a panel of experts. Several rounds of questionnaires made and circulated, and the anonymous responses are aggregated and shared with the group after each round.

- Consensus means that each person must agree to support the implementation of the decision, regardless of whether or not he/she agrees with the decision.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
3. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding – Answers

- 1) a)
- 2) b)
- 3) d)
- 4) c)
- 5) a)
- 6) d)
- 7) a)
- 8) b)
- 9) b)
- 10) a)
- 11) c)
- 12) a)
- 13) b)
- 14) c)
- 15) e)

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Notes**Unit - 2.2: Quantitative Forecasting****Objectives**

At the end of this unit, you will be able to:

- Linear regression
- Moving average
- Weighted moving average
- Exponential smoothing

Introduction

Quantitative forecasting is a data-based mathematical process that sales teams use to understand performance and predict future revenue based on historical data and patterns. Forecasting results give businesses the ability to make informed decisions on strategies and processes to ensure continuous success.

Forecasting future data as a function of past data is done using quantitative forecasting models. When prior numerical data is accessible and it is acceptable to believe that some of the data patterns will persist into the future, they are appropriate to employ. These techniques are typically used to make short- or intermediate-term judgments. Time-series methods include causal (econometric) forecasting methods, last period demand (nave), simple and weighted N-period moving averages, and simple exponential smoothing, which are all examples of quantitative forecasting methods. Models of quantitative forecasting are frequently compared by comparing their accuracy performance measures. Mean Absolute Deviation (MAD), Mean Squared Error (MSE), and Mean Absolute Percentage Error are examples of these metrics (MAPE).

2.2.1 Linear Regression

- Linear regression is a statistical technique for predicting future values based on past data. It's a popular quantitative method for determining the underlying trend and when prices have been overextended.
- A linear regression trendline plots a straight line through prices using the least squares approach to minimise the gaps between the prices and the resulting trendline. The trendline value for each data point is plotted using this linear regression indicator.
- An approach for determining the relationship between two variables is linear regression analysis. The dependent variable, which responds to the change, and the independent variable are the two variables involved. It is important to note that we are not calculating the dependency of the dependent variable on the independent variable just the association.
- For example, if a company invests a particular amount of money in the marketing of a product and collects sales data over time, we may forecast next year's sales by analysing the correlation between the marketing budget and sales data. The principle of prediction outlined above may appear magical, yet it is based only on

data. Fitting a straight line to our dataset in order to predict future events is what linear regression is all about.

The best fit line would be of the form:

$$Y = B_0 + B_1X$$

Where, Y – Dependent variable

X – Independent variable

B₀ and B₁ – Regression parameter

- Linear Model Assumptions

Six key assumptions underpin linear regression analysis:

1. The slope and intercept of the dependent and independent variables have a linear relationship.
2. There is no randomness in the independent variable.
3. The residual (error) is equal to zero.
4. The residual (error) value remains constant across all observations.
5. The residual (error) value is not consistent across all observations.
6. The residual (error) values are distributed normally.

2.2.2 Moving Average



A moving average is a technical indicator used by investors and traders to evaluate which way the market is trending.

By analysing the price movements of an item, analysts utilise the moving average to look for support and resistance. A moving average indicates a security's prior price action/movement. The information is then used by analysts or investors to predict the asset's price movement. It is regarded as a lagging indicator because it produces a signal or shows the direction of a given trend by trailing the price movement of the underlying item. It's calculated by summing all the data points from a given period and dividing the total by the number of periods.

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- Types of Moving Averages

The following are the two basic forms of moving averages:

1. Simple Moving Average (SMA)

The simple moving average (SMA) is a simple technical indicator that is calculated by adding up all of the recent data points in a collection and dividing the total by the number of time periods. The SMA indicator is used by traders to generate tips about when to enter or quit a market. Because it is based on prior price data for a certain period, a SMA is backward-looking. It can be calculated for a variety of price kinds, including high, low, open, and close.

Analysts and investors in financial markets use the SMA indicator to establish buy and sell signals for stocks. The SMA is used to determine where to start and exit a trade by identifying support and resistance values. Traders must first compute the SMA by adding prices over a specific time period and dividing the total by the entire number of periods. After then, the data is plotted on a graph.

The formula for Simple Moving Average is written as follows:

$$\text{SMA} = (A_1 + A_2 + \dots + A_n) / n$$

Where:

- A is the average in period n
- n is the number of periods

Example of a Simple Moving Average

By looking at the stock's closing prices over the last five days, John, a stock trader, wishes to compute the simple moving average for Stock ABC. The latest five days' closing prices for Stock ABC are as follows: \$23, \$23.40, \$23.20, \$24, and \$25.50. The SMA is then calculated as follows:

$$\text{SMA} = (\$23 + \$23.40 + \$23.20 + \$24 + \$25.50) / 5$$

$$\text{SMA} = \$23.82$$

2. Exponential Moving Average (EMA)

The other type of moving average is the exponential moving average (EMA), which gives more weight to the most recent price points to make it more responsive to recent data points. An exponential moving average tends to be more responsive to recent price changes, as compared to the simple moving average which applies equal weight to all price changes in the given period.

When calculating the exponential moving average, the following three steps are used:

a. Calculate the Simple Moving Average for the Period

The EMA has to start someplace, and the previous period's EMA is the simple moving average. It's calculated by multiplying the total of the security's closing prices for the period in question by the number of periods.

b. Calculate the Multiplier for Weighting the Exponential Moving Average

The formula for calculating the multiplier is as follows:

$$\text{Multiplier} = [2 / (\text{Selected Time Period} + 1)]$$

For example, if the time period in question is 10, the multiplier will be calculated as follows:

$$\text{Multiplier} = [2 / (10+1)] = 0.1818$$

c. The Last Step is to Calculate the Current Exponential Moving Average

The current EMA is calculated in the final step by using the price, multiplier, and the prior period's EMA value to calculate the duration from the beginning point of EMA to the most recent time period. It is computed using the following formula:

$$\text{Current EMA} = [\text{Closing Price} - \text{EMA (Previous Time Period)}] \times \text{Multiplier} + \text{EMA (Previous Time Period)}$$

The weighting given to recent price data is higher for a longer-period EMA than a shorter-period EMA. A multiplier of 18.18% is applied to the recent price points of a 10-period EMA, whereas a 9.52% multiplier is applied for the recent price points of a 20-period EMA.

- Exponential Moving Average vs. Simple Moving Average

The key distinction between the two technical indicators is the degree of sensitivity to price movements. The exponential moving average is more sensitive to recent price point changes than the simple moving average. As a result, the EMA is more responsive to recent price fluctuations.

Although the technique for computing the EMA is complex, most charting software makes it simple for traders to follow an EMA. The SMA, on the other side, gives equal weight to all total study sample. It is simple to compute, as it is obtained simply calculating the arithmetic mean of prices during the specified time period.

2.2.3 Weighted Moving Average



- A technical indicator that identifies trend direction is the weighted moving average. It provides trade signals by giving recent data points more weight and past data

Notes

points less weight. Asset closure prices are frequently used as data points. It goes a step further and is more accurate than the simple moving average (SMA), which determines market movement by giving all the numbers in a data set the same weight.

- The WMA calculates the average of supplied input values over specified time periods by giving recent data more weight. This is accomplished by multiplying a weighting factor by the price of each bar in a given set and then adding the results. It is then used to smooth the data series in order to help traders spot data trends by filtering out market noise.
- When the price approaches or crosses above the WMA line on the chart, for example, it may indicate that a short-term price decline is on the way. As a result, traders should close their positions. If the price falls near or slightly below the WMA line, however, it may indicate that the price will soon rise. As a result, it is a good moment to get into the business.

How to Calculate the Weighted Moving Average

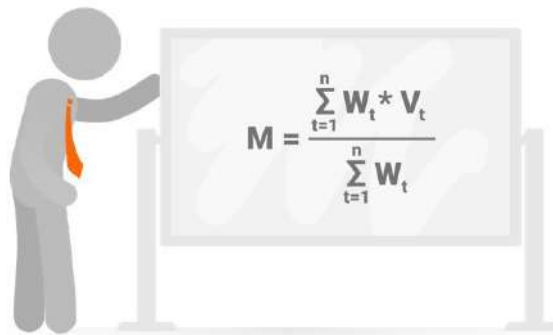
When calculating the weighted moving average, recent data points are given a higher weighting than previous data points. When all of the weights are added together, the overall value should equal 100 percent, or 1.

As previously stated, the weighted moving average calculation differs from the standard moving average calculation, which assigns equal weighting to all data points. You need to simply add all the data points within a given period and divide the total by the number of periods to get the SMA. The WMA, on the other hand, analyses the significance of each data piece, as evidenced by the 'weight' assigned to it.

As a result, the WMA is more descriptive and ultimately more detailed than the SMA. Additionally, it is more responsive to price action.

Weighted Moving Average Formula

The WMA formula is expressed as follows:



M = Average value

V = Actual value

W = Weighting factor

n = Number of periods in the weighting group

Example 1

Let's assume, for the past 3 days the closing stock of XYZ stock has been \$50, \$45, and \$60. Now, Calculate the weighted moving average.

Step 1 – Identify the Numbers to Average

So here we have the three numbers provided, which were recorded over a three-day period — \$50, \$45, and \$60.

Step 2 – Assign the Weights to Each Number

Because there are three periods, the most recent day (\$60) gets the highest weight of 3, the second most recent day (\$45) gets a weight of 2, and the last day (\$50) gets only one. Of course, this is only a simplified illustration.

Step 3 – Multiply Each Price by the Assigned Weighting Factor and Sum Them

The calculation will go as follows: $[(3 \times \$60) + (2 \times \$45) + (1 \times \$50)] = 320$, using the WMA formula.

Step 4 – Divide the Resulting Value by the Sum of the Periods to the WMA

$1+2+3 = 6$ is the sum of the periods. As a result, the three-period weighted average is $(180 + 90 + 50) / 6 = 53.33$. The WMA value of 53.33 is higher than the 51.67 calculated by the SMA. This step's division by 6 resulted in the weightings totalling $6 / 6 = 1$.

Remember, they should add up to 100% or 1.

How the Weighted Moving Average Works

Let's look at how you can utilise the weighted moving average of a group of variables in your trades now that you know how to compute it:

- To Help Establish Trend Direction - When the price is above its weighted MA line, it usually means the asset is trading higher on average than it has been over the time period being studied. This confirms the upward trend. When the price is below the WMA line, however, a downtrend is confirmed.
- To Highlight Areas of Support and Resistance – A rising weighted moving average can signal price support. A dropping WMA, on the other hand, can imply price resistance over a specific time period. Traders frequently employ this approach to make buy orders when the price is near the rising WMA and sell orders when the price is near the dropping WMA.
- To Confirm the Market's Pricing Strength and Momentum — Because the most recent prices are now higher than the average, price activity above its moving average indicates that the market is becoming stronger relative to where it was previously. Price activity below its moving average, on the other hand, indicates that the market is weakening in comparison to where it was previously.
- To Assess Price Movement - The weighted moving average is more sensitive

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to price movement than the simple moving average, making it a better gauge and a better predictor of trends than the SMA. However, because the WMA is more volatile than the related SMA, this might be a double-edged sword.

- To Generate Trade Signals - The WMA can be used in conjunction with other technical indicators, such as Keltner Channels, to identify perfect trading signals. Traders might enter the market at the WMA when there is a price pullback from the Keltner Channel's peak during an upswing.

2.2.4 Exponential Smoothing



- Exponential smoothing (Brown, 1959; Holt, 1957; Winters, 1960) was suggested in the late 1950s and has inspired some of the most successful forecasting approaches. Exponentially smoothed forecasts are weighted averages of previous observations, with the weights decaying exponentially as the observations get older.
- In other words, the larger the related weight, the more recent the observation. This framework produces accurate forecasts fast and for a wide range of time series, which is of significant benefit and major importance for industrial applications. Simple exponential smoothing is the most basic of the exponentially smoothing approaches (SES). This strategy works well when there is no evident trend or seasonal pattern in the data.

Exponential smoothing has been widely employed in forecasting applications at the strategic, tactical, and operational levels for more than half a century, such that:

- At Strategic Level: The forecasting method is used for planning on investment and growth, as well as the impact of innovations.
- At Tactical Level: The forecasting is used to determine spending, inventory concerns, and customer satisfaction.
- At Operational Level: The forecasting is used to set goals, predict quality, and confirm compliance with standards.

- Types of Exponential Smoothing

Broadly, there are three types of exponential smoothing techniques that rely on trends and seasonality. They are:

1) Simple Exponential Smoothing (SES)

When the data, in particular, does not support any of the above, SES is utilised for time series anticipation.

- Trend: A slope that is sloped either uphill or downward.

- b) Seasonality: The occurrence of a given pattern as a result of seasonal characteristics such as hours, days, years, and so on.

Only the level component is estimated by a single exponential smoothing. Weighted averages are used by SES (the average of previous level and current observation). Again, the biggest weights correspond to recent data, whereas the smallest weights correspond to previous observations. Smoothing parameter α (alpha - single parameter/hyperparameter) is always used to determine the weight of each parameter or the decrease in weight.

The value of α (alpha) lies between 0 to 1 such that;

- α (alpha)=0: indicates that future anticipated values are the average of previous data (giving more weights to historical data)
- α (alpha)=1: indicates that the future forecast values are based on the most recent observation (giving more weights to recent observations).

A value around 1 suggests quick learning (i.e., just the most recent values influence forecasts), whereas a value near 0 indicates slow learning (past observations have a large influence on forecasts).

Hyperparameter for single exponential smoothing is:

- Alpha: Smoothing factor for the level

The formula for the single exponential smoothing is given by:

$$S_t = \alpha x_t + (1 - \alpha) S_{(t-1)} = S_{(t-1)} + \alpha (x_t - S_{(t-1)})$$

where α is smoothing parameter, and t is given time period.

2) Double Exponential Smoothing (DES)

DES adds support for patterns in univariate time series in particular. Holt's linear trend model is the name given to it when it is combined with additive trends. The name comes from the name of the method's creator, Charles Holt.

This strategy assists in modifying trends through time in many ways, either additively or multiplicatively, depending on whether the trend is linear or exponential, i.e.

- Additive Trend: DES with a linear trend
- Multiplicative Trend: DES with an exponential trend

The alpha parameter is utilised to control/determine the smoothing factor for a specific level, and an extra smoothing factor, beta, is included to govern the decay of the impact of trend changes.

When projections include a longer time horizon or more forecasting phases, the trend tends to become unrealistic, which helps to attenuate the trend over time.

The term "dampening" refers to the process of reducing the magnitude of a trend over time until it becomes a straight line (no trend).

Damping the trend is done in the same way as modelling it, with additive or multiplicatively applied dampening effects for linear and exponential dampening effects, respectively.

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A damping coefficient ϕ is utilised to determine and control the dampening rate in this scenario, i.e.,

- Additive Dampening: Make a damped trend linear.
- Multiplicative Dampening: Make a dampened trend exponential.

Hyperparameters are following;

- Alpha: Level smoothing factor
- Beta: Trend smoothing factor
- Trend Type: Multiplicative or Additive
- Dampening Type: Multiplicative or Additive
- Phi: Damping coefficient

The formulas for double exponential smoothing are given by:

$$S_1 = x_1$$

$$B = x_1 - x_0$$

For $t > 1$,

$$S_t = \alpha x_t + (1 - \alpha) (S_{(t-1)} + b_{(t-1)})$$

$$\beta_t = \beta (S_t - S_{(t-1)}) + (1 - \beta) b_{(t-1)}$$

where,

- $S_{t=s}$ smoothed statistic, it is the simple weighted average of recent observation x_t
- $S_{(t-1)}$ = previous smoothed statistic
- α = smoothing factor of data; $0 < \alpha < 1$
- t = time period
- b_t = best estimate of trend at time t
- β = trend smoothing factor; $0 < \beta < 1$

3) Triple Exponential Smoothing (TES)

TES explicitly supports seasonality in univariate time series. It is also known as Holt-Winters Exponential Smoothing after two contributors, Charles Holt and Peter Winters.

The Holt-Winters exponential smoothing approach permits the level, trend, and seasonality patterns to evolve and change since it is an adaptable technique. In addition to the two smoothing components, alpha and beta, a third factor, gamma, is included to control/determine the impact on the seasonal aspect.

Seasonality can be treated in terms of additive or multiplicative processes for linear and exponential variation in seasonality, in accordance with the trend.

- Additive Seasonality: TES with a linear seasonality.
- Multiplicative Seasonality: TES with an exponential seasonality.

TES can create single and double exponential smoothing models through configuration since it is an advanced variant of exponential smoothing. Furthermore, the number of time steps in a seasonal period should be specified in order to create and verify the correctness of seasonality models.

Hyperparameters are following:

- Alpha: Smoothing factor for the level.
- Beta: Smoothing factor for the trend.
- Gamma: Smoothing factor for the seasonality.
- Trend Type: Additive or multiplicative.
- Dampen Type: Additive or multiplicative.
- Phi: Damping coefficient.
- Seasonality Type: Additive or multiplicative.
- Period: Time steps in seasonal period

The formulas for triple exponential smoothing are given by:

$$\begin{aligned}
 s_0 &= x_0 \\
 s_t &= \alpha \frac{x_t}{c_{t-L}} + (1 - \alpha)(s_{t-1} + b_{t-1}) \\
 b_t &= \beta(s_t - s_{t-1}) + (1 - \beta)b_{t-1} \\
 c_t &= \gamma \frac{x_t}{s_t} + (1 - \gamma)c_{t-L}
 \end{aligned}$$

where,

- $S_{t=s}$ smoothed statistic, it is the simple weighted average of recent observation x_1
- $S_{(t-1)}$ = previous smoothed statistic
- α = smoothing factor of data; $0 < \alpha < 1$
- t = time period
- b_t = best estimate of trend at time t
- β = trend smoothing factor; $0 < \beta < 1$
- c_t = sequence of seasonal correction factor at the time t
- γ = seasonal change smoothing factor: $0 < \gamma < 1$

Check Your Understanding

- 1) At _____ the forecasting method is used for planning on investment and growth, as well as the impact of innovations.
 - a) Strategic level
 - b) Operational Level
 - c) Tactical Level
 - d) Managerial Level
 - e) All the above
- 2) An _____ tends to be more responsive to recent price changes.

Notes

Notes

- a) Weighted Moving Average
 - b) Simple Moving Average
 - c) Exponential moving average
 - d) None of the above
 - e) All the above
- 3) The _____ is a simple technical indicator that is calculated by adding up all of the recent data points in a collection and dividing the total by the number of time periods.
- a) Multiplier
 - b) Simple moving average
 - c) Linear Aggression
 - d) Dependable Variable
 - e) None of the above
- 4) _____ is a popular quantitative method for determining the underlying trend and when prices have been overextended.
- a) Linear Regression
 - b) Moving Average
 - c) Weighting method
 - d) Exponential Smoothing
 - e) None of the above
- 5) A _____ is a technical indicator used by investors and traders to evaluate which way the market is trending.
- a) Swift Polling
 - b) Moving average
 - c) Process documentation
 - d) Strategic Planning
 - e) None of the above
- 6) A _____ trendline plots a straight line through prices using the least squares approach to minimise the gaps between the prices and the resulting trendline.
- a) Linear regression
 - b) Delphi Method
 - c) Historical Analogy
 - d) Consensus
 - e) None of the above
- 7) _____ is a data-based mathematical process that sales teams use to understand performance and predict future revenue based on historical data and patterns.

- a) Quantitative forecasting
- b) Qualitative forecasting
- c) Naïve forecasting
- d) Demand forecasting
- e) None of the above

Summary

- Time-series methods include causal (econometric) forecasting methods, last period demand (naïve), simple and weighted N-period moving averages, and simple exponential smoothing, which are all examples of quantitative forecasting methods.
- The key distinction between Exponential Moving Average and Simple Moving Average is the degree of sensitivity to price movements.
- A linear regression trendline plots a straight line through prices using the least squares approach to minimise the gaps between the prices and the resulting trendline.
- Exponentially smoothed forecasts are weighted averages of previous observations, with the weights decaying exponentially as the observations get older.
- A technical indicator that identifies trend direction is the weighted moving average. It provides trade signals by giving recent data points more weight and past data points less weight.

Activity

1. Write a Case study of any organization of your choice, to show how they have used Quantitative forecasting methods and how these forecasts have helped their organization to grow well.

Questions and Exercises

1. What is Weighted moving average? How do you calculate weighted moving average?
2. What is Quantitative Forecasting? Explain the methods of making Quantitative Forecasting?

Glossary

- Linear regression is a statistical technique for predicting future values based on past data.
- A moving average is a technical indicator used by investors and traders to evaluate which way the market is trending.
- The simple moving average (SMA) is a simple technical indicator that is calculated by adding up all of the recent data points in a collection and dividing the total by the number of time periods.

Notes

- Exponentially smoothed forecasts are weighted averages of previous observations, with the weights decaying exponentially as the observations get older.
- The term “dampening” refers to the process of reducing the magnitude of a trend over time until it becomes a straight line (no trend).

Further Readings

1. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
2. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check your understanding - Answers

- 1) a)
- 2) c)
- 3) b)
- 4) a)
- 5) b)
- 6) a)
- 7) a)

Unit - 2.3: Analytical Tools for DSS for Operations Management

Notes

Objectives

At the end of this unit, you will be able to:

- Learn analytical tools for DSS for operations management
- Understand types of Decision Support System
- Learn management requirements of the Decision Support Systems

Introduction

- Over the last three decades, decision support systems have progressed from simple model-oriented systems to complex multi-function entities. Most Decision Support Systems of the 1960s were based on powerful (and expensive) mainframe computers that generated organised, monthly reports for managers.
- Decision Support Systems evolved into increasingly complex computer-based systems that supported production, promotion, pricing, marketing, and some logistical activities as MIS theory evolved during the 1970s. Academics were more interested in Decision Support Systems in the early 1980s, and the framework for Decision Support Systems was substantially enlarged by the end of the decade.
- Decision Support Systems underwent a paradigm change in the 1990s, with increasingly complex systems including advanced database technology and client/server capabilities arising from a variety of business processes.
- Object oriented technology and data warehousing began to make an impact on Decision Support Systems as many businesses upgraded their network infrastructure. The fast spread of the Internet expanded the reach of Decision Support Systems, resulting in the development of many new innovative systems such as OLAP and other web-drive systems.
- Types of DSS

As mentioned earlier, DSS can be classified in many different categories. Amongst the common ones are the following:

a) Data Driven DSS

These DSS emphasise access to and manipulation of massive quantities of structured data and include file drawer systems, data analysis systems, analysis information systems, and data warehousing.

b) Model Driven

Accounting models, finance models, representation models, optimization models, and other disciplines or areas of specialisation may be used to drive the DSS's underlying model. Model drive DSS focuses on model access and manipulation rather than data, i.e., it leverages data and parameters to assist decision makers in evaluating

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a scenario. These systems are often not data-intensive, and as a result, they are not linked to huge databases.

c) Knowledge Driven

These systems offer user-friendly recommendation and/or suggestion schemes to help them choose the best solution to an issue. Management expert systems or intelligent decision support systems are terms used to describe knowledge-driven DSS. They concentrate on knowledge and provide recommendations to management based on an examination of a knowledge base. Furthermore, it contains specialised problem-solving skills that are strongly connected to data mining, which is the process of filtering through enormous volumes of data to find relevant correlations.

d) Document Driven

By integrating a range of storage and processing technologies to enable complete document retrieval and analysis, these systems assist managers in retrieving and managing unstructured documents and web pages. It also has access to documents like business rules and procedures, product specifications, catalogues, corporate history papers, minutes of meetings, critical communications, corporate records, and so on, which are generally accessed through a task-specific search engine.

e) Communication Driven

This type of DSS is commonly referred to as group decision support systems (GDSS). They are a form of hybrid DSS that stresses the use of communications and decision models to help decision makers solve problems as a group. GDSS uses technologies including two-way interactive video, bulletin boards, and e-mail to enable electronic communication, scheduling, document sharing, and other group productivity and decision-making tasks.

- **Management Requirements of the Decision Support Systems**

At each management level, the purposes and requirements for information are different. Higher levels of management place a higher emphasis on external environmental data and, to some extent, less emphasis on internally created data. Top management frequently uses information to make long-term planning decisions, and hence analyses long-term trend data to do so (Gore et al, 1984). Coupling analysis is also important for top-level decision-making, since managers want more knowledge regarding risk and uncertainty, alternative outcomes (and their likelihood/probability, etc.), and various actions to take. Internally produced information is prioritised at lower management levels, and short-term goals are prioritised.

Lower management reports and analyses, on the other hand, are frequently utilised in top management decision-making, therefore it's critical that the DSS supports lower management just as much as it does top management. In summary, information requirements for different levels of management are focused on supervisory responsibilities for lower management, tactical decision making for middle management, and strategic decision making for top management.

Furthermore, because the DSS is used to strengthen management control, it should focus on the key management control duties. These are the following:

- Assigning resources to certain tasks.
- Creating budgets that account for both resource expenditures and projected achievement goals and objectives.
- Examining the outcomes obtained in exchange for the resources expended.
- Assessment of the outcomes.
- Adjustment of activities and resource allocation in response to outcomes evaluation
- Evaluation of an appropriate Decision Support Systems.

Every area of the business process that will be affected within the organisation is involved in the DSS design and development process. Business leaders and managers should get familiar with DSS so that they can more properly explain their needs and requirements, making the formulation of deliverables, overview of capabilities, results, and demands, as well as what decisions the proposed DSS should support, simpler.

The following steps can be used as a methodology to evaluate, design and implement a DSS:

1. Firm's Strategic Focus

First and foremost, a company that wishes to implement a DSS must clearly define its strategic focus within the industry, with a particular emphasis on future ambitions and the firm's path. This will serve as the foundation for the DSS's information needs and scalability, as well as a framework for the project team responsible for developing and executing the system.

2. Creating an Evaluation Team

A project team should be formed based on backgrounds and experience using a multi-disciplinary team approach with experts from database, networking, communications, accounting, and other fields. This group will do an initial analysis that will serve as the foundation for the remainder of the project. Team formation (forming, norming, storming, and performing) and group dynamics are important factors in ensuring the team's overall performance. As a result, the following concerns should be taken into account during this critical stage of the project:

- Specify roles and delegate responsibilities
- Anticipate and manage time commitment
- Choose appropriate leadership
- Create a charge/direction for the team with a technique that is well-defined.

3. Defining Requirements

The more the team understands DSS categories, the more they'll be able to fine-tune the system's specs and choose the right type of Decision Support System. A multidisciplinary team approach is required, including input from all key stakeholders involved in the DSS implementation. Team members should be held accountable for communicating with other members of the company who will be impacted by the chosen DSS, and they should make every effort to articulate all relevant needs from all stakeholders. Because non-compliance by certain parties may dilute the system's

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power, it's critical to have "buy-in" from all persons who will use or be affected by Decision Support Systems.

- Issues to consider:
 - What functions will managers perform, now and in the future, with the DSS?
 - What controls and security are needed?
 - What are the operational performance requirements?
 - What is the design concept
 - How does it fit into the current architecture and network layout

4. System Design and Evaluating Products/Alternatives

- Clearly defines deliverables.
- Include a preliminary assessment of both products and vendors available.
- "off-the-shelf" DSS are hard to come by, and most businesses must adopt a specific DSS to meet their unique business demands.
- Functional screening and thorough review based on predefined criteria and specifications:
 - Issues of cost, design, support, and installation should all be addressed independently and collectively.
 - Is the "package" consistent with the company's requirements?
 - Requirements for hardware and software.
- Includes performance evaluation:
 - Can the system handle heavier loads?
 - What is the scalability (horizontal and vertical)?
 - What are the direct and indirect costs?
 - Stability, reliability and consistency of system?
 - Quality of information provided: timely, accurate, reliable, unbiased?
 - Restrictions of systems
 - Capabilities of system
 - After implementation support? Costs involved?

5. Negotiating with Vendors

Review vendors according to product offerings, reputation in industry, financial stability, experience in similar projects/systems. Also clarify issues such as

- Service contract
- Price of package
- Time frame of deliverables
- Review vendors offerings and compare with other vendors

- o Vendor's involvement with implementation/deployment of system
- o Training of users and after implementation support

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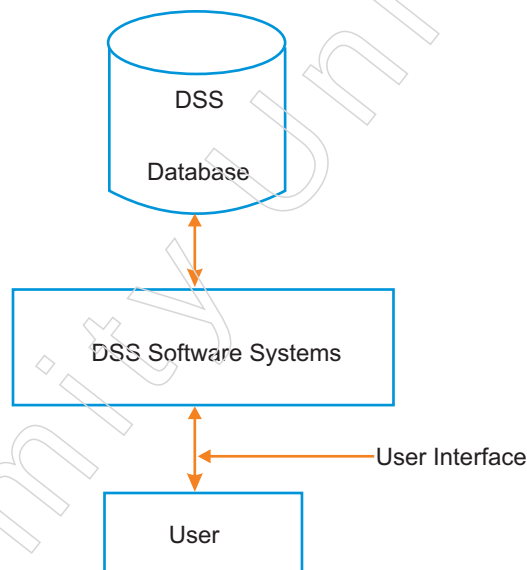
2.3.1 Analytical Tools for DSS for Operations Management

A decision support system consists of three main components, namely database, software system and user interface.

1. **DSS Database:** This database holds data from a variety of sources, including internal data from the company, data created by various applications, and external data mined from the Internet, among others. The database for decision support systems might be a small database, a standalone system, or a large data warehouse that provides support to an organization's information needs. The DSS database generally contains a duplicate of the production database to prevent the decision support system from interfering with the functioning of operational systems.
2. **DSS Software System:** It is made up of a good number of mathematical and analytical models that are used to evaluate complicated data and provide the desired results. A model predicts the result based on a variety of inputs or conditions, or determines the conditions and inputs that are necessary to create the intended output.

A decision support system may have many models, each of which performs a specialised purpose. The models that must be included in a decision support system family are determined by the needs of the users and the goals of the DSS.

It's worth noting that the DSS programme has predefined models (or routines) that may be used to create new models to support certain types of choices.



Some of the commonly used mathematical and statistical models are as follows: -

- **Statistical Models:** They include a variety of statistical functions, such as mean, median, mode, and deviations, among others. These models are used to create connections between event occurrences and other factors associated with that event. It can, for example, link product sales to variances in location, income, season, or other variables. They include software that can evaluate a set of data and project future results, in addition to statistical operations.

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- **Sensitivity Analysis Models:** These are used to offer solutions to commonly recurring what-if scenarios in an organisation. Several times during the study, the value of one variable is modified, and the effects on other variables are noted. Product sales, for example, are influenced by a variety of factors such as pricing, advertising costs, sales staffing, and manufacturing. The price of a product can be adjusted (raised or lowered) frequently using a sensitivity model to determine the sensitivity of different parameters and their influence on sales volume. For this type of study, Excel spreadsheets and Lotus 1-2-3 are frequently utilised.
 - **Optimization Analysis Models:** They're used to figure out what the best value for a target variable is in a given situation. They're commonly utilised to make judgments about how to make the most use of a company's resources. During optimization analysis, the values for one or more variables are modified frequently while keeping the constraints in mind until the optimal values for the target variable are discovered. They can, for example, calculate the greatest level of output that can be reached by altering worker task assignments, keeping in mind that certain workers are competent and cannot have their job assignments modified. For this type of study, linear programming techniques like the Solver function in Microsoft Excel are commonly employed.
 - **Forecasting Models:** They employ a variety of forecasting tools and techniques, including as regression models, time series analysis, and market research methodologies, among others, to make predictions about the future. They give data that aids in the analysis of company circumstances and the formulation of future strategies. For sales forecasting, several methods are commonly employed.
 - **Backward Analysis Sensitivity Models:** The approach used in these models, also known as goal seeking analysis, is the polar opposite of the technique used in sensitivity analysis models. Instead of constantly altering the value of a variable to observe how it impacts other variables, goal seeking analysis establishes a target value for a variable and then adjusts other variables until the target value is met. For example, to increase production by 40% using backward sensitivity analysis, the target value for the production level can be set first, and then the required changes in other factors, such as the amount of raw material, machinery and tools, number of production staff, and so on, to achieve the target production level.
3. **DSS User Interface:** This is a graphical platform that allows the DSS and its users to interact more easily. It presents the analysis' results (output) in a variety of formats, including text, tables, charts, and graphics. The user can choose the suitable choice for seeing the result based on his needs.

A manager, for example, might prefer to see comparable sales data in a tabular style, but an architect working on a design plan would prefer to see the analytical results in a graphical format. The modern decision support system, which is developed utilising a Web-based interface, offers its customers several unique features, such as improved interaction, facility for customization and personalization, and more ease of use.

• Classification of DSS Systems

Different authors suggest different classifications. In the paper will be applied classification of DSS to

- Classification of DSS systems according to user level,
- Classification of DSS systems according to conceptual level,
- Classification of DSS systems according to system level,
- Classification of DSS systems according to according to way of realization
- Classification of DSS systems according to according to concrete realization and etc.

Classification of DSS Systems According to Way of Realization

DSS systems, according to way of realization split to:

- EDSS (Expert Decision Support System),
- KB-DDS (Knowledge-Based DSS) or IDSS (Intelligent DSS),
- GDSS (Group Decision Support System),
- MADSS (Multi-Attribute Decision Support System),
- MCDSS (Multi-Criteria Decision Support System),
- MDSS (Multi-Participant Decision Support System),
- NSS (Negotiation Support System),
- ODSS (Organizational Decision Support System),
- PDSS (Planning Decision Support System),
- TDSS (Team Decision Support System),
- WB-DDS (Web-Based Decision Support System).

Classification of DSS Systems According to Concrete Realization.

Concrete realized DSS systems, according to sort of given services, for wider geographical area are:

- CDSS (Consumer Decision Support System),
- EDSS (Environmental Decision Support System),
- GSDSS (Group Spatial Decision Support System),
- IDSS (Intelligent Environmental Decision Support System),
- LADSS (Land Allocation Decision Support System),
- MC-SDSS (Multi-Criteria S Decision Support System),
- SDDS (Spatial Decision Support System),
- TDSS (Tactical Decision Support System),
- WebSDDS (Web-Based Spatial Decision Support System) and etc.

Software for Decision Support (DSS)

Most known software DSS tools and package are:

- Software for strategic decision,
- Software for financial simulation and modelling (FINANSEER, Budget Expres, Micro SIMPLAN etc),
- Statistical and econometric software (SAS, Forecast Master Plus, ESP etc.),

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- Software for building matrices and tree of decision (Expert Choice, Decision Aide, Decision Pad etc.)
- Software for special applications (PROMCALC, GAIA, TACDSS, TAPS) etc.
- TAPS

TAPS (The Assessment Profiling System) is a multi-level self-reporting visualisation that may be used to assist any environmental management programme. TAPS is a decision-making tool that provides real-time analysis, as well as status and situational awareness.

TAPS is a command-level visualisation that uses a multiple attribute logic engine to get started. The TAPS system combines powerful graphical representation with the area status of a logical engine to create a Web-based informative service for real-time evaluation and decision-making. The TAPS system's goal is to enable decision-making in centralised situational awareness (Situational Awareness - SA) of the primary with pertinent data, which are needed for effective processing of complex situations.

TAPS access for high-level organisation mission evaluation is an operational strategy and method of decision-making analysis for expressing efficiency measurements in terms of quantifiable measures of characteristics and other key factors. The TAPS Web-based graphical service translates the status of the data model into a dynamic display that may be updated frequently. Figure below depicts a graphical representation of the TAPS system's menu.

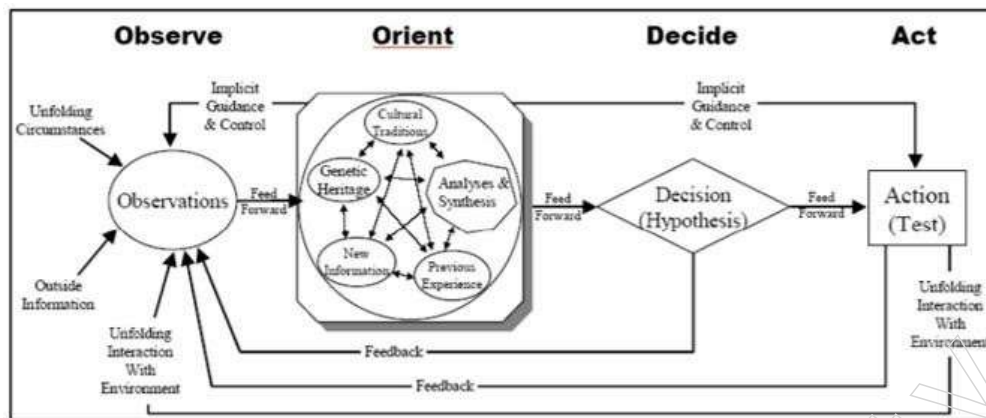
Benefits of TAPS system is seen in:

- User defined data model;
- Goals;
- Difficulty of attributes;
- Information sources;
- Real-time results;
- Reporting format that is permanent;
- Web-based data processing and reporting (TAPS Web service provides rapid data collecting, processing, and distribution from many sources);
- Flexible modes of data collection and,
- Adjusted frequency of assessment.

The objective of the TAPS method is to assist managers in achieving desired goals for any complicated operation or project in a timely manner. The steps for developing a functioning software for managing a decision system are outlined.

- Development and addition of specific programme management information;
- Creation and processing of a state-of-the-area data model for a given project;
- Loading and executing programme management checks;
- During the start-up phase of implementation, administration and management of TAPS Web-based programme service.
- Providing full support in terms of training conditions, paperwork, and technical assistance.

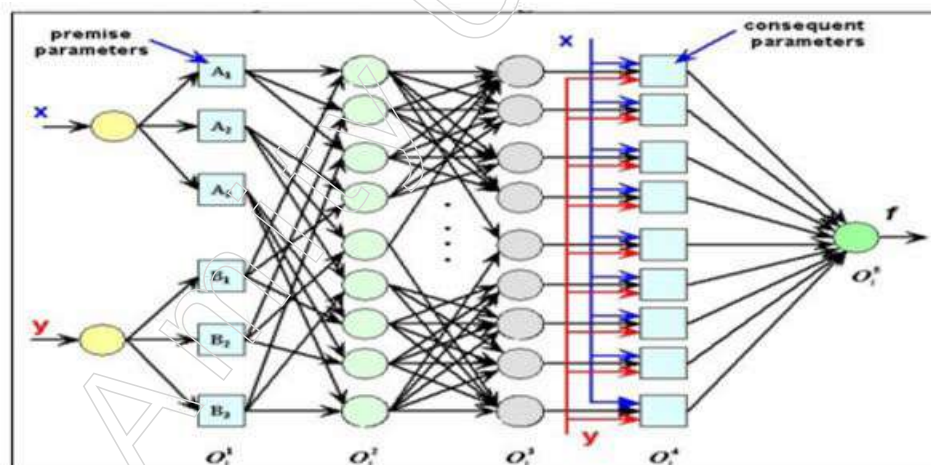
The OODA (Observe, Orient, Decide, and Act) flowchart of decision making in the TAPS system is illustrated in figure and is based on the information control issue (Col John R. Boyd's OODA Loop, 1995).



TACDSS

TACDSS (Tactical Air Combat Decision Support System) refers to the construction of a reliable decision-making system that comprises two key elements: the collecting and analysis of prior data and the evaluation of the solution. Data can be in the form of an image, a real number, a binary code, or natural language textual data, depending on the object area. The factor of choice is also known as the object of a decision issue.

This item may be stated mathematically as a universal set within a decision problem. A decision factor is a subset of a problem decision. To make the best decisions, the system should be able to adapt to information provided by foreign words or descriptions of problem areas in any natural language. The ANFIS (Adaptive Network Based Fuzzy Inference System – adaptive network-based phase inherent technology) system was utilised to build the TACDSS system based on the Takagi-Sugeno fuzzy interface system. Figure depicts the ANFIS system's six-layered design.



Check Your Understanding

1) _____ have progressed from simple model-oriented systems to complex multi-function entities.

a) Decision support systems

Notes

- b) Data Warehousing
 - c) Management Information System
 - d) Operations Management
 - e) None of the above
- 2) _____ holds data from a variety of sources, including internal data from the company, data created by various applications, and external data mined from the Internet, among others.
- a) Accounting Models
 - b) DSS Database
 - c) Financial Models
 - d) Document driven Models
 - e) None of the above
- 3) _____ include a variety of statistical functions, such as mean, median, mode, and deviations, among others.
- a) Knowledge driven Models
 - b) Communication Driven Models
 - c) Evaluation Team
 - d) Statistical Models
 - e) None of the above
- 4) _____ are used to offer solutions to commonly recurring what-if scenarios in an organisation.
- a) Sensitivity Analysis Models
 - b) Performance Evaluation
 - c) Statistical Models
 - d) Forecasting Models
 - e) None of the above
- 5) _____ are commonly utilised to make judgments about how to make the most use of a company's resources.
- a) Backward Analysis Sensitivity Models
 - b) Forecasting Models
 - c) Optimization Analysis Models
 - d) DSS User Interface
 - e) None of the above
- 6) What is the full form KB-DDS?
- a) Knowledge-Based Decision Support System
 - b) Knowledge-Bits Decision Support System

- c) Kilo-Bytes Decision Support System
d) Know-How Based Decision Support System
e) None of the above
- 7) What is the full form of GDSS?
a) General Decision Support System
b) Group Decision Support System
c) Group Division Support System
d) Group Development Support System
e) None of the above
- 8) Full form of LADSS:
a) Learn Allocation Decision Support System
b) Land Allotment Decision Support System
c) Land Allotment Division Support system
d) Language Analysis Decision Support System
e) Land Allocation Decision Support System
f) None of the above
- 9) Triple Exponential Smoothing is also known as _____.
a) Holt-Winters Exponential Smoothing
b) Swift Polling
c) Mean Absolute Deviation
d) Absolute Deviation
e) None of the above
- 10) _____ is a decision-making tool that provides real-time analysis, as well as status and situational awareness.
a) Adaptive Network Based Fuzzy Inference System
b) Tactical Air Combat Decision Support System
c) The Assessment Profiling System
d) Forecast Master Plan
e) None of the above

Summary

- Over the last three decades, decision support systems have progressed from simple model-oriented systems to complex multi-function entities.
- Decision Support Systems evolved into increasingly complex computer-based systems that supported production, promotion, pricing, marketing, and some logistical activities as MIS theory evolved during the 1970s.

Notes

- Management expert systems or intelligent decision support systems are terms used to describe knowledge-driven DSS.
- The DSS database generally contains a duplicate of the production database to prevent the decision support system from interfering with the functioning of operational systems.
- The objective of the TAPS method is to assist managers in achieving desired goals for any complicated operation or project in a timely manner.

Activity

1. Understand the Decision support system of any organization and what software they use to support the same. Also, how DSS helps the managers of the organization to take decisions effectively and efficiently.

Questions & Exercises

1. What is Decision Support System? Describe its components and how to implement a Decision Support System?

Glossary

- Sensitivity Analysis Models are used to offer solutions to commonly recurring what-if scenarios in an organisation.
- Forecasting Models: They employ a variety of forecasting tools and techniques, including as regression models, time series analysis, and market research methodologies, among others, to make predictions about the future.
- Backward Analysis Sensitivity Model, also known as goal seeking analysis, is the polar opposite of the technique used in sensitivity analysis models.
- DSS User Interface is a graphical platform that allows the DSS and its users to interact more easily. It presents the analysis' results (output) in a variety of formats, including text, tables, charts, and graphics.
- The Assessment Profiling System is a multi-level self-reporting visualisation that may be used to assist any environmental management programme.
- TACDSS (Tactical Air Combat Decision Support System) refers to the construction of a reliable decision-making system that comprises two key elements: the collecting and analysis of prior data and the evaluation of the solution.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor

Check your understanding – Answers

- | | | | |
|-------|--------|-------|-------|
| 1) a) | 2) b) | 3) d) | 4) a) |
| 5) c) | 6) a) | 7) b) | 8) d) |
| 9) a) | 10) c) | | |

Module - 3: Facility Location and Facility Layout

Notes

Structure:

- 3.1 Facility Location
 - 3.1.1 Factors Affecting Location Decisions
 - 3.1.2 Factor Rating Method
 - 3.1.3 Centre Of Gravity Method
 - 3.1.4 Load Distance Method
 - 3.1.5 Break Even Analysis Method
 - 3.1.6 Utilization Of GIS In Plant Location
- 3.2 Facility Layout
 - 3.2.1 Principles Of Facility Layout
 - 3.2.2 Process Layout
 - 3.2.3 Product Or Line Layout
 - 3.2.4 Fixed Position Layout
 - 3.2.5 Concepts Of Line Balancing
 - 3.2.6 Merits And Demerits Of Product Layout
 - 3.2.7 Merits And Demerits Of Process Layout

Notes

Unit - 3.1: Facility Location

Objectives:

At the end of this unit, you will be able to:

- Learn about facility location
- Understand factors affecting location decision
- Learn about factor rating method
- Explain centre of gravity method
- Brief long-distance method
- Understand break even analysis
- Explain utilization of GIS in plant location

Introduction

Facility



A facility is an installation, contrivance, or other item that enables anything; a location where something may be done. A hotel, resort, school, office complex, sports arena, or conference centre, as well as a medical facility, are examples of commercial or institutional buildings.

A facility is a location where certain activities take place. For example, sports and military activities take place at sports and military facilities. In a nuclear research centre, experts work on nuclear research. We also use the phrase to refer to structures, equipment, and services that individuals give for a certain purpose. For Example: Shopping facilities, are places where we can buy things. Medical facilities may refer to medical equipment or a building.

- Plant Location

The choice of area and the selection of a specific site for establishing a business or factory is referred to as plant placement. However, the decision is taken only after weighing the costs and advantages of several other places. It's a strategic choice

that can't be reversed once made. The site should be chosen based on its own requirements and conditions if it can be altered at all at a significant cost. Each plant is unique in its own way. A businessperson should strive for the best or ideal location.

- **Ideal Location**

An ideal site is one where the product's cost is maintained to a minimum, the market share is big, the risk is low, and the social benefit is high. It is the location with the greatest net benefit or the lowest unit cost of production and delivery. Small-scale entrepreneurs can utilise locational analysis to accomplish this goal.

- **Locational Analysis**

Locational analysis is a dynamic process in which an entrepreneur evaluates and analyses the suitability of several locations with the goal of selecting the optimal location for a specific business. It consists of the following components:

- Demographic Analysis:** This entails examining the area's demographic in terms of total population (in number), age composition, per capita income, educational level, occupational structure, and so on.
- Trade Region Analysis:** This is a study of the geographic area that provides the firm with repeat customers. He'd also look into the possibility of reaching the trading area from other locations.
- Competitive Analysis:** This method aids in determining the type, location, scale, and quality of competition in a particular industry.
- Traffic Study:** To provide an approximate estimate of the number of prospective consumers passing by the proposed location during business hours, the traffic analysis evaluates alternative sites in terms of pedestrian and vehicle traffic.
- Site Economics:** Under this, alternative locations are assessed in terms of establishment costs and operational costs. Establishment costs are incurred for permanent physical structures, whereas operational costs are incurred for day-to-day company operations and are also known as operating costs.

Two sites A and B are evaluated in terms of above mentioned two costs as follows:

Cost	Site A (Rs)	Site B (Rs)
Cost of establishments:		
Land and Buildings	350000	230000
Equipment	60000	60000
Transport facilities	20000	30000
Cost of operations:		
Materials, freight and carriage	34000	24000
Taxes and insurance	10000	7500
Labour	100000	70000
Water, power and fuel	10000	8000
Total	584000	429500

Notes

Although site A has lower expenses in some ways, the following cost statement suggests that site B is superior than site A based only on economic factors. Site B would be preferable if the criterion of optimal location is a place with a maximum net benefit or the lowest unit cost of production and distribution.

The process of determining a geographic location for a company's activities is known as facility location. When evaluating the appropriateness of a given location, managers of both service and manufacturing firms must consider a variety of criteria, including accessibility to customers and suppliers, labour expenses, and transportation costs.

Location conditions are complicated, and each one has its own set of concrete (freight rates, production expenses) and intangible (reliability, frequency security, quality) characteristics

It's difficult to assess the conditions in a given location. Tangible costs such as, wages and product expenses, may be measured exactly into what makes places better to compare. Non-tangible properties, such as reliability, availability, and security, may only be quantified on an ordinal or even nominal scale. Other intangible characteristics, such as the percentage of unionised staff, can also be measured. To summarise, non-tangible elements have a significant role in business location decisions.

On the basis of the type of the organisation, it is reasonable to separate the variables that impact the plant or facility location as follows:

- General locational factors, for all type of organizations which include controllable and uncontrollable factors.
- Specific locational factors in particular required for manufacturing and service organizations.

The following are the two types of location factors: Dominant variables are those that have a particularly significant influence on sales or expenses and are derived from competitive priorities (cost, quality, time, and flexibility). Secondary variables are also essential, but if other factors are more important, management may minimize or even disregard some of them.

The Way in which Facility Works

A facility is an arrangement between a corporation and a public or private lender that permits the company to borrow a specific amount of money for a limited period of time for various reasons. The loan is for a certain sum and does not need any kind of security. The borrower makes interest-bearing monthly or quarterly instalments until the loan is completely paid off.

A facility is especially crucial for businesses that wish to prevent problems like layoffs, halting growth, or closing shop during quiet seasons when sales are low.

For example, if a jewellery store runs out of cash in December when sales are down, the owner can ask a bank for a \$2 million loan, which would be repaid in full by July when business resumes up. The funds are used to keep the jeweller in business, and the loan is repaid in monthly instalments by the agreed-upon date.

3.1.1 Factors Affecting the Location Decisions

General Locational Factors

Following are the general factors required for location of plant in case of all types of organisations.

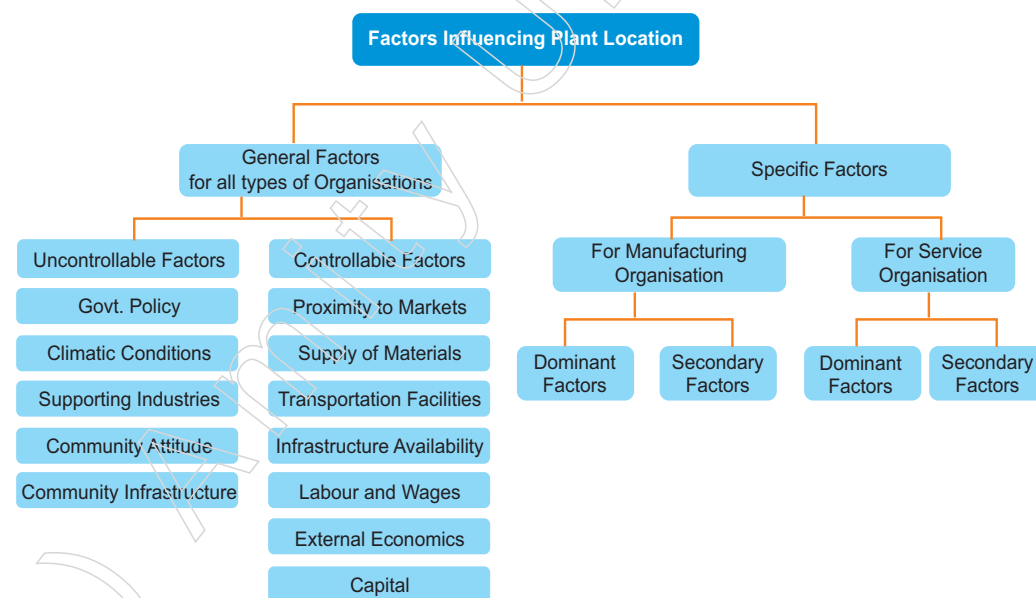
Controllable Factors

1. Proximity to markets.
2. Supply of materials
3. Transportation facilities
4. Infrastructure availability
5. Labour and wages
6. External economies
7. Capital

Uncontrollable Factors

1. Government policy
2. Climate conditions
3. Supporting industries and services
4. Community and labour attitudes
5. Community Infrastructure

Factors Influencing Plant Location



a) Controllable Factors

1) Proximity to Markets:

Notes

Notes

Every firm is required to serve its consumers by delivering goods and services when and where they are needed at a reasonable price. Depending on the product, companies may opt to place facilities near or far from the market. When the product's customers are concentrated, it's best to place the facilities near the market. It is preferable to be closer to the market if:

- The items are fragile and prone to spoiling.
- After-sales services are frequently required.
- Transportation costs are expensive and considerably raise the cost.
- The product has a short shelf life.

The closeness to the market ensures that clients have a constant supply of goods while also decreasing transportation expenses.

2) Supply of Raw Material:

In order to maintain continuous manufacturing, the organisation must get raw materials in the appropriate quality and at the appropriate time. If the supplies are perishable and shipping costs are considerable, this issue becomes critical.

The following are some general principles for the influence of raw materials on plant location:

- Locate the factory at the raw material supply, the market, or wherever in between when a single raw material may be used without losing weight.
- When a weight-losing raw material is required, situate the factory near the source of the raw material.
- Choose a location near a market with plenty of raw supplies.
- If the raw materials are processed from a number of different places, the facility may be placed to reduce total transportation costs.

Proximity to raw resources is essential in industries such as sugar, cement, jute, and cotton textiles.

3) Transportation Facilities:

Rapid transportation facilities guarantee that raw materials and completed items are delivered on time to the firm and to the consumers. The transportation facility is required for the plant's location. Air, road, train, water, and pipeline are the five fundamental means of physical transportation. Goods destined primarily for export need a location near a port or a big airport. The transportation mode used, and therefore the site, will be determined by relative costs, convenience, and appropriateness. As a result, one of the factors for plant site is transportation cost to value added.

4) Infrastructure Availability:

Basic infrastructural amenities such as power, water, and garbage disposal, among others, become important considerations when choosing on a location. Certain sectors, such as aluminium and steel, require a lot of electricity and should be placed near a power plant or somewhere where there is a reliable power supply all year. Such enterprises may face a survival dilemma if power isn't available. Continuous operation

is required in process industries such as paper, chemicals, and cement. Water supply in vast quantities and of excellent quality, as well as mineral content of water, become essential factors. The placement of the plant is influenced by a waste disposal facility for process industries.

5) Labour and Wages:

During plant location, the challenge of finding an appropriate number of labourers with certain capabilities is a concern that should be considered both at the territorial and community levels. Importing labour is generally expensive and complicated administratively. A potential community's labour relations history will be investigated. A prospective community will be investigated. Labor productivity is another essential element to consider. Prevailing salary patterns, cost of living, industrial relations, and union negotiating power are all essential factors to consider.

6) External Economies of Scale:

Urbanization and locational economies of scale are examples of external economies of scale. The first relates to a company's benefits from establishing operations in a big city, while the second alludes to "settling down" among other firms in comparable industries. In the case of urbanisation economies, companies benefit from locating in larger cities rather than smaller ones in order to access a larger pool of labour, transportation infrastructure, expand their markets for selling their products, and get access to a much broader choice of business services.

In the industrial industry, location economies of scale have grown over time, resulting in higher competition owing to production facilities and reduced production costs due to lower transportation and logistics expenses. This resulted in manufacturing districts, where several firms from related industries are clustered together. Large businesses have sought to reduce inventory costs by establishing "Just in Time" manufacturing systems after realising that stocks and warehouses have become a key cost issue (the so-called Kanban System). One of the major reasons for the Japanese automobile industry's success was its very efficient production method.

Just in time delivery guarantees that spare parts are delivered from suppliers within a few hours after placing an order. To meet these requirements, businesses must be located in the same region, therefore expanding their market and providing better service to major businesses.

7) Capital:

It's crucial to separate the physiology of fixed capital in buildings and equipment from financial capital by looking at capital as a location condition. Building and construction expenses vary by area, thus fixed capital costs are necessary. Buildings can be hired, and existing factories can be enlarged, on the other hand. Financial capital is extremely mobile and has little impact on decision-making. Large multinational corporations like Coca-Cola, for example, operate in a variety of nations and can obtain money where interest rates are lowest and circumstances are best.

When it comes to venture capital, money is a major consideration. In this situation, we're talking about new, fast-growing (or not) high-tech companies with few fixed

Notes

assets. These businesses require a large number of financial resources as well as highly educated staff.

b) Uncontrollable Factors

1) Government Policy:

The policies of state governments and municipal governments concerning labour laws, construction standards, and safety, among other things, are considerations that must be considered. Both the federal and state governments in our nation provide a package of incentives to entrepreneurs in certain regions in order to achieve a balanced regional growth of industries. The incentive package might include a sales tax and excise duty exemption for a certain length of time, a soft loan from a financial institution, a discount on power prices, and an investment subsidy. Some of these incentives may entice people to locate the facility in order to take use of the services provided.

2) Climatic Conditions:

The geology of the location, as well as the climatic circumstances, must be examined (humidity, temperature). Climate has a significant impact on human productivity and behaviour. Some businesses, such as textile mills, require special climatic conditions, such as humidity.

3) Supporting Industries and Services:

Nowadays, the manufacturing company does not produce all of the components and pieces in-house and instead outsources the job to suppliers. As a result, one of the variables influencing the location will be the supplier of component parts.

Various services such as communications, banking, professional consulting, and other civil amenity services will all play a role in the location selection process.

4) Community and Labour Attitudes:

The attitude of the community about their job and the potential industries may make or break the industry. Support for trade union activity in the community is an essential criterion. Even when all other criteria are favourable, the location of a facility in a given area is undesirable due to labour attitudes against management, which frequently result in strikes and lockouts.

5) Community Infrastructure and Amenity:

All industrial operations require community infrastructure, including economic overhead capital such as roads, railroads, port facilities, power lines, and service facilities, as well as social overhead capital such as schools, universities, and hospitals.

These issues must also be taken into account when making site selections, as infrastructure is extremely expensive to construct, and for most manufacturing operations, the existing infrastructure places physical constraints on location options.

c) Specific Locational Factors for Manufacturing Organization

- Dominant Factors

Factors dominating location decisions for new manufacturing plants can be broadly classified in six groups. They are listed in the order of their importance as follows:

1. Favourable labour climate
2. Proximity to markets
3. Quality of life
4. Proximity to suppliers and resources
5. Utilities, taxes, and real estate costs

1) Favourable Labour Climate:

For labour-intensive businesses like textiles, furniture, and consumer electronics, a favourable labour climate may be the most significant consideration in location selection. Wage rates, training requirements, attitudes toward work, worker productivity, and union power are all factors in the labour climate. Many executives see weak unions or a low likelihood of successful union organising activities as a unique benefit.

2) Proximity to Markets:

Management must choose a site for the facility that will fulfil that need after evaluating where the highest demand for goods and services exists. When the finished items are large or heavy, and outgoing transportation charges are high, being close to markets is very crucial.

3) Quality of Life:

Quality of life is enhanced by good schools, recreational facilities, cultural events, and an appealing lifestyle. This is a little issue in and of itself, but it may make a big impact in site selections.

4) Proximity to Suppliers and Resources:

Many firms rely on other facilities for management and personnel assistance, or plants supply parts to other facilities. These activities need regular coordination and communication, which becomes more difficult as the distance between them grows.

5) Utilities, Taxes and Real Estate Costs:

Utility expenses (telephone, energy, and water), local and state taxes, financial incentives given by municipal or state governments, relocation costs, and land prices are all essential considerations.

- Secondary Factors

Other factors to consider include expansion space, construction costs, accessibility to multiple modes of transportation, the cost of moving people and materials between plants, workforce competition from other companies, community attitudes, and many others. Firms are prioritising local personnel skills and education, as well as local infrastructure, for global operations.

Notes

d) Specific Locational Factors for Service Organization

- Dominant Factors

The same considerations that apply to manufacturers apply to service providers, with one notable exception: the influence of location on sales and customer satisfaction. Customers generally assess how near a service location is, especially if the procedure necessitates a lot of client interaction.

1) Proximity to Customers

The convenience with which consumers may conduct business with a company is determined by its location. Few individuals, for example, would want to go to a faraway dry cleaner or supermarket if another option is more convenient. As a result, the impact of location on income is often the most important aspect.

2) Transportation Costs and Proximity to Markets

Transportation costs and accessibility to markets are critical for warehousing and distribution businesses. Many businesses may keep goods closer to the consumer by having a warehouse nearby, decreasing delivery times and increasing sales.

3) Location of Competitors

The impact of rivals is one problem in evaluating the sales potential at various locations. Management must not just evaluate rivals' existing locations, but also try to predict how they will respond to the firm's new location. It pays to stay away from regions where rivals are already well-established. However, in other businesses, such as new-car dealerships and fast-food restaurants, being close to competition might be helpful. The goal is to achieve critical mass, when a cluster of competing businesses in one area attracts more consumers than the total number of people who would shop at the same establishments in other locations. Recognizing this impact, several companies employ a "follow the leader" approach while looking for new locations.

Secondary Factors

Retailers must also evaluate the volume of retail traffic, residential density, traffic flow, and visibility of the location. The importance of retail activity in the region is significant, since customers frequently decide to go shopping or dine at a restaurant on the spur of the moment. Because most consumers arrive in automobiles, traffic flow and visibility are critical. Distance from the roadway and the size of surrounding buildings and signage are factors in visibility. When the population in the region matches the firm's competitive priorities and target market group, high residential density assures night time and weekend business.

- Criteria of a Good Layout

The layout of a plant or facility cares with the physical placement of resources like equipment and storage facilities, which should be designed to facilitate the efficient flow of consumers or materials through the manufacturing or service system. The layout design is extremely important and will be taken very seriously because it can have a big impact on the value and efficiency of an operation and may involve substantial investment in time and money. the choices crazy regards to the power layout will have

an immediate influence on how efficiently workers are going to be ready to perform their jobs, what proportion and the way fast goods are often produced, how difficult it's to automate a system, and the way the system in place would be ready to answer any changes with regards to product or service design, product mix, and volume of the demand.

In many operations the installation of a replacement layout, or redesign of an existing layout, are often difficult to vary once they're implemented thanks to the many investments required on items like equipment. Therefore, it's imperative to form sure that the policy decisions concerning the organisation, method and work flow are made before the facilities are laid out instead of trying to suit these three into the layout. This is often a crucial area of production and operations management since it's handling the capital equipment of the organisation which, generally, is difficult to relocate once it's been put into position.

1. **Maximum Flexibility:** A good layout should be planned in such a manner that changes can be made quickly to meet changing conditions, and it should be created with the operation's future needs in mind.
2. **Maximum Co-ordination:** The layout should be built in such a way that entering and exiting any department or functional area is as simple as possible for the issuing and receiving departments.
3. **Maximum Use of Volume:** The facilities should be seen as cubic devices, with the volume provided being utilised to the fullest extent possible. This idea is beneficial in shops, where products may be kept to great heights without generating problems.
4. **Maximum Visibility:** The writers also argue that all employees and materials should be visible at all times, and that there should be no concealed areas where items or information may be left or forgotten. When using partitioning or screening, organisations should be cautious since they may generate unwanted segregation, reducing the efficient use of floor space.
5. **Maximum Accessibility:** Service and maintenance areas should always be accessible, regardless of the presence of machinery, equipment, or other installations. Certain service points, such as electricity and water mains, may become blocked, slowing down the manufacturing process.
6. **Minimum Distance and Material Handling:** All movements within the plant should be both essential and direct. Because handling labour increases the cost but not the value of the product, any superfluous movement should be minimised and, if necessary, removed. It is preferable to avoid handling the material and information, but if this is unavoidable, it should be kept to a minimum by employing appropriate equipment.
7. **Inherent Safety:** All procedures that might endanger employees or consumers should be made inaccessible to unauthorised personnel. Pathways should be properly defined and uncluttered, and fires should be clearly marked with unrestricted access.
8. **Unidirectional Flow:** All materials employed in the manufacturing process should flow in one direction, beginning in storage and continuing through all processes and facilities to the finished product, which is then shipped for storage or sold straight to the client.

Notes

9. Management Co-ordination: The placement of employees and communication equipment inside the specified plan should aid supervision and communication.

- Key Factors in Facility Location Decision-making

There are several elements that might influence where an organization's facilities are located. Some aspects become more significant than others in how facility location impacts an organization's success in every given circumstance. When a firm has to build a new production facility, for example, numerous criteria must be considered to decide which site would lower operating costs while delivering a high level of market responsiveness.

- Proximity to Sources of Supply:

To save money on transportation, companies that handle bulk raw materials generally locate near the source of supply. Paper mills are built near woods, canneries are built in farming regions, and fish processing factories are built near harbours where fishing boats dock.

- Proximity to Customers:

There are a variety of reasons why a company may want to be near to its consumers. Grocery stores, petrol stations, fast food restaurants, and hospitals, for example, must be close to their consumers in order to be handy. In other cases, such as concrete manufacture, transportation costs may necessitate proximity to clients. Bakeries and fresh flowers, for example, require perishable items to be produced near to the ultimate market.

- Community Factors:

To encourage businesses, communities may give a variety of incentives, such as tax exemptions or reductions, as well as access roads, water and sewage connections, and utilities. Attitudes in the community might also influence a company's decision on where to locate. Some communities may deliberately reject businesses that might add to the area's pollution, noise, or traffic. Some communities may object to a jail being built on their land. Other towns may embrace such businesses because they promise employment, tax money, and economic variety.

- Labour Factors:

Because labour is a crucial element for many businesses, research reveals that the majority of site selections are primarily based on labour variables. The prevailing salary rate in a town for similar jobs, the quantity of competent employees, and the average education level of the local population are all labour considerations to consider (percentage of high school graduates, etc.). Various labour variables might include the degree of unionisation and a community's overall work ethic, as well as other indicators of absenteeism, and worker duration in a job can play a significant influence when a company decides where to locate.

- Other Factors:

Many additional variables, such as quality of life (crime rates, good schools, climate, and recreation choices), availability to key transportation arteries, building costs, competitive nearby, and future growth prospects, can all influence the location selection. Hence, an important consideration for selecting a suitable location must be given to factors given below:

Notes

- a) Environmental or climatic factors.
- b) Availability and proximity to raw material suppliers.
- c) Transport costs - both in terms of procuring raw materials and in terms of distributing or marketing completed products to end consumers.
- d) Access to market: small retail, wholesale, and service firms should be situated in heavily populated regions.
- e) Access to infrastructural facilities such as established industrial sheds or sites, connecting roads, proximity to railway stations, airports, or sea ports, power, water, public utilities, civil amenities, and means of communication are all crucial, particularly for small enterprises.
- f) Technically qualified and trained management, as well as skilled and non-skilled labour.
- g) Banking and financial institutions are within walking distance.
- h) Locations with links: Cost savings and reductions in transportation overheads and other expenses arise from the development of industrial areas or business centres.
- i) Concerns about safety and security should be prioritised in strategic planning.
- j) Government influences: There are both positive and negative incentives for an entrepreneur to choose a certain location. Electricity, banking transportation, tax reduction, subsidies, and liberalisation are all examples of low overhead luxuries. Limits on placing firms in metropolitan areas for pollution control and industrial decentralisation are examples of negative incentives.
- k) Residents of small company owners seek to establish themselves close to their homes.

According to one research on small-scale unit locational factors, the entrepreneur's native place or homelands was the most relevant element. The strong preference for home implies that small-scale business is not easily transportable. Concessions and incentives cannot compensate for inadequate infrastructure, as seen by the low preference for government incentives. The table below also shows that locational preferences alter as the amount of development in different locations varies (hills and plains).

Considerations	Entrepreneur's Response					
	Hills		Plains		Total	
	No.	%	No.	%	No.	%
Homeland	15	67	11	39	26	52
Government Incentives	3	14	1	4	4	8
Availability of Raw Material	0	0	1	4	1	2
Availability of labour	2	9	0	0	2	4
Availability of market	0	0	5	18	5	10
Availability of Infrastructure Facilities	1	5	9	32	10	20
Others	1	5	1	4	2	4
Total	22	100	28	100	50	100

Notes

3.1.2 Factor Rating Method

It is the most commonly utilised method of analysis. It is the most often used approach since it allows for the inclusion of a wide range of factors in the study. It may be used in service and industrial settings. This approach of combining quantitative and qualitative elements assigns weights to factors based on relative significance, and a weight age score for each site is generated using a preference matrix. The best option is determined by the site with the greatest weighted score.

Steps of Factor Rating Method

The process of selecting a new facility location involves a series of following steps:

- 1) Determine the most critical location factors.
- 2) Give each element a rating based on its relative relevance; higher ratings indicate a more important factor.
- 3) Assign each place to each category based on the qualities of the site.
- 4) Multiply the factor allocated to each site by the fundamental variables taken into account to calculate the rating for each location.
- 5) Add up the total product for each criterion and choose the best place with the greatest overall score.

Illustration:

Assume that a new medical facility called Health-care would be built in Delhi. The following table shows the location factors, factor ratings, and scores for two prospective sites.

Which is the best location based on factor rating method?

Sl. No.	Location factor	Factor rating	Rating	
			Location 1	Location 2
1.	Facility utilization	8	3	5
2.	Total patient per month	5	4	3
3.	Average time per emergency trip	6	4	5
4.	Land and construction costs	3	1	2
5.	Employee preferences	5	5	3

Solution:

Sl. No.	Location factor	Factor rating (1)	Location 1		Location 2	
			(Rating) (2)	Total = (1) . (2)	(Rating) (3)	Total = (1) . (3)
1.	Facility utilization	8	3	24	5	40
2.	Total patient per month	5	4	20	3	15
3.	Average time per emergency trip	6	4	24	5	30
4.	Land and construction costs	3	1	3	2	6
5.	Employee preferences	5	5	25	3	15
			Total	96	Total	106

The total score for location 2 is higher than that of location 1. Hence location 2, is the best choice.

3.1.3 Center of Gravity Method

The Center of Gravity Method is a mathematical approach for determining the most cost-effective site for a distribution centre. When determining the optimal site for a distribution centre, the technique considers the location of markets, the volume of items transported to those markets, and transportation costs.

Steps of Center of Gravity Method

The placement of the places on a coordinate system is the first stage in the centre of gravity technique. The coordinate system's origin and scale can be whatever you choose, as long as the relative distances are accurately represented. Placing a grid over an ordinary map is a simple way to accomplish this.

Equations:

The centre of Gravity is determined using equations

$$\text{x-coordinate of the centre of gravity} = \frac{\sum d_{ix} Q_i}{\sum Q_i}$$

$$\text{y-coordinate of the center of gravity} = \frac{\sum d_{iy} Q_i}{\sum Q_i}$$

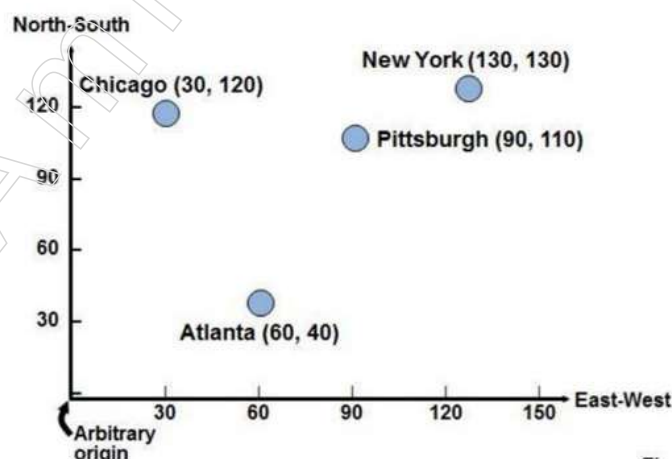
where,

- d_{ix} = x-coordinate of location i
- d_{iy} = y-coordinate of location i
- Q_i = Quantity of goods moved to or from location i

Distance alone should not be the primary consideration because the quantity of containers carried each month has an impact on prices. The centre of gravity approach assumes that both distance and volume transported are directly related to cost. The optimal site is one that reduces the weighted distance between the warehouse and its retail outlets, which is calculated based on the number of containers delivered.

For Example:

Center-of -Gravity Method



Notes

Centere-of-Gravity Method

Store Location	Number of Containers Shipped per Month
Chicago (30, 120)	2,000
Pittsburgh (90, 110)	1,000
New York (130, 130)	1,000
Atlanta (60, 40)	2,000

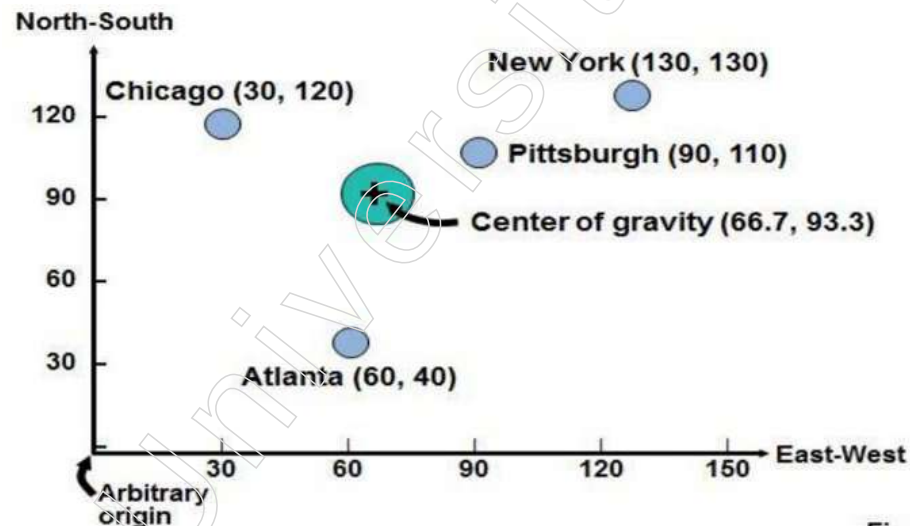
$$x\text{-coordinate} = \frac{(30)(2000) + (90)(1000) + (130)(1000) + (60)(2000)}{2000 + 1000 + 1000 + 2000}$$

$$= 66.7$$

$$y\text{-coordinate} = \frac{(120)(2000) + (110)(1000) + (130)(1000) + (40)(2000)}{2000 + 1000 + 1000 + 2000}$$

$$= 93.3$$

Centere-of-Gravity Method



Result:

The firm will be located at:

- X coordinate 66.7
- Y coordinate 93.3

Center of Gravity Method in Distribution Center Location

When it comes to your storage strategy, you must consider the What, Where, and How questions. In warehousing, there are at least three essential elements that may help you cut costs and improve service levels:

1. Strict and Required Warehousing Procedures.
2. Do you have any warehouse metrics in place to help you evaluate if your operations are running well or not?
3. Your warehouse location is ideal to satisfy service level obligations in the first place.
4. All of these things together can help you save money in your warehouse.

Where to create a distribution center, fulfilment center, or warehouse may be a strategic question. The solution to “where” are drivers to costs and therefore the customer experience. Where to create your next warehouse is an incredibly important question to answer. After you answer the “where” question, then we’d like to debate what warehouse processes and warehouse metrics ought you determine.

Of course, there are several things to consider:

- What is your inventory profile forecasted to be?
- What is your plan for geographic distribution?
- Incentives from the state and local governments? Incentives for state and local governments?
- Getting Closer to the Logistics Carrier (such as UPS World Hub or Ground Transport)
- The accessible pool of local talent in the area.
- Transportation expenses, both inbound and outbound.
- Access to interstate highway systems or ports of entry for inbound marine imports.
- Crime rate, housing, and educational facilities are all factors to consider.
- Nexus, State Taxes.
- Zoning limitations and building construction codes.

The factors listed above are essential and should be taken into account. However, in order to answer the issue of “where,” a more quantitative method must be utilised in conjunction with the aforementioned principles. The Center of Gravity Strategy is a popular method for determining where to locate the next single fulfilment centre site.

The Center of Gravity Method is a method for calculating geographic coordinates for a single new facility with the lowest possible cost. It’s a method that takes into account the following primary inputs:

- Markets
- Volume of goods shipped
- Shipping costs

This method is beneficial because it’s (1) Simple to compute, (2) Considers existing facilities, (3) and minimizes the costs.

Let’s take another example of how to use center of gravity method

Step 1:

- Create a coordinate grid with the locations of existing warehouses, fulfilment centres, and distribution centres.
- Glue the grid on a regular map.
- It’s important to keep track of the relative distances.

Step 2:

Then, using the equations below,

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- $C_x = \sum d_{ix} V_i / \sum V_i$
- $C_y = \sum d_{iy} V_i / \sum V_i$

These equations are used to compute the X and Y coordinates for the new facility, where C_x is the X (horizontal axis) coordinate. C_y is the new facility's Y (vertical axis) coordinate, d_{ix} is the old location's X coordinate, d_{iy} is the existing location's Y coordinate, and V_i is the volume of products transported to or from the i th site.

Step 3:

Once you have obtained the X and Y coordinates place that location on the map.

But it does not end there. There are additional variables to consider, as I said before. This technique provides a point of departure – or, to put it another way, a beginning point for where you may expand your fulfilment or logistics network based on longitude and latitude.

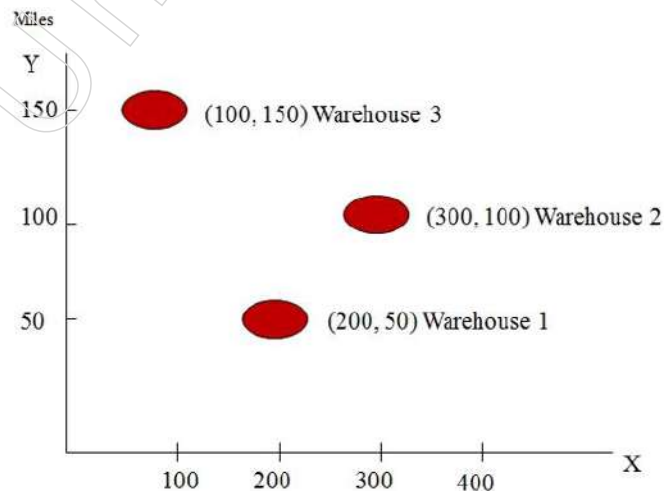
Center of Gravity Example

Let's say your firm wishes to extend its logistics network by adding a facility to an existing network of three. What are the coordinates for the new prospective site based on the following assumptions?

Let's assume the following:

- Warehouse 1 has a daily outbound goods volume of 2,500 units
- Warehouse 2 has a daily outbound goods volume of 1,300 units
- Warehouse 3 has a daily outbound goods volume of 5,000 units

And the current coordinates of the existing facilities:



Given the assumptions and grid coordinates above, we get the following:

- $d1x = 200$
- $d2x = 300$
- $d3x = 100$
- $d1y = 50$
- $d2y = 100$

- $d3y = 150$

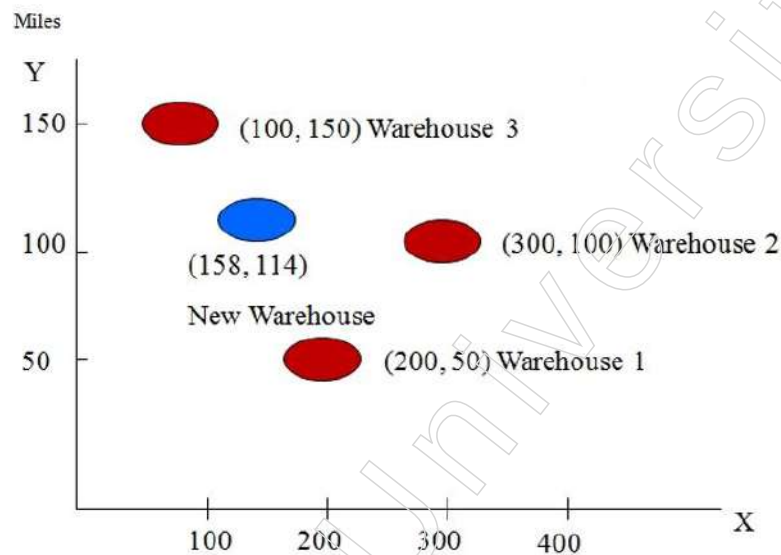
And,

- $V1 = 2,500$
- $V2 = 1,300$
- $V3 = 5,000$

Which gives us,

- $Cx = (200 \times 2,500) + (300 \times 1,300) + (100 \times 5,000) / (2,500 + 1,300 + 5,000) = 158$
- $Cy = (50 \times 2,500) + (100 \times 1,300) + (150 \times 5,000) / (2,500 + 1,300 + 5,000) = 114$

If we were to plot the new suggested warehouse coordinates on the grid, we would get:



Of course, the debate isn't over yet. Once a point or multiple coordinates have been discovered in the context of the present logistics network, further considerations must be addressed. This is only the beginning — a starting point.

3.1.4 Load Distance Method

The load-distance approach is a mathematical model for evaluating places based on their closeness to one another. The goal is to choose a site that reduces the total weighted loads entering and leaving the facility. When two points are assigned to grid coordinates on a map, the distance between them is represented. Instead of using distance, time can be used as an alternative.

Distance Measures

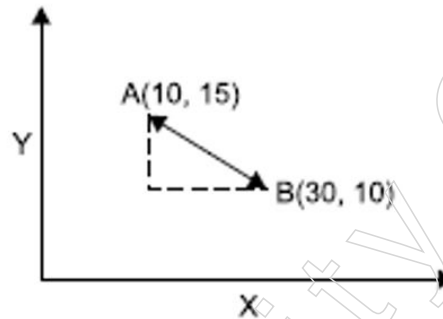
Assume a new warehouse is being built to service Delhi. Several vendors, including one in Ghaziabad, will send incoming goods. What would be the distance between the two facilities if the new warehouse were in Gurgaon? The distance travelled by truck is determined by the highway system and the route followed. The

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exact distance between any two sites in the same county may be calculated using computer software. However, a crude estimate that is either Euclidean or rectilinear distance measure can be employed for the load-distance technique. The straight-line distance, or shortest feasible path, between two locations is known as Euclidean distance.

Distance between point A and point B



The supplier's location in Ghaziabad is represented by point A on the grid, while the alternative warehouse site in Gurgaon is represented by point B. The length of the hypotenuse of a right triangle is the distance between points A and B, or

$$d_{AB} = \sqrt{(X_A - X_B)^2 + (Y_A - Y_B)^2}$$

where d_{AB} = distance between points A and B

X_A = x-coordinate of point A

Y_A = y-coordinate of point A

X_B = x-coordinate of point B

Y_B = y-coordinate of point B

Rectilinear distance is a measurement of the distance between two locations that are separated by a sequence of 90° turns, similar to city blocks. This distance is the sum of the two dashed lines that indicate the triangle's base and side. The absolute value of the change in x-coordinates is the distance travelled in the x-direction. When you multiply this result by the absolute value of the difference in y-coordinates, you get

$$D_{AB} = |X_A - X_B| + |Y_A - Y_B|$$

Calculating a Load-distance Score

Assume that a company planning a new location wishes to choose a location that minimises the distances that loads, especially bigger ones, must travel to and from the facility. A load might be shipments from suppliers, between facilities, or to customers, or it can be customers or staff travelling to or from the facility, depending on the industry. The company aims to reduce load-distance by finding a site where heavy loads may travel short distances.

To compute a load-distance for each prospective site, we simply multiply the loads flowing to and from the facility by the distances travelled, using any of the distance measurements. These burdens might be represented in terms of tones or weekly

travels. This needs a practical illustration in order to grasp the concept's significance. Let us take another trip to a new health-care institution.

Illustration:

Seven census areas in Delhi will be served by the new health-care facility. The coordinates for the centre of each census tract, as well as the estimated populations in thousands, are shown in the table below. Customers will travel to the new hospital from the seven census tract centres when they want medical attention. The centres of census tracts C and F are (5.5, 4.5) and (7, 2), respectively, and are being considered for the new facility. The following table lists the population of seven census tract centres, as well as their co-ordinate distances. Which place is superior in terms of its total load-distance score if we use the population as the loads and utilise rectilinear distance?

Sl. No.	Census tract	(x, y)	Population (I)
1	A	(2.5, 4.5)	2
2	B	(2.5, 2.5)	5
3	C	(5.5, 4.5)	10
4	D	(5, 2)	7
5	E	(8, 5)	10
6	F	(7, 2)	20
7	G	(9, 2.5)	14

Solution:

Calculate each location's load-distance score. Calculate the load-distance score for each tract using the coordinates from the preceding table.

Using the formula $D_{AB} = |X_A - X_B| + |Y_A - Y_B|$

Census tract	(x, y)	Population (I)	Locate at (5.5, 4.5)		Locate at (7, 2)	
			Distance (d)	Load-distance	Distance (d)	Load-distance
A	(2.5, 4.5)	2	$3 + 0 = 3$	6	$4.5 + 2.5 = 7$	14
B	(2.5, 2.5)	5	$3 + 2 = 5$	25	$4.5 + 0.5 = 5$	25
C	(5.5, 4.5)	10	$0 + 0 = 0$	0	$1.5 + 2.5 = 4$	40
D	(5, 2)	7	$0.5 + 2.5 = 3$	21	$2 + 0 = 2$	14
E	(8, 5)	10	$2.5 + 0.5 = 3$	30	$1 + 3 = 4$	40
F	(7, 2)	20	$1.5 + 2.5 = 4$	80	$0 + 0 = 0$	0
G	(9, 2.5)	14	$3.5 + 2 = 5.5$	77	$2 + 0.5 = 2.5$	35
Total				239	Total	168

When the scores for all tracts are added together, a total load-distance score of 239 is obtained when the facility is placed at, compared to a load-distance score of 168 when the facility is located at (7, 2). As a result, the site in census tract F is preferable.

3.1.5 Break-even Analysis

Break-even analysis might help you figure out what amount of output you need or what sales mix you want. Because the metre and computations are not utilised by

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external parties such as investors, regulators, or financial institutions, the research is exclusively for the purpose of a company's management. The break-even point is calculated in this sort of study (BEP). The break-even point is derived by dividing total fixed costs by the price per individual unit minus variable expenses. Fixed costs are costs that do not change regardless of the number of units sold. Break-even refers to a situation in which a firm does not earn a profit or a loss, but instead recovers all of the money spent.

The amount of fixed expenses is compared to the profit made by each extra unit produced and sold in a break-even analysis. A firm with smaller fixed expenses will, on average, have a lower break-even point.

Importance of Break-even Analysis

- **Manages the Size of Units to be Sold:** Using break-even analysis, the firm or the owner may determine how many units are required to pay the costs. The break-even analysis requires the variable cost and selling price of each individual product, as well as the overall cost.
- **Budgeting and Setting Targets:** Because the company or the owner understands when a business can break even, it is simple to create a goal and a budget for the business. This study may also be used to determine a company's realistic objective.
- **Manage the Margin of Safety:** When a company's financial situation deteriorates, its sales tend to decline. The break-even analysis assists the firm in determining the minimum number of sales necessary to break even. The management can make a high-level business choice based on the margin of safety reports.
- **Monitors and Controls Cost:** Fixed and variable costs can have an impact on a company's profit margin. As a result, management may use break-even analysis to see whether any impacts are affecting the cost.
- **Helps to Design Pricing Strategy:** Any change in a product's pricing might influence the break-even point. If the selling price is increased, the amount of the product that must be sold to break even is lowered. Similarly, if a company's selling price is decreased, it must sell more to break even.

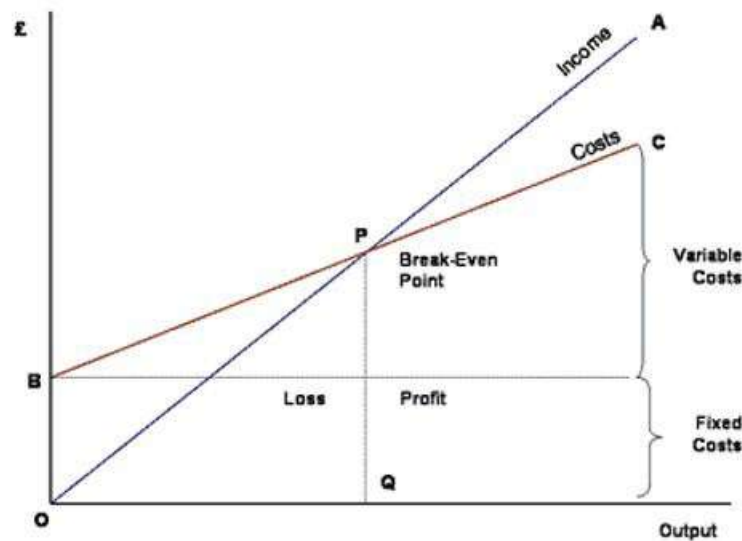
Production managers and management accountants frequently employ break-even analysis. It is based on separating production expenses into "variable" (costs that fluctuate when the manufacturing output varies) and "fixed" (costs that remain constant) (costs not directly related to the volume of production).

To identify the level of sales volume, sales value, or output at which the firm earns neither a profit nor a loss, total variable and fixed expenses are compared to sales revenue (the "break-even point").

• The Break-even Chart

The break-even chart, in its most basic form, is a graphical depiction of expenses at various levels of activity displayed alongside the change in income (or sales, revenue) with the same variation in activity. The "break-even point," which is shown on the chart below by the junction of the two lines, is the point at which neither profit nor loss is realised.

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The OA line in the diagram above illustrates the variance in revenue at different levels of production activity ("output"). The entire fixed costs of the firm are represented by OB. Variable costs rise as output rises, resulting in an increase in overall costs (fixed + variable). Costs are larger than income at low levels of output. Costs are precisely equal to revenue at the point of intersection, P, thus there is no profit or loss.

- **Fixed Costs**

Fixed costs are expenses that are not proportional to the amount of output or production. In other words, regardless of whether the firm produces nothing or a large amount of product, the fixed expenses will remain about the same. Fixed costs can change over time, for example, as a result of increased production capacity (e.g., adding a new factory unit) or increased overheads necessary to run a larger, more complicated firm.

Examples of fixed costs:

- Rent and rates
- Depreciation
- Research and development
- Marketing costs (non- revenue related)
- Administration costs

- **Variable Costs**

Variable costs are those that change in direct proportion to the amount of output. They represent payment inputs such raw supplies, direct labour, and gasoline, as well as revenue-related expenditures like commission.

A difference is often made between "Direct" variable costs and "Indirect" variable costs.

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Direct variable costs: These are those that may be linked to the production of a certain product or service and are assigned to a specific cost centre. Raw materials and pay paid to individuals who labour on the assembly line are two examples.

Indirect Variable Costs: Although indirect variable costs are not directly related to production, they do fluctuate with output. Depreciation (when it is computed in relation to production, such as machine hours), maintenance, and some labour expenses are examples.

- **Semi-variable Costs**

While the difference between fixed and variable costs is a useful way of categorising company costs, certain costs are constant in nature yet grow as production hits specific thresholds. These are mostly determined by the business's overall "size" and/or complexity. When a company's productivity or sales are low, it may not need to spend money on things like human resource management or a fully staffed finance department.

However, when the firm increases in size (e.g., production, number of employees, number and complexity of transactions), more resources are needed. If output unexpectedly increases, a temporary increase in warehousing and/or transportation may be necessary. In this case, we say that portion of the cost is variable and part of the cost is constant.

Components of Break-even Analysis

- **Fixed Costs:** These expenses are sometimes referred to as overhead expenses. Once a business's financial activity begins, these expenses become apparent. Taxes, wages, rentals, depreciation costs, labour costs, interest costs, energy costs, and so on are all included in the fixed pricing.
- **Variable Costs:** These expenses change with the amount of production and will drop or rise accordingly. These expenses include packaging, raw material prices, gasoline, and other production-related items.

Uses of Break-even Analysis

- **New Business:** A break-even study is critical for a new enterprise. It provides pricing strategy guidance to management and is cost-effective. This study also determines whether or not the new firm is profitable.
- **Manufacture New Products:** If an established business wants to introduce a new product, it must first do a break-even analysis to see whether the product would contribute required spending to the company.
- **Change in Business Model:** The break-even analysis works even if the company model is changed, such as switching from retail to wholesale. This study will aid the firm in determining whether or not a product's selling price has to be adjusted.

Break-even Analysis Formula

Break-even point = Fixed cost/Price per cost – Variable cost

Example of Break-even Analysis

Company X sells a pen. The company first determined the fixed costs, which include a lease, property tax, and salaries. They sum up to ₹1,00,000. The variable cost linked with manufacturing one pen is ₹2 per unit. So, the pen is sold at a premium price of ₹10.

Therefore, to determine the break-even point of Company X, the premium pen will be:

$$\begin{aligned}\text{Break-even point} &= \text{Fixed cost} / \text{Price per cost} - \text{Variable cost} \\ &= ₹1,00,000 / (₹12 - ₹2) \\ &= 1,00,000 / 10 \\ &= 10,000\end{aligned}$$

Therefore, given the variable costs, fixed costs, and selling price of the pen, company X would need to sell 10,000 units of pens to break-even.

When to Use a Break-even Analysis?

There are four common scenarios when it helps to do a break-even analysis.

1. Starting a New Business

A break-even study is essential if you're considering establishing a new firm. It will not only assist you in determining whether your company concept is practical, but it will also require you to conduct research, be realistic about costs, and consider your pricing approach.

2. Creating a New Product

Even if you currently have a firm, you should conduct a break-even analysis before investing in a new product, especially if it will incur substantial costs. Even if your fixed expenses, such as office rent, remain constant, you'll need to calculate the variable costs of your new product and establish prices before you begin selling.

3. Adding a New Sales Channel

Your expenses will vary whenever you add a new sales channel, even if your pricing don't. If you've been selling online and are considering a pop-up store, for example, you'll want to make sure you at least break even. Otherwise, the financial burden might jeopardise the rest of your company.

This is true for introducing new online sales channels, such as shoppable Instagram posts. Will you be incurring any additional expenditures, such as Instagram advertisements, to promote the channel? Those expenses must be factored into your break-even calculations.

4. Changing the Business Model

You should perform a break-even analysis if you're considering about altering your company strategy, such as transitioning from drop shipping to holding inventory. Your expenses may vary dramatically, and this will assist you determine whether or not your rates should alter as well.

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Break-even analysis is useful for the following reasons:

- It helps to determine remaining/unused capacity of the company once the breakeven is reached. This will help to show maximum profit on a particular product/service that can be generated.
- It helps to determine the impact on profit on changing to automation from manual (a fixed cost replaces a variable cost)
- It helps to determine the change in profits if the price of a product is altered.
- It helps to determine the number of losses that could be sustained if there is a sales downturn.

Additionally, break-even analysis is very useful for knowing the overall ability of a business to generate a profit. In case of a company whose breakeven point is near to the maximum sales level, this signifies that it is nearly impractical for the business to earn a profit even under the best of circumstances.

Therefore, it's the management responsibility to monitor the breakeven point constantly. This monitoring certainly reduces the break-even point whenever possible.

Ways to Monitor Break-even Point

- **Price Analysis:** Minimize or eliminate the use of coupons or other price reductions offers, since such promotional strategies increase the breakeven point.
- **Technology Analysis:** Implementation any technology that can enhance the business efficiency, thus increasing capacity with no extra cost.
- **Cost Analysis:** Reviewing all fixed costs constantly to verify if any can be eliminated can surely help. Also, review the total variable costs to see if they can be eliminated. This analysis will increase the margin and reduce the breakeven point.
- **Margin Analysis:** Push sales of the highest-margin (high contribution earning) items and pay close attention to product margins, thus reducing the breakeven point.
- **Outsourcing:** If an activity consists of a fixed cost, try to outsource such activity (whenever possible), which reduces the breakeven point.

Benefits of Break-even Analysis

- 1) **Catch Missing Analysis:** When you're considering starting a new company. It's very probable that you'll overlook a few charges. As a result, a break-even analysis can assist you in determining your break-even point by reviewing all financial obligations. This research significantly reduces the frequency of unexpected events in the future, or at the very least prepares a corporation for them.
- 2) **Set Revenue Targets:** After you've completed the break-even analysis, you'll know how much you need to sell to break even. This will assist you and your sales team in establishing more specific sales objectives.
- 3) **Make Smarter Decisions:** Entrepreneurs frequently make business decisions based on their emotions. Emotion, or how you feel, is crucial, but it isn't sufficient. To be a successful entrepreneur, you must make judgments based on facts.

- 4) **Fund Your Business:** A vital component of every business strategy is a financial analysis. If you want to get money from investors, it's usually a must. You must demonstrate the viability of your company strategy in order to receive funding. Furthermore, if the analysis appears to be sound, you will feel confident enough to shoulder the financial load in a variety of methods.
- 5) **Better Pricing:** Finding the break-even threshold will aid in improved product pricing. This tool is frequently used to determine the ideal pricing for a product that will provide the most profit without raising the current price.
- 6) **Cover Fixed Cost:** Break-even analysis aids in the coverage of all fixed costs.

Limitations of Break-even Analysis

Break-even analysis plays an important role in making business decisions, but it's limited in the type of information it can provide.

1. Not a Predictor of Demand

It's crucial to remember that a break-even analysis isn't a demand forecast. It won't tell you how much money you'll make or how many people will want to buy what you're selling. All it will tell you is how many units you need to sell to break even. It's also worth noting that demand isn't consistent. The number of individuals ready to buy your stuff will fluctuate as your pricing changes.

2. Dependent on Reliable Data

Costs can sometimes be classified as both fixed and variable. This can make computations difficult, and you'll almost certainly have to fit them into one of the two. For example, you may have a fixed labour cost and an extra labour cost that varies depending on how much goods you sell.

Correct data is required for your break-even point to be accurate. You won't obtain a trustworthy result if you don't enter good data into the algorithm.

3. Simplistic

The formula for calculating the break-even point is straightforward. Many firms sell a variety of items at different rates. That subtlety won't be picked up by it. You'll probably have to deal with one product at a time or make an average pricing estimate based on all of the items you'll be selling. If this is the case, it's a good idea to practise a few different scenarios so you're ready.

Costs change in lockstep with pricing. Only one item changes at a time, according to this paradigm. Instead, lowering your pricing and selling more items may cut your variable expenses because you have more purchasing power or are able to operate more effectively. In the end, it's only a guess.

4. Ignores Time

The break-even analysis overlooks temporal variations. The time frame you choose to compute fixed expenses will determine the time frame you use (monthly is most common). Although you'll be able to see how many units you need to sell over the course of a month, you won't be able to see how things alter if sales shift week to week

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or seasonally over the course of a year. You'll need strong cash flow management and maybe a decent sales prediction for this.

It also neglects to consider the future. Break-even analysis solely considers the present. If the cost of your raw materials increases next year, unless you boost your pricing, your break-even point will be much higher. If you boost your rates, you could lose customers. This delicate balance is always in flux.

5. Ignores Competitors

As a newcomer to the market, you will have an impact on rivals and vice versa. They might modify their pricing, affecting demand for your goods and forcing you to adjust your prices as well. If they expand fast and a raw resource that you both use becomes scarce, the price may rise.

Finally, break-even analysis will provide you with a strong knowledge of the prerequisites for success. It's a must-have. However, it isn't the only study you should conduct before establishing or changing a firm.

Strategies to Lower Your Break-even Point

What if you finish your break-even analysis and discover that the number of units you need to sell is far more than you anticipated? Don't be alarmed if the figure appears unreasonable or unreachable. You might be able to decrease your break-even point by making certain changes.

1. Lower Fixed Costs

Check to see if you can reduce your fixed costs. You'll need to sell fewer units to break even if you can acquire them cheaper. If you're considering launching a retail business but the economics don't add up, try selling online instead. What impact does this have on your fixed costs?

2. Raise Your Prices

You won't need to sell as many units to break even if you boost your prices. The marginal contribution will be greater per unit sold. Consider what the market is prepared to pay and the expectations that come with a price when considering boosting your rates. You won't need to sell as many units, but you will need to sell enough—and customers may demand a better product or better customer service if you price more.

3. Lower Variable Costs

The most challenging choice, especially if you're just starting out, is to reduce your variable expenses. However, the more your company grows, the easier it will be to cut variable expenditures. It's worthwhile to try to save expenses by negotiating with vendors, switching suppliers, or altering your method. For example, you could discover that packing peanuts are less expensive than bubble wrap when shipping sensitive items.

3.1.6 Utilization of GIS in Plant Location

The most significant and far-reaching strategic choice faced by industry managers

is where to put industrial facilities. Operational research, geographic information systems, and computer science are some of the answers to these problems. Historically, industrial site analysis was seen through the lens of operational research. Location analysis is increasingly relying on intelligent technologies and geographic information systems.

The macro-location analysis and the micro-location analysis are the two phases of the industrial location analysis. The macro site is a geographic area that can satisfy the fundamental criteria for industrial building and development while maintaining low operating costs. The micro location is a specialised site inside the macro location that satisfies technological, infrastructural, and operational criteria.

Industrial decision-making is based on a large number of facts about the situation. According to estimates, 80 percent of data utilised by managers and decision makers is geographical (spatial). Geographical or spatial decision issues are decision problems that require geographical data. Under limitations, spatial decision making is a very complicated process of selecting among options to achieve an aim or a collection of objectives. Not only do they frequently entail a plethora of technical needs, but they may also include economic, social, and environmental issues with competing goals.

The problem is determining how to conduct a comprehensive geographical industrial location study: geographic information systems appear to be the best instrument for spatial analysis and strategic decision making. Geographic information systems (GIS) are sophisticated tools for spatial analysis that allow you to record, store, query, analyse, display, and export geographic data. As a result, they have a significant impact on the spatial decision-making process.

Recent advancements in the field of spatial analysis have resulted in significant enhancements to GIS's location analysis capabilities. Distance analysis, attribute data analysis, multi-criteria analysis, and other methods are used to examine these developments. To address all of the aforementioned issues, we examine GIS functionalities in industrial location analysis in the Vojvodina area of Serbia as a study case.

Background

The location has a significant influence on the total risk and profitability of the industry. The location has a direct impact on transportation expenses, taxes, wages, raw materials, leasing prices, and other expenditures, which may account for up to 25% of the buy product price. When all expenditures are added together, the location may result in a 50 percent increase in overall operating costs. The efficiency of the industry is influenced by spatial conditions, with certain sites being better than others for a particular output.

A huge number of decisions have a geographic (spatial) component, and engineers and scientists are challenged to come up with a successful formula for selecting the best industrial site. The collection of data allows for the development of prospective industrial locations, while the use of selection criteria reduces the selection over time, as shown in the diagram below.

One of the most important aspects of industrial location analysis is the enormous number of choice possibilities. There are generally a variety of prospective locations for

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the facility in the area of interest. Bank Hofer recommends taking a two-step strategy, starting with a macro location and then moving on to a micro-location. Alternatives that are theoretically possible are pre-selected in each stage (scanned). Before doing a more comprehensive study, compliance with minimal requirements is evaluated for those options that passed the screening. Autschbach splits the evaluation procedure into country preselection, rough analysis, and comprehensive study of options, similar to BankHofer.

Concept creation, country preselection, macro-analysis, micro-analysis, and decision are the five steps Goette proposes. Finding the best fit position in a place that has the most appropriate combination of desired characteristics is a spatial challenge that necessitates comparing the attributes of different places and determining the best fit location in a place that has the most suitable combination of desired attributes. The ability of Geographic Information Systems to allow a decision maker to analyse geographic space as well as the information (attributes) associated with it at the same time.

This feature makes GIS a must-have tool for resolving complicated geographical issues like industrial placement. Geographic Information Systems (GIS) are a set of processes that assist an organization's decision-making operations by providing data input, storage and retrieval, mapping, and geographic analysis for both spatial and attribute data. Geographical data is geo-referenced data that has been processed into a form that is relevant to the recipient decision-maker and has actual or perceived value in the decision-making process.

In GIS, there are a variety of ways for analysing geographic data. Techniques for analysing geographic data and methods for analysing attribute data exist. GIS spatial analysis includes attributive (tabular) data analysis, overlapping layers (i.e., spatial data query), distance analysis, and nonparametric approaches like Multi-Criteria Decision Analysis (MCDA), Genetic Algorithms (GA), and so on. One thematic layer's attribute data can be analysed using SQL queries against a table with attribute data, various arithmetic operations (addition, subtraction, multiplication, and so on), logarithmic functions, trigonometric functions, multicriteria decision methods, and artificial intelligence methods. When we talk about geographic data analysis, we're talking about both vector and raster data.

GIS tools and processes are critical in analysing choice issues as part of a spatial decision support system for industrial site study, particularly during the screening phase. The function of GIS in the screening phase is to analyse plausible options that will be considered later in the assessment phase. The function of GIS in the review phase is to create suitability maps based on the findings of multi-criteria decision analysis and decision makers' value judgements. Unfortunately, there isn't enough scientific study in the industrial location field on spatial analysis. As a result, we provide instances of GIS-based spatial industrial location analysis.

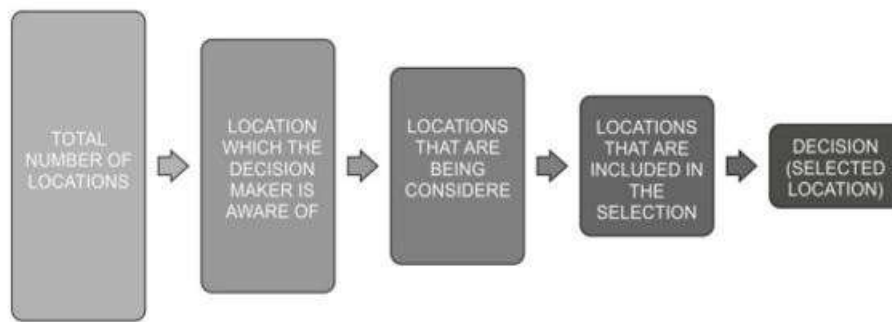


Fig: Potential Industrial Location

Application of GIS in Land/Location Suitability Analysis

GIS is a computer system that makes it easier to capture, verify, store, integrate, analyse, and display data about the planet in a geographically referenced manner. GIS is a cross-disciplinary tool that has been embraced and used in a variety of disciplines of research and engineering. Land suitability analysis is the process of determining if a land resource is acceptable for some intended purposes and determining its suitability level, as well as the search for the optimum site of one or more facilities to serve some desired function. The intrinsic appropriateness of the land for particular specified, long-term usage is referred to as land suitability.

The location of a retail store to the positioning of several ambulance dispatch locations are all examples. Land suitability study or evaluation is carried out by taking into account specific land factors such as hydrology, geography, topography, geology, and so on in a GIS-enabled environment.

GIS technology has been applied by various researchers in biogas plant location suitability assessment. A few examples are given here: The final suitability map from the study provided constraint map and the energy potential grid, synthesized in the form of a raster GIS file, to develop potential for collective biogas plants in France. GIS was used to geo-reference the bio-resources potential and also to locate the optimal sites at both national and regional scales for the country.

The project was effective in producing a suitability map by analyzing aerial photos and Landsat imagery to determine the specific geo-location of farms, which was employed in the identification of agricultural leftovers. Similarly, research was conducted in Southern Finland to evaluate the best sites, sizes, and number of biogas plants. This study looked at the geographical distribution and amount of possible biomass feedstock for bio-methane production in the study region. A GIS-based spatial data mining approach was used to model the optimal location for distributed biomass power generation facilities in Tumkur district, India, in addition to numerous studies on site suitability analysis for biogas plants. These studies asserted that GIS is an appropriate and recognized spatial tool for location analysis.

Check Your Understanding

- 1) A _____ is an installation, contrivance, or other item that enables anything; a location where something may be done.
 - a) Facility

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- b) Warehouse
 - c) Location
 - d) All the above
 - e) None of the above
- 2) _____ is a dynamic process in which an entrepreneur evaluates and analyses the suitability of several locations with the goal of selecting the optimal location for a specific business.
- a) Demographic Analysis
 - b) Trade Region Analysis
 - c) Locational analysis
 - d) Competitive Analysis
 - e) None of the above
- 3) The process of determining a geographic location for a company's activities is known as _____.
- a) Facility location
 - b) Site Economics
 - c) General Location
 - d) Specific Location
 - e) None of the above
- 4) _____ and _____ are examples of external economies of scale.
- a) Urbanization and locational economies of scale
 - b) Urbanization and Just-In-Time
 - c) Infrastructure and locational economies of scale
 - d) Just-In-Time and locational economies of scale
 - e) None of the above
- 5) The _____ is a mathematical approach for determining the most cost-effective site for a distribution centre.
- a) Factor Rating Method
 - b) Load Distance Method
 - c) Break-Even Analysis
 - d) Center of Gravity Method
 - e) None of the above
- 6) The _____ is a mathematical model for evaluating places based on their closeness to one another.
- a) Load-distance approach
 - b) Centre of Gravity Method

- c) Centre Location
d) Distance Approach
e) None of the above
- 7) _____ are costs that do not change regardless of the number of units sold.
a) Variable Costs
b) Fixed costs
c) Indirect Variable Costs
d) Direct Cost
e) None of the above
- 8) _____ refers to a situation in which a firm does not earn a profit or a loss, but instead recovers all of the money spent.
a) Break-even
b) Centre of Gravity
c) Catch Missing Analysis
d) Macro location Analysis
e) None of the above
- 9) _____ are those that change in direct proportion to the amount of output.
a) Fixed Costs
b) Indirect Fixed Costs
c) Administration Costs
d) Variable costs
e) None of the above
- 10) What is the Full Form of MCDA?
a) Multi-Criteria Decision Analysis
b) Macro-Criteria Decision Analysis
c) Micro-Criteria Decision Analysis
d) Multi-Concept Decision Analysis
e) None of the above

Summary

- A facility is a location where certain activities take place. For example, sports and military activities take place at sports and military facilities.
- An ideal site is one where the product's cost is maintained to a minimum, the market share is big, the risk is low, and the social benefit is high.
- Locational analysis is a dynamic process in which an entrepreneur evaluates and analyses the suitability of several locations with the goal of selecting the optimal location for a specific business.

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- Rapid transportation facilities guarantee that raw materials and completed items are delivered on time to the firm and to the consumers.
- In the case of urbanisation economies, companies benefit from locating in larger cities rather than smaller ones in order to access a larger pool of labour.
- Quality of life is enhanced by good schools, recreational facilities, cultural events, and an appealing lifestyle.
- The layout of a plant or facility cares with the physical placement of resources like equipment and storage facilities.
- A good layout should be planned in such a manner that changes can be made quickly to meet changing conditions, and it should be created with the operation's future needs in mind.
- The Center of Gravity Method is a method for calculating geographic coordinates for a single new facility with the lowest possible cost.
- The break-even point is derived by dividing total fixed costs by the price per individual unit minus variable expenses.
- Fixed costs are expenses that are not proportional to the amount of output or production. In other words, regardless of whether the firm produces nothing or a large amount of product, the fixed expenses will remain about the same.
- GIS is a cross-disciplinary tool that has been embraced and used in a variety of disciplines of research and engineering.

Activity

1. Visit an organization and find out whether is it located at a location where all the requisites and warehouse are available at a near distance, so that production process is accomplished timely.

Questions & Exercises

1. Explain the complete concept of Break-even analysis.
2. What is Plant Location? Explain the factors affecting facility location.
3. Describe the methods effectively determining cost-effective site for a facility location.

Glossary

- Demographic Analysis entails examining the area's demographic in terms of total population (in number), age composition, per capita income, educational level, occupational structure, and so on.
- Trade Region Analysis is a study of the geographic area that provides the firm with repeat customers. He'd also look into the possibility of reaching the trading area from other locations.
- Competitive Analysis aids in determining the type, location, scale, and quality of competition in a particular industry.
- Traffic study provides an approximate estimate of the number of prospective consumers passing by the proposed location during business hours, the traffic analysis evaluates alternative sites in terms of pedestrian and vehicle traffic.

- Factor Rating Method is the most commonly utilised method of analysis. It is the most often used approach since it allows for the inclusion of a wide range of factors in the study.
- The Center of Gravity Method is a mathematical approach for determining the most cost-effective site for a distribution centre.
- The load-distance approach is a mathematical model for evaluating places based on their closeness to one another. The goal is to choose a site that reduces the total weighted loads entering and leaving the facility.
- Break-even refers to a situation in which a firm does not earn a profit or a loss, but instead recovers all of the money spent.
- Land suitability analysis is the process of determining if a land resource is acceptable for some intended purposes and determining its suitability level, as well as the search for the optimum site of one or more facilities to serve some desired function.
- GIS is a computer system that makes it easier to capture, verify, store, integrate, analyse, and display data about the planet in a geographically referenced manner.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor

Check your understanding – Answers

- 1) a)
- 2) c)
- 3) a)
- 4) a)
- 5) d)
- 6) a)
- 7) b)
- 8) a)
- 9) d)
- 10) a)

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Unit - 3.2: Facility Layout

Objectives:

At the end of this unit, you will be able to:

- Learn about principles of facility layout
- Understand process layout
- Identify product or line layout
- Explain fixed position layout
- Learn concepts of line balancing
- Understand merits and demerits of product layout
- Identify merits and demerits of process layout

Introduction

The layout of the facility becomes the next priority in production planning after settling on a site location. The goal is to determine which design is most efficient and effective for the particular production process. A corporation might choose a U-shaped production line over a long, straight one to allow things and personnel to move more quickly from one site to another. Layout is important to service firms as well, but they are more concerned with how it influences consumer behaviour. A hospital's freight elevators may be more convenient at the heart of the building, but this may obstruct the movement of patients, visitors, and medical workers between floors and departments.

There are four main types of facility layouts: process, product, fixed-position, and cellular. The workflow is organised around the production process in the process layout. Workers who do comparable duties are assigned to the same group. Products are moved from one workstation to the next (but not necessarily to every workstation). All grinding, for example, would be done in one location, assembly in another, and inspection in still another. Firms that produce modest quantities of a wide range of products, often employing general-purpose equipment that can be quickly converted to new processes for varied product designs, benefit from this process arrangement. A process layout, for example, might be used by a custom machinery maker.

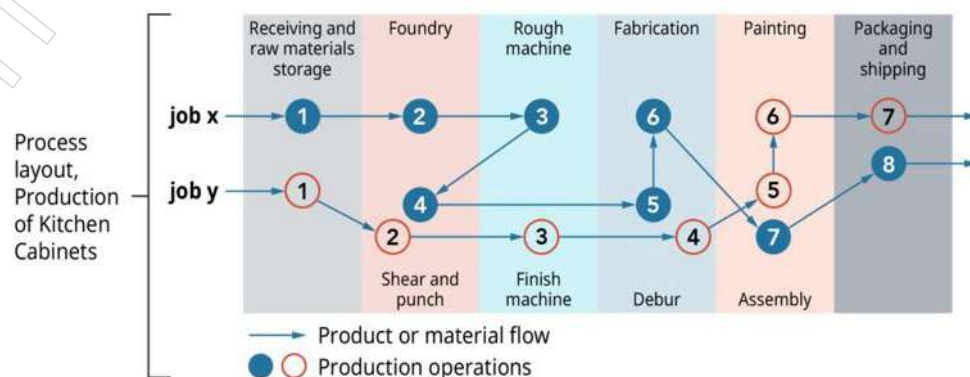


Figure 1: An Example of a Process Facility Layout. Source: Adapted from Operations Management, 9th edition, by Gaither/Frazier.

Products that need endless or repetitive production process use the merchandise (or assembly-line) layout. When large quantities of a product must be processed on an ongoing basis, the workstations or departments are arranged during a line with products moving along the road. Automobile and appliance manufacturers, also as food-processing plants, usually use a product layout. Service companies can also use a product layout for routine processing operations.

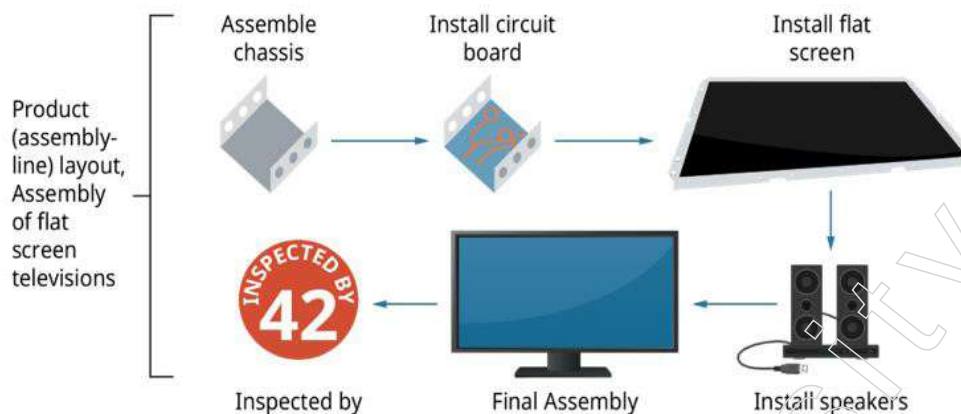


Figure 2. An Example of a Product Facility Layout. Source: Adapted from Operations Management, 9th edition, by Gaither/Frazier.

Plant Layout

It is the efficiency of production depends on how well the varied machines; production facilities and employee's amenities are located during a plant. Only the properly laid out plant can make sure the smooth and rapid movement of fabric, from the staple stage to the top product stage. Plant layout encompasses new layout also as improvement within the existing layout. it's going to be defined as a way of locating machines, processes and plant services within the factory so on achieve the proper quantity and quality of output at rock bottom possible cost of producing. It involves a judicious arrangement of production facilities in order that workflow is direct.

Definition

A plant layout is often defined as follows: Plant layout refers to the arrangement of physical facilities like machinery, equipment, furniture etc. within the factory building in such a fashion that have quickest flow of fabric at rock bottom cost and with the smallest amount of handling in processing the merchandise from the receipt of fabric to the shipment of the finished product.

According to Riggs, "the overall objective of plant layout is to design a physical arrangement that most economically meets the required output – quantity and quality."

According to J. L. Zundi, "Plant layout ideally involves allocation of space and arrangement of equipment in such a manner that overall operating costs are minimized.

Importance

Plant layout is a crucial decision because it represents long-term commitment. a perfect plant layout should provide the optimum relationship among output, floor area

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and manufacturing process. It facilitates the assembly process, minimizes material handling, time and price, and allows flexibility of operations, easy production flow, makes economic use of the building, promotes effective utilization of manpower, and provides for employee's convenience, safety, comfort at work, maximum exposure to natural light and ventilation. It's also important because it affects the flow of fabric and processes, labour efficiency, supervision and control, use of space and expansion possibilities etc.

- **Objectives of Plant Layout**

Plant layout aims at an appointment of labour areas and equipment which will be the foremost economical to work and yet be safe and satisfactory to all or any parties concerned, i.e., the workers and management, and therefore the products are produced at a price low enough to sell at a profit during a competitive market. There are variety of objectives a layout has got to meet. Essentially, facility planning may be a combination of objectives and considerations. The ultimate planning rests on a compromise of varied isolated benefits and limitations which successively are modified by time, degree of relative importance, management attitude and policy.

Few basic objectives, a planner must keep in view while planning a layout are:

1. Increased output/turnover.
2. Fewer production delays.
3. Greater utilisation of man and machinery.
4. Minimum capital investment.
5. Full utilisation of floor and cubic space.
6. Reduced inventory in process.
7. Minimum material handling.
8. No congestion and confusion.
9. Easier adjustment to changing conditions.
10. Easier and better supervision.
11. Satisfaction and safety of workers.

1. Increased Output:

Layout should be such production is administered in most effective manner. Machines, equipment and work areas should be arranged such material moves smoothly along a line. Effort should be made to avoid or eliminate any delay at any point during the method. It's being said that 80 per cent of the time a component/part either moves or is stored and only 20 per cent of the time the part is processed. Therefore, layout should be such a worker has to not walk long distances for collecting materials, tools or instructions. Flow of the work should be planned such the work passing through any area are often easily identified, counted or checked with little possibility of being mixed with other parts or batches in adjacent areas.

2) Fewer Production Delays:

In the manufacturing schedule, interruptions for whatever cause should be avoided.

Unbalanced machine capabilities, congestion at specific workstations, or malfunctions may all contribute to the disruption. Delays in a layout may be minimised by designing the material to flow progressively from one workstation operation/process to the next, with the least amount of stoppage and interference.

3) Greater Utilisation of Man and Machinery:

The goal of a good layout is to make the most of the available resources, such as manpower and machines. At no time should a man or machine be idle. Proper planning for each work station and function ensures that facilities are fully used. Improved manpower and machinery utilisation in the factory boosts production, encourages direct and indirect labour, and boosts employee morale.

4) Minimum Capital Investment:

Unnecessary capital investments should be avoided as much as possible and maintained to a bare minimum. Selection of appropriate machinery and departmental layout can significantly reduce the amount of equipment required. In addition to saving money on machine investment, foresight in selecting processing methods helps to save money. By using idle time on existing equipment, capital investment may be reduced.

If it is discovered that one item, when processed, only occupies a portion of a machine's capacity, such as a broaching machine, capital investment should be avoided by processing the part on existing drilling and reaming equipment or outsourcing the task. It's pointless. Purchasing and installing any equipment is of no use, if proper load is not there for the equipment.

5) Economical Use of Floor Area and Cubic Space:

In a factory, every square foot of floor area costs money in terms of both capital investment and care. This cost is significantly lowered by minimising the built-up area. Floor space is required not just by equipment and storage, but also by service departments and the mobility of man and material.

Unoccupied, wasteful, or idle floor space is a burden, whereas floor space occupied by equipment in operations and other important activities pays its own way. Stock containers on just one level, disorganised retail areas, huge distances between equipment or superfluous pathways, and barriers or partitions all indicate inefficient space use.

After examining all of the essential supplies for any activity as well as the mobility of man and material, proper architecture mandates minimal distance between two workstations or equipment. Furthermore, consideration should be given to the prospect of personnel or material moving in any of the three directions, therefore saving a significant amount of floor space.

6) Reduced Inventory in Process:

Despite the fact that this is the duty of the production planning department, a good layout has a significant impact on the inventory in process. When the architecture is designed so that material flows straight from one operation to the next, the material moves quicker, requiring less inventory in the process. If superfluous storage, waiting

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time, or material delays at workstations are eliminated, inventory levels can be minimised. Shortening the distances between workstations, machines, and departments can also help reduce inventory.

Shortening the distances between workstations, machines, and departments can also help reduce inventory. The inventory in the process is also minimised if the layout is structured such that back tracking of materials or components during the manufacturing cycle is kept to a minimum, if at all feasible. The items should be moved as quickly as possible using appropriate material handling equipment for the conditions.

7) Minimising of Material Handling:

It is true that the majority of production activity is material handling rather than processing, as is commonly assumed, and the pace of output at a plant is mostly influenced by the adequacy of its material handling facilities. The material's handling starts at the receiving point, where it comes as raw material, and then progresses through the process, travelling in the form of worked material from department to department, machine to machine, and operation to operation.

The working material is passed through inspection, assembly lines for sub assembly, final assembly, and packing as it flows in and out of temporary storage. The finished goods are stored at the shipping department, which then loads them for delivery to clients. Material handling at the plant, from receipt through shipment, costs a significant amount of money to management but adds little value to the output.

The amount of money and time spent on material handling at a factory is rarely disclosed. Material handling is estimated to account for 40-60% of overall production costs; as a result, one should endeavour to prevent material transportation and, if that is not possible, organise the layout such that handling is kept to a minimum, both in terms of distance and time.

If at all feasible, manual handling of materials should be avoided, and mechanised handling should be used wherever practicable. All motions should be coordinated with the goal of moving the part to the shipping location. While in transportation, the part should be "in process" as much as feasible, such as painting, backing, degreasing, and so on.

8) No Congestion and Confusion:

Unnecessary material movement or handling, as well as crossing transportation channels, produce confusion and congestion. Moving material directly and keeping it moving decreases direct and indirect costs, as well as almost all confusion and congestion. Provision of enough space for all activities and storage of material/parts required throughout the production process aids in cost reduction. It also aids in keeping the facility nice and tidy.

9) Safety, Satisfaction and Comfort to Workers:

The safety, contentment, and comfort of the workers are the most important considerations while designing a layout. Light, heat, ventilation, humidity, moisture removal, filth, dust, and other factors must all be considered in order to achieve these goals.

Excessive-noise-producing equipment should be separated as much as feasible or housed in a room with sound-absorbing walls and ceiling. Vibratory equipment should be positioned on pads to prevent vibrations from being transmitted to the floor or nearby items. Machines and auxiliary production equipment must be situated in such a way that they do not injure employees or cause material or equipment damage.

A detailed examination of work place arrangement, material handling procedures, storage strategies, ventilation, lighting, fire protection, and all other aspects involved in plant operations may be used to incorporate safety into the layout. When workers are happy and there are no dangers or fears of accidents, production rises, operational expenses fall, and employee morale rises.

10) Easier Adjustment to Changing Conditions:

Because of rapid advancements in science and technology, a factory or department that was designed to produce a specific quantity of a product may be forced to modify the product design, adopt a new method, or install new equipment. It may be required to improve manufacturing capacity, switch to different items, or eliminate a few products on occasion.

Plants that are unable to swiftly modify or restructure their production facilities in response to changing conditions (which are all too common these days) suffer significant setbacks and lose their markets. The required needs for such adjustments should be foreseen and allowances made in the early planning, so that facility rearrangement may be done without wasting a lot of time, causing a lot of trouble, and spending a lot of money in direct and indirect costs, or losing output.

11) Easier and Better Supervision:

The layout should allow supervisors to quickly monitor the job and therefore save time. This may be accomplished by lining up workstations in a straight line, eliminating the use of needless partition walls and strange angles. If the task is done on a single floor rather than many floors, supervision is still beneficial.

It may be tempting to include all of the objectives in the layout while planning it, but this may not be achievable. In reality, some of the objectives are virtually mutually exclusive; for example, one may desire flexibility while minimising floor space, or one may desire a mechanised material handling system while minimising capital expenditure, and so on. Regardless, each target represents a significant goal for the layout engineer to achieve.

When goals appear to be at odds in a given context, an equitable compromise must be made for the best possible results, taking into account the circumstances and relevance of each goal.

- Types of Layouts

As previously said, the plant architecture allows for the organised placement of machinery, equipment, and other physical amenities inside the manufacturing grounds. To build out an appropriate layout for new or existing factories, an entrepreneur must have the necessary competence. It differs from one facility to the next, from one area to the next, and from one industry to the next. However, the fundamental principles that determine plant layout are much the same. Small businesses, on the other hand,

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require a lesser area or space and may be situated in any type of structure as long as space is accessible and handy. The factory building and built-up area are inextricably associated with the plant layout for small businesses.

From the point of view of plant layout, we can classify small business or unit into three categories:

1. Manufacturing units
2. Traders
3. Service Establishments

1. Manufacturing Units

In case of manufacturing unit, plant layout may be of four types:

- (a) Product or line layout
- (b) Process or functional layout
- (c) Fixed position or location layout
- (d) Combined or group layout

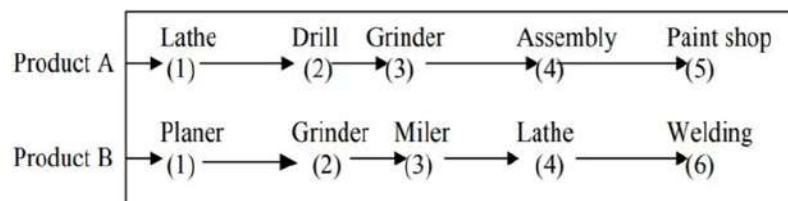
(a) Product or Line Layout:

Machines and equipment are grouped in a single line according to the product's needed sequence of processes. The supplies are transported from one workstation to the next in a straight line, with no backtracking or variation. Machines are placed together in this manner. As a result, raw materials are fed into the first machine, and finished goods are automatically transported from one machine to the next, with the output of one machine becoming the input of the next. For example, in a paper mill, bamboos are fed into one end of the machine, and paper emerges from the other. With minimal work in progress storage and material handling, raw material passes quickly from one workstation to the next.

The grouping of machines should be done keeping in mind the following general principles.

- a) All machine tools and other equipment must be put at the point specified by the operation sequence.
- b) There should be no locations where two lines intersect.
- c) Materials can be supplied wherever they're needed for assembly, but not all at once.
- d) The line must cover all processes, such as assembly, testing, and packing.

A line layout for two products is given below.



Advantages:

Product layout provides the following benefits:

- a) Due to the direct and short path and the lack of retracing, material handling costs are low.
- b) Operations are smooth and uninterrupted.
- c) Workflow that is constant
- d) Lower inventory and work-in-progress expenditures
- e) Optimal floor space use
- f) Faster output or shorter processing time
- g) Work in the process is less congested.
- h) Work inspection that is simple and effective, as well as production control that is streamlined.
- i) Lower per-unit manufacturing costs

Disadvantages:

Product layout suffers from following drawbacks:

- a) A large initial capital expenditure in a special-purpose equipment
- b) Exorbitant overhead costs
- c) A single machine failure will halt the entire manufacturing process.
- d) Limited flexibility due to the fact that the product was designed specifically for it.

Suitability:

Product layout is useful under following conditions:

- 1) Production of standardised items in large quantities.
- 2) A straightforward and repeatable production procedure.
- 3) The time it takes for different processes to run is more or less the same.
- 4) A reasonable level of product demand.
- 5) Consistent material supply.

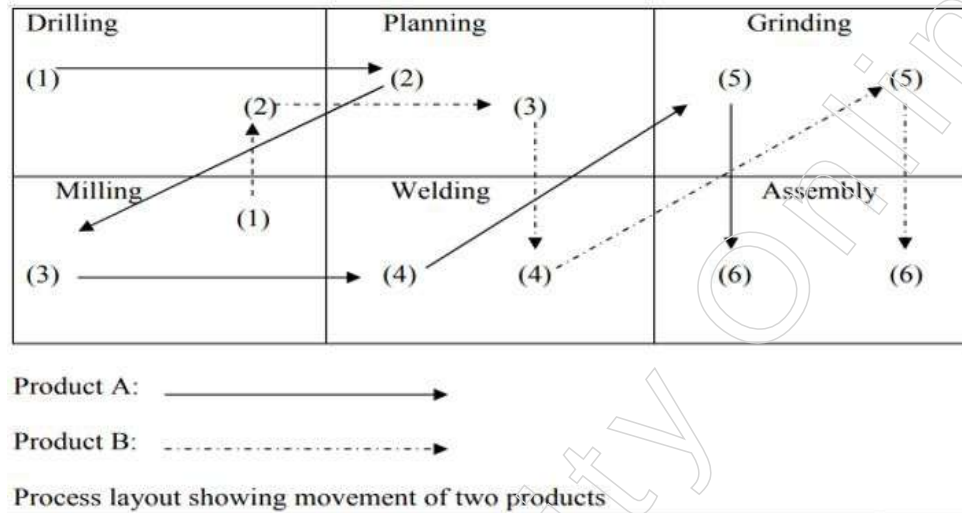
As a result, manufacturing units involving continuous manufacturing processes, producing few standardised products continuously on the firm's own specifications in anticipation of sales, such as chemicals, sugar, paper, rubber, refineries, cement, automobiles, food processing, and electronics, would prefer product layout.

(b) Process Layout:

In this form of architecture, related machines are grouped together in one location. Machines that do drilling activities, for example, are organised together in the drilling department, whereas machines that perform casting operations are grouped together in

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the casting department. As a result, the equipment is put in facilities that adhere to the process layout.



As a result, such layouts commonly include drilling, milling, welding, heating, and painting departments, among other things. From the beginning of time, the process or functional arrangement has been followed. It came about as a result of the handcraft technique of manufacturing. The work must be assigned to each department in such a way that no machines are picked to perform multiple tasks, i.e., the emphasis is on general purpose machines.

The work that has to be done is distributed among the machines according to loading schedules, with the goal of completely loading each machine. The following diagram depicts the process arrangement.

The following considerations must be kept in mind while grouping machines according to procedure.

- Departments should be as close together as possible to minimise long-distance material transfer.
- Departments should be in order of activities.
- The setup should allow for easy inspection and supervision.

Advantages:

Process layout provides the following benefits:

- Lower initial machine and equipment capital investment. Because a machine is not barred for a particular product, there is a high level of machine usage.
- Overhead expenses are cheap.
- Changes in output design and volume are easier to adapt to the production of a variety of items.
- A single machine failure does not need a total shutdown of operations.
- Supervision can be more specific and effective.

- f) There is more flexibility in terms of expansion options.

Disadvantages:

Product layout suffers from following drawbacks:

- a) Due to backtracking, material handling expenses are significant.
- b) More skilled labour is required, resulting in a higher cost.
- c) Production has a longer time gap or lag.
- d) The amount of work in progress inventory is considerable, necessitating more storage space.
- e) More regular inspection is required, resulting in expensive monitoring.

Suitability:

Process layout is adopted when:

- a) The products are not uniform.
- b) The quantity generated is little.
- c) Product design and style are always changing.
- d) Work is done at a job shop.
- e) Machines are quite costly.

As a result, process planning or functional layout is appropriate for job order manufacturing including non-repetitive processes, customer demands, and non-standardized items, such as tailoring, light and heavy engineering products, made-to-order furniture, and jewellery.

c) Fixed Position or Location Layout

The main product being produced is fixed at one location in this configuration. The labour and components for the equipment are relocated to that place. All of the equipment is brought in and set around a single workstation. This style of layout is unsuitable for small business owners. The diagram below depicts a fixed position configuration for shipbuilding.

Advantages:

Fixed position layout provides the following benefits:

- a) It reduces the amount of time and money spent moving work from one workstation to another.
- b) The layout is adaptable, allowing for changes in work design and operating sequence.
- c) It is more cost-effective to execute many orders in various levels of completion at the same time.
- d) By altering the order of activities, adjustments can be made to satisfy material shortages or labour absences.

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Disadvantages:

Fixed position layout has the following drawbacks:

- a) Due to the protracted production time, capital investment is high.
- b) Material and equipment storage near the product need a substantial amount of area.
- c) Because many processes are frequently required

Suitability:

The fixed position layout is followed in following conditions:

- a) Production of bulky and heavy goods such as locomotives, ships, boilers, generators, waggon construction, and aircraft production, among other things.
- b) Buildings, flyovers, and dams are being constructed.
- c) The patient is taken to the hospital, where the drugs, physicians, and nurses are delivered (product).

(d) Combined Layout

Certain manufacturing units, such as job shops, mass production shops, and representative process combination process [i.e., assorted shops], may require all three processes.

Only a product layout, process layout, or permanent location layout exists in the vast majority of industries. As a result, integrated layout is used in industrial operations when many items are produced in large quantities with no chance of continuous production. In actuality, a mix of the product and process layout or another combination is commonly observed, for example, in businesses combining part fabrication and assembly, fabrication tends to use the process plan, while assembly regions frequently use the product layout. The machinery for making soap is organised on a product line basis in a soap manufacturing facility, but supporting services such as heating, glycerine production, power generation, and water treatment are organised on a functional basis.

4. Traders

When two stores provide almost identical items, buyers choose to shop at the one that appeals to them the most. As a result, good layout, such as strong lighting, appealing colours, adequate ventilation, air conditioning, modern design and organisation, and even music, attracts and retains clients. All of this translates to better consumer convenience, attractiveness, and company volume.

Service, efficiency, and quality leave a lasting impression on the consumer. As a result, the layout is critical for handling products, which is organised according to the available space, the type and volume of items to be sold, and consumer comfort.

There are three kinds of layouts in retail operations today.

1. Self-service or modified self-service layout
2. Full-service layout

3. Special layouts

Self-service layouts save time for sales clerks and allow clients to choose their own items. Customers should be guided around a business in such a way that they see as much of the display space as possible, such as at grocery stores or department stores. The needs or convenience items should be positioned near the back of the store in those establishments. Colour and lighting are critical for drawing attention to interior exhibits and making the most of the store's layout.

Self-service does not apply to all processes. Specialty businesses, for example, sell to a smaller number of clients or sell a higher-priced product. Clothing, office machinery, recreational goods, fashion items, hardware, top quality shoes, jewellery, luggage and accessories, furniture and appliances are just a few examples of products that demand time and personal attention to sell. These full-service designs give the essential space and equipment in such situations.

Some layouts are solely dependent on the sort of specialty store to be established, such as TV repair shops, soft ice cream shops, and drive-in soft drink shops, to name a few. As a result, a successful retail plan should be one that saves money in terms of rent, time, and labour.

5. Services Centres and Establishment

Client convenience, quality of service, efficiency in delivering services, and a pleasant office environment are all factors that must be considered by service establishments such as motels, hotels, and restaurants. Clients expect ease of access to various departments of a service organisation in today's climate, therefore layout 106 should be created in a way that provides clients rapid and simple access to the services provided by a service business.

Factors Influencing Layout

A small-scale businessman should consider the following aspects while choosing on his factory, unit, facility, or store:

- a) **Factory Building:** The amount of floor space available for layout is determined by the nature and size of the structure. Special requirements, such as air conditioning, dust management, and humidity control, must be considered while planning.
- b) **Nature of Product:** Product layout is more suited to homogenous products, whereas process layout is better suited to custom-made items.
- c) **Production Process:** Product layout is superior in assembly line businesses. Process layout, on the other hand, is beneficial in work order or intermittent manufacturing.
- d) **Type of Machinery:** General-purpose machines are frequently organised by process, whereas special-purpose machines are organised by product.
- e) **Repairs and Maintenance:** machines should be positioned in such a way that sufficient space is available between them for the movement of equipment and workers needed to fix the machines.

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- f) Human Needs: Appropriate arrangements for cloakrooms, washrooms, lockers, drinking water, toilets, and other staff amenities should be created, as well as suitable provisions for the disposal of effluents, if necessary.
- g) Plant Environment: Heat, light, noise, ventilation, and other factors should all be taken into account; for example, paint shops and plating sections should be housed in a separate hall so that harmful vapours may be evacuated by sufficient ventilation. It is also necessary to provide adequate safety arrangements. As a result, the layout should promote employee health and safety. It should guarantee that persons and materials move freely and efficiently. When designing a manufacturing layout, future development and diversification should be taken into account.

Dynamics of Plant Layout

Plant layout is a dynamic rather than a static idea, which means that once completed, it cannot be improved or revised in the future without keeping up with the development of new machines or equipment, improvements in production processes, changes in materials handling devices, and so on. However, any layout adjustment should only be undertaken when the savings from the revision outweigh the expenses of the revision.

The following factors may necessitate a redesign of the plant layout:

- a) Increase the output of an existing product
- b) Introduce a new product and diversify the company's offerings
- c) Technological advances in machinery, materials, processes, product design, and fuel, among other things.
- d) Layout flaws that were overlooked by the layout engineer at the start.

Applicability of Plant Layout

Plant layout may be used in a variety of sectors or plants. Certain plants need unique arrangements, which, when used, distinguish the layout from the varieties already mentioned. Plant layout's applicability in manufacturing and service sectors is examined further down.

In the instance of detergent powder production, a multi-story structure is created specifically to house the boiler. On separate floors, materials are kept and injected into the boiler at different stages. Other amenities are available at several points around the boiler.

The production of talcum powder is another use for this architecture. The machinery is placed vertically, from top to bottom, in this case. As a result, material is put into the first machine on the top level, and powder is discharged from the bottom of the ground-floor machinery.

Another application of this style is in the newspaper industry, where the time aspect is crucial, with the completion measured in seconds. To avoid distance, delay, and confusion, the plant layout must be basic and direct. Because materials must never fail, flawless coordination of all departments and machinery or equipment is required.

Plant layout may also be used to five-star hotels. Lodging, a bar, a restaurant, a kitchen, stores, a swimming pool, a laundry, shaving salons, shopping arcades, a conference centre, and parking places, among other things, should all have a place in the design. Cleanliness, elegance, convenience, and compact appearances are all important factors in attracting clients.

A similar plant architecture may be used to a movie theatre, where the focus is on the comfort and convenience of the audience. The projector, screen, sound system, firefighting apparatus, and atmosphere, among other things, should be of the highest significance.

A plant layout includes not only the grouping of machines, but also the placement of additional amenities. Receiving and dispatching stations, inspection facilities, staff amenities, and storage are examples of such facilities.

Receiving and dispatching departments should, in general, be located at opposite ends of the facility. To save handling expenses, the storeroom should be close to the manufacturing, receiving, and dispatching centres. Inspections should be done last, before dispatch, therefore they should be immediately adjacent to other dispatch departments.

The maintenance department, which includes lighting, safety devices, fire protection, rubbish collection and disposal, scrap, and other items, should be placed in a convenient location for all of the plant's other departments. Other employee amenities, such as restrooms, drinking water, a first-aid room, and a cafeteria, may be a bit more away from other departments, but they should be within easy reach of the employees. As a result, there are various industries or factories where plant layout can be used.

3.2.1 Principles of Facility Layout

1. Principle of flexibility
2. Principle of flow
3. Principle of safety
4. Principle of economy
5. Principle of optimum utilization of floor space
6. Principle of comfort and convenience

Plant layout plays an important role in factory organization and planning.

Hence, at the time of designing layout of the plant the following principles should be kept in mind:

1. Principles of Flexibility – Plant layout should be adaptable in nature. It should be able to incorporate new equipment, for example, without causing large changes to the facility. In truth, it's a terrible plant layout that doesn't allow for any changes.
2. Principle of Flow – A good plant structure should result in a smooth flow. As a result, the structure of the plant should aid in the regular and unhindered

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movement of production. There should be no bottlenecks in the process, and backtracking should be kept to a minimum.

3. Principle of Safety - The layout of a plant should assure the workers' safety. It should guarantee that there are no dangers or accidents.
4. Principle of Economy- Plant layout should be cost-effective. It should include a lower amount of money spent on the construction and upkeep of the company's building.
5. Principle of Optimum Utilization of Floor Space—A good plant architecture should provide the most efficient and effective use of available floor space.
6. Principle of Comfort and Convenience – On a plant layout, correct arrangement for comfort and convenience should be established. Worker working conditions should be decent. Water, restrooms, toilets, power, and other necessities should all be available.

Techniques of Plant Layout

1. Process flow chart
2. Process flow diagram
3. Template layout
4. Model layout
5. Machine Data card

Technique of plant layout is referred as tool of the plant layout. It is also known as visual aid to plant layout. It is used for depicting the internal set-up of industrial unit.

1. Process Flow Chart - A process flow chart is a map that is displayed as a graph. As a result, it's known as a plant layout visual map. It graphically depicts the sequence of the cycle elements involved in product manufacture. It summarises all of the tasks and operations that must be completed in a plant from start to finish.
2. Process Flow Diagram – Process flow Diagram is presented in the form of a floor plan sketch of the buildings. It comprises information on the location of machines, internal transportation, the space occupied by each machine, and several other operations related to product manufacture. It so aids in visualising the whole flow of materials, hence assisting in the identification of flaws and the improvement of the plan.
3. Template Layout – The use of templates in plant planning is a common practise. Thick paper is utilised in this technique. It's made by cutting out multiple templates to size from this thick paper. The space covered by machines, racks, furniture, equipment, and components is depicted.
4. Model Layout - Another name for this pattern is model layout. Three-dimensional models of equipment, furnishings, and other items are created for such a plan. These models are so simple to look at that even a layperson can get a sense of how the plant is laid up.
5. Machine Data Card – This is one of the plant layout strategies. Several cards are prepared when this procedure is utilised. These cards provide all of the important

information. This knowledge mostly pertains to the unique properties of machines, among other things. The card is connected to the appropriate machinery and equipment.

Objectives of Plant Layout

- a) Minimize material handling and transportation. This will result in more cost-effective material management.
- b) To make the best use of persons, space, and machines.
- c) To reduce the amount of work-in-progress and increase inventory turnover. To have a low level of inventory, the material should move quickly through the plant and areas of congestion should be minimised.
- d) To establish sufficient and conveniently situated service centres.
- e) To guarantee that equipment and processes are planned and controlled efficiently.
- f) To lessen the likelihood of production system delays and bottlenecks. This will result in a smooth work flow.
- g) To guarantee that the layout is adaptable to changing demands.
- h) To guarantee effective worker supervision.
- i) To improve staff morale by ensuring that they are comfortable and satisfied.

3.2.2 Process Layout

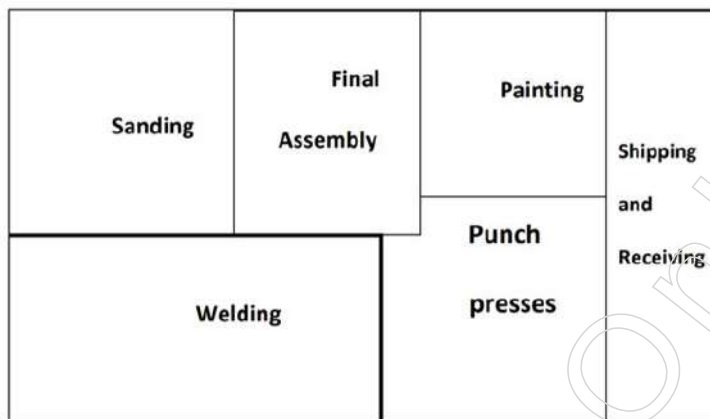
Fixed position layout or product layout are the other two main alternatives for building out facilities to create things or supply services, and process (or functional) layout is one of them. The CELL LAYOUT, the fourth option, is a hybrid facility architecture that incorporates some of the ideas of fixed position and product layouts.

All related production processes are placed together in the same department or region, according to the phrase “process layout.” This method of facility layout can be used for component manufacture or assembly. Different manufacturing methods such as milling, drilling, turning, grinding, plastic moulding, and so on may be used in component manufacture. A process layout in assembly could have distinct locations for creating different subassemblies, final assembly, testing, packaging, and other tasks.

Process layouts are most commonly used in batch operations, where batches of parts (or, in the case of services, customers) are routed from one process area to another, where a single production operation, or a limited number of operations, is performed. In service provision, examples are more difficult to find, although in retailing, the arrangement of stores on a high street may be called a process layout because they all offer the same things (bread, vegetables, hardware, etc.). The relative advantages and drawbacks of process architectures are a source of dispute.

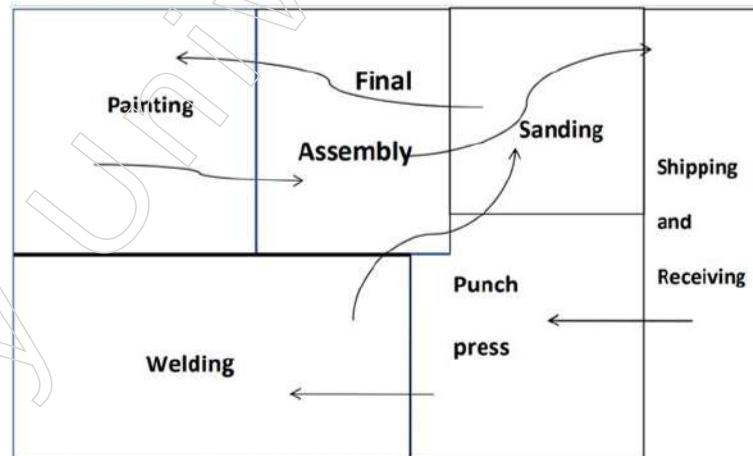
— A process layout groups together similar devices in one part of the facility. Products are transported to the next step once procedures are completed. Each of the regions depicted in the facility diagram below indicates a cluster of similar machines or operations.

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It is required to assess the flow of materials or work in progress from one process to the next in order to enhance the layout of a facility. This flow study may reveal that the layout is faulty, resulting in inefficient human utilisation and an excessive number of material and work-in-process movement. Take a look at the illustration below. This example depicts a process arrangement that is comparable but not as efficient. Examine how materials and work in progress flow through the facility. This research indicates that the arrangement may not be as efficient as it may be, necessitating more material handling and transportation than is required or desirable.

Work in progress is moving across several process sectors in this arrangement. This isn't a very efficient design. The purpose of a process layout analysis is to discover the best plan that meets the layout design objectives.

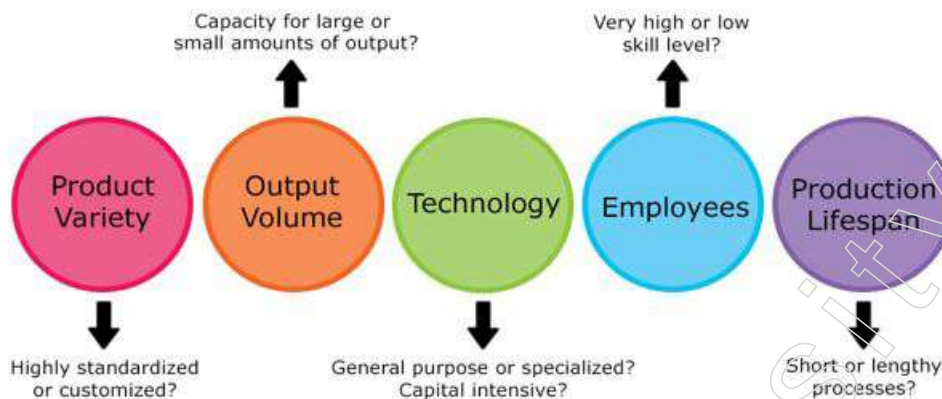


The timing, synchronisation, and balance of different steps in the process are the primary concerns of the analysis involved in the design of production lines and assembly lines. Because of the high quantity of transportation and handling required, the relative organisation of departments and equipment is a crucial component in process design.

Every company that makes a product or provides a service employs a process. This process will make use of the company's resources to convert the major inputs into some sort of output. Several essential criteria must be addressed while creating the actual process, notably the number and sequencing of phases.

1. Variety of product - Is the product highly standardised or highly customised?

2. Output volume – Is the company designed to generate vast quantities or a limited amount of output?
3. Will the technology be utilised for broad or specialised purposes? Is it a capital-intensive business?
4. Is the staff skill level very high or very low?
5. How long do you think it will take?



Make-to-order and Make-to-stock

It is useful to categorize processes as either make-to-order or make-to-stock.

The customer's order is not made until it is received in a make-to-order firm. This enables for customisation to the customer's exact needs. It's also known as "build-to-order" construction. This kind of manufacturing is known as a pull system. When there is a demand for the job, it is "pulled" through the process.

The drawback of this technique is that it takes time for the company to get the necessary materials and components, as well as arrange and create the customer's order. Because the goods are created in tiny quantities, they may be more expensive.

The advantage of this method is that inventory is kept to a minimum, as opposed to a traditional make-to-stock technique. There is no doubt about what the consumer wants, and there is no stock that has to be rid of. Dell Computer has had a lot of success with this sort of technology when it comes to producing personal computers.

A make-to-stock method involves producing items in advance of client demand, generally based on a sales prediction. These items are typically produced in bigger quantities and stored in anticipation of consumer requests. Although greater production quantities may result in cheaper unit costs, there may be losses owing to prediction mistake, excess inventory, obsolescence, and theft.

- Process Types

Project

A project is a one-time occurrence, such as the construction of an apartment building, the deployment of a new ERP system, or the authoring of a book. Each of these initiatives is highly customised, makes extensive use of resources, and involves a complicated set of tasks. At the end of the project, there is only one output.

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Job Shop

A job shop procedure is used by many firms. When the product being made is unique for each buyer, this is the most typical method. It's a make-to-order company that produces on a sporadic basis (i.e., rather than one entire product being completed at a time, work will continue on multiple products as time permits). Frequently, each customer's product has distinct qualities. Workers in this sort of firm are extremely knowledgeable about their profession or trade. They are frequently referred to as "craftsmen" or "makers." In a work shop, the amount of output is low.

The equipment that was utilised was rather common in nature. A tiny bakery that makes exquisite bespoke wedding cakes, or a company that builds custom guitars or bicycles depending on the clients' measurements and preferences for materials and components, are two examples.



Batch

Some firms are in the habit of producing large batches of similar items on a regular basis. A batch is the term for these groups. From beginning to completion, the batch will go through a series of phases that must be performed. Multiple batches may be coming through the process at different phases for an organisation. This sort of processing is likewise only done on a sporadic basis. (Begin, stop, begin) When opposed to a work shop, this sort of business offers less variation, and the equipment employed will be more general-purpose and tailored to the industry. Employees must have the necessary skills and expertise to operate the equipment and produce the items. Baked goods, aeroplane components, apparel, and vaccinations are all examples of

batch manufacturing items. The size of the batch is a crucial consideration for these businesses.



Repetitive

This sort of company creates items that are more uniform in character. The production is usually rather high. Due to the standardised nature of the items, the equipment utilised is usually fairly specialised and frequently highly tailored for that procedure. Because the stages are extremely standardised, the employees' skill level is generally low. Although these occupations don't always necessitate a trade or a lot of expertise, they do necessitate abilities like multitasking, focus, problem solving, and cooperation. These procedures frequently include flexible automation that enables modification, such as the inclusion of improved functionality. Examples of a repetitive process include assembly lines such as assembling automobiles or electronics, a carwash, or a cafeteria line.



Continuous

When a large volume of standardised product is produced, it is referred to as a continuous process. The product being manufactured is classified as non-discrete. This indicates that these companies don't make distinct items, but rather a product that is frequently a liquid or a product like sugar, steel or fuel. An oil refinery is an example of this sort of procedure. The product does not flow from one phase to the next inside the system, hence there are no separate individual workstations. In this sort of procedure,

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the equipment is extremely complicated and was created specifically for that product at that location. Except for those in charge of process monitoring, maintenance, and cleaning, there are relatively few employees.



Hybrids

Many businesses use a combination of process types. The Mass Customization model of production is one such common exception. When a corporation uses mass customisation, it combines low-cost, high-volume manufacturing, yet each client order is personalised to the customer's needs. This personalization is usually made possible by the use of computer-aided manufacturing techniques. Furniture designers, for example, may wait to construct the precise model of couch based on the customer's measurements and fabric preference, or a car manufacturer may provide dozens of modification packages and colour options, making each vehicle unique to the buyer. A modular design is a must for effective mass customisation since it allows for quick and smooth transitions from one product to the next.



- 5 Key Objectives of Process Design

Process design's overarching purpose is to ensure consistency in performance throughout a given business operation. Brainstorming and capturing the many phases in a process is a powerful approach to plan it out, maintain consistency, and consolidate outcomes.

However, there are a few more goals associated with process design that can help you succeed in every project or assignment you do.

Here are five that are worth pointing out.

1) Normalize Outcomes

Process design helps you identify particular actions that are critical to achieving a certain objective, which normalises workflow outputs. The design process, with its organised approach, involves extensive investigation into each new technique, resulting in a simplified, stable, and enhanced workflow. Process design, for example, can be used to control outcomes in the manufacturing of products, the delivery of services, or the shipment of items.

2) Encourage Efficiency

Process design allows you to include and increase efficiency in a variety of company processes. The less time it takes for teams or individuals to complete a task, the more time they will save. Process design also allows you to get the most out of your resources, lowering your expenses.

Process design, for example, helps avoid process-dependent projects from stalling since it guarantees that enough time, money, employees, and materials are available for each activity you frequently participate in.

3) Avoid Chaos

Certain procedures, especially in bigger firms, may be split among numerous teams or departments. A disconnected approach to day-to-day operations is avoided by systematising a recurrent set of tasks through process design. Example: Process design is used by many businesses to develop best practises and standardise training programmes.

4) Ensure Accountability

Building in accountability checkpoints at various phases of a process, whether internal or external, helps maintain a particular degree of quality. More professional interactions arise from process design, which boosts client confidence. Example: Use business process mapping to improve your process architecture, identify who is accountable for what, and define workflow norms.

5) Streamline Collaboration

One of the most crucial – and frequently underestimated – goals of process design is to promote improved communication and cooperation. Visually laying out a new process makes it much easier for everyone involved to participate to its design, put it to use, and teach it to others as necessary.

Example: Using a flowchart to develop and record new work processes encourages idea sharing, issue resolution, and openness in communications with customers and other stakeholders.

Remember, the easier it is for you and the people with whom you work to recognise the issue that needs to be solved, the question that needs to be answered, or the goal that needs to be achieved, the easier it will be to develop a process that consistently produces excellent outcomes. Using a visual approach to process design will not only aid in defining requirements and evaluating execution, but it will also aid in determining the best approach to any new business process you establish.

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- Process Flowcharting

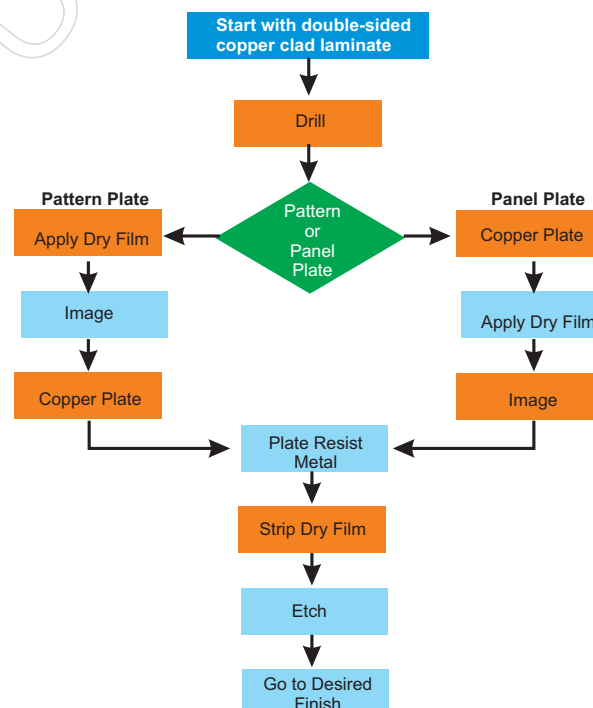
Any endeavour to enhance a process will always start with a visual depiction of the current process. This is vital to ensure that everyone on the team understands how the present process works. All phases and flows must be recognised and set down in the correct order. It's critical that the right people are involved in this activity!

Flowcharts come in a variety of shapes and sizes, and there are several software programmes that may help with this task. Rather of utilising a computer projected onto a screen, most practitioners choose to use a huge roll of paper on the wall where the group may participate. Before starting this assignment, go out into the workplace and walk through the procedure.

This is a pretty common practise that takes place in businesses all over the world. As an operations manager, it's quite likely that you'll be involved with this at some time in your career. A standard set of symbols is used to map a process. Because there are so many distinct symbols, it's important not to get too wrapped up in them all. The following are examples of standard symbols:

Symbol	Meaning
	Start / Stop
	Operation
	Decision
	Storage
	Flow

For Example:



- **5 Popular Process Design Tools**

Many technologies are available to assist you in streamlining design and process documentation. Examples include the simple but restricted (think paper forms or whiteboards), the more advanced but laborious (think text documents or spreadsheets), and the genuinely dynamic (think interactive whiteboards) (like digital flowcharts and visual diagrams).

As a minimum, all of the process design tools listed here will assist you in creating and recording workflow procedures.

The more complex choices, on the other hand, will allow you to effortlessly communicate, oversee, and enhance your operations by allowing you to:

- Assist with company-wide communication, cooperation, and integration of work platforms.
- Use a shared visual environment to track deliverables, due dates, resources, and dependencies.
- Use visualisation of priorities and filtering of different criteria to test multiple scenarios.

For each process design tool listed below, we've provided a brief process documentation example.

6) Microsoft Tools

Did you know that Microsoft Word, PowerPoint, and Excel can all be used to design simple process workflows?

They're perfect for collecting, storing, and in certain circumstances analysing process data because of their flat, linear shape. They're especially excellent for working on a new process design on your own time, with a tool you're likely already familiar with.

However, Microsoft technologies are fairly restricted when it comes to understanding your process needs in a bigger perspective – or drawing key connections between different areas of information.

Workflow Example: Simply enter the information for that new sales process you've been experimenting with into Excel's flowchart template and export it to create shapes and connectors automatically (or use MindManager's advanced integration features to quickly and easily import all your process data directly from your spreadsheets).

Watch this webinar to learn more about MindManager's powerful Microsoft connections.

7) Digital Flowcharts

The most extensively used process design tools are digital flowcharts. Flowcharts are a simple method to graphically lay out the phases of your process, with no text or spreadsheets to worry about. They use:

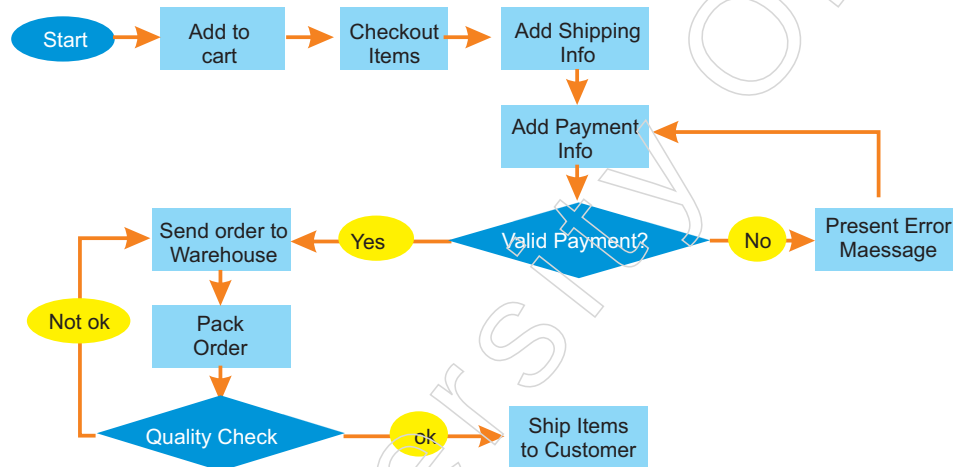
- A set start and finish point
- Symbols to symbolise events or activities (usually circles, squares, and diamonds), and

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- Workflow and task linkages are shown by directional arrows.

Flowcharts are very useful for designing and describing processes without the use of jargon or technical phrases that might be confusing.

Workflow Example: Digital flowcharts are perfect for optimising and documenting processes of all sorts, including procedures that must meet regulatory standards, because they guarantee that everyone on your team follows best practises (like your various accounting practices, for example). A simplified flowchart illustrating a common eCommerce workflow process is shown below.



8) Visual Workflow Diagrams

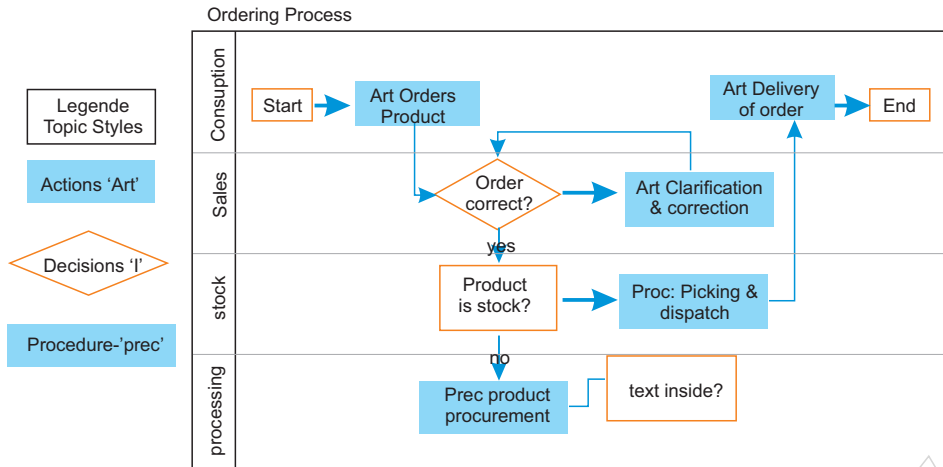
Visual process diagrams exist in all kinds and sizes, from swim lanes to tree diagrams, because workflow is the foundation of knowing exactly what your organisation does and how it does it.

Workflow diagrams may be used to: With a variety of shapes representing different jobs and a variety of branch connection configurations to pick from, you can use them to:

- Determine where process steps are not being adequately handled or where parallel procedures are taking too long to finish.
- Create the best combination of manual and automated jobs.
- Implement process controls to ensure that the process is carried out correctly.

Interactive workflow diagrams allow you to map out complicated information so that teams and individuals may simply, efficiently, and consistently follow prescribed procedures.

Workflow Example: Visual workflow diagrams may assist boost customer satisfaction with your product or service by offering your support team a step-by-step troubleshooting method since they resemble the way our minds take in and evaluate information. The flowchart and swimlane tools in MindManager were used to generate this process diagram.



9) Process Maps

Process maps are similar in design to flowcharts and workflow diagrams and are commonly used to envision and plan corporate operations. They exist in a variety of designs that serve different objectives. Idea maps, for example, are useful tools for designing new processes because they allow participants to pull broad thoughts and concepts out of their minds and arrange them using colours and graphics in a progressive manner. No more misplacing notes or sifting through pages of scribbles.

Strategy maps, on the other hand, are perfect for researching, documenting, and controlling strategic planning processes in order to better achieve your objectives. They may assist you in fully illustrating a process utilising a hierarchical approach that organises all of your team's thoughts on one page.

Workflow Example: To more correctly represent complicated processes, use MindManager's process maps in conjunction with integrated timeline diagrams, budgeting, and forecasting capabilities. You won't have to guess at the chronological sequence of events or estimate their worth in terms of cost and return on investment if you need to conduct a combined marketing campaign across departments, for example. A screenshot of a simple strategic planning template in MindManager is shown below to assist you in designing a new process map for your team.



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10) Process Documentation Software

Process documentation software makes it simple to visualise and collaborate across shared business procedures by combining all of the benefits of the process design tools outlined above under one roof.

Software like MindManager, for example, acts as a virtual whiteboard, allowing groups and individuals to easily document their understanding of a process.

You may graphically collect information and thoughts, then organise and contextualise them by:

- Dragging events around
- Connecting them, and
- Including notes, links, files, and other resources that are relevant to the topic.

A variety of flowchart and swim lane diagram templates provide a starting point for easily and dynamically generating, changing, and reorganising your business processes. These versatile mapping choices can be used as stand-alone process design tools or combined for process method documentation.

You may also use auxiliary lines, grid alignment, and item groups to greatly speed up your documentation by creating and integrating your own process design forms.

As a process design tool, MindManager may be used for a variety of tasks:

- As a planning tool, MindManager assist you to define your requirements and objectives for the process you need to design.
- As an analysis tool, MindManager makes it easy for your team to play out and evaluate alternative procedures.
- As an improvement tool, MindManager lets you flexibly integrate new ideas to optimize an established workflow.

Visual clarity and the freedom to select the best process design tool aren't just advantageous when it comes to unlocking a fantastic user experience; they're also critical when it comes to assisting new employees or remote workers in becoming comfortable with established processes.

Process documentation software overcomes problems like data gaps, misunderstanding, and task redundancy by combining all information relevant to a given process from across people, locations, and platforms onto a cohesive dashboard. You'll also be able to utilise all of the most common process design tools, as well as many various forms of process design, as a MindManager user.

3.2.3 Product or Line Layout

The arrangement of machines in a line (not always straight) or in a sequence in which they would be employed in the manufacturing process of a product is known as product or line layout. This sort of arrangement is best suited for continuous industries where raw materials are fed at one end and completed products are removed at the other. A separate production line will need to be maintained for each type of product.

Product layout is suitable where:

- (i) a significant number of standardised products are created;
- (ii) the standardised products are to be processed repeatedly or constantly on the supplied production facilities;
- (iii) the volume of goods processed must be adequate to keep the production line actively engaged;

In enterprises that employ assembly lines or production lines, product (line) layouts are used. An assembly line may be the ideal solution if the product has a high volume. The equipment in these sorts of layouts is frequently quite capital-intensive, and it is planned out in accordance with the process's progressive phases. Each work station is positioned along the line and may include a person and equipment, as well as robots. Adding components (assembly line) or altering a product is common at each work station (production line). It's crucial to notice that it's not always a straight line; many assembly lines zigzag or are shaped to make the most of the available space. Some services, such as making hospital meals or a cafeteria line, may employ a line arrangement. Due of the high expense of setting up an assembly line, a huge quantity of goods must be produced. A continuous and consistent demand is desirable.

The commodities produced in a line plan are often relatively uniform, as are the labour procedures. A fixed route material handling system, such as a conveyor belt, is employed because each product follows the same set of steps. It's possible that this conveyor belt is either manual or automated. It might run at a set speed or at the pace of the workers. It can operate constantly or intermittently. The amount of product produced every shift will be determined by the conveyor's speed.

In contrast to a process layout a product layout is very efficient. There are a number of reasons for this.

1. There is less variability in the work performed due to the division of labour and repetition.
2. There is no inventory build-up and no waiting. When a project is finished at one workstation, it is immediately moved to the next. The system only keeps track of inventory that is currently being processed. When goods are finished, they are usually sent.
3. Goods travel fast and not too far thanks to the material handling system.
4. Because changeovers aren't required, switching between items takes extremely little time.

It's critical to have a well-balanced assembly line. The amount of time spent at one work station should be roughly equivalent to the amount of time spent at the next work station.

Challenges in a product (line) layout include:

1. Because the line may be vulnerable to stop downs due to equipment failures, preventative maintenance is essential. Preventative maintenance include inspecting and replacing any parts with a high likelihood of failure, as well as keeping enough of spare parts on hand and following a specific maintenance

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schedule for each piece of equipment.

2. Employee training and job rotation are important actions to ensure that workers are competent of executing work responsibilities and that each position can be filled by numerous persons.
3. When it comes to repeated standardised duties, appropriate ergonomic work design is crucial. Organizations that ensure the health, safety, and well-being of people.

Cellular Layout

Cellular layouts are referred to as “hybrid” layouts, because it has elements of both a Process and a Product (Line) structure. It's very uncommon for a company to have many product lines with much too much diversity to fit on a single manufacturing line. Frequently, these enterprises used a process layout, which included all of the product flow and waiting periods. The Cellular layout type is an alternative that gained popularity in the late 1980s. Today, this manufacturing layout is still widely used.

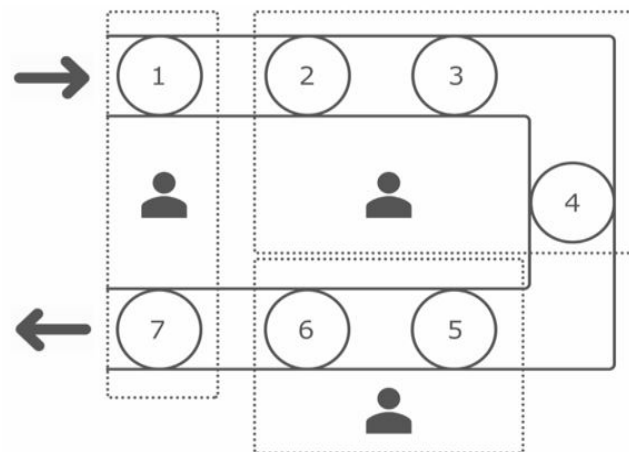
This layout starts with a comprehensive investigation of the company's goods and determining which items are comparable to one another in terms of geometry and processing needs, as well as equipment, machinery, technology, and personnel abilities. In a work cell, these goods are clustered together and manufactured. Group technology is the term for this.

In an assembly line structure, each work cell will have its own set of equipment to create this family of items. The equipment is organised in a U form, with equipment near together so that jobs do not have to travel long distances.

Advantages of a cellular layout include:

1. Shorter set-up periods for each piece of equipment Because each machine produces fairly identical items, set-up times are generally minimal or non-existent.
2. Speed is substantially improved since tiny batches may now be used, and commodities entering the system will continue to be processed until they are finished. Small batches offer shorter wait times and faster run times.
3. Inventory investment has been decreased as a result of the small batch sizes afforded by the quick set up times.
4. Employees only work on a limited variety of items within that cell, which improves quality. Employees are cross-trained to ensure that they have a complete understanding of the whole manufacturing process.
5. Employee morale improves as a result of working as part of a team that is responsible for the cell's throughput and quality. The U-shaped form encourages workers to collaborate more.
6. Because machines are close together and product movement is minimal, less floor space is required.

An example of a U-shaped layout can be found below:



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Other Layouts Include

- 1) Office Layout: Office spaces in 2020 are vastly different from those of previous generations. The amount of floor area per employee has decreased dramatically. The flow of paperwork is significantly less of a concern than in the past. Employees are frequently divided into groups based on the jobs they do and the work teams in which they participate. Workspaces are becoming more flexible, with less paper and furniture used to store files and data. Many businesses place a higher value on having collaborative areas that are comfortable. The layouts are considerably more open concept, with lower walls to increase workstation visibility.
- 2) Retail Layout: When designing a retail space, the ultimate objective is to maximise the number of sales per square foot in the space. This is accomplished by carefully analysing store traffic patterns in order to optimise the quantity of merchandise to which each client is exposed. As a result, the milk is frequently located at the store's far end, forcing shoppers to trek through all other departments to get to it.
- 3) Warehouse Layout: The goal of a good warehouse plan is to make the most of the entire amount of space available. The link between incoming products reception, storage space, and outgoing goods selecting, packaging, and shipment is thoroughly examined. The location of inventory items is crucial in order to reduce the distance that commodities and staff must travel. Many warehouses have unique storage needs, such as freezers, cold storage, and high-security zones.

3.2.4 Fixed Position Layout

Fixed position layout is one of the four basic layout types, the other three layout categories are functional layout (also known as process layout), cell layout, and line layout (also known as product layout) (Slack, Brandon-Jones, and Johnston, 2011).

The changed resources stay (more or less) stable in a fixed position arrangement, whereas the transforming resources (equipment, machinery, plant, and people) who execute the processing move as needed (Groover, 2007). Many times, the changed

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resource remains in one location because it is either too huge to transfer easily, too fragile to transport, or may protest to being moved.

For example, the transformed resource in shipbuilding or highway construction is too enormous to move; the patients in open-heart surgery are too fragile to transfer; and the client in a high-end restaurant would object to being relocated to where the food is made. Furthermore, the volume and diversity of the operation will impact the decision to choose a fixed-position arrangement.

When the volume is low and the variety is high, “flow” isn’t a big deal. Because each product is unique and goods “flow” through the operation seldom, a fixed-position arrangement is likely to be useful in telecommunications satellite manufacturing. As a result, arranging facilities to reduce the flow of components through the operation is pointless.

As a result, fixed position layouts are common in project, jobbing, and, in certain situations, batch manufacturing. In certain cases, there is a choice of layout type, and the fixed position option is chosen according to the relative benefits it provides. For example, in sailboat assembly, a fixed position arrangement combined with group work is sometimes used since it allows for greater flexibility in producing a wide range of completed goods. Furthermore, workforce multiskilling and increased autonomy, as well as a focus on the full product rather than a tiny element of it, can give motivation to enhance quality and labour efficiency.

Fixed position layouts, on the other hand, have a variety of drawbacks. First, as compared to alternative layout possibilities, this type of style tends to result in somewhat high unit prices. Second, space and activity scheduling might be tricky. Finally, fixed position layouts cause extensive mobility of changing resources such as personnel, machinery, equipment, and plants.

Advantages Offered by Fixed Position Layout:

- (i) There is a reduction in material mobility.
- (ii) Capital investment is kept to a minimum.
- (iii) Because the work is normally completed by a group of operators, operations are guaranteed to run smoothly.
- (iv) Production centres are autonomous from one another. As a result, effective planning and loading may be accomplished. As a result, the total cost of manufacturing will be lowered.
- (v) It gives for more flexibility in terms of product design, product mix, and manufacturing volume.

Limitations of Fixed Position Layout:

- (i) Highly trained personnel are necessary.
- (ii) Moving machinery and equipment to the manufacturing centre may take some time.
- (iii) For work and tool placement, complicated fixtures may be necessary. The cost of production may rise as a result of this.

3.2.5 Concepts of Line Balancing

Line balancing is a flow-oriented production approach for increasing productivity and lowering costs in large-scale manufacturing. For the creation of a specific product, an appropriate time period is set. Tasks are then allocated evenly across workers and workstations to guarantee that each operation in the line is completed within the time period set.

Line balancing is a production approach that includes matching the production rate to the takt time by balancing operator and machine time.

Takt time can be defined as the rate at which parts or products must be produced in order to meet customer demand.

If production time is precisely equal to takt time for a certain manufacturing line, the line is perfectly balanced. Otherwise, bottlenecks and surplus capacity should be removed by reallocating or rearranging resources. To put it another way, the number of personnel and machines allocated to each task in the line should be rebalanced to achieve the highest possible production rate.

Criteria In Line Balancing

There are some criteria which should be considered in a line balancing process.

1) Takt Time

Takt time, according to Heizer and Render (2010), is a required technique while doing line balancing tasks. Takt time is the rate at which production is coordinated with consumer demand. It demonstrates how quickly products must be manufactured in order to meet client demands. (2006, Vome lean Briefs) Producing faster than the takt time results in waste, but producing slower than the takt time leads in bottlenecks, where client orders may not be met in a timely manner.

$$\text{Takt Time} = \frac{\text{Available time per day}}{\text{Customer demand per day}}$$

Simply said, takt time is the average rate at which buyers purchase items, and hence the rate at which products should be made. It is measured in seconds. Takt Time is defined as available work time split by daily client demand. It may increase productivity by making processes easier to control and ensuring that the output of each process meets consumer demand.

The following are some of the benefits of employing takt time (Heizer and Render, 2010):

- a) Achieve a consistent and continuous flow of production.
- b) Eliminate overproduction waste by meeting real customer demand.
- c) Enhances planning precision.
- d) Encourage the creation of standardised work instructions, which will improve quality and efficiency.
- e) Establish real-time production objectives that indicate operators where their work output should be at any given time.

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- f) Create a what-if scenario based on flexible manpower for client demand.

Takt time, on the other hand, has its own limitations. Takt time is suitable for basic cells, but not for work shops or other low-volume, high-variability tasks. If designers simply consider takt time, they may decide that certain goods cannot be manufactured utilising cellular technology, and their firms may lose out on the benefits of doing so. It can also lead to inefficient process designs.

Takt time is usually documented on video since it allows for a visual record of the entire exercise. It's also a reliable way to keep track of things.

(Heizer and Render, 2010) A modern way to establishing method:

- a) Capture a representative sample of the process.
- b) With the operators present, go through the video.
- c) Break down the 'components' of work and time each one separately.
- d) Determine which elements are value-added and which are not.

2) Cycle Time

Gaither and Fraizer (2001) define cycle time as the number of times a production line can create a product given its present resources and workforce. It's a precise representation of how the line is now set up to run. The total production quantities are taken into account while calculating cycle time. The composite cycle time is smaller than the actual lapse time of any particular line when numerous lines are generating the same output.

In other terms, the estimated average total production time per unit produced is known as cycle time. Each operator on an assembly line or in a work cell with numerous operators will have his or her own time to do the task. Normally, we focus on the individual cycle period that is the longest. To shorten a line's cycle time, all we have to do is concentrate on the operation that determines the speed.

Takt time and cycle time are not the same thing. Takt time refers to the maximum amount of time an operator has to satisfy a customer's demand, whereas cycle time refers to the amount of time it takes an operator to complete one cycle of his operation. Customer demand determines both takt time and cycle time.

$$\text{Cycle Time} = \frac{\text{Production time per day}}{\text{Required output per day (unit)}}$$

By eliminating the non-value activities through continuous improvement and waste reduction, the cycle time can be reduced.

$$\text{Cycle time} = \text{value added activities} + \text{non-value-added activities.}$$

3) Downtime

Downtime can be defined as the time that is non value added. (Chase, et al. 2000). It is often related with the 7 wastes as below:

a) Defects

A company's direct expenses are defects. Defects, according to the Toyota philosophy, should be viewed as chances to improve rather than as a trade-off for a less-than-optimal arrangement.

b) Overproduction

One of the major wastes obstructs the smooth flow of products and services, perhaps resulting in unnecessary lead and storage times. As a result, faults will not be noticed sooner, and the items will degrade. It will also result in an overabundance of work-in-progress inventories.

c) Waiting

When the items aren't moving or being worked on, this happens. It impacts both commodities and people, since the time spent waiting may be better spent on value-added activities like training and maintenance.

d) Transportation

In a manufacturing, any movement might be considered waste. The distance between processes is related to the time it takes to feedback reports of poor quality and take remedial action, therefore double handling and excessive movements are likely to cause damage and degradation.

e) Unnecessary Inventory

Inventory helps to hide problems. Inventory will lengthen the lead time, making it more difficult to spot problems and taking up more space. Significant storage expenses are squandered, lowering the value stream organization's competitiveness.

f) Unnecessary Motion

Involve the ergonomics of production where employees may be required to stretch, bend, or pick up items that might be avoided. It not only exhausts the workers, but it also reduces productivity.

g) Inappropriate Processing

Overcomplicating a process discourages ownership and pushes staff to overproduce in order to recoup the high cost of the complicated machinery. It promotes bad quality and fails to fix it.

Benefits of Line Balancing**1. Reduce Waiting Waste**

One of the eight categories of waste in Lean manufacturing is waiting waste. Any idle time that happens when operations are not perfectly synchronised is referred to as this. Waiting waste, for example, happens while operators are waiting for resources or someone else to finish their duty. Another form of waiting waste is equipment downtime, which is defined as time when the equipment is not in use.

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Line balancing guarantees that all operators and machinery are working in harmony. No machine or operator should be overworked or idle. Line balancing decreases waiting waste by decreasing downtime.

2. Reduce Inventory Waste

Another sort of waste is inventory waste. It refers to a surplus of raw resources, unfinished things, or finished goods. Inventory waste shows poor capital deployment. Line balance standardises production, making it considerably simpler to minimise inventory build-ups. Line balancing guarantees that there is little work in process by decreasing idle time. Finally, it ensures on-time delivery by bringing production time closer to takt time.

3. Absorb Internal and External Irregularities

Line balancing is a technique for reducing variances in a manufacturing line. A well-balanced manufacturing line is both steady and adaptable to changes.

For example, if consumer demand shifts—and therefore the takt time shifts—operations may be promptly adjusted via line balancing. The effects of making modifications to a well-balanced manufacturing line are predictable. As a result, modifying the line to adapt the output rate is significantly easier.

4. Reduce Production Costs and Increase Profits

Worker and machine performance are totally synced as a result of perfect line balance. Standing idle is not compensated in any way. Every machine is put to its greatest capability. In other words, the capacity of both labour and machines is maximised. Process efficiency translates to lower costs and higher earnings.

How Can You Achieve Line Balancing?

1. Calculate Takt Time

Knowing your takt time is critical since the purpose of line balancing is to match production rate to takt time. Tulip comes with an out-of-the-box Takt Time App and Takt Time Dashboard that make calculating and tracking takt time a breeze.

$$\text{Takt Time} = \frac{\text{Available working time per shift}}{\text{Rate of customer demand per shift}}$$

2. Perform Time Studies

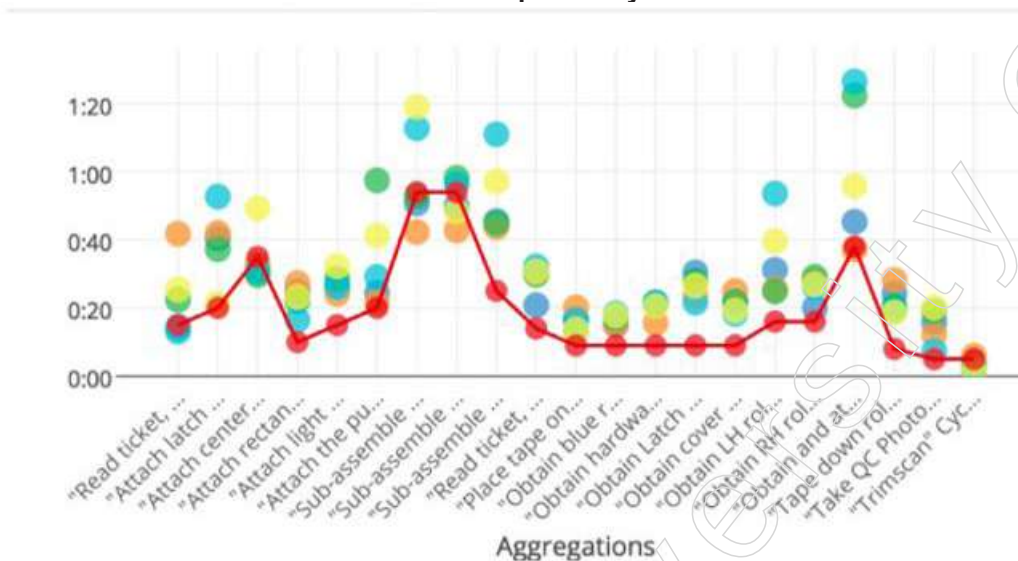
We published a comprehensive guide on how to perform time studies. Time studies are used to determine how long it takes to perform each activity on a manufacturing line. To put it another way, you want to know how much time workers and robots spend on each step of a process.

Keep in mind that while using a stopwatch and a clipboard to conduct time studies is still doable, there are now many better solutions. Data collecting and storage have been changed by IoT connection and cloud computing. Manufacturers may now undertake autonomous and continuous time studies using sensors and production software, avoiding human bias and the sample size effect.

3. Identify Bottlenecks and Excess Capacity

When it's time to examine the data from your time studies, look for areas where the process takes longer than takt time. Late delivery, hefty shipping expenses, and disgruntled customers are all consequences of exceeding the takt time. Also, keep track of which components are taking less time than takt. In some locations, there is a surplus of capacity.

Table Build - Step Time by User



4. Reallocate Resources

To begin, think about task precedence, or the order in which tasks must be completed. If a step requires a certain part, for example, you must ensure that that item is complete before proceeding to the next step. A Precedence Diagram may be very useful.

Then restructure jobs to eliminate bottlenecks and extra capacity. Move employees and equipment from portions of the line with extra capacity to bottlenecks, for example. To put it another way, try to reduce workload where there are bottlenecks and shift it to areas where surplus capacity may be filled by taking on more work. This will lower the amount of time people waste waiting in situations where there was an overabundance of capacity. It will also aid in the improvement of production flow when bottlenecks exist.

Organize elementary tasks into groups to reduce operator idle time and increase machine and equipment use. Distribute the workload among the operators in the most rational manner possible, taking into consideration the data you've obtained on operator performance. To achieve synchronisation, each group of tasks should be finished in the same amount of time.

Consider if you have an excessive number of or insufficient number of workstations. Line balancing may increase process efficiency to the point where your line has extra capacity. Removing workstations or combining operations might be useful.

You should attempt to decrease the imbalance between employees and workloads anywhere you have numerous operators doing consecutive activities and operating as a

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unit. In production lines, proper work layout and allocation assist optimise output at the appropriate time.

5. Make other Improvements

Analysing the quantitative data on your lines will almost certainly uncover more areas where you may enhance the balance of your lines. Three parameters may be changed to optimise the process: operator time, machine time, and setup time. You may, for example, provide additional training to employees who take longer to finish jobs or make transitions easier to decrease changeover times. You may also upgrade equipment or ensure that personnel follow standard operating procedures for machine setup and maintenance (SOPs).

3.2.6 Merits and Demerits of Product Layout

Machines and equipment are organised in the order in which they will be employed in the manufacturing process for the product or set of items. The machines in this sort of outlet can be arranged in a U form or in a line configuration, as seen below:

Machines are arranged in such a way that one machine's output becomes the input of the next.

Suitability of Product Layout

This arrangement is ideal for industrial facilities that operate on a continuous basis. At one end, raw materials are fed, while at the other, completed goods are delivered. Long manufacturing runs of identical items with a high level of automation are required. It may be used in the production of cement, steel, sugar, cigarettes, fertilisers, vehicles, and metal extraction, among other things.

Advantages of Product Layout

The following are the advantages of product layout:

1. **Smooth Flow of Production:** The entire manufacturing process is linked. As a result, the danger of manufacturing halting at various phases of production is removed. As a result, the production process runs smoothly.
2. **Lower Material Handling Costs:** There is no backtracking (moving back and forth) or criss-crossing of materials since machines are organised according to the order of activities. As a result, the cost and time spent managing supplies is reduced. The expense of moving materials from one machine to another is no longer an issue.
3. **Lesser Work-in-progress:** Work-in-progress is reduced since the production process is continuous and unbroken. The amount of money spent on work-in-progress is reduced, as is the amount of storage space required.
4. **Optimum Space Utilization:** Because all machines are organised in a logical order, the available space may be efficiently utilised. Even in a cramped location, machine placement is not an issue because the machines may be stacked in a U form.

5. **Effective Utilization of Resources:** Product layout ensures efficient utilization of resources by ensuring:
 - a. minimum movement of workers
 - b. continuous production process and reduced wastage
 - c. less work-in-progress and
 - d. materials handling mechanization
6. **Effective Supervision:** Since the manufacturing process is integrated and continuous, it is simple to supervise and regulate it. Inspection points are built into the manufacturing process.
7. **Production Control:** Production control is simplified since the manufacturing process is continuous. Management can prepare for operations and take steps to ensure that the job is completed in accordance with the plans.
8. **Savings in Time:** The amount of time spent carrying goods can be reduced. Mechanized equipment, such as conveyor belts, may be used to move goods since machines are organised based on the sequence of actions. As a result, a large amount of time is saved.

Disadvantages of Product Layout

The following are the disadvantages of product layout

1. **Rigidity:** The layout isn't flexible. Because the procedures are carried out in a sequential order, no changes can be made throughout the manufacturing process.
2. **Expansion is Difficult:** It is not easy to expand production beyond the capacity of each line of production.
3. **Costly:** Where machines are arranged on the basis of sequence of operations and not according to functions, this type of layout is costly. Therefore, it results in duplication of similar type of machines needed for different lines of production.
4. **Supervision Difficult:** There are no departments for different sorts of work in this layout. As a result, specialising in supervision is challenging.
5. **Complete Stoppage during Breakdown:** Because the output of one machine is the input of the next, any breakdown of one machine causes the entire process to come to a halt.
6. **Monotony:** Workers experience monotony as a result of the repetitious nature of their employment. Workers may get dissatisfied with their jobs. The labour force has little opportunities to showcase its abilities.
7. **High Labour Cost:** As people are trained to operate on certain equipment, they are unable to work on other machines. As a result, if a worker assigned to a certain task is missing, the entire workflow may be disrupted. It's possible that the company will need to hire and train excess personnel who can work on any equipment.

3.2.7 Merits and Demerits of Process Layout

Similar machinery and equipment of the same functional kind are grouped together

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in a process layout. The processes are separated, with each process' machinery maintained together and each process kept separate. For example, the milling department would house all milling equipment, the grinding department would house all grinding machines, and the welding department would house all welding machines. It necessitates a huge number of particularly built machines that are operated by limited personnel.

Suitability of Process Layout

The process plan is appropriate for job order manufacturing, in which numerous variations of items are manufactured in small quantities in response to client demands. It is appropriate for catalytic crackers used in the refinement of crude oil into gasoline, kerosene, wax, rolling mills, wire drawing, and chemical facilities, among other applications.

Advantages of Process Layout

The important advantages offered by a process layout are as follows:

1) Flexibility

When compared to a product plan, it is more flexible. Without disrupting the present arrangement, changes in operations and their order may be implemented. Any new operation may be added to the system.

2) Lower Investment

General-purpose machines are utilised, which are often low-cost. Machine duplication is avoided. Furthermore, general-purpose machines do not degrade as quickly as special-purpose computers.

3) No Stoppage of Production

In the event that a machine fails, the entire operation does not come to a halt. The work of a machine that has broken down can be shifted to other machines.

4) Scope for expansion

Expansion of different capacity lines can be done under this type of layout. New machines and labour can be added without upsetting the existing order of arrangement.

5) Full Utilization of Equipment

The layout of the process allows for maximum equipment use. Each department employs general-purpose machinery that can handle a wide range of tasks. There is no requirement for each product line to have its own machine.

6) Better Supervision

Due to specialization in operation, an efficient and better supervision is possible.

Disadvantages of Process Layout:

The following are the disadvantages of process layout:

1. Inefficient Material Handling

Materials must be transported forth and backward on a regular basis. Material handling mechanisation becomes challenging. Material handling equipment with fixed paths, such as conveyor belts and chutes, cannot be employed, thus material handling costs are low.

2. High Space Requirement

Space requirement is more than product layout. Around machines, more storage space will be supplied for awaiting material to be processed.

3. High Investment in Inventory

As a result of continuous flow of production there is high in-process inventory. Frequently materials have to be transported back and forth. As a result, there are delays and therefore the investment in inventory is high.

4. High Supervision Cost

Cost of supervision is high because the number of employees per supervisor is less resulting in reduced span of control. Further, the work is to be checked after each operation.

5. Longer Production Time

In case of product layout, time required for production is more.

6. Skilled Labour Required

In order to perform variety of operations in general purpose machines, skilled labour needs to be employed.

Check Your Understanding

- 1) _____ ideally involves allocation of space and arrangement of equipment in such a manner that overall operating costs are minimized.
 - a) Plant layout
 - b) Plant Location
 - c) Plant Utilization
 - d) Line Balancing
 - e) None of the above
- 2) Which of the following is not the technique of Plant Layout?
 - a) Process flow chart
 - b) Process flow Diagram
 - c) Template layout
 - d) Process Layout
 - e) All the above

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- 3) A process flow chart is also known as _____.
a) Plant Layout Visual Map
b) Plant Location Visual Map
c) D
d) D
e) None of the above
- 4) Product layout suffers from drawback due to _____ and material handling expenses.
a) Capital Investment
b) Backtracking
c) Mass Production
d) Combined Processes
e) None of the above
- 5) _____ is graphically depicts the sequence of the cycle elements involved in product manufacture. It summarises all of the tasks and operations that must be completed in a plant from start to finish.
a) Process Flow Chart
b) Machine Data Card
c) Process Flow Diagram
d) Fixed Position Layout
e) None of the above
- 6) _____ Process flow Diagram is presented in the form of a floor plan sketch of the buildings.
a) Process Flow Chart
b) Template Layout
c) Process flow Diagram
d) Model Layout
e) None of the above
- 7) A _____ involves producing items in advance of client demand, generally based on a sales prediction.
a) Make-To-Stock Method
b) Quantitative Method
c) Capacity Planning Method
d) Lag Method
e) None of the above
- 8) The _____ employs smaller incremental modifications to the manufacturers'

capacity rather than boosting demand ahead of time or increasing demand after current capacity is depleted.

- a) Lag Strategy
 - b) Dynamic Strategy
 - c) Lead Strategy
 - d) Match Strategy
 - e) None of the Above
- 9) _____ is normally done at the master schedule level and is used for short-term planning, which might last anywhere from a week to two months.
- a) Capacity Planning
 - b) Plant Capacity
 - c) Rough-cut planning
 - d) Balancing the Capacity
 - e) None of the above

Summary

- Plant layout is a crucial decision because it represents long-term commitment. a perfect plant layout should provide the optimum relationship among output, floor area and manufacturing process.
- Plant layout aims at an appointment of labour areas and equipment which will be the foremost economical to work and yet be safe and satisfactory to all or any parties concerned.
- Layout should be such production is administered in most effective manner.
- The safety, contentment, and comfort of the workers are the most important considerations while designing a layout.
- Light, heat, ventilation, humidity, moisture removal, filth, dust, and other factors must all be considered in order to achieve these goals.
- Process design allows you to include and increase efficiency in a variety of company processes.
- Takt time, according to Heizer and Render (2010), is a required technique while doing line balancing tasks. Takt time is the rate at which production is coordinated with consumer demand
- Cycle time is defined as the number of times a production line can create a product given its present resources and workforce.
- An efficient plant layout is one that aims at achieving various objectives like efficient utilization of available floor space, minimizes cost, allows flexibility of operation, provides for employee's convenience, improves productivity etc.

Activity

1. Visit a restaurant that you can classify as being in the small sector. Comment

Notes

upon how has the layout of facilities added to or detracted from the ease of movement? Could you suggest some alternative layout?

Questions & Exercises

1. What is a Plant Layout? Elaborate the factors that a planner must keep in mind while planning a layout.
2. How to develop an effective planning process? Elaborate the challenges that businesses face while capacity planning.

Glossary

- Plant layout refers to the arrangement of physical facilities like machinery, equipment, furniture etc. within the factory building.
- Plant layout is a dynamic rather than a static idea, which means that once completed, it cannot be improved or revised in the future without keeping up with the development of new machines or equipment, improvements in production processes, changes in materials handling devices, and so on.
- A process flow chart is a map that is displayed as a graph. As a result, it's known as a plant layout visual map.
- Process flow Diagram is presented in the form of a floor plan sketch of the buildings. It comprises information on the location of machines, internal transportation, the space occupied by each machine, and several other operations related to product manufacture.
- Flowcharts are a simple method to graphically lay out the phases of your process, with no text or spreadsheets to worry about.
- Process maps are similar in design to flowcharts and workflow diagrams and are commonly used to envision and plan corporate operations.
- Cellular layouts are referred to as "hybrid" layouts, because it has elements of both a Process and a Product (Line) structure. It's very uncommon for a company to have many product lines with much too much diversity to fit on a single manufacturing line.
- Line balancing is a flow-oriented production approach for increasing productivity and lowering costs in large-scale manufacturing.
- Takt time can be defined as the rate at which parts or products must be produced in order to meet customer demand.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor

Check your understanding – Answers

- | | | | |
|-------|-------|-------|-------|
| 1) a) | 2) d) | 3) a) | 4) b) |
| 5) a) | 6) c) | 7) a) | 8) d) |
| 9) c) | | | |

Module - 4: Production Planning and Control

Notes

Structure:

- 4.1 Concepts Of Capacity Planning, Aggregate Planning, Master Production Scheduling
 - 4.1.1 Concepts Of Capacity Planning
 - 4.1.2 Concepts Aggregate Planning
 - 4.1.3 Concepts of Master Production Scheduling
- 4.2 Concepts Of Push System, Pull System
 - 4.2.1 Concepts Of Push System
 - 4.2.2 Concepts Of Pull System
- 4.3 Job Shop Scheduling And Its Criteria
 - 4.3.1 Mean Flow Time
 - 4.3.2 Mean Tardiness
 - 4.3.3 Number Of Tardy Jobs
- 4.4 Single Processor Job Shop Scheduling
 - 4.4.1 Single Processor Job Shop Scheduling - Due Date Method
 - 4.4.2 Shortest Processing Time Method
- 4.5 Concepts Of Two Machines Scheduling
 - 4.4.3 Concepts Of Two Machines Scheduling - Johnson's Rule

Notes**Unit - 4.1: Concepts Of Capacity Planning, Aggregate Planning, Master Production Scheduling****Objective:**

At the end of this unit, you will be able to:

- Learn about capacity
- Learn what is capacity planning
- Understand concepts of capacity planning
- Explain procedure for capacity planning
- Identify importance of capacity planning
- Learn types of capacity planning
- Understand how to start capacity planning
- Learn what is capacity planning strategy and its types
- Understand capacity requirement
- Identify optimum plant capacity
- Understand balancing the capacity
- Learn factors influencing effective capacity
- Explain factors favouring over capacity and under capacity
- Identify adjusting capacity
- Understand capacity constraints

Introduction

Production planning and control is one of the most critical parts of production management; in fact, it is a manufacturing organization's nervous system. It is critical in a manufacturing company that production is carried out in the best possible way at the lowest possible cost, and that the items are of correct quality and delivered on time. This can only be secured by adequate production planning; yet, just planning production will not fix the problem because production plans are not self-actuating and do not result in automatic fulfilment.

For this, the production manager must take efforts such as regulating work assignments, reviewing work progress, and devising techniques to bring conformity between real and planned performance — so that plans are followed and the standards established during the planning stage are fulfilled and enhanced. The role of production control is to do just that.

As a result, production control is a directing function that entails the coordination and integration of operations and activities of various production components in order to maximise efficiency. To ensure smooth operation of the enterprise, proper work planning, laying out of exact routes that operations will follow, correct fixing of time-tables within which productive operations will start and end, uninterrupted releasing

of orders and work facilities, and timely initiation of appropriate follow-up steps are all necessary. Production control, in other words, entails planning, routing, scheduling, dispatching, and follow-up.

Production Planning and Control

Production planning is the planning of production and manufacturing modules in a company or industry. It utilizes the resource allocation of activities of employees, materials and production capacity, in order to serve different customers.

Production control looks to utilize different type of control techniques to achieve optimum performance out of the production system as to achieve overall production planning targets. Therefore, objectives of production control are as follows:

- Regulate inventory management
- Organize the production schedules
- Optimum utilization of resources and production process
- Steps in Production Planning and Control



1. Planning

The careful preparation of production plans is the first important step in production planning and control. Production plans determine what will be produced and where, as well as what type of product will be produced, who will produce it, and how it will be produced. The required information for thorough operations planning may be collected from a variety of sources inside the company. Customers' orders and the sales budget can provide information about the number and quality of items to be created, while management and the engineering department can provide information regarding manufacturing facilities. As a result, the planning function develops production plans and converts them into manpower, machinery, and material needs.

Production planning, regardless of the planning period, aids in avoiding randomness in production, providing a regular and steady flow of production activities, maximising the use of production facilities to reduce operating costs and meet delivery schedules, coordinating various departments of the enterprise to maintain proper balance of activities, and, most importantly, providing the foundation for enterprise control.

2. Routing

The determination of the path (i.e., route) of raw material flow through various equipment and operations in the factory is the next key function of production planning and control.

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“Routing” is defined as “planning where and by whom work will be done, determining the path that work will take, and determining the necessary sequence of processes.” To discover this path, the focus is on defining operating data, which often comprises planning of “where” and “by whom” work should be done, as well as the path that work should take and the required sequence of activities.

These operational data are provided in the standard process sheet, which aids in the creation of a routing chart that shows the sequence of activities and the equipment to be employed. Alternate route may be put on the routing chart if the machine loan chart shows that some machines are unavailable. The most efficient route may have to be compromised due to machine availability at a certain moment. “Routing creates the processes, their course and order, as well as the suitable class of machines and persons necessary for these tasks,” to put it another way.

Because many production control activities are closely connected processes and are dependent on routing functions, it may be deduced that routing is one of the most important parts and primary concerns of production control. As a result, it's critical to resolve issues like hiring the right people, getting the most out of machinery, and calculating the time necessary in the manufacturing process to a precise degree.

3. Scheduling

Scheduling is the process of planning the time aspect of production – that is, deciding ahead of time “when work will be done.” It includes the start and finish times for the different procedures to be completed. In other words, the scheduling function decides when an operation or job is to be finished; the distinction is in the specifics of the scheduling technique. In order to fulfil the targeted delivery dates, the scheduling function, which is part of the production control function, sets the time when each operation on the route sheet is to be performed on the designated machine.

The good control function not only indicates when each operation should begin, but also the progress of each manufacturing part, the quantity of work that each machine has ahead of it, and the availability of each machine for new work assignments. There are two sorts of schedules: master schedules and detailed schedules. If activities were recorded on a plant-by-plant basis, the master schedule would be prepared, but only detailed schedules are used to plan the production and assembly procedures necessary for each product.

4. Dispatching

Dispatching is the portion of production control that converts paper — work into real-world production. It's the team in charge of coordinating and translating plans into reality. The dispatching function follows the details established during the routing and scheduling processes. As a result, dispatching ensures that the material is delivered to the proper work location, that tools are available for the specific tasks, and that the job is completed in accordance with the routing instructions. Dispatching is in charge of carrying out the physical tasks that are proposed by scheduling. As a result, dispatching entails the creation of work orders. These work instructions denote production authority.

These orders contain the following information:

- The product name;

- The name of the part to be produced, sub-assembly or final assembly;
- The order number;
- The quantity to be produced;
- Descriptions and numbers of the operations required and their sequence,
- The departments involved in each operation
- The tools required for particular operation; and
- Machines involved in each operation and starting dates for the operations.

5. Expediting

The final stage of the production control process is expedition or follow-up. This function is used to keep track of the amount of work that has been done. The goal is to guarantee that what has been intended and planned is carried out. "Expediting entails providing production statistics and looking into deviations from set timelines." The major goal of the mission is to ensure that promises are fulfilled." It comes with the following features:

1. Verify that all materials, tools, component parts, and accessories are available in required quantities at all work locations in order to begin and complete production activities.
2. Review the status of work-in-progress and finished work at different work stations. This involves gathering data on the start and finish times of work, as well as the date of completion, the state of work-in-progress in relation to scheduled completion dates, the location of materials, component parts, and sub-assemblies within the plant, and inspection findings.
3. Maintaining up-to-date control boards and preparing progress reports.
4. Reporting any major deviations to manufacturing management so that remedial action can be implemented. It also includes reporting to the production planning section in order to adapt future plans.

Thus, by completing the above-mentioned processes, production planning and control assures the creation of items of the proper quality, quantity, and at market-competitive prices. One thing to keep in mind is that production planning and control is a never-ending process with interdependent functions.

- Production Planning in 5 Steps



Notes

Are you a business that produces goods? If that's the case, you'll need a production plan to guarantee that you have all of the necessary manufacturing inputs on hand at the proper moment to satisfy demand.

According to the Business Dictionary, production planning is "the administrative process that occurs within a manufacturing business and that involves ensuring that sufficient raw materials, staff, and other necessary items are procured and ready to create finished products according to the schedule specified."

Your company's manufacturing operations are guided by a production plan. It sets and sequences actions that must be completed in order to meet a production goal, ensuring that all employees are aware of who is responsible for what, when, where, and how.

By enhancing process flow, eliminating waiting time between processes, and maximising the usage of plant, equipment, and inventory, a production plan may help you satisfy product demand while minimising production time and expense. To do this, you must connect your production plan with your company strategy and business plan, as well as collaborate with other departments such as procurement, finance, and marketing to assist production planning.

The diagram above shows the production planning and control process divided in five steps:

- Step 1: forecast the demand of your product
- Step 2: determine potential options for production
- Step 3: choose the option for production that use the combination of resources more effectively
- Step 4: monitor and control
- Step 5: Adjust

STEP 1. Forecast the Demand of Your Product

Estimate your demand to figure out how many things you'll need to make in a certain time frame. You may already have some confirmed orders for the next several months, but you must also forecast how many more will arrive.

There are several approaches for forecasting product demand. A typical method for estimating product demand is to use historical data (e.g., orders placed by your customers in the past). While this is a standard strategy, you must take into account external and internal developments in your company environment that may cause prior trends to change.

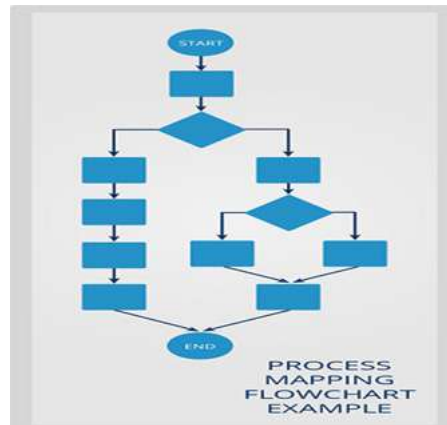
New market trends, a slowing economy, or a new marketing campaign, for example, might raise or reduce demand for your goods compared to previous years.

STEP 2. Determine Potential Options for Production

Determine the various manufacturing alternatives available to satisfy your product's expected demand. If you want to make 100 shirts, for example, you'll need a particular amount of equipment, people, supplies, and time. Production timelines and prices might vary depending on how these inputs are combined.

Notes

- a. Begin by Mapping out all of the Processes in Your Manufacturing Process. Consider if tasks are sequential or reliant on one another, and whether they occur concurrently or separately. An example of a simple process-mapping flowchart is shown below. Each box indicates a step in the manufacturing process. Each company's manufacturing process map will be individual and distinct. Consider how you may increase process flow by removing bottlenecks.



- b. Determine the Resources Needed to Complete Each Task Involved in Your Production Process.

Examine how different resource combinations result in varying production times and costs:

- **Human Resources.** Determine how many people will be engaged in each stage of the manufacturing process, as well as their availability and cost. Ascertain that their time is well spent.
- **Machinery and Equipment.** It is a broad term that encompasses a wide range of items. Determine the machines that are required and their availability, as well as any necessary repair or replacement.
- **Materials.** Make a list of all the items you'll need for the project, as well as how you'll get them. Examine your vendors' dependability, including delivery time. It's critical for the manufacturing process to have supplies on hand when they're needed.
- **Inventory.** It's critical that you think about ways to make your inventory more efficient. Maintaining a big inventory is costly, while maintaining a low inventory is dangerous if demand swings often. A strong inventory control system may assist your company in accommodating changes in demand and mitigating potential difficulties or delays that may arise throughout the manufacturing process. Check out the video for additional information on how to manage your inventory.

STEP 3. Choose the Option for Production That Uses the Combination of Resources More Effectively

Compare the costs and timelines of each conceivable production option and pick the one that leverages the most economical mix of resources while still allowing you to fulfil demand. The choice you choose should enhance your company's operating capabilities.

Notes

Always double-check that you'll be able to afford the charges of the manufacturing procedure (purchase of materials, office rent, payment of staff salary, leasing, etc.)

You must communicate your production plan with all departments and employees involved in the manufacturing process, such as human resources, procurement, finance, marketing, and so on. Operations will go more smoothly if everyone knows what to do and what supplies and equipment to utilise for each duty in the manufacturing process.

STEP 4. Monitor and Control

You want to make sure that your strategy is operating as it should. Monitoring and regulating entails comparing what is taking place to what should be taking place. Having a control system in place allows you to discover issues as soon as they arise, giving you more time to rectify problems before they become too serious.

STEP 5. Adjust

Prepare to make changes to the plan if necessary. To meet variations in client demand, the manufacturing strategy must be adaptable (e.g. an important order that gets cancelled). You must also consider any risks that may happen throughout the manufacturing process (for example, a machine breaking down, a worker being ill, or a supplier failing to deliver on time) and develop a risk mitigation strategy.

- Characteristics of Production Planning and Control

The forgoing discussion brings out the following traits of production planning and control:

1. It is the planning and control of a company's production process. What is to be made, for example? When is it going to be made? How to keep the production schedule, for example? —are decided and implemented in order to get good outcomes.
2. All forms of inputs, such as materials, workers, and equipment, are employed efficiently to keep the production process running smoothly.
3. Various production elements are combined in order to make the most efficient and cost-effective use of them.
4. The production process is designed to ensure that none of the work centres are overworked or underworked. The division of labour is done with great care to ensure that every available resource is used to its greatest potential.
5. The work is regulated from the beginning of the procurement of raw materials to the end of the final product.

- Objectives of Production Planning and Control

The overall objectives of production planning and control include the following:

- Continuous Flow of Production - The continuous flow of production through the manufacturing plant is one of the key goals of production planning and management. One of the fundamental ideas of lean manufacturing, six sigma, and overall production enhancement is continuous flow and process improvement. By eliminating bottlenecks and waste from your

facility, production planning and control will strive to produce a smooth and continuous production process, allowing you to take your production to the next level in terms of productivity.

- **Planned Requirements** - Production planning is an important part of production planning and control because it ensures that your manufacturing facility has what it needs, in the appropriate quantity, at the right time. A production plan guarantees that you have adequate material, machinery, tools, equipment, and labour to complete the job.
- **Optimal Inventory Levels** - Inventory is sometimes regarded as an useless resource in manufacturing processes. This is because inventory-related expenditures are among the highest for many businesses. This is why, in order to effectively save expenses within your operations, you should strive to reduce inventory levels as much as possible. Production planning and control prepares and executes procedures that allow for a continuous flow of production, bringing manufacturing operations closer to JIT (just-in-time) manufacturing and lowering inventory levels.
- **Increased Productivity** - Another important goal in production planning and control is increased productivity. Production planning and control strives to boost output by improving efficiency while remaining cost-effective. Increased productivity is accomplished by maximising the utilisation of current manufacturing and labour resources while reducing waste and squandering of materials.
- **Customer Satisfaction** - Every business wants to improve customer happiness. This is because consumers drive demand, and happy customers imply more future business. Defective items and delays in shipment days are both detrimental to the firm. The primary goal of production planning and control is to ensure that the manufacturing process completes orders on time and reduces lead times. This guarantees that client orders are filled on schedule and in a timely manner, resulting in increased consumer satisfaction with your business.
- **Importance of Production Planning and Control:**

The nervous system of a plant is the production planning and control system. It serves as a coordinating body for the engineering, buying, manufacturing, sales, and stock control departments. An effective production planning and control system aids in the delivery of better and more cost-effective items to clients at a reduced cost. It is required by all plants, regardless of their size or species.

The principal advantages or importance of production planning and control are summarized below:

1. Better Service to Customers:

Production planning and control, via effective scheduling and job expediting, aid in offering better services to clients in terms of higher-quality items at reasonable rates on time. On-time delivery and correct quality go a long way toward gaining clients' trust, boosting customer relations, and encouraging profitable repeat orders.

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2. Fewer Rush Orders:

Production activities in an organisation with an effective system of production planning and management run smoothly, following the original plan and meeting the specified delivery dates. As a result, there will be fewer rush orders in the factory and less overtime than there would be in the same business if production planning and control were not appropriate.

3. Better Control of Inventory:

A well-designed production planning and control system aids in maintaining correct inventory levels and, as a result, reducing inventory investment. To provide effective service to consumers, a reduced inventory of work-in-progress and finished product is required. It also aids in gaining a better handle on raw-material inventory, which leads to more efficient purchasing.

4. More Effective Use of Equipment:

The most efficient utilisation of equipment requires an efficient system of production planning and management. It informs management on a regular basis on the current status of all orders in progress, as well as equipment and staff requirements for the next weeks. If retrenchment, layoffs, transfers, or other changes are expected, workers can be informed well in advance. Unnecessary equipment and material purchases can also be avoided. As a result, optimal exploitation of equipment and other resources is feasible.

5. Reduced Idle Time:

Because it guarantees that supplies and other facilities are accessible to employees on time according to the production schedule, production planning and control helps to reduce idle time, or time spent by workers waiting for materials and other facilities. As a result, there are fewer lost man-hours, which lowers production costs.

6. Improved Plant Morale:

The actions of all departments participating in the manufacturing activity are coordinated by an effective system of production planning and control. It provides a smooth workflow and prevents hurried orders. It prevents workers from becoming “sped up” and ensures that the plant is kept in good operating order. As a result, plant morale improves as a result of this.

7. Good Public Image:

A well-designed production planning and control system aids in the organization's systematisation. Such a company is capable of completing orders on time and to the satisfaction of its consumers. Higher customer satisfaction leads to increased sales, earnings, workplace harmony, and, eventually, a positive public image for the company.

8. Lower Capital Requirements:

Everything related to production is planned far in advance of activities under

a good system of production planning and management. Before the actual manufacturing process begins, it is understood where, when, and what type of input is necessary. Inputs are sent on time, ensuring that production runs smoothly and without bottlenecks. Facilities are better used, and inventory levels are maintained in accordance with the timetable, neither too high nor too low. As a result, production planning and control aid in the reduction of capital expenditures on equipment and inventory.

- **Limitations of Production Planning and Control**

Undoubtedly, the system of production planning and control is a must for efficient production management; but in, practice, sometimes, it fails to achieve the expected results because of the following limitations.

The principal disadvantages or limitations of production planning and control are summarized below:

- 1. Lack of Sound Basis:**

Production planning and control are based on certain assumptions or forecasts about the availability of inputs like materials, power, equipment, etc. and customers' orders. In case these assumptions and forecasts do not go right, the system of production planning and control will become ineffective.

- 2. Rigidity in Plant's Working:**

Production planning and control may be responsible for creating rigidity in the working of the plant. Once the production planning has been completed, any subsequent change may be resisted by the employees.

- 3. Time-consuming Process:**

Production planning is a time-consuming process. Therefore, under emergencies, it may not be possible to go through the process of production planning.

- 4. Costly Device or Machine:**

Production planning and control is not only a time-consuming process but is a costly process also. Its effective implementation requires services of specialists for performing functions of routing, scheduling, loading, dispatching and expediting. Small firms cannot afford to employ specialists for the efficient performance of these functions.

- 5. External Limitations:**

The effectiveness of production planning and control is sometimes limited because of external factors which are beyond the control of production manager, Sudden break-out of war, government control, natural calamities, change in fashion, change in technology, etc. are factors which harm the implementation of production planning and control.

- **Decision-making and its Importance in Production Planning**

Aligning processes with an organization's strategic goals, building and executing process architectures, establishing process measurement methods that correspond

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with organisational goals, and teaching and organising managers to manage processes effectively are all examples of process management.

Production should be thought of as a system having inputs and outputs. In the system, inputs are converted into outputs through the use of appropriate technological, organisational, and management procedures. Alá and Raner (2004) define feedback as “potential adjustments in outputs, technology, administrative choices, and input selection.” It is a response to client complaints and requests, and it must result in acceptable answers and decisions (Figure below) (Demoč 1996).

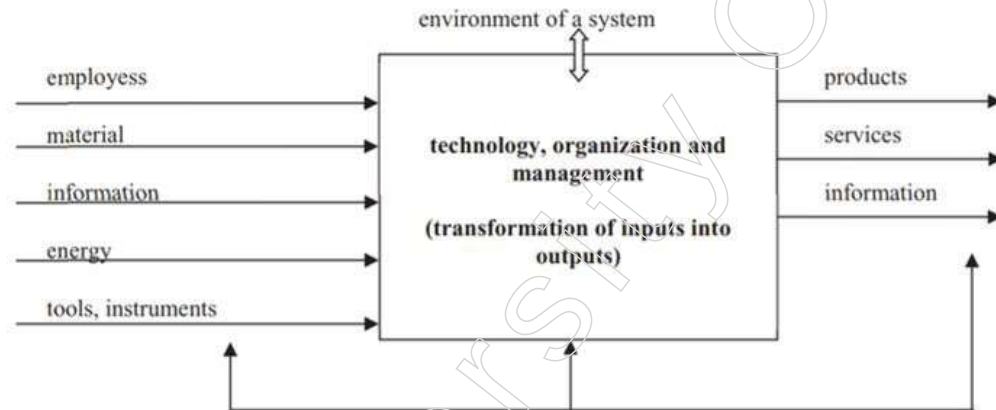


Figure: Production in a system (Demoč 1996)

The purpose of production is not to produce any product or service, but to produce only those that will be effectively implemented in the market, that will meet consumer needs, and that will generate appropriate (optimal) profit and market share. The transformation process should be the most efficient since it operates at the lowest cost and consumes the least amount of resources (Matuzsewski 2003).

Each component must be properly identified and described in order to select the best one for a certain type of production and set of objectives.

What does have an Impact on Production?

There are several criterions which have decisive impact on each production process.

For the specification of production type, it is necessary to evaluate:

- Universal character of machinery
- Amount of producing products for particular time period
- Number of products types
- Demands for employees' qualification
- Time of production cycle (process)
- Character of labour division etc.

Material and Methods

Planning as a Managerial Activity

Planning is one of the oldest human activities, as well as one of the most fundamental and crucial managerial actions for defining a company's goals and activities that contribute to achieving those goals.

Production planning (or plans), personal planning, investment planning, R&D planning, marketing planning, finance planning, and so on are examples of firm departments. Basic management functions are depicted in the diagram below.

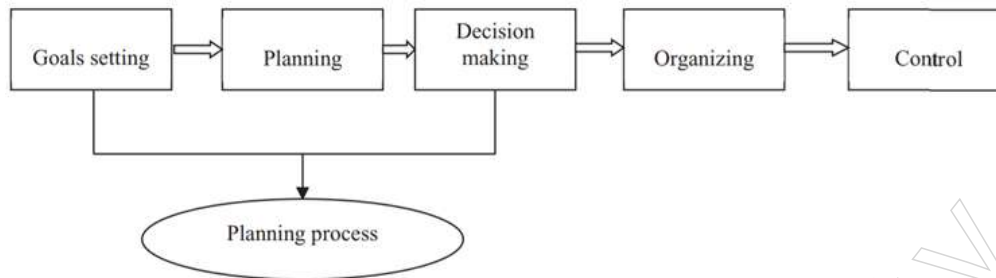


Figure: Managerial functions and specification of planning process (Robbins, Coulter, 2004).

Goals must be defined, and effective ways to attain those goals must be determined. The need for planning stems from the nature of organisations as goal-oriented entities. When making plans, we need numerous pieces of information from the environment, the market, and even the inside of a corporation. More up-to-date and thorough data means more precise plans and objectives (Kokles 2000 and Molnár 1992).

Some interesting ideas about planning:

- Those who do not plan will never know where they failed since planning is not a description of what will happen, but rather of what we desire to happen.
- Accidental incidents are more likely when the planning process is merely generic and not specific, while at the same time...
- If the preparation is excessively precise and exact, the influence of unplanned occurrences is reduced.
- The planning process should be as comprehensive as is required, not as detailed as is possible.

The content of the planning process connects not only to the departments that plan, but also to the specific goals that must be reached. Each action, each process, should be carried out at the lowest possible cost, on time, and with the highest possible quality.

As a result, strategies should adhere to the following managerial goals (Rajnoha 2005):

- Customer Satisfaction: Satisfying customer requests is essential for profit and growth. The product must be manufactured to the customer's specifications while maintaining high quality standards, and it must be delivered on schedule and at the agreed-upon price.
- Consistent Material Flow: If planned production schedules are disrupted due to a lack of materials, staff, or any other cause, costs and production time rise.

Notes

- **Optimum Inventory Levels:** The minimal inventory levels required to maintain continuous material flow may not be the most cost-effective. Ordering costs fall as purchased order numbers increase (demand variables and lead times stay same), while inventory costs rise. When ordering costs fall faster than inventory expenses rise, the ideal inventory level rises.
- **Increased Productivity:** The working process must be designed and regulated in such a way that production time and expenses are kept within preset limitations (levels). Shortening manufacturing periods or increasing the volume of semi-products can boost productivity.

8 Steps of Planning Process

In general, and in a nutshell, the planning process estimates - what should be done, in what order, and at what cost. The depth and specifics of planning are determined by the specific anticipated circumstance, as well as its complexity and scope. However, there should be eight essential phases in any planning process:

1st Step – What is a Goal of Planning Process? What, Why and How Should be Reached?

The effectiveness of each planning process is determined by two essential steps: precisely stated goals and a strategy for achieving them. It is vital to re-evaluate appraised goals and strategy if responsible managers or workers cannot discover relevant and appropriate answers to the above-mentioned questions (what, why, how?).

2nd Step – What Must Be Done?

All actions must be recognised and described (characterised) in this stage (for example by brainstorming method). The tasks must then be organised in a logical order, such as by technology line or material flow. Last but not least, the link between activities must be considered.

3rd Step – Who Will Perform (Realise) Given Activities?

This process includes function planning and definition, as well as job task descriptions for specific personnel. Each action must be assigned to a certain employee (or technological device) who will be in charge of its execution.

4th Step – Who and for What Will Be Responsible?

Each employee should be capable and accountable for his or her work. In a so-called matrix of responsibility, these abilities and obligations may be plainly portrayed.

5th Step – When Will Particular Activities Be Realized?

All activities should be organised in this phase of the planning process according to a logical order and linkages. It is also given the time it will take to complete the task, and a time plan is created.

6th Step – Which Costs and Resources are Demanded?

Appropriate expenses and resources should be planned for each task. Cost and

resource amounts can be derived from previous periods and anticipated using various analytical and calculating methods. Standards of consumed materials and standards of consumed work performance can be used to estimate and plan resources and working time.

7th Step – How to Control?

It must be determined how frequently and by what means activities will be monitored and managed. It is also vital to identify communication mechanisms (meetings, e-mail) for effective data collection, transmission, and assessment during this phase.

8th Step – What Will Happen If...?

The planning process should also include alternate outcomes that may develop as a result of accepted decisions. As a result, in order to avoid undesirable consequences, it is vital to examine both the external and internal environments. Within the planning process, this final stage might be viewed as feedback.

Managerial Tools and Methods in Planning and Decision-making

It should be noted that the activities and responsibilities of Production Planning may be split among numerous departments or personnel in any given firm, and that they are not always structured into a Production Planning division. The consolidation of the planning function into a Production Planning division enhances and streamlines the coordination and execution of planning operations. Production Planning's functional responsibilities will often involve (Matuzsewski 2003):

- Active engagement in sales forecasting or sales forecasting
- Determining the volume of output required (specification and quantification of raw materials, other inputs, machinery in order to effectively utilise capacities in a company and to meet sales forecast)
- Inventory control (both input and output stock optimization)
- Workforce requirements (criteria for the choice of most suitable employees and necessary number of employees).

Managers must make decisions every day without knowing what will happen in the future. Making decisions necessitates making predictions about the future, therefore managers must frequently plan based on their subjective feelings and best projections. Managers will be more prepared if their sentiments are more accurate (Alá and Raner 2004). Managers' judgements and abilities to foresee events tend to improve with experience. Managers can use a variety of quantitative methodologies to foresee occurrences. Intuition and quantitative methods are combined by the finest managers.

Basic quantitative tools:

- Break - even analysis
- Time series analysis
- Causal modelling (regression analysis)

Forecasting qualitative tools aid in the generation of information, ideas, and judgments that managers require for planning and decision-making. Quantitative

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approaches focus on picking the most desired choice from a group of alternatives, whereas qualitative tools place a greater emphasis on identifying options.

The following is a list of some qualitative tools:

- Decision tree
- PERT
- Petri Nets
- Decision matrix
- Brainstorming
- Delphi technique
- Nominal group technique

The definition of the whole supply chain, as well as the identification of decision-making knots and scenarios using proper managerial tools and approaches, is an objective of this debate. All of this should lead to not only real improvements in a company's goals, but also to the discovery of potential difficulties and the evaluation of appropriate processes, criteria, and choices for remedies.

Results and Discussion

It all starts with determining what clients want. As a result, it is vital to identify and obtain the resources, materials, and semi-products required for manufacturing. The identification of a client's wants or the identification of a specific customer can only reflect a given circumstance (stochastic perspective) or a decision-making process (deterministic view).

If the corporation wants to fulfil the expectations of all prospective customers, the first knot (step) will be stochastic (of course taking into account structure and capacity of own production and technology). In this situation, the firm accepts and records each customer's order, and it makes every effort to fulfil all of them. The second circumstance (the first knot) will arise when the corporation selects a certain consumer for whom its items will be manufactured, leaving other customers to the competition. It is possible to argue that not every consumer should be interested in the firm meeting his or her needs.

In this situation, the first knot could be deterministic. This is mostly governed by manufacturing capacity, costs of adding capacity to fulfil rising demand for a certain product or to launch a new product on new assembly lines, and expenses of logistics and other services. Of course, the cost of research and development (R&D) for new items must also be considered.

Purchasing inputs is represented by the supply chain's following knot. This knot should be deterministic, and two fundamental options were stated – the first is purchasing from external providers, and the second is self-supply using one's own capacity. This decision-making is extremely reliant on specific input. There are some common “make or purchase” queries and answers. More than 80% of all inputs with main costs are covered by external inputs. The remaining 20% of inputs originate from internal capacity and are compensated by secondary charges.

Costs, quality, flexibility, and the required volume of inputs are the deciding factors

when deciding between own internal inputs and external sources. “Make or purchase” can be utilised as an optimal approach for such a decision.

The tangle “Supplier selection” is a critical element of the supply chain because if this step fails, the ramifications can be seen in subsequent processes and, ultimately, in poor final product quality, resulting in customer unhappiness. Aside from quality, another crucial element that influences the overall period a product spends on the market is adaptability. The number of inputs available has an impact on storage and transportation expenses. Along with the selection of a provider, consideration must be given to the selection and assessment of transportation methods. When inputs originate from one’s own ability, this decision-making is unnecessary. External inputs can be transported in one of three ways: by personal vehicle, by provider vehicle, or by transport ordered from a third party.

Furthermore, these options may be classified into three categories: rail, road, and mixed transportation. Costs, flexibility, transport capacity, conveyed goods, warehouse technical equipment, and route disposal are all taken into account as criteria. It’s a good idea to keep track of your inputs. Because technology, manufacturing processes, and delivery conditions make it impossible to reduce inventory to zero, warehouses, their capacity, and management must be addressed. You have two options: utilise your own warehouses or outsource it. The following are the primary criteria for deciding between these two options: costs, supply flexibility, and warehousing capacity.

The supply chain’s next phase (knot) is “production.” This knot has the potential to be both deterministic and stochastic. Of course, one of the main assumptions in management work is to think in terms of options and to be prepared for a variety of them, so analysing this knot as deterministic is the best option. There are three options here: in-house manufacturing, outsourced production in its whole, and outsourced production in part. It is debatable, and it must be determined whether or not the final two options are lucrative for the firm in issue. The profitability of a production, expenses, capacity utilisation, time demand, flexibility of a production, quality, and loss of know-how should all be considered. The last link in the supply chain is the sale. Several years ago, the corporation used two methods of sales: its own sales department and external sales organisations. For this knot, we may list the following criteria: capacity to expand or at least maintain current market share, sales costs, and customer feedback (market). The supply chain comes to a close where it began: with a consumer.

However, the supply chain does not end with the sale of a specific product to a buyer. Nowadays, not only the goods but also the after-sale services and customer communication associated to that product are paid a lot of attention. The proposed supply chain model is shown in Figure 1 using the decision tree technique. Following this chart concept, all knots (nodes) and transitions must be described.

Conclusion

The company’s flexibility to satisfy market needs is influenced by both production planning and decision-making. And it’s not only about final product planning and forecasting; it’s also about manufacturing capacity and technology assembly issues and decisions. The objective of planning is to ensure that all production factors are spent in the supply chain at the lowest possible cost and in the shortest possible period. It is a dynamic process that must take economic cycles into consideration. Value chain

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analyses should be focused on identifying so-called bottlenecks, which are activities that prevent businesses from increasing their profit margins. Simultaneously, these evaluations reveal the inefficiency caused by the oversizing of some operations, which results in a lower degree of assurance and worse performance of other business activities. The importance of multicriteria decision-making approaches for evaluating alternatives does not lay in a specific increase in certain indicators, but rather in the use of numerous models and procedures that lead to that. It allows managers to organise options based on a large set of criteria, as well as explain specific phases in the solution and their logical sequence.

This whole solution process is visible, repeatable, and there are clear beginning assumptions as well as how these assumptions, conditions, criteria, and incidents influence the final outcome. Actual management activity in specific organisations demonstrates the subjective assessment of criterion relevance. This value depends on the value patterns of the decision-maker (management) or the value system of the firm. The strategies listed above should aid in reducing subjective evaluation and, as a result, subjective outcomes from planning and decision-making processes in management work.

4.1.1 Concepts of Capacity Planning

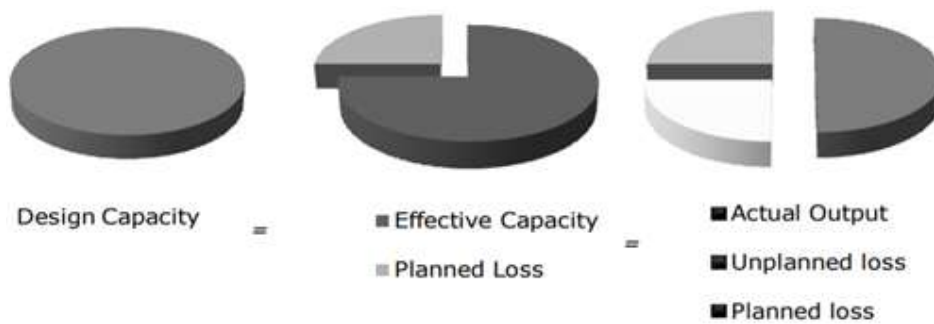
Capacity

Capacity refers to a company's maximum production level for producing a product or providing a service. Capacity planning necessitates management's acceptance of production process constraints. Capacity can refer to a manufacturing process, human resource allocation, technological thresholds, or a variety of other related ideas, depending on the type of business. No system can function at full capacity for an extended length of time due to inefficiencies and delays, making it impossible to achieve a theoretical production level over time.

Design capacity, effective capacity, and actual capacity are the three types of capacity. These may then be used to calculate the operations' resource utilisation and process efficiency.

- **Design Capacity:** This is a theoretical figure that isn't used in the day-to-day operations of a company. The output that an operation can create continuously, at maximum rate, without halting for shift changes, maintenance, or other delays is referred to as design capacity. Under ideal conditions, what the process is capable of producing. This might be construed as maximum capacity in some circumstances.
- **Effective Capacity:** This addresses how the operation will function in the long run, as well as how it will be manned and maintained. All planned stoppages that fall within the typical working time limit are taken into account. This is also known as capacity available. Shift changes, lunch breaks, set-up delays, and a variety of other operational issues can all cause stoppages.
- **Actual Capacity:** This is the same as effective capacity, only it includes both unexpected and planned losses. Poor productivity, absenteeism, or new employee training are just a few examples.

The figure below shows the proportions of Capacity measures



Actual output plus unplanned losses is the same as effective capacity. Therefore, the operation which is working its assets efficiently is minimising unplanned losses.

Capacity Planning

The most critical task of production and operations management is to effectively manage capacity. The goal of capacity management, also known as capacity planning and control, is to match the level of operations to the level of demand. Capacity planning is focused with determining solutions to basic capacity problems such as:

- (i) What kind of capacity is needed?
- (ii) How much capacity is needed?
- (iii) When this capacity is needed?

Capacity planning must take into account future growth and expansion plans, market trends, sales projections, and other factors. A facility's capacity is the pace at which it can produce goods. Capacity is generally measured in terms of production volume per unit of time.

Capacity planning is required for the following:

- Sufficient capacity is required to meet the customers demand in time,
- Capacity affects the cost efficiency of operations,
- Capacity affects the scheduling system,
- Capacity creation requires an investment,
- Capacity planning is the first step when an organisation decides to produce more or new products.

Capacity planning is mainly of two types:

- i) Long-term capacity plans, which are concerned with new facility and equipment investments. These plans are for a period of at least two years.
- ii) Short-term capacity plans, which consider labour force size, overtime budgets, inventory, and other factors.

The greatest load that a functioning unit can handle is referred to as capacity. A plant, a department, a machine, a store, or a worker might be the operating unit. The greatest rate of output (goods or services) that a plant can generate is known as its capacity. A facility's or a business's production capacity is the greatest rate of production that the facility or firm is capable of. It's commonly measured in terms of

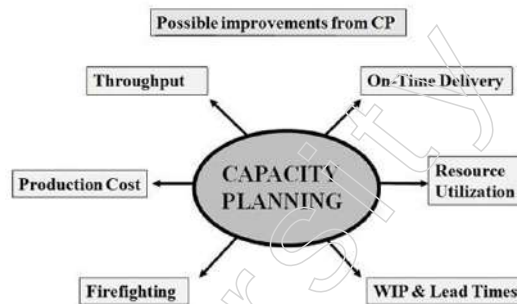
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production volume per unit of time (i.e., hour, day, week, month, quarter etc.). Capacity refers to a company's capacity to fulfil present and future market demand.

The way in which companies or teams match personnel with the demands of a project is referred to as capacity planning. Capacity is the maximum amount of work that can be completed in a given amount of time. It is usually assessed in terms of the amount of work completed by personnel. In addition, planning entails allocating work hours to a specific quantity of labour.

For example, if a company has 20 employees who each work 20 hours per week, the company's weekly working capacity is 400. Without factoring in overtime, this means the company can handle a maximum of 400 hours a week.



- Concepts of Capacity Planning

- Design Capacity

It refers to the maximum amount of output that may be obtained in a given amount of time.

- Effective Capacity

Given changes in product mix, equipment maintenance, programming and operational challenges, labour concerns, and so on, it refers to the best level of output. It is frequently smaller than the entire capacity of the design.

- Actual Output

It is, in fact, the degree of output that is attained. Because of machine failures, workforce absenteeism, inconsistency in raw material supply, abnormal delays in equipment delivery, and power outages, it cannot be more than sufficient capacity.

- Procedure for Capacity Planning

1. **Assessment of Present Capacity**

The output or inputs of a department can be used to determine its capacity. Manufacturing enterprises such as car plants (number of vehicles), iron and steel plants (tonnes of steel), breweries (barrels of beer), canneries (tonnes of processed foods), power companies (megawatts of energy), and others are authorised to use output measures.

Hospitals (number of beds), airports (number of flights), theatres (number of seats), restaurants (number of tables and chairs), universities (number of students), warehouses (spaces), and a variety of other service sectors may all be used to quantify capacity in terms of inputs.

2. Estimating Future Capacity Needs

Forecasting product demand at various phases of the product life cycle can be used to anticipate short-term capacity needs. Because of market and technical uncertainty, forecasting long-term capacity requirements is increasingly difficult.

Capacity forecasting aids in determining the difference between existing and expected capacity, allowing for appropriate modifications. For example, a firm that manufactures two items may discover that one has low demand in the summer (e.g., coffee or tea) while the other has low demand in the winter (e.g., cold drink).

3. Identifying Alternative ways of Modifying Capacity

In the event that current capacity is insufficient to meet expected demand, an expansion will be required to address the deficit. To increase capacity, extra shifts or overtime will be required. In a similar vein, the expansion will provide scale and aid in fulfilling demand forecasts, but it will necessitate further investment and risk falling short of future demand estimates.

There is a requirement to reduce surplus capacity when current capacity exceeds anticipated capacity. Staying on top of this might be accomplished by developing new goods, selling existing facilities, laying off staff, or obtaining more positions from other businesses.

4. Evaluation of Alternative

From an economic, technological, and other perspective, many options for capacity improvement or decrease are considered. To obtain the right analysis, the reactions of employees and locals should be taken into account throughout the examination. Cost-benefit analysis, queuing theory, decision theory, and other assessment approaches are among the most common.

5. Choice of Suitable Course of Action

After completing a cost-benefit analysis of several options for increasing or decreasing capacity, the optimum option has been selected.

- Importance of Capacity Planning

Capacity planning is essential because of the following:

1. It Ensures Availability:

Are you sure you have the team ready to work on the next contract before you put pen to paper to seal it? Capacity planning discusses what you'll need to do and how you'll accomplish your goals. For example, you may require individuals to manage duties or determine whether you require extra hands to assist you in taking over projects. Capacity planning assists you in determining all of these factors and ensuring that you do not disappoint your customers.

2. Manage Your Skills

Capacity planning is also important for determining your team's skill inventory. A talent inventory is a list of people who can do a given work and what abilities they have.

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This is particularly significant in a technological environment, such as a programming or software firm. You might, for instance, make a list of each team member's main competencies, such as:

- Mobile app development
- Website development
- Cloud computing
- Cybersecurity

When you give someone a task, you can immediately see if they are capable of doing it. Whether you want to construct a mobile app or a gorgeous and functional website, you already know who to contact, which will make things much simpler. When a worker gains new abilities or knowledge, you may quickly update the skill inventory.

3. It Helps to Identify Easily Any Skill Gap

A common example of this is when your company has just received a large contract that requires a large amount of work and a variety of abilities that your company may not have.

Capacity planning has the advantage of allowing you to spot any problems early on. You don't have to start working on the project until you've determined who you'll need to assist you. When you prepare well, you'll be able to identify who fits in and what they can contribute to the cause.

You can train your team in advance for future chances that may occur from that type of work if you can identify the skill gap. For example, if you don't have a mobile app developer on staff this year for a mobile app development project, capacity planning might help you plan ahead for next year.

• 3 Types of Capacity Planning

The three forms of capacity planning ensure that you have enough of three primary resources in the long and near term, but not too much. You should plan weeks, months, or even a year ahead of time.

1. Product Capacity Planning

A product capacity plan guarantees that you'll have enough items or materials to meet your deadlines. Flowers, vases, and cards are examples of what a florist could sell. This would include materials like chlorine that are essential to accomplish the work for a pool maintenance firm.

2. Workforce Capacity Planning

Workforce capacity planning guarantees that you have adequate team members and work hours to finish projects. This form of planning will also help you convey general business, resource, and personnel needs to key stakeholders, identify when you need to acquire more people, and establish how far ahead of time you should begin recruiting based on the length of your onboarding process.

3. Tool Capacity Planning

Tool capacity planning ensures that you have the right tools for the job. This includes any vehicles, assembly line parts, or machinery you'll need to make and distribute your goods.

- How to Start Capacity Planning

There are three basic steps to capacity planning.



1. Measure

To begin, you must assess your resource capacity. In a particular period, how many deliveries can each of your drivers make? How many orders can each of your vehicles accommodate? How long does your fleet manager spend planning 50 deliveries? Because the remainder of your strategy will be dependent on these statistics, it's critical to answer these questions as correctly as possible.

2. Analyze

Once you have reliable measurements, you may spend time reviewing the data to see if you have insufficient capacity, excess capacity, or if all available resources are being completely utilised. Making graphs will assist you in comprehending the statistics and making demand forecasting more straightforward.

3. Formulate

Taking all of the information you've acquired and making a strategy is the final stage. You can do the math to figure out how much it will cost to fund new initiatives or recruit a full-time employee vs hiring seasonal part-time staff. You might also evaluate the return on investment (ROI) for modernising machines or adding assembly lines to your manufacturing facility. The formulation step allows you to see what the expected outcomes of various alternatives are so that you can make the best selection possible.

- Capacity Planning Strategy

The process of determining the resources required by manufacturers to fulfil demand for their products or services is known as capacity planning strategy. The quantity of output in the form of products and services that manufacturers can generate to meet consumer demand is directly related to capacity.

Capacity planning methodologies may help businesses figure out how much raw materials, equipment, labour, and facility investment they'll need over time to satisfy

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future product demand. Clients' demands are not met immediately when capacity planning is lacking, and these customers may be lost to competitors.

A effective capacity planning approach aids in the efficient use of industrial resources. Excess capacity suggests that the manufacturer's money is being spent inefficiently, when it could be better spent elsewhere. Low capacity, on the other hand, refers to the manufacturer's incapacity to create exactly what the client needs at any given time.

Types of Capacity Planning Strategies

1. Lead Strategy



One of the more aggressive tactics employed is the Lead Strategy, which entails an upfront investment in more capacity that is required. Manufacturers are planning to expand their capacity ahead of time, even before demand grows. This takes care of any projected increases in demand. This approach is used by several firms to increase market share against competitors. When rivals are prone to inventory shortages, especially when demand skyrockets, this is also employed. The risk associated with the Lead Strategy is that if actual demand does not meet expected demand, producers will be left with surplus inventory to store.

2. Lag Strategy



The Lag Strategy is more cautious than the Lead Strategy in that it waits until present capacity is fully utilised before adding additional capacity. Manufacturers use this method to respond to a real increase in demand by increasing capacity after the existing operation has reached full capacity. Manufacturers escape the difficulty of holding surplus inventory in this scenario, but they risk losing consumers to competitors.

3. Match Strategy



The Match Strategy is often a hybrid of the Lead and Lag methods. This technique employs smaller incremental modifications to the manufacturers' capacity rather than boosting demand ahead of time or increasing demand after current capacity is depleted. This is done in response to the changing market conditions. Despite its complexity, most producers will find this to be a safer bet than the other Capacity Planning Strategies since it is considerably more risk-averse.

4. Dynamic Strategy



This strategy is significantly safer because it is based on forecasts. It entails increasing significant or small amounts of capacity ahead of time, based on real demand and sales projection numbers. Because it is data-driven, producers can plan their capacity objectives with more accuracy, avoiding wastage and capacity shortages. However, strong capacity planning tools that can produce accurate estimates are required for this technique.

- Capacity Planning Strategy Benefits

1. Monitor Operating Costs

Personnel, facilities, finances, production schedules, and supplies are all factors to consider while developing capacity planning strategies. This can help producers keep a close eye on all production expenses, especially during booms and busts. Manufacturers can effectively budget for forthcoming changes and deploy financial resources where they are required when they can forecast projected capacity demands. This can also aid in the development of essential supply delivery dates as well as shipping timetables for finished goods.

2. Ensure Adequate Availability

Manufacturers may guarantee they have the resources they need to deliver work even before a contract is signed by using a Capacity Planning Strategy. The Capacity Planning Strategy advises manufacturers on the scope of new projects that may be undertaken, as well as adequate resources to meet the demands. Manufacturers can acquire critical data points that properly estimate the probability of overtime based on existing work schedules using actionable analytics.

3. Maintain Production Cycles

With a competent Capacity Planning Strategy, manufacturers can maintain adequate production levels in accordance with projected business cycles. Seasonal demand swings may be forecasted using past data, and production capacity can be quickly increased to meet demand. The Capacity Planning Strategy also predicts when the economic cycle will decline, allowing seasonal labour to be hired and wasteful costs to be avoided.

4. Identify Skill Gaps

Adequate capacity planning may aid in identifying the appropriate skills needed to complete critical projects and preparing for any talent shortages ahead of time. Manufacturers may plan work appropriately, predict skill requirements, and decide whether to utilise in-house or outsourced resources. Manufacturers may simply plan staff training requirements and determine how future projects will be delivered.

5. Plan New Production Facilities

As your business expands, you may need to expand your manufacturing facilities. You may generate a more accurate prediction of demands for facilities and people levels, as well as the type of output that can be expected from the new location, by using capacity planning information from your existing locations. This is a useful tool for creating a business plan and budget for your company's expansion.

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6. Meet Operations Budget

Manufacturers can satisfy demand with the least amount of waste and boost utilisation rates when they employ the right capacity planning tools. It also assists them in meeting their financial needs based on expected sales or demand forecasts, as well as reducing additional costs.

Capacity Planning techniques can aid in improving operational performance and meeting production objectives. You may find yourself in a problem if your Capacity Planning Strategies are not tailored to your company's business model.

- Industries That Utilize Capacity Planning

Capacity planning is typically used across three primary industries-

- Production planning and manufacturing capacity
- Planning service capacity
- Planning human capacity

1. Manufacturing Industries

Capacity planning principles are used by many medium and large-scale industrial enterprises to determine overall production capacity. This is utilised in a variety of sectors, including automotive, appliance manufacture, process industries, pharmaceuticals, and semiconductor production.

2. Service Industries

Because service sectors have particular issues due to the fact that they cannot be kept like products, capacity planning is utilised to match supply and demand. Customers will receive the proper degree of service at the right time as a result of this. Cloud computing services, aircraft seat capacity, and fast-food restaurants are just a few examples.

3. Human Capacity

Organizations that offer unique abilities of their team members fall under the category of human capacity. Project management, technical service technicians, and data centres are examples of this. Financial auditing firms, legal firms, and engineering project firms are among the many enterprises that leverage human capability.

- Ways to Develop an Effective Planning Process

The nine major steps associated with implementing a sound capacity planning process.

1. Select an appropriate capacity planning process owner.
2. Identify the key resources to be measured.
3. Measure the utilizations or performance of the resources.
4. Compare utilizations to maximum capacities.
5. Collect workload forecasts from developers and users.

6. Transform workload forecasts into IT resource requirements.
7. Map requirements onto existing utilizations.
8. Predict when the shop will be out of capacity.
9. Update forecasts and utilizations.

Step 1: Select an Appropriate Capacity Planning Process Owner

Selecting a properly competent individual to act as the process owner is the first step in building a comprehensive capacity planning process. This individual is in charge of the process's design, implementation, and maintenance, and has the authority to negotiate and delegate with developers and other support groups.

First and foremost, this person must be able to communicate successfully with developers, as accurate information and constructive feedback from developers to infrastructure planners is critical to the success and credibility of a capacity plan. This individual must also be familiar with systems and network software and components, as well as software and hardware configuration.

In choosing the capacity planning process owner, some other medium- and lower-priority attributes are advised (see table below). These characteristics, as well as their priorities, clearly differ from one store to the next, based on the sorts of apps and services available.

Step 2: Identify the Key Resources to be Measured

One of the first duties for the process owner after being chosen is to identify the infrastructure resources that need to have their utilisation or performance assessed. This decision is based on current information regarding which resources are most important for achieving future capacity requirements. In many companies, these resources concentrate around network bandwidth, server processor count and performance, or the number, size, and density of centralised secondary storage disc volumes.

The following is a more comprehensive list of potential resources:

1. Network bandwidth
2. Centralized disk space
3. Centralized processors in servers
4. Channels
5. Tape drives
6. Centralized memory in servers
7. Centralized printers
8. Desktop processors
9. Desktop disk space
10. Desktop memory

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Step 3: Measure the Utilizations or Performance of the Resources

The usage or performance of the resources indicated in Step 2 should now be monitored. Two important pieces of information are provided by these measures.

1. A usage baseline that may be used to forecast and assess future patterns.
2. The amount of available extra capacity for each component.

A crucial server, for example, may be used at 60% of its capacity on a daily basis during peak hours. To do trending analysis, these daily statistics may be averaged and displayed on a weekly and monthly basis.

The usage of resources is often measured using a variety of approaches. To the total usage matrix, each tool adds a separate component. Processor and disc channel utilizations might be provided by a single tool. Another may offer statistics on disc space consumption, while a third may reveal how much of that space is really used by databases.

This last gadget has the potential to be really useful. Database managers frequently pre-allocate databases to a size that they believe will enable expansion over a reasonable amount of time. Knowing how full certain databases are and how fast they are filling up gives you a more realistic view of disc space use. This information is frequently only known by database administrators in contexts where computers are utilised as database servers. In these situations, it's critical to create an open line of communication between capacity planners and database managers, as well as to have access to a tool that can give this critical data.

Step 4: Compare Utilizations to Maximum Capacities

The goal is to figure out how much more capacity is available for specific components. Each component's usage or performance should be compared to its maximum useable capacity. It's worth noting that the maximum useable is virtually never the maximum possible. For example, the maximum useable server capacity is generally only 80 to 90%. Network bandwidth and cache store hit ratios are both limited in the same way. The process owner should now be able to forecast when a specific resource's surplus capacity will be exhausted by extrapolating usage trends statistics and comparing them to the maximum useable capacity.

Step 5: Collect Workload Forecasts from Developers and Users

This is one of the most important elements in the capacity planning process, yet it's also the one you have the least influence over. Developers are frequently requested to assist users with the completion of IT workload projections. The outcome is only as good as the input in this case, as it is in many others. Working with developers and a few select pilot users to create a basic yet effective worksheet can help make this process go more smoothly. A sample user workload forecast spreadsheet may be found here. This should be as personalised as feasible to match the specific needs of your specific area.

Step 6: Transform Workload Forecasts into IT Resource Requirements

Following the collection of workload projections, the expected changes must be

converted into IT resource needs. Advanced measuring tools or the skills of a senior analyst can aid in converting expected transaction loads, for example, into greater server processing capacity. The spreadsheets also allow you to project the time ranges in which workload increases are expected to occur. It's a good idea to use the performance centres that major vendors of servers, database software, and corporate applications now provide for big application workloads.

Step 7: Map Requirements onto Existing Utilizations

The estimated resource requirements resulting from the users' workload estimations in Step 6 are now shown on the excess utilisation charts from Step 4. This diagram depicts the amount of new capacity required by each component to fulfil predicted demand.

Step 8: Predict When the Shop Will Be out of Capacity

The mapping of the amount of extra capacity necessary to satisfy expected workload needs also pinpoints the time range in which these improved resources will be needed.

Step 9: Update Forecasts and Utilizations

Capacity planning is not a one-time event, but rather a continuous process. The plan's full usefulness comes from keeping it up to date and revising it on a regular basis. At least once a year, the strategy should be modified. The shops that do the best with this system are those who alter their strategy every quarter. When assessing resource needs for new applications, the production acceptance process employs a type of capacity planning.

- **Capacity Planning Challenges**

Most manufacturers go through five common capacity planning challenges, irrespective of the operational strategy used. These challenges could impact the production flow no matter the level or scale or the complexity of the organization.

1. Siloed Data and Collection Method

To plan production capacity, traditional industrial procedures have primarily depended on siloed data. Separate, disconnected systems need human data reconciliation prior to consumption. As a result, more time is used, and data may become obsolete before it can be applied and used. A disconnected system can be problematic because it increases reliance on manual reporting and human competence to recognise and manage trends, since most capacity planning systems rely on inputs from demand predictions, supply chain, warehouse management, and other sources.

2. Inferior Data Quality

Manufacturing activities frequently generate capacity data in the form of records and reports, which must then be manually aggregated before the final data can be used. After that, planners include the critical supply and demand data and devise a formula to determine available capacity. If any of these various data inputs are faulty or obsolete at any point throughout the process, or if they exist in several forms, they must

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be formatted and standardised anew before they can be used for planning reasons. Because the majority of these data points are unconnected, any subsequent rounds must repeat the process.

3. Complex Formulae and Calculations

To arrive at the final capacity plan, planners apply a variety of complicated formulas and computations. This can contain things like material availability, work centre load, substitute sourcing, attribute-based planning criteria, and so forth. Manual spreadsheets have long been the “go-to” for planners to compute various aspects to fit all of this, as correct calculations are critical. If data input mistakes or inaccurate data are present throughout this process, the overall capacity design may be incorrect. Furthermore, there is a lag depending on the time it takes to gather the data, and new information and modifications must be entered into many sources, delaying the time it takes to create a plan and increasing the chance of mistakes.

4. Multi-level Planning

Capacity planning occurs at many levels in most production contexts. Rough-cut planning is normally done at the master schedule level and is used for short-term planning, which might last anywhere from a week to two months. A 12- to 18-month planning window is used in medium or aggregate planning to give a longer view that helps the company to guarantee that demand can be satisfied in the long run. It also aids in the smoothing of supply chain issues by reducing manufacturing costs. Because they are utilised for many decision-making activities, each of these tiers requires greater data sets and longer time periods. As a result, the problems of data collecting, data quality, equations, and computations are compounded in complexity, thereby increasing the risk of mistakes.

5. Gap in Communication

Because the capacity planning process comprises so many dynamic and moving pieces, and because so few of them are interconnected, there is a danger of a communication breakdown, which might jeopardise the capacity plan's integrity. This is true both with suppliers and internally, where isolated buying systems may not connect with those in production or scheduling. This can result in a reduction in collaboration, leaving capacity planners vulnerable to new data, missing data, or inaccuracies in current data.

● Capacity Requirement

Capacity Requirement Planning: CRP (capacity requirement planning) is a strategy for determining what equipment and labour/personnel capabilities are needed to satisfy production targets (i.e., product volume) according to the master production schedule and material requirement planning (MRP-I).

Capacity Requirement Planning Strategies: Two types of capacity planning strategies used are:

- (i) “Level capacity” plan and
- (ii) “Matching capacity with demand” plan.

The “level capacity” strategy is based on “produce-to-stock and sell” techniques, in which production systems are run at consistent levels and completed products inventories grow and decrease depending on whether production exceeds demand or vice versa from period to period (say every quarter).

“Matching Capacity with Demand” Plan: Production capacity is matched to demand in each time in this strategy (weekly, monthly or quarterly demand). To meet demand, material flows and machine capacity are usually altered from quarter to quarter. Low quantities of completed products inventory result in lower inventory carrying costs, which is one of the key benefits. In addition, the cost of back-ordering is decreased. Because of the frequent changes in the workforce, the drawbacks include high labour and material expenses (hiring, training and lay-off costs, overtime or idle time cost or subcontracting costs).

Optimum Plant Capacity:

Plant capacity has a significant impact on production costs. As production volumes increase, economies of scale emerge, resulting in lower average costs per unit produced. There is an ideal amount of output per year for a specific manufacturing facility that results in the lowest average unit cost. The plant's “best operating level” is defined as this level of output. Average unit costs decrease as the volume of output in a given production facility expands outward from zero.

These declining costs are because of the following reasons:

- i) Fixed costs are spread over more units produced,
- ii) Plant construction costs are less,
- iii) Reduced costs of purchased material due to quantity discounts for higher volume of materials purchased and
- iv) Cost advantages in mass production processes.

Longer production runs (i.e., bigger batch quantities of items produced) have lower setup costs per unit of product produced, less scrap, and so on, resulting in cost savings per unit. This is referred to as “scale economies.” However, the cost per unit will be reduced only up to a specific amount of manufacturing. Additional output volumes over this level result in a rising average unit production cost.

Higher congestion of supplies and personnel reduces production efficiency, as well as other variables such as scheduling difficulties, damaged goods, lower employee morale owing to severe work pressure, increased usage of overtime, and so on, resulting in “diseconomies of scale.” As a result, the plant capacity should be such that the optimal level of output is attainable, resulting in the lowest average cost of production per unit. The term “optimum plant capacity” refers to this plant capacity.

- **Balancing the Capacity**

Due to variations in product-mix, the load on different machinery and equipment varies in companies that manufacture a variety of items (a product line or a product-mix). There will be an imbalance between the work loads of different machines if the output rates of different machines do not meet the required output rate for the items to be produced.

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As a result, certain machines or equipment will become a “bottleneck work centre,” restricting plant capacity and, as a result, increasing production costs per unit. To address the issue of machine imbalance, extra machines or equipment are introduced to the bottleneck work centre to boost its capacity and bring it in line with the capacity of other work centres.

Adding new machines or equipment to bottleneck work centres to alleviate capacity imbalances between work centres has been proven to be more cost-effective than giving workers working in bottleneck centres excessive overtime, which raises production costs. Subcontracting surplus work load of bottleneck centres to outside suppliers or subcontractors is another way to eliminate imbalance. Another technique to balance capacity is to adjust the product mix by changing sales for various goods to arrive at an appropriate product-mix that evenly loads all work centres.

- **Factors Influencing Effective Capacity**

The effective capacity is influenced by –

- (1) Forecasts of demand,
- (2) Plant and labour efficiency,
- (3) Subcontracting,
- (4) Multiple shift operation,
- (5) Management policies.

Forecasts of Demand: The capacity plan will be heavily influenced by the demand prediction. As a result, accurately forecasting demand is challenging due to the fact that it varies greatly depending on the stage of the product life cycle and the quantity of items available. When opposed to products with a shorter lifespan, products with a longer lifecycle generally see stable demand growth. As a result, prediction accuracy has an impact on capacity planning.

Plant and Labour Efficiency: Plant and equipment efficiency of 100 percent is challenging to achieve. Because of forced idle time owing to equipment breakdowns, scheduling delays, and other factors, efficiency is less than 100 percent. The efficiency of a plant varies from one piece of equipment to the next, as well as from one organisation to the next. Overall capacity utilisation is aided by labour efficiency. The industrial engineer's standard time is for a representative or typical worker. However, the pace and efficiency of the real labour varies. When assessing efficiency, the real efficiency of the labour should be taken into account. As a result, plant and labour efficiency are critical for achieving realistic capacity.

Subcontracting: Subcontracting is the process of offloading part of the work to outside suppliers in order to hire the capacity needed to satisfy the organization's needs. It is necessary to conduct a thorough study to choose whether to make or buy. To make the decision, an economic comparison between the cost of making the component vs the cost of buying the component must be done.

Multiple Shift Operation: The firm's capacity utilisation will be improved by operating in several shifts. However, the rejection rate is greater in the third shift. Multiple shifts are suggested, especially in process sectors where investment is substantial.

Management Policy: The capacity planning will be influenced by management policies such as subcontracting, multiplicity of shifts (determination about how many shifts to operate), which work stations or departments to run for third shift, machine replacement policy, and so on.

- **Factors Favouring Over Capacity and Under Capacity**

Demand forecasting is extremely challenging since demand is usually accompanied with uncertainty. Demand forecasts will be either greater or lower than actual demand. As a result, there is always a risk when building capacity based on expected demand. This results in either an overabundance or an underabundance of capacity.

The over capacity is preferred when:

- (a) The capacity's fixed cost is not too high.
- (b) Due to design confidentiality and/or quality requirements, subcontracting is not possible.
- (c) Increasing capacity takes a long period.
- (d) The business cannot afford to miss the delivery deadline or lose the customer.
- (e) The plant's economic capacity size is below which it is not cost-effective to operate.

The under capacity is preferred when:

- (a) Fixed cost of the capacity is very high.
- (b) Shortage of products does not affect the company (i.e., lost sales can be compensated).
- (c) The technology changes fast, i.e., the rate of obsolescence of plant and equipment are high.
- (d) The cost of creating the capacity is prohibitively high.

Illustration:

A department works on 8 hours shift, 250 days a year and has the usage data of a machine, as given below:

Product	Annual demand (units)	Processing time (standard time in hours)
X	300	4.0
Y	400	6.0
Z	500	3.0

Determine the number of machines required.

Solution:

Step 1: Calculate the processing time needed in hours to produce product x, y and z in the quantities demanded using the standard time data.

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Step 2: Annual production capacity of one machine in standard hours = $8 \times 250 = 2000$ hours per year

Step 3: Number of machines required

$$= \frac{\text{Work load per year}}{\text{Production capacity per machine}} = \frac{5100}{2000} = 2.55 = 3 \text{ machines}$$

- **Adjusting Capacity**

Inefficiencies emerge from discrepancies between an organization's capacity and its customers' requests, resulting in underutilised resources or disgruntled consumers. The former may be a significant expense, but the latter may result in lost revenue, customers, and reputation. These differences will be minimised if the operation is able to alter essential resources.

Some capacity adjustments can be implemented practically instantaneously, while others may take longer. An operation's capacity is a sophisticated blend of resources. These resources are process inputs that allow capacity to be increased or decreased by altering the process's inputs. The resource's flexibility is determined by how rapidly it can be changed.

Capacity may be improved by a variety of strategies that entail altering an organization's resources and inputs, such as:

- Introducing new methodologies and materials
- Expanding the number of service providers or machines available
- Increasing the number of hours available for operation
- Acquiring additional facilities.

Decreasing capacity can be more difficult or expensive; it tends to rely on the operations ability to sell or reduce resources as cost effectively as possible. There are usually costs involved in reducing resources, for example if the resource is staff there may be redundancy costs, or closing facilities may incur significant costs.

Therefore, the decision to alter capacity has to be taken carefully in line with future predictions of demand.

- **Capacity Constraints**

A capacity limitation is a resource that is less competent than other sections of the operation of increasing throughput during a particular time period. On a production line, a number of machines may be in order, but one may not be able to process as many units per hour as the others. This underproducing equipment will limit capacity, perhaps causing a 'bottleneck' in the process. The overall capacity of the facility will be increased by expanding the capacity of this machine.

The lowest generating element of the process constantly limits capacity. An operation will 'always go at the pace of the slowest walker,' in layman's words. The total capacity of the operation will be improved by identifying a constraining portion of the process and adding resources to raise its capacity.

The following are examples of resource combinations that might be restricting to an operation:

- **Staff/Skill Levels:** Over time, employees can be educated to be more flexible in their contributions to the process. The operation can benefit from the learning curve, in which a new person becomes more efficient at a specific procedure and hence faster at their work, increasing the operation's capacity.
- **IT Facilities/Technology:** This might be a little or major enhancement to a process. ICT investment may shorten process times or perhaps change the character of the process entirely. By decreasing the number of employees necessary to conduct a transaction and thereby dramatically boosting the bank's ability to interact with its clients, online banking has been a big development in the financial sector.
- **Materials Availability:** A change in raw material supply might boost an operation's capacity potential. The capacity might be enhanced if there is a constraint in the supply of materials or a timing problem that is resolved.
- **Product or Service Mix:** Adjustments to the facility's other goods or services might limit the operation's capacity. Because various goods and services utilise varying amounts of resources per unit, a change in the product mix might lead to a change in capacity.
- **Storage:** If there is a resource limitation in the process that is impacted by time, this might have an impact on the operation's capacity. In the near term, if the operation has the ability to store work in progress or finished items, it can increase the capacity of the process. The ability to hold items and enable the full capacity of the business to flow can help to alleviate demand swings and variations.
- **Working Schedules and Access to Facilities:** This can potentially affect the capacity's full availability. A lecture theatre with a capacity of 100 students might function outside of normal business hours; however, employees and students may have a problem with 6 a.m. lectures!

To determine their capacity to change over time, these elements should be assessed in a short-term, medium-term, and long-term time frame. To cope with the extra demand, a short-term strategy for expanding capacity in a cafe would be to add a few extra tables outside or extend staff working hours. In the medium term, the cafe owner would have more options available to increase capacity, such as hiring more staff or having additional cooking facilities in the kitchen. Long-term, the possibilities are significantly bigger; the premises might be enlarged, better equipment installed, and more people hired, among other things. The more time an operation has to plan, the more possibilities are available to it.

4.1.2 Concepts Aggregate Planning

Aggregate Planning

The act of establishing, assessing, and maintaining a preliminary, estimated timetable of an organization's total operations is known as aggregate planning. Targeted sales estimates, production levels, inventory levels, and customer backlogs are all part of the overall strategy. This timetable is designed to meet the demand predicted

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at the lowest possible cost. When done correctly, aggregate planning should reduce the impact of myopic, day-to-day scheduling, in which little amounts of material are ordered one week, with people laid off, and bigger amounts are bought the next week, with workers rehired. This longer-term view on resource consumption can assist reduce short-term need fluctuations and save money.

Aggregate planning, to put it simply, is an attempt to balance capacity and demand in such a way that expenses are kept to a minimum. Because planning at this level covers all resources “in the aggregate,” such as a product line or family, the word “aggregate” is employed. The entire number of people, hours of machine labour, or tonnes of raw materials are all examples of aggregate resources. Gallons, feet, and pounds of output, as well as aggregate units occurring in service sectors such as hours of service supplied, number of patients seen, and so on, are examples of aggregate units of output.

Aggregate planning makes no distinction between sizes, colours, features, or other factors. For example, aggregate planning in the automobile industry would evaluate the overall quantity of automobiles anticipated rather than specific models, colours, or extras. When determining aggregate units is problematic (for example, when output variation is severe), comparable units are generally used. These comparable units might be determined by value, cost, or labour hours, among other factors.

Aggregate planning is classified as intermediate-term (rather than long- or short-term). As a result, most aggregate plans are for three to 18 months. Future short-term planning, such as production scheduling, sequencing, and loading, is built on the basis of aggregate plans. The aggregate plan has been characterised as “disaggregated” in the master production schedule (MPS) used in material requirements planning (MRP).

The process of creating an aggregate plan begins with determining demand and determining current capacity. Capacity is defined as the total number of units that can be produced in a given time period (this requires that an average number of units be computed since the total may include a product mix utilising distinctly different production times). The total number of units required is indicated as demand. If the two aren't balanced (equal), the company must determine whether to raise or decrease capacity to meet demand or demand to meet capacity. A variety of approaches are available to do this.

Options for situations in which demand needs to be increased in order to match capacity include:

1. Pricing. Changing price to boost demand at times when demand is lower than peak. For example, matinee movie theatre costs, off-season hotel rates, weekend telephone service rates, and pricing for commodities with seasonal demand.
2. Promotion. Demand is shifted via advertising, direct marketing, and other types of promotion.
3. Back Ordering. Demand is transferred to a period when capacity is not completely used by deferring delivery on present orders. This is only a method of demand smoothing. In order to discourage walk-in clients, service sectors might calm demand by accepting reservations or arranging appointments.

This is referred to as “partitioning” demand by some.

4. **New Demand Creation.** A new, but complementary, demand for a product or service is developed. When clients are forced to wait at a restaurant, they are typically steered to the bar, which is a complimentary (but not complimentary) service. The installation of video arcades into movie theatres and the development of services at convenience shops are two further examples.

Options which can be used to increase or decrease capacity to match current demand include:

1. **Hire/Lay off.** Firms can maintain a balance between capacity and demand by recruiting extra workers as needed or laying off workers who are not currently required to fulfil demand.
 2. **Overtime.** Firms can produce a temporary boost in capacity without the added expenditure of recruiting new workers by requesting or forcing people to work extra hours per day or an extra day per week.
 3. **Part-time or Casual Labour.** Using casual labourers or temporary workers (workers who are considered permanent but only work when needed, on an on-call basis, and typically without the benefits given to full-time workers).
 4. **Inventory.** In periods of low demand, finished-goods inventory can be built up and then utilised to satisfy demand during periods of strong demand. No additional employees are required, no temporary or casual labour is required, and no overtime is required.
 5. **Subcontracting.** Frequently, companies prefer to have a different manufacturer or service provider offer the product or service to the subcontracting company's clients. Additional capacity is temporarily achieved by subcontracting work to a different supplier.
 6. **Cross-training.** Employees that have been cross-trained may be able to do jobs in many operations, giving them more flexibility when it comes to scheduling capacity.
 7. **Other Methods.** While well-known options such as adjusting labour size and utilisation, inventory build-up/backlog, and subcontracting are well-known, there are other, more creative approaches that are used in business. Sharing staff with counter-cyclical organisations and striving to identify exciting and relevant tasks for employees to complete during slow periods are two of these approaches.
- **Aggregate Planning Strategies**

There are two pure planning strategies available to the aggregate planner: a level strategy and a chase strategy. Firms may choose to utilize one of the pure strategies in isolation, or they may opt for a strategy that combines the two.

1. Level Strategy

A level strategy aims to create an overall plan that maintains a constant pace of output and/or employment. In order to meet changing client demand, the company must increase or decrease inventory levels in anticipation of higher or lower predicted

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demand. When demand is low, the company maintains a consistent staff and output pace. This permits the company to stockpile more goods than is currently required. As demand rises, the company is able to maintain a constant production rate/level of employment while allowing the inventory excess to absorb the additional demand.

The utilisation of a backlog or backorder is another option. A backorder is just a commitment to deliver the goods when it becomes more easily accessible, generally when capacity begins to catch up with falling demand. In essence, a backorder is a mechanism for shifting demand from one period to the next, preferably one with lower demand, so smoothing demand requirements across time.

A level approach allows a company to keep its output constant while yet meeting demand. From the aspect of employee relations, this is ideal. Excess inventory expenses, subcontracting or overtime charges, and backorder costs, which are often the cost of expediting orders and the loss of consumer goodwill, are all negative outcomes of the level approach.

2. Chase Strategy

In a chase approach, demand and capacity are matched period by period. This might result in a large number of people being hired, fired, or laid off; insecure and disgruntled personnel; higher inventory carrying costs; labour union issues; and unpredictable plant and equipment use. It also shows that the company is willing to be flexible. The main benefit of a chase strategy is that it allows inventory to be retained at the lowest level feasible, which may save a lot of money for some businesses. When it comes to aggregation planning, most companies that embrace the just-in-time production idea use a chase strategy.

Most businesses find it beneficial to combine the level and chase strategies. A combination strategy (also known as a hybrid or mixed strategy) can be utilised to better accomplish company goals and policies while also lowering expenses as compared to either of the pure techniques employed alone.

- Importance of Aggregate Planning

Aggregate planning is a tried-and-true method for adding predictability and stability to production. It assists management in achieving a company's long-term goals.

The following are some of the benefits of aggregate planning:

- Creates a satisfied and happy staff
- Reduces changes in the workforce's levels
- Assists in determining short-term resources
- Assists in maximising space use
- Achieves a company's overall aims and objectives
- Assists in capacity adjustment to meet demand
- Cuts inventory stocking expenses to the minimum.
- Decrease investments in diverse inventory.
- Aligning demand with supply and decreasing client wait times to improve customer service.

- Provides a higher level of consumer value.
- Appropriate use of production facilities
- Getting the most out of various types of equipment
- Reducing the pace of change in production.
- Reduces variable costs and improves the financial statement's bottom line, allowing a company to achieve its business objectives.
- Techniques for Aggregate Planning

Aggregate planning strategies range from informal trial-and-error methods that often use basic tables or graphs to more structured and complicated mathematical methods. An informal but helpful trial-and-error technique for aggregate planning is described in outline form in William Stevenson's textbook Production/Operations Management.

The steps in this broad method are as follows:

1. Calculate the demand for each time period. Each period's need should be determined.
 2. Calculate capacity for each time period. This capacity should be able to meet demand, which may include overtime or subcontracting.
 3. Identify any relevant business, departmental, or union policies. Maintaining a certain level of safety stock, maintaining a reasonably stable workforce, backorder policies, overtime policies, inventory level policies, and other less explicit rules such as the nature of employment with the individual industry, the risk of a bad image, and the loss of goodwill are just a few examples.
 4. Calculate the unit costs for the units produced. The fundamental production expenses are usually included in these prices (fixed and variable costs as well as direct and indirect labour costs). Changes in capacity and their related expenses are also covered. Inventory holding expenses, as well as storage, insurance, taxes, spoiling, and obsolescence charges, must all be addressed. Finally, expenses for backorders must be calculated. While this is difficult to quantify, it often involves expediting expenses, customer goodwill loss, and income loss from cancelled orders.
 5. Create alternate plans and cost estimates for each.
 6. If a suitable plan emerges, choose the one that best meets the goals. This happens a lot.
- Mathematical Approaches to Aggregate Planning

The following are some of the better-known mathematical techniques that can be used in more complex aggregate planning applications.

1. Linear Programming

Linear programming is an optimization approach that allows a user to discover the most profit or income or the lowest cost depending on the availability of limited resources and specific constraints. The Transportation Model is a sort of linear programming that may be used to generate aggregate plans that balance capacity

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and demand while minimising costs. However, only a small percentage of real-world aggregate planning decisions are consistent with linear programming's linear assumptions.

2. Mixed-integer Programming

Mixed-integer programming may be beneficial for aggregate plans generated on a product family basis, where the plan is effectively the total of the plans for separate product lines. A technique for calculating the number of units to be produced in each product family may be found in mixed-integer programming.

3. Linear Decision Rule

Another strategy for optimization is the linear decision rule. It uses a collection of cost-approximating functions (three of which are quadratic) to generate a single quadratic equation in order to reduce overall production costs (labour, overtime, hiring/layoff, inventory carrying cost). The quadratic equation may then be deduced into two linear equations, one to plan output for each period and the other to plan workforce for each period, using calculus.

4. Search Decision Rule

Some of the constraints of linear programming's linear cost assumptions are solved by the search decision rule approach. The user can specify cost data inputs in very broad terms using the search decision rule. It necessitates the creation of a computer software that will explicitly assess the cost of any manufacturing plan. It then compares other plans to find the one with the lowest cost. Unlike linear programming, however, there is no guarantee of optimality.

5. Simulation

For aggregate planning, a variety of simulation models can be utilised. It is possible to evaluate an aggregate plan in the context of a simulation model under a range of scenarios in order to discover suitable plans for consideration. These models may also be used as part of a decision support system to help design and evaluate different control plans. These models can use diverse quantitative measurements of productivity, customer service, and flexibility to combine the numerous conflicting objectives inherent in industrial strategy.

• Problems in Aggregate Planning

Aggregate planning is a short- to medium-term strategy for a certain department inside a company. It isn't part of a larger planning approach for achieving company objectives. Rather, it's about balancing demand and supply by adjusting the workforce, output rate, and inventory levels. As a result, it faces a number of issues, including:

1. Smoothing

The expense of altering inventories, relying on backorders, and temporarily employing or laying off personnel is referred to as smoothing. Anything that isn't typical of day-to-day operations has a price tag attached to it. Aggregate planning might be an extra expense for the company.

2. Planning Horizon

Aggregate planning is tied to a three- to twelve-month time frame. As a result, it's critical to identify the exact time frame ahead of time or well in advance in order to keep track of what has to be done. Organizations must define the levels at which they must be maintained in terms of staff and inventory. This might necessitate a lot of preparation on their side.

3. Bottleneck

It's not always black and white when it comes to shifting demand. Even if businesses estimate demand at a specific level, this may not be correct. As a result, bottleneck planning refers to the inability to meet demand owing to capacity constraints.

Forecasting and assumptions are used to develop aggregate planning methods. This necessitates an understanding of problem logic. There are various methods to break down problems into smaller bits by analysing them.

4.1.3 Concepts of Master Production Scheduling

Master Production Scheduling is a method of assisting manufacturers in determining which items and quantities to create during specific time periods. In terms of what is made and what resources are obtained, MPS is proactive in that it directs the manufacturing process.

The three parts of inventory planning tripod are Bills of Materials (BOM) and Material Requirements Planning (MRP) and Master Production Schedule (MPS).

Master Production Schedule

Definition

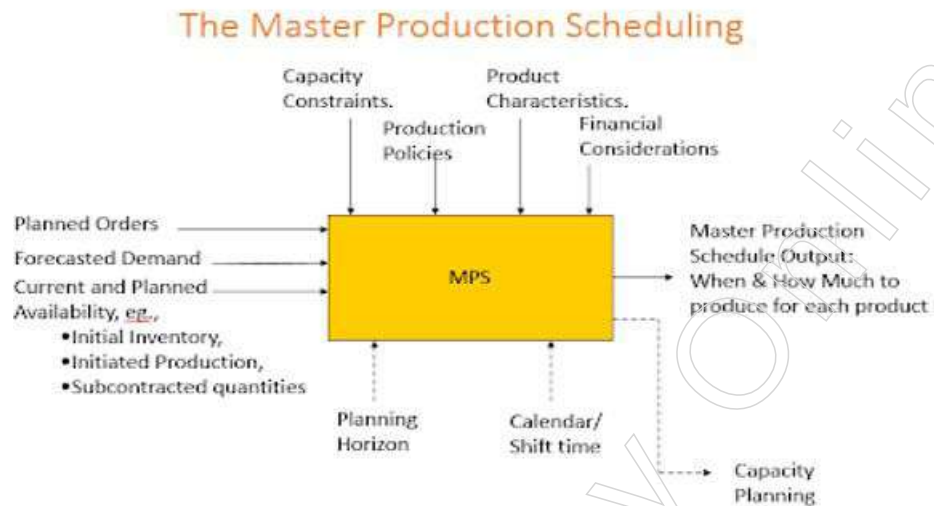
The MPS, as the name suggests, is in charge of deciding what items are created and when. The completed goods BOM then identifies the needed raw materials, and the data is combined with current inventory data to form the MRP for raw materials procurement.

The Master Production Schedule provides the foundation for sales and manufacturing collaboration. When sales and production use the MPS as a contract, sales may make genuine order guarantees. The MPS is not a set-in-stone strategy. MPS is a flexible plan that may be adjusted in response to changes in demand or capacity.

MPS is a Crucial Planning Function

MPS often provides a critical planning function as part of a fully integrated ERP system, extracting real supply and demand data, as well as projections, to create accurate and timely production plans that assist manufacturers meet their output goals and save procurement costs. In its estimates, MPS also considers the plant's manufacturing capacity. The MRP process is started when the production orders have been examined and authorised, and purchase orders may be created. MPS also protects against resource shortages, unforeseen scheduling snafus, and poor resource distribution.

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- **Different Master Production Scheduling Techniques**

The MPS must plan for what will be made at the highest degree of efficiency. It will be difficult to put the plan into action efficiently if the MPS includes too many items. The output, on the other hand, will suffer if the strategy is not comprehensive enough. As a general rule, master scheduling should take place when there are the fewest number of product options.

- In a make-to-stock environment, a small number of things, such as video recorders or computers, are constructed from a greater number of components. In this situation, the MPS should be a list of finished products items.
- In a make-to-order environment, many distinct finished items can be created or produced from a relatively small number of raw materials. The manufacturing of automobiles is a fantastic illustration of this procedure. Many automotive subassemblies will be the same, however various types of cars will have distinct choices such as colour, audio, sunroof, and electronic windows. In a make-to-order scenario, the MPS is a list of real client orders.
- Assemble to order environments raw materials are used to form basic components and complete subassemblies. A variety of final products are made out of these components and subassemblies. As a result, the Master Production Schedule should be created at the subassembly level.

Each of these choices demands a somewhat different focus on the Master Production Schedule. In each situation, the timetable will be based on the fewest amount of product selections possible, as shown below:

Supply Demand Options	MPS Focus
Make-to-Stock (MTO)	Schedule finished goods
Make-to-Order (MTO)	Schedule raw materials
Assemble-to-Order (ATO)	Schedule module production

- **Relationship between MPS and MRP**

- The Material Requirements Plan is driven by the Master Production Schedule (MPS) (MRP). MPS can identify which components are required

from production and which components must be acquired using the bills of material. The MPS is a manufacturing priority plan.

- While MPS and MRP have several features in common, such as the capacity to produce planned manufactured goods and purchase and transfer orders, they differ in a key way. Unlike MRP, MPS, for example, focuses its planning capabilities on the creation of completed items, components, or parts that create the most profit for the manufacturer and hence account for the majority of the resources required for production.
- Another distinction is that the MPS can only be used at one level of an item's BOM, but MRP may be used at any level. MRP is primarily concerned with fulfilling demand for component parts or subassemblies, whereas MPS is more concerned with generating production plans to match real demand for finished goods as well as expected customer delivery dates.
- MPS plans are often based on real-time sales orders, service orders, available resources, inventory levels, and capacity restrictions or predictions. Manufacturers can use these projections to predict product demand and change production schedules depending on seasonality, promotions, and fluctuating demand for certain components and/or completed goods. The quantities of an item to be produced, due dates, and quantities accessible to promise are all included in MPS' output.
- Benefits of a Master Production Schedule

There are multiple benefits to introducing an MPS in a manufacturing business:

- It gives a strong foundation on which to construct, improve, and track the sales forecast.
- It gives you a strong foundation for establishing the inventory levels you want.
- As part of the next step of Material Requirements Planning, it offers a firm foundation for determining the quantities of parts, subcomponents, or raw materials to acquire or make.
- It gives you a strong foundation for determining the quantity of work and shifts you'll need.
- It enables the plant's installed capacity to be optimised and the load to be balanced.
- The manufacturing department can calculate the expenses of production and maintenance for the workstations.
- The company's finance department may use the MPS to estimate projected revenues and costs and provide a cash flow projection. This, among other things, will aid in the development of investing strategies.
- The MPS can be used by the human resources department to forecast labour shortages.

The MPS should be as near to the business strategy as feasible. This necessitates a continuous updating from all corporate areas.

If the marketing department is planning a sales campaign, for example, the increased demand must be accounted for in the MPS and forecast. The MPS and

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projection should be updated if the sales staff discourages the sale of one product line in favour of another.

- **Functions of Master Production Scheduling (MPS)**

Master Production Scheduling's automated software solution has much too many features. It is typically regarded as a good practise to study the important components of this computerised system that aid production before adopting the automated system of Master Production Scheduling.

Here are the critical functionalities of the Master Production Scheduling automated system:

1) Transforming Plans

This section of the programme estimates how much material, labour, and specialised equipment will be needed to reach the manufacturing target.

2) Provide Alternative Production Routes

The automated Master Production Scheduling system generates a trial-and-error schedule that shows a different way to complete the production.

That would deal with any unanticipated mishaps that could occur throughout the production process.

3) Utilization of Available Resources

Automated software of Master Production Scheduling establishes links between the load and utilization for the machinery and equipment used in the manufacturing process. This enables for the most efficient use of all available resources and a smoother production flow.

4) Establish Capacity Requirement

Capacity planning is made easier with the use of automated tools for Master Production Scheduling. Manufacturers can use Master Production Scheduling to get the information they need regarding capacity requirements.

5) Aid in Information Processing

MPS chooses the most appropriate delivery time. It communicates with the various management information systems in a consistent manner.

- **Master Production Schedule Problems**

Master production schedules come with unique problems. The most notable include:

1) Disputed MPS

A master production schedule is subject to debate. Some stakeholders (various departments) can disagree with the data that was utilised to create the schedule. A deadlock or other complications, such as creating the wrong parts and quantities, are likely if there isn't a strategy that everyone can agree on.

2) Poor Sales and Operational Planning

Master production schedules are only as good as the data that went into creating them. The greatest MPS won't mean much if sales and operations aren't balanced and done correctly.

3) Rigid MPS

Master production schedules must also be flexible. Because demand and supply fluctuate on a regular basis, the MPS must be adaptable to such changes.

4) Poor Execution/Lack of Execution

To be efficiently executed, a master production schedule needs the assistance of an expert custodian. The implementers need to be well-versed in planning approaches and principles. Furthermore, a production schedule is useless if it is not followed.

- Objectives of Master Production Scheduling (MPS)

The main objectives of Master Production Scheduling (MPS) include the following:

- Inventory Control - MPS manages inventory and strives to maintain a certain amount of inventory inside the facility. This is accomplished by making optimal use of existing resources and reducing total expenses and waste in the operation. One of MPS' key goals is inventory control, which may be extremely beneficial to a manufacturing organisation.
- Setting up Due Dates - This goal entails establishing deadlines for finished items availability as well as giving information on resources and supplies. Within the manufacturing plant, this goal serves as the foundation for aggregate planning.
- Customer Service - Customers want to receive their things on schedule and shortly after they have been ordered, therefore delivery speed is critical in assuring customer happiness. Customer delivery commitments may be kept by using master production scheduling to deliver in a timely and cost-effective manner.
- Scheduling - MPS establishes specific production schedules for parts and components that are used as inputs to material needs planning. This facilitates the formulation of an optimal and effective production schedule that assures on-time delivery and the completion of the product within the timeframe specified with the resources available.
- Stabilize Production - A Master Production Plan guarantees that everyone in the firm is informed of what will be produced, eliminating surprises and the need for last-minute alterations to the schedule. To allow the shop floor to focus on what's coming next without worrying about modifications, there's typically a frozen window of time during which production orders aren't updated.

- Common 'MPS' Output Reports

Some of the standard reports created by a MPS include:

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- Available-to-promise – Each MPS line item's available-to-promise amounts are shown. The report is broken into "buckets" of time, which are usually weekly or monthly.
- Demand Tracking Report – Uses historical data to compare actual shipments and order bookings to management predictions.
- Forecast Data Report – A statistical analysis of previous demand activity that highlights the significance of forecast and reality variances.
- Period Summary Forecast – Forecasts are provided for each period through the following 12 periods for each line item within a product group, as well as summaries by period for the group and yearly for line items.
- Item Demand and Forecast – It shows several years of historical data (user-specified) as well as forecast demand for the next 12 months for each item. YTD totals, total yearly demand, and quarterly totals, as well as percent comparisons between items and their total product group, are all typical data elements.
- Build Schedule Report – Reports the build schedule for all or one assembly.
- Schedule vs. Actual Output – Reports the actual output compared with the scheduled output at a particular work center.
- Where Used Report – Lists all parts/tools used at each work center/machine.

Check Your Understanding

- 1) _____ is defined as planning where and by whom work will be done, determining the path that work will take, and determining the necessary sequence of processes.
 - a) Dispatching
 - b) Enhancing Process Flow
 - c) Routing
 - d) Forecasting
 - e) None of the above
- 2) _____ is the process of planning the time aspect of production – that is, deciding ahead of time "when work will be done."
 - a) Scheduling
 - b) Dispatching
 - c) Planning
 - d) Expediting
 - e) None of the above
- 3) The final stage of the production control process is _____.
 - a) Supply
 - b) Demand
 - c) Mapping
 - d) Expedition

- e) None of the above
- 4) _____ focus on picking the most desired choice from a group of alternatives, whereas qualitative tools place a greater emphasis on identifying options.
- a) Managerial Approaches
 - b) Decision-making Approach
 - c) Quantitative Approaches
 - d) Qualitative Approaches
 - e) None of the above
- 5) _____ refers to a company's maximum production level for producing a product or providing a service.
- a) Capacity
 - b) Decision Tree
 - c) Petri Nets
 - d) Delphi Techniques
 - e) None of the above
- 6) The "level capacity" strategy is based on _____ techniques, in which production systems are run at consistent levels and completed products inventories grow.
- a) Procure-to-stock and sell
 - b) Produce-to-stock and sell
 - c) Produce-to-stock and buy
 - d) Procure-to-stock and buy
 - e) None of the above
- 7) _____ is the process of offloading part of the work to outside suppliers in order to hire the capacity needed to satisfy the organization's needs.
- a) Subcontracting
 - b) Capacity Requirement Planning
 - c) Level Capacity
 - d) Matching Capacity
 - e) None of the above
- 8) _____ to put it simply, is an attempt to balance capacity and demand in such a way that expenses are kept to a minimum.
- a) Short-term Planning
 - b) Optimum Plant Capacity
 - c) Capacity Planning
 - d) Aggregate Planning

Notes

- e) None of the above
- 9) _____ is an optimization approach that allows a user to discover the most profit or income or the lowest cost depending on the availability of limited resources and specific constraints.
- a) Mixed- Integer Programming
 - b) Linear Decision Rule
 - c) Linear programming
 - d) Simulation
 - e) None of the above
- 10) The _____ is a sort of linear programming that may be used to generate aggregate plans that balance capacity and demand while minimising costs.
- a) Simulation Model
 - b) Transportation Model
 - c) PULL Model
 - d) PUSH Model
 - e) None of the above

Summary

- Production control is a directing function that entails the coordination and integration of operations and activities of various production components in order to maximise efficiency.
- Production planning, regardless of the planning period, aids in avoiding randomness in production, providing a regular and steady flow of production activities, and providing the foundation for enterprise control.
- The continuous flow of production through the manufacturing plant is one of the key goals of production planning and management.
- The purpose of production is not to produce any product or service, but to produce only those that will be effectively implemented in the market, that will meet consumer needs, and that will generate appropriate (optimal) profit and market share.
- Planning is one of the oldest human activities, as well as one of the most fundamental and crucial managerial actions for defining a company's goals and activities that contribute to achieving those goals.
- One of the more aggressive tactics employed is the Lead Strategy, which entails an upfront investment in more capacity that is required. Manufacturers are planning to expand their capacity ahead of time, even before demand grows.
- The Match Strategy is often a hybrid of the Lead and Lag methods. This technique employs smaller incremental modifications to the manufacturers' capacity rather than boosting demand ahead of time or increasing demand after current capacity is depleted.

- Rough-cut planning is normally done at the master schedule level and is used for short-term planning, which might last anywhere from a week to two months.
- A capacity limitation is a resource that is less competent than other sections of the operation of increasing throughput during a particular time period.
- The act of establishing, assessing, and maintaining a preliminary, estimated timetable of an organization's total operations is known as aggregate planning.
- The Master Production Schedule provides the foundation for sales and manufacturing collaboration. When sales and production use the MPS as a contract, sales may make genuine order guarantees.

Activity

1. Visit some manufacturing company in your neighbourhood, study their forecast for the next year for different products. Suggest a suitable aggregate plan based on it.

Questions & Exercises

1. what is aggregate planning? Explain the strategies, techniques and approaches used to perform aggregate planning.
2. Elaborate the concept of Master Production Scheduling.
3. What is Capacity Planning. Define the strategies used for capacity planning and also throw light on the challenges faced in the due course of capacity planning.

Glossary

- Production planning is the planning of production and manufacturing modules in a company or industry. It utilizes the resource allocation of activities of employees, materials and production capacity, in order to serve different customers.
- Production control looks to utilize different type of control techniques to achieve optimum performance out of the production system as to achieve overall production planning targets.
- Routing is defined as planning where and by whom work will be done, determining the path that work will take, and determining the necessary sequence of processes.
- Scheduling is the process of planning the time aspect of production – that is, deciding ahead of time “when work will be done.”
- Dispatching is the portion of production control that converts paper — work into real-world production.
- Expediting is a function which is used to keep track of the amount of work that has been done. The goal is to guarantee that what has been intended and planned is carried out.
- Capacity refers to a company's maximum production level for producing a product or providing a service.
- Capacity planning necessitates management's acceptance of production process constraints.

Notes

- Long-term capacity plans are concerned with new facility and equipment investments. These plans are for a period of at least two years.
- Short-term capacity plans consider labour force size, overtime budgets, inventory, and other factors.
- Capacity requirement planning is a strategy for determining what equipment and labour/personnel capabilities are needed to satisfy production targets (i.e., product volume) according to the master production schedule.
- A level strategy aims to create an overall plan that maintains a constant pace of output and/or employment. In order to meet changing client demand, the company must increase or decrease inventory levels in anticipation of higher or lower predicted demand.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check your understanding – Answers

- 1) c)
- 2) a)
- 3) d)
- 4) c)
- 5) a)
- 6) b)
- 7) a)
- 8) d)
- 9) c)
- 10) b)

Unit - 4.2: Concepts of Push System, Pull System

Notes

Objectives

At the end of this unit, you will be able to:

- Concepts of push system
- Concepts of pull system
- Adapting logistics operations to the push or pull strategy
- Attributes of a “push” system
- Attributes of a “pull” system

Introduction

“Push type” refers to “make-to-stock” production, which is not based on real demand. “Pull type” refers to “make-to-order” production, which is based on real demand.

The Push System is a method of producing things based on our best forecasts of what the market wants. The manufacture of things is essentially scheduled and based on a plan with deadlines. The commodities are subsequently pushed to the market. Push techniques are particularly popular in any Mass Production or industrial scenario where the firm is managed by Enterprise or Material Resource Planning (ERP/ MRP) systems. They are also frequent in most service-based firms, as customers are scheduled around the company’s timetable. A dinner buffet at a restaurant (where much of the food is not eaten and is thrown away) is another example of push, as is any aircraft, rail, or bus timetable where the firm sets the schedule and hopes enough people buy tickets to profit.

A pull system is one in which the individual or group that consumes the good initiates the creation of that good. Push defines a situation in which inventory is pushed to the consumer of the good, whereas pull depicts a situation in which inventory is withdrawn from the producer by the customer. Methods of “pull” manufacturing are founded on the idea that consumers should be able to get their products or services when they want them. As a result, pull techniques are more adaptable and built to “Build to Order.” Giving the consumer control over the timetable results in less waiting (and waste) in the customer’s perspective. It also increases demand variance and pushes a company to build flexible and responsive procedures.

In push systems, merchandise is pushed from the producer to the customer, whereas in pull systems, product is withdrawn by the consumer as needed. Pull systems are used in Lean Manufacturing to eliminate excess inventory and to much more easily deal with adjustments that need to be made to that inventory.

4.2.1 Concepts of Push System

Inventory control with the push approach entails anticipating inventory requirements to satisfy customer demand. Companies must forecast which things customers will acquire as well as the quantity of goods they will buy. The corporation

Notes

will then make enough product to fulfil anticipated demand and sell, or push, the items to consumers.

Materials Requirements Planning, or MRP, is an example of a push system. MRP blends financial, operational, and logistics planning calculations. It's a computer-based data system that manages scheduling and ordering. Its goal is to ensure that raw materials and other production-related items are accessible when they are needed.

- **Disadvantages of the Push System**

Forecasts are frequently inaccurate with the push inventory control planning since sales are uncertain and fluctuate from year to year. Another issue with push inventory management systems is when there is an excessive number of items in stock. This raises the expense of storing these commodities for the firm. The push system has the benefit of assuring the firm that it will have enough goods on hand to fulfil client requests, avoiding the product's incapacity to meet demand.

4.2.2 Concepts of Pull System

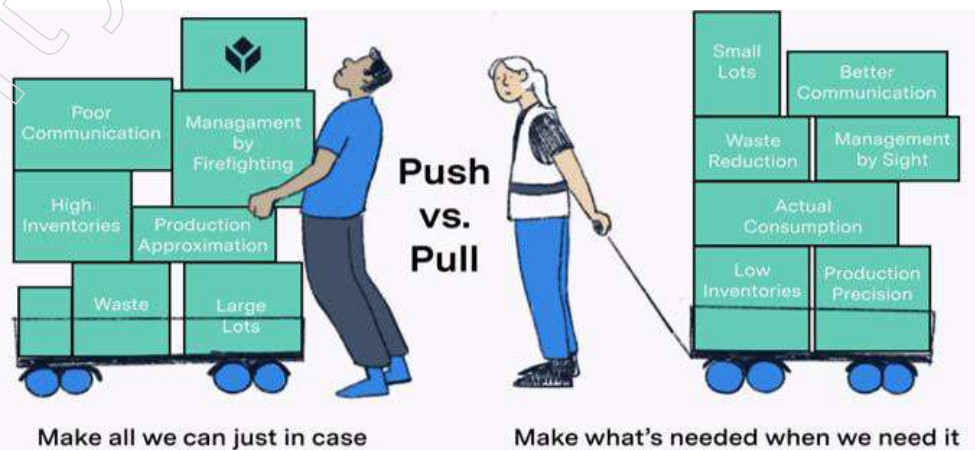
The pull inventory control system begins with a customer's order. With this strategy, companies only make enough product to fulfil customer's orders. One advantage to the system is that there will be no excess of inventory that needs to be stored, thus reducing inventory levels and the cost of carrying and storing goods.

The just-in-time, or JIT, inventory management system is an example of a pull inventory control system. The idea is to maintain inventory levels low by just keeping enough inventory to match consumer demand, not more or less. The JIT approach decreases waste by minimising the quantity of inventory storage space required and the expenses associated with holding items.

Disadvantages of the Pull System

One key downside of the pull approach is that it is more probable for a corporation to run into ordering issues, such as a supplier's inability to deliver on time. As a result, the firm is unable to complete the order, which leads to consumer discontent.

Differences between Push vs Pull



Push System – No dependency on Demand nor limits on WIP

The push method allows a corporation to estimate demand and use the Material Requirements Planning (MRP) process to manufacture products and services ahead of schedule. This is connected to the notion of Just-in-Case.

Although this estimate may not always be correct and may necessitate inventory hoarding, it is still a viable option for items with a lot of work in progress (WIP) or extended lead times.

The push approach is especially beneficial for items with little demand uncertainty or where economies of scale are critical for cost reduction.

Example

Since there are no prior WIP constraints, the Material Requirements Planning (MRP) indicated above is a push system. Goods are manufactured in accordance with the master production schedule, regardless of present condition.

Pull System – Dependent on Demand and Limitations to WIP

The Pull System is a lean manufacturing approach that employs the Just-in-Time strategy, which prevents things from being produced until an order has been received. The pull mechanism generates 'as required' rather than projecting demand.

This is especially effective for businesses dealing with significant demand volatility, a limited product mix, and a low value placed on economies of scale.

Example

The classic Kanban is a pull system since there are a fixed number of cards available, and this limits the WIP.

- Adapting Logistics Operations to the Push or Pull Strategy

In terms of logistics, the choice of a push or pull system has an effect on the different characteristics and operations of a warehouse:

- **Storage Capacity:** if you choose a push system, you'll need more storage space because all of the inventory you generate to meet demand will be housed in your facility. Pallet racks with drive-in/drive-thru access are perfect for maximising warehouse space.
- **Order Preparation:** while adopting the pull strategy, this should be agile and efficient in order to supply the manufacturing lines with the necessary commodities at the correct moment.
- **Movement Flows:** Automated storage systems and handling equipment, such as pallet conveyors and powered monorails, smoothly connect the warehouse and the manufacturing facility, which is critical in any pull model.
- **Inventory Management:** Regardless of the technique used, it's critical to keep track of all movements in the facility while also assessing the number of items in stock. In addition to an ERP and an MRP (material requirements planning) system, you should install a warehouse management system (WMS) to do this.

Notes

The logistics operations of all businesses are inextricably linked to the manufacturing strategy chosen. As a result, it's critical to alter that plan in order to improve the supply chain's efficiency.

- Examples of the Push and Pull System

The type of product created and its demand level are the most important factors in determining the production method to use.

The push approach is typically employed in sectors or businesses where demand can be predicted more easily, such as those with highly seasonal items. Schaal Chocolatier, a famous French gourmet chocolate manufacturer, employs the push production strategy to anticipate demand spikes at various periods of the year, particularly during the Easter and Christmas seasons. Schaal Chocolatier features a sophisticated, high-tech automated storage and retrieval system (AS/RS) for pallets installed by the Mecalux Group to maintain an efficient supply chain.

The pull approach is particularly well suited to enterprises that offer items with a high production cost, as keeping them in stock anticipating a sale would be unfeasible. This method is also used for items that allow for a high degree of personalisation in order to cater to client requests as much as feasible. This is the situation with Forma 5, a furniture company. All of the company's logistics are just-in-time. That is, the Interlake Mecalux Easy WMS warehouse management system connects the production facility with the 35 sites of its suppliers, allowing Forma 5 to get the semi-finished items it requires on time.

There are many industries that employ a push-pull management technique. For example, the automobile sector produces standard vehicles with certain specifications (push system). Others, on the other hand, are created on demand with personalised traits (pull system). The Mecalux Group built and implemented a miniload system (AS/RS for boxes) for Israeli vehicle maker Zbeda Group, which supplies the production lines while making both on-demand and mass-produced cars.

- Attributes of a "Push" System

Manufacturing activities are planned based on a market forecast rather than actual customer demand.

Implicitly this means:

- There is an emphasis placed on a central planning function
- Increasing or reducing completed products inventory levels ensures service levels.
- By level loading the plant, the system optimises "efficiency" rather than "effectiveness."
- Materials are routed through the plant in batches according to a work order's routing sheet.
- To compensate for the inherent complexity of the optimization issues faced, a substantial reliance on heuristics is used.
- Based on the forecast, "planning horizons" and "fences" are used to change the production plan on a weekly or monthly basis.

- There is a difference between dependent and independent demand.
- Attributes of a “Pull” System
- Manufacturing plan is based on actual customer demand “pull”.

Implicitly this means:

- Manufacturing execution is controlled at the worker level.
- Service levels are maintained by adjusting kanban levels between workstations (WIP).
- The system is optimised for “effectiveness,” which is attained through improving “efficiency” over time.
- Material moves through the production in visible queues driven by “pull” from the final kanban by customers.
- The system necessitates a hands-on management approach.



Check Your Understanding

- _____ is a computer-based data system that manages scheduling and ordering.
 - Control Planning
 - Capacity Planning
 - Materials Requirements Planning
 - Enterprise Resource Planning
 - None of the above
- A _____ guarantees that everyone in the firm is informed of what will be produced, eliminating surprises and the need for last-minute alterations to the schedule.

Notes

Notes

- a) Master Production Plan
 - b) Scheduling
 - c) Inventory Control
 - d) Stabilize Production
 - e) None of the above
- 3) Inventory control with the _____ entails anticipating inventory requirements to satisfy customer demand.
- a) Pull Approach
 - b) Push Approach
 - c) Inventory Control Approach
 - d) Just-In-Time Approach
 - e) None of the above
- 4) Full form of MRP is:
- a) Module Requirement Planning
 - b) Material Reversal Planning
 - c) Management Requirement Planning
 - d) Materials Requirements Planning
 - e) None of the above
- 5) What is the full form of WiP?
- a) Work In Progress
 - b) Workable-In-Progress
 - c) Work-In-Process
 - d) Window-In-Process
 - e) None of the above
- 6) Push type means _____ in which the production is not based on actual demand.
- a) Make-To-Stock
 - b) Made-To-Stock
 - c) Manage-To-Stock
 - d) Made-To-Sell
 - e) None of the above
- 7) Pull type means _____ in which the production is based on actual demand.
- a) Manage-To-Order
 - b) Make-To-Order
 - c) Made-To-Order

- d) Make-To-Organize
- e) All the above

Notes

Summary

- Inventory control with the push approach entails anticipating inventory requirements to satisfy customer demand.
- The pull inventory control system begins with a customer's order. With this strategy, companies only make enough product to fulfil customer's orders.
- The Pull System is a lean manufacturing approach that employs the Just-in-Time strategy, which prevents things from being produced until an order has been received.
- One key downside of the pull approach is that it is more probable for a corporation to run into ordering issues, such as a supplier's inability to deliver on time.
- The Just-In-Time approach decreases waste by minimising the quantity of inventory storage space required and the expenses associated with holding items.
- The push approach is typically employed in sectors or businesses where demand can be predicted more easily, such as those with highly seasonal items.

Activity

1. Write a Case study or visit an organization and understand whether the push strategy is used in Production planning or Pull strategy and how does it affect the Logistics Operations.

Questions & Exercises

1. Explain the concept of "Push and Pull system" and what is the difference between the two.

Glossary

- The just-in-time, or JIT, inventory management system is an example of a pull inventory control system.
- Push System is a system in which we produce goods based on our best projections of what the market wants.
- Pull System is a system in which the production of goods is initiated by the person or organization who consumes that good.
- Methods of "pull" production are based on the concept that customers should have the product or service delivered when they want it. Pull methods are, therefore, more flexible and designed to "Build to Order."
- "Push type" means "make-to-stock," in which the production is not based on actual demand. "Pull type" means "make-to-order," in which the production is based on actual demand.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh

Notes

2. Production and Operations Management; Prepared and Compiled By:
Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
3. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding – Answers

- 1) c)
- 2) a)
- 3) b)
- 4) d)
- 5) a)
- 6) a)
- 7) b)

Unit - 4.3: Job Shop Scheduling and Its Criteria

Notes

Objectives

At the end of this unit, you will be able to:

- Learn about mean flow time
- Understand mean tardiness
- Learn about number of tardy jobs

Introduction

A Job Shop is a work environment that has a number of general-purpose workstations that are used to accomplish a range of tasks. Job shop scheduling has become a must-have for modern firms looking to increase production efficiency and decrease waste in their manufacturing operations.

The job shop is a frequent scheduling issue in which many jobs are handled on various machines. Each job consists of a series of activities that must be completed in a specified order, and each work must be completed on a specific machine. The task may, for example, be the production of a single consumer item, such as a car. The goal is to arrange the work on the machines in such a way that the length of the schedule—the time it takes for all of the activities to be completed—is kept to a minimum.

A plan or timetable is vital for a manufacturing business that wants to get visibility into its operations. This is directly applicable to job shop manufacturing, where the purpose is to balance total supply and demand. Supply considerations include capacity, resource, labour, and material availability in the job shop, whereas demand refers to real task orders inside the manufacturing facility based on the resources and capacities required to perform the project effectively. The purpose of job shop scheduling is to integrate these internal supply and demand components in the most effective and best method feasible. This is why job shop managers must adhere to a strict timetable.

Scheduling

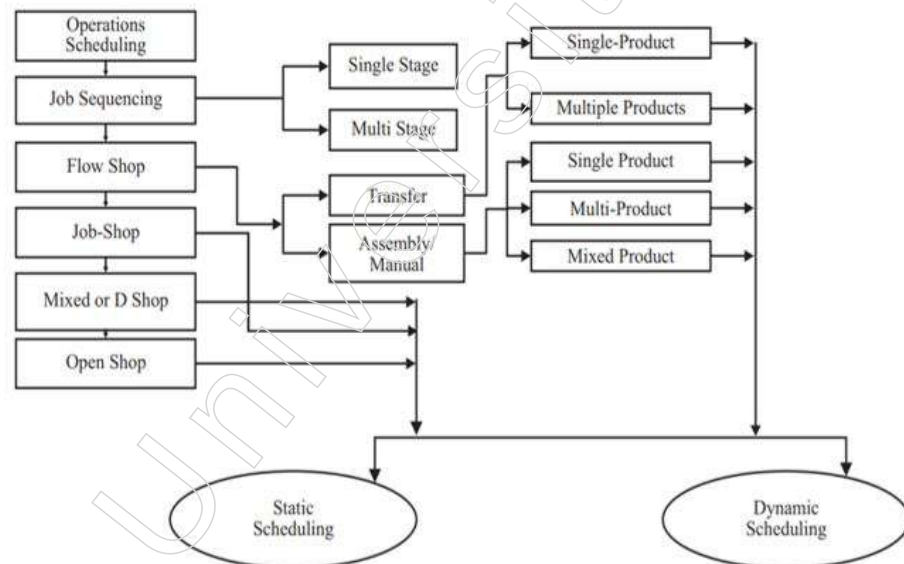
Scheduling may be described as “the allocation of resources across time to accomplish a collection of activities” or “the process of allocating a set of tasks to resources over a period of time.” Scheduling issues are easy to explain and articulate in its simplest static and deterministic forms, but they are challenging to solve because they need complicated combinatorial optimization. If n jobs are to be completed on m machines, for example, there are possibly $(n!)^m$ sequences, albeit many of them may be infeasible owing to different limitations.

Manufacturing industry operational scheduling is critical for cost-effective and competitive operations, as well as market survival. Operational scheduling involves a large range of scheduling models, and while numerous attempts have been made to address them using analytical, heuristic, and artificial intelligence-based techniques, no reliable solution for dealing with the vast range of scheduling models has yet to be developed. Because of their adaptability to the combinatorial character of the scheduling issue, Genetic Algorithm (GA)-based solutions have recently become prominent.

Notes

A GA-based scheduling system is presented in this paper, in which a novel combination of a recently introduced crossover scheme for job-based chromosome representation with Stochastic Universal Selection (SUS), simple mutation of gene exchange, and a heuristic for chromosome evaluation is proposed, which is easy to code and robust enough to deal with flow shop as well as job shop scheduling problems. Case studies from real-world scheduling challenges are used to validate the system.

Job operational scheduling on the shop floor is critical for effective and efficient resource use, resulting in cost-effective production and increased productivity. "The assignment of jobs to be processed on the associated equipment, in a particular time frame, for a workshop consisting of numerous machines or production facilities comprising operational employees," according to operation scheduling. As indicated in Figure below, there are several types of scheduling models, including work sequencing, flow shop scheduling, job shop scheduling, mixed shop scheduling (D-Shop), and open shop scheduling.



The job sequencing model specifies the sequence in which jobs waiting to be processed on a particular machine or facility are processed. If N components are to be scheduled on a single machine, there exist $N!$ permutation schedules (sequences) from which a work sequence is chosen for maximising or minimization of specific objective functions. The sequence of machines in the flow shop schedule is fully similar for all jobs to be produced, according to multiple-stage manufacturing.

This is a common flow pattern in mass manufacturing. There are two types of flow shops: permutation flow shops (PShop) and ordinary flow shops (F-Shop). Passing is not permitted in the PShop (which follows the first in, first out rule), but it is permitted in the normal flow shop. When the number of machines in the P-Shop is limited to one, the problem becomes a simple work scheduling problem.

The sequence of machines in a work shop varies depending on the job. Flow shop models, including permutation and generic flow shops, are constrained job shop scheduling models. The job shop is representative of most jobbing kinds and batch types when it comes to diverse production.

The mixed shop combines the benefits of both a flow shop and a work shop in one location. In such a scenario, certain tasks have fixed machine orders (as in the F Shop), but other jobs' activities can be executed in any sequence (as in the Job Shop). In other words, work in a mixed shop must be processed in a sequence that corresponds to a partial order of machines.

The sequence of machines in an open shop (O –Shop) is unconstrained, and the task can be handled in any order through the equipment. All of the following models are sub models of the O-Shop scheduling model, which may be expressed notationally as PFJDO, which indicates P-Shop, F-Shop, J-Shop, and D-Shop models are sub models of O-Shop. It clearly demonstrates that the scheduling domain is quite broad and embraces a wide range of models².

Different attempts to solving the problem of operational scheduling have resulted in valuable analytical, heuristic, and artificial intelligence (AI) based solutions, but no stable solution to deal with the wide range of scheduling models is currently available. Genetic Algorithm (GA)-based solutions are becoming increasingly popular due to their adaptability to the combinatorial character of the scheduling problem.

The GA is based on the natural reproduction process, which involves chromosomal selection through crossover and mutation from one generation to the next. An issue that will be handled by a GA must be correctly encoded utilising several encoding techniques^{7,8}. Crossover and mutation occur after encoding. For crossover and mutation, many strategies are employed. The fitness values produced by evaluating the chromosome against specific pre-defined goal functions are utilised for further selection of the chromosome in the future generation, based on the survival of the fittest principle.

Besides probability of crossover and mutation, population size, and generation size, the quality of a GA solution is determined by a combination of different techniques for chromosome encoding, crossover, mutation, and chromosome selection for reproduction, which results in a variety of GA-based algorithms.

A novel combination of recently introduced crossover scheme for job based chromosome representation with Stochastic Universal Selection (SUS), simple mutation of gene exchange, and a heuristic for chromosome evaluation is proposed in one of these varieties of GA, which is easy to code and robust enough to deal with flow shop as well as job shop scheduling problems.

- Job Shop Scheduling (JSS)

Job Shop Scheduling (JSS) or Job Shop Problem (JSP) is a popular optimization problem in computer science and operational research. This focus on assigning jobs to resources at particular times.

Factors to Describe Job Shop Scheduling Problem

Based on the researches and experiments done in Job Shop Scheduling, following variables affects mostly.

1. Arrival pattern
2. Number of machines (work stations)

Notes

3. Work Sequence
4. Performance evaluation criterion

Types of Arrival Patterns

The arrival pattern of jobs to machines are of two forms, either static or dynamic.

- Static — n jobs arrive at an idle shop and must be scheduled for work
- Dynamic — intermittent arrival (this is often stochastic)

Types of Work Sequence

- Fixed, repeated sequence — flow shop
- Random sequence — All patterns possible

Performance Evaluation Criterion on Job Scheduling

The performance criteria that most researches are based on following optimal job scheduling heuristics.

- Makespan — total time to completely process all jobs
- Average time of jobs in shop
- Lateness
- Average number of jobs in shop
- Utilization of machines
- Utilization of workers
- Job Shop Scheduling Benefits

The benefits of job shop scheduling benefits include the following:

- Visibility and Transparency - A task shop manufacturer cannot avoid introducing appropriate scheduling software. Any scheduling tool's overall purpose is to gain transparency over the production process. An advantageous job shop schedule, on the other hand, provides insight into not only current orders, but also labour and machine capacity, waiting periods, due dates, and production dependencies. The schedule must include all of the specific details required to completely comprehend the components that impact the existing and future production processes. It may also be advantageous to have a system that is accessible to a larger number of individuals. This allows the general plan to be visible to more than one person in the work plant.
- Control and Precision over Planning - You'll be able to take complete control once you've gained visibility into the manufacturing process and a thorough understanding of ongoing procedures. Manufacturers may anticipate and direct future orders, as well as adjust current orders, with precise production schedule. You can shift from a reactive to a proactive approach to influencing output.
- Identification of New Opportunities - Identifying new possibilities can result in significant productivity gains. New possibilities can arise in the same manner that bottlenecks can be identified through enhanced visibility. It could be discovered that a specific capacity is usually idle and might be put to better use

elsewhere, or a production line could be changed to improve throughput time. Identifying new possibilities is critical for supply chain expansion in general.

- **Real-time Feedback** - An advantageous job shop schedule will give a manufacturer with notifications, such as error or other production-related alerts. These notifications enable you to make decisions based on real-time difficulties or roadblocks in the production line, such as when on-time delivery is in jeopardy or production is running behind schedule. This will help you prevent mistakes and increase your productivity.
- **What-if Scenarios** - What-if scenarios can provide you a clear picture of which production schedules will benefit your manufacturing process the best. This tool enables you to easily add or remove tasks while also allowing you to see the impact of simple changes to other processes, resources, materials, staff, and other capacity. This aids producers in locating rapid answers to a variety of issues.

All of these benefits of job shop scheduling can aid your production with ease.

- **The Job Shop Scheduling Problem**

The Job-shop Scheduling Problem (JSP) is one of the most critical issues in manufacturing, because of its impact on the whole company and supply chain performance. The JSP entails scheduling a group of tasks, each with its own set of processes, to be processed in a set of machines for a specific amount of time. Due to the tremendous complexity of job-shop setups, finding a perfect solution to this challenge in a reasonable length of time is difficult.

JSPs are classified as intractable numerical problems, or NP-hard, due to the factorial explosion of alternative solutions. Exact or full methods that yield optimum answers may take longer to calculate in this class of problems, but approximation approaches might provide approximately ideal solutions in a shorter amount of time. As a result, several techniques and strategies for dealing with JSP exist, each having a distinct influence on job-shop scheduling. The following are some of the techniques and methods:

Job-shop Scheduling Problem	Approaches for dealing with the JSP	Heuristic Rules			
		Classical Optimization			
		Artificial Intelligence (A.I.) approaches			
	Scheduling Flexibility	Classical Approach			
		Dynamic approach	Rule used for creating the scheduling	Completely reactive scheduling	
				Predictive-reactive scheduling	
				Predictive-reactive robust scheduling	
				Robust pro-active scheduling	
		Decision modes		Simulation based approaches	
				Artificial intelligence based approaches	
				Agent-based approaches	
		Rescheduling policies		Event-based rescheduling	
				Periodic rescheduling	
				Hybrid rescheduling	
			Updating an existing scheduling		Right-shift rescheduling
					Complete rescheduling
Partial rescheduling					

Fig: Job-shop Scheduling Problem approaches and methods

Notes

The volume, frequency, and intensity of data interchange in the job shop, as well as the scheduling quality, are all directly influenced by the various techniques and methodologies. These qualities have an impact on MAS setup and implementation, as well as a company's production performance. As a result, the subsections that follow address various techniques and procedures.

- **Approaches to Deal with JSP**

There are three main approaches used for handling the JSP:

1. Heuristic Rules,
2. Classical Optimization and
3. Artificial Intelligence (A.I.) approaches.

All of them can be used both for creating a scheduling in advance (classical approach) as for dealing with scheduling dynamically (dynamic approach).

Heuristic rules, also known as dispatching rules, priority rules, or scheduling rules, are used to determine which work will be chosen next from a list of jobs waiting to be made on a certain machine, without first seeing the results of the selections. A range of dispatching rules are recognised for a variety of scheduling criteria, in addition to being one of the most popular ways used for scheduling. A heuristic is a quick and simple method for solving issues or making judgments. Heuristic assessment is a method used by professional evaluators to determine the usability of a design or product. As specialists, they run through a checklist of criteria to find flaws which design teams overlooked.

The Nielsen-Molich heuristics state that a system should:

1. Keep users correctly and immediately informed about its condition.
2. Present information in ways that consumers can comprehend based on how the real world works, and in their own language.
3. Give people power and make it simple for them to undo mistakes.
4. Be consistent so that consumers don't become confused about what varied phrases, images, and other symbols represent.
5. Prevent errors - A system should either avoid situations where problems occur or alert users before they do potentially dangerous actions (for example, "Are you sure you want to do this?" notifications).
6. Make visible information, instructions, and so on so that users can recognise alternatives, actions, and so on rather than having to rely on memory.
7. Be adaptable so that seasoned users may achieve their objectives more quickly.
8. Keep your workspace clutter-free by just storing information that is relevant to your present responsibilities.
9. Provide easy-to-understand assistance with mistakes and solutions.
10. Using lean, searchable documentation, provide a list of short actions for solving problems.

- Classical optimization methods, the branch and bound method, and dynamic programming method, in turn, are separated into two basic classical methods, both based on enumeration techniques. Branch and bound is a problem-solving technique that uses lower limits of the best solution to guide a branching process that breaks the problem into smaller, mutually exclusive sub-problems until the best solution is identified. Dynamic programming, on the other hand, generates a collection of different schedules by counting each work as the last and evaluating how well each option would aid in maximising a given criterion. By selecting the smallest of this collection, the best schedule is determined.
- Meta-heuristics are high-level heuristics that guide closest neighbourhood search algorithms, which follow the principle of exploring neighbourhoods for escape from local optima, according to this description. Simulated annealing, tabu search, and evolutionary algorithms are examples of Meta-heuristic approaches, which may tolerate inferior answers for escaping local optima or focus on discovering more appropriate beginning solutions (rather than selecting them up at random) for the local search.
- Finally, artificial intelligence methods also have broad application to handle the JSP in a variety of ways and combinations, including its use in heuristics and optimization methods as well.
- Scheduling Job Shops

In order to combat the existing uncertainty in a work shop, some type of organisation, visualisation, and planning tools must be included. However, manufacturers face a hurdle in creating a good work shop schedule because of the prevalent instability. Simply expressed, the ultimate purpose of job shop scheduling is to distribute jobs and partial activities of a job to different working stations in an effective manner. Clearly, this is a difficult and ambitious endeavour.



- How can a work shop manufacturer implement a structured strategy when job shops lack structure the most?
- How can abrupt and frequent changes be effectively accounted for?
- How can operations be optimised and improved when procedures and predictions are scarce?

Nobody claimed introducing a timetable would be “easy,” but knowing certain key insights and using some helpful tools may greatly enhance one’s ability to correctly plan

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a job shop manufacturing facility. A job shop manufacturer may use this information to boost productivity, minimise overall production durations, and increase On-Time-Delivery by preventing or decreasing delays.

- Constraints for a Job Shop Schedule

The so-called sequencing problem, faced by job shops, involves several constraints and challenges:



1. A single job may be handled at a time by a working station, such as a machine.
2. Workstations or machines may have different capacity levels.
3. A new subtask in a job's production line can only begin when the previous stages have been completed successfully. The task sequence of each individual work must be rigorously accounted for in the timetable.
4. A subtask must always be finished in its entirety.
5. Job-to-job routing varies.
6. Priorities shift all the time (e.g., if you have to squeeze in a rush order).

Job shop scheduling seeks to overcome these limitations by allocating shared resources to jobs and giving a visual depiction of the allocation. Ultimately, the goal is to make the most effective use of the work shop's capabilities and resources.

- Characteristics of Job Shop

- 1) Layout: In a process layout, comparable equipment or functions are placed together in the job shop, such as all drill presses in one area and grinding machines in another. Material handling, cost, and work-in-process inventories are all minimised by the arrangement. Rather than specialised, product-specific equipment, job shops employ general-purpose equipment. Digital numerically controlled equipment is frequently used in job shops to allow them to swiftly alter set-ups on numerous machines. Job shops compete on product quality, delivery speed, customisation, and new product launch, but they are unlikely to compete on price because to the lack of scale efficiencies.
- 2) Routing: When an order arrives at the job shop, the part being worked on is routed through several sections in a predetermined order. Not every job in the facility will require the utilisation of every equipment. Jobs are frequently routed haphazardly and may return to the same computer for processing many times. This arrangement may also be seen in services such as department shops or hospitals, where specific

zones are dedicated to a specific product (men's apparel) or a specific service (maternity ward).

- 3) **Employees:** Employees at a job shop are often highly competent craftspeople who are capable of operating a variety of machines. Because of their greater skill levels, these workers get paid more. Job shop personnel require less supervision due to their high skill level. Workers may be paid a set hourly rate or be compensated through an incentive scheme. Management is in charge of bidding on tasks and setting prices for client orders. The processing of information is the most important activity in a work shop.
- 4) **Information:** The most important part of a work shop is information. To quote a price, bid on a task, route an order through the shop, and describe the particular work to be done, you'll need information. Before the task is given to the floor, the information begins with quotation, followed by the preparation of a job sheet and blueprint. Employees fill work sheets and time cards on the production floor to calculate labour costs and update records for quoting future tasks where variations exist.

While it is frequently simple to bid on jobs that the shop has previously made, new works need proper costing of labour, materials, and equipment, as well as appropriate overhead assignment. Each work is tracked via tickets throughout the shop, where time and activity are logged. The work shop competes on quality and customer service rather than pricing since it manufactures unique, bespoke goods.

Customers come in the components and materials to be worked on, hence the job shop has very minimal raw materials inventory. While tasks are being performed, the job shop has work-in-process inventory, but because the client is often waiting for the order and expects rapid delivery, there is no finished goods inventory in this make-to-order setting. Like many small enterprises, some workshops thrive on cash flow management. They may work on little works in order to finish them by the end of the month so that they may bill consumers.

- 5) **Scheduling:** The route, processing needs, and priority of a job all define it. In a job shop, the product mix is a critical factor in determining how and when to schedule work. To save the expense of machine set-ups and change-overs, jobs may not be performed according to their arrival schedule. Work may also be arranged according to the quickest turnaround time.

In the job shop, capacity is difficult to quantify since it is affected by lot sizes, task complexity, the mix of projects previously scheduled, the ability to schedule work properly, the number of machines and their condition, worker input quantity and quality, and any process enhancements.

- **Job Shops as An Early Form of Manufacturing Organization**

The majority of today's enterprises originated as job shops and expanded into other production processes when volume allowed. The job shop gives entrepreneurs the most freedom in creating a wide range of items that match consumer quality and service expectations. The job shop may organise equipment into work cells to process batches of similar jobs when clients request repeat jobs and volume increases.

In the process life cycle, job shops are one of the earliest buildings for a manufacturing. Structures move from the work shop to a batch flow to an assembly

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line, and finally to a continuous flow as volume grows and firms cut or standardise their product offerings. Due to large volume and uniformity, flexibility reduces with time, while unit prices fall. Assembly lines or continuous flow activities are arranged in a product layout, and the work shop is structured by process. The equipment or labour processes are placed in the latter arrangement according to the exact stages in which the product is created, and each part's path resembles a straight line.

- **Different Approaches to Job Shop Scheduling:**

Many techniques to production control and management are being used nowadays, such as lean manufacturing, finite capacity scheduling, rapid response manufacturing (QRM), CONWIP, the theory of constraints (TOC), and so on. All of these methods provide a rough or comprehensive production schedule in real time or ahead of time. A brief explanation of a few productions scheduling approaches, including manual efforts, is shown below.

1. Manual Scheduling:

A scheduler's job description is frequently limited to checking task progress on the shop floor and reporting it to management. The following approach is used in various job shops to simplify production scheduling:

- Customer orders have due dates that are set based on average lead times (for example, three weeks), regardless of workload or resource availability.
- The time it takes to load a job is dependent on material availability and a pre-defined manufacturing lead time. Some individuals calculate the workload at crucial work centres using a few basic calculations and alter job loading times accordingly.
- Following the conclusion of an operation, each job is moved from one work centre to the next, with a dispatch rule in place at each work centre for choosing jobs (for processing) from the relevant waiting queue.
- If a work is in danger of missing its deadline, it will be accelerated on a high priority basis, maybe requiring overtime.
- When all of the work's operations are done, the job is delivered.

With this practise, job shop production may be managed, but it can also lead to significant WIP, long lead times, poor on-time delivery, and frequent firefighting. Push scheduling, as the method is known, is always discouraged by lean manufacturing specialists. While attempting to achieve deadlines, they engage in heavy real-time scheduling, or firefighting, without fully understanding the impact of their real-time decisions on the production plan.

Typically, firefighting breeds and remains in the shop for extended periods of time. Chaos, poor planning, and a lack of complete grasp of the dynamic nature of a work shop might all be signs of firefighting. It may erode management and consumer trust in the work shop's ability to meet deadlines. Job shops won't be able to increase their performance without compromising productivity unless the requirement for firefighting is reduced. Manual scheduling is ineffective for complicated job shops that manage a large number of orders with varied due dates at the same time.

Scientific scheduling can help HMLV workshops optimise productivity, on-time delivery, and resource efficiency. Scheduling software vendors frequently claim that their products use finite capacity scheduling (FCS) logic. FCS simply refers to scheduling that does not violate capacity limits, i.e., does not overburden work centres or resources. FCS is not a specialised scheduling method or algorithm, and there is no literature on FCS or practically useful models for production scheduling to aid software developers.

As a result, there might be a lot of variety in how FCS is implemented across scheduling applications. It's likely that the FCS logic utilised in certain scheduling software is too weak and limited to be useful in a wide range of workshops. Due to the security of proprietary scheduling methods, customers never get an opportunity to examine the underlying FCS logic.

2. Scheduling on Whiteboards and Excel Spreadsheets:

Whiteboards are still used by many planners for production scheduling. For certain small production systems, this simple, manual technique may be sufficient. However, it is difficult and ineffective for scheduling a large number of operations for various projects on limited resources. It's also inconvenient for rapid workload rescheduling when the real work flow differs greatly from the whiteboard timetable. Many software companies have expanded on the whiteboard concept by providing an electronic version that allows for manual schedule creation and Gantt chart adjustments (on computer screen). For dynamic scheduling of job shops, low-cost software solutions that permit operational level production scheduling through easy drag-and-drop operations on a computer screen will be significantly more effective than whiteboards.

Excel spreadsheets are popular in job shops for a variety of reasons, including:

- Almost everyone who is comfortable with computers can readily handle data in Excel spreadsheets.
- A majority of ERP / MRP solutions can readily export data necessary for scheduling into Excel spreadsheets.
- Many data activities, such as editing, reformatting, and transformation, are simple to accomplish in Excel spreadsheets.
- VBA programming is useful for complex Excel data manipulations.
- Some people believe that Excel scheduling software are less expensive and adequate for small businesses.

They are suitable for small and basic production systems in job shops. Despite their numerous benefits, Excel programmes have some significant drawbacks when it comes to organising the creation of complicated job shops. Regardless of shop size, Excel (with VBA) cannot readily provide a diverse scheduling paradigm and robust logic to adequately manage the scheduling complexity of various HMLV task shops.

Excel programmes, for example, are inefficient when dealing with (a) individual weekly calendars and calendar exceptions for resources, (b) changes in work priorities, (c) various resource requirements of operations, (d) quick and detailed what-if analysis, and so on. They also lack robust and user-friendly visual capabilities, such as drag-and-drop functionality, and provide minimal assistance in preventing manual data entry

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mistakes. Despite the ease with which users may modify Excel files, the editing process is prone to human mistake.

Excel scheduling solutions tend to become enormous, unwieldy, and inflexible as production size and complexity grow. In HMLV j, these apps take a long time to schedule hundreds of jobs.

Most of these disadvantages may be eliminated, however, if a best-of-breed scheduling software can receive data straight from Excel spreadsheets. Users of Optisol's Schedlyzer scheduling software, for example, may take advantage of both the benefits of Excel and the scheduling capability of Schedlyzer because the programme can receive data from Excel spreadsheets in a sensible manner.

If Excel data is not accessible in the proper format, a tiny add-in software component may be required. Pure Excel programmes with appropriate scheduling capacity may or may not offer a lower total cost of ownership (including implementation and training) than some best-of-breed scheduling systems like Schedlyzer for small task shops.

3. Scheduling by Project Management Software Tools:

Production management of job shops resembles resource-constrained multi-project management in many respects. However, the major differences include:

- In many job shops, task dependence relations among job tasks may be defined using a task sequence, but task dependency relations in a project are often described using a directed network.
- In the majority of workshops, a task can only be begun until all of its predecessors have been completed. In a project context, the need of completing all previous tasks before beginning a new work may not be essential. The work, however, cannot begin until all of its previous duties have completed.
- Tasks in job shops are less likely to be effort driven, which means that devoting extra resources may not be enough to shorten the time of a work shop activity.
- Tasks in projects have longer durations with more variation.
- Projects are more likely to be affected by major uncertain events.

For job shop scheduling, many schedulers utilise project management (PM) software. ETO manufacturers and shipyards, for example, employ well-known PM technologies to schedule a diverse task. When jobs create queues while vying for finite resources, the critical path approach (a project scheduling method used by PM software) might produce useless production schedules, which schedulers must accept hopelessly.

The great majority of PM programmes are ineffective at automatically resolving resource conflicts, making them unsuitable for resource-constrained scheduling of thousands of tasks. When all resource conflicts in the schedule need to be manually addressed by drag-and-drop operations as required by project management packages, it is frequently arduous to plan and reschedule several jobs (with hundreds or thousands of actions) in a realistic manner.

These tools don't provide production systems with powerful, trustworthy, or quick what-if analysis or a capacity planning mechanism. Popular project management solutions, on the other hand, are superb information systems with an exquisite visual user interface that displays project / production schedules extremely well. For resource-constrained, complicated production scheduling involving thousands of activities, most project management solutions are insufficient. The majority of users of such tools in a production setting are unaware that sophisticated solutions for resource-constrained scheduling are available.

4. Scheduling Modules of ERP Systems:

Enterprise resource planning (ERP) systems, which are complex enterprise-wide, integrated information systems, are being implemented at a rapid pace in manufacturing today. With the aid of these systems, businesses can effortlessly manage all of their data and increase the efficiency of their operations. The integration of shop floor data gathering systems with a central database has expanded the value of ERP systems in industrial execution. From a customer service standpoint, these technologies allow companies to provide real-time job status to their consumers. However, many ERP systems do not fully justify the term "resource planning" because they lack strong planning and intelligent decision support modules.

There are various ERP and shop management software programmes on the market that claim to eliminate all of the disadvantages of the approaches discussed above. Although ERP scheduling modules are generally superior to whiteboards and Excel spreadsheets, they are nevertheless ineffective or inappropriate for complicated task shops. Many job shops are now using ERP scheduling modules because (1) they have replaced the laborious manual scheduling and rescheduling that is done on whiteboards and Excel spreadsheets, (2) there is no need to provide job information separately for planning and scheduling purposes, and (3) the level of scheduling knowledge on the shop floor is low.

Modules based on MRP scheduling logic are unsuitable for scheduling in a job shop. Many of them may frequently fail to automatically generate a feasible schedule that satisfies all relevant constraints. Some of them estimate job queueing times at various work centres and utilise those estimates to generate schedules. On the shop floor, such components are nearly hard to deploy. They do infinite capacity scheduling and determine the required capacity, which may or may not always be feasible. In certain modules, an electronic whiteboard may be accessible to drag-and-drop operations to fix an infeasible timeline.

Some ERP scheduling systems produce an infeasible plan with resource conflicts and export it to a project management tool, forcing a scheduler to manually fix the schedule for hours using Gantt chart drag-and-drop operations. It is extremely time consuming to reschedule a large number of operations and conduct what-if studies in response to big changes in production using this method. Modules that need a long time to run in order to provide a single workable schedule are inefficient for doing a rapid and comprehensive what-if analysis. Some job shops, interestingly, collect production data from ERP systems and schedule production via an Excel spreadsheet or MS Access programme.

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5. Scheduling by Theory of Constraints (TOC):

The Drum-Buffer-Rope (DBR) approach of the theory of constraints is one of the basic methods offered for production scheduling in complicated job shops (TOC). Eli Goldratt created it in the early 1980s, when businesses lacked sophisticated information systems, shop floor data gathering mechanisms, powerful and low-cost computers, and reasonable scheduling tools. After working on the creation of software tools for production scheduling, Goldratt recognised how difficult it was to schedule complicated production systems under such adverse conditions, and he saw the need to simplify those complex systems so that the scheduling issue could be removed.

The DBR scheduling approach is based on the premise that a production system has a single resource limitation and that the other resources have the capacity to completely support any realistic schedule on the constraint resource. If this assumption is false, factory management must choose a resource to be the constraint and enhance the capacity of other resources to ensure that the basic DBR scheduling approach may be used. The resource that is being limited may fluctuate over time. DBR is a simple concept to grasp and put into reality, and a scheduler may quickly execute it with the help of an Excel spreadsheet. DBR involves a procedure called buffer management that addresses the system variation, from the execution point of view.

DBR's basic premise of a single constraint resource allows us to plan jobs on the constraint resource (in increasing order of due dates) while disregarding all non-constraint resources' capacity. Job loading times are calculated using the schedule created on the constraint resource, with the effect of uncertainty taken into consideration in a simplistic manner. The main concept is to keep the limited resource occupied at all times while limiting WIP in the system by releasing jobs at the appropriate moments. DBR isn't viable unless the underlying premise is correct. If the assumption is incorrect, task queues generated by constrained capacity resources will have a negative influence on DBR buffer management.

Changing a complicated job shop to enable the deployment of the basic DBR scheduling approach can be costly, and it necessitates strategic planning because such modifications might result in surplus resource capacity. In complicated systems, it makes sense to keep production scheduling simple, but it's also vital to consider the cost of doing so.

6. Scheduling by Lean Manufacturing Practices:

Toyota's production system (TPS) has grown over decades into an inspiring world-class manufacturing system thanks to Toyota employees' tireless efforts. Many concepts, beliefs, practises, procedures, and a work culture that are particularly effective in manufacturing were adopted and developed by Toyota. Some ideas and approaches, such as heijunka, one-piece flow, takt time, and kanban-based production control, are less useful in job shops that produce a variety of low-quantity projects at the same time. Toyota's kanban system is extremely effective in controlling repeated output. It may be thought of as a real-time scheduling system for manufacturing. By managing material flow through the production system, it avoids starvation of bottleneck work centres and the buildup of enormous inventories at any work centre in repeated production.

The kanban system absorbs uncertainty, natural fluctuation, and variances in work centre output rates, but it works best when demand is consistent and predictable. However, no kanban system can give the predictability of workflow, WIP, lead times, resource usage patterns, and task completion times in job shops with variable demand and diversified jobs that pass through distinct sequences of work centres. In complicated job shops with a diverse workload, such predictability is critical for capacity planning and production management. Job shops require what-if analysis of dynamic production, which a Kanban system cannot provide.

Also, using a kanban system, determining the best time to load a project into the shop based on the due date, amount, and route of the job, resource availability, and present shop floor workload is difficult. Kanban management does not provide you a complete picture of how job shops work in real time. Similarly, TPS heijunka scheduling is irrelevant for job shops that create a large number of low-quantity works at the same time, each with its own set of due dates, priorities, routings, process times, and resource and material needs.

Since Toyota proved the efficacy of heijunka scheduling and kanban control in the management of its repetitive production, there have been several efforts to forcefully install them in complicated job shops with false confidence. In job shops when several works are being produced at the same time, a kanban system will be inconvenient and have minimal influence on throughput and on-time delivery. Another shop floor control method is quick response manufacturing (QRM), which is also known as real-time job shop scheduling. To solve the complexity involved with various work routings, the POLCA approach of QRM is more suited than kanban control. POLCA, on the other hand, also cannot provide the required what-if analysis and predictability as required by job shops.

7. Scheduling Algorithms:

Researchers provided a significant amount of job shop scheduling material, which may be found in books and research publications. They came up with innovative ideas and evaluated fascinating work shop scheduling objectives and limits. WIP, production lead time, work lateness, and other concerns are addressed in the objectives. However, the scheduling models and algorithms they created are mainly inadequate for work shop scheduling challenges.

This is primarily due to the fact that (1) existing mathematical techniques are insufficient to deal with practical scheduling problems, (2) many assumptions are made in order to reduce a problem to a level where it can be solved by a non-trivial method, (3) developing simple methods (heuristics) for fairly good solutions to complex practical scheduling problems is not an exciting activity for researchers, and (4) researchers have a natural tendency to take a complex, rigorous view of the problem. Researchers' groundbreaking ideas are hidden behind a thicket of mathematical treatment. When these researchers resist the impulse to publish papers, they may come up with compelling and effective answers to real-world challenges.

8. Finite Capacity Scheduling (FCS):

For work shop scheduling, finite capacity scheduling (FCS) techniques are also available. The main disadvantages of FCS have been that (1) users must adjust

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numerous tool parameters, (2) users find the tools cumbersome, unfriendly, and less intuitive, (3) users require extensive training on the tools, (4) input data is either unavailable or incorrect, and (5) the tools require a significant amount of computing power and time. However, owing to quick improvements in hardware and software, these tools vastly improved and overcome the drawbacks over time. The Windows-based tools provide users with a great deal of flexibility, convenience, and context guidance. The usage of FCS tools is substantially facilitated by high-speed personal computers. The data quality and availability will progressively increase as upper management begins to look at the schedule output from these technologies.

Job shops can watch job status and make seemingly sensible decisions in real time to fulfil job due dates because to the availability of strong information systems and sophisticated shop floor data gathering systems. To create effective production schedules and depict the dynamic workflow and bottleneck formations over time, FCS tools rely on the same technologies. Fast workload rescheduling, prediction of future bottlenecks and project completion timeframes, and effective capacity planning are just a few of the advantages that FCS technologies provide to job shops. FCS has certain flaws as well. Every FCS tool has a different scheduling paradigm.

If a scheduling problem does not fit into the underlying paradigm, the FCS solution, no matter how complex it looks, may be worthless. FCS solutions created only on the basis of hours available for each resource in each time interval may be hard to apply on the shop floor (day, week, month, etc). The sequence of works to be loaded at each work centre may not be specified in such schedules. The inability to deal with variance and precise schedule output are two significant criticisms of FCS. At various points, FCS logic can be adjusted to establish time buffers (as in TOC approaches) for jobs to absorb the effect of schedule deviation. The output of the schedule may be examined at the necessary degree of detail. When deploying these sophisticated tools, schedulers must employ common sense and domain expertise. They can engage with the tools to find the optimum schedules.

- Scheduling Job Shop – A Case Study

Abstract

Job shop scheduling is crucial for effective machine utilisation in the manufacturing business. There are several methods for scheduling jobs that rely on equipment, tools, indirect consumables, and jobs that must be processed. In this paper, a case study for work scheduling when parts are handled on available equipment is presented. Time and motion studies are used to quantify setup and operation time as a total processing time for a variety of commodities using various manufacturing processes. Jobs are given different levels of priority based on their due dates, and jobs are scheduled according to their priority. Based on the measured processing time, the timings for processing some additional jobs are calculated, and an algorithm is presented and evaluated for efficient usage of the machines available.

1. Introduction

In manufacturing industries discrete products are manufactured through processing of raw materials on different machines. Different methodologies are used to organize the processing of jobs in different industries. The ideal handling areas are known as

job shop, cellular manufacturing, project shop, continuous manufacturing, flow line, flexible manufacturing and reconfigurable manufacturing systems. The efficient use of the manufacturing facility depends upon the proper scheduling of jobs which is dependent upon product's nature, its lot size, the transfer of energy and material and implementation of intangible functions like production planning, process planning, inventory control and supply chain management. The scheduling of jobs in industries is mostly done by experienced workers with pen, paper and graphical approach and also through some industrial databases. For immediate solution of problems, simple dispatching rules are typically used like sequencing at a level of work center. This can create confusions in scheduling, where it is difficult to predict the completion dates, also some time the jobs with higher priority are chased down by the management.

Many guidelines for scheduling jobs based on due dates, criticality of operation, processing time, resource usage, and critical ratio may be found in the literature. The crucial ratio is the ratio of a job's processing time to its due date. The shortest processing time (SPT) and longest processing time (LPT) rules are based on processing time (LPT). Tasks with a shorter processing time are processed first in SPT, whereas jobs with a longer processing time are handled first in LPT. Jobs with earlier due dates are processed first in order of priority. The other dispatching rules are first come, first serve (FCFS), which processes jobs in the order they arrive, and last come, first serve (LCFS), which processes jobs in the order they arrive.

Single machine scheduling, flow shop scheduling, and job shop scheduling are the most well-known scheduling methods in the manufacturing industry. All jobs are processed on a single computer, and jobs are planned in increasing order of time for the best results. Every work must be processed through each machine in a flow shop, and the processing time for each job varies per machine. In a flow shop, jobs are grouped in several orders and the optimal order is chosen based on the shortest batch completion time. In a job shop, each work may or may not be processed on all machines, and the time it takes to process each job varies per machine. It's one of the most difficult problems to solve, and optimum scheduling is difficult to compute. Many methods have been developed from time to time for optimal job shop scheduling.

A work shop is made up of machines that all do the same thing when it comes to material processing. The goal of the scheduling challenge is to sequence production choices in low-volume, high-variability systems where one component's path is set but varies from part to part. For example, a problem given by Muth and Thompson in 1963 involving 10 tasks and 10 machines remained unanswered for 20 years until it was solved by Carlier and Pinson in 1985.

Takefuji was the first to use a neural network for JSPs. Different neural architectures were later designed and successfully employed to tackle JSP. For job shop scheduling, genetic algorithms are created and employed.

For job shop issues, many heuristic approaches have been developed. Johnson, Gupta, bottle neck, Palmer branch and bind, and others are well-known. These methods have been refined through time, but changes are still needed. The main issue with these methods is that some are based on processing time while disregarding the due date, while others take the due date into account while ignoring the processing time.

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This study aims to provide a new algorithm for the job shop problem that takes into consideration projects with several machine operations and due dates, therefore accounting for lead time.

2. The Problem Formulation

Take a look at a typical job shop with n jobs and m machines. Each task will only be processed via one machine at a time, and all jobs will not be processed through all machines. Every work will be processed in a specified sequence. Each job's processing time is set and predictable.

To schedule jobs the following parameters, have to be identified first:

- I. The facility that is available on the shop floor.
- II. The total number of jobs in the system.
- III. Time spent setting up and operating each equipment for each work.
- IV. The amount of time it will take to perform all of the tasks.

Following the identification of the parameters, the jobs are scheduled using the limitations listed below.

- I. Every job will be processed through respective machines ($j = 1, 2, 3, \dots, m$).
- II. A single job can be processed by a single machine.
- III. The setup time for distinct processes is independent of the order.
- IV. The set up times for various operations are separate from the sequences and are factored into the processing time.
- V. Job scheduling will be done on a first-come, first-served basis (due dates). Three degrees of priority will be allocated to jobs. Priority 1 is the highest priority, priority 2 is the middle, and priority 3 is the lowest. The jobs with the highest priority will be scheduled first, followed by the jobs with the lowest priority.

4. Analysis of the Manufacturing Facility and the Jobs Processing

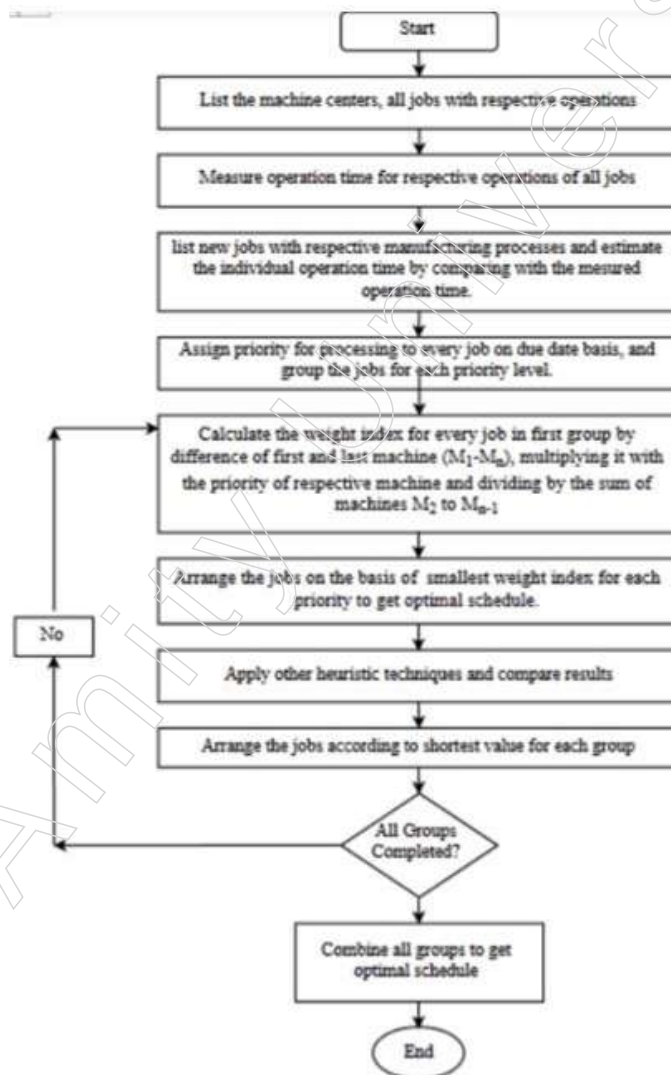
The availability of machines and the activities to be done on each machine are examined at the production facility. The production facility is split into machine centres for Shapers (S), Lathes (L), Milling (M), Boring (B), Welding (W), and Grinding (G) based on similarities. The jobs to be processed are chosen once the facility has been analysed. The time it takes to do each job on each machine centre is measured using a stop watch, and the job is completed by the best worker. Table 1 shows the order of operations and processing time for various jobs on different machine centres, as well as their priority order.

Table 1. Processing time for jobs (1, 2 ... m-1, m) on machines (1, 2... n-1, n)

Job Name	Jobs No	S	L	M	B	W	G	Priorit y
Nut	1	0	17	0	0	0	0	2
Bolt	2	0	19.5	0	0	0	0	2
Gear	3	60	18.5	30	15	0	4	1
Hexagonal Gear	4	65	17.5	45	4	0	6	1
Flange Pipe	5	0	50	35	35	27	25	2
Screw	6	62	58	15	0	0	0	1
Lead Screw	7	60	34.5	20	0	0	0	2
Universal Coupling	8	60	17.5	20	10	0	6	3
Lathe Center	9	63	29.5	0	0	0	5	3
Piston Connecting Rod	10	0	0	17	11	0	7	3
Sprocket	11	0	0	15	55	0	5	2
Tie Rod End	12	0	26	27.5	27	0	6.5	3
Gear Pump Flange	13	0	27	15	31	17	5	3

5. The Solution Methodology

In the proposed algorithm the jobs are scheduled on the basis of priority. The algorithm is described as following flow chart: Figure below

**Figure: Algorithm Flow Chart**

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5. Results and Discussions

Different well-known heuristics of job shop scheduling are applied to the 13 jobs and 6 machines problem in Table 1 to ensure that the suggested heuristic produces optimal results. Table 2 shows the results of using several heuristics, along with the computed sequence and total makespan.

Table 2. Comparison of algorithms

Algorithm	Sequence													Make Span (Minutes)
SPT	1	2	10	11	12	13	9	8	7	3	6	4	5	559.5
LPT	5	4	6	3	7	8	9	13	12	11	10	2	1	580
Johnsons	1	2	10	11	12	13	5	6	4	3	7	8	9	404.5
Gupta	1	2	10	11	12	13	5	6	4	3	7	8	9	404.5
CDS	1	2	11	13	12	10	8	4	5	9	3	6	7	424.5
Palmer	11	5	10	13	12	1	2	8	3	9	4	7	6	443
WI	10	5	12	11	13	1	2	6	3	4	7	8	9	404.5

Table 2 illustrates that the WI approach may produce optimal outcomes, and that it is similar to well-known heuristic strategies such as Johnson, Gupta, Palmer, and CDS. For the sequences obtained using different algorithms in Table 2, the effect of lead time is ignored. As a result, the goal of this research is to find the best sequence by taking into account both lead time and processing time. The processes outlined in solution approach are now applied to the lead time. Jobs with a one-day lead time are prioritised above jobs with a two- or three-day lead time. So, start with the tasks with a “1 day” lead time, calculate their WI, then sort them in ascending order of WI value. Apply the same approach to groups “2” and “3” as stated in Table 3 in combined form:

Table 3. Sequence based on WI

Machines/Jobs	S	L	M	B	W	G	Lead Time	WI
6	62	58	15	0	0	0	1	0.849315
3	60	18.5	30	15	0	4	1	0.88189
4	65	17.5	45	4	0	6	1	0.887218
5	0	50	35	35	27	25	2	-0.13605
1	0	17	0	0	0	0	2	0.058824
7	60	34.5	20	0	0	0	2	0.061135
11	0	0	15	55	0	5	2	0.085714
2	0	19.5	0	0	0	0	2	0.102564
10	0	0	17	11	0	7	3	-0.25
12	0	26	27.5	27	0	6.5	3	-0.08075
13	0	27	15	31	17	5	3	-0.05556
8	60	17.5	20	10	0	6	3	1.136842
9	63	29.5	0	0	0	5	3	1.966102

Makespan calculated for the schedule obtained in Table 3 is 498 minutes. By matching the results to, LPT, SPT, Gupta, Palmer, and Johnson rule it gives optimum results as shown in Table 4. If the priority of the jobs were changed the total makeup span will also vary.

Notes

Table 4. Comparison of results obtained from various algorithms

Algorithms	Total Make Span (Minutes)
SPT	503.5
LPT	525
Johnson	503.5
Gupta	503.5
CDS	522
Palmer	510
WI	498

The new method divides jobs into groups depending on lead time. Jobs with shorter lead times are prioritised, and the sequence is determined using the WI approach. Similarly, tasks with a medium lead time are placed after jobs with a shorter lead time, and their sequence is determined using the WI approach. Finally, tasks with a long lead time are put after jobs with a medium lead time, and their order is determined using the WI approach. As a result, taking the lead time into account yields a new optimum sequence. For new items, processing time is calculated based on available data. As a result, a new problem with six jobs and six machines is formed, along with the necessary lead time. Its goal is to see if the new heuristic produces optimal results when applied to a different group of problems. Table 5 summarises the findings:

Table 5. Processing time with priorities for new jobs

Machines / New Jobs	S	L	M	B	W	G	Priority
P1	65	10	0	0	5	0	1
P2	0	65	60	55	0	20	2
P3	85	60	40	60	25	0	3
P4	80	55	30	20	0	20	3
P5	80	20	30	0	10	10	2
P6	65	15	0	20	10	0	1

Neglecting lead time effect makespan is calculated based on SPT, LPT, Johnson, Gupta, CDS, Palmer, heuristic as shown in Table 6:

Table 6. Comparison of algorithms

Algorithm	Sequence						Make Span
SPT	P1	P6	P5	P2	P4	P3	560
LPT	P3	P4	P2	P5	P6	P1	435
Johnsons	P2	P3	P4	P5	P6	P1	390
Gupta	P2	P3	P4	P5	P6	P1	390
CDS	P2	P6	P5	P1	P6	P3	545
Palmer	P3	P4	P5	P1	P6	P2	560
WI	P2	P3	P4	P5	P6	P1	390

Table 6 shows that the makespan estimated using the WI approach produces the best results, which is equivalent to the Johnson and Gupta heuristic.

Now, taking into account the lead time, follow the methods outlined in the methodology section to determine the best sequence. Table 7 indicates the best job sequence on available machines:

Notes

Table 7. Sequence based on WI method

Machines / Jobs	S	L	M	B	W	G	Priority	WI
P6	65	15	0	20	10	0	1	1.444444
P1	65	10	0	0	5	0	1	4.333333
P2	0	65	60	55	0	20	2	-0.111111
P5	80	20	30	0	10	10	2	1.166667
P3	85	60	40	60	25	0	3	0.459459
P4	80	55	30	20	0	20	3	0.571429

Makespan computed is 500 minutes for sequence obtained based on WI method. Table 8 shows a comparison of WI to other algorithms:

Table 8. Comparison of Algorithms

Algorithms	Total Make Span (Minutes)
SPT	560
LPT	500
Johnson	505
Gupta	505
CDS	500
Palmer	560
WI	500

6. Conclusion

For job shop scheduling, LPT, SPT, Johnson, Gupta, and Palmer scheduling strategies are explored. A novel heuristic method called “weighted index (WI)” is devised that produces optimal outcomes that are equivalent to these four strategies. In comparison to existing heuristics, the new WI technique is simple and has a lower computing cost. The addition of assumption improves the new heuristic as well as the other four heuristics. The assumption is based on categorising jobs according to their lead times. Earlier techniques were solely dependent on processing time, with no consideration for lead time. Both processing time and lead time aspects both are included in these new modified methods.

4.3.1 Mean Flow Time

The mean flow time of a schedule provides a measure of the average time that a task spends within a computer system, and also the average number of incomplete activities in the system. A schedule's mean flow time is equal to the sum of all tasks in the system's completion times.

The mean Flow time of a schedule provides a measure of the typical time that a task spends within a computing system, and also the typical number of unfinished tasks within the system. A schedule's mean flow time is defined as the total of all tasks in the system's completion times. For establishing minimal mean flowtime schedules for n independent activities on a set of identical processors, $O(n \log n)$ methods exist. Generally, there'll be large class C of schedules, of widely different lengths, which will minimize mean flow time. the matter of finding the shortest schedule in C is NP-complete. We give heuristics that find schedules in C that are not any quite 25% longer than the shortest schedule in C . The advantage of a brief schedule is that processor utilization is high.

Each subsequent flow time is calculated by adding the present job's operating time to the previous job's flow time. When taking the typical flow time, the lower number the higher. A job's lateness is calculated by subtracting the maturity from the operating time.

4.3.2 Mean Tardiness

One of the most researched subjects in single machine difficulties is reducing overall tardiness. To handle this NP-hard problem, researchers propose a variety of optimization and heuristic strategies. The subject of minimising overall tardiness in a learning effect setting is investigated in this study. Learning effects refers to the reduction in processing times that occurs as a result of process repetition. To overcome the problem, a 0–1 integer programming paradigm is created. A random search, tabu search, and simulated annealing-based approaches are also presented for the problem, and these methods are used to find solutions to large-scale problems with up to 1000 tasks. To our knowledge, there are no works on the overall tardiness problem with a learning impact that has been addressed in this area.

The majority of research on classical scheduling issues assumes that all tasks have constant processing times, and that work processing times are independent of their position in the scheduling sequence. However, in many cases, this assumption is incorrect. Recent empirical investigations in a variety of sectors have confirmed that when companies create more of a product and gain knowledge or expertise, unit costs decrease. In the literature, this phenomenon is referred to as the “learning effect.” Since its discovery by Wright over half a century ago, the learning effect has been frequently used in management research. However, there have been very few research on production scheduling in general. These researches were conducted on single and multi-machine systems.

The principle of group technology is the manufacture of a family of related components or procedures utilising identical equipment and tools. A group or a cell in a group production and/or cellular manufacturing system will process a finite number of diverse but comparable jobs in a finite production cycle.

As a result, the machine operators will do a series of comparable procedures over and over again. As a result, operators will gain experience, progress up the learning curve, and create higher-quality components faster over time. As a result, small batch and job production systems use a lot of group technology and cellular manufacturing. It is self-evident how important it is to include learning effects when scheduling comparable projects for production.

In this section, we look at the overall tardiness problem in a single machine with a learning effect. The term “single machine scheduling” does not always imply that just one machine is involved. It may also refer to a collection of machines or a system that can be regarded as if it were a single unit.

One of the most essential objectives in single machine scheduling is to minimise overall tardiness, which is an NP-hard problem. For the static total tardiness problem, many implicit enumeration strategies have been proposed, including branch-and-bound, dynamic programming, dominance rules, and decomposition.

Table 1 summarises the major research breakthroughs in the quest to solve the static total tardiness problem to optimality. Branch and bound methods, on the

Notes

other hand, have significant computational needs, whereas dynamic programming approaches have high memory requirements, making them inapplicable for large-scale issues. The dynamic form of the overall tardiness problem, on the other hand, is frequently addressed by referring to priority rules. Two cost allocation strategies were provided for a dynamic total tardiness problem originating from a decentralised job acquisition by many agents. Because the total tardiness problem in a single machine is an NP-hard problem, the problem addressed here is likewise an NP-hard one. To our knowledge, there are no works on total tardiness with a learning impact.

Table 1
Optimizing techniques for the total tardiness problems [47]

Year	Method	Complexity	Reference
1961	BB ^a		Schild and Fredman [23]
1962	DP ^b		Held and Karp [29]
1964	DP	$O(n2^n)$	Lawler [30]
1968	Network model		Elmaghraby [48]
1969	Dominance rules		Emmons [38]
1971	DP-based hybrid		Srinivasan [31]
1972	BB	30 jobs, ≤ 4.13 s	Shwimer [24]
1975	BB	20 jobs, > 300 s	Rinooy Kan et al. [25]
1976	BB/ Lagrangian relaxation		Fisher [26]
1977	Decomposition/DP	$O(n^4 \sum p_i)$	Lawler [32]
1978	Decomposition/DP	50 jobs	Baker and Schrage [33]
	BB	20 jobs, ≤ 12.8 s	Picard and Queyranne [27]
	DP	50 jobs, 0.844 s	Schrage and Baker [34]
1982	Decomposition/DP	$O(n^7/e)$	Lawler [35]
	Lawler decompositions extension		Potts and Van Wassenhove [40]
1983	MINIT		Sen et al. [49]
1987	DP	≤ 100 jobs	Potts and Van Wassenhove [36]
1991	DP		Sen and Borah [37]
1995	Decomposition extension		Chang et al. [41]
1996	Extension of Emmons' dominance		Yu [39]
	Decomposition/BB	100–150 jobs	Szwarc and Mukhopadhyay [28]
1997	Decomposition		Tansel and Sabuncuoglu [42]

^a BB: Branch-and-bound.

^b DP: Dynamic programming.

1. Problem Description

In the problem under consideration, we assume that a set $N = \{1, 2, \dots, n\}$ of n concurrently available tasks is to be processed on a single machine. p_j stands for job j 's processing time, while d_j stands for the due date of the job j . Once a project is started on the machine, it must be finished without interruption (i.e., no preemption is allowed). At any one time, the machine cannot do more than one operation.

The tardiness for this work j happens when the completion time of any job j is longer than the due date. That is, $\sum T = \sum_{j=1}^n \max \{C_j - d_j, 0\}$. C_j is the completion time, while the job's tardiness is T_j respectively. In other words, if p_{jr} is the actual processing times of job j scheduled in position r in a sequence, then $p_{jr} = p_j a^r$ (where $a \leq 0$ is the learning index, given is the logarithm to the base 2 of the learning rate). The goal is to find a schedule that minimizes the total tardiness on single machine (the problem is denoted as $1/LE/\sum T$, where LE is learning effect).

Heuristic Methods

The majority of scheduling issues are NP-hard. As a result, the branch-and-bound, dynamic programming, and integer programming models can only tackle small-scale problems optimally. However, dealing with large-scale challenges in real-world applications in business may be required. The researchers looked at developing heuristic approaches for tackling challenges of this nature.

The proposed integer programming approach may handle the discussed single machine scheduling with learning effect problem optimally for small size problems with up to 25 tasks. That is why, in order to address large-scale issues, we used meta heuristic approaches. Three tabu search methods, three simulated annealing approaches, and a random search method are among the heuristics. The next sections go through these approaches in detail.

a) Tabu Search Heuristic

In combinatorial optimization, tabu search has been frequently employed. The main idea is to gently tweak a known (current) solution in a certain way (referred to as neighbourhood structure) and use the best change as the new current solution. Neighbours of the present solution are such changed solutions. A move is an operation that results in a neighbour. To prevent becoming caught in a local optima, the best neighbouring solution that is worse than the current solution is permitted to become the new current solution. Certain movements are labelled as tabu to prevent cycling. If an ambition requirement is met, a tabu move may be permitted. This process is repeated until a requirement is satisfied. An experimental approach was used in this work to determine appropriate tabu search parameters. Table 2 shows the chosen tabu search parameters in relation to the results. In addition, the tabu search strategies were divided into three categories based on the first solutions employed (TS-I, TS-II, TS-III). These are the EDD, E&G, and random solutions. The E&G heuristic process is briefly described here.

b) E&G Heuristic

The E-G heuristic uses a process similar to the NEH procedure. Jobs are sorted by their due dates in descending order. The best partial sequence of the first two tasks is discovered in terms of reducing overall tardiness with learning effect processing times, and the third job is incorporated into the best two-job partial sequence, resulting in the best three-job partial sequence. In this fashion, the entire sequence is built.

Table 2
Parameters of the tabu search

Parameters	TS-I	TS-II	TS-III
Initial solutions	Random	EDD	E&G
Tabu list length	$2\sqrt{n}$	$2\sqrt{n}$	$2\sqrt{n}$
Neighborhood search strategy:	API	API	API
Stopping criterion	n Iteration no improvement	n Iteration no improvement	n Iteration no improvement

c) Simulated Annealing

Simulated annealing (SA) is a meta-heuristic local search technique capable of escaping local optima. It has been a popular approach over the last two decades due to its ease of implementation, convergence qualities, and use of hill-climbing motions to escape local optima.

SA is a neighbourhood search approach that has shown to be effective in solving combinatorial issues. An initial solution is generated at random in a normal SA approach. At each stage, if a new solution in the neighbourhood of the current solution has a lower or equal cost (in our study objective function value), it is accepted as the new current solution; if it has a higher cost, it is accepted with a probability that

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decreases as the cost difference grows and the method's temperature decreases. This temperature, which is merely a positive integer, is gradually decreased by some temperature scheme as the technique advances, moving from a relatively high value to near zero.

As a result, most deteriorating movements are approved at the start of SA, but only improving moves are likely to be allowed by the conclusion. As the temperature approaches zero, the technique converges to a local optimum, but because SA has done several perturbations at a high temperature, pushing the search path into new locations, a better local optimum solution should hopefully be found. Figure 1 depicts a conventional SA algorithm.

To solve any problem using the SA algorithm, some parameters and rules must be established. These are (1) the temperature parameter's beginning value, (2) the number of iterations to be conducted at each temperature, (3) a cooling function, and (4) a stopping condition to end the process. Table 3 lists the chosen simulated annealing parameters in the problem based on experimental design findings. In addition, the simulated annealing methods were divided into three groups based on the initial solutions employed (SA-I, SA-II, SA-III). These were the EDD, E&G, and random solutions.

Step 1.
Given an initial solution $i \in S$;
Given simulation temperature values $T_0 > T_f > 0$;
Setting an iterative counter $k = 0$.

Step 2.
Generating an initial solution $j \in N_i$;
Evaluating $\Delta f = f(j) - f(i)$.

Step 3.
If 'Metropolis criterion' is satisfied, i.e., $\min\{1, \exp(-\Delta f / T_k)\} > \eta \in [0,1)$, then $i = j$.

Step 4.
If 'Metropolis equilibrium' under T_k is realized, then go to Step 5; Else go to Step 2.

Step 5.
If 'stop criterion' is not satisfied, i.e., $T_k > T_f$, then reducing the temperature $T_{k+1} = T_k - \Delta T_k, T_k > 0$, and $k = k + 1$, go to Step 2; Else output $i_{opt} = i$.

End.

Fig. 1. A standard SA algorithm.

Table 3
Parameters of the simulated annealing

Parameters	TS-I	TS-II	TS-III
Initial solutions	Random	EDD	E&G
Neighborhood search strategy:	API	API	API
Initial temperature	500	500	500
Final temperature	1	1	1
Cooling rate	0.995	0.995	0.995
Maximum iteration length at a particular temperature	1	1	1

C) Random Search

Random search is a technique for randomly selecting a set of solution points (a sample size) from the solution space. Random search compares the objective function values of the selected points (sequences) and finds the best sequence in the sample.

The optimal sequence is memorised, and the procedure is then repeated. The old sequence is updated with the new sequence if the new best sequence is better than the prior one; else, it remains unchanged.

After a certain number of iterations, the search is terminated. In the random search technique, there are two parameters: one is the sample size specification, and the other is the stopping criteria. Stopping can be done after a specified number of iterations or when no improvement is shown after a certain number of iterations. We used (n 1) as the sample size in the experimental investigation, and no improvement for n repetitions as the stopping criteria for the random search of the issue.

Experimental Results

All of the experiments in this study were carried out on a personal computer with a Pentium IV/2 512 RAM processor. Using the Hyper LINDO/PC 6.01 software package, the integer programming paradigm is employed to determine the best solutions to the problem at hand. C++ Builder 5 was used to create the heuristic algorithms utilised in this article. The precise answers for four distinct work sizes were discovered: $n = 10$, 15, 20, and 25. For each issue size, ten repetitions were performed.

According to Heizer and Render, various products have distinct learning curves, and the rate of learning changes based on the method and product's potential. They claim that between 1920 and 1988, learning rates for various industrial businesses in the United States ranged from 70% to 90%. For example, 79 percent and 80 percent of those employed in the steel and aeroplane assembly industries, respectively. As a result, we used an 80 percent learning rate in our investigation.

In this study, experimental analyses are realized at two stages:

1. Obtaining exact findings for small-scale issues and solving them using heuristic approaches, then comparing the answers to the exact results.
2. Using heuristic approaches to solve large-scale challenges.

4.3.3 Number of Tardy Jobs

Managers' success is frequently measured by the number of late jobs or, equivalently, the proportion of on-time shipments. This is because manufacturing executives in developed countries regard timely and accurate delivery as critical because it affects costs, profitability, and customer satisfaction, all of which affect a company's long-term financial performance by increasing repurchase intentions and strengthening customer loyalty.

When schedulers are dealing with various work sets belonging to distinct agents or clients, multi-agent scheduling models are used. With only two agents, the two-agent scheduling model is a specific instance of multiagent scheduling models. Researchers should look at two-agent scheduling since it lays the groundwork for building methods to handle multi-agent scheduling difficulties.

Agents compete with each other for the allocation of common resources in two-agent scheduling models. The two agents' goals are frequently at odds with one another. As a result, organisations that handle tasks from these agents must find a balance between the various agents' goals. Because it contains several objectives,

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the two-agent scheduling issue is significantly more difficult than the single-agent scheduling problem.

Artificial intelligence, decision theory, operations research, and other fields all benefit from two-agent scheduling applications (Agnētis et al., 2004). Schultz et al. (2002) discussed how a two-agent scheduling issue might be used in telecommunication to transfer data between agents (e.g. voice service, file transfer). The implementation of a two-agent scheduling model in machine maintenance between the maintenance department and the production department was studied by Perez-Gonzalez and Framinan (2014).

The emphasis of this research is on a two-agent scheduling issue with a weighted number of late work objectives. This kind of optimization issue involving late tasks might be used in a variety of businesses. Improving client satisfaction in the service business requires reducing the number of late tasks. Failure to finish projects on time is frowned upon in most sectors. Lowering the frequency of late jobs is an indication of better resource use and cross-departmental cooperation in supply chain management.

Each agent in a two-agent scheduling problem has its own objective function. Thus, depending on how the individual objective functions are treated as objectives and constraints in the corresponding optimization problem (Yin et al. (2020)), there are three different optimization problems associated with the two agents scheduling problem. Linear combination optimization model, restricted optimization model, and Pareto optimization model are the three optimization models that result. The constraint optimization model of the two-agent scheduling issue is used in this work.

This paper considers a two-agent single machine scheduling issue. Each agent has a distinct set of responsibilities. These jobs are all processed on the same computer and released at the same time. Each job's processing time, deadline date, and weight are unique. The goal of this challenge is to decrease the first agent's weighted number of late jobs while keeping an upper bound on the second agent's weighted number of tardy tasks.

Moore (1968) was a forerunner in the field of tardy job research. The intricacy of the problem was handled by the author, whose goal is to reduce the number of late jobs on a single system. A fully-time polynomial method was also proposed by the author. The minimization of the weighted number of tardy tasks on a single machine is an NP-hard issue, according to Karp (1972).

To overcome this problem, M'Hallah and Bulfin (2003) presented a branch and bound method. Baker and Smith (2003) and Agnētis et al. (2003) were the first to propose two-agent scheduling models (2004). They worked on the issues with two agents and two task sets that belonged to each of these agents. They worked on challenges involving a variety of scheduling objectives, such as the makespan objective, completion time objective, maximum lateness objective, and so on.

These goals are non-decreasing functions that are proportional to the amount of time it takes to complete them (Mor and Mosheiov, 2010). The addition of an agent having a list of jobs for individual agents brings multi-agent scheduling models closer to real-life circumstances than bi-criteria or multi-objective scheduling issues. The method the objectives are optimised differs significantly between Baker and Smith (2003) and Agnētis et al. (2004). Baker and Smith applied weights to each aim and reduced the

total objective to the smallest possible value. Agnetis et al. (2004) applied the concept of constrained optimization, in which the first agent's objective function is minimised while the objective functions of the remaining agents are kept within a predetermined range.

Ng et al. (2006) investigated the two-agent scheduling issue with the goal of minimising the first agent's weighted completion time while keeping an upper constraint on the number of tardy tasks. The author established the problem's difficulty and devised a pseudo polynomial method to solve it. Cheng et al. (2011) expanded on Ng et al. (2006)'s research by taking into account the learning impact. To overcome the problem, they presented a branch and bound approach as well as three simulated annealing algorithms. Wu (2013) investigated the topic of reducing overall tardiness for the first agent while the second agent has no tardy jobs. To tackle this problem, the author gave both an exact and a genetic algorithm. Xintong and Yong (2017) investigated the challenge of minimising total weighted late work for the first agent while keeping an upper constraint on the second agent's maximum regular function cost.

An extensive literature on due date related two-agent scheduling problem can be found in the work of Li et al. (2020). For a feasibility model, Cheng et al. (2006) developed a dynamic-programming-based precise method. The feasibility model's dynamic programming approach is enhanced to tackle the challenge in this research. However, there are two major drawbacks to this precise approach. The algorithm's initial restriction is that it could only tackle issues involving integral weights.

The second drawback is that it could only solve a limited number of cases. As a result, meta-heuristics were created to handle larger problems. Researchers can use meta-heuristics to construct heuristics to tackle their issues since they give a generic framework and methodologies. To find better answers, the meta-heuristics go on a single stroll or numerous walks throughout the neighbourhood.

Problem Description

The dilemma revolves around the roles of two agents, namely agent A and agent B. Each agent has a set of jobs (JA, JB), each of which contains nA / nB jobs respectively. Each work has a processing time, a deadline date, and a weight assigned to it. At time zero, jobs are ready to be processed. Jobs are processed on a single machine, therefore one job at a time is processed. Preemption of jobs is not permitted. The jobs of agents A and B can be completed in any order. A late job is one that completes its processing beyond its specified due date.

$$n^A / n^B$$

Number of jobs in agent A/ agent B

$$n$$

Total number of jobs, $n = n^A + n^B$

$$i/j$$

Index for jobs

$$J^A / J^B$$

Job set belonging to agent A/ agent B,

$$J_A = \{J_1^A, J_2^A, J_3^A, \dots, J_{n^A}^A\}$$

$$J_B = \{J_1^B, J_2^B, J_3^B, \dots, J_{n^B}^B\}$$

Notes

p_i^A / p_i^B	Processing time of each job from agent A/agent B respectively
D_i^A / D_i^B	Due date of each job from agent A/agent B respectively
W_i^A / W_i^B	Weight of each job from agent A/agent B respectively
σ	Ordered set of jobs already scheduled
$C_i^A(\sigma) / C_i^B(\sigma)$	Completion time of each job from agent A/agent B respectively
$U_i^{A/B}$	Binary variable takes the value of 1 if job is tardy, or 0 otherwise
Q	Upper bound of weighted number of tardy jobs of agent B..

Given the notations above, the problem can be represented as follows: the problem can be represented as minimization of weighted number of tardy jobs of agent A (i.e. Minimize $\sum_{1 \leq i \leq n^A} w_i^A U_i^A$), while keeping the weighted number of tardy jobs of agent B within a specific limit (i.e. $\sum_{1 \leq j \leq n^B} w_j^B U_j^B \leq Q$)

Check Your Understanding

- 1) What is the full form of FCS?
 - a) Finite Capacity Scheduling
 - b) Infinite Capacity Scheduling
 - c) Production Scheduling
 - d) Real-Time Scheduling
 - e) None of the above
- 2) The _____ is based on the premise that a production system has a single resource limitation and that the other resources have the capacity to completely support any realistic schedule on the constraint resource.
 - a) Heijunka scheduling
 - b) DBR scheduling approach
 - c) Job-Shop Scheduling
 - d) Work-Shop Scheduling
 - e) None of the above
- 3) A _____ is equal to the sum of all tasks in the system's completion times.
 - a) Minimal Mean Flow
 - b) Schedule's Mean Flow Time
 - c) Minimize mean Flow
 - d) Mean Tardiness
 - e) None of the above
- 4) _____ also known as dispatching rules.
 - a) Priority Rules
 - b) Scheduling Rules

- c) Heuristic rules
d) Dynamic Approach
e) None of the above
- 5) There are three main approaches used for handling the JSP are _____, _____ and Artificial Intelligence (A.I.) approaches.
- a) Dispatching Rules and Scheduling Criteria
b) Classical Approach and Dynamic Process
c) Heuristic Rules and Scheduling Job Shops
d) Heuristic Rules and Classical Optimization
e) None of the above
- 6) What is the full form of GA?
- a) Genetic Algorithm
b) Greater Algorithm
c) Generate Algorithm
d) Geographical Algorithm
e) None of the above
- 7) Full form of TOC is:
- a) Theory of Concepts
b) Theory of Constraints
c) Time of Constraints
d) Theory of Combination
e) None of the above
- 8) Drum-Buffer-Rope involves a procedure called _____ that addresses the system variation, from the execution point of view.
- a) Kanban Management
b) Factory Management
c) Buffer Management
d) Supply Chain Management
e) None of the above

Summary

- Enterprise resource planning (ERP) systems, which are complex enterprise-wide, integrated information systems, are being implemented at a rapid pace in manufacturing today.
- The Drum-Buffer-Rope (DBR) approach of the theory of constraints is one of the basic methods offered for production scheduling in complicated job shops (TOC).
- Flow shop models, including permutation and generic flow shops, are constrained job shop scheduling models.

Notes

- One common scheduling problem is the job shop, in which multiple jobs are processed on several machines.
- Job shop scheduling has become a necessity for modern-day manufacturers that are seeking to boost production efficiency and eliminate waste within their manufacturing operation.

Activity

1. Write a case study on Job Shop Scheduling.

Questions & Exercises

1. What is Job Shop Scheduling? Write down the different approaches to Job Shop Scheduling.
2. What is number of tardy jobs? Explain the concept of mean tardiness and the methods employed to reduce tardiness.

Glossary

- Scheduling may be described as “the allocation of resources across time to accomplish a collection of activities” or “the process of allocating a set of tasks to resources over a period of time.”
- The mean flow time of a schedule provides a measure of the average time that a task spends within a computer system, and also the average number of incomplete activities in the system.
- Heuristic rules, also known as dispatching rules, priority rules, or scheduling rules, are used to determine which work will be chosen next from a list of jobs waiting to be made on a certain machine, without first seeing the results of the selections.
- A Job Shop is a work location in which a number of general-purpose work stations exist and are used to perform a variety of jobs.
- DBR is a simple concept to grasp and put into reality, and a scheduler may quickly execute it with the help of an Excel spreadsheet.

Further Readings

1. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
2. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your understanding – Answers

- | | |
|-------|-------|
| 1) a) | 2) b) |
| 3) b) | 4) c) |
| 5) d) | 6) a) |
| 7) b) | 8) c) |

Unit - 4.4: Single Processor Job Shop Scheduling

Notes

Objectives

At the end of this unit, you will be able to:

- Learn single processor job shop scheduling - due date method
- Understand the shifting bottleneck procedure
- Learn about shortest processing time method

Introduction

The job shop is a frequent scheduling issue in which many jobs are handled on various machines. Each job consists of a series of activities that must be completed in a specified order, and each work must be completed on a specific machine. The task may, for example, be the production of a single consumer item, such as a car. The goal is to arrange the work on the machines in such a way that the length of the schedule—the time it takes for all of the activities to be completed—is kept to a minimum.

4.4.1 Single Processor Job Shop Scheduling - Due Date Method

The due date is the date by which a task must be finished in a job shop. When jobs (client orders) are received, judgments on due date assignment are made. In order to sustain high delivery performance, good due date assignments are required (delivery speed and delivery reliability). Due dates can be set in two ways: (1) exogenously or (2) endogenously.

In the first situation, due dates are set by third-party organisations (sellers, buyers). The current study focuses on the second scenario, in which due dates are defined internally based on task and shop characteristics in order to optimise job shop delivery performance. The endogenous due date assignment is particularly beneficial when manufacturers need to “promise” a delivery date to clients, as well as for improved shop floor activity management. We look at a variation of the Job Shop Scheduling (JSS) Problem in this study. The suggested model includes all of the elements of the well-known Job Shop model, as well as release dates and deadlines for all jobs. The goal remains the reduction of the makespan. This model is referred to as the Job Shop Scheduling Problem with Deadlines (JSSD).

The introduction of deadlines is for practical reasons. The first is that the Job Shop model does not capture an important component of a manufacturing department on its own. In actuality, there is never a single production run; instead, new work arrives on a regular basis, requiring any timetable to be adjusted. Of course, the Job Shop concept may be used again whenever new tasks become available, but it does have certain disadvantages. In most cases, existing projects must be finished within previously agreed timeframes, while new jobs must be performed as quickly as feasible. Finding the best potential makespan for the new projects gives an indicator for defining their deadlines in case they're needed later for schedule calculations or to agree on delivery dates with the client. To deal with this production paradigm, it is clear that deadlines must be added to the standard Job Shop structure.

Notes

A second issue is that some devices may become unavailable at times. This can be due to machine failure or maintenance, as well as the fact that a machine is already assigned to a higher priority project for a set period of time. To put it another way, the department manager may choose which jobs are more important and schedule them first. Then the additional work would be scheduled, with the caveat that the machines would no longer be accessible at all times. This may be achieved by treating all higher priority work activities as dummy jobs with release dates and deadlines that match to their start and finish timeframes.

Release dates and deadlines are a useful tool for controlling the final schedule in general. A proposed schedule is frequently required to be revised due to factors beyond the model's control. It is frequently feasible to adjust the solution in the direction of a more acceptable timetable by modifying certain release dates and/or deadlines a posteriori. Despite this, and most likely because of the intricacy of the topic, the JSSD has never been examined in the literature to our knowledge.

Because the Job Shop Issue is NP-hard and the JSSD is a particular instance of it, the latter problem is also NP-hard.

In this study, we propose a heuristic strategy for the JSSD based on Adams, Balas, and Zawack's Shifting Bottleneck approach (1988). This method iteratively detects a bottleneck machine and optimises its sequence while keeping the task sequence constant on the machines that have previously been handled and ignoring the others. This is accomplished by resolving a single-machine challenge involving release periods and deadlines. Despite the fact that this issue is strongly NP-complete, it can be handled fast in practise because of Carlier's ingenious branch-and-bound approach (1982). After all of the machines have been sequenced, each one is liberated and re-sequenced in a cyclical manner.

This technique has recently undergone two significant enhancements, both of which may be applied to JSSD. The first one deals with the following problem. When sequencing a machine, it's possible to set restrictions on the sequence of another machine, such as requiring task i to come before job j by a certain amount of time. These are referred to as delayed precedence limitations (DPC). Taking these DPC into account necessitates a significant revision of Carlier's approach. Balas, Lenstra, and Vazacopoulos (1995) proposed a modified method for addressing the one-machine issue using DPC, which was demonstrated to greatly enhance the performance of the Shifting Bottleneck Procedure.

The second enhancement involves combining the Shifting Bottleneck approach, which can be thought of as optimization over the neighbourhood defined by arbitrary changes in the sequence of any single machine, with optimization over the neighbourhood defined by interchanging certain pairs of jobs anywhere in the overall job sequence. This is accomplished by adopting the Shifting Bottleneck technique as a basic framework and employing DPC to solve a series of one-machine issues, but replacing the cyclic re-optimization phase with a guided local search procedure based on pairwise job exchange. This method, as reported by Balas and Vazacopoulos (1994), has resulted in considerable improvements.

Both of the aforementioned Shifting Bottleneck technique improvements are included in our process for the JSSD problem. We proposed an algorithm for solving

the one-machine issue with DPC to optimality in the presence of task deadlines. The backbone of our whole heuristic technique for the JSSD problem is this precise algorithm. We also adjust the guided local search when deadlines are present.

Leon and Wu (1992) studied an important particular case of the one-machine problem with deadlines: the one-machine problem (without DPC) with unavailability across certain time intervals. In the manner explained above, this problem may be included into the framework of release dates and deadlines. Our algorithmic method differs from that of Leon and Wu (1992) and performs better in terms of computing.

We use the continuous production model described at the outset to test our procedure for the JSSD. First, we run the procedure on data from a factory with a production environment that closely resembles that of a job shop. Second, we use the approach to analyse data from a well-known benchmark case. The findings indicate that we have created a valuable tool for addressing the issue at hand. We appear to have struck a fair balance between computing time and solution quality in particular.

Problem Description

The Job Shop Scheduling Problem with Deadlines is defined as follows: a set $J = \{1, \dots, |J|\}$ of jobs have to be processed on set M of machines within the shortest possible time, subject to the constraints that (i) the sequence of machines for each job must be prescribed, (ii) each machine can only process one job at a time, and (iii) jobs must begin after given release dates and end before given deadlines. An operation is the process of processing a work on a machine for a set amount of time.

We denote by

Problem JSSD

$$\begin{aligned} \hat{t}_n &:= \min t_n \\ \text{s.t.} \quad & t_0 = 0 \quad (1) \\ & t_j - t_i \geq p_i \quad (i, j) \in A \quad (2) \\ & t_j - t_i \geq p_i \quad \vee \quad t_i - t_j \geq p_j \quad (i, j) \in E_k \quad k \in M \quad (3) \\ & t_{\alpha(j)} \geq r_j \quad j \in J \quad (4) \\ & t_{\omega(j)} + p_{\omega(j)} \leq d_j \quad j \in J \quad (5) \end{aligned}$$

We get a normal Job Shop Problem by dropping the limitations (5) and (4) into Problem JSSD. We point out that the limitations (4) may be easily included into a conventional Job Shop Problem by simply introducing dummy operations of duration r_j . As a result, we'll presume that release dates have already been determined in this manner. The addition of deadlines distinguishes the problem from a conventional JSS and makes it more complex.

A disjunctive graph $G = (N, A, E)$ with node set N , directed arc set A , and undirected edge set E can be used to illustrate the problem. Because the edges in E are orientable, they are referred to as disjunctive, whereas the arcs in A are referred to as conjunctive. The length of an arc $(i, j) \in A$ is p_i , whereas the length of an edge $\{i, j\} \in E$ is either p_i or p_j depending on its orientation (if we choose (i, j) then it is p_i , otherwise it is p_j). A disjunctive clique is formed of a set N_k of nodes (operations) and a set E_k

Notes

of edges for each machine. Let $D = (N, A)$ represent the directed graph formed by deleting all the disjunctive edges of G . A machine selection S_k is a set of arcs obtained by orienting each edge in E_k . If S_k is acyclic then it induces a complete ordering on the operations on machine k .

The Shifting Bottleneck Procedure

The Shifting Bottleneck Procedure is generalised to the JSSD in this section. The objective behind the Shifting Bottleneck Procedure is to sequence the machines one by one. The current most essential machine receives priority on a dynamic basis. The following is a quick summary of Adams, Balas, and Zawack's (1988) Shifting Bottleneck Procedure (referred to as SB1) for the Job Shop Problem (without deadlines). Let M' be the set of machines that have previously been sequenced, i.e., the machines for which a selection has been computed ($M' = \emptyset$ at the start).

Step 1. Identify a bottleneck machine k among the unscheduled machines $M \setminus M'$ and put it in the best possible order. Set $M' \leftarrow M' \cup \{k\}$ and go to step 2.

Step 2. On the machines in M' , reoptimize the sequence. If $M' = M$, then stop; otherwise go to 1. Step 2's resequencing step is carried out on a single machine at a time. In the initial version of the SB1, this job, as well as the one of finding the bottleneck machine, was accomplished by solving a one-machine problem using Carlier's method (1982). Balas, Lenstra, and Vazacopoulos (1995) improved on this by formulating and solving a one-machine problem with delayed precedence constraints (DPCs).

Balas and Vazacopoulos (1994) suggested a variation of the Shifting Bottleneck Technique in which, instead of calculating a sequence of one machine problems, Step 2 is carried out by a novel local search procedure based on pairwise interchange. This local search is based on a specific notion of a solution's neighbourhood and the qualities of critical pathways in the disjunctive graph DS . SB2 is a novel method that has produced better computational results than SB1.

To address the Job Shop Problem with Deadlines, we'll employ an approach similar to SB2. The Balas, Lenstra, and Vazacopoulos (1995) method for the one machine problem and a local search for a JSS Problem given by Balas and Vazacopoulos are two essential tools in SB2 (1994). In order to meet the deadlines, we must change both the one machine problem and the local search. In terms of the one machine problem, we've devised a step-by-step approach for resolving a one-machine problem with deadlines. In the following part, we'll go through how to use this approach. In terms of local search, we have somewhat modified Balas and Vazacopoulos' (1994) local search by depending on auxiliary job shop difficulties.

A general scheme of the resulting procedure, denoted SBD, is as follows:

Let M' be the set of machines that have previously been sequenced, i.e., the machines for which a selection has been computed ($M' = \emptyset$ at the start).

Step 1. Locate a bottleneck machine k among the unscheduled machines $M \setminus M'$ and order it in the best possible order by solving a one machine problem with deadlines. Set $M' \leftarrow M' \cup \{k\}$ and go to step 2.

Step 2. Improve the present partial selection through the modified local search procedure. If $M' = M$, stop; otherwise go to 1.

In Step 1, the criterion for identifying a bottleneck machine is as follows: if all one-machine problems are feasible, the bottleneck machine with the largest makespan is the bottleneck machine; if there are infeasible one-machine problems, the bottleneck machine with the largest tardiness, that is, the largest deadline violation is the bottleneck machine.

For a variety of reasons, single machine scheduling (SMS) issues appear to have gotten a lot of attention. These sorts of challenges are significant both in terms of their intrinsic significance and as building blocks for more broad and sophisticated problems. In a multi-processor system, a single processor schedule can be used to avoid bottlenecks or to manage job assignment to a costly processor; in other cases, a whole manufacturing line can be viewed as a single processor for scheduling reasons.

SMS issues are also more theoretically tractable than multiple processor scheduling. As a result, additional problem classes may be solved in polynomial time, and a wider range of model parameters can be examined, such as different types of cost functions or the introduction of change-over cost. Thus, single processor difficulties are of a fundamental nature, allowing for some insight and development of concepts when dealing with more general scheduling issues. The relative simplicity of single-processor scheduling on the one hand, and its essential nature also for multiprocessor scheduling issues on the other, motivates a more thorough examination of the single-processor case.

4.4.2 Shortest Processing Time Method

To tackle one and two machine job shop challenges, job shop scheduling models are utilised. Shortest processing time, first come, first served, due date scheduling, Moore's technique, slack time, slack per operation, longest processing time, and critical ratio are all options for the one machine problem. Johnson's approach is used to decrease the makespan for two machines.

Methods. The rules available for scheduling include:

1. Shortest processing time (SPT)
2. First come first serve (FCFS)
3. Earliest due date (Due Date)

Shortest Processing Time (SPT): The quickest processing time is (SPT) The shortest processing time rule prioritises jobs with the shortest processing times first. When a machine is released, it will begin processing the shortest work available at the moment. This approach is best for determining the shortest total and weighted completion times.

It works on a priority sequencing rule that states that the work that requires the least amount of processing time gets processed first.

Analysis

Method.

Notes

Job scheduling data provided by Positrol director of operations is listed below in table 1:

Table 1: Job Scheduling Data

Job	Date Order Received	Operating Time	Due Date
XC	Mar 3	5.1	12
EFF	Mar 20	3.2	20
ABC	Feb 2	4.5	10
RBP	Feb 11	6.0	16
SDS	Mar 8	2.9	18

The average job flow time and average job lateness are used to assess the efficacy of the existing FCFS regulation and two alternative sequencing rules, SPT and EDD. The rule is more effective if the average flow time and average lateness are as low as possible. The total flow time for the jobs divided by the number of jobs equals the average flow time for a set of jobs. The total lateness for a group of jobs is divided by the number of jobs to get the average lateness. The time it takes to process a job comprises both the time it takes to process it and the time it takes to wait for it to be processed. The discrepancy between the job flow time and the due date is known as job lateness. We presume that the data supplied by the director of operations was reasonable and reliable, that the work set is understood, and that the due date is not always at the front of the manager's mind.

SPT is an abbreviation for Strategic Planning (Shortest Processing Time). According to the SPT sequencing rule, the work that takes the least amount of time to process is completed first. According to Table 2, under the SPT premise, jobs ABC, XC, and RBP were not finished by the deadline.

Average flow time is 11.4 days and

The average lateness is -3.8 days.

Table 2: Short Processing Time (SPT)

	1	2	1-2
Job Sequence	Flow Time	Due Date	Lateness
SDS	2.9	18	-15.1
EFF	6.1	20	-13.9
ABC	10.6	10	0.6
XC	15.7	12	3.7
RBP	21.7	16	5.7
Totals	57		-19

Average Flow Time: $57/5 = 11.4$ days

Average Lateness: $-19/5 = -3.8$ days

FCFS (First Come, First Served). FCFS sequencing rule is that "jobs are processed in the order in which they arrive at a machine or work center."¹ This is the sequencing rule that Positrol is currently using. Table 2 shows the job flow time, lateness, and averages. Job ABC and RBP both have a negative lateness value,

indicating that they were finished ahead of schedule under the FCFS system. The average flow time is 14.16 days and the average lateness is -1.04 days.

Table 3: First come First Served (FCFS)

FCFS	1	2	1-2
Job Sequence	Flow Time	Due Date	Lateness
ABC	4.5	10	-5.5
RBP	10.5	16	-5.5
XC	15.6	12	3.6
SDS	18.5	18	0.5
EFF	21.7	20	1.7
Totals	70.8		-5.2

Average Flow Time: $70.8/5 = 14.16$ days

Average Lateness: $-5.2/5 = -1.04$ days

EDD (Earliest Due Date). EDD sequencing rule is that "jobs are processed according to the due date, earliest due date first." Table 4 shows that both SDS and EFF were not completed before the due date under EDD rule. The average flow time is 13.98 days and the average lateness is -1.22 days.

Table 4: Earliest Due Date (EDD)

	1	2	1-2
Job Sequence	Flow Time	Due Date	Lateness
ABC	4.5	10	-5.5
XC	9.6	12	-2.4
RBP	15.6	16	-0.4
SDS	18.5	18	0.5
EFF	21.7	20	1.7
Totals	69.9		-6.1

Average Flow Time: $69.9/5 = 13.98$ days

Average Lateness: $-6.1/5 = -1.22$

Three rules are compared. Table 5 shows that the SPT sequencing rule has the shortest average flow time and the shortest average lateness when compared to the average values for FCFS and EDD. As a result, the SPT rule is the most effective method for scheduling client orders. Why is it that the rule is more effective when the average numbers are lower? The rationale for this is that a shorter average flow time means fewer in-process inventory. Furthermore, decreasing average lateness might lead to improved customer service. The FCFS and EDD regulations turn out to be the least effective of the rules in this scenario.

Table 5: Comparison of the three rules

Rule	Average Flow Time (days)	Average Lateness (days)
FCFS	14.16	-1.04
SPT	11.4	-3.8
EDD	13.98	-1.22

Notes

Notes

Limitations

In the actual world of business, new positions may arrive at any time once processing begins, or one of the five existing jobs may be eliminated. There may be some processing delays due to mechanical problems, accidents, or worker sickness. Instead of being deterministic, processing times might be variable. Another disadvantage of this study is the tiny sample size. A broader sample of work would be more typical of real-life situations. Either of these constraints would have an impact on our findings and might lead to the discovery of a new sequencing rule.

Conclusion

- The shortest processing time (SPT) is the optimum sequencing rule for scheduling Positrol client orders because it has the lowest average flow time (11.4 days) and average lateness (-3.8) when compared to the other two sequencing rules.
- A bigger work sample would be more typical of real-world situations, resulting in a shift in the optimum sequencing rule.

Check Your Understanding

- 1) _____ is used to decrease the makespan for two machines.
 - a) Johnson's approach
 - b) Graphical Approach
 - c) Heuristic Approach
 - d) Annealing-Based Approach
 - e) None of the above
- 2) What is the full form of JSDD?
 - a) The Job Scale Scheduling Problem with Deadlines
 - b) The Job Shop Scheduling Problem with Deadlines
 - c) The Job Shop Scheduling Problem with Demand
 - d) The Job Shop Scheduling Process with Deadlines
 - e) None of the above
- 3) To tackle one and two machine job shop challenges, _____ are utilised.
 - a) Mathematical Model
 - b) Continuous production Models
 - c) Scheduling Models
 - d) Job Shop Scheduling Models
 - e) None of the above
- 4) What is the full form of SPT?
 - a) Shortest Processing Time
 - b) Same Processing Time

- c) Shortest Part Time
 - d) Similar Processing Time
 - e) None of the above
- 5) What is the full form of FCFS?
- a) First come Finish Serve
 - b) Fast Come Fast Serve
 - c) First come first serve
 - d) Fast Come First Serve
 - e) None of the above
- 6) Is the approach which is best for determining the shortest total and weighted completion times.
- a) Shortest processing time
 - b) Sampling Approach
 - c) Heuristic Approach
 - d) Supply Chain Management Approach
 - e) None of the above
- 7) The elapsed time between the start and finish of a sequence of operations in a set of machines is _____.
- a) Holding costs
 - b) Storage Costs
 - c) Makespan
 - d) Opportunity Cost
 - e) None of the above

Summary

- Moore's technique, slack time, slack per operation, longest processing time, and critical ratio are all options for the one machine problem.
- The objective behind the Shifting Bottleneck Procedure is to sequence the machines one by one.
- One common scheduling problem is the job shop, in which multiple jobs are processed on several machines.
- Each job consists of a sequence of tasks, which must be performed in a given order, and each task must be processed on a specific machine.
- The due date is the date by which a task must be finished in a job shop. When jobs (client orders) are received, judgments on due date assignment are made.
- In most cases, existing projects must be finished within previously agreed timeframes, while new jobs must be performed as quickly as feasible.

Notes

- A proposed schedule is frequently required to be revised due to factors beyond the model's control.

Activity

1. Write a case study on worst case performance of Shortest Performance Time.

Questions & Exercises

1. What is single Processor job shop scheduling? Explain the rules available for scheduling.

Glossary

- The quickest processing time is (SPT) The shortest processing time rule prioritises jobs with the shortest processing times first.
- FCFS (First Come, First Served). FCFS sequencing rule is that "jobs are processed in the order in which they arrive at a machine or work center.
- EDD (Earliest Due Date). EDD sequencing rule is that "jobs are processed according to the due date, earliest due date first.
- Job Shop is one common scheduling problem, in which multiple jobs are processed on several machines.
- The elapsed time between the start and finish of a sequence of operations in a set of machines is makespan.

Further Readings

- 1 Production and Operations Management; S. Anil Kumar/ N Suresh
- 2 Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor

Check your understanding – Answers

- 1) a)
- 2) b)
- 3) d)
- 4) a)
- 5) c)
- 6) a)
- 7) c)

Unit - 4.5: Concepts of Two Machines Scheduling

Notes

Objectives

At the end of this unit, you will be able to:

- Understand concepts of two machines scheduling - Johnson's Rule
- Learn about Johnson's Algorithm

Introduction

Johnson's principle is a method for shortening the time it takes to accomplish a set of jobs which must be performed on two distinct machines or at two different work centres.

Johnson's approach is used to sequence 'n' jobs via two work centres. The objective is to reduce equipment idle time as well as the total time required to accomplish all activities. Because there are no priority constraints and all activities are equal in significance, arranging jobs based on the amount of time they take may minimise the length of time jobs stay idle on machines. This minimises the overall time required.

4.5.1 Concepts of Two Machines Scheduling - Johnson's Rule

Johnson's rule is a strategy for reducing the time it takes to complete a collection of jobs that must be handled on two machines or at two different work centres.

For sequencing 'n' jobs via two work centres, Johnson's method is utilised. The goal is to decrease machine idle time and the overall time it takes to complete all of the operations. Because there are no priority rules because all activities have equal importance, ordering jobs according to the amount of time they require may reduce the amount of time jobs remain idle on machines. This reduces the total time taken.

The algorithm can be fulfilled in the following steps.

- Step 1: For each work, find the shortest time between the times taken by machines 1 and 2.
- Step 2a: Place the related work in the first available position in the final sequence if machine 1 requires the shortest processing time to finish the job. Then proceed to step 3 to complete the process. (In the event of a tie, you may apply the above rule to any of them.)
- Step 2b: Place the linked job in the last available slot in the final sequence if machine 2 requires the least processing time to finish the work. Then proceed to step 3 to complete the process. (In the event of a tie, you may apply the above rule to any of them.)
- Step 3: Remove the assigned job from consideration and return to step 1 until all the positions in the sequence are filled.

Consider the following example:

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Parts			
Machines	P1	P2	P3
M1	30	14	30
M2	25	28	19

The minimum time taken is 14, by P2, on M1. So P2 will be first in the sequence; since P2 is assigned, we remove it from consideration.

The next least time is 19, taken by P3 on M2. Since it is on M2, it is assigned is the last position in the sequence.

The remaining middle position is then taken up by P1. Sequence: P2 P1 P3

- Johnson's Algorithm

In a manufacturing or production system, Johnson's algorithm is used to determine the best sequence or shortest pathways. Johnson's rule is a technique of scheduling tasks in two work centres in operations research and production management, with the primary goal of determining the best sequence of jobs to decrease makespan. It also cuts down on idle time between the two work locations, however the outcomes aren't always ideal, particularly for a small set of projects.

Johnson's method is one strategy that may be used to reduce the total time elapsed between the start of a task on one machine and the completion of a job on another. The heuristic algorithm can be used to schedule jobs. Johnson's algorithm is a simple method for determining the best order (Taha, 2012). Johnson Solution is a great algorithm for the shortest path issue, according to Sanan et al. (2013). (SPP). They went on to say that the Johnson Algorithm can solve combinatorial issues and can be used in a variety of situations, particularly when processing n jobs across two or more computers is necessary.

According to Johnson E. B. (1977), it is vital to look at the flow process of jobs through the available machines when addressing the shortest-path problem. The argued that understanding the possible pathways job flow process and estimating the minimal length from source to destination is critical. Johnson method, unlike other algorithms that only evaluate positive values, may be applied in real-value problems where the distance between two places is considered, especially scenarios where weights are allocated for positive and negative weights.

Other areas of research where Johnson's method has been successfully shown include the work of (Black 2004; Cormen et al. 2001; Russel and Norvig 2003, Pogorilyy et al. 2017). In a machine shop, scheduling is the process of allocating start and end times to each individual order and project. As a result, scheduling can increase shop floor efficiency by giving a timetable for processing a collection of activities. It's nothing more than arranging numerous jobs on a set of resources (machines) in order to maximise specific performance metrics.

There are ' n ' tasks in the flow shop scheduling issue, each of which requires processing on ' m ' separate machines. The process sequence of a task refers to the order in which the machines must process the work. All of the occupations have the same procedural sequences. However, the processing times for different jobs on a

system may vary. When an operation is missing from a task, the processing time of that operation is deemed to be 0. A work shop is a sort of manufacturing facility that produces small quantities of a wide range of specialised items. Most of the goods produced in a work shop require a particular set-up and sequencing of processing stages. In a process layout, similar equipment or functions are placed together, such as all drill presses in one area and grinding machines in another. Material handling, cost, and work-in-process inventories are all minimised by the arrangement.

Related Researching on Sequencing or Scheduling of Jobs

In his book “Scheduling Algorithms,” Bruker (1998) stated plainly that the Johnson Algorithm is an excellent approach of finding solutions to the problem when considering a two-server, one-queue system. According to him, it is critical to keep track of server processing times. If the first server has the shortest processing time, the work is placed in the first available spot in the sequence.

The work is placed in the final available spot in the sequence if the processing time is on the second server. When there is a tie in processing, various factors are used to determine the minimum processing timeframes. He went on to show this by solving the problem using a two-machine, one-queue, no-passing configuration. Many innovative and heuristic algorithms use this as a starting point. Glaser and Sinha (2010) looked at the topic of reducing the total time necessary to accomplish a set of related tasks.

The conventional heuristic method was adopted, and Johnson's algorithm remains a dependable method or technique that may be used to handle complicated combinatorial problems. The method is simple and clear, making it simple to comprehend and apply. In Group Technology, Sandor et al. (2003) used Johnson's approach to solve the scheduling problem (GT).

They considered two automated machine systems, Machine A (MA) and Machine B (MB). They structured the machines in such a manner that MA is very much accessible to start or accomplish the available work at a specified moment, and MB is also very much ready to engage the jobs flowing out of MA while solving the problem using the Johnson Algorithm. In addition, the workpieces that will flow from MA to MB are organised and well-positioned in billets for simple feeding into MA and MB.

The pattern of organisation is done in groups so that the workpiece may flow from one place to another and the setup time for all groups of operations is reduced. The Johnson method was used to solve the problem of setup time, and the authors were pleased with the results. Xiong et al. (2014) investigated the challenges surrounding task scheduling in cloud data centres by examining current restrictions and developing a mathematical model for efficient job scheduling. To overcome the problem, a hybrid algorithm was created by crossing Johnson's Rule (JR) with Genetic Algorithm (GA) to generate Johnson's rule base genetic algorithm, where flawless coding was accomplished and simulations were employed for sensitivity analysis.

The hybrid algorithm outperformed the classic scheduling method in terms of accuracy and speed. Zotkiewicz et al. (2016) solved the challenge of scheduling in energy efficiency systems by assigning variable resources in a reasonable manner to allow for successful job scheduling. Hu and Veeravalli (2013) considered a dual-stage task arrangement to solve the challenge of scheduling a bag of tasks. Cheng et al.

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(2013) investigated a minimization process scheduling model for a two-machine flow shop issue with a decreased learning utility or task where a job's definite processing time is a function of its location at a given moment.

Dong et al. (2013) proposed a heuristic solution for solving a scheduling problem in a dualstage hybrid flow shop that integrates two hybrid algorithms. By examining a scheduling problem with multiple product flow via dual-stage assembly system, Liao et al. (2015) created a heuristic model that was addressed as a mixed integer problem. Nikzad et al. (2015) developed a model for the scheduling problem using a two-stage hybrid flow shop to reduce product makespan. Other related studies on schedule optimization may be found in the work of the writers listed below (Mansouria et al. 2016; Bellanger et al. 2013; Hekmatfar et al. 2011).

Materials and Methods

An investigation was conducted on the Iterlene bottled water facility in Effurun, Nigeria. This study looks at a two-machine system: the filling machine (FM1) and the capping machine (CM2). When the machine reaches two (2), Johnson's method can be extended, but it must also meet the criteria ($\min t_{i1} > \max t_{i2}$, $\min t_{i3} > \max t_{i2}$). The same approach may be used to reduce a three-machine problem to a two-machine problem by combining "Machine 1 and Machine 2" and then "Machine 2 and Machine 3." Machine 1 and Machine 2 will be viewed as one machine, Machine 1, and Machine 2 and Machine 3 will be considered as Machine 2. The jobs are then processed using Johnson's method, which is a heuristic. There might be a large number of processed jobs. Five tasks are processed on two machines in this study. Summarisation of Johnson's rule is as follows:

- In a table with the processing time of the machine on the left hand side column and the number of jobs on the right hand side column, jobs are listed along with their processing times on each machine.
- On the machines that were available, the processing time was investigated. Using the two computers FM1 and CM2, the shortest processing time was documented in each column of the table, i.e. (t_{1j}, t_{2j}) for all j .
- Select the work in the first available place in the sequence if the least processing time is on the first machine (FM1). If the processing time on the second machine (M2) is longer than the processing time on the first computer (M1), the work should be placed last in the sequence. Breaking bonds arbitrarily;
- Check out the jobs you've been allocated from the table and go on to the next one;
- For machines FM1 and CM2, determine the idle time. Idle time for FM1 = (Total elapsed time) minus (Total elapsed time) (Time required for the last job to be completed on FM1). Idle time for CM2 = (Time when the first job on FM1 is done) + (Time the j th job in a sequence commenced on CM2)
- (The time when the $(j-1)$ th job in a series on CM2 was done)
- Total elapsed time for all jobs = the time necessary to perform the n th job on CM2.

Symbols

i – index representing number of jobs

t_{ij} – time required for job i on machine j

T – Total elapsed time for processing all jobs on machine, it includes idle time if any exist

I_{ij} – Idle time on machine j for the end of job $(i-1)$ to the start of job i

J_i – jobs to be moved from one machine to another

a_i – average processing time for job to pass through the filling machines in seconds

b_i – average processing time for job to pass through the capping machines in seconds

S – Sequencing of Job

Acronyms

FM – Filling machine

CM – Capping Machine

IISL – Iterlene Industrial Services Limited

Processing of n Jobs through 2 Machines

Assume there are n jobs to be handled on two computers, A and B. Passing is not permitted since each task must go through the same series of actions in the same order. A work is allocated to machine B once it has been completed on machine A. If machine B is not available at that time, the work is placed in the waiting line. According to the FIFO (first in, first out) principle, each job from the waiting queue is assigned to machine B.

Let A_i = processing time for i th job on machine A

B_i = processing time for i th job on machine B

T = total elapsed time

The goal is to establish the order in which n tasks should be handled by machines A and B in order to minimise total elapsed time (T). Following is a description of a method devised by Johnson and Bellman for identifying an optimal solution.

Johnson's Algorithm

Step 1 Out of all A_i 's and B_i 's, choose the minimum processing time. If it is A_r then do the r th job first. If it is B_s then do the s th job at last.

Step 2 If there is a tie in determining the shortest of all processing times, the problem is handled in one of three ways:

If the minimum of all the processing times is A_r , which is also equal to B_s , that is, $\min(A_i, B_i) = A_r = B_s$, then do the r th job first and s th job at last.

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- (ii) If $\min(A_i, B_i) = A_r$, but $A_r = A_k$, i.e., there is a tie for minimum among A_i 's, then select anyone.
- (iii) If $\min(A_i, B_i) = B_s$, but $B_s = B_t$, i.e., there is a tie for minimum among B_i 's, then select anyone.

Step 3 Remove the work that has already been allocated from consideration and repeat steps 1 and 2 until the best sequence is identified.

Note: The Johnson's approach focuses on reducing machine idle time. It is demonstrated that the ideal sequence of n tasks to be processed on machines A and B in the order AB requires the identical job ordering on each machine. When the work sequence is the same for both computers, the total elapsed time is the shortest.

Example: Suppose that there are five jobs, each of which has to be processed on two machines A and B in the order AB. Processing times are given in the following table:

Job	Machine A	Machine B
1	6	3
2	2	7
3	10	8
4	4	9
5	11	5

Determine a sequence in which these jobs should be processed so as to minimize the total processing time.

Solution: The minimum time in the given table is 2, which corresponds to job 2 on machine A. So, the allocation of jobs will start as . Now, we eliminate job 2 from further consideration. The reduced set of processing times is as follows:

Job	Machine A	Machine B
1	6	3
3	10	8
4	4	9
5	11	5

Now, the minimum time is 3 for job 1 on machine B. Therefore, this job would be done at last. The allocation of jobs till this stage would be . After deletion of job 1, the reduced set of processing times is as follows:

Job	Machine A	Machine B
3	10	8
4	4	9
5	11	5

Similarly, by repeating the above steps, the optimal sequence is obtained as . On the basis of this optimal sequence, the minimum elapsed time is obtained from the following table as 36 hours.

Job	Machine A		Machine B	
	Time in	Time out	Time in	Time out
2	0	2	2	9
4	2	6	9	18
3	6	16	18	26
5	6	27	27	32
1	27	33	33	36

Further, idle time for machine A = total elapsed time – time when the last job is out of machine A = $36 - 33 = 3$ hours. Idle time for machine B = time at which the first job in a sequence finishes on machine A + (time when the i th job starts on machine B) – (time when the $(i - 1)$ th job finishes on machine B). Therefore, idle time for machine B = $2 + (9 - 2) + (18 - 9) + (26 - 18) + (32 - 26) + (36 - 32) = 4$ hours

Check Your Understanding

- 1) _____ is one strategy that may be used to reduce the total time elapsed between the start of a task on one machine and the completion of a job on another.
 - a) Inventory Management Method
 - b) Inventory Control Method
 - c) Johnson's method
 - d) Statistical Method
 - e) None of the above
- 2) What is the full form of GA?
 - a) Genetic Algorithm
 - b) General Algorithm
 - c) Genetic Analysis
 - d) General Analysis
 - e) None of the above

Summary

- In a manufacturing or production system, Johnson's algorithm is used to determine the best sequence or shortest pathways.
- Johnson's rule cuts down on idle time between the two work locations, however the outcomes aren't always ideal, particularly for a small set of projects.
- According to Johnson E. B. (1977), it is vital to look at the flow process of jobs through the available machines when addressing the shortest-path problem.
- Johnson Solution is a great algorithm for the shortest path issue, according to Sanan et al.
- Johnson method, unlike other algorithms that only evaluate positive values, may be applied in real-value problems where the distance between two places

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is considered, especially scenarios where weights are allocated for positive and negative weights.

Activity

1. Write a case study on Johnson rule Of two machine scheduling.

Questions & Exercises

1. How can you solve the problem of job shop scheduling using Johnson Algorithm.

Glossary

- Johnson's rule is a strategy for reducing the time it takes to complete a collection of jobs that must be handled on two machines or at two different work centres.
- Johnson's rule is a technique of scheduling tasks in two work centres in operations research and production management, with the primary goal of determining the best sequence of jobs to decrease makespan.
- In a process layout, similar equipment or functions are placed together, such as all drill presses in one area and grinding machines in another.

Further Readings

1. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding – Answers

- 1) c)
- 2) a)

Module - 5: Inventory Management, Supply Chain Management & Logistics Management and Foundations of Quality

Notes

Structure:

- 5.1 Inventory Management
 - 5.1.1 Basic Concepts Of Inventory Management
 - 5.1.2 Lead Time
 - 5.1.3 Safety Stock
 - 5.1.4 Elements Of Inventory Costs
 - 5.1.5 ABC Analysis
 - 5.1.5 Basic EOQ Model
 - 5.1.6 Just In Time System
- 5.2 Basic Concepts of Supply Chain Management
 - 5.2.1 Basic Concepts Of Supply Chain Management
- 5.3 Basic Concepts Of Logistics Management
 - 5.3.1 Basic Concepts Of Logistics Management
- 5.4 Foundations Of Quality
 - 5.4.1 Foundations Of Quality
 - 5.4.2 Introduction To TQM
 - 5.4.3 Concepts Of Statistical Quality Control
 - 5.4.4 X (Bar), R Chart; C Chart, P Chart
 - 5.4.5 Introduction Of Acceptance Sampling
 - 5.4.6 Concepts Of Kaizen
 - 5.4.7 5S
 - 5.4.8 ISO Standards
 - 5.4.9 6-Sigma
 - 5.4.10 Lean Manufacturing
 - 5.4.11 TPM
- 5.5 Issues In Operations Management
 - 5.5.1 Issues In Operations Management

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Unit - 5.1: Inventory Management

Objectives

At the end of this unit, you will be able to:

- Understand basic concepts of inventory management
- Learn about lead time
- Understand safety stock
- Understand elements of inventory costs
- Learn about ABC analysis
- Identify basic EOQ model
- Learn about just-in-time system

Introduction



Inventory management is a method for acquiring, keeping, and selling inventory, including both raw materials (components) and completed commodities. Inventory management, in business terms, refers to having the correct stock, at the right levels, in the right place, at the right time, and at the right cost as well as price.

Inventory management aids businesses in determining which goods to order and when to order it. It keeps track of merchandise from purchase to sale. The technique recognises and responds to patterns to guarantee that there is always adequate stock to fulfil client orders and that shortages are properly warned about. Inventory becomes income if it is sold. Inventory ties up cash before it sells, despite the fact that it is represented as an asset on the balance sheet.

As a result, having too much inventory costs money and lowers cash flow. Inventory turnover is one indicator of successful inventory management. Inventory turnover is an accounting metric that shows how frequently stock is sold over time. A company does not want to have more inventory than it can sell. Deadstock, or unsold inventory, can result from low inventory turnover.

5.1.1 Basic Concepts of Inventory Management

- Inventory Management Formulas

If you're new to inventory, you'll likely come across a number of formulae that

are first perplexing. These calculations, with a little study, may be quite effective for maintaining stock levels optimum.

Here's an overview of some of the most common inventory formulas...

1. Economic order quantity (EOQ) formula
2. Day's inventory outstanding (DIO) formula
3. Reorder point formula
4. Safety stock formula

1. Economic Order Quantity (EOQ) Formula

Your EOQ is the minimum amount of goods you should order to keep your overall cost of purchasing or keeping stock as low as possible. Identifying your EOQ has the potential to save you a lot of money.

$EOQ = \sqrt{(2DK / H)}$, or the square root of $(2 \times D \times K / H)$

Where:

D = Setup or order costs (per order, generally includes shipping and handling)

K = Demand rate (quantity sold per year)

H = Holding or carrying costs (per year, per unit)

D = FIXED COST PER YEAR
K = DEMAND IN UNITS PER YEAR
H = CARRYING COST PER UNIT PER YEAR

$$EOQ^* = \sqrt{\frac{2DK}{H}}$$

2. Days Inventory Outstanding (DIO) Formula

The number of days it takes for inventory to transform into sales is referred to as days inventory outstanding (DIO) or days sales of inventory (DSI). The average inventory days outstanding varies by industry, however a lower DIO is typically preferable.

$$DIO = \left(\frac{\text{COST OF AVERAGE INVENTORY}}{\text{COST OF GOODS SOLD}} \right) \times 365$$

The average for your industry, your company plan, the sorts of items you offer, and other factors all play a role in determining whether your DIO is high or low.

3. Reorder Point Formula

The reorder point formula provides an answer to the age-old question of when to order extra goods,

There are three steps to calculating your reorder point:

1. Calculate the number of days you'll need to meet your lead time need.

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2. Determine the number of days in your safety stock.
3. Add your desire for lead time and your safety stock.

Reorder Point = Lead Time Demand + Safety Stock

4. Safety Stock Formula

As previously said, safety stock serves as an emergency buffer that you may use if it appears that you're about to sell out. You want enough safety stock to fulfil demand, but not so much that additional carrying costs put your finances under strain.

While this may appear to be plain sense, the problem is determining how much safety stock to keep on hand:

1. Multiply your maximum daily consumption by the number of days in your maximum lead time.
2. Take your average daily consumption and divide it by the average lead time in days.
3. Determine your safety stock by calculating the difference between the two.

$$\text{SAFETY STOCK} = \left(\frac{\text{MAXIMUM DAILY USAGE} \times \text{MAXIMUM LEAD TIME IN DAYS}}{\text{MAXIMUM LEAD TIME IN DAYS}} \right) - \left(\frac{\text{AVERAGE DAILY USAGE} \times \text{AVERAGE LEAD TIME IN DAYS}}{\text{AVERAGE LEAD TIME IN DAYS}} \right)$$

- Ways to Improve Inventory Management

1. Focus on Your Needs

It might be difficult to manage a warehouse full of merchandise. Identifying the most essential issues and focusing on them first might help you manage everything better. It's improbable that every item in your warehouse will see the same level of client demand. If you maintain the best-selling things in stock, you'll be well on your way to keeping your consumers satisfied.

2. Engage with Suppliers

It's critical to handle supplier relationships properly in any stock-based firm. It's critical to build positive connections with your company's major suppliers if you want to ensure consistent supply, get competitive pricing, and stay on top of developing trends that might affect your business.

3. Develop an Inventory Management System

It's critical to consider how your company handles order volumes, replenishment cycle durations, safety stock, forecasts, seasonality, and other factors. Make adjustments to each activity based on your individual business, keeping note of what works and what doesn't. Making a significant improvement in one area might be more beneficial than making a few minor improvements all around.

4. Utilise Real-time Data

Information may be a powerful tool, but only if it is accurate and current. Real-

time data and analytics may make a big impact in your organisation, from multilayer inventory management to predicting data, autonomous ordering, and customised safety stock. Consider adopting perpetual inventory management software for the most accurate data, since it is the best method to ensure that the information you need is always at your fingertips.

5. Go Mobile

Inventory management has been transformed by mobile technologies. Barcode scanning, for example, speeds up the receipting and tracking of items while also reducing mistakes. Meanwhile, sales applications provide inventory data to salespeople on the go. You don't have to be linked to a computer in your warehouse any more. You can monitor critical company activities from your home, on vacation, or anywhere you are.



6. Develop an Inventory Management System

Inventory management on an ad hoc basis can only go so far. You'll need an inventory management system to stay on top of your supply. Every organisation has its own set of requirements, so finding a system that fits your needs is critical. Sam, for example, may have been able to manage her inventory using spreadsheets in the early days of her business. A worldwide stock-based firm like Amazon, on the other hand, requires a customised, complex system that can handle the massive volume of orders handled every day.

- **Inventory Management Techniques**

Inventory management employs a variety of techniques to have the proper number of items on hand to meet consumer demand while being profitable. When dealing with thousands of stock-keeping units (SKUs) spread across numerous warehouses, this process becomes even more difficult. Among the approaches are:

- **Stock review**, it is the most basic way of inventory management and is, in general, more appealing to small enterprises. A frequent examination of stock on hand vs estimated future demands is known as stock review. It mostly relies on manual labour, while automated stock reviews can be used to establish minimum stock levels, allowing for frequent inventory inspections and reordering of goods to fulfil the minimums. Stock review can provide you some control over your inventory management process, but it's time-consuming and prone to mistakes.
- **Just-in-time (JIT) methodology**, based on studying client behaviour, in which things come as ordered by customers. This method entails analysing purchasing habits, seasonal demand, and location-based characteristics to provide a precise image of which commodities are required at certain times

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and locations. JIT has the benefit of being able to meet consumer demand without having to maintain significant volumes of items on hand and in near-real time. However, there is a danger of misreading market demand or having distribution challenges with suppliers, which can result in out-of-stock situations.

- ABC analysis methodology, which divides inventory into three groups based on the item's inventory value and cost relevance. High-value and low-quantity items are classified as category A, moderate-value and moderate-quantity goods are classified as category B, and low-value and high-quantity goods are classified as category C. An inventory management system may handle each category independently. To maintain adequate buffer stock on hand, you need to know which goods are the greatest sellers. More costly category A products, for example, may take longer to sell but don't need to be held in big numbers. ABC analysis has the advantage of providing greater control over high-value products, but it also has the downside of requiring a significant amount of resources to continuously assess the inventory levels of all categories.
- Economic Order Quantity (EOQ) methodology, in a warehouse management system, a formula calculates the best time to restock goods. The purpose is to determine how many things may be ordered at any particular moment. As a result, money that might otherwise be held up in surplus inventory is freed up, and costs are reduced.
- Minimum Order Quantity (MOQ) methodology, which determines the least number of items a supplier will sell. The supplier will not sell to a company that cannot meet the minimum buy requirements. Suppliers benefit from this strategy because it allows them to swiftly get rid of inventory while screening out bargain hunters.
- First in, First out (FIFO) methodology, to help keep product fresh, the oldest stuff is sold first. This is a particularly significant strategy for organisations dealing with perishable goods that would spoil if not sold within a certain amount of time. It also keeps products from becoming outdated before they may be sold by a company. This usually entails maintaining older things at the front of the shelves and transferring fresh items to the back.
- Last in, First out (LIFO) methodology, in this case, the most recent inventory is usually reported as sold first. When there is a problem with inflation and prices are rising, this is an excellent technique to follow. Because the most recent inventory has the greatest manufacturing costs, selling it before older inventory results in lower earnings and taxable revenue. LIFO also means that inventory is reported at the lower cost of older items left on the shelf. However, because older objects that hang around have a danger of becoming outdated or perishing, this is a tough strategy to put into reality.
- Safety Stock Methodology, when a company sets aside inventory in case of an emergency, this is known as contingency planning. Before diving into the safety stock, the safety stock strategy also serves as a notice that it's time to replenish items. Businesses should incorporate safety stock into their warehouse management plan in the event that their supply chain is disrupted.

- Functions or Characteristics of Inventory

From the definition of inventory, it is clear that it is related to stock of raw materials, semi-finished and finished products/items maintained by the enterprise/business/organization.

The following points will explain the concept and functions of inventory:

(i) Inventories Serve as Cushions:

It isolates multiple industrial procedures from one another and makes them autonomous to protect against demand/supply changes, allowing each activity to be completed cheaply.

For example, a company may interact with a number of customers and vendors, and because of their unpredictable behaviour, there are constantly swings in demand or supply of items, disrupting the company's schedule.

Inventories absorb these swings and assist in sustaining uninterrupted production, i.e., we successfully isolate manufacturing operations from consumers and vendors through stock cushions.

Furthermore, raw material purchasing/ordering may be done separately from completed product distribution, and both of these activities can be made low-cost processes, such as ordering raw material in bulk and distributing products in large batches rather than tiny quantities. As a result, not only does it improve the use of personnel and equipment, but it also improves the economics.

(ii) Inventory, a Necessary Evil for Any Enterprise:

Maintaining inventories necessitates precious space, capital, and other overheads. Until the stocks are used, the investment cash sits dormant. On the other hand, without inventory, the organization's smooth operation is impossible, hence it is a must. It has also been discovered that the costs of not having inventory (stock out circumstances) are frequently higher than the expenses of having it. As a result, inventory is an unavoidable evil.

(iii) Inventory Provides Production Economies:

Purchases in desired amounts cancel out the impact of price or supply fluctuations. Stocks enable the purchasing of multiple inputs at a lower cost due to bulk buying discounts.

(iv) Maintenance of Smooth and Efficient Production Flow:

Maintains a smooth and efficient production flow, allowing a process to run continuously.

(v) Creation of Motivational Effect in Decision-making:

Creates a motivational effect in decision and policy making, for example, if inventory are exhibited in quantity, a person may be persuaded to acquire more.

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- Major Objectives/Reasons for Holding Inventory:

Any inventory control system that covers all forms of inventory is designed to achieve a wide range of goals, either directly or indirectly.

Most of them are as follows:

(i) Financial Objectives:

The main financial goal of keeping inventory is to retain the investment within the cash position of the company so that the working capital does not become substantially out of balance.

(ii) To Create a Buffer Stock between the Input and Output:

As a result, the exiting flow of goods is as unaffected as feasible by the input material characteristics.

(iii) To Ensure against Delay in Deliveries:

Holding inventory stock of finished items prevents a delay in delivery of the finished product to the buyer.

(iv) To Allow for a Possible Increase in Output if so Required:

Market demands may cause the company's production programme to be disrupted. Stocks must be kept and supplied in accordance with production needs.

(v) To Ensure Against Scarcity of Materials in the Market:

When there are huge swings in output and demand for raw materials, they might become rare and difficult to get. For effective production processes, a reserve supply of raw materials is required.

(vi) To Make Use of Quantity Discounts:

Because of quantity discounts and lower shipping costs/charges, input materials and components/parts may be cheaper when acquired in bulk quantities.

(vii) To Utilize to Advantage Price Fluctuations:

While price fluctuations can have a significant impact on an enterprise's or organization's procurement policy, if these fluctuations are to be used to the unit's advantage, materials must be purchased in sufficient quantities when prices are at their lowest.

All of the reasons or goals listed above have a monetary component. Inventory control is primarily concerned with making the best judgments possible about the factors listed above. Inventory is an unused resource that may be used to create value.

- Types of Inventories

Inventory can be classified into three main categories, namely,

1) Raw Materials

Raw materials, also known as bought components, are the raw resources that will be utilised to create completed goods. Raw materials, on the other hand, have many additional aspects, such as different sorts of raw materials, why they are significant, and how to determine how much raw material you require.



Every other thing we use was made with some combination of raw components. Because it is the most important component of inventory for any production unit, maintaining raw material inventory is also critical. Natural latex, for example, is used to manufacture a variety of finished items such as mattresses, gloves, balloons, rubber bands, tennis shoes, and so on.

“A raw material, also known as a feedstock, unprocessed material, or main commodity, is a fundamental substance used to manufacture commodities, completed products, energy, or intermediate materials that are utilised as feedstock for future finished products.”

Types of Raw Materials

Despite the fact that all raw materials are obtained organically, they may be classified into three groups based on their origin.

- Plant/Tree-based — Plants and trees provide resources such as vegetables, fruits, flowers, timber, resin, and latex.
- Animal-based – animal-based materials include leather, flesh, bones, milk, wool, and silk.
- Mining-based – minerals, metals, crude oil, coal, and other resources are mined from the soil.

Apart from this, a manufacturing unit divides the raw materials into 2 main categories.

- Direct Raw Materials: Direct raw materials are the fundamental components used to create a final product. Wood, for example, is a direct raw material used to make furniture such as chairs, tables, and beds. Another example is leather, which is used to make handbags, shoes, and bags, among other things.
- Indirect Raw Materials: Indirect raw materials, on the other hand, are materials that are used to complement direct raw materials in the production of a completed product. For example, glue, nails, varnish, and other materials required in the construction of wooden furniture such as chairs, tables, and beds are all indirect raw materials. Buckles, metal hoops, zips, adhesive, lining fabric, colours, and

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other components used in the manufacture of leather purses and shoes are similar.

Importance of Raw Materials

It is evident from the preceding description and meaning that making any finished product necessitates the use of raw materials, which are subsequently processed and transformed into finished items.

Inventory management enters the scene at this point. Because the completed items we sell are manufactured from these components, it is critical to keep raw materials inventory in the appropriate number and location to avoid any catastrophes.

Raw materials calculation formula:

Now we'll look at some basic formulas for calculating raw materials. We may use the formula below to figure out how much raw material is needed in the manufacturing process:

Raw material used = Opening raw material + raw material purchases – closing raw material

2) 'Work in Process' Goods

'Work in Progress' is a phrase used to describe a piece of work that Items that are half completed and, on their way, to becoming finished goods are referred to as goods.



The process of creating final items begins after the producer has the raw ingredients in-house. Work in progress inventory refers to inventory that is in the process of being transformed from raw materials into completed products.

It is critical for any manufacturing organisation to understand the exact quantity of inventory it has, whether it be raw materials or work in progress. Inventory management aids in the counting and maintenance of all types of stock. Regularly counting inventory will provide the producer a good sense of how much needs to be produced and will also aid in projecting production in accordance with demand. Aside from that, one of the most significant financial management jobs is determining work in process expenditures. Work in progress inventory is included as an asset on the financial balance sheet when documenting inventory.

Now for calculating this one must refer to the balance sheet of the previous quarter, month or year to get the required details. You might have always been thinking about how to find work in process inventory.

To calculate the WIP inventory i.e., work in process inventory, you need to use the following work in process inventory formula:

Formula: Beginning work in process amount + manufacturing costs – cost of manufactured goods

3) Finished Goods

The finished product created by industries or businesses that is ready to be sold to wholesalers, retailers, and other businesses. Finished products are items that have finished the production process and are ready to be sold to merchants or end consumers. The finished items, on the other hand, are referred to as merchandise once they have been sold.



Manufacturers use the terms 'finished goods' and 'finished product' to describe their products. A shopkeeper does not need to classify his inventory into raw materials, work in progress, or completed items because he stores all of the finished goods in his store. As a result, a retailer's wares are referred to as merchandise. Manufacturers, on the other hand, must generate and maintain inventory of all commodities, regardless of their state of production - raw materials, work in progress, or finished goods. A "finished good" is one that has been completed in terms of manufacture but has not yet been sold or delivered to the final user.

This is the final stage of the commodities processing process. The products are ready for consumption or distribution. After this step, the vendor is not obligated to process the products in any way.

However, in the supply chain management flow, one supplier's finished goods might be used as a raw material by another, making finished goods a relative word.

Calculating the value of finished goods may be a difficult undertaking. However, there is a simple formula that you can use to make things easier for yourself.

Finished goods at the beginning of year + Finished goods produced – Finished goods sold

= Finished goods at the end of a year

4) Transit Inventory, aka Pipeline Inventory

Transit inventory refers to inventory that is in transit from a manufacturer's location to a retailer's or wholesaler's location. Inventory transportation can sometimes take days, if not weeks, to complete.

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It is the inventory that is always moving from one point in the supply chain to another. It is made up of orders that have been placed but have not yet been received, and it is therefore counted as inventory.



As the name implies, transit inventory is goods that has been sent by the seller but has not yet arrived at the buyer's destination. Because the inventory is in transit, it is also known as pipeline inventory, and it is an important aspect of inventory management, believe it or not.

For example, a lot of crude oil is transferred from Saudi Arabia to Europe, and we all know how far apart these two countries are and how long it takes for the shipment to arrive. As a result, any crude oil that is sent is referred to as transit inventory, and it is the seller's obligation to pack and store the product properly while it is in transit, as it is a hazardous item.

However, under such circumstances, the issue of ownership emerges. Because neither the vendor nor the buyer owns the inventory. Who owns the inventory while it is in transit?

To address this, multiple forms of sales agreements were established, and as a result, various types of transit inventory were established based on the ownership stated in the sales terms and conditions.

5) In-transit Inventory Terms and Conditions

Freight on Board (FOB) is a commonly used abbreviation for freight on board. Ownership of the items passes to the customer as soon as the inventory is boarded on the ship under a Freight on Board Shipping arrangement.

- **FOB Destination**– Freight on Board Destination signifies that the seller retains control of the merchandise until the shipment arrives at the buyer's destination.
- **CIF (Cost, Insurance and Freight)** is a short abbreviation for Cost, Insurance and Freight. The seller retains ownership of the items in a Cost Insurance Freight shipment since he pays for the cargo's cost, insurance, and freight. As soon as it arrives at the target port, the buyer takes possession of it.
- **eCommerce Shipping** — Due to the massive growth in the world of online shopping, this is one of the most popular shipment kinds these days. Under the eCommerce shipping terms of agreement, an eCommerce merchant owns the inventory until it reaches the end-customer.

6) Buffer Inventory, aka Safety Stock

To minimise customer service issues and the costs connected with them, safety stock is deployed. It guards against demand fluctuations, lead times, and supply variances. The larger the safety supply, the greater the service level you wish to give. Similarly, the bigger the lead time and the greater the demand variances, the greater the safety stock.



The definition of safety stock is: Every merchant holds inventory based on the average demand of its customers. Now and again, there may be a sales rush. That is, you will quickly be out of stock before you can restock your inventory. This is for those times when we need a backup plan. A stock order is a type of request generated by retail software when fresh suppliers are needed to supply refilled shelves and inventory.

Safety stock is essentially a buffer for your on-hand inventory, or surplus product that isn't being consumed by customers. Sellers and merchants are familiar with this word, therefore it's critical for Jon and the rest of them to know what it means and how much should be stored as safety stock.

As stated above, safety stock is a buffer inventory. The question arises, what should be the size of this on hand buffer inventory?



The graph above demonstrates how demand forecasting may be used to predict the quantity of safety stock needed. As a company's revenues increase, so does demand. As a safety stock, the difference between the maximum and average demand is calculated. You can simply assess the needed safety stock size for your firm if you can precisely estimate your average and maximum demand based on historical data.

Safety Stock Calculation

Let's first understand the terminologies used in the safety stock formula. Our retailer Jon advises retailers of today to know about these terminologies which goes like this:

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The formula of safety stock goes as follows:



3 (Jon-ey) Methods for Calculating Safety Stock:

Every business has its own methods of calculating safety stock. However, these methods are basically classified into three different styles; statistical-based calculation, time-based calculation, and fixed safety stock.



a) Statistical Based Calculation

It's a mathematical strategy (which Jon despises but has filled his coffers) that employs mathematical theories of probability to compute the level of safety stock in order to avoid stock-out situations. The advantages and disadvantages of using statistical data to calculate safety stocks are discussed. "Although this statistical strategy is simply based on precise mathematics, business prediction is not always as accurate," Jon notes. Forecasting isn't a fool proof method of predicting the future. Even if you've done your research on assessing the needs, problems might happen that

you can't anticipate. Statistical approaches, for example, have no influence on market demand. Statistics may indicate that you should preserve a safety supply of things that no longer have a market worth.

b) Time-based Calculation

This technique of estimating safety stock is based on the passage of time. The stocks necessary for a certain time period are estimated based on the time period and are evaluated on a regular basis to verify accuracy and that the demand is met. Jon predicted that 100 cases of component A would be eaten each day. In such situation, Jon's weekly safety stock count would be 700 cases. Demand forecasting for a certain substance in the next day's employs a time-based technique in this manner. The biggest downside of employing a time-based computation for safety stock is that it reduces your financial liquidity. This is due to the fact that a large portion of your wealth is invested in safety stock. If your product is a sluggish seller, this will be much more difficult because your safety supply will now become excess stock and will be sitting in your warehouse, consuming valuable shelf space.

c) Fixed Safety Stock

Companies use this strategy to set a fixed level of safety stock for their products. Rather than depending on a number based on a statistical computation, your production manager/planner would set that level or value for the safety stock. Until the production manager manually adjusts the value of safety stock, it remains constant. Both cycle stocks and safety stocks are important.

When it comes to safety stock, merchants should be aware of two key points that will help them retain stock in their inventory more efficiently:

- Economic Order Quantity (EOQ)
- Reorder level

Safety inventory formula – $(\text{Maximal daily usage} * \text{Maximal lead time in days}) - (\text{Average daily usage} * \text{Average lead time in days})$.

7) Anticipation Inventory

It's a type of inventory that's utilised to absorb fluctuations in consumer demand over a certain length of time. Some firms that have a known seasonal demand build up inventory during low demand times so that production rates are not hindered when high demand periods arrive.



Anticipation inventory refers to inventory held in anticipation of predicted customer demand. It's comparable to safety stock, but it's different in that this stock is normally retained only seasonally, when product demand varies substantially. With an automated

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inventory management system, demand forecasting for items can be done easily. Anticipation Stock is a stock of components, resources, or commodities held on hand by a firm or business to satisfy demand or to make up for irregular production shortfalls. It's also known as build stock, anticipation inventory, seasonal inventory, or seasonal stock.

For example, a merchant may stock up on ice cream just as the summer season is about to begin, anticipating increased demand throughout the hot season. As the winter season approaches, the same shopkeeper cuts his ice cream supplies.

Advantages of Anticipation Inventory

- Anticipation inventory is similar to additional inventory stored by retailers, allowing a firm to fulfil shifting consumer demands. Pumpkins, for example, are in great demand during the Halloween season. So, if more pumpkins are supplied during this period, the businessperson will benefit.
- A company may provide constant service to its customers by avoiding running out of supply during periods of fluctuating demand for a certain product. As a result, it assures that the shop can maintain a steady production and staff throughout the year, avoiding stock outs.
- One of the most significant advantages of storing additional quantities of items by forecasting future demand is the protection against uncertainty. The future is uncertain, but if you are fortunate enough to own things that have increased in demand due to unanticipated events, you may profit from the situation.

Disadvantages of Anticipation Inventory

- Making a mistake in demand forecasting is one of the most difficult problems. It might be an issue if you expect a higher demand and store more things, but the demand actually decreases. In the other case, you may not have enough inventory when demand rises.
- Anticipated inventory has a longer lasting effect than usual inventory errors. In normal situations, if an error is made in managing inventory, the effect lasts just until the stock runs out, but if things are purchased in anticipation, you may have a difficult time disposing of them if they are in large quantities.
- Apart from that, a merchant or businessperson has significant carrying costs since he or she must acquire normal stock as well as additional stock in case demand for the items increases. This implies the merchant will have to buy as well as store the extra inventory, which will be pricey.

8) Decoupling Inventory

This is the inventory retained in manufacturing facilities, where one component of the product's stock is kept in a certain number so that processing of the other part of the product is not hampered. It's similar to safety stock, except that this is safety stock for internal use, so that demand for a specific replacement component may be addressed while the completed product is being manufactured.



Decoupling inventory is today's buzzword. Manufacturing companies commonly employ it as a backup system. Inadequate raw material availability and malfunctions at some stage in the manufacturing process are common problems for businesses. This has the potential to sabotage the entire manufacturing process. We'll look at how it can save the production process and function as a safety cushion in a variety of difficult scenarios.

Understanding the Need of Decoupling Inventory

Companies with mass production are susceptible to tons of tricky areas, which may bring the assembly line to a halt. These setbacks arise from both within and out of doors the organization. I am putting down the main points between which mishaps can occur to know the vulnerability below:

- Demand
- Suppliers
- Inventory
- Production Line
- Delivery

As you'll see, all of those aspects, except the inventory and assembly line, are externally controlled.

The manufacturing can get derailed thanks to the breakdown of a specific machine or thanks to changes in production rate at some points. Discrepancies and lack of proper inventory management also are other internal problems. When a selected element of the chain stops functioning as per the assembly plan, the whole ecosystem is affected. Under such conditions, having additional stock of unfinished products at each stage of the assembly cycle provides a cushioning effect. The power can still give an equivalent output while the troubleshooting gets done. Subsequent station can continue with the regular production without requiring the previous one to deliver.

Having such an appointment is important when the corporate is using an equivalent component during a multiple product mix. As a result, they will focus manufacturing on completing critical orders while halting less important operations. This may help in coping up with the demand in order that the corporate doesn't suffer losses thanks to such issues. Here, the stress is laid on preventing loss of your time.

Fluctuations in demand, the lack of suppliers to supply goods in time, and changes in delivery conditions are caused from outside the organization but ultimately put pressure on its infrastructure. The decoupling stock also acts as a cushion against these shortcomings.

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9) Cycle Inventory

A lot-sized inventory is also known as cycle inventory. It is proportional to the size of the lot. The quantity changes depending on the period between orders; the longer the interval between orders, the greater the cycle stock. If you order every two weeks, for example, your cycle supply will be equal to two weeks' demand.



Cycle counting is one among the foremost crucial aspects of inventory management in any organization, especially in recent times. I even have seen tons of budding businesses struggle with finances thanks to mismanaged inventories, and it all starts with improper record keeping. Either they completely overlook the necessity for counting stocks, or they don't use the right methods. During this article, I'm getting to cover the whole cycle count process, and its technicalities intimately. So, hang on to explore and find out how to manage your inventory audits accurately.

The Effects of Scaling up on the Inventory Management

Every organization undergoes multiple changes with the rise in its size. Usually, smaller firms have small inventories, and managing them isn't a drag. It's done by typical visual inspection. But, because of the operations proportion, the corporate starts to stock raw materials and consumables to satisfy business requirements and increase profitability. The matter arises once they lose track of what they need available and whether the worth of their inventory is perfect or not. Within the absence of inventory audits, they find themselves denting their profitability. This is often because they can't make crucial decisions like EOQ (Economic Order Quantity), make inappropriate purchases, suffer a shortage of warehouse space, and derail the manufacturing process.

The reason behind these problems is that the increase within the sorts of raw materials and consumables required. Also, scaling up will end in an increased need for land for storage purposes. The workers will access the storage facility more often, leading to higher chances of inventory shrinkage. Failing to supply raw materials thanks to miscalculations cause stalling of production lines or hasty purchases. Thus, the increased volume of stocked items has direct financial implications on the business.

Hence, the necessity for tracking the components in both pre-production and post-production stages arises. To satisfy this requirement, many systematic approaches were developed, which are listed down below. You'll later analyse and devour the proper one consistent with the business requirements.

What is Cycle Count?

A perpetual inventory management approach is a kind of cycle counting. It is

employed in the auditing process. These audits are also necessary for financial accounting or taxes compliance, in addition to internal processes. In a cycle count, a small fraction of the overall inventory is tallied at a time to get total stock numbers. It is a sampling approach that employs data from a tiny sample of the population to quantify the larger picture. During elections, this approach is also utilised to estimate the general public's viewpoint.

Cycle Counting in Inventory Management

Cycle counting is a process of checking and balancing so that the companies confirm the physical counts that match the inventory records.

- **Popular Approaches to Cycle Count**

There are many ways in which you can count the piled stocks in your warehouses. All of them differ on the basis of which portion of the inventory is considered for audit. Learn more about these approaches below:

a) Cycle Counting by Consumption

This is one of the most basic types of cycle counting since it just counts the objects that are regularly utilised. The items are arranged in order of how frequently workers access them. You don't need to designate specific staff-hours because the storage racks are positioned appropriately. Priority is given to high-consumption products, and the level of intricacy is minimal. It does not, however, distinguish between the costs of the stored components. It also ignores the impact of any errors on the manufacturing process or financial losses. Because of these drawbacks, it is only ideal for low-value, comprehensive inventory-carrying businesses.

b) Objective Counting by Surface Area

For auditing purposes, the warehouse is separated into smaller parts using this strategy. As part of these sections, all shelves and racks are dispersed. The person who is counting is only allowed to work in the area that has been assigned to them. As a result, when the exercise is completed, the organisation will be able to update the records with the specific position of the stored goods. It also aids in recovery and the resolution of any disputes, as the patterns for any given area/material are immediately detected.

c) Control Group Cycle Counting

Have you ever wondered how SOPs are created? They are created with the assistance of experienced employees, engineers, and tech professionals who repeat, investigate, and infer conclusions on a certain activity. A control group is established to specify the cycle count standard operating procedure (SOP). Within a short amount of time, the workers will count the items in this set of articles multiple times. This is done in order to determine the optimum technique for auditing that specific type of item and create a standard operating procedure (SOP) for it. It also aids in the detection of any flaws in the procedure that might later lead to more serious issues. The documentation can potentially be used as a source of inspiration and improvisation in the future.

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d) Random Sample

The staff selects things at random in plants with a large number of comparable items. The counting is done on a regular basis for a certain set of components. This is one of the most convenient approaches since, although covering the full inventory, other operations are not harmed as a result of the planning. The remainder of the surgeries go off without a hitch. The objects are dispersed using the following two techniques to complete this process:

- **Constant Population Counting**

This strategy is less efficient since it concentrates on a single group and audits it on a regular basis. The remaining stock is completely neglected. Because of this feature, I believe that it would be useless to everyone but a few enterprises. The worth of things is totally ignored, much like objective counting by surface area. As a result, it is not recommended unless you have a limited number of distinct goods with minor price differences.

- **Diminished Population Counting**

It's a better option since the corporation will count the people before excluding them. These groupings are stored separately once they have been excluded. They'll be tallied again after the whole audit of the warehouse or firm is completed, which happens about four times a year. The remaining groups are audited on a regular basis according to the company's management's selection criteria.

10) Maintenance, Repair and Operating Supplies (MRO) Inventory

These include items like lubricants, screws and ball bearings, gloves and packing supplies that aid in the creation of completed goods.

Staples, pens, pencils, papers and other office stationery products are also included in the operational supplies.



What Does MRO Mean, or What is MRO?

MRO is an acronym for maintenance, repair, and operations used by companies to create an end product.

What is MRO Inventory?

MRO inventory includes any item that is used in the manufacturing process but is not visible in the final product. MRO items may include:

- **Gloves**

- Safety equipment
- Computers
- Industrial equipment (valves, compressors, pumps)
- Consumables (office supplies)
- Repair tools
- Janitorial Supplies

Need of MRO Inventory

According to studies, a scarcity of spare parts accounts for up to 50% of factory downtime. Despite this, MRO inventory mobility across the company is as low as 10%.

- Importance of Inventory Management

“The motto of inventory management is to minimize holding costs while optimizing inventory.”

To put it another way, the goal is to lower stock levels while simultaneously increasing product availability. This goal can only be reached if inventory is maintained in a methodical manner.

Here are some other salient reasons for the importance of inventory management:

1) Enables You to Do the Planning and Forecasting Accurately

Managing your inventory makes all of the facts around you make sense. Inventory management can assist you in analysing and distinguishing between high-performing and shelf-eating items. This boosts income creation while also freeing up cash flow.

It would also be fantastic if you could predict your customers' purchase patterns and refill your inventory accordingly. Inventory management also allows you to accomplish things like this.

For example, prior inventory figures indicate that your predicted product “x” would be in high demand during the holiday season. You saved money by stockpiling product “x” in your inventory. You will not only create cash, but you will also be able to restock your inventory depending on your clients' daily sales patterns.

2) Improves Your Delivery Time

Nowadays, customers demand no but the simplest within the market, whether it's going to be regarding the standard of the merchandise delivered or the time taken to ship the merchandise to its end customer.

Late delivery thanks to out-of-stock or inventory mismanagement cannot once be the explanation for one sales loss because it affects your brand reputation.

Properly arranged and tracked inventory helps to locate when the order arrives and hence enables you to deliver the merchandise quickly.

Opting for inventory management can protect you from mismanagement of inventory and also enables you to extend customer satisfaction by a speedy delivery.

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3) Controls Your Inventory Costs

Inventory management is all about managing the flow of inventory through your organization. Stock management helps you better understand which stocks do well and which stocks are just eating up your shelf space. The key is to order the inventory the maximum amount as required, not insufficient and not an excessive amount of.

Due to accurate forecasting through with the assistance of inventory management systems, you'll order the products within the correct quantity and hence save the prices involved in purchasing extra products and storing them.

This also protects you from keeping a less demanding inventory, backorders, excessive inventory, etc. Also, better inventory management helps you understand which products got to be reordered and the way frequently from your suppliers. this will assist you crack deals together with your supplier, thus saving money.

4) Increases Your Business Efficiency and Productivity

If you're spending tons of your time checking out the products your customers order, then maybe it's time to take a position time into inventory management. Inventory management includes allocating specific locations to specific products, and thus , you'll track the products and their quantities in no time!

Systematically managing products not only saves time and efforts but also allows you to divert your human resources towards more pressing matters regarding your business, hence, improving overall productivity.

In short, companies need inventories to work , and having the proper products available at the proper time within the right quantities to satisfy customer requirements is that the key to achieving the company's objectives. And inventory management and control help in doing just that, that's why it's crucial.

- Inventory Management Process

5 key stages

Inventory management is keeping track of and regulating goods as it flows from suppliers through warehouses to consumers. The following are the five major stages to follow:

1. Purchasing: This might include purchasing raw materials to transform into goods or purchasing ready-to-sell products that don't require any assembly.
2. Production: Putting your completed product together from its component pieces. Not every business will engage in manufacturing; wholesalers, for example, may choose to skip this phase completely.
3. Holding Stock: Keeping raw materials on hand before they're made (if necessary) and completed items on hand before they're sold.
4. Sales: Getting your inventory into the hands of clients and collecting payment.
5. Reporting: Companies must know how much they are selling and how much money they are making on each transaction.

Using Sam's firm as an example, here's how each stage of the process is used by Sam's company:



- **Inventory Management Challenges**

The primary challenges of inventory management are having too much inventory and not being able to sell it, not having enough inventory to fulfill orders, and not understanding what items you have in inventory and where they're located. Other obstacles include:

- **Getting Accurate Stock Details:**

If you don't have accurate stock details, there's no way to know when to refill stock or which stock moves well.

- **Poor Processes:**

Outdated or manual processes can make work error-prone and slow down operations.

- **Changing Customer Demand:**

Customer tastes and needs change constantly. If your system can't track trends, how will you know when their preferences change and why?

- **Using Warehouse Space Well:**

Staff wastes time if like products are hard to locate. Mastering inventory management can help eliminate this challenge.

- **12 Ways to Take Control and Optimise Inventory Flow**

1. Employ Intelligent Demand Forecasts

How trustworthy is your data? Keep meticulous records. To develop more accurate projections, have systems that use dependable historical data and sales information. For the greatest outcomes, include and integrate strategies from cross-functional teams.

The measures below should be done to aid in best-practice demand forecasting.

Demand forecasting best practises:

- Establish monthly processes that can be repeated. Improved demand forecasting accuracy necessitates a standardised, timely procedure that compares past projections to actual market results. The statistics will show you if your forecasts were correct or not, as well as what the market demand

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was at the time. You may reduce future errors by following a monthly routine and assessing the outcomes.

- Determine what you'll measure and how frequently you'll measure it. You should concentrate on the most important facts to correctly anticipate demand. POS data, stockout frequency, any amounts of old stock, shipping and dispatch, and even competition sales information are all things to consider.
- Make sure to aggregate data from each specific product for all channels and to integrate data from all sales channels. After you've done this for all SKUs, you'll be able to determine which channels provide the best return on investment for each product. As a result, better inventory control decisions will be made.
- Measure forecast accuracy at the SKU, location, and customer planning levels, since enhanced consumer requirements are a primary driver of demand volatility.

Data is essential; without correct data, forecasting demand is difficult. Demand forecasting best practises rely on exact, up-to-date sales data and the use of salespeople's first-hand expertise.

2. Know What You Have and Where It Is Kept

How many widgets does one have available in stock, in what varieties/size and where are you able to find them? A well-managed warehouse or storage system should make it easy for workers to understand quantities available, to quickly locate items and accurately record stock movements.

One way of doing this is often deploying mobile technology within the warehouse. This makes it possible for warehouse staff to continuously update inventory and logistics information as product is received and dispatched – while equipping staff within the field with a mobile device and internet access means they will log orders faster, preventing stock-outs and reducing lead times. Unleashed Software features a mobile sales app that your business' sales team can use to look at inventory and make sales on the go.

3. Use Your Space Efficiently

Is it easy to access high turnover or perishable goods? Improve the layout of your store to optimise the inward/outward flow of inventory. Have a fixed-location storage system with clearly marked positions for stock keeping units.

4. Rethink Order Cycles and Quantities

Is it possible to decrease the amount and frequency of stock orders? Smaller, more frequent deliveries will free up space while also reducing the danger of waste. Reducing the size of an order has the potential to save money on shipping costs.

5. Keep Good Records

Are your widgets on the shelf or on the loading bay? make sure that records are often trusted to properly reflect physical inventory. Product barcodes, information, stock counts and movement should be timely and accurate.

6. Set Par Levels

Setting par levels, or the minimum on-hand amounts of inventory products for your firm, helps you minimise stock outs while reducing the danger of overstocking.

When inventory supply drops below certain pre-set levels, you know it's time to buy new inventory stock. Minimum stock levels will vary every product and should be determined by how rapidly the item sells and how long it takes to restock.

Because conditions might change over time, check par levels a few times over the business year to make sure they're still appropriate and modify up or down as needed.

7. Match the Inventory Control System to the Type of Product

Inventory control policies should be prioritised round the nature of the inventory moving through the availability chain. Products with a brief time period, like perishables, got to turnover at a way higher rate than manufactured goods or textiles. In such cases, a First-In First-Out (FIFO) policy is adopted to make sure that goods move down the availability chain in accordance with their expiry date.

Of course, without the power to trace, trace and account for inventory in real-time by using a web internal control system, errors so as quantities thanks to inaccurate data-driven purchases won't be ready to be rectified by a FIFO system alone.

Manufactured goods or goods that contains sub-assemblies and multiple components require more complex inventory management systems. Here it's vital to label, itemise and store goods accurately and efficiently. a listing management software solution is usually the sole way businesses handling more complex inventory management challenges can cope. In this manner every component utilized in the assembly of a product can accurately be traced, tracked and accounted for in real-time.

8. Monitor Lead Times

The time it takes for a product to travel from order placing through manufacture to final delivery to your warehouse or facility is known as lead-time. Varying suppliers have different lead times, so it's a good idea to do some research and ask around to see what their lead times are before deciding on the best one for your company.

In general, the shorter the lead-time, the better for your business; but, if the quality is noticeably increased, it may be worthwhile to wait a little longer from a supplier. Knowing your suppliers' lead times makes the job of your buying managers a lot easier.

This ensures that items are reordered in a timely manner to fulfil fluctuating client demand. A competent inventory management software system may be set up to notify managers when inventory levels are low or excessive, and to promote timely reordering depending on supplier lead times.

9. Deploy Cloud-based Technology

Many small firms are hesitant to invest in a complex software solution to manage their inventory control requirements. What they don't account for is their company's ability to scale as it expands. Today's cloud-based inventory management software service providers are inexpensive for organisations of all shapes and sizes, whereas

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top-of-the-line inventory management software used to need a significant upfront financial commitment. This is because the focus has switched from providing on-site inventory software that needs a significant financial commitment to selling inventory software as a service. Small company owners may now seamlessly combine their operations with a sophisticated, continually improving inventory management software system that can manage their inventory with pinpoint precision and complete functionality.

10. Train Your Employees

Inventory control isn't possible without a strong team to back it up. To function efficiently, a group of personnel need sufficient training. Employees must get training in order to comprehend the consequences and necessity of inventory control. It also enables employees to grow with the company and follow best practises.

There are several advantages to a company that receives proper formal training. Inventory control training educates staff and keeps them informed. The system's subtleties are detailed in training so that employees have a better knowledge of the overall picture and how it pertains to the whole firm. It allows workers to connect with a variety of online system tools and capabilities, which upskills personnel and benefits both the organisation and the trainee. Furthermore, training keeps staff on the same page and establishes employers' expectations.

Increased corporate efficiency are another advantage of staff inventory control training. It has the ability to make internal processes more efficient. Tasks may be transmitted more quickly throughout the warehouse, office, and to external vendors. Things can proceed more quickly if personnel have been taught on the proper systems.

Employee morale can also be improved through training. These programmes can help to ensure job stability, which can contribute to increased work satisfaction. Employees are more likely to be fulfilled in their jobs if they are engaged and given a purpose through training. Employees that are happy stay motivated, and their desire to succeed does not wane as quickly. Since training gives them purpose, they are more likely to come to work and less likely to quit.

Once the training has been completed and supervisors are satisfied with the level of performance from the employees, less supervision is required overall. Less supervision means that the supervisor will have more time to spend working on other important tasks across the business. This keeps efficiency high and skill levels continue to improve.

11. Manage Relationships

The ability to adapt quickly is a key component of effective inventory management. Whether you need to return slow-selling goods, immediately refill hot-selling items, or debug production concerns, having a solid connection with your suppliers is critical.

With your supplier, you may create a win-win situation. Clear, proactive two-way communication is required for effective interactions. Allow suppliers enough time to modify production if you anticipate an increase in sales. Allow them to alert you if a product is running late so you may suspend promotions or find a temporary replacement.

12. Strive for Continuous Improvement

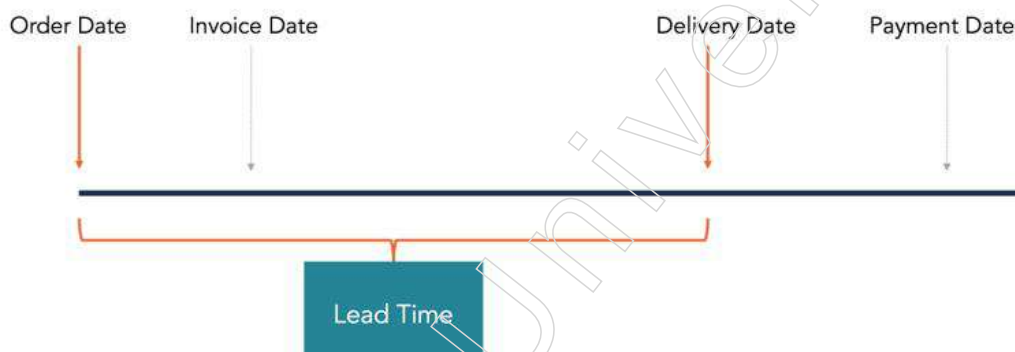
Examine every aspect of your organisation on a regular basis to identify areas where continuous improvement may be implemented and implemented: are your sales, production, inventory management, and customer support operations completely optimised and running as efficiently as they can be?

An important area to explore when attempting to reduce possible bottlenecks is your capacity to handle any additional business. You'll need trustworthy technology to manage your day-to-day operations as well as the technology to provide the goods your company sells.

Do you have the resources to keep up with any new business that comes your way? Prepare for any eventuality to aid in anticipating and removing any dangers to the organization's growth. This entails having the appropriate technologies in place to support business functions, as well as established processes and well-trained, qualified personnel to carry out the required tasks.

5.1.2 Lead Time

Lead time is the amount of time that passes from the start of a process until its conclusion.



During the pre-processing, processing, and post-processing stages, companies evaluate lead time in manufacturing, supply chain management, and project management. They can detect where inefficiencies exist by comparing outcomes to established standards.

Reducing lead time can help to simplify processes and boost production, resulting in increased output and income. Longer lead times, on the other hand, have a detrimental impact on sales and production operations.

Lead Time?



How to Calculate Lead Time

When predicting inventories, don't forget to account for lead time. The method you use to compute this may vary depending on the situation. Distinct processes have different components, which impacts the total length of the process.

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In inventory management, lead time is defined as the period between placing a purchase order to refill items and receiving the order in the warehouse. Order lead times vary per supplier; the larger the chain of suppliers, the longer the lead time is likely to be.

The formula for lead time calculation:

Depending on the conditions, the formula you apply to determine lead time will change. Pre-processing (planning), processing (procurement and manufacturing), and post-processing (dispatch) lag times must all be included when calculating manufacturing lead time. As a result, we may summarise this as follows:

Manufacturing lead time = pre-processing time + processing time + post-processing time.

There are two major aspects to consider when estimating lead time in inventory management: supply delay (the time between ordering and receiving product) and reordering delay (the period you have to wait before ordering supplies). The time it takes for suppliers to process and fulfil orders is covered by these two elements.

Therefore, the formula for calculating inventory lead time is as follows:

Lead time = reordering delay + supply delay

Why Lead Time is Central to Inventory Management

Lead time is a central pillar of inventory management. Businesses must ensure that they have a solid understanding of lead time and that they can keep it under control. Otherwise, there could be serious knock-on effects, with ensuing delays for customers.

Indeed, if lead times start to get out of control, this can come to form a vicious circle, with lead times growing steadily worse as the supply chain seizes up thanks to the increased number of units required. This can then lead to longer delays for consumers, causing lead times to deteriorate further.

Poorly-managed lead time can mean that stocks run out and customers can't get their orders fulfilled. Inadequate order management can do a great deal of damage to businesses' reputations, causing them to lose custom to competitors. It is essential, therefore, that you do everything you can to control lead times as effectively as you can.

• Types of Lead Time

The length of time between the start of a procedure and its conclusion is known as lead time. Lead Period refers to the time between a confirmed client order and its planned pick-up or delivery, as defined by our terms and conditions. These changes depending on the goods and the consumer.

There are various sorts of lead time, but for our needs in a manufacturing or assembly setting, there are four basic categories of lead time.

1. Customer Lead Time — the time it takes for an order to be confirmed and fulfilled (either pick up or delivery depending on the agreement with the customer).

2. **Material Lead Time** - the moment it takes to place an order with a supplier and receive it, from the time the order is confirmed to the time it is received.
 3. **Factory/Production Lead Time** - the time it takes to create and transport a product if all of the necessary ingredients are on hand.
 4. **Cumulative Lead Time** - the moment it would take to order all of the materials from the time the order is confirmed to the time the product is delivered (if none were on hand). It's the result of adding material and manufacturer lead times.
- **Components of Lead Time**
 1. **Pre-processing Time:** This is also known as planning time, and it refers to the time it takes to receive a replenishment request, analyse it, and prepare a purchase order (when purchasing an item) or a task (in the case of a manufacturing business).
 2. **Processing Time:** The processing time is the time it takes to obtain or produce an item after receiving a purchase order.
 3. **Waiting Time:** The interval between obtaining essential components and the start of the production process is referred to as the waiting time.
 4. **Storage Time:** The amount of time that things are stored at a warehouse or factory before being delivered is referred to as storage time.
 5. **Transportation Time:** The transportation time is the time it takes for a manufactured item to go from a warehouse or factory to a client.
 6. **Inspection Time:** The inspection time is the time spent by the customer checking the product to see if it meets the specifications. It also refers to the time required to deal with any non-conformity with the order request.
 - **Ways to Reduce Lead Time**

The following are some of the ways that a company can reduce lead time:

1) Reduce Non-value-added Activities

Value stream mapping should be used to identify non-value-added activities that cause lead times to increase. Make a list of these tasks and cross out those that the firm can do without while keeping the ones that improve product quality.

2) Change Shipping Methods

The firm can also arrange for faster shipping options or more frequent shipments than are currently available. Suppliers may favour slower delivery methods that save money in the long run, which might influence lead times. Even if it comes at a higher cost, switching to a more flexible shipping option can help cut lead time over time.

3) Source Locally

If the firm's imported raw materials are accessible locally, the company can switch to local suppliers as long as the quality of the goods is not compromised. Purchasing things locally rather than from foreign vendors cuts down on lead time since goods are shipped across shorter distances.

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4) Vertical Integration

Vertical integration might entail merging the processes of two suppliers or the company's own production operations. When a business manufactures and assembles components in different places, for example, it may combine the two operations internally. This reduces the time it takes for components to travel from one area to another.

5) Automate the Process

Human mistakes can cause lead time delays when the person in charge of ordering new supply takes too long to contact suppliers. When the product runs out, the firm can utilise a Vendor-Managed Inventory (VMI) or Vendor-Owned Inventory (VOI) system to automatically refill it. Because the supplier receives a request early enough before the firm runs out of stock, this technique decreases lead time.

- Lead Time Formula

Discovering your lead time requires a simple formula. There are two options depending on if you're a manufacturer or a retailer.

For manufacturers, the lead time formula is:

$\text{Total Lead Time} = \text{Manufacturing Time} + \text{Procurement Time} + \text{Shipping Time}$

For retailers, the lead time formula is:

$\text{Total Lead Time} = \text{Procurement Time} + \text{Shipping Time}$

5.1.3 Safety Stock

A merchant or a manufacturer's safety stock is simply excess inventory maintained in case demand spikes unexpectedly. This signifies you have extra stock over and above the acceptable inventory level for day-to-day operations.

Preventing stockouts is one of the key reasons that retailers and manufacturers use a safety stock strategy. The following are the most common causes of stockouts:

- Changes in consumer demand
- Incorrect stock forecasts
- Variability in lead times for raw materials

It might be challenging to plan for these factors while maintaining a goal inventory level. A safety stock formula, on the other hand, comes into play here.

Because there are several formulas for calculating safety stock, it might be confusing, but this article will walk you through the different safety stock calculations and how they relate to the ordering process.

- Need of Safety Stock

The reason for having a safety stock strategy is to protect you against two external factors over which you have little to no control: demand uncertainty and lead time uncertainty.

1) Demand Uncertainty

Every store and producer will have things that sell well all year and those that are in high demand at certain times of the year. Products like razor blades, for example, are purchased all year, making reorder quantities easy to determine. You may not need a huge amount of safety stock if supply and demand are stable.

A product like desk fans, on the other hand, is significantly more difficult to foresee. In general, you'll sell more over the summer, but how can you prepare for an unexpectedly strong demand during a heat wave? To cover an unknown situation like this, you'll need a lot more desk fans than you need razor blades.

2) Lead Time Uncertainty

Lead time is a significant issue in determining minimum inventory and safety stock requirements for manufacturers and firms that assemble goods utilising various components.

In the car business, just-in-time manufacturing relies on parts arriving at the facility just hours before they need to be used on the production line. Supplier production delays and customer delivery difficulties may have a major impact, leading whole lines to shut down.

This, in turn, may have an impact on your own lead times. If deliveries are arriving sooner or later than planned, a safety stock formula can help you offset unforeseen delays and demand fluctuations so that your output remains steady.

Need of Safety Stock Formula

Running out of Stocks have a direct influence on your day-to-day operations. As a result, understanding the importance of safety stock in the ordering process necessitates factoring in the cost of inventory stockout.

Some of the direct impacts on your business include:

- Loss of revenue
- Loss of gross profit
- Loss of customers
- Reduced market share
- Poor efficiency
- Strained supplier and retailer relationships

To put it simply, if you can't meet the demand of your customers, they will find someone else who will, be it online or a store down the street.

- 4 Risks Related to Safety Stocks



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1. Setting Safety Stock to Zero

While having safety stock appears to be a win-win situation, it's crucial to be aware of the hazards associated with it. Stock levels might be over-optimized, which isn't always the ideal method. Here are four major safety stock concerns to be aware of and account for when calculating your safety stock.

2. Using a Standard Safety Stock Formula

This is a typical blunder made by supply chain management professionals. The goal is to minimise inventory and stock-on-hand levels. This may be accomplished by setting the safety stock to zero. However, there is a danger that service levels would suffer as a result, resulting in a cost that exceeds the cost of carrying excess inventory.

3. Letting Safety Stock Decline as Supplier Lead Time Reduces

Supplier lead time should not affect your safety stock, only your cycle stock, because safety stock helps to handle fluctuation in your supply chain and demand. Always keep an eye on the standard deviation time when it comes to your inventory. For example, if your supplier has a five-day lead time and a two-day standard variation, you'll require two days of safety stock (which is the variable).

4. Using Safety Stock to Prevent All Stock-outs

Stock-outs are unavoidable, no matter how hard you try to avoid them. The safety stock formula is in place to prevent the vast majority of stock-outs, but not all. When assessing safety stock, which is what you're attempting to keep, service level is crucial. It's crucial to strike the correct balance between inventory holding costs and service level. A 100 percent service level would imply that you always have product on hand, which is a risk in and of itself, as demand changes.

- Safety Stock with EOQ (Economic Order Quantity)

Companies that make purchasing decisions rather than production decisions typically employ safety stock in conjunction with economic order quantity.

The Wilson formula, commonly known as the Economic Order Quantity (EOQ), is a computation used to calculate the least expensive number of units to order. The goal is to keep ordering and keeping stock costs as low as possible while yet satisfying demand and service level standards.

The cost of ordering things includes the cost of placing your purchase, the cost of shipping and transportation, and the cost of getting your order. The financial expenses of paying for items in advance, warehousing and storage fees, and depreciation costs are all included in holding costs.

EOQ = economic order quantity in units

Q = estimated annual quantity used in units (can be found in the annual purchases budget)

O = estimated cost of placing one order

C = estimated cost to carry one unit in stock for one year

EOQ's ultimate goal is to have as minimal inventory as possible while still being able to meet service needs. To achieve this delicate balance, you must have a clear understanding of your safety stock requirements.

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5.1.4 Elements of Inventory Costs

There are four main components to the carrying cost of inventory:

1. Capital cost
 2. Storage space cost
 3. Inventory service cost
 4. Inventory risk cost
- Capital Cost

The capital cost of carrying inventory is the expense that a firm incurs as it expands. It accounts for the majority of the entire cost of inventory management.

A company's capital cost is expressed as a proportion of the total monetary value of its inventory.

For instance, if a corporation claims that capital expenses are 35% of total inventory costs and total inventory held is \$6000, the capital cost is \$2100.

Although corporations will disclose a proportion of their capital cost, this amount might be factual, based on a calculation, or subjective, based on experience or industry norms.

- Storage Space Cost

The cost of storage space is made up of the warehouse rent or mortgage, lighting, heating, and air conditioning, as well as the price of transferring products into and out of the warehouse.

Some expenditures are set, such as rent or mortgage, but others are variable, such as material handling, which varies depending on inventory levels.

When a 3PL or a private warehouse is employed, all of the expenditures may be rolled into a monthly fee, making the cost of storage space irrelevant when calculating the cost of carrying inventory.

- Inventory Service Cost

Inventory service charges will be included in the cost of carrying inventory. These expenses include inventory insurance and local government taxes.

The amount of insurance a firm must pay is determined by the sort of items in the warehouse as well as the amount of inventory on hand. The insurance price will be greater the larger the level of goods in the warehouse.

Many municipal governments charge the amount of inventory in a warehouse, thus having more inventory means paying more taxes and paying more for inventory service.

- Inventory Risk Cost

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Inventory management entails a certain amount of risk. This risk is included in the cost of maintaining inventories. When a corporation stores products in a warehouse, the danger of the items losing actual value throughout the storage time is always there.

An object might, for example, become obsolete or be replaced by a new model or version.

If a corporation stores components for its work centres or equipment, but then replaces those parts with a new version, the items in the warehouse may be worth significantly less than the initial purchase price. Because finished products may be seasonally particular, the risk is substantially larger in the retail business.

If the things are kept in the warehouse for an extended period of time, their value may be reduced to a fraction of their initial value.

Other components of inventory risk include the likelihood that the things being held will expire, particularly if they have a sell-by or use-by date. If the things reach their expiration date, they may become useless and must be discarded. Water damage, heat damage, and inappropriate storage can all cause items in the warehouse to deteriorate. The cost of inventory risk should also include the cost of pilferage and theft.

- **Types of Inventory Costs**

Ordering, holding, and shortage costs make up the three main categories of inventory-related costs. These groupings broadly separate the many different inventory costs that exist, and below we will identify and describe some examples of the different types of cost in each category.

1) Ordering Costs

Ordering expenses, also known as setup charges, are the costs you experience each time you place an order with your supplier. Here are several examples:

- Clerical costs of preparing purchase orders — clerical costs include invoice processing, accounting, and communication.
- Cost of finding suppliers and expediting orders — they are likely to be uneven, but they are significant expenses for the company.
- Transportation costs — the costs of transporting products to a warehouse or shop. These expenses vary a lot depending on the industry and the goods.
- Receiving costs — These expenses include the costs of unloading products at the warehouse and checking them to ensure that they are proper and defect-free.
- Cost of electronic data interchange (EDI) — These are techniques that major firms, particularly retailers, employ to lower the expenses of the ordering process.

No matter how modest your order is, there will be some sort of ordering fee. The more the number of orders placed, the higher the ordering expenses. If you place a large purchase and plan to utilise the items over a lengthy period of time, the expense of ordering can be stretched out. However, if your company gets raw materials just as needed and keeps minimal inventory on hand, you may be able to bear higher ordering expenses as long as the overall holding cost is lower.

2) Holding Costs

Also known as carrying costs, these are costs involved with storing inventory before it is sold.

- Inventory financing costs — this comprises all costs associated with an inventory investment, such as interest on working capital. Depending on the business, financing charges might be complicated.
- Opportunity cost of the money invested in inventory — This is calculated by subtracting the lost options for locking money up in inventories, such as term deposits or mutual funds.
- Storage space costs — these are costs associated with the site where the inventory is kept and will vary by location. The cost of the storage facility itself, or lease payments if it is not owned, will be incurred. Then there's the expense of facility upkeep, such as lighting, heating, and ventilation.
- Inventory services costs — This comprises the cost of handling the items physically, as well as insurance, security, and IT hardware and software, if applicable. Inventory control and cycle counting expenses are two other examples.
- Inventory risk costs — Shrinkage is a significant cost, since it refers to the loss of items between the time of purchase from the supplier and the time of final sale, which can occur for a variety of reasons, including theft, vendor fraud, shipping problems, and damage in transit or storage. Dead stock is another prominent example.

3) Shortage Costs

These costs, also called stock-out costs, occur when businesses become out of stock for whatever reason.

- Disrupted production — When a company produces items in addition to selling them, a shortfall means the company will have to pay for things like idle staff and industrial overhead, even though no goods are being created.
- Emergency shipments — for merchants, stock-outs may mean paying a premium to receive a supply on time or switching suppliers.
- Customer loyalty and reputation — in addition to losing money from consumers who shop elsewhere, the company suffers a blow to customer loyalty and reputation when customers are dissatisfied.

4) Spoilage Costs

If perishable goods are not sold quickly enough, it can decay or spoil, hence inventory control is critical to avoid spoiling. Many businesses, including food and beverage, pharmaceutical, healthcare, and cosmetics, are concerned about perishability since their goods' expiration and use-by dates influence them all. Spoilage not only costs money, but it also implies that you won't get a return on your investment.

Inventory spoiling and waste are no longer isolated incidents of poor inventory management; spoilage has become a global environmental issue. When you think that an estimated \$200 billion is spent in the United States alone on farming, preparing, shipping, and disposing of food that is never eaten.

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Inventory control is the first line of defence against spoilage and waste. You can enhance forecasts, boost productivity, and access real-time inventory data and up-to-date information on the lifetime of your stock with the correct inventory system, allowing employees to rotate and manage stock to ensure older goods sell first.

In the supermarket and FMCG industries, this strategy is utilised to move items with shorter expiration dates to the front of the shelf. Items that are about to expire are frequently significantly reduced in order to clear inventory.

5.1.5 ABC Analysis

ABC Analysis



ABC analysis is derived from the term “The Pareto Principle” named after an Italian economist Vilfredo Pareto, commonly known as 80/20 rule. This principle suggests that 80% of the total output is generated only by 20% of the valuable efforts.

When it comes to stock or inventory management, ABC analysis divides inventory into three groups based on revenue and control requirements: A represents 20% of products with 80% of total revenue and so requires strict supervision; B represents 30% of items with 15% revenue; and C represents 50% of items with at least 5% income and thus requires the most permissive treatment.

The metrics for any firm may differ, but they all follow a consistent trend. This study tries to focus managers’ attention on the crucial few (A-items) rather than the insignificant many (C-items), and to concentrate inventory management efforts on the things that will have the greatest impact.

Understanding your sales over time as a purchasing manager will help you evaluate and categorise which products belong in which category, i.e., A, B, or C. This will also aid the purchasing manager in determining what to purchase and in what quantities.

- Category A is the smallest and is always designated for the most successful gamblers. It is a symbol of your highest-quality, most valued items and consumers. Such items will contribute significantly to total profit while consuming little of the seller’s resources.
- Category B products are less critical than those in Category A, but more critical than those in Category C. As a result, stay in the middle. This category, in particular, has the potential to be moved into category A if sales are strong, or even to be relegated to category C if sales are low

- Category C items are just somewhat valued. These items will let you operate your business on a consistent basis with a predictable revenue, but they don't add much value to the seller or the firm as a whole.

Example 1: ABC Analysis in Customer Segmentation

ABC analysis may also be used to analyse the market and design your company strategies for your consumer or client base. Revenue potential, support expenses, sales revenue, and contribution margin are four major variables that may be used in an ABC analysis for customers.



- Implementing the ABC Analysis for Customers

Create charts based on the four major categories defined above to begin using ABC analysis for your consumers. Compare the charts once they're finished, particularly the sales revenue and contribution margin charts. Sort your consumers into three categories: A, B, and C.

Customers in Category A will be the most valuable to you. Now, because these clients will generate a significant amount of money and account for a significant share of the contribution margin, they should be near to the revenue ceiling.

Category B includes a subset of clients who are of lower value but must not be overlooked. These are loyal consumers who will spend a significant amount of money with you on a regular basis. However, you should never expect them to spend a lot of money.

The remaining consumers are placed in Category C by default. Customers are those who come in on a regular basis and make a purchase. On the other hand, frequent clients who make a lot of minor transactions may fall into this category. These clients will undoubtedly spend money in your store, but they will have little effect on your sales and profits. These are consumers that have a limited amount of potential.

When it comes to assessing your clients, sales data alone may be quite deceiving. Seeing a customer who only buys once or twice a week for a tiny amount might easily lead you to believe that they are valued when they are not. Instead, then focusing on sales statistics, shift your focus to actionable data. This will allow you to make more informed judgments that will increase your earnings.

Example 2: ABC Analysis in Inventory Management

Lisa has a knitwear company that has grown tremendously in popularity over time. Lisa has filled in 180 sorts of sweaters instead of just 50 in her hunt for variety and additions to her sweater collection. She now knows that this is a seasonal enterprise in which she has already invested much. Lisa utilised the ABC inventory classification

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in her company model after conducting extensive study on how to simplify the difficulty outlined above. She was now able to picture the regions of her firm where more profit was predicted.



Now, because Lisa had too many possibilities, she divided them into three categories: Category A, B, and C, based on their selling ratio, price, demand, and so on.

- Category A – The most crucial sweaters for the firm are those that fit within this group. They might be the ones that are in great demand, the ones that generate the most income, or the ones that follow the season's trendiest trend.
- Category B – These sweaters are necessary for the firm, but not to the same extent as those in category A. Some of the factors include a smaller market or lower comparable demand. Christmas sweaters, for example, are largely sold during the holiday season and not much before or after. However, sales do increase over the holidays. Lisa can't afford to ignore such sweaters, and she can't sell them for much money when they're out of season. As a result, we have category B.
- Category C -- This category contains goods that do not fit into either category A or B. Because these sweaters aren't of great value to the firm, Lisa doesn't want to put in any more effort to sell them. Lisa may have placed these sweaters in category C because of their unusual size, colour combinations, or patterns.
- Need of ABC Analysis

There are 6 reasons why a business should opt for ABC analysis...

1) End-of-life Management

There is always a lifecycle for a product: debut, growth, maturity, and decline. When a product achieves maturity, it will almost certainly be rejected, sooner or later. Customer demands determine the product's whole lifecycle. ABC analysis enables a seller to assess client demand for a certain product category and then manage inventories accordingly. Every product has a different maturity and decline time. With more client demand, the risks of declination increase. The smaller the market, the faster it will decline.

2) Supplier Negotiation

When it comes to sourcing category A products, it's common knowledge that 70-80 percent of the budget will be spent on those vendors. Negotiation is a must at all times. There's a chance a supplier will say yes to your offer, but don't back down. You may

still make concessions, such as lowering the down payment or giving free shipment, for example. If you can make it a win-win situation for both you and the supplier, there's a good chance you'll win the transaction. Saving more on category A items can help your company make more money.

3) Inventory Optimization

The most important reason why many firms chose ABC inventory analysis is inventory optimization. to classify and categorise the inventory's items according to their income and relevance. The items are also divided into categories based on client need.

4) Strategic Pricing

When the pricing for the items is set wisely, the ABC analysis of the inventory adds a lot more value to the organisation. The corporation will be able to simply plan the expenses of the items and then align them into distinct categories using ABC analysis. The greater the demand for a product, the more likely the supplier will raise the price.

5) Resource Allocation

Category A is closely watched while the resource distribution procedure is underway. If it is discovered that a product from category A is underperforming or is no longer desired by clients, it might be shifted to a lesser category, such as category B or C.

6) Customer Service Levels

The cost of the item, the amount sold, and the margin established are all elements that might influence the worth or relevance of a product. There's no reason to overstock your warehouse with products that have a poor profit margin and aren't sold regularly. The ABC inventory system analysis allows planners to define service levels based on product classification in such instances. This smooths out the supply chain even further by eliminating the need for extra safety stockpiles.

- **Ways to Perform ABC Analysis**

Identifying the goal, you're aiming to achieve is the first step in a full ABC analysis. After that, gather the information you'll need to classify the products. Once the classes are in place, keep a careful eye on the data and make judgments based on it.

Here's how to perform an ABC analysis step-by-step:

1. **Identify the Objective:** An ABC analysis can assist you in achieving one of two goals: lowering procurement costs or increasing cash flow by managing inventory levels of the correct goods depending on client sales or production.
2. **Collect Data:** The most typical information to gather is the annual cost on each item. This information is presented in raw purchasing dollars. You can gather the weighted cost, comprising gross profit margin, ordering, and carrying cost data, if it's simple to compute.
3. **Sort by Decreasing Order of Impact:** Using the ABC analysis technique, rate each inventory item's cost order from highest to lowest impact.
4. **Calculate the Sales Impact:** Calculate the impact of each inventory item on

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sales as a percentage by yearly cost by the total cost of all products spent. This is the percentage (or fraction), you will use to compare items in the list. Here's the formula:

$$\% \text{ Impact} = (\text{annual item cost}) / (\text{aggregated total of all items spent}) \times 100$$

5. Sort Items into Buy Classes: Work on contract renegotiation, vendor consolidation, strategic sourcing technique shifts, or e-procurement implementation once you've defined the classes. Making adjustments in these areas can result in considerable cost savings or ensuring that Class A products are available in stock. Rather of sticking to the 80/20 guideline, take a more comprehensive approach.
 6. Analyse Classes: Schedule reviews to track the success or failure of actions after categories and strategic cost management have been set.
- ABC Analysis Best Practices

Best practises in ABC analysis include consistency, sales, and paying attention to events that might alter stock levels or value. Managing inventory using technology is a best practise that streamlines the process from beginning to end.

Apply these best practices when doing an ABC analysis:

- Keep Classifications Simple: Sort products into categories depending on how frequently they pass through your company. Stockouts are more likely to occur with fast-moving products. You may also group goods according to their worth or gross profit margin. Class A would contain the costliest things, Class B would contain the average-priced items, and Class C would have the least expensive items.
- Assign Service and Labour Levels at the Same Time: Assign service levels to items based on their class. The greatest objectives are set for Class A items, while the lowest standards are set for last-class products. Managers might spend ten hours examining 100 Class A things and ten hours reviewing 10,000 Class C items, for example. Schedule cycle counting by categorization, ensuring that Class A items (those that have the greatest and most important influence on sales success) are cycle counted more frequently than Class B and C items.
- Segment KPIs by Class: Create distinct KPIs, corresponding reports and dashboards for each class.
- Establish Performance Reviews: When executing comprehensive inventory maintenance or around timetables and rules that rely on ABC classifications, do performance reviews.
- Review Surplus Stock: Determine if your existing surplus stock levels are appropriate for your business. Surplus stock may cause extra risk and holding costs in the global, just-in-time economy. If it seems sense to keep this inventory, properly categorise it.
- Manage Across Locations: Supply chain managers must be able to handle inventory in multiple locations.

- **Count Inventory in Transit:** Keep track of the time between shipping and receipt when supplies are moved between locations. These types of audits maintain inventory records in order and guarantee that any damage or loss is recorded.
- **Reclassify Purposefully:** Reclassify objects in a variety of ways and at different times. Because of market changes, changes in your client base or their buying patterns, new goods that become popular, or a modification in your KPIs or company plan, you may need to reclassify inventory on a regular basis.
- **Consider Sales and Inventory in Tandem:** Recognize the link between inventories and sales. As sales rise, inventory turns rise, and you'll need to refill according to a predetermined timetable. A market slump, on the other hand, may necessitate a re-evaluation of item classifications and stock levels. Based on the categorisation, review pricing and advertising initiatives.
- **Leverage Technology and Resulting Data:** Automated systems are used by inventory managers to complete replenishment procedures, identify demand spikes, and avoid fulfilment issues. Data may be used to control lead times and demand forecasting.
- **ABC Analysis Benefits**

A long list of benefits can result from applying ABC analysis to inventory management, including:

- **Increased Inventory Optimization:** The study determines which items are in high demand. A corporation can then use its valuable warehouse space to effectively store such commodities while keeping Class B and C stock levels low.
- **Improved Inventory Forecasting:** Increasing the accuracy of sales forecasting by monitoring and collecting data on goods with strong consumer demand. Managers might use this data to adjust inventory levels and pricing in order to boost the company's total income.
- **Better Pricing:** A rise in sales for a certain item indicates that demand is growing, and a price increase may be justified, resulting in increased profitability.
- **Informed Supplier Negotiations:** Because Class A goods account for 70% to 80% of a company's income, it makes sense to negotiate better terms with suppliers for such items. Negotiate post-purchase services, down payment reductions, free delivery, or other cost savings if the seller would not agree to decrease prices.
- **Strategic Resource Allocation:** ABC analysis is a method of regularly evaluating resource allocation to ensure that Class A goods are aligned with consumer demand. Reclassify the item as demand falls to make better use of staff, time, and space for the new Class A items.
- **Better Customer Service:** Many factors influence service levels, including amount sold, item cost, and profit margins. Offer better service levels for the most profitable things once you've identified them.

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- **Better Product Life Cycle Management:** Knowing where a product is in its life cycle (launch, growth, maturity, or decline) is crucial for estimating demand and effectively stocking inventory levels.
- **Control Over High-cost Items:** The success of a corporation is inextricably linked to its Class A inventory. Monitor demand and maintain healthy stock levels as a top priority to ensure you always have enough of the most important goods on hand.
- **Sensible Stock Turnover Rate:** Maintain an adequate stock turnover rate through thorough inventory control and data collection.
- **Reduced Storage Expenses:** By carrying the right percentage of stock based on A, B, or C classes, you may save inventory carrying costs associated with maintaining excess inventory.
- **Simplified Supply Chain Management:** Use an ABC analysis of inventory data to see if it's time to combine suppliers or go to a single source to save money and streamline operations.

- **ABC Analysis Limitations**

Despite its many advantages for inventory management and maintenance, ABC analysis is not a one-size-fits-all solution. Customer demand patterns, classifications, systems, and other challenges vary every business, affecting the utility of an ABC study.

ABC analysis has two drawbacks: an emphasis on the financial worth of inventory and the large amount of time and discipline required to execute the procedure.

Here are a few more challenges:

- **Parameter Instability:** Managers frequently allocate up to 50% of items to a new category every quarter or year as a consequence of ABC analysis. Companies are frequently unaware of changes until there is a demand problem, and the necessity to evaluate can waste time and compromise customer happiness.
- **Limited Pattern Consideration:** Factors like as new product releases and seasonality are not taken into consideration by the typical ABC technique. A new product, for example, may have low sales volume due to a lack of purchasing history. ABC analysis offers a rather static view of demand and will result in inventory inefficiencies anytime demand shifts or is uncertain.
- **Low Information Extraction:** Information from ABC classes may not have all of the statistical data or detail required to make well-informed strategic management decisions.
- **High Resource Consumption:** Bike shedding is the practise of giving excessive weight to minor concerns, which can be an unintended result of ABC analysis. Staff may interject their ideas or request their own versions because ABC analysis is simple to understand, making it a resource-intensive procedure rather than a time-saving tool.
- **Value Blindness:** Although ABC analysis assigns product importance based on revenue or frequency of use, some goods may defy this logic. A shop display item, for example, may sell seldom yet attract a large number of consumers (who will buy other things) due to its novelty. In the aerospace industry, a

- specialised item for a plane may be infrequently used and have low market value, yet it may provide a critical safety role.
- **System Incompatibility:** ABC inventory analysis is incompatible with typical costing techniques and does not comply with GAAP rules. Labour expenses will grow in tandem with inefficiency if you have to run numerous costing systems.
 - **Undersupply or Oversupply Issues:** One downside of ABC analysis is that it looks at dollar-based values rather than inventory amount, which means there's a chance of running out of Class B or C products. It's also possible for the reverse to happen. If you reorder low-class products without reviewing them on a regular basis, you may end up with surplus inventory.
 - **Loss Risk:** Just because B and C things aren't as valuable as Class A items doesn't imply, they're worthless. Excess inventories are constantly in danger of obsolescence or damage, which is one of the limits of ABC analysis. As a result, merchandise that is routinely uncounted or unmonitored may be vulnerable to theft.
 - **Mandatory Standardization:** The ABC technique is only effective if all items are subjected to material standardisation, which involves naming, storing, and routinely rating and monitoring them.
 - **Arbitrary Categorization:** Because there are no fixed boundaries or agreed-upon norms for each category, the manager's professional judgement is required to categorise commodities. As a result, this might be a subjective procedure.
 - **Business Limitations:** Companies with an equable yearly consumption value of inventory goods by type will not benefit from ABC analysis. A firm that offers the identical version of an item, such as candies, nails, or socks, may not be able to use the Pareto Principle to sort stock.
 - **High Resource Consumption:** To control inventory using ABC classification, companies with a large number of inventory items may need to engage more people or purchase specific equipment.

5.1.6 Basic EOQ Model

The Economic Order Quantity (EOQ) is the number of units a firm should add to inventory with each order in order to reduce overall inventory expenses such holding charges, order costs, and shortfall costs. The EOQ is part of a continuous review inventory system in which the inventory level is constantly checked and a set amount is ordered if the inventory level hits a predetermined reorder threshold. The EOQ is a model for determining the proper reorder point and optimal reorder amount in order to ensure that inventory is replenished instantly and without shortages.

It may be a useful tool for small company owners who need to decide how much inventory to have on hand, how many things to order each time, and how frequently to reorder to keep prices down.

The EOQ model implies steady demand and inventory depletion at a constant pace until it reaches zero. A particular number of things arrive at that point, restoring the inventory to its original state. There are no inventory shortages or associated expenses

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because the model assumes instantaneous replenishment. As a result, under the EOQ model, the cost of inventory is a trade-off between inventory holding costs (the cost of storage as well as the cost of locking up cash in inventory rather than investing or using it for other reasons) and order costs (any fees associated with placing orders, such as delivery charges).

Ordering a big quantity at once raises a small business's holding expenses, whereas ordering fewer products more frequently lowers holding costs but raises order costs. The EOQ model determines the quantity that reduces the total cost.

- **EOQ Formula**

Economic Order Quantity (EOQ) is a way of estimating the quantity of stock that needs to be re-ordered by factoring in demand for that item/product as well as your inventory holding costs.

The solution to the question, "How much stock should you reorder to refill your inventory?" is the EOQ. It's essentially the optimal quantity of inventory that a corporation should acquire.

Jon may simply reduce the cost of inventory and safety stock by determining the economic order quantity.

The following is the economic order quantity formula:

Economic Order Quantity	$Q = \sqrt{\frac{2DS}{H}}$	where: Q = EOQ units D = Demand in units (typically on an annual basis) S = Order cost (per purchase order) H = Holding costs (per unit, per year)
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The best case scenario is to use economic order quantity to calculate a certain re-order level for a specific amount of inventory. When this amount is reached, it will necessitate placing a new order for more units, preventing a stock out situation. Consider the following scenario: you own a retail store that sells women's tops, and your annual sales total around 2000 items. It costs \$5 per year to keep those units of the top in your inventory, and the fixed cost of placing an order is \$2.

The EOQ equation then would be,

$$\sqrt{\frac{(2 * 2,000 \text{ top}) * (\$2 \text{ order cost})}{\$5 \text{ holding cost}}} = 40 \text{ units}$$

Hence the ideal order quantity to minimize costs and meet customer demand would be 40.

Reorder Point = Lead time demand + Safety Stock

- **Need to Calculate EOQ**

Calculating the EOQ for your company has a number of advantages that affect your bottom line. It's a wonderful approach to figure out how much merchandise you'll need to keep your ecommerce supply chain running smoothly while keeping expenses down.

Here are the top benefits of calculating EOQ.

1) Minimize Inventory Costs

Excess inventory may easily drive-up storage expenses. Inventory expenses might also rise as a result of how you order, what gets damaged, and what never sells. If you're continually reordering low-volume items, EOQ can assist you figure out how much to order in a given time frame.

2) Minimize Stockouts

EOQ can assist you figure out how much and how often you need to reorder. You can avoid stockouts by determining how much you need based on how much you sell in a specific period of time. This way, you don't have too much inventory on hand for too long. You could be shocked to learn that ordering in lower quantities is more cost-effective for your company, or that the converse is true – calculating EOQ might help you figure this out.

3) Improve Overall Efficiency

Overall, calculating EOQ can assist you in making better inventory storage and management decisions. The fact is that many ecommerce businesses place purchases based on their "gut feelings" about how much to order rather than ordering the exact amount of merchandise required. Using EOQ to properly define how much you need based on key cost drivers is a sensible move.

- 3 Factors You Will Need to Calculate EOQ

The EOQ formula is made up of three variables: holding costs, demand, and order cost. We break down each variable below.

1. Holding Costs (H)

The overall cost of keeping inventory (also known as carrying expenses) is referred to as holding cost. Inventory cost reduction is a crucial retail supply chain management approach. How much do you spend per unit, per year, on inventory keeping and storage? You must first identify your holding cost in order to accurately calculate EOQ. To do so, you can refer to the simple formula below:

$$\frac{(\text{Storage Costs} + \text{Employee Salaries} + \text{Opportunity Costs} + \text{Depreciation Costs})}{\text{Total Value of Annual Inventory}} = \text{Inventory Carrying/Holding Cost}$$

2. Annual Demand (D)

How much demand for a product do you get each year? You can figure out how much stuff you sell year over year by looking at past order data.

3. Order Cost (S)

How much does an order cost per purchase, also known as 'setup cost'? This is done per-order and includes both shipping and handling fees.

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5.1.7 Just-in-time System

JIT is a type of inventory management that necessitates tight collaboration with suppliers to ensure that raw materials arrive at the time production is set to begin, but no earlier. The idea is to keep as little inventory on hand as possible to fulfil demand.

Just-in-Time (JIT) is an inventory management method in which materials and goods are manufactured and obtained just a few hours before they are used. Firms use the Just-in-Time approach to eliminate the unneeded strain of inventory management when demand is less than the inventory available.

The goal of just-in-time is to enhance inventory turnover while lowering holding costs and other associated expenditures. This approach is promoted once again by Japanese companies, which make material orders on the same day that the product is to be manufactured.

As a result, the JIT approach reduces the need for the producer to hold big inventories and incur substantial carrying expenses and other related expenditures. The distribution of materials and the manufacturing cycle must be properly synchronised in order to reap the benefits of this strategy. The JIT system necessitates a clear agreement between the maker and the supplier in terms of delivery and material quality.

Just-in-success time's is determined on how well you manage your suppliers. They are under a lot of strain since they must be able to offer the producer with appropriate quality material as soon as the requirement arises.

The JIT Process Diagram and Steps



The specifics of how organisations deploy JIT in particular environments may differ, but the essential stages remain the same. The cycle of continual improvement in JIT inventory management is depicted in this diagram.

Steps in Cycle of Continuous Improvement for JIT Inventory

1. Design: Product design, process design, people, and manufacturing planning are all reviewed as part of the JIT process. Then strategies are made to minimise disturbance, save waste, and create a flexible system.
2. Manage: A Total Quality Management (TQM) review guarantees that the process is continually improved. A management review establishes and measures statistical

quality control, stabilises schedules, and verifies load and capacity schedules and levels.

3. Pull: Using signalling systems such as Kanban, educate the team on production and withdrawal processes. Review your lot size policies and try to lower them.
4. Establish: Vendor relationships are critical to JIT's success. Examine the vendor listings. Decide on preferred vendors, negotiate contracts, and talk about lead times, delivery expectations, and usage metrics and KPIs. Learn how to use them in the supply chain to your advantage.
5. Fine-tune: Determine inventory requirements, regulations, and controls, as well as inventory movement.
6. Build: Educate your team on the skills and talents they'll need to fulfil their tasks, and hold team education and empowerment meetings.
7. Refine: By refining, standardising, and analysing the whole process, you can reduce the number of parts and stages in manufacturing.
8. Review: Define and execute quality measurements and metrics, as well as do a root cause analysis on any issues that arise. To enhance every component of JIT, emphasise improvements and track trends.

Advantages of JIT Inventory Management

JIT inventory management boosts a company's ROI by lowering inventory carrying costs, increasing efficiency and decreasing waste.

- Waste Reduction: Overordering and excess of any sort are eliminated using the JIT inventory management strategy.
 - Reduce Obsolete Inventory and Dead Stock: Low items levels reduce the danger of inventory going unsold and becoming obsolete in the warehouse.
 - Reduce Defective Product Loss: When production levels are low, it is simpler to discover and repair defective inventory items, lowering scrap costs.
- Improved Efficiency: JIT cuts down the expenses of unused raw materials, excess inventory, and product storage.
 - Raise Inventory Turnover Ratios: Higher inventory turnover is a result of increased efficiency.
 - Minimal Inventory Obsolescence: A high inventory turnover rate prevents things from becoming outdated by allowing them to linger in your facility for too long.
 - Minimize Raw Materials on Hand: Receiving supplies in the lowest quantities possible—often many times per day—virtually eliminates raw material inventories.
 - Local Sourcing: The shorter distances between suppliers and a company's manufacturing site help to ensure timely delivery. The requirement for safety stock is reduced when items are delivered on time and reliably.
- Greater Productivity: JIT boosts productivity by lowering the amount of time and resources spent on manufacturing operations.

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- o Faster Product Turnaround: Manufacturers can create items more rapidly.
 - o Shorter Production Runs: JIT allows manufacturers to swiftly and easily supply new items.
 - o Simplify Change Orders: With less raw material stock to draw down before product modifications, engineering change orders to current goods are easier to implement.
- Smoother Production Flow: Across the whole manufacturing process, JIT may remove bottlenecks and delays.
 - o Shorter Production Cycles: JIT reduces production time, reducing customer lead times.
 - o Reduce Product Defects: Production errors may be detected and repaired more quickly, resulting in fewer damaged goods.
 - o Shorter Production Runs: Quick setup of equipment Reduce production runs to lessen final goods investment.
 - o More Functional Production Cells: Employees in a work cell carry individual components through the production processes, reducing scrap. Work-in-process queues that develop up at more specialised workstations are likewise eliminated using cell models.
 - o Compressed Operations: The quantity of work-in-process inventory travelling between cells is reduced by placing production work cells close together.
- Lower Costs: Receiving items on a need-to-know basis saves money on inventories.
 - o Reduce Working Capital: JIT's low inventory levels reduce the amount of working capital required.
 - o Lower Holding Costs: Because less space is utilised, inventory holding costs (such as those for warehousing) are low.
 - o Lower Cash Investment: Because JIT does not need having a large inventory on hand, companies invest less money on inventory.
 - o Reduce Large Raw Material Spends: JIT allows organisations to order raw materials just when they are needed, freeing up capital for other useful purposes.
 - o Reduce Labour Costs: Because the number of person-hours necessary to fulfil orders is often less than full-time manufacturing, labour costs are cheaper.
- Improve Quality: A flexible workforce can concentrate on producing high-quality, low-defect goods. Better results boost customer satisfaction while lowering manufacturing costs.
 - o Reduce Work-in-progress Goods: With fewer things on the shop floor, teams can concentrate on producing high-quality goods.
 - o Less Damage: Because there is less goods on hand, storage-related mishaps are reduced.

- o Certified Quality: Suppliers make a quality guarantee in advance. As a result, instead of being kept at receiving for inspection, goods are sent directly to production zones.

To support these goals, you can invest in new technology or update existing solutions that will link your system with your suppliers to coordinate the delivery of parts and materials.

- JIT Inventory Methodology

The JIT inventory technique incorporates a variety of strategies to ensure that operations function smoothly. Organization, attention to detail, small lot sizes, improved transparency, cell manufacturing, and the use of a pull (rather than a push) approach are all emphasised in the lean strategy.

Techniques Involved in JIT Inventory Methodology

- Order: Maintain a high level of physical and organisational discipline to keep everything in order.
- Better Quality: Eliminate flaws by paying close attention to detail and making continual improvements.
- Reduced Setup Time: When installations need to modify to match consumer demand, create flexible changeover techniques.
- Small Lot Size: One is the best lot size in JIT. The tiny size cuts down on in-process inventory, transportation expenses, storage space, and makes inspection and rework easier.
- Load Uniformity: Levelling is a control method that results in a consistent, level daily routine.
- Flow Balance: Flow scheduling organises throughput to ensure that energy and labour are distributed evenly.
- Diversified Skills: Cross-trained workers can be deployed to different areas to keep production moving.
- Visibility for Control: Using communication tools, like those found in Kanban, keeps the entire team informed of inventory levels.
- Ongoing Maintenance: Ongoing oversight and focus on detail, including the machinery and tools the business uses every day, helps maintain a low defect, low problem environment.
- Use Fitness: JIT spaces designed to fit each process speeds up production. One workstation pulls output from the one before it, as needed, based on a master schedule or customer demand.
- Logical Plant Layout: Product-oriented design makes assembly easier and more efficient.
- Strong Supplier Network: JIT inventory is most successful when vendors have strong ties.
- Worker Immersion: To attain JIT goals, every team member should be committed to the process and their co-workers.

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- **Cell Manufacturing:** Create an atmosphere in which teams may work as swiftly as possible to produce as many goods as feasible while minimising waste.
- **Pull System:** When a product is only replaced after it has been utilised in production, it is referred to as a pull system.
- **Disadvantages of Just-in-time Inventory Techniques**

Precision forecasting and strong connections with key suppliers are critical components of JIT inventory management. When any of those fails, it's an issue because there are no backup choices.

For example, a single supplier who is unable to produce for an extended length of time might cause the entire supply chain to break down and cause your business to shut down. Furthermore, organisations that use tight JIT inventory management are unlikely to have excess stock on hand to meet unexpected demands.

For example, if an organization's forecasting can't account for an increase in demand, it won't have enough stock to meet those orders. This might result in a loss of money and, possibly, consumers.

- **Potential Risks of Just-in-time Inventory**

The primary risk of JIT comes from its philosophy. JIT inventory management requires everyone in an ecosystem and supply chain to commit and work cohesively. If any part of that arrangement breaks down, it risks the entire infrastructure.

- **Supply Chain Disruptions:** Supply chain disruptions can cause the manufacturing process to stop.
- **Missed Opportunities:** A corporation may not be able to satisfy big and unexpected requests instantly if it has few or no finished items on hand.
- **Unexpected Price Changes:** The cost of components is fixed in JIT. Profit margins shrink as costs grow.
- **Overreliance on Forecasts:** Because of the dependence on predicting, adapting to abrupt spikes or dips is challenging.
- **Order Issues:** Shortages and stock-outs can cause inventory systems to malfunction.
- **Local Sourcing Costs:** JIT relies on local procurement, which might be more expensive for a variety of reasons. In the quest of dependability, this reliance can have a negative impact on profitability.
- **Time Pressure:** Scheduling may raise the cost of goods sold (COGS) since a firm cannot always expect to get the cheapest price for raw materials from a supplier.
- **Undisciplined Staff:** Team members that aren't on board with JIT might have a negative impact on production, quality, and other factors.
- **Supplier Dependence:** When a supplier fails to deliver items on schedule and in the correct quantities, the entire manufacturing process might be disrupted.
- **Acts of Nature:** A natural disaster that disrupts a vendor's supply chain can put a stop to production.

- History of Just-in-time Inventory Management

The “Toyota Production System” (TPS) or just-in-time production was the original name for the just-in-time concept. The method was created in post-World War II Japan, when vehicle manufacturers faced resource constraints and needed to reduce resource usage to stay competitive.

When Toyota Motor Company (TMC) saw that American carmakers of the time were surpassing their Japanese competitors, Japanese industrial engineers Eiji Toyoda and Taiichi Ohno devised the method. They created the Toyota manufacturing method after considerable experimentation and bridged the gap between 1945 and 1970. JIT has grown in popularity over the world. The main underlying concept of this method is to reduce the use of resources that bring no value to a product.

Check Your Understanding

- 1) What is full form of EOQ?
 - a) Economic order quantity
 - b) Economic Organize Quantity
 - c) Encompass Order Quantity
 - d) Economic Order Quality
 - e) None of the above
- 2) A lot-sized inventory is also known as _____.
 - a) Kaizen Process Cycle
 - b) Kaizen Events
 - c) Cycle Inventory
 - d) Kanban System
 - e) None of the above
- 3) _____ is a process of checking and balancing so that the companies confirm the physical counts that match the inventory records.
 - a) Just-In-Time
 - b) Cycle counting
 - c) Order Preparation
 - d) Logistics Operations
 - e) None of the above
- 4) _____ includes any item that is used in the manufacturing process but is not visible in the final product.
 - a) Maintenance, Repair and Operations inventory
 - b) Product Inventory
 - c) Raw material Inventory
 - d) Work-In-Process Inventory

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- e) None of the above
- 5) _____ is the amount of time that passes from the start of a process until its conclusion.
- a) Lead time
 - b) Completion Time
 - c) Real Time
 - d) Process Time
 - e) None of the above
- 6) A company's _____ is expressed as a proportion of the total monetary value of its inventory.
- a) Current Cost
 - b) Capital cost
 - c) Inventory Cost
 - d) Production Cost
 - e) None of the above
- 7) Shortage Costs are also called _____, occur when businesses become out of stock for whatever reason.
- a) Computing Costs
 - b) Change-Over Costs
 - c) Material Handling Costs
 - d) Stock-Out Costs
 - e) None of the above
- 8) _____ has two drawbacks, which includes an emphasis on the financial worth of inventory and the large amount of time and discipline required to execute the procedure.
- a) Economic Order Quantity
 - b) ABC analysis
 - c) Reorder Point
 - d) Safety Stocks
 - e) None of the above
- 9) The _____ is the number of units a firm should add to inventory with each order in order to reduce overall inventory expenses such holding charges, order costs, and shortfall costs.
- a) Days Inventory Outstanding
 - b) Stock Review
 - c) Economic Order Quantity
 - d) First In First Out

- e) None of the above
- 10) _____ is an inventory management method in which materials and goods are manufactured and obtained just a few hours before they are used.
- a) Just-in-Time
 - b) Buffer Stock
 - c) Work-In-Process Goods
 - d) Transit Inventory
 - e) None of the above

Summary

- Inventory management aids businesses in determining which goods to order and when to order it.
- Safety stock is essentially a buffer for your on-hand inventory, or surplus product that isn't being consumed by customers.
- Anticipation Stock is a stock of components, resources, or commodities held on hand by a firm or business to satisfy demand or to make up for irregular production shortfalls.
- Freight on Board Destination signifies that the seller retains control of the merchandise until the shipment arrives at the buyer's destination.
- In a cycle count, a small fraction of the overall inventory is tallied at a time to get total stock numbers. It is a sampling approach that employs data from a tiny sample of the population to quantify the larger picture.
- Stock management helps you better understand which stocks do well and which stocks are just eating up your shelf space.
- Reducing lead time can help to simplify processes and boost production, resulting in increased output and income.
- The motto of inventory management is to minimize holding costs while optimizing inventory.
- ABC analysis is derived from the term "The Pareto Principle" named after an Italian economist Vilfredo Pareto, commonly known as 80/20 rule. This principle suggests that 80% of the total output is generated only by 20% of the valuable efforts.
- Identifying the goal, you're aiming to achieve is the first step in a full ABC analysis. After that, gather the information you'll need to classify the products. Once the classes are in place, keep a careful eye on the data and make judgments based on it.

Activity

1. Visit an organization and understand how do they conduct inventory management and which method they adapt to do the same.

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Questions & Exercises

1. What is Inventory management? Explain the types and basic formulas of inventories.
2. Explain the characteristics of Inventory and the techniques followed by Inventory management to meet consumer demand while being profitable.
3. Elaborate the concepts of MRO Inventory and Cycle count inventory management and the ways to optimise inventory flow.
4. What is Lead Time and ways to reduce Lead time.
5. Elaborate the concept and need of ABC Analysis.
6. Why do businesses need to perform Just-in-Time and the techniques used to do the same. Also elaborate the potential risks involved in Just-In-Time inventory.

Glossary

- Inventory management is a method for acquiring, keeping, and selling inventory, including both raw materials (components) and completed commodities.
- ABC analysis methodology divides inventory into three groups based on the item's inventory value and cost relevance, these are High-value, moderate-value and low-value category.
- Just-in-time (JIT) methodology is based on studying client behaviour, in which things come as ordered by customers.
- Minimum order quantity (MOQ) is a methodology which determines the least number of items a supplier will sell.
- First in, first out (FIFO) is a methodology which helps to keep product fresh, the oldest stuff is sold first. This is a particularly significant strategy for organisations dealing with perishable goods
- Last in, first out (LIFO) methodology is the most recent inventory is usually reported as sold first. When there is a problem with inflation and prices are rising, this is an excellent technique to follow.
- Raw materials, also known as bought components, are the raw resources that will be utilised to create completed goods.
- 'Work in Progress' is a phrase used to describe a piece of work that Items that are half completed and, on their way, to becoming finished goods are referred to as goods.
- Transit inventory refers to inventory that is in transit from a manufacturer's location to a retailer's or wholesaler's location.
- Ordering expenses, also known as setup charges are the costs you experience each time you place an order with your supplier.
- Holding Costs, also known as carrying costs are costs involved with storing inventory before it is sold.
- The EOQ is a model for determining the proper reorder point and optimal reorder amount in order to ensure that inventory is replenished instantly and without shortages.

Further Readings

- 1 Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
3. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding – Answers

- 1) a)
- 2) c)
- 3) b)
- 4) a)
- 5) a)
- 6) b)
- 7) d)
- 8) b)
- 9) c)
- 10) a)

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Unit - 5.2: Basic Concepts of Supply Chain Management

Objectives

At the end of this unit, you will be able to:

- Importance of supply chain management
- Benefits of effective supply chain management
- Key features of effective supply chain management
- Principles of supply chain management

Introduction

Supply chain management refers to the administration of the flow of products and services, and it includes all operations that convert raw materials into completed commodities. It comprises aggressively streamlining a company's supply-side procedures in order to boost customer value and gain a market competitive advantage.

Through supply chain management, suppliers try to design and run supply networks that are as effective and cost-effective as possible (SCM). Manufacturing and product development are all part of supply chains, as are the information systems needed to manage these operations.

SCM often seeks to centralise or link the manufacture, transport, and distribution of a product. By optimising the supply chain, businesses may save costs and deliver things to clients more quickly. Internal inventories, internal production, distribution, sales, and business vendor stockpiles are all being closely monitored.

The administration of the movement of goods and services is referred to as supply chain management, and it encompasses all processes that turn raw materials into finished products. It entails actively simplifying a company's supply-side processes in order to optimise customer value and obtain a competitive edge in the marketplace.

5.2.1 Basic Concepts of Supply Chain Management

Six components of Supply Chain Management



The following are the six components of supply chain management.

1. Planning

The first and most important aspect of supply chain management is planning. The planning component's goal is to manage and plan all of the organization's resources in order to deliver products and services that fulfil consumer demand.

An organization's supply chain design must be planned ahead of time. Then they decide which measure they should employ to guarantee the organization's supply chain is efficient and effective. The purpose of planning is to stay on track with the organization's objectives while also staying ahead of the competition. The firm ensures that it provides value to its consumers by using an action plan.

2. Sourcing

Sourcing is the second aspect of supply chain management. Suppliers have a significant impact on an organization's performance. Choosing the correct suppliers to offer the materials and services that an organisation requires to develop final goods is a critical task.

The organisation enters into a contract with the suppliers, and the suppliers furnish the organisation with products and services. To manage its connections with suppliers, an organisation employs a variety of techniques. Sourcing may be done in a variety of ways.

The primary operations involved in sourcing include ordering goods, receiving material, inventory management, maintaining communication with suppliers, and authorising payment to suppliers.

3. Making

The manufacturing component is the next component of supply chain management. Coordination of production operations is the job of the supply chain manager. The supply chain management process includes a variety of actions.

Accepting raw materials, producing goods, checking the quality of final products, rejecting and recycling items that do not meet quality requirements, packing final products, and arranging final delivery are examples of these processes.

Organizations monitor the quality, staff productivity, and output of production on a regular basis to ensure that the company is producing items that satisfy the quality requirements.

4. Delivering

Logistics is another term for the delivery component of supply chain management. The distribution procedure is complicated, and customers pick your items based on the quality and speed with which you supply them.

Coordination of client orders, scheduling delivery, mode of payment, sending deliveries, invoicing customers, and receiving payment are all part of the supply chain management process. The speed with which you transport things is determined by the number of cars in your fleet.

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Many firms outsource their delivery services to specialist organisations in order to lessen their workload and when the delivery procedure necessitates special handling and extra effort.

5. Returning

The company returns things that are undesirable, faulty, or in excess quantity in this process. The provider should be prepared to accept the returned items and discard or recycle the faulty ones. If the received items are in acceptable condition but have an excess of volume, they should be returned to the warehouse for sale.

6. Enabling

To keep track of the information flowing through the supply chain, many support processes are required. Regulations must be followed by supply chain management. Human resource, finance, information technology, portfolio management, facilities, product design, quality assurance, and sales are all critical processes in the enabling process.

- Why is Supply Chain Management Important?

In the production cycle, effective supply chain management solutions reduce cost, waste, and time. A just-in-time supply chain, in which retail sales instantly transmit replenishment requests to manufacturers, has become the industry norm. Retail shelves can then be restocked almost as quickly as product is sold. One way to further improve on this process is to analyse the data from supply chain partners to see where further improvements can be made.

The CIO.com post¹ cites three instances where efficient supply chain management adds value to the supply chain cycle by evaluating partner data:

- Identifying Potential Problems. When a consumer requests more merchandise than the manufacturer can supply, the buyer has the right to complain about unsatisfactory service. Manufacturers may be able to predict a scarcity through data analysis before the consumer is disappointed.
- Optimizing Price Dynamically. Seasonal items have a finite shelf life. These goods are often destroyed or sold at substantial discounts towards the end of the season. Airlines, hotels, and other businesses that sell perishable “items” usually modify their rates dynamically to match demand. Similar forecasting approaches, especially for hard commodities, can boost margins by applying analytic tools.
- Improving the Allocation of “Available to Promise” Inventory. Analytical software solutions assist in dynamically allocating resources and scheduling work based on sales forecasts, actual orders, and promised raw material delivery. Manufacturers may confirm a product delivery date when an order is submitted, lowering erroneously completed orders dramatically.
- Benefits of Supply Chain Management

Effective supply chain management provides three primary benefits to an organization:

1. Lowered Costs

Organizations can reduce operational expenses by responding more flexibly to consumer requirements by integrating suppliers and utilising technology. Managing based on demand, for example, prevents firms from over-producing, which not only decreases labour and raw material expenses, but also inventory management and transportation costs.

2. Increased Revenue

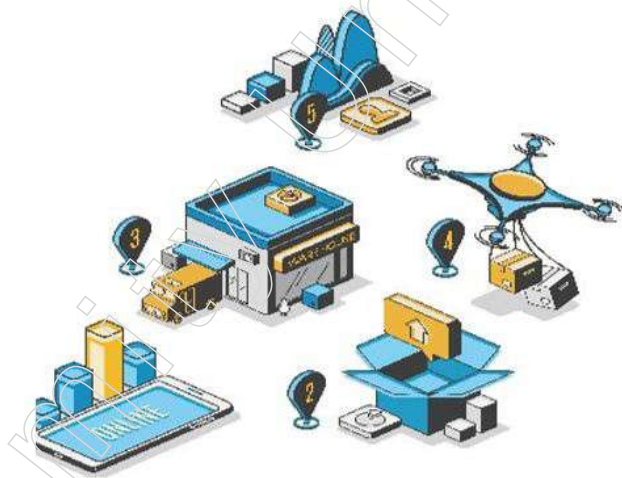
When firms utilise technology to stay closer to consumer demand and respond more rapidly (as in the Walmart example of keeping shelves stocked), it increases the likelihood that items will be available for purchase by customers. When production is simplified to manufacture just what is needed, manpower and materials may be focused to inventing new things to offer customers and expanding the product mix. Outside of the product sector, this might imply providing clients with extra services.

3. Asset Utilization

Organizations that utilise good supply chain management may make the most use of capital assets such as manufacturing or transportation equipment. Rather of causing unnecessary wear and tear on production equipment, firms may manufacture in response to demand.

Supply chain management enables firms to deliver more rapidly, assure product availability, decrease quality concerns, and easily navigate returns, eventually creating value both inside the organisation and for consumers.

- Key features of Effective Supply Chain Management



One of the most important aspects of any firm is supply chain management. A competent and effective supply chain management assists firms in enhancing their business and building the company's brand.

A corporation with a high reputation can stay in business for a long time and build long-term connections with their consumers. If you want to study about supply chain management in depth, you must first understand about the characteristics of a good supply chain management.

Notes

1. Comprehensive

The data gathered through supply chain management is compared to the data gathered in real-time. This approach produces results that are both quick and comprehensive. Supply chain management aids in the prevention of future delay.

2. Connected

Supply chain management not only connects suppliers and manufacturers, but it also allows manufacturers to engage with their consumers. A manufacturer can place orders based on the availability of raw materials for the manufacturing process.

Businesses, for example, can gather data on shifting patterns and utilise this knowledge to prepare for future manufacturing activity. This enables the firm to provide better service to its clients.

3. Cognitively Enabled

Collecting and coordinating information is how supply chain management works.

This enables the firm to execute successfully and make supply chain decisions. Many firms have advanced supply chains that allow them to participate in automated learning.

4. Collaborative

Supply chain management facilitates coordination among suppliers and producers. An active engagement with suppliers enables an organisation to better serve its consumers.

5. Cyber Aware

An appropriate supply chain management should not only assist the company in boosting their business but also in making the organization's system safe and intrusion-proof.

- Principles of Supply Chain Management



The following are the seven key supply chain management concepts that can be applied to every organization's supply chain, regardless of the sort of product it produces.

1. Segment

Customers can be segmented based on their need and the service they desire. Then, utilise this data to make changes to your supply chain so that you can service each client group effectively and profitably.

Having a thorough understanding of your consumers' expectations and purchase behaviours allows you to appropriately meet their demands.

2. Customization of Logistics Based on the Segments

Once you've tailored sections of your customers' and their wants, you'll need to adapt your logistics network to match the service requirement and efficiently serve each segment.

Curate your offerings in such a way that you can suit your consumers' particular needs. You will not only be able to satisfy your consumers, but you will also be able to remain profitable.

3. Forecast Demands

Marketers must pay attention to market demands so that they may organise their supply chain properly. Make certain that your prediction is consistent and that you apply optimal resource allocation. Make your Supply Chain Management (SCM) efforts as adaptable as possible so that you can be ready for any unforeseen occurrences that may arise in the organisation.

4. Products Differentiation

Make your items stand out deeper down the supply chain. This will assist you in identifying precise demand measurements. You can be ready with customised items that your clients may require at any moment in the future.

5. Minimize Costs

You may manage your resources in such a manner that your manufacturing costs are kept to a minimum. The entire cost of owning material and making things can therefore be minimised. You may reduce raw material costs by working with your suppliers. You can gain from the relationship with the suppliers in this way.

6. Use Technology

Utilize technology and supply chain management tools to assist you in decision-making at multiple levels and to give a clear perspective of products, services, and associated information.

7. Analyse the Performance

Finally, you must evaluate the effectiveness of all of your efforts. This will assist you in deciding where you need to spend more effort in order for your processes to work smoothly.

Notes

Check Your Understanding

- 1) The _____ component's goal is to manage and plan all of the organization's resources in order to deliver products and services that fulfil consumer demand.
 - a) Organizing
 - b) Designing
 - c) Planning
 - d) Controlling
 - e) None of the above
- 2) _____ is responsible for planning, implementing, and controlling the effective movement of storage, goods, services, and related information from the point of origin to the site of consumption.
 - a) Supply chain management
 - b) Logistics Management
 - c) Just-In-Time Inventory Management
 - d) Delphi Method
 - e) None of the above
- 3) A _____ supply chain, in which retail sales instantly transmit replenishment requests to manufacturers, has become the industry norm.
 - a) Cyber Aware
 - b) Product Differentiation
 - c) Logistics
 - d) Just-In-Time
 - e) None of the above
- 4) Which of the following is not the benefit of supply chain management?
 - a) Lowered Costs
 - b) Increased Revenue
 - c) Asset Utilization
 - d) All the above
 - e) None of the above
- 5) What is the full form of SCM?
 - a) Mkd
 - b) Supply Chain Management
 - c) S
 - d) Sd
 - e) None of the above

- 6) _____ is the management of the flow of goods and services and includes all processes that transform raw materials into final products.
- a) Supply chain management
 - b) Inventory Management
 - c) Strategic Management
 - d) Total Quality Management
 - e) None of the above

Summary

- Logistics is another term for the delivery component of supply chain management. The distribution procedure is complicated, and customers pick your items based on the quality and speed with which you supply them.
- In the production cycle, effective supply chain management solutions reduce cost, waste, and time.
- Supply chain management enables firms to deliver more rapidly, assure product availability, decrease quality concerns, and easily navigate returns, eventually creating value both inside the organisation and for consumers.
- An appropriate supply chain management should not only assist the company in boosting their business but also in making the organization's system safe and intrusion-proof.
- Suppliers have a significant impact on an organization's performance. Choosing the correct suppliers to offer the materials and services that an organisation requires to develop final goods is a critical task.

Activity

1. Write a case study to explore a few companies that have managed to sustain their supply chain cost-reduction efforts and keep expenses under control.

Questions & Exercises

1. What is Supply chain management. Explain its components and why is it important to have a proper supply chain?

Glossary

- Logistics management is a component of supply chain management that is used to satisfy customer expectations through the effective transportation and storage of connected information, commodities, and services from origin to destination.
- Logistics management improves process strategy, planning, and implementation by conforming to customer demands and industry norms.
- Analytical software solutions assist in dynamically allocating resources and scheduling work based on sales forecasts, actual orders, and promised raw material delivery.

Notes

- Supply chain management enables firms to deliver more rapidly, assure product availability, decrease quality concerns, and easily navigate returns, eventually creating value both inside the organisation and for consumers.
- Supply chain management is defined as the management of the flow of goods and services and includes all processes that transform raw materials into final products.

Further Readings

1. Production and Operations Management; S. Anil Kumar/ N Suresh
2. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor

Check Your Understanding – Answers

- 1) c)
- 2) a)
- 3) d)
- 4) e)
- 5) b)
- 6) a)

Unit - 5.3: Basic Concepts of Logistics Management

Notes

Objectives

At the end of this unit, you will be able to:

- Learn basic concepts of logistics management
- Understand different types of logistics management
- Distribution network of logistics management
- Importance of logistics management
- Logistics operations: 4 key process

Introduction

Logistics management is a part of supply chain management which is used to meet customer expectations by efficiently transporting and storing related information, commodities, and services from point of origin to point of destination. Logistics management helps firms cut expenses while increasing customer service. The logistics management process starts with the collection of raw materials and finishes with the delivery of things to their final destination.

5.3.1 Basic Concepts of Logistics Management

Logistics management improves process strategy, planning, and implementation by conforming to customer demands and industry norms. Logistics management is responsible for the management of resources ranging from physical products (such as materials, equipment, and supplies) to food or other consumable things. Logistics management is concerned with combining the flow of information and its management tools, materials handling, production packing, inventory, transportation, warehousing, and, in certain cases, security.

A specific simulation programme is frequently used to simulate, analyse, visualise, and optimise this complicated logistical issue. A logistician is someone who works in this sector.

- Relation to Supply Chain Management

Logistics management is a subset of supply chain management in general. Supply chain management is responsible for planning, implementing, and controlling the effective movement of storage, goods, services, and related information from the point of origin to the site of consumption. This is done in order to suit the needs of the clients.

Logistics management in business is applicable to all sectors. Its goal is to manage project life cycles, supply networks, and the resulting efficiency. Business logisticians have grown into supply chain logisticians as firms have become more complicated and expanded into a worldwide marketplace. The aim of corporate logistics management is twofold: inbound logistics for internal activities and outbound logistics for the external flow from point of origin to point of consumption. Inventory management, purchasing, shipping, warehousing, consultancy, and the organisation and mapping of these activities are all priorities for logisticians.

Notes

Different Types of Logistics Management

There are four main types of logistics management, each emphasizing a different aspect of the supply process.

1. Supply Management and Logistics

This entails planning, buying, and arranging items that are required at a specific time and location for the completion of a work. This includes both transportation and a location to keep the supplies. Furthermore, reviewing the amount of supply at various stages of the process is essential to ensure that the customer's expectations are satisfied, such as providing resources to a building site or components to a manufacturing facility.

2. Distribution and Material Movement

This delivers resources that have been stored to their final destination. Moving items, including loading, unloading, and transportation, as well as keeping track of stock and how it is utilised, are all difficulties in this case. This sort of management oversees the transfer of goods from a central warehouse to the retail outlets where the product is sold to the general public.

3. Production Logistics and Management

This oversees the processes of assembling scattered materials into a product, such as organising what is required to create or assemble something. This entails preparing materials for use in the construction of a product at the appropriate moment. This sort of logistics management is related to product management.

4. Reverse Logistics and Product Return

This is about the management of reclaiming materials and supplies from production. For example, on a construction site it involves the removal of excess material and returning those materials to one's stock. It can also refer to the return of unwanted or unused products from the end customer seeking a refund.

- The Distribution Network of Logistics Management

The various links and points of distribution in a logistics management network include the following:

- Factories that manufacture products
- Warehouses that store products
- Distribution centres to receive and return items for clients
- Transport to deliver product
- Retail locations, from small to larger stores to sell product

These are the major hubs for the logistics of a product, though there can be vendors and intermediaries operating between these points.



Tips for Smart Logistics Management

Logistics management necessitates extensive planning: the more stages, the better. You are more likely to boost efficiency and profits if you analyse every stage of the product, its distribution, and the return of resources and supplies.

The logistics management becomes more complex and challenging as the organisation grows in size. As a result, the more important it is to have a solid logistics management plan. The following are some pointers to keep in mind if you want to be prepared and have the finest strategy possible.

- **Have a Strong Plan:** Like any management, it succeeds or fails based on the strength of the strategy. The more detailed your strategy, the less you'll have to think on your feet. There will always be challenges, and there are only so many possible dangers you can plan for, but planning ahead of time and in depth may assist alleviate delays and other blockages to the free flow of materials and supply.
- **Make a Plan B:** No matter how brilliant your initial plan is, something will always come up that it cannot handle. That is why, in order to respond to unanticipated situations, you must have a contingency plan for each part of your logistics strategy. But it's also crucial to know when to abandon the initial plan in favour of a backup.
- **Hire a Manager:** It is vital that this process has a leader who is knowledgeable and capable of working with a wide range of diverse parties who are all engaged in the logistics of the goods and supplies. As a result, interpersonal skills are essential. They should also have strong industry relationships to cope with last-minute logistics changes in suppliers, etc.
- **Automate:** Automation is, of course, a built-in approach to improve workflow efficiency. There are several activities that may be aided by task automation, ranging from tracking to delivery monitoring to fleet and inventory management software.
- **Learn from Mistakes:** This applies to almost everything. On your logistics management journey, you will make mistakes. That is self-evident. What is not certain is that you will learn from your errors and avoid doing them again. As a result, take some time to reflect on what you've done, where it worked and where it didn't, and solicit feedback from your team.

Notes

- Importance of Logistics Management

Obviously, the goal of logistics management is to discover more efficient and effective ways to transport resources and products from conception to completion and, eventually, to the client. However, the driving reason behind these measures is to fulfil consumer demand and deliver the finest service possible in order to retain clients and keep them satisfied by meeting their needs.

Customers want better service, which necessitates the need to ship faster, more precisely, and with a high degree of quality. Customer satisfaction is gained via logistics management.

But it isn't the only advantage of logistics management. It also aids in increasing visibility in the company's supply chain. Logistics managers can improve the flow of materials and anticipate future interruptions by reviewing historical data and watching the real-time movement of items.

As a result, logistics management contributes to increased income. It enhances customer service and contributes to the company's strong reputation and brand, resulting in new and increased business. With greater insight into the supply chain, there is the potential to reduce operational costs by limiting inbound money, maintaining appropriate inventory levels, and coordinating the reverse flow of commodities.

- Logistics Operations: 4 Key Process

Logistics operations are a collection of activities that work together to optimise the supply chain, from transporting items from the producer to the seller through last-mile delivery.

1. Suppliers and Manufacturers

The initial step in supply chain management is obtaining raw materials, which is handled by the manufacturer or supplier. A trustworthy manufacturer is in charge of managing the work-in-process inventory phase (i.e., the flow of raw materials being processed into finished goods) to ensure that you receive a high-quality, sellable product on time and at the lowest possible cost. It is critical to select the correct supplier; manufacturers that are unreliable in producing a quality product and/or transporting inventory to the seller can stymie operations from the start.

2. Distributed Fulfilment Centres

Fulfilment centres are warehouses that keep goods close to the end customer, so that each order is chosen, packed, and dispatched as soon as it is placed to ensure the quickest possible delivery. Many direct-to-consumer (DTC) firms rely on ecommerce fulfilment services provided by a third-party logistics (3PL) provider. They won't have to worry about fulfilling and delivering orders themselves this way.

Another significant advantage of working with a 3PL is the flexibility to disperse goods across many logistics locations. ShipBob, for example, is a retail fulfilment firm that provides a powerful worldwide fulfilment centre network that enables merchants to distribute inventory between locations in order to be closer to their consumers. This lowers transportation costs and expedites delivery.

3. Warehousing

Warehousing is a critical stage in the logistics operations process that involves the storage and management of physical items prior to their sale. Warehouse management consists of numerous critical components, including the safety and security of stored items, inventory control methods that aid in the optimization of inventory storage, and much more.

These crucial operational warehouse management systems help to track where items are located, when they arrived, and how long they have been there.

4. Shipping

Orders that are dispatched to customers precisely and swiftly demonstrate how successfully your logistics operations are managed. Although the carrier is ultimately responsible for delivering goods to their final destination, you may pick which shipping options to give your clients (e.g., standard through ground, expedited via air, etc.) and to which regions (domestic vs. international shipping).

You also have the option to ship orders in-house, most often which involves standing in line at the post office. Otherwise, you can outsource fulfilment to a company that has access to discounted shipping rates from major carriers and can take care of the shipping process (and all other logistics operations) for you.

Check Your Understanding

- 1) _____ consists of numerous critical components, including the safety and security of stored items, inventory control methods that aid in the optimization of inventory storage.
 - a) Inventory Management
 - b) Warehouse management
 - c) Stock Management
 - d) Supply Management
 - e) None of the above
- 2) What is the full form of DTC?
 - a) Direct-To-Consumer
 - b) Detail-To Consumer
 - c) Direct-To-Consumption
 - d) Demand-To-Consume
 - e) None of the above
- 3) _____ is responsible for planning, implementing, and controlling the effective movement of storage, goods, services, and related information from the point of origin to the site of consumption.
 - a) Consumer Chain Management
 - b) Supply Chain Management

Notes

- c) Supply chain Management
 - d) Project Management
 - e) None of the above
- 4) _____ is concerned with combining the flow of information and its management tools, materials handling, production packing, inventory, transportation, warehousing, and, in certain cases, security.
- a) Project Management
 - b) Process Management
 - c) Lead Time Management
 - d) Logistics management
 - e) None of the above
- 5) Logistics management is a component of _____ that is used to satisfy customer expectations through the effective transportation and storage of connected information, commodities, and services from origin to destination.
- a) Supply Chain Management
 - b) Inadequate Order Management
 - c) Strategic Cost Management
 - d) Sensible Stock Turnover Rate
 - e) None of the above

Summary

- Logistics management is a component of supply chain management that is used to satisfy customer expectations through the effective transportation and storage of connected information, commodities, and services from origin to destination.
- Logistics management is responsible for the management of resources ranging from physical products (such as materials, equipment, and supplies) to food or other consumable things.
- Logistics management is concerned with combining the flow of information and its management tools, materials handling, production packing, inventory, transportation, warehousing, and, in certain cases, security.
- Logistics management contributes to increased income. It enhances customer service and contributes to the company's strong reputation and brand, resulting in new and increased business.
- Operational warehouse management systems help to track where items are located, when they arrived, and how long they have been there.

Activity

1. Write a logistics management process of any organization and find out how its Logistics performance affects its supply chain and customer satisfaction.

Questions & Exercises

1. What is logistics Management? Write down its importance and how to make logistics management effective for a business?

Glossary

- Logistics management assists businesses in lowering costs and improving customer service. The logistics management process begins with the gathering of raw materials and ends with the delivery of items to their destination.
- Supply management entails planning, buying, and arranging items that are required at a specific time and location for the completion of a work.
- Fulfilment centres are warehouses that keep goods close to the end customer, so that each order is chosen, packed, and dispatched as soon as it is placed to ensure the quickest possible delivery.
- Direct-to-consumer (DTC) firms rely on ecommerce fulfilment services provided by a third-party logistics (3PL) provider.
- Warehouse management consists of numerous critical components, including the safety and security of stored items, inventory control methods that aid in the optimization of inventory storage, and much more.

Further Readings

1. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K. SET, B. ED Assistant professor
2. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding – Answers

- 1) b)
- 2) a)
- 3) c)
- 4) d)
- 5) a)

Notes

Notes**Unit - 5.4: Foundations of Quality****Objectives**

At the end of this unit, you will be able to:

- Learn about foundations of quality
- Learn about introduction to TQM
- Understand concepts of statistical quality control
- Explain X (Bar), R chart; C chart, P chart
- Understand introduction of acceptance sampling
- Learn concepts of Kaizen and 5S
- Understand ISO standards
- Understand 6-sigma
- Learn about lean manufacturing
- Identify total productive maintenance

Introduction

Quality is the ongoing process of building and sustaining relationships by assessing, anticipating, and fulfilling stated and implied needs. Quality means “Performance upon expectations” and “fit for functions.” A product is said to be of good quality if it satisfies the customer requirements in terms of performance, grade, durability, appearance and intended use/purpose, etc.

Quality Assurance

QA contains all those planned and systematic actions necessary to supply adequate confidence that a product or service will satisfy/fulfil given requirements for Quality. QA includes QC, but it emphasizes quality in the design of products and processes.

Quality Control

QC is the process through which we measure the actual quality performance and compare it with the set standards. And if there is a deviation then we take corrective actions. The typically quality control program is based on the periodic inspection, followed by feedback of the results and change or adjustment whenever required.

Quality System

A Quality System is the company-wide and plant-wide operating work structure, responsibilities, procedures, and processes for guiding the coordinated actions of the workforce, the machines, and the information of the organization in the best and most practical ways to assure customer satisfaction. Quality Assurance and Quality Control both are part of the quality system.

5.4.1 Foundation of Quality

- The Foundations of Quality, Deming

Many individuals today still believe that quality is about 'incremental improvement' rather than innovation and rapid dynamic change, and those who focus on six sigma, in particular, believe that it is exclusively about 'defect reduction/elimination.' It's always a good idea to go back to the source, and for quality, that means Deming. Consider these factors when running your business. Deming believed that quality begins with top management and cannot be outsourced.

He believed that loyal clients were insufficient since they did not develop or generate anything and just knew what you led them to anticipate. We must question ourselves, "What are we in business for?" Is it, for example, constructing carburetors or providing fuel and air in a combustion chamber? What will be of more assistance to our customers? What will we be doing in 5-10 years? Our duty is to meet needs, not to provide a product or service.

He mentioned the buggy whip producer. If they had merely used incremental progress, they would have made the greatest whip in the world for the next 100 years. But how is the market for whips these days? It is not enough to focus on small improvements; we must also consider the big picture. It is the interdependence or links that are crucial - the system as a whole. Synergy, synergy, holistic thinking, and collaboration are central to the system. In terms of defect reduction, the absence of faults will not keep a firm running; quality is about much more than that.

In terms of culture, in order to have joy on the work, we must eliminate fear and worry.

Here are Deming's 14 points:

- 1) Create consistency of purpose toward product and service improvement in order to become competitive, stay in company, and offer jobs.
- 2) Adhere to the new philosophy. We are living in a new economic era. Western management must face the challenge, learn their duties, and take on change leadership.
- 3) Reduce reliance on inspection to attain quality. Eliminate the need for mass inspections by designing quality into the product from the start.
- 4) Put an end to the practise of awarding business based on pricing. Instead, aim to reduce total cost. Move toward a single supplier for any one item, based on a long-term loyalty and trust relationship.
- 5) Consistently and indefinitely enhance the production and service systems in order to increase quality and productivity and, as a result, constantly reduce costs.
- 6) Implement on-the-job training.
- 7) Institute management. The goal of supervision should be to assist people, machines, and devices in doing better. Management supervision, like manufacturing worker supervision, has to be revamped.

Notes

- 8) Remove fear from the workplace so that everyone can work efficiently for the organisation.
- 9) Break through departmental borders. People in research, design, sales, and production must work together to anticipate production and use difficulties that may arise with the product or service.
- 10) Remove slogans, exhortations, and objectives for the workforce that call for zero faults and increased production. Such exhortations merely foster antagonistic relationships, because the majority of the causes of low quality and low productivity are systemic and hence outside the control of the workforce.
- 11) Remove impediments that deny the hourly worker the right to be proud of his or her job. Supervisors' responsibilities must shift from raw quantity to quality.
- 12) Remove impediments that deny employees in management and engineering the right to be proud of their job. This includes, among other things, the abolition of the yearly or merit rating, as well as management by aim.
- 13) Implement a robust educational and self-improvement programme.
- 14) Put everyone in the firm to work on the transition. The transformation is everybody's job.

<https://www.probuilder.com/blog/foundations-quality-deming>

5.4.2 Introduction to TQM

Quality

What is quality? 'Is this a quality watch?' Pointing to my wrist, I ask this question of a class of students – undergraduates, postgraduates, experienced managers – it matters not who. The answers vary:

- 'No, it's made in Japan.'
- 'No, it's cheap.'
- 'No, the face is scratched'
- 'How reliable is it?'
- 'I wouldn't wear it.'

My watch has been mocked in cities throughout the world, including London, New York, Paris, Sydney, Brussels, Amsterdam, and Leeds. Clearly, the quality of a watch is determined by what the user expects from it – possibly a piece of jewellery to give the appearance of riches; a timepiece that provides the needed data, including the date, in digital form; or one that can work at depths of 50 metres? The quality is determined by these standards.

Quality is frequently used to denote a product's or service's 'excellence' — people talk about 'Rolls Royce quality' and 'top quality.' In some manufacturing firms, the term may be used to signify that a piece of material or equipment adheres to certain physical dimensions qualities, which are sometimes specified in the form of a very 'tight' specification. In a hospital, it might be used to denote 'professionalism.' If we are to define quality in a way that is beneficial in its management, we must acknowledge

the necessity to integrate the genuine requirements of the 'customer' - the needs and expectations – in the evaluation of quality.

Quality is simply addressing the needs of the consumer, as described by different writers in a variety of ways:

- 'Fitness for purpose or use' – Juran, an early doyen of quality management.
- 'Quality should be aimed at the needs of the consumer, present and future' – Deming, another early doyen of quality management.
- 'The total composite product and service characteristics of marketing, engineering, manufacture and maintenance through which the product and service in use will meet the expectation by the customer' – Feigenbaum, the first man to publicize a book with 'Total Quality' in the title.
- 'Conformance to requirements' – Crosby, an American consultant famous in the 1980s.
- 'Degree to which a set of inherent characteristics fulfils requirements' – ISO (EN) 9000:2000 Quality Management Systems – fundamentals and vocabulary.

Another term that has to be defined properly is dependability. 'What makes you purchase a BMW?' The solution is 'quality and dependability.' The two terms are frequently used interchangeably, often in very different contexts. Clearly, part of a product's or service's acceptability is dependent on its capacity to work properly over time, and it is this component of performance that is referred to as dependability.

It is the product's or service's ability to continue to fulfil the needs of the client. Because it is a major consideration in many purchase choices where alternatives are being explored, reliability ranks alongside quality in importance. Many of the typical management difficulties that arise when attempting to achieve product or service quality also apply to dependability.

It is critical to understand that the concept of quality as "filling the client expectations" is not limited to the functional qualities of products or services. Anyone who has children understands that the quality of some of the things they buy is more related with ownership satisfaction than with some utilitarian attribute. This is also true for a wide range of objects, from antiques to specific pieces of apparel. The sale of some executive vehicles, some bank accounts and charge cards, and even hospital beds is accounted for by the demand for status symbols! In determining the quality of any product or service, the requirements are critical.

By continually addressing client needs, we may reach a higher level of satisfaction — delighting the customer. There is no question that many firms have so thoroughly organised their competence to satisfy their customers' objectives that they have earned a reputation for 'excellence.' Customer loyalty is a result of this way of thinking about customers and their happiness, and it is a crucial factor in an organization's success. According to research, focusing on client loyalty can give various commercial benefits:

- Retaining customers is less expensive than acquiring new ones.
- The longer the customer connection, the better the profitability.
- A committed consumer will spend more money with its preferred supplier.

Notes

- Approximately half of new consumers are referred by current customers (indirectly reducing acquisition costs).

Companies such as 3M utilise customer loyalty measurements to identify customers who are 'totally happy,' would 'absolutely refer,' and would 'certainly repurchase.'

- Quality Management

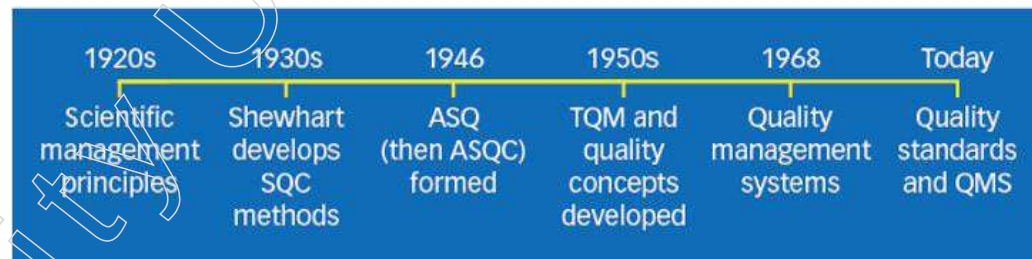
Total quality management (TQM) is the ongoing process of identifying and reducing or eliminating production defects, optimising supply chain management, enhancing the customer experience, and ensuring that staff are properly trained. Total quality management seeks to make all stakeholders engaged in the manufacturing process responsible for the overall quality of the finished product or service.

William Deming, a management consultant whose work had a significant influence on Japanese industry, established TQM. While TQM has many similarities to the Six Sigma improvement approach, it is not the same. TQM is concerned with ensuring that internal rules and process standards eliminate mistakes, whereas Six Sigma is concerned with reducing defects.

- History of Total Quality Management

Total quality management (TQM) emerged as a term developed by the Naval Air Systems Command to define its Japanese-style quality improvement management methodology. It is an umbrella technique for continuously improving the quality of all processes, and it is based on understanding of the ideas and practises of:

- The behavioural sciences
- The analysis of quantitative and nonquantitative data
- Economics theories
- Process analysis



History of Total Quality Management (TQM)

1920s	As the ideas of quality management swept across American business, some of the earliest seeds of quality management were sown.
	Businesses clearly separated the processes of planning and carrying out the plan, and union resistance grew as workers were denied a say in the circumstances and functions of their job.
	The Hawthorne trials in the late 1920s demonstrated how involvement may affect worker productivity.
	Walter Shewhart created the methodologies for statistical analysis and quality control.

Notes

1950s	W. Edwards Deming taught Japanese engineers and executives' statistical analysis and quality control methodologies. TQM can be traced back to this point.
	Joseph M. Juran taught the ideas of quality control and management breakthrough.
	Armand V. Feigenbaum's book Total Quality Control, a forerunner for the present understanding of TQM, was published.
	Philip B. Crosby's promotion of zero defects paved the way for quality improvement in many companies.
1968	The Japanese dubbed their entire quality method "companywide quality control." Around this period, the phrase quality management systems are coined.
	Kaoru Ishikawa's philosophical synthesis aided Japan's rise as a quality leader.
Today	Total Quality Management (TQM) is the name given to the notion of a wide and holistic approach to managing organisational quality.
	TQM concepts and methods are specified by quality standards such as the ISO 9000 series and quality award programmes such as the Deming Prize and the Malcolm Baldrige National Quality Award.
	As worldwide quality management standards have been created, the word "TQM" as a phrase to define an organization's quality policy and practise has gone out of favour. For additional information, please read our quality management system pages.

- Key Principles of TQM



Total quality management is comprised of a number of essential principles that, when used collectively, may propel any firm toward business success.

Notes

1) Customer Focus

The concept that quality is established by the customer is central to all effective TQM systems. Customer feedback, whether in the form of reviews, return rates, or satisfaction surveys, is the only way to know if the efforts you implement to improve the quality of your products and services have been effective.

2) Involvement

Everyone in a business, from entry-level employees to management, is responsible for the quality of its goods and services. Employees, on the other hand, can only be invested if they feel empowered to make their own decisions, which is dependent on management providing the ideal workplace environment.

3) Centred on Process

Without a strong focus on processes and process-led thinking, a TQM system will fail. Most problems are ultimately caused by a process flaw, which is why efficient monitoring of each and every stage is a crucial aspect of assessing, maintaining, and improving quality.

4) Integrated System

A company should have an integrated system in place that enables successful comprehensive quality management. This might be a custom-built system or one based on a quality standard like ISO 9001, but it must be understood and implemented across all activities and departments.

5) Strategic and Systematic Approach

Critical to quality management is the existence of a strategic plan that outlines how an organization intends to achieve its mission and business goals. It goes without saying that quality should be a core component of such a plan.

6) Decision-making Based on Facts

Only accessible statistics, such as sales numbers, income figures, and customer retention rates, may be used to evaluate business performance. Customers', employers', and suppliers' views should never be utilised to make choices.

7) Communication

When an organisation implements big changes for the sake of business improvement, effective communication is critical. Every employee should be informed on the plan, the timeframes involved, and the reasons for adopting it.

8) Continuous Improvement

Using DMAIC and Lean Six Sigma concepts will establish a culture of continuous improvement in a firm, motivating all workers to always explore new methods to be more competitive and offer high-quality goods to all stakeholders.

- Objectives of Total Quality Management

Total quality management ensures that every single employee is working towards the improvement of work culture, processes, services, systems and so on to ensure long term success.

Total Quality management can be divided into four categories:

- Plan
- Do
- Check
- Act

Also referred to as PDCA cycle.

Planning Phase

The most important aspect of complete quality management is planning. Employees must come up with concerns and questions that need to be addressed at this phase. They must identify the many problems they experience in their daily operations as well as the fundamental cause of the problem. Employees must do essential research and collect relevant data to assist them discover answers to all difficulties.

Doing Phase

Employees provide solutions to the challenges described in the planning phase during the doing phase. Employee difficulties are met by strategies that are designed and implemented. This stage also assesses the efficacy of solutions and initiatives.

Checking Phase

The checking step is where individuals actually compare before and after data to confirm the effectiveness of the procedures and assess the results.

Acting Phase

Employees document their findings and plan to handle further issues during this phase.

- Benefits of TQM

The benefits of TQM include:

- Less Product Defects. One of the TQM concepts is that goods and services be created correctly the first time. This implies that fewer problems are shipped with items, resulting in fewer product recalls, future customer support costs, and product repairs.
- Satisfied Customers. High-quality products that satisfy the demands of customers result in increased customer satisfaction. Increased market share, revenue growth through upsells, and word-of-mouth marketing launched by customers can all result from high customer satisfaction.
- Lower Costs. Companies save money on customer assistance, product replacements, field service, and the development of product fixes as a

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consequence of fewer product failures. The cost savings are passed on to the bottom line, resulting in improved profit margins.

- Well-defined Cultural Values. TQM-practicing organisations establish and foster fundamental principles centred on quality management and continuous improvement. TQM penetrates all elements of a business, from hiring to internal operations to product creation.
- TQM Implementation and Systems

There is no single approach for every circumstance or workplace when it comes to creating and executing a whole quality management system or quality management strategy.

Each business has its own culture, management techniques, and methods for creating and delivering products and services. Quality management strategies differ from one another; nonetheless, a set of essential elements should be included in some manner.

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- General Strategy Model for Implementing TQM System
 1. Top management becomes aware of TQM and decides to commit to it. TQM is identified as a strategy for the organisation.
 2. The company evaluates its present culture, customer satisfaction, and quality management methods.
 3. Top management identifies and conveys the basic values and concepts to be applied.
 4. Steps 1, 2, and 3 are used to create a TQM master plan.
 5. The company analyses and prioritises client wants, then aligns goods and services to satisfy those demands.
 6. Management diagrams the main processes through which the firm serves the demands of its consumers.
 7. Management is in charge of forming teams for process improvement activities.
 8. The steering committee manages the momentum of the TQM endeavour.
 9. Individual managers contribute to the endeavour through hoshin planning, training, mentoring, or other means.
 10. Daily process management and standardisation occur.
 11. Progress is assessed, and the strategy is changed as necessary.
 12. Continuous employee awareness and status feedback are offered, and a reward/recognition procedure is formed.

5.4.3 Concepts of Statistical Quality Control

- Statistical Quality Control

Controlling and enhancing quality is an essential strategy for a firm since it gives

a competitive edge. The firm that delights its consumers by enhancing and regulating quality will be able to outperform its competition. As a result, the primary goal of a corporation is to accomplish quality assurance so that it may not only satisfy but also excite its clients.

With the arrival of the industrial revolution in the nineteenth century, mass production supplanted small-scale production by trained craftsmen and artisans. While in small businesses, each individual worker was totally accountable for the quality of his or her job, this was no longer the case in mass production, because each individual worker's contribution to the completed product was only a little part of the overall process. Quality control was performed in a huge company with the assistance of quality inspectors who were in charge of checking the product quality with a 100% inspection of all critical qualities.

Many statistical approaches have been developed that are effective for controlling the quality of items in relation to specifications or standards. The technique of utilising statistical methods to control product quality versus requirements is known as 'statistical quality control' (SQC). SQC should be considered as a set of methodologies that can affect decisions in the specification, manufacturing, and inspection functions.

SQC is described as the approach of utilising statistical methods based on probability theory and sampling to set and maintain quality standards in the most cost-effective manner. It is a cost-effective and efficient approach for maintaining and enhancing output quality throughout the specification, production, and inspection processes, based on continuous testing using random samples.

Quality control encompasses the quality of the product or service provided to the client, leadership, management commitment, continuous improvement, rapid reaction, actions based on facts, employee engagement, and a quality-driven culture. SQC's goal is to control (i) material receipt, internal rejections, and client claims, (ii) evaluations of the same, and (iii) remedial measures and their follow-up. All quality operations are guided by these systems and procedures. The creation and application of performance indicators is inextricably tied to customer requirements and satisfaction, as well as management.

SQC is one of the scientific management strategies. The statistical quality control principle applies to practically all elements of manufacturing, including material quality, labour quality, machine quality, and management quality.

SQC refers to the application of statistical methods in the monitoring and maintenance of product and service quality. It is the phrase used to define the collection of statistical approaches used by quality experts to assess quality. SQC has three terms, which are detailed below.

Statistics - It is a branch of science concerned with the gathering, categorization, analysis, and inference of data or information. It denotes the availability of a sufficient amount of data to achieve accurate results. Statistics manipulates data in order to derive conclusions using a variety of methodologies.

Quality - Quality is a relative concept that is usually discussed in relation to the final usage of the product. As a result, quality is defined as fitting for purpose.

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Control – A control system is a method of measuring, checking, or examining a phenomenon. It recommends when to inspect, how often to inspect, and how much to inspect. Control determines the quality features of an item, compares them to the required quality standards, and distinguishes faulty items from non-defective ones.

The data for SQC can be collected in several ways namely (i) direct observation, and (ii) indirect observations. Data on quality characteristics are categorized as (i) continuous variable data which can assume any number on continuous scale within a range, and (ii) discrete attributes which are the count of an event and which assume a finite or countable infinite number of values.

Dr. Walter A. Shewhart, a researcher at Bell Telephone Laboratories (an American Telephone and Telegraph Company Research and Development Unit), is known as the “Father of Quality Control Analysis.” He created the SQC ideas. In 1924, he created charting tools and statistical processes for regulating in-process manufacturing activities to regulate quality. His statistical approaches are based on the idea of randomly distributed random variables that are independent and identically distributed. Based on these notions, he distinguished between chance causes, which produce random variation inherent in the process, and assignable causes, which must be identified and corrected. He suggested 3 sigma (standard deviation) boundaries above and below the variable’s mean value to discover assignable causes. Observations outside of these parameters were then utilised to determine the assignable causes. This is commonly seen as the start of SQC.

Variation is a natural occurrence, and industrial processes are no exception. Variation can occur (i) inside the product, (ii) between items manufactured at the same time, and (iii) between products manufactured at different times. The probability of an event describes the likelihood of that event occurring. The theory of probability serves as the foundation of SQC.

The size of components created on the same machine and in the same batch (if measured precisely) vary from component to component, according to probability theories. This might be due to the intrinsic qualities of the equipment or the external circumstances. The likelihood or condition that a sample reflects the full batch or population is derived from probability theory. SQC measures batch quality and manages the quality of processes and products using probability theory.

SQC employs three scientific techniques: (i) sampling inspection, (ii) data analysis, and (iii) control charting. It is classified into three major categories: (i) descriptive statistics, (ii) statistical process control, and (iii) acceptance sampling.

Descriptive statistics are those that are used to characterise specific quality aspects of observed data, such as central tendency and variability. It’s also a description of the connection. Statistics such as the mean, standard deviation, range, and a measure of data distribution are examples of descriptive statistics.

Statistical process control (SPC) refers to statistical procedures that entail analysing a random sample of a process’s output and determining whether or not the process produces products with characteristics that fit within a predefined range. SPC provides a response to the issue of whether or not the process is working well. These strategies are critical for a process since they aid in recognising and correcting quality issues during the manufacturing process.

Acceptance sampling is the sort of inspection process used when inspecting a product with the goal of accepting or rejecting it based on conformance to a standard. Acceptance sampling aids in determining whether there is a quality problem and whether desired quality is being attained for a batch of goods. Acceptance sampling is the process of assessing a random sample of items and choosing whether or not to accept the full lot depending on the results. This sample is used to determine if a batch of items should be approved or rejected.

Terms Used in SQC

The following are some of the terms used in the SQC technique.

Mean – It is a useful statistical technique for determining the central tendency of a set of data. The mean is calculated by adding all of the data together and dividing by the total number of observations.

Range and Standard Deviation – The variability of the data is provided by this information. It describes how the data is distributed around the mean. The range of a collection of data is the distance between the greatest and smallest observation, whereas standard deviation is a statistic that indicates the degree of data dispersion around the mean. Small range and standard deviation values indicate that the data are closely grouped around the mean, whereas big range and standard deviation values indicate that the observations are scattered about the mean.

Distribution of Data – It is a metric for determining quality qualities. When the data distribution is symmetric, the number of observations below and above the mean is the same. When just normal variance is present in the data, this is what is generally discovered. A skewed distribution occurs when a disproportionate number of observations are either above or below the mean.

Population – The population is the collection of all objects that share a common trait of interest. A parameter is a property of a population that describes it. A sample is a subset of a larger population.

Statistic – It is a sample feature that is used to draw conclusions about population characteristics that are generally unknown.

Accuracy and Precision – Precision is the degree of uniformity of observations around a desired value, whereas accuracy is the degree of uniformity of observations around a desired value. The degree of variability in the observations is referred to as precision.

- **Basic SQC Techniques**

Quality control employees can use a variety of diagnostic procedures to investigate quality issues. SQC employs seven fundamental approaches. These fundamental approaches include (i) check sheets, (ii) histograms, (iii) Pareto analysis, (iv) control chart, (v) cause and effect diagram, (vi) stratification, and (vii) scatter diagram.

The term “seven fundamental methods of quality” refers to a predetermined collection of graphical approaches regarded as being most useful in troubleshooting quality concerns. These strategies are referred to be basic since they are appropriate for those with minimal formal experience in statistics and may be used to tackle the vast majority of quality-related concerns.

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The seven fundamental SQC issue solving strategies are commonly used to discover improvement opportunities and to aid in the reduction of unpredictability and waste. They may be utilised in a variety of ways throughout the DMAIC (define, measure, analyse, improve, and control) problem-solving process, which is a data-driven quality method for process improvement.

1) Check Sheets

It is usually important in the early phases of process improvement to collect either historical or current operational data regarding the process under consideration. This is a popular action in the DMAIC measure stage. In this data gathering task, a check sheet might be quite beneficial. An aerospace firm engineer created the check sheet while analysing issues on one of the business's tanks. The check sheet was created by the engineer to assist summarise all of the historical defect data available on the tanks. Because just a few tanks were built each month, it seemed reasonable to summarise the data monthly and identify as many different sorts of problems as feasible. The time-oriented summary is very useful for identifying trends or other relevant patterns.

It is critical to explicitly describe the type of data to be gathered, the part or operation number, the date, the analyst, and any other information relevant in determining the reason of poor performance when constructing a check sheet. If the check sheet is used as the foundation for additional computations or as a worksheet for data entry into a computer, it is critical to ensure that it is suitable for this purpose. A trial run to test the check sheet layout and design might be useful in some instances.

Check sheets are simple data collection tools. They are used to collect data in an effective and efficient manner. They are in charge of preparing data for future analysis. A check sheet is a predefined format that is used to gather data in a systematic and real-time way at the site where the data is created. It can collect either quantitative or qualitative data. When the information is numerical, the check sheet is frequently referred to as a tally sheet. A check sheet is distinguished by the fact that data is recorded by making markings (checks) on it. A standard check sheet is divided into regions, and the importance of marks made in different parts varies. The position and number of markings on the sheet are used to read data. Various types of check sheets are (i) process distribution check sheet, (ii) defect cause check sheet, (iii) defect location check sheet, and (iv) defect cause-wise check sheet.

2) Histogram

A histogram is a graphical depiction of data distribution (a bar chart). Karl Pearson developed it as an estimate of the probability distribution of a continuous quantity. A histogram is a depiction of tabulated frequencies that is displayed as contiguous rectangles erected at defined intervals (bins) with an area equal to the frequency of the observations in the interval. The frequency density of the interval, i.e. the frequency divided by the width of the interval, is also equal to the height of a rectangle. The whole area of the histogram equals the total number of data points.

A normalised histogram can also be used to represent relative frequencies. The proportion of instances that fit into each of various categories is then shown, with the total area equaling one. Normally, the categories are given as non-overlapping, sequential intervals of a variable. The categories (intervals) must be contiguous and,

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in most cases, of the same size. A histogram's rectangles are constructed in such a way that they contact each other to demonstrate that the original variable is continuous. Histograms are commonly used to visualise data density and for density estimation, which is the estimate of the probability density function of the underlying variable. The total area of a histogram used for probability density is always normalized to one.

A histogram is a graphical representation of data. To create a histogram for continuous data, divide the data range into intervals, which are commonly referred to as class intervals, cells, or bins. To optimise the visual information in the histogram, the bins should be of identical width if feasible. To create a good presentation, the number of bins must be chosen with some discretion. The number of bins is determined by the number of observations as well as the degree of scatter or dispersion in the data. A histogram that has too few or too many bins is uninformative. In most circumstances, 5 to 20 bins are deemed sufficient, and the number of bins grows with the number of observations. In practise, selecting a number of bins that is nearly equal to the square root of the number of observations works well.

After determining the number of bins and the bottom and upper boundaries of each bin, the data is sorted into the bins and the number of observations in each bin is counted. The horizontal axis is used to show the measuring scale for the data, while the vertical axis is used to represent the counts, or frequencies. The frequencies in each bin are sometimes divided by the total number of observations, and the vertical scale of the histogram shows relative frequencies. Rectangles are drawn over each bin, and their heights are proportionate to frequency (or relative frequency).

Histograms are used to analyse the pattern of variation in a measured attribute with regard to location and spread. They provide information on the setup of a process as well as its variability. Histograms show the process's capacity to satisfy the requirements as well as the level of the process's nonconformance. Figure 1 depicts many histogram patterns.

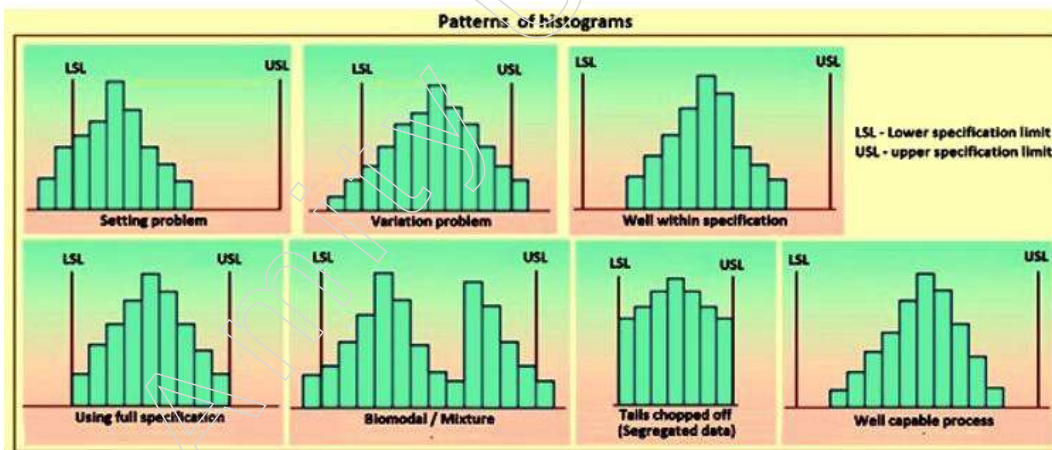


Fig 1 Patterns of histogram

3) Pareto Chart

In general, the Pareto chart is one of the most helpful SQC approaches. Its applicability to quality improvement are solely limited by the analyst's imagination. The

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Pareto chart is nothing more than a frequency distribution (or histogram) of attribute data organised by category. Pareto charts are widely used in DMAIC's measure and analyse stages.

People would want to concentrate their efforts on resolving the most pressing issue. The Pareto chart is a simple and formal approach for identifying the top percentage of the reasons that must be addressed in order to solve the majority of issues. It is a decision-making strategy that statistically distinguishes a small set of input parameters as having the most effect on a desired or unwanted outcome. In its most basic form, a Pareto chart indicates that a disproportionate improvement may be obtained by ranking numerous sources of a problem and focusing on those solutions or things having the greatest influence. The fundamental assumption is that not all inputs have the same, or even proportional, influence on a particular outcome.

The Pareto chart is also known as the '80/20 rule.' According to this concept, it is thought that addressing 20% of the causes yields 80% of the results. The Pareto chart approach is used to identify the 20% of those causes that, when addressed, fixes 80% of the issues. This ratio is only a useful rule of thumb and should not be regarded as a natural law.

4) Control Chart

A control chart is a graph that displays whether a sample of data falls within the normal or common range of variance. The upper control limit (UCL) and lower control limit (LCL) of a control chart differentiate common from assignable sources of variation. The UCL is the maximum acceptable variance from the mean for a controllable process, whereas the LCL is the smallest acceptable variation from the mean for a controllable process. Control chart limits are used to establish the common range of variance. Control limits are derived from process output data and are not specification limitations. When a data plot demonstrates that one or more samples are outside the established control boundaries, the process is out of control. As a result, by utilising control charts, one can see graphically how production is progressing and where corrective action is and is not required.

The control chart is a highly effective approach for process monitoring. When there are exceptional causes of variability, sample averages are shown outside of the control bounds. This indicates that the process will be investigated and remedial action will be done to eliminate these unique causes of unpredictability. The systematic application of a control chart is a great method for reducing variability.

Walter A. Shewhart presented the basic theory of control charts, and control charts constructed in accordance with these principles are commonly referred to as Shewhart control charts. The control chart is a graphical representation of a quality parameter measured or computed from a sample versus the sample number or time. The figure includes a centre line that reflects the average value of the quality attribute associated with the in-control condition. That is, only random causes exist. The figure also shows two more horizontal lines, the UCL and the LCL.

Figure 3 depicts a basic control chart. This graph depicts the averages of measurements of a quality attribute in process samples vs time (or the sample number). The chart includes a centre line (CL) as well as upper and lower control boundaries. If

there are no unexpected causes of variability, the centre line depicts where this process characteristic should lie. Simple statistical considerations are used to calculate the control limits.

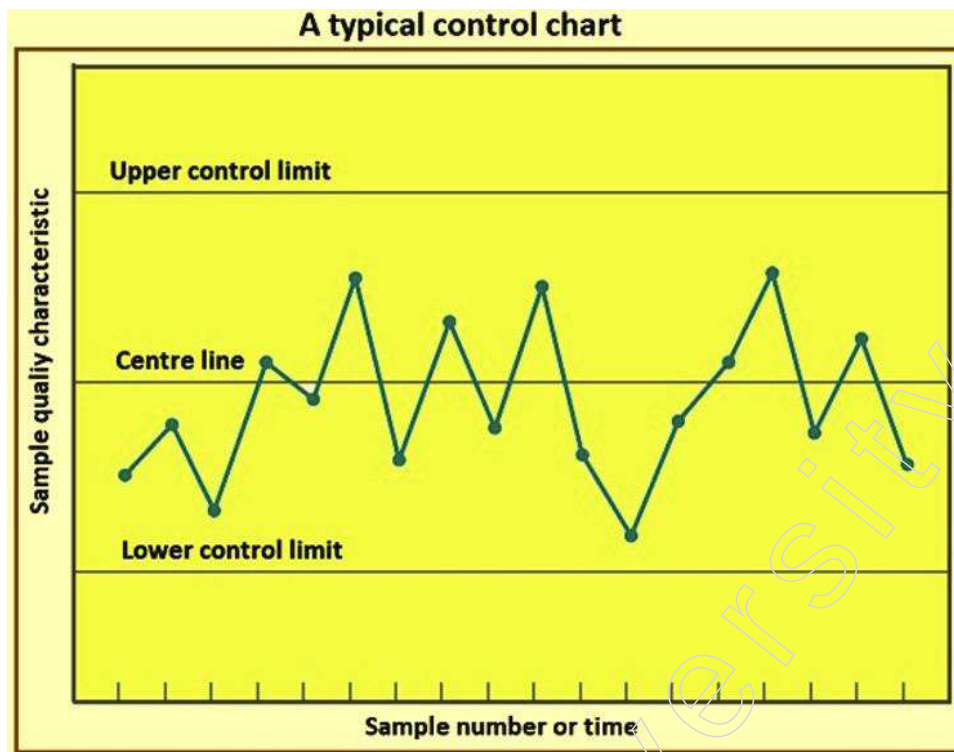


Fig 3 A typical control chart

The control limits are selected so that virtually all of the sample points fall between them if the process is under control. The process is deemed to be in control as long as the points plot inside the control boundaries, and no action is required. A point plotting outside of the control boundaries, on the other hand, is read as evidence that the process is out of control, and more study and corrective action are required to identify and remove the assignable cause or reasons responsible for this behaviour. It is common practise to connect the sample points on the control chart with straight-line segments to make it simpler to see how the point sequence has grown over time.

Various advantages of using the control charts are given below.

- Control charts are a tried-and-true method for increasing production. A effective control chart programme decreases scrap and rework, which are the key productivity killers in any manufacturing process. When scrap and rework are eliminated, productivity rises, costs fall, and production capacity rises.
- Control charts are useful for fault avoidance. The control chart aids in the control of the process, which is congruent with the 'do it correctly the first time' principle. It is never cheaper to separate 'good' from 'poor' items later than it is to develop it correctly the first time. People are paying someone to manufacture a non-conforming product if they do not have good process control.
- Control charts eliminate the need for needless process adjustments. A control chart helps tell the difference between natural fluctuation and background noise. No other instrument, including a human operator, is as effective at distinguishing

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between the two. When process operators make modifications based on periodic tests that are unconnected to a control chart programme, they usually overreact to background noise and make unnecessary changes. Unnecessary modifications might significantly degrade process performance. In other words, the control chart adheres to the adage “if it is not broken, don’t repair it.”

- Control charts give diagnostic data. Often, the arrangement of points on the control chart offers diagnostic information for an experienced operator or engineer. This information enables the process to be changed in order to increase its performance.
- Control charts give information on the capabilities of the process. It gives information on the value of key process parameters as well as their stability over time. This provides for an estimation of process capabilities. This data is extremely useful to product and process designers.

Control charts are one of the most essential methods of management control. They are just as vital as cost and material controls. Because data collecting and analysis may be conducted in real-time, on-line at the work centre on a microcomputer or a local area network terminal, modern computer technology has made it simple to integrate control charts in any sort of process.

5) Stratification

Stratification is a strategy that is used in conjunction with other data analysis methods. When data from many sources or categories is combined, the meaning of the data might get obscured. The approach isolates the data in order to reveal the pattern. Stratification can be used i) before collecting data, (ii) when data comes from multiple sources or conditions such as shifts, days of the week, suppliers, materials, products, departments, equipment, or population groups, and (iii) when data analysis requires separating data from multiple sources or conditions. The stratification technique consists of the three phases listed below.

- Before gathering data, examine which information about the data’s sources may have an impact on the outcomes. Data pertaining to this information must also be acquired during data collection.
- Different markers or colours must be used to identify between data from different sources when plotting or graphing the collected data on a scatter diagram, control chart, histogram, or other analytic approach. Data that is identified in this way is referred to as ‘stratified.’
- The stratified data subsets are evaluated independently.

An example of data stratification is given in Fig 6.

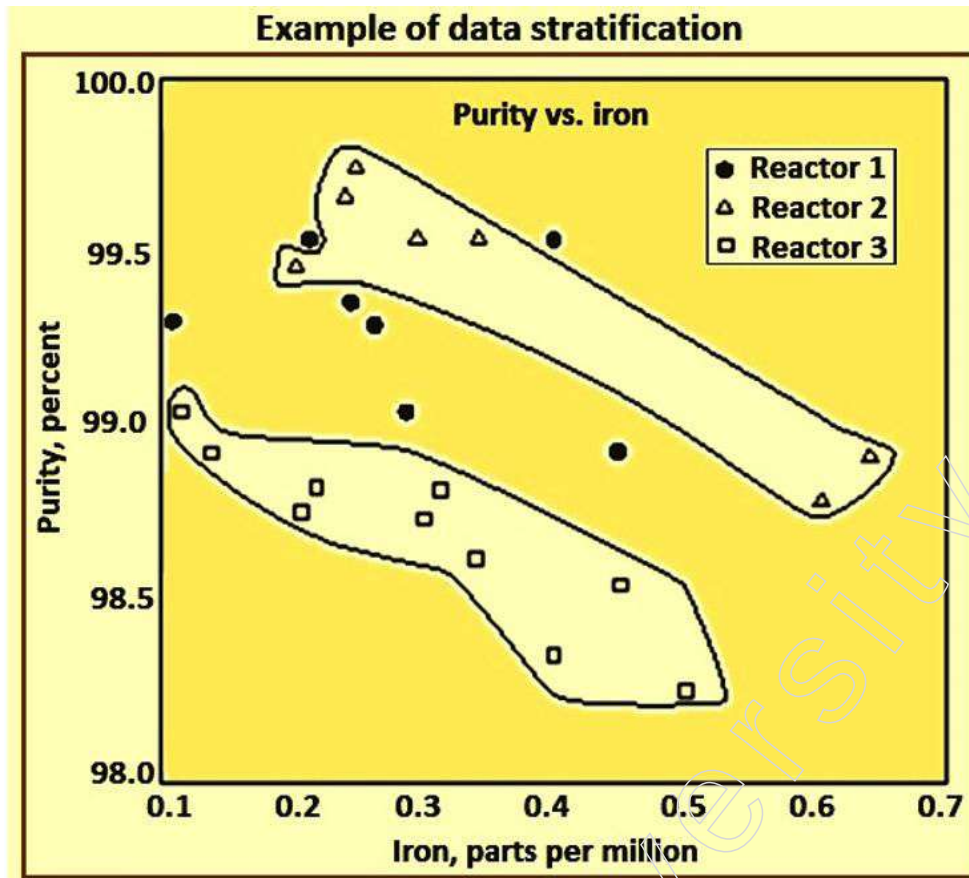


Fig 6 Example of data stratification

6) Scatter Diagram

The scatter diagram is made up of graphs that are produced using pairings of numerical data, having one variable on each axis, to show the relationship between them. The points fall along a line or curve if the variables are correlated. A scatter diagram is a form of mathematical graphic that displays values for two variables in a collection of data using Cartesian coordinates. The data is represented as a collection of dots, with the value of one variable dictating the location on the horizontal axis (x axis) and the value of the other variable defining the position on the vertical axis (y axis).

When a variable exists that is outside the operator's control, a scatter diagram is employed. If there is a parameter that is regularly incremented and/or decremented by the other, it is referred to as the control parameter or independent variable and is often displayed along the x axis. Traditionally, the measured or dependent variable is plotted along the y axis. If there is no dependent variable, any kind of variable can be displayed on either axis, or a scatter diagram can be used to show merely the degree of correlation (not causation) between two variables.

The scatter diagram is an effective tool for determining a possible association between two variables. Data are collected in pairs on the two variables, say, (y_i, x_i) – for $i = 1, 2, \dots, n$. Then y_i is plotted against the corresponding x_i . The scatter diagram's form typically reveals the sort of relationship that might exist between the

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two variables. Scatter diagrams are in handy when it comes to regression modelling. Regression is a highly important approach in the DMAIC analyse stage. Figure 7 depicts typical scatter diagrams.

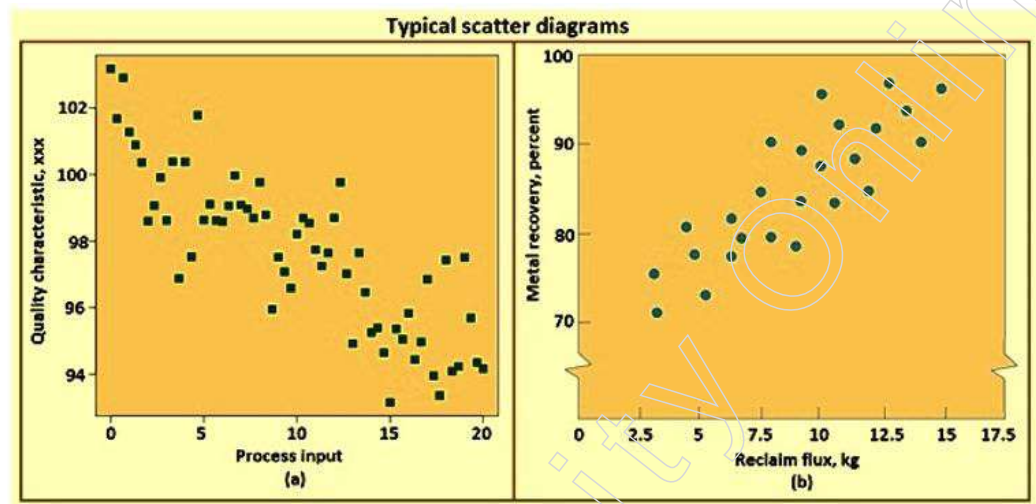


Fig 7 Typical scatter diagrams

A scatter plot linking metal recovery (in percent) from a magna-thermic smelting process for magnesium to matching levels of reclaim flux fed to the crucible is shown in Fig 7b. The scatter plot shows a significant positive connection between metal recovery and flux quantity, indicating that as the amount of flux injected increases, so does metal recovery. It's easy to think that the relationship is one of cause and effect. One may always assure good metal recovery by increasing the amount of reclamation flux employed. This kind of thinking is risky since correlation does not always imply causation.

This apparent association might be the result of something entirely else. For example, these variables can be connected to a third, such as the temperature of the metal prior to the reclamation pouring process, and this relationship can account for the results seen in Fig 7. If higher temperatures result in more metal recovery and the norm is to increase reclaim flux in proportion to temperature, then adding additional flux when the process is running at a low temperature has no effect on yield. The scatter diagram can be used to detect potential correlations. Experiments will be designed to test causality.

- Benefits of SQC

Unwanted variations in quality can be detected during the 100 percent inspection after a large number of defective items have already been produced, whereas in the SQC technique, the moment a sample point falls outside the control limits, it is regarded as a danger signal and the necessary corrective measures are implemented. There are various advantages to SQC versus 100% checks.

The advantages of SQC include: (i) it takes less time than a full inspection and hence promotes efficiency; and (ii) it is less expensive than a full inspection. (ii) it allows for the detection of errors during inspection; (iii) it leads to more consistent production quality; (iv) it enhances customer relations; and (v) it decreases inspection expenses

since only a percentage of the output is reviewed. (vi) it minimises the number of rejects and saves material costs; (vii) it offers a foundation for attainable standards; (viii) it identifies bottlenecks and difficulty spots; and (ix) it gives a way of assessing the manufacturing process's capabilities. (x) specifications can be easily predicted for the future, which is not possible even with 100 percent inspection; (xi) in cases where product destruction is required for inspection, 100 percent inspection is not possible (because it spoils all the products); sampling inspection is used; (xii) it promotes the understanding and appreciation of quality control; and (xiii) once the SQC plan is established, it is easy to apply even by man who does not have extensive experience.

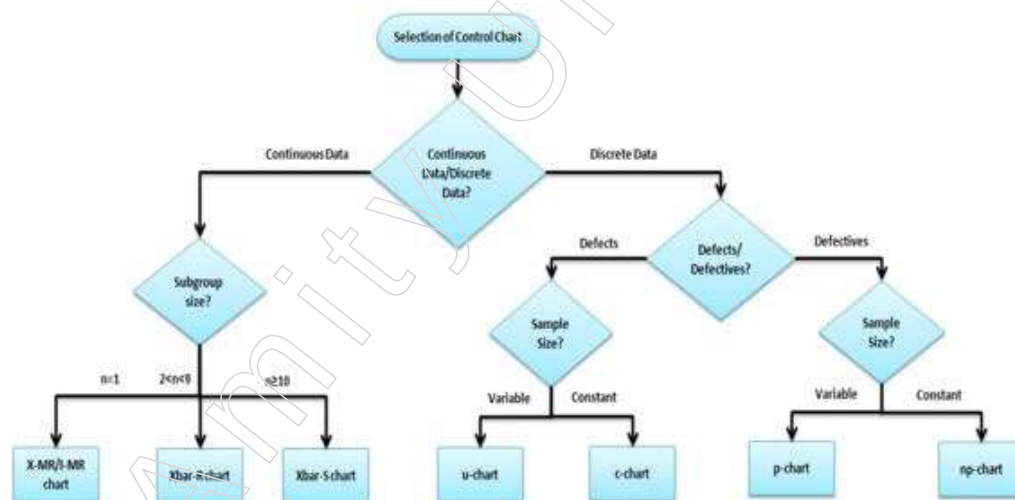
5.4.4 X (Bar), R Chart; C Chart, P Chart

- Quality Control Charts

Quality control charts are an excellent tool for engineers to use in determining whether or not a process is statistically controlled. They aid in the visualisation of variation, the detection and correction of issues, the prediction of predicted ranges of outcomes, and the analysis of patterns of process variation caused by unique or common sources. Quality control charts are one of the seven core quality tools for process improvement and are frequently used in Lean Six Sigma and DMAIC projects during the control phase.

In many businesses, X Bar R charts are the most commonly used control chart for variable data to monitor process stability (such as hospital patients' blood pressure over time, customer call handle time, length of the part in the manufacturing process, and so on).

The selection of an appropriate control chart is critical in control chart mapping; otherwise, the data will have inappropriate control limits.



The X bar R chart is used to assess the process performance of continuous data and data acquired in subgroups at predetermined time intervals. It is a two-plot example of statistical process control that monitors the process mean and process variance across time. These combination charts aid in understanding process stability and detecting the existence of special cause variation.

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The cumulative sum (CUSUM) and exponentially weighted moving average (EWMA) charts also monitor the mean of the process, but unlike the X bar chart, they take into account the previous value means at each point.

X Bar R Control Chart Definitions

X-bar Chart: The mean or average change in process from subgroup values across time. The X-control Bar's limits take the sample's mean and centre into account.

R-chart: The range of the process over the time from subgroups values. This monitors the spread of the process over the time.

Make use of X Bar R Control Charts. When:

- Even extremely stable processes can have slight deviations that create process instability. The X bar R chart will aid in identifying process variation over time.
- When it is assumed that the data is regularly distributed.
- The X bar R chart is for subgroup sizes more than one (the I-MR chart has just one subgroup size) and is often used when logically collecting measurements in subgroup sizes ranging from two to ten observations.
- When the subgroup size exceeds 10, the X Bar S Control chart should be considered
- When the acquired data is continuous (e.g., length, weight), and it is captured in temporal order.

How to Interpret the X Bar R Control Charts

- Always study the R chart first when interpreting an X bar R chart.
- The X bar chart control limits are determined from the R bar (average range) values; if the values in the R chart are out of control, the X bar chart control limits are inaccurate.
- If the points on the R chart are out of control, halt the procedure. Determine the unique reason and fix the problem. Take those subgroups out of the computations.
- Once the R bar chart is under control, examine the X bar chart and compare the points to the control boundaries.
- All points must be interpreted in relation to the control limitations but not the specification limits. Control limits are determined from the average and range values of the subgroups, whereas specification limits are given by the customer or management.
- If any point gets out of hand in the X bar chat. Determine the unique reason and fix the problem.
- Process capability studies can only be undertaken if both the X bar and the R chart values are within the control limits. It is pointless to conduct process capacity assessments on an unstable process.

Steps to Follow for X Bar R Chart

Objective of the chart and subgroup size

- Determine the objective of the chart and choose the important variables
- Choose the appropriate subgroup size and the sampling frequency
- Collect minimum 20 to 25 sets of samples in the time sequence

Example: In a manufacturing industry the plate thickness is one of the important CTQ factor, during Measure phase, project team performed the process capability study and identified that the process is not capable (less than 2 sigma). In Analyse phase collected 20 sets of plate thickness samples with a subgroup size of 4.

Sample	Measured values			
	1	2	3	4
1	44	26	24	34
2	50	48	51	43
3	32	28	26	22
4	52	55	56	44
5	16	16	21	26
6	36	36	35	31
7	21	22	18	21
8	29	21	23	22
9	26	46	44	14
10	24	22	22	44
11	18	24	24	49
12	24	20	26	23
13	19	21	27	28
14	8	11	12	12
15	24	18	27	24
16	56	52	56	50
17	32	22	18	25
18	8	12	11	17
19	51	54	52	49
20	30	28	35	22

Compute X bar and R values

- Measure the average of each subgroup i.e., X bar, then compute grand average of all X bar value, this will be center line for X bar chart
- Compute the range of each subgroup i.e., Range, then measure grand averages of all range values i.e., R bar and this will be the center line for R chart

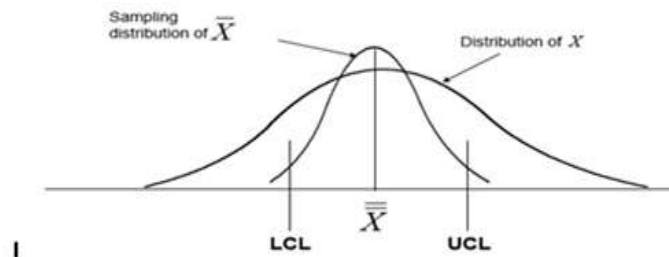
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Sample	Measured values				Sample Average ($\bar{X}$ bar)	Sample Range-R
	1	2	3	4		
1	44	26	24	34	32	20
2	50	48	51	43	48	8
3	32	28	26	22	27	10
4	52	55	56	44	51.75	12
5	16	16	21	26	19.75	10
6	36	36	35	31	34.5	5
7	21	22	18	21	20.5	4
8	29	21	23	22	23.75	8
9	26	46	44	14	32.5	32
10	24	22	22	44	28	22
11	18	24	24	49	28.75	31
12	24	20	26	23	23.25	6
13	19	21	27	28	23.75	9
14	8	11	12	12	10.75	4
15	24	18	27	24	23.25	9
16	56	52	56	50	53.5	6
17	32	22	18	25	24.25	14
18	8	12	11	17	12	9
19	51	54	52	49	51.5	5
20	30	28	35	22	28.75	13
Total					597.5	237
Average $\bar{X}$ bar					29.9	
R bar						11.85

Determine the Control Limits

- The first set of subgroups is responsible for determining the process mean and standard deviation; these data will be used to create control limits for both the range and mean of each subgroup.
- If the individual value (x) assumed to be normal then the distribution of average of reading in a sample ($\bar{X}$ -bar) will be normal.
- The standard deviation around the sample means is smaller by a factor of $\frac{1}{\sqrt{n}}$



- $\hat{\sigma}$ = estimated standard deviation of X
- $\frac{\hat{\sigma}}{\sqrt{n}}$ = estimated standard error of $\bar{X}$

- Controlling the process in the early stages of manufacturing. If any of the points

are out of control during the initial phase, special causes must be found, and the subgroup must be deleted for computation.

- It is sometimes beneficial to have a few points out of control on the x-bar segment during the early phase. Otherwise, if all of the readings are inside the control limits, it might be due to slippage in the measuring system, and the team will not focus on it. Determine the most suitable Measurement System Evaluation (MSE).

Walter Shewhart mentioned that control limits should be 3 times standard deviation from the center line in order to reduce the probability of error happening in detecting the assignable causes of variation.

$$\text{X bar chart : } UCL_{\bar{x}} = \bar{\bar{x}} + 3 \frac{\hat{\sigma}}{\sqrt{n}} = \bar{\bar{x}} + A_2 \bar{R}$$

$$LCL_{\bar{x}} = \bar{\bar{x}} - 3 \frac{\hat{\sigma}}{\sqrt{n}} = \bar{\bar{x}} - A_2 \bar{R}$$

$$\begin{aligned} \text{R chart : } UCL_{\bar{R}} &= D_4 \bar{R} \\ LCL_{\bar{R}} &= D_3 \bar{R} \end{aligned}$$

- Where,
 - X is the individual value (data)
 - n is the sample size
 - X bar is the average of reading in a sample
 - R is the Range, in other words the difference between largest and smallest value in each sample
 - R bar is the average of all the ranges.
 - UCL is Upper control limit
 - LCL is Lower control limit

The below control chart constants are approximate values to measure the control limits for X bar R chart and other control charts based on subgroup size

Subgroup	X bar chart		Sigma estimate	R chart		S chart	
	A ₂	A ₃	d ₂	D ₃	D ₄	B ₃	B ₄
2	1.880	2.659	1.128	-	3.267	-	3.267
3	1.023	1.954	1.693	-	2.574	-	2.568
4	0.729	1.628	2.059	-	2.282	-	2.266
5	0.577	1.427	2.326	-	2.114	-	2.089
6	0.483	1.287	2.534	-	2.004	0.030	1.970
7	0.419	1.182	2.704	0.076	1.924	0.118	1.882
8	0.373	1.099	2.847	0.136	1.864	0.185	1.815
9	0.337	1.032	2.970	0.184	1.816	0.239	1.761
10	0.308	0.975	3.078	0.223	1.777	0.284	1.716
11	0.285	0.927	3.173	0.256	1.744	0.321	1.679
12	0.266	0.886	3.258	0.283	1.717	0.354	1.646
13	0.249	0.850	3.336	0.307	1.693	0.382	1.618
14	0.235	0.817	3.407	0.328	1.672	0.406	1.594

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15	0.223	0.789	3.472	0.347	1.653	0.428	1.572
16	0.212	0.763	3.532	0.363	1.637	0.448	1.552
17	0.203	0.739	3.588	0.378	1.622	0.466	1.534
18	0.194	0.718	3.640	0.391	1.608	0.482	1.518
19	0.187	0.698	3.689	0.403	1.597	0.497	1.503
20	0.180	0.680	3.735	0.415	1.585	0.510	1.490
21	0.173	0.663	3.778	0.425	1.575	0.523	1.477
22	0.167	0.647	3.819	0.434	1.566	0.534	1.466
23	0.162	0.633	3.858	0.443	1.557	0.545	1.455
24	0.157	0.619	3.895	0.451	1.548	0.555	1.445
25	0.153	0.606	3.931	0.459	1.541	0.565	1.435

Control Chart Constants

Refer common factors for various control charts

Example cont.: In the above example $n=4$

n	A_2	D_3	D_4	d_2
2	1.88	0	3.27	1.13
3	1.02	0	2.57	1.69
4	0.73	0	2.28	2.06
5	0.58	0	2.11	2.33
6	0.48	0	2.00	2.53
7	0.42	0.076	1.92	2.70

Find the control limits

$$X \text{ bar chart : } UCL_{\bar{x}} = \bar{\bar{x}} + A_2 \bar{R} = 29.87 + 0.73 * 11.85 = 38.5$$

$$LCL_{\bar{x}} = \bar{\bar{x}} - A_2 \bar{R} = 29.87 - 0.73 * 11.85 = 21.2$$

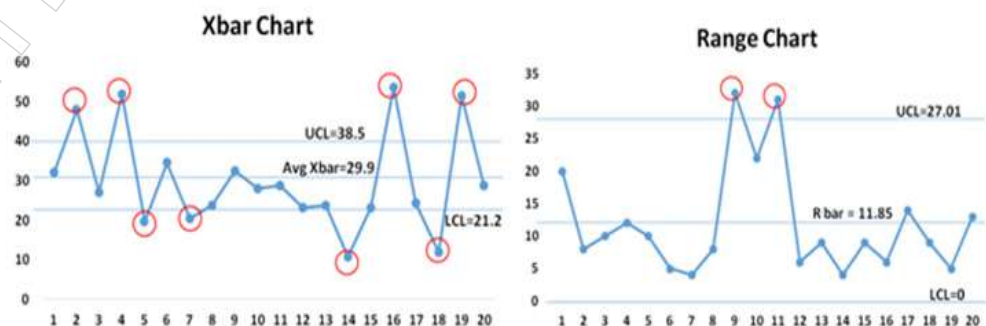
$$R \text{ chart : } UCL_{\bar{R}} = D_4 \bar{R} = 2.28 * 11.85 = 27.01$$

$$LCL_{\bar{R}} = D_3 \bar{R} = 0$$

Interpret X Bar and R Chart

- Plot both X bar and R chart and identify the assignable causes

Example Cont.: Use the above values and plot the X bar and Range chart



From the both X bar and R charts it is clearly evident that most of the values are out of control, hence the process is not stable

Monitor the Process After Improvement

Once the process has stabilised and control limits have been established, monitor the process performance over time.

Example cont.: Control Phase- Once the process has been enhanced and matured, the team identified the X bar R chart as one of the control methods in the Control plan to monitor the process performance over time.

Following are the measurement values in Control phase of the project

Sample	Measured values			
	1	2	3	4
1	23	25	24	26
2	22	26	24	25
3	28	28	22	23
4	25	25	26	36
5	22	22	25	26
6	26	24	23	22
7	29	24	24	24
8	26	25	25	22
9	22	25	24	24
10	25	22	26	24
11	24	24	24	23
12	24	25	26	23
13	22	28	22	26
14	23	24	25	26
15	24	25	29	24
16	24	22	28	26
17	24	25	25	25
18	22	24	25	26
19	26	25	22	24
20	26	22	24	25

Compute X bar and Range

Sample	Measured values				Sample Average (X bar)	Sample Range R
	1	2	3	4		
1	23	25	24	26	24.5	3
2	22	26	24	25	24.25	4
3	28	28	22	23	25.25	6
4	25	25	26	36	28	11
5	22	22	25	26	23.75	4
6	26	24	23	22	23.75	4
7	29	24	24	24	25.25	5
8	26	25	25	22	24.5	4
9	22	25	24	24	23.75	3
10	25	22	26	24	24.25	4

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11	24	24	24	23	23.75	1
12	24	25	26	23	24.5	3
13	22	28	22	26	24.5	6
14	23	24	25	26	24.5	3
15	24	25	29	24	25.5	5
16	24	22	28	26	25	6
17	24	25	25	25	24.75	1
18	22	24	25	26	24.25	4
19	26	25	22	24	24.25	4
20	26	22	24	25	24.25	4
Total					492.5	85
Average X bar					24.6	
R bar						4.25

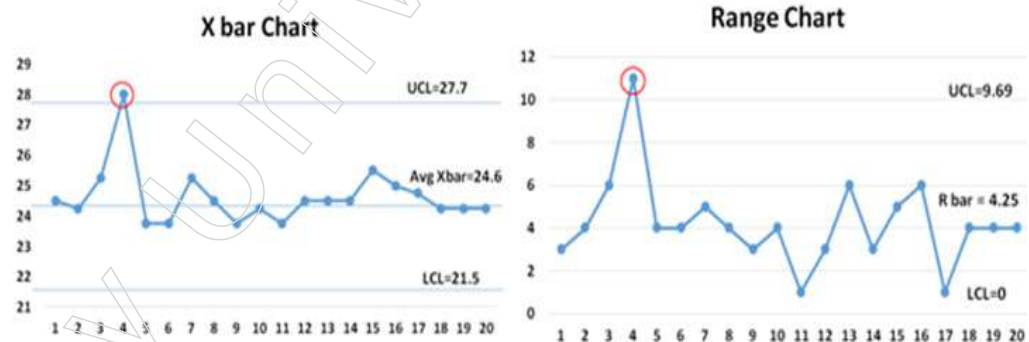
Find the control limits

$$X \text{ bar chart : } UCL_{\bar{x}} = \bar{\bar{x}} + A_2 \bar{R} = 24.6 + 0.73 * 4.25 = 27.7$$

$$LCL_{\bar{x}} = \bar{\bar{x}} - A_2 \bar{R} = 24.6 - 0.73 * 4.25 = 21.5$$

$$R \text{ chart : } UCL_{\bar{R}} = D_4 \bar{R} = 2.28 * 4.25 = 9.69$$

$$LCL_{\bar{R}} = D_3 \bar{R} = 0$$



Both the X bar and R charts plainly show that the process is nearly steady. During the initial phase, one value is out of control; the team must do a root cause analysis for the particular cause, while also smoothing out the process from data set number 16. If this persisted, the chart would require new control limits from that point on.

- After the process stabilised, any point that went outside of control limits shows an assignable cause in the process that needs to be addressed. This is a continuous procedure for monitoring the process's performance.

p-chart

What is p-Chart?

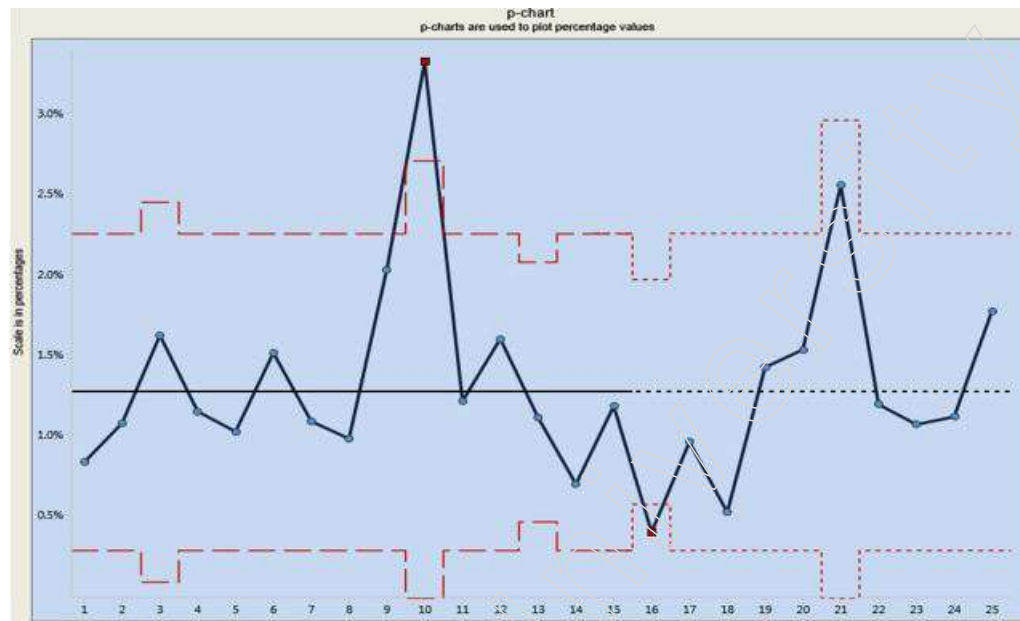
A p-chart is attributes control chart that is used with data that has been collected in subgroups of differing sizes. Because the size of the subgroup varies, it displays a

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proportion of nonconforming items rather than the exact count. P-charts depict how the process evolves over time. The process property (or characteristic) is usually expressed as a yes/no, pass/fail, go/no go. Use a p-chart, for example, to illustrate the proportion of incomplete insurance claim forms received each week. The subgroup would be determined by the total number of claims received each week. P-charts are used to assess if a process is stable and predictable, as well as to track the results of process improvement ideas. P-charts can be created using software programs like SQCpack.

What Does It Look Like?

The p-chart depicts the proportion of nonconforming units in different size groupings.



When is it Used?

Use a p-chart when you can answer “yes” to all these questions:

1. Do you need to evaluate the system's stability?
2. Is the data made up of counts that can be transformed to percentages?
3. Is there just one possible outcome for each given check?
4. Has the plotted attribute been operationally specified prior to data collection?
5. Is the subgroups' time order preserved?

c-chart

What is it?

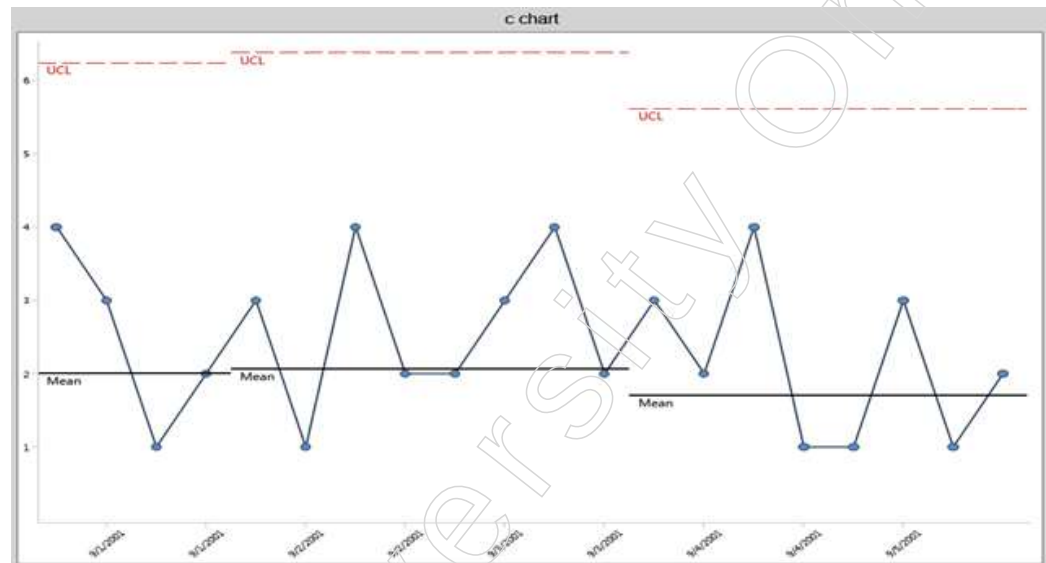
A c-chart is attributes control chart that is used with data gathered in equal-sized subgroups. C-charts depict how the process varies over time as measured by the number of nonconformities per item or set of items. Nonconformities are flaws or occurrences that were discovered in the sampled subgroup. They can be defined as any feature that is there but should not be, or any feature that is absent but should

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be. Nonconformities include things like a scratch, dent, bubble, blemish, missing button, and a rip. C-charts are used to assess if a process is stable and predictable, as well as to track the results of process improvement ideas. C-charts can be created using software products like SQCpack.

What Does It Look Like?

The c-chart shows the number of nonconformities in subgroups of equal size.



5.4.5 Introduction of Acceptance Sampling

Acceptance Sampling

Acceptance sampling is a statistical measure used in quality control. It allows a company to determine the quality of a batch of products by selecting a specified number for testing. The quality of this designated sample will be viewed as the quality level for the entire group of products.

A company cannot test every one of its products at all times. There may be too many to inspect at a reasonable cost or within a reasonable timeframe. Also, comprehensive testing might damage the product or make it unfit for sale in some way. Testing a small sample would be indicative without ruining the bulk of the product run.

A History of Acceptance Sampling

Acceptance sampling dates back to the early 1940s in its present industrial form. It was first used by the United States military to test bullets during World War II. Harold Dodge, a veteran of Bell Laboratories' quality assurance department who was engaged as a consultant to the Secretary of War, devised the concept and approach.

While the bullets had to be examined, Dodge reasoned those choices regarding entire lots could be made based on samples chosen at random. He developed a detailed sampling strategy to be used as a standard, together with Harry Romig and other Bell colleagues, establishing the sample size, the number of allowable flaws, and other criteria.

Acceptance sampling techniques were prevalent during and after World War II. Acceptance sampling, however, is not the same as acceptance quality control, as Dodge himself said in 1969. It pertains to certain lots and is an instantaneous, short-term test—sort of like a spot check—depending on specific sampling strategies. Acceptance quality control, on the other hand, relates to the complete product line in a broader, more long-term meaning; it acts as an essential element of a well-designed manufacturing process and system.

- Classification of Acceptance Sampling:

Depending upon the type of inspection acceptance sampling may be classified in two ways:

- (i) Acceptance sampling based on qualities, such as GO and NOT GO gauges, and
- (ii) Acceptance sampling based on variables.

In acceptance sampling by characteristics, no real measurements are taken, and the inspection is performed using GO/NOT GO gauges. If the product meets the specifications, it is accepted; otherwise, it is rejected. In this scenario, the amount of the inaccuracy is unimportant.

For example, if cracks are the inspection criteria/items with cracks will be rejected and goods without cracks will be approved, the form and size of the cracks will not be assessed and evaluated.

Actual dimensions are measured or physical and chemical testing of the qualities of a sample of materials/products is performed in acceptance sampling by variables. If the findings are as expected, the lot is approved; otherwise, it is rejected.

- Advantages of Acceptance Sampling:

- (i) The approach is appropriate in industries that have mass production and follow a predetermined manufacturing procedure.
- (ii) The procedure is inexpensive and simple to learn.
- (iii) Reduces weariness and boredom.
- (iv) The computational labour involved is fairly minor.
- (v) Training may be simply provided to those participating in inspection.
- (vi) During inspection, products of a destructive character can be conveniently inspected by sample.
- (vii) Scheduling and delivery times are improved as a result of the rapid inspection procedure.

- Limitations of Acceptance Sampling:

- (i) Because it does not provide 100 percent guarantee for specification validation, there is always the possibility/risk of drawing incorrect conclusions about the batch/quality. lot's
- (ii) The system's success is reliant on sample randomness, quality parameters to be assessed, batch size, and lot acceptance criteria.

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5.4.6 Concepts of Kaizen

Kaizen is a Japanese phrase that means “improvement for the better” or “constant improvement.” It is a Japanese corporate concept that focuses on methods that constantly improve operations while including all personnel. Productivity development, according to Kaizen, is a slow and meticulous process.

“Kaizen” is a Japanese term that signifies “improvement” or “better transformation.” Kaizen is described as a continual effort by all employees (from the CEO to the field crew) to improve all processes and systems of a specific firm. Working for a Japanese firm will quickly make you know how important the Kaizen process is to them. The Kaizen method assists Japanese businesses in outperforming all other rivals by sticking to particular specified principles and norms to remove flaws and assure long-term better quality and, ultimately, customer happiness.

Kaizen works on the following basic principle.

“Change is for good”.

Kaizen is defined as “the continual improvement of an organization’s processes and functions via change.” In layman’s terms, Kaizen is a constant little improvement in the overall operations that ultimately aims towards the success of the firm. The Japanese believe that numerous tiny continual adjustments in systems and policies provide better results than a few significant ones.

The Kaizen process strives to continuously improve processes not just in the manufacturing sector but also in all other areas. Implementing Kaizen tools is the duty of every person who is directly connected with the organisation, not just one. Every individual, regardless of title or position in the hierarchy, must participate by adding modest enhancements and adjustments into the system.



- Five S of Kaizen

The “Five S” of Kaizen is a methodical technique that results in flawless systems, standard policies, rules, and regulations, resulting in a good work culture within the firm. A representative of a Japanese corporation is unlikely to be unhappy or unsatisfied. Employees in Japan never speak poorly of their employer. Yes, the Kaizen method plays a key role in employee and customer happiness by implementing tiny continual adjustments and reducing flaws. Kaizen tools promote a well-organized workplace, which leads to increased productivity and better results. It also results in personnel that have a deep attachment to the organisation.

Let us understand the five S in Detail:

1. Seiri - SEIRI stands for Sort Out. Employees, according to Seiri, should go through and arrange their belongings. Label the things as "Necessary," "Critical," "Most Important," "Not required right now," "Useless," and so on. Toss everything that isn't useful. Set away anything that isn't required right now. Items that are crucial and important should be kept in a secure location.
2. Seition - The word seition denotes "organisation." According to research, employees squander 50% of their valuable time looking for goods and vital papers. Every object should have its own space and should only be stored in its designated location.
3. Seiso - The term "SEISO" translates to "shine the workplace." The workplace should be kept clean at all times. Clear out your workspace. Documents must be stored in appropriate folders and files. Store your stuff in cabinets and drawers.
4. Seiketsu-Seiketsu refers to Standardization. To maintain exceptional quality, every firm must follow certain basic norms and procedures.
5. Shitsuke or Self Discipline - Employees must respect the organization's policies and follow all rules and regulations. Self-control is crucial. Do not come to work in your slacks. Follow work procedures and bring your identification cards to work. It instils pride and respect in you for the company.

- The Duality of Kaizen

Kaizen is essentially a two-part idea, with action accounting for the majority of the notion and philosophy or method of thinking accounting for the remainder.

First, let's look at the activity. While it may be done on a much smaller scale, kaizen is frequently event-centric. Most firms that want to implement kaizen have internal events that focus on aspects of the business that can or should be improved. The events' goal is to determine the best strategies and tools to accomplish necessary reforms. Teams of employees from all levels of the organisation participate in such activities since the changes are intended to influence and benefit everyone.

The idea of kaizen is all about creating a certain sort of business culture and way of thinking that encourages all employees to be actively involved in the firm's success and productivity, as well as looking out for one another's best interests. The purpose of kaizen philosophy is to create a work atmosphere that fosters happiness, which is then reflected in how people work together and alone, as well as how devoted they are to their workplace.

- Kaizen Events

A typical Kaizen event has a process that goes something like this:

1. Establish goals and offer any background information that is required.
2. Evaluate the present situation and devise a plan for improvement.
3. Carry out improvements.
4. Go over what doesn't work and correct it.

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5. Report outcomes and identify any follow-up actions.

This form of Kaizen process cycle is commonly known as PDCA (Plan, Do, Check and Act). The PDCA method takes a scientific approach to creating improvements:

- Plan: develop a hypothesis
- Do: run experiment
- Check: evaluate results
- Act: refine your experiment; then start a new cycle
- Kaizen events adhere to the PDCA cycle (Plan, Do, Check, Act). To elaborate, a typical kaizen event begins by collecting all workers, or at least a few employees, from each department of the organisation and creating goals and standards that will benefit everyone and the company as a whole. During these occasions, it is critical to conduct a thorough analysis of the company's present procedures in order to identify areas that require major change or to promote new or reinforced goals that aren't being met.
- Once the planning part of a kaizen event is complete, it is time to put the improvements that were agreed upon during the planning stage into action. Employees may actively engage in establishing new processes and seeing how they affect the work environment, morale, and if they are successful in assisting in the achievement of newly stated goals.
- Following first deployment, management seeks employee feedback to determine which new processes aren't functioning and which need to be adjusted. This series of actions and checks should be repeated until the optimal balance is attained or it is judged that new goals must be established and met.



• Benefits of Kaizen

The Kaizen methodology is useful for a wide range of company structures and operational philosophies. It may enhance work processes, minimise waste, improve quality, and raise your company's profitability. While it is not a fast cure, applying Kaizen may result in steady and long-term progress. Let's have a look at how Kaizen may help to enhance the workplace and overall corporate success.

1) Continual Improvement for Your Product and Services

When Japanese philosophy is applied to business, it results in better products and services. You can solve the little flaws that degrade product quality, resulting in returns and replacement expenses. Customer satisfaction rises as a result of quality improvements since they encounter fewer unfavourable concerns. This improves your brand's reputation and generates precious word of mouth, resulting in increased sales volumes and revenues.

2) Delivers a Competitive Advantage

It might be difficult to keep a distinct lead over your competition in modern company. You may struggle to keep ahead of the pack if they have more funding behind them or are more adaptable when it comes to innovation. However, lean kaizen may provide you a significant advantage over your competitors and position you as the dominating leader in your sector. You can determine the core reason of how to boost productivity. You may extend a lead that others cannot match by boosting your quality, productivity, and efficiency (no matter how much money they invest in R&D or marketing). This process improvement attitude also leads in a more competent staff, which adds long-term value to your organisation.

3) Encourages Grassroots Thinking

Kaizen isn't only a management-focused practise. It is applied by frontline personnel in the organisation, regardless of their function or level of seniority. Employees gradually learn to think differently about how they approach their regular responsibilities. Instead of feeling helpless in their jobs, it encourages employees to take the initiative and offer improvements to how things are done. They begin to understand the worth and influence of their involvement when these ideas are implemented and the outcomes are realised. This encourages frontline staff to submit additional input that benefits the organisation. Employees at the ground level frequently understand difficulties better than senior management because they deal with or solve them on a daily basis. Because of this, they are more qualified to creating practical answers, thus it is critical to involve them.

4) A Culture of Continual Improvement

Kaizen-enabled businesses often have a highly engaged staff. They develop the culture through enhancing the company's operations and products. Employees are empowered to make improvements and are aware that their input is valued. They recognise that their contributions may result in real change, which contributes to a sense of fulfilment. This permeates the organisation, resulting in a more positive shared culture and a more fun place to work.

5) Avoids Information Silos

Siloed thinking is something that many businesses suffer with. Information is not transmitted across departments or supervisory levels, resulting in irritation on both sides. Communication breakdowns may be costly, both financially and morale-wise. However, Kaizen enables more productive connections amongst teams by opening up these channels. People are encouraged to express their thoughts and ideas so that it

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becomes a normal part of their everyday work life. This results in better collaboration and confidence across departments.

6) Boosts Workplace Productivity

Productivity is scrutinised in practically every industry. When things go rough, we're motivated to get more done with less resources. It might appear to be an uphill fight and a continual source of stress. However, Kaizen allows you to increase your production by enhancing your efficiency. Fewer quality concerns and product returns increase the organization's total production. Staff are naturally motivated to put more effort into their job when corporate culture and employee engagement improve. This results in productivity that cannot be coerced with reprimands or purchased with bonuses.

7) Enhances Creativity

Companies are frequently searching for more innovative approaches to business difficulties. However, because creativity has no rule book or set of instructions to follow, cultivating it in the workplace may be difficult. The Kaizen work method, on the other hand, pushes employees to think creatively on a regular basis. It pushes students to solve unanticipated problems as they emerge and trains them to think beyond the box. This fosters a culture of innovation in which individuals are encouraged to come up with new ideas and freely discuss them.

8) Improves Teamworking

Improvements to organisational culture, the elimination of silos of thought, and the encouraging of grassroots initiative all lead to increased collaboration. As they work to enhance the firm, employees are more likely to interact and collaborate with one another. Rather of depending on team-building days twice a year, Kaizen provides a continuous collaboration endeavour that achieves genuine outcomes. You will build a robust and resilient workforce by using the potential of cross-functional cooperation.

5.4.7 5S

5S is a method of visual management that employs anything from floor tape to operations manuals. It's not only about keeping things clean and organised; it's also about increasing efficiency and profit. 5S is a framework that stresses the application of a certain mentality and tools in order to increase efficiency and value. It includes watching, evaluating, cooperating, and looking for garbage, as well as the practise of waste removal.

What Does 5S Stand For?

5S, often known as 5s or Five S, refers to five Japanese phrases used to define the processes of the 5S visual management system. Each word begins with a S. Seiri, Seiton, Seiso, Seiketsu, and Shitsuke are the five S's in Japanese. Sort, Set in Order, Shine, Standardize, and Sustain are the five S's in English,

Origin of 5S

Toyota Production System

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The 5S system may be traced back to post-war Japan, when executives of a rising Toyota Industries (Toyoda Automatic Loom Works) sought to decrease industrial waste and inefficiencies. Their approach, known as the Toyota Production System, incorporates a variety of methods that would go on to become well-known in their own right, including as Just-In-Time production, Jidoka, and the basic notion of the visual workplace.

The Toyota Production System was initially a closely guarded secret, but the massive economic boom Japan experienced in the 1980s piqued the interest of foreign corporations, who wondered how Toyota, the gleaming jewel in Japan's manufacturing crown, was able to build so many products, so quickly, and at such a high-quality level. As a progressive worldwide interchange of ideas took shape, author and ULVAC Inc. executive Hiroyuki Hirano conceived the five pillars of the visual workplace, a notion that would evolve into the 5S technique as we know it today.

What is the 5S System?

The 5S method is a lean manufacturing technique for increasing workplace productivity and reducing waste. The system consists of five stages, each beginning with the letter S:

1. Sort 2. Set in Order 3. Shine 4. Standardize 5. Sustain



5S helps facilities minimise lost production due to delayed work or unexpected downtime by offering a systematic foundation for organisation and cleanliness.

- The Steps of 5S

Because 5S was developed in Japan and the original “S” terminology were in Japanese, the English translations for each of the five steps may differ. The fundamental concepts and their interrelationships, on the other hand, are simple to grasp.

Step Name	Japanese term	Explanation
1. Sort	Seiri (tidiness)	Remove unnecessary items from each area
2. Set in Order	Seiton (orderliness)	Organize and identify storage for efficient use
3. Shine	Seiso (cleanliness)	Clean and inspect each area regularly
4. Standardize	Seiketsu (standardization)	Incorporate 5S into standard operating procedures
5. Sustain	Shitsuke (discipline)	Assign responsibility, track progress, and continue the cycle

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Because these processes are interconnected, the order is critical. Clearing away unneeded things in step 1 (Sort) will make room for organising the key stuff in step 2. (Set In Order). After the work area has been decluttered and arranged, filth and grime may be eliminated in step 3. (Shine). These changes in workers' job activities and work environment should be reflected in revised procedures through step 4 of the process (Standardize).

Finally, unless responsibility is given and progress is recorded — as required by step 5 — the new processes will be ineffective (Sustain). And, with accountability and tracking, workers will continue to follow the processes, eventually returning to step 1.

Step 1: Sort

Sort, or “seiri,” which translates to “tidiness,” is the first phase in the 5S process. The Sort step's purpose is to minimise clutter and free up space by eliminating items that do not belong in the area.

- **Clearing the Work Area**

Examine the equipment, tools, and materials in a work space for this stage. Items that are required or valuable for the task being done in that place should be retained at that location. Everything else should be taken out.

Some of the objects removed will have to be discarded or repurposed. Other things may belong to a different work procedure or area and should be returned to their “homes.” However, you may come across certain goods that you are unsure about.

If you come across an object that you can't identify or whose ownership is unknown, it's time to use a red tag. “Red-tagging” briefly adds a highly visible tag to the object, noting where and when it was discovered. Then, red-tagged objects from all work areas are gathered in one place: a “lost and found” for tools, supplies, and equipment.

Whether an important tool is missing from a work area, examine the red tag collecting area to see if it was located someplace else. Supervisors for each work space should check the red tag collecting area on a regular basis to ensure that nothing has been missed. Anything that belongs in the workplace should be returned there.

- **Reassigning Tagged Items**

Items may be kept in the red tag collecting area for an extended period of time. In such situation, the original work area (from which that item originated) does not appear to require it any more. However, it may be advantageous in other situations.

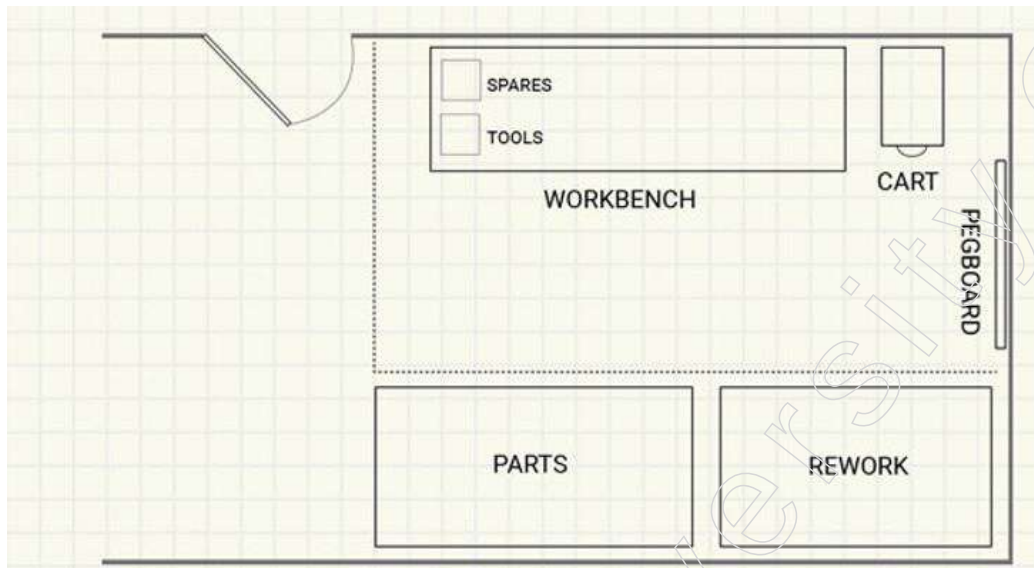
Items may be left in the red tag collecting place for thirty days in one typical technique. The item may then be claimed by any supervisor for their own work area. If no one wants it after another week, the item can be removed totally from the facility. It can be sold, recycled, or thrown away.

If an item will undoubtedly be required by the organisation but is not required right now, it may be preferable to keep it for later use. Before putting something away, make sure it will be required again. Make a plan to pull that thing out of storage again at a certain time. Don't save items “just in case” unless there's a legitimate reason to, and keep track of what you've got.

Step 2: Set in Order

Set In Order, the second phase, was originally named “seiton,” which translates to “orderliness.” In English, other titles have been used, such as “Systematic Organization,” “Straightening Out,” and “Simplify.” The purpose of this stage, whatever it’s named, is to arrange the work space. Each object should be simple to locate, use, and return: there should be a place for everything and everything in its place.

- Building a 5S Map



Tools that are regularly used should be kept close to where they are utilised. Spare equipment, supplies, and other things that are used infrequently can be housed in a single area and shared by various teams. Items that are often used together (such as drills and drill bits) should be kept close together. Each of these selections will make sense on its own, but keeping track of everything may become challenging. As part of this procedure, it may be beneficial to build a 5S map.

A 5S map is a graphic or floor plan that depicts a work area, procedure, or station. It serves as a visual reference to demonstrate where the tools, supplies, employees, and travel pathways are located, as well as how they connect to one another. A good map will also provide a description of the job that is done in the region shown.

Depending on your facility’s requirements, one solution may be easier than another:

- Create a plan and then put it into action.
- Arrange the workspace physically first, then plan it out.
- Create a map as you go, testing concepts and noting what works best.

Regardless of the method used to build it, the resultant 5S map should be preserved as a training tool, used for reference in subsequent phases of 5S, and updated over time as the work area changes.

- Communicating the Plan

After assigning storage places, each storage area should be labelled. Label the exterior of cabinet doors to assist workers in immediately identifying what is within each

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one. Then, mark any inside shelves to indicate where certain items belong. The same concepts apply to rack labels, bins, and other types of storage solutions.

Many facilities employ a “shadow board” for tool storage to guarantee that each instrument is easily returned to its proper storage location. With this method, a label that matches the form and size of the instrument is placed where it belongs. Workers can quickly identify where each item belongs and determine whether or not the item is present. No more wasting time sorting through drawers and boxes.

The floor can also benefit from organisation. Floor marking tape can be used to indicate work zones, circulation lanes, and storage locations for supplies and finished items.

Step 3: Shine

Shine, or “seiso,” which means “cleanliness,” is the third phase in the 5S method. While the first and second stages cleaned out space and organised the area for efficiency, this step targets the filth and grime that unavoidably accumulates beneath the clutter and attempts to prevent it from returning.

- Routine Cleaning

Shine is more than simply a broom that is pushed around every now and again. It entails cleaning every element of the work environment on a regular basis — usually a daily wipe-down followed by a more comprehensive cleaning once a week.

It is important to note that the Shine phase is not intended to be a duty for the maintenance or cleaning crew. Each person is responsible for cleaning their own work space as well as the equipment they use. This method has various advantages:

- Workers who are familiar with the region will detect any difficulties that develop;
- Hazards or challenging circumstances will be recognised and accounted for; and
- Items that are out of place or missing will be identified.
- During typical operations, workers will tend to maintain their personal workplaces tidy.

Everyone should be concerned about the overall cleanliness of the workplace, and be willing to pick up rubbish, for example. However, in order for 5S to provide the optimum outcomes, each worker must take personal responsibility for their own working environment.

- Shine as Preventative Maintenance

Keeping work environments tidy will provide several benefits. One significant advantage is that leaks, fractures, and misalignments are easier to detect. If the employees who keep the place clean are the same people who work there on a daily basis, they will notice any of these issues right away.

If those issues go unrecognised and unsolved, they can lead to equipment failure, safety concerns, and lost productivity. The system can feed into a preventative maintenance programme with the continual cleaning and inspections employed in the

Shine stage of 5S. As a result, 5S can assist to extend the operating life of equipment and decrease emergency downtime.

Step 4: Standardize

The first three stages of 5S address the fundamentals of clearing, organising, and cleaning a work area; these actions will bring short-term advantages on their own. Standardize, or “seiketsu,” which simply means “standardisation,” is the fourth phase. You may incorporate new practises into standard work procedures by documenting what is done, where it is done, and who is doing it. This opens the door to long-term reform.

- The Power of Writing Things Down

“If it isn’t written down, it didn’t happen,” as the saying goes. Writing down your 5S programme selections can assist guarantee that your effort does not just vanish. If you created a 5S map during the Set In Order stage, that map may be included in your new standard for the area. Similarly, the method for red-tagging objects may be documented and incorporated in the standards.

However, just because you’ve written down your decisions doesn’t mean you can’t alter your mind. The goal of 5S is to improve your workplace, not to make it unchangeable. You are creating the standards for your facility, which you may modify to reflect new facts or business needs.

- Tools for Standardizing

Once you’ve decided how to improve your work procedures, you must convey those decisions to your employees. This communication is a critical component of the Standardize process. The following are examples of common tools used in this process:

- a) 5S checklists - Listing the specific phases of a procedure makes it easier for employees to finish that process entirely. It also includes a basic auditing tool for subsequent monitoring of progress.
- b) Job cycle charts - Identify each job that must be completed in a work area and establish a timetable or frequency for each of those tasks. Then, delegate authority to a specific employee (or job duty). The generated graphic might be displayed in order to answer queries and promote responsibility.
- c) Procedure labels and signage — Place operating instructions, cleaning procedures, and preventative maintenance procedures where they will be required.

Step 5: Sustain

Sustain, or “shitsuke,” which literally means “discipline,” is the fifth phase in a 5S programme. The concept here is one of ongoing dedication. It’s critical to follow through on your decisions — and to repeatedly return to the earlier phases of 5S in an ongoing cycle.

Never “Once and Done”

The 5S technique was never intended to be a one-time event, but rather a continuous cycle. This is critical since early 5S triumphs might pave the way for

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difficulties. If vacant space becomes available during the Sort process, but tools and materials are then allowed to progressively fill it without any organisation, the ultimate outcome might be an even greater mess. The solution is to repeat the 5S concepts as a routine component of daily work. That is why Sustain is so crucial.

5.4.8 ISO Standards

Meaning of ISO Certification

Standardization is the action of creating facilities for common and repeated usage in response to real or anticipated difficulties, with the goal of achieving the highest degree of order in a particular setting. NOTE 1: The activity consists mostly of the procedures of developing, issuing, and implementing standards. NOTE 2: Significant advantages of standardisation include improved appropriateness of goods, processes, and services for their intended uses, elimination of trade obstacles, and promotion of technological collaboration (ISO/IEC Guide 2).

A standard is a document that is produced by consensus and authorised by a recognised authority that provides for common and repeated usage, rules, guidelines, or characteristics for actions or their outcomes, with the goal of achieving the highest degree of order in a specific setting. NOTE: Standards should be based on the best available research, technology, and experience, and should strive to maximise community benefits. (ISO/IEC Guide No. 2)

A product is an organization's output that can be produced without any interaction between the organisation and the customer. NOTE 1: The production of a product occurs without any transaction taking place between the provider and the customer, but it frequently includes this service element upon delivery to the customer. NOTE 2: The most important feature of a product is that it is typically tangibly. NOTE 3: Hardware is a tangible item, and its cost is a measurable attribute (e.g. tyres). Processed materials are palpable, and their quantity is a consistent feature (e.g. fuel and soft drinks). Goods are hardware and processed materials. Software consists of information regardless of delivery medium (e.g. computer programme, mobile phone app, instruction manual, dictionary content, musical copyright, and a driving licence) (ISO 9000:2015)

- **Benefits of Standardization**

There are many benefits of standardization for the private sector, the public sector and consumers. The use of standards in international and local trade and everyday life is now commonplace and well embedded; so much so, that socio-economic systems without standards are not conceivable

a) Standards and the Private Sector

The many benefits that standards bring to the private sector include:

- **Reduction in Production and Transaction Costs:**

This is accomplished mostly through the elimination of variation and the rationalisation of processes, which leads to economies of scale. Another advantage is that a greater range of component and sub-assembly sub-suppliers are available, all of whom must adhere to the established standards.

- Increased Market Opportunities:

Many commodities have been standardised in a variety of marketplaces. As a result, compliance with such standardised items allows the producer to access multiple markets rather than being confined to one where a distinct standard is anticipated.

- Competitive Positioning:

Manufacturers and suppliers to anticipate market trends. Progressive firms who are at the forefront of technical changes in the industry may see their corporate standards become the foundation for national or even international standards, further strengthening their market position.

- Improved Risk Management:

The risks connected with the implementation of technical requirements or the introduction of new technologies can be reduced by the proper use of standards. When a new technical rule is established, technical regulations based on recognised, proven standards reduce the chance of noncompliance. The introduction of new technologies into the market has unavoidable expenses; however, these costs can be shared with other manufacturers if the new technology is codified in standards that all manufacturers can use.

As well as specifying standards for government purchases, the public sector can benefit substantively from the use of standards, namely:

- A Basis for Regulation:

Standards are accepted methods of implementing the health and safety standards for regulated items. Legislators, for example, acquire stakeholder support for regulations because they were involved in the establishment of the standards that serve as the foundation for regulation. Standards are kept up to current through five-year review programmes; hence, regulations that use them as a foundation can also be up to date, depending on the manner of referencing.

- Efficient Regulation Through Participatory Processes:

Regulations that are based on standards might be utilised to offer direction on key requirements. This has the advantage of allowing market forces to be used to reduce compliance expenses to a minimal. The government may select conformity-assessment service providers who meet with standards such as ISO/IEC 17021 and ISO/IEC 17025 to conduct such services. If more than one service provider is approved, market forces will keep costs in check, and the nation will not be required to offer such services.

- Contribution to Socio-economic Development:

Regulations based on standards provide the country with access to the most recent state-of-the-art requirements for items that have been agreed upon by a broad stakeholder group. Furthermore, local businesses can get easier access to international markets where the rules are based on the same standards; this means that such businesses do not need to manufacture goods that must conform with a distinct set of requirements.

b) Standards and the Consumer

The consumer benefits from standardization in several ways, such as:

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- Consumer Involvement:

The customer is asked to participate in the standards-development process, and so has a say in the process. Standards are important in a legal sense when unsafe or non-compliant consumer goods must be discovered after items and processes have been sold, regardless of whether they must conform with technical laws or not.

- Consumer Safety:

Products from a variety of providers that adhere to the same underlying safety or compatibility criteria give the buyer with options in terms of pricing and design. As a result, buyers may be confident that the product will be safe and will work with common systems such as power supply, telecommunications standards, the internet, and common types of machinery.

- Sustainability:

Standards are widely employed in occupational health and safety systems, environmental protection, and, increasingly, in deciding the social elements of sustainable development, all of which benefit society.

- Definition of a Standard

A standard is a document developed by agreement and authorised by a recognised authority that specifies, for common and repeated usage, rules, norms, or characteristics for actions or their outcomes, with the goal of achieving the highest degree of order in a specific setting.

Organization for International Standardization (ISO)

- International federation of national standards bodies representing around 150 nations, with one delegate from each.
- Switzerland-based non-governmental organisation (NGO) founded in 1947.
- ISO Certification

ISO certification verifies that a management system, manufacturing process, service, or documentation technique meets all standardisation and quality assurance standards. ISO (International Organization for Standardization) is a non-governmental organisation that creates standards to assure the quality, safety, and efficiency of products, services, and systems.

ISO certifications are available in a wide range of industries, including energy management and social responsibility, as well as medical devices and energy management. To maintain uniformity, ISO standards are in place. Each certification has its own set of requirements and criteria and is categorised numerically. For instance, the ISO certification we currently hold at Mead Metals is ISO 9001:2015.

- ISO 9001 Certification

If a company advertises itself as "ISO 9001 Certified," it signifies that it has satisfied the criteria outlined in ISO 9001. (Which you can read in full here). ISO 9001 demands enterprises to establish and implement a quality management system that is both acceptable and effective, as well as to identify and address areas for improvement.

As a result, it is often assumed that a business claiming ISO 9001 accreditation has goods and services that fulfil quality requirements.

ISO Certification Explained

Being ISO 9001 accredited is necessary to sell to specific businesses, such as the automobile industry. We received ISO certification in 1998, and it has aided our firm in a variety of ways.

ISO standards have laid out what has to be done on each order. We understand the requirements and have the resources in place to assure quality, consistency, and safety. ISO specifies what we do, how we verify for quality, and what is expected of us.

ISO certification standards also contribute to the relevance of our goods and services. When the standards change, we adjust as well. When we start working with a new client, we offer them our certification information so they know they can anticipate ISO standards to be followed.

Popular Standards

ISO 9000 Quality management

ISO 14000 Environmental management

ISO 3166 Country codes

ISO 26000 Social responsibility

ISO 50001 Energy management

ISO 31000 Risk management

ISO 22000 Food safety management

ISO 27001 Information security management

ISO 20121 Sustainable events



5.4.9 “6-Sigma”

Six Sigma is a collection of quality-control technologies that firms may employ to

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reduce errors and optimise processes in order to increase profitability. It was created in the 1980s by a scientist while working at Motorola 1.

Six Sigma is a statistical and data-driven method that operates by analysing and reducing errors or flaws. It focuses on improving cycle times while limiting manufacturing faults to no more than 3.4 occurrences per million units or events. This indicates that a six-standard deviation event from the mean is likely to create an error since only 3.4 out of a million occurrences along a bell curve would fall outside of six standard deviations.

- **Five Steps of the Six Sigma Process**

The abbreviation DMAIC refers to the most critical processes and procedures in the Six Sigma methodology. When using the Six Sigma method, it is critical to comprehend this procedure.

- **Define:** The define stage is the first and most basic level of the Six Sigma process. The team begins by taking a high-level perspective of corporate procedures in order to assess client demands and identify a problem or opportunity for improvement.
- **Measurement:** The second stage focuses on data collection, which is crucial throughout the project's lifespan. To begin, the team maps out the present process to establish a baseline and investigates what is generating difficulties. Throughout the project, it empirically charts potential improvements.
- **Analysis:** The third step is analysis, which is a continuous process in which the team examines data and focuses on the root cause of a problem.
- **Improvement:** The fourth phase is enhancement. This is the phase of solution development in which the team implements a solution and validates that it functions as planned.
- **Control:** The fifth and final phase comprises control, where the emphasis is on sustaining the solution and progressing the team as achieved.

- **Sigma Belt Rankings**

Individuals can receive Six Sigma certification, as previously said, which reinforces and certifies their professional talents. These certificates are conferred using a belt system similar to that used in karate training. These are the belt levels:

- **White Belt:** A person with a white belt has not received any formal instruction or certification. This belt provides professionals with a fundamental foundation that allows them to participate in quality control and waste reduction programmes.
- **Yellow Belt:** This level goes above and beyond the white belt level in terms of training. Yellow belts can become valuable members of project teams. Managers with higher belts may benefit from their assistance.
- **Green Belt:** Those who pass this level must complete a comprehensive course that teaches them how to develop process improvement techniques. Individuals who work in certain industries, such as project or financial management, as well as health care, will benefit from the green belt certification. Graduates are frequently appointed as project managers.

- **Black Belt:** After completing the green belt level, students can advance to the black belt level. Successful graduates can deconstruct and manage more complicated jobs and projects. Through lean Six Sigma initiatives, they are taught how to deal with large-scale changes that may have an impact on their organisations.

People holding a black belt can advance to the ranks of masters and champions. A master black belt is regarded an adept and powerful leader with exceptional problem-solving abilities. A champion is a Six Sigma lean leader who can optimise profitability by eliminating waste and faults. You can obtain certification by taking courses offered by certain colleges or businesses. However, keep in mind that there is no consistent standard for the courses. As a result, the curriculum for Six Sigma certification varies, which implies that individual courses may differ based on the company providing them.

- **Principles of Six Sigma:**

Six Sigma deployment success is dependent on five fundamental principles:

1. Emphasis on customer requirements
 2. Using reliable measurements to do error-free statistical analysis in order to comprehend improvement prospects for constructing a solution for identifying the underlying cause of problems (variations)
 3. Being efficient and capable of removing variance in order to continuously improve the process.
 4. Including human resources in Six Sigma cross-functional teams.
 5. Being thorough and adaptable
- **What are the Main Concepts in Six Sigma?**

1. DPMO (Defects Per Million Opportunities)

DPMO is an abbreviation for defects per million opportunities. This is a concept we used to hear in manufacturing, and it is most often used by manufacturing who are involved in the production process. DPMO is one of the important methods for determining the likelihood of faults occurring during the manufacturing process, allowing us to make better use of available resources and manage problems that occur during the manufacturing process. The DPMO used to be aware of the number of possibilities for faults to occur during the manufacturing process.

The formula for DPMO –

$$\text{DPMO} = \frac{\text{Total Number of Defects found in Sample}}{(\text{Sample Size} * \text{Number of Defects Opportunities per Unit in the Sample})} * 1000000$$

2. Critical to Quality:

One of the primary concepts in Six Sigma techniques is critical to quality, because quality may impact the whole company purpose or goal.

The quantifiable features of a product or service that must fulfil the customer's need of the product or service are referred to as critical to quality.

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3. Defect:

A fault is defined as a breakdown in product or service attributes that are absolutely unacceptable.

4. Process Capability:

Process capacity is the ability of your business processes to supply or generate a service or product that fulfils the needs of your customers.

5. Variation:

In Six Sigma approach, variation implies that after providing service and product, the control person may measure the difference between what the client sees and what they actually feel after accepting the product or service.

6. Stable Operations:

Stable operations include not only stabilising the process, but also ensuring that the business process is consistent in order to narrow the gap between what the consumer sees and experiences.

- Customer and Product Driven Features of Six Sigma Methodology:
 1. Six Sigma is intended to eliminate waste and inefficiency, hence boosting customer satisfaction by delivering exactly what the client expects.
 2. Six Sigma is a highly organised and disciplined process that is tailored to the unique participants.
 3. Six Sigma is a statistical data-driven approach that demands precise or error-free data to analyse the process perfectly.
 4. Six Sigma is about more than simply improving quality and increasing sales; it is also about altering financial results.
 5. By implementing Six Sigma in your firm, you will be able to effectively reach the following milestones, often known as business success facts:
 - o Improving Process
 - o Lowering Defects
 - o Increased Profit
 - o Increased Customer Satisfaction
 - o Reducing Costs
 - o Reducing Process Variability
- Benefits of Six Sigma to Businesses:

1. Lowering Defects:

Six Sigma is based entirely on a single principle: minimising faults and, as a result, increasing the overall quality of a product or service. So, when a firm implements Six Sigma in their process, the business will be completely focused on that one principle-based statistical data-driven strategy.

2. Reducing Cost:

This is one of the last and long-term benefits that organisations receive when they adopt Six Sigma in their process since Six Sigma implementation ultimately leads to defect reduction, which saves time, resources, power, personnel efforts, and ultimately money.

3. Increased Customer Satisfaction:

Six Sigma and DMAIC implementation begin with measuring variance, which is the difference between what the customer sees and experiences. If the Six Sigma deployment is completed successfully and is sustainable for company, it will eventually minimise variance in the services or products, which will inevitably boost customer satisfaction and customer loyalty.

Benefits of Six Sigma Certification**1. Helps to Protect Your Organization to Avoid Errors:**

If you work in a business that needs to improve, your Six Sigma certification and expertise will assist your firm in producing a better product or providing a better service with fewer faults.

With Six Sigma certification and expertise, you will be able to improve processes and boost your organization's profit. It will eventually lead to you being a valuable asset to the firm.

2. Higher Salary:

According to a Payscale.com poll, Six Sigma certified experts earn the best salaries and work for some of the world's largest corporations.

3. Increase Your Managerial And Leadership Ability:

When you operate as a Six Sigma expert in a company, you must consider all aspects such as money, personnel, and resources before making any decisions or implementing any changes in any process, service, or product.

Those who have earned the Six Sigma Black Belt Certification serve as Change Agents in their organisations since they have a greater awareness and knowledge of the process and how to improve it.

To operate as a leader in the Six Sigma industry, you must have expertise and awareness of financial management and risk assessment.

4. Large Scope for Six Sigma across All the Industries:

Many people believe that Six Sigma is just useful in the manufacturing area, but it is truly relevant to many industries. Service industries such as IT, healthcare, and hospitality must improve their services to provide 100 percent customer satisfaction while meeting business objectives. As a result, all service industries require six sigma professionals to improve their overall business processes and meet business objectives.

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5. Increase Your Productivity and Capability:

After earning six sigma certifications, you will be accomplishing responsibilities with less effort, resulting in greater productivity and the ability to manage complex projects quickly and sustainably.

5.4.10 Lean Manufacturing

Lean manufacturing is a manufacturing method that aims to maximise productivity while minimising waste within a manufacturing operation. According to the lean concept, waste is everything that does not offer value that consumers are willing to pay for.

Lean manufacturing, or simply “lean,” is a systematic strategy for reducing waste in a production system while maintaining efficiency. Lean manufacturing, which originated in Japan with the Toyota Production System (TPS), aims to reduce waste within a manufacturing process, with the goal of clearly displaying what contributes value while eliminating what doesn't. As your organisation considers lean manufacturing, keep in mind that the process of going lean takes time - much like turning a cruise ship around.

There are several lean methodologies available, allowing each firm to incorporate lean manufacturing principles into its own unique production process. We'll go through the eight forms of waste that lean manufacturing aims to remove, as well as the five most prevalent lean principles, tools, and strategies that businesses across the world have integrated into their production processes.

- **Advantages and Disadvantages**

Lean manufacturing carries several advantages and disadvantages depending on how and where it is implemented.

Advantages:

1. Saves Time and Money

The most obvious advantage of lean manufacturing is cost savings. Improved processes, resource allocation, production, and storage may all help enterprises of any size or output. Time savings not only provides for shorter lead times and better service in delivering items to clients, but it may also assist save money by allowing for a more streamlined staff.

2. Environmentally Friendly

Reducing waste in time and resources, as well as eliminating superfluous procedures, may save money on energy and gasoline. This has a clear environmental advantage, as does the adoption of more energy-efficient equipment, which may also save money.

3. Improved Customer Satisfaction

Improving the delivery of a product or service to a consumer at the proper price increases customer happiness. This is critical for company success since satisfied clients are more likely to return to your store or promote your product or service to others.

Disadvantages:**1. Employee Safety and Wellbeing**

Lean critics contend that it can disregard employee safety and well-being. It is tempting to disregard the demands put on individuals who are given minimal tolerance for mistake in the workplace by focusing on reducing waste and simplifying operations. Lean has been likened to nineteenth-century scientific management approaches, which were battled against by labour reforms and were considered antiquated by the 1930s.

2. Hinders Future Development

The inherent focus on waste in lean manufacturing might cause management to remove aspects of a firm that are not regarded important to current strategy. These, however, may be critical to a company's heritage and future progress. Lean can cause an over-emphasis on the present while ignoring the future.

3. Difficult to Standardise

Some detractors argue that because lean manufacturing is a culture rather than a fixed procedure, it is hard to develop an uniform lean production model. This might give the impression that lean is a flimsy and ill-defined strategy rather than a solid one.

- What are the 5 Principles?

Value, the value stream, flow, pull, and perfection are the five key concepts of lean manufacturing. These are currently utilised as the foundation for implementing lean.

1. Value: Value is assessed from the customer's point of view and pertains to how much they are willing to pay for items or services. This value is then generated by the manufacturer or service provider, who should strive to decrease waste and expenses in order to deliver the best price to the consumer while still maximising profits.
2. Map the Value Stream: This approach entails analysing the materials and other resources necessary to manufacture a product or service in order to discover waste and opportunities for improvement. The value stream encompasses a product's complete lifespan, from raw materials through disposal. Each stage of the manufacturing cycle should be inspected for waste, and everything that does not add value should be eliminated. Chain alignment is frequently advocated as a method of accomplishing this phase.

Modern manufacturing streams are frequently complicated, necessitating the collaborative efforts of engineers, scientists, designers, and others, with the actual manufacture of a physical product being only one element of a larger stream of work.

3. Create Flow: Creating flow entails reducing functional obstacles in order to reduce lead times. This guarantees that operations run smoothly and with little delay or unnecessary waste. Interrupted and discordant production processes incur expenses, and generating flow entails assuring a continuous stream for production or service delivery.
4. Establish a Pull System: A pull system operates by only starting work when there is a demand for it. Push systems, which are utilised in manufacturing resource planning

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(MRP) systems, are the inverse of this. Push systems plan stocks ahead of time, with output scheduled to meet sales or production projections. However, because many projections are inaccurate, this can result in either too much or not enough of a product being produced to fulfil demand. This might result in higher storage expenses, timetable disruptions, or poor customer satisfaction. A pull system only operates when there is a demand for it, and its success is dependent on flexibility, communication, and efficient operations.

The pull system can entail teams only going on to new tasks once the previous stages have been completed, allowing the team to adjust to obstacles as they emerge while knowing that the earlier work is still primarily useful to providing the product or service.

5. Perfection: The pursuit of perfection via continuous process improvements is known as 'Kaizen,' which was coined by Toyota Motor Corporation founder Kiichiro Toyoda (see 'When and Who Invented Lean Manufacturing?' above). Lean manufacturing necessitates continuous review and improvement of processes and procedures in order to remove waste and identify the best solution for the value stream. To create a real and long-term change, the concept of continuous improvement must be embedded into an organization's culture and involves the assessment of metrics such as lead times, production cycles, throughput, and cumulative flow.

It is critical for a culture of continuous improvement to permeate all levels of an organisation, from team members and project managers all the way up to the executive level, in order to foster a sense of collaborative responsibility for development and value creation.

- Why is Lean Manufacturing Important and How Can it Help?

Waste in industry, whether it's idle personnel, inefficient processes, or underutilised resources, is a drain on production, and lean manufacturing strives to remove it. Depending on who you ask, the motivations for this range from raising revenues to giving benefits to customers. Regardless of the overarching motivations, there are four fundamental benefits of lean manufacturing:

- Eliminate Waste: Waste is a detriment to costs, timelines, and resources. It adds no value to goods or services.
- Improve Quality: Improved quality enables businesses to remain competitive while meeting the changing demands and desires of their consumers. Designing procedures to match these expectations and wants keeps you ahead of the competition while putting quality improvement first.
- Reducing Costs: Overproduction or having more materials than needed results in storage costs, which may be decreased by improving procedures and materials management.
- Reducing Time: Wasting time with inefficient working processes is also a loss of money, whereas more efficient practises reduce lead times and allow goods and services to be supplied more quickly.

5.4.11 TPM

Total Productive Maintenance (TPM) is the process of maintaining and improving

the integrity of production and the quality of systems by employing machines, equipment, workers, and supporting procedures.



What Is Total Productive Maintenance (TPM)?

Total productive maintenance (TPM) is a technique that operates on the premise that everyone in a facility, rather than just the maintenance personnel, should engage in maintenance. This strategy makes advantage of all employees' talents and attempts to include maintenance into a facility's day-to-day operations.

- 4 benefits of Total Productive Maintenance

1. Less Unplanned Maintenance

Equipment is kept in excellent condition thanks to well-planned and scheduled maintenance. Furthermore, TPM encourages all plant staff to take ownership of their machines, making maintenance a profitable investment. Assets are better cared for because TPM makes upkeep personal.

2. Reduced Equipment Downtime

Fewer failures result from more scheduled maintenance. Downtime incidents are reduced by conducting proactive maintenance actions.

3. Lower Manufacturing Costs

Production expenses are reduced when overall equipment efficiency (OEE) improves. Better productivity results in higher earnings and lower spending on equipment downtime and maintenance.

4. Strengthened Workplace Safety

Because TPM is based on the 5S principles, the workplace is meticulously ordered and sanitised. The 5S method's processes - sort, straighten, shine, standardise, and sustain - reveal underlying issues and barriers to sustaining the work environment.

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- 8 Pillars of Total Productive Maintenance



1) 5S - Sort, Straighten, Shine, Standardize and Sustain

Building a robust TPM process, like constructing a physical structure, needs a solid foundation in the form of 5S principles. This is a workplace organising strategy that is broken down into five simple steps:

- Sort tools, equipment, and materials to determine which may be discarded.
- Straighten and arrange items in correct order to eliminate needless motion and transit effectively between working groups and locations.
- Shine denotes doing the necessary housekeeping to tidy up the work place.
- Standardize and plan activities in order to deliberately develop habits that will keep the workplace orderly.
- Maintain the methodology and concepts for long-term application.

The 5S approach provides a systematic approach to cleaning the workplace, thereby uncovering underlying problems and challenges.

2) Autonomous Maintenance

Maintenance and equipment care should begin with the individuals who use the equipment. The ability of operators to concentrate on minor maintenance jobs effectively frees up maintenance crews to focus on more specialised activities.

3) Continuous Improvement

Continuous Improvement, also known as Kaizen in Japan, encourages the mindset of striving for zero losses and zero faults. The overall efficacy and efficiency of the organisation are built through tiny but continuous changes to procedures.

4) Planned Maintenance

Preventing equipment breakdown requires planned maintenance activities. Planned maintenance entails evaluating the condition of equipment on a regular basis in order to prevent deterioration and mechanical failures.

5) Quality Maintenance

Manufacturing processes strive for zero-defect production to ensure customer satisfaction. Superior quality standards, as well as inspections to ensure that the requirements are met, should be in place. The purpose of quality control is to discover any potential sources of deviations from zero-defect manufacturing.

6) Training

The premise behind TPM is that everyone contributes to the overall productivity of the manufacturing process. Proper training is essential to provide each member with the theoretical and practical know-how of working with machines and equipment in order to achieve optimal performance and build each member's competency.

7) Office TPM

The administrative department, which works behind the scenes, plays an important function that is sometimes underestimated. Management and administrative activities, like the rest of the production teams and processes, are subject to productivity enhancement. This involves discovering and reducing losses as well as contributing to the plant's overall performance.

8) Safety, Health and Environment

The eighth and final pillar focuses on providing a safe workplace. When actively applied to each of the other pillars, the essence of this pillar is manifested. This pillar's successful implementation will lead to a safe and hazard-free workplace.

- Steps in Introduction of TPM in an Organization
 - Stage A - Preparatory Stage:

STEP 1 - Announcement by Management to All about TPM Introduction in the Organization:

This level requires proper understanding, commitment, and active participation from senior management. Senior management should conduct awareness campaigns, following which an announcement should be issued to the whole organisation. Put it in the home magazine and on the notice board. If necessary, send a letter to all parties involved.

STEP 2 - Initial Education and Propaganda for TPM:

Training will be provided as needed. Some require significant training, while others only require awareness. Take important individuals to sites where TPM has previously been successfully applied.

STEP 3 - Setting up TPM and Departmental Committees:

TPM encompasses improvement, autonomous maintenance, quality maintenance, and so on. When committees are formed, they should address all of these issues.

Notes

STEP 4 - Establishing the TPM Working System and Target:

Each area has now been benchmarked, and a target for achievement has been set.

STEP 5 - A Master Plan for Institutionalizing:

The next phase is execution, which leads to institutionalisation, in which TPM becomes an organisational culture. Achieving the PM award demonstrates that you have reached a sufficient level.

- **STAGE B – Introduction Stage**

This is a ceremony, and everyone should be invited. Suppliers should be aware that we expect high-quality supplies from them. Related and related firms that may be our clients, sister concerns, and so forth. Some may learn from us, while others may assist us, and consumers will get messages from us indicating that we value excellent output.

- **STAGE C - Implementation**

At this level, eight actions are carried out, which are referred to as the “eight pillars” in the growth of TPM activity.

These four actions are for creating the system for production efficiency, one for the initial control system of new goods and equipment, one for enhancing administrative efficiency, and one for safety and sanitation control.

- **STAGE D - Institutionalising Stage**

All of their activities would indicate that they had entered the mature stage. Now is the time to apply for the PM prize. Consider the most difficult degree to which you can take this movement.

Check Your Understanding

- 1) _____ is concerned with ensuring that internal rules and process standards eliminate mistakes, whereas Six Sigma is concerned with reducing defects.
 - a) Total Quality Management
 - b) Inventory Management
 - c) Supply Management
 - d) Logistics Management
 - e) None of the a above
- 2) _____ should be considered as a set of methodologies that can affect decisions in the specification, manufacturing, and inspection functions.
 - a) Work-In-Progress
 - b) Statistical Quality Control
 - c) Kaizen System

- d) Sampling
e) None of the above
- 3) _____ is a useful statistical technique for determining the central tendency of a set of data.
a) Range
b) Standard Deviation
c) Mean
d) Statistic
e) None of the above
- 4) A _____ is a depiction of tabulated frequencies that is displayed as contiguous rectangles erected at defined intervals (bins) with an area equal to the frequency of the observations in the interval.
a) Histogram
b) Check Sheets
c) Pareto Analysis
d) Stratification
e) None OF The above
- 5) The upper control limit (UCL) and lower control limit (LCL) of a _____ differentiate common from assignable sources of variation.
a) Scatter Diagram
b) Analytical Approach
c) Control Chart
d) X Bar Chart
e) None of the above
- 6) _____ is a strategy that is used in conjunction with other data analysis methods. When data from many sources or categories is combined, the meaning of the data might get obscured.
a) Control Chart
b) Stratification
c) Mapping
d) Sampling Strategy
e) None of the above
- 7) The _____ is made up of graphs that are produced using pairings of numerical data, having one variable on each axis, to show the relationship between them.
a) Visual Workflow Diagram
b) Process Flow Diagram

Notes

- c) Scatter Diagram
 - d) Facility Diagram
 - e) None of the above
- 8) What is the full form of EWMA?
- a) Enterprise Weighted Moving Average
 - b) Exponentially Workflow Moving Average
 - c) Exponentially Weighted Moving Average
 - d) Exponentially Weighted Moving Analysis
 - e) None of the above
- 9) A _____ is attributes control chart that is used with data that has been collected in subgroups of differing sizes. This chart depicts how the process evolves over time.
- a) P-chart
 - b) Q-chart
 - c) C-chart
 - d) X- Bar Chart
 - e) None of the above
- 10) _____ depict how the process varies over time as measured by the number of nonconformities per item or set of items.
- a) P-Chart
 - b) C-charts
 - c) R-Chart
 - d) R -Control Chart
 - e) None of the above
- 11) _____ is a statistical measure used in quality control. It allows a company to determine the quality of a batch of products by selecting a specified number for testing.
- a) Sampling Frequency
 - b) Statistical Sampling
 - c) Acceptance sampling
 - d) Probability Sampling
 - e) None of the above
- 12) Kaizen is a Japanese phrase that means _____.
- a) Make Decisions
 - b) Improvement for the better
 - c) Well Informed Decisions

- d) Gain Accurate Information
e) None of the above
- 13) _____ is a non-governmental organisation that creates standards to assure the quality, safety, and efficiency of products, services, and systems.
- a) International Organization for Standardization
b) International Organization for Stabilization
c) Indian Organization for Standardization
d) International Origination for Standardization
e) None of the above
- 14) _____ is a collection of quality-control technologies that firms may employ to reduce errors and optimise processes in order to increase profitability.
- a) Pareto Optimisation
b) Six Sigma
c) Machine Scheduling
d) Shifting Bottleneck
e) None of the above
- 15) What is the full form of DPMO?
- a) Defects Per Million Opportunities
b) Detect Per Million Opportunities
c) Division Per Million Opportunities
d) Divide Per Million Opportunities
e) None of the above

Summary

- Total quality management seeks to make all stakeholders engaged in the manufacturing process responsible for the overall quality of the finished product or service.
- Total quality management (TQM) emerged as a term developed by the Naval Air Systems Command to define its Japanese-style quality improvement management methodology.
- Total quality management ensures that every single employee is working towards the improvement of work culture, processes, services, systems and so on to ensure long term success.
- Controlling and enhancing quality is an essential strategy for a firm since it gives a competitive edge.
- To create a histogram for continuous data, divide the data range into intervals, which are commonly referred to as class intervals, cells, or bins.

Notes

- Quality control charts are an excellent tool for engineers to use in determining whether or not a process is statistically controlled.
- Kaizen is described as a continual effort by all employees (from the CEO to the field crew) to improve all processes and systems of a specific firm.
- The Kaizen process strives to continuously improve processes not just in the manufacturing sector but also in all other areas.
- Siloed thinking is something that many businesses suffer with. Information is not transmitted across departments or supervisory levels, resulting in irritation on both sides.
- 5S is a method of visual management that employs anything from floor tape to operations manuals. It's not only about keeping things clean and organised; it's also about increasing efficiency and profit.
- ISO certification verifies that a management system, manufacturing process, service, or documentation technique meets all standardisation and quality assurance standards.
- Six Sigma is intended to eliminate waste and inefficiency, hence boosting customer satisfaction by delivering exactly what the client expects.

Activity

1. Through a case study or visit to an organization see how Quality management affects the success of the organization, along with "Quality Control".

Questions & Exercises

1. Describe Kaizen concept and 5S of kaizen.
2. Elaborate the quality control charts.
3. What is Statistical Quality Control? Put a light on fundamental approaches of Statistical Quality Control.
4. Define Total quality management and provide a historical overview of TQM. Also, write down the essential principles that, when used collectively, may propel any firm toward business success.
5. What is ISO Certification and elaborate the steps to acquire the same.
6. What is the concept of Six sigma and what six sigma rankings can individuals acquire which reinforces and certifies their professional talents?
7. How can you introduce Total Productive Maintenance in an organization Throw light on 8 Pillars of Total Productive Maintenance.

Glossary

- Quality is frequently used to denote a product's or service's 'excellence'.
- Total quality management (TQM) is the ongoing process of identifying and reducing or eliminating production defects, optimising supply chain management, enhancing the customer experience, and ensuring that staff are properly trained.

- The technique of utilising statistical methods to control product quality versus requirements is known as 'statistical quality control'.
- Statistics denotes the availability of a sufficient amount of data to achieve accurate results. Statistics manipulates data in order to derive conclusions using a variety of methodologies.
- Check sheets are simple data collection tools. They are used to collect data in an effective and efficient manner.
- Pareto chart is a simple and formal approach for identifying the top percentage of the reasons that must be addressed in order to solve the majority of issues. It is a decision-making strategy that statistically distinguishes a small set of input parameters as having the most effect on a desired or unwanted outcome.
- X-bar chart: The mean or average change in process from subgroup values across time. The X-control Bar's limits take the sample's mean and centre into account.
- R-chart: The range of the process over the time from subgroups values. This monitors the spread of the process over the time.
- Nonconformities are flaws or occurrences that were discovered in the sampled subgroup. They can be defined as any feature that is there but should not be, or any feature that is absent but should be.
- Kaizen is defined as "the continual improvement of an organization's processes and functions via change."
- 5S, often known as 5s or Five S, refers to five Japanese phrases used to define the processes of the 5S visual management system. Each word begins with a S. Seiri, Seiton, Seiso, Seiketsu, and Shitsuke are the five S's in Japanese.
- ISO (International Organization for Standardization) is a non-governmental organisation that creates standards to assure the quality, safety, and efficiency of products, services, and systems.
- Six Sigma is a statistical and data-driven method that operates by analysing and reducing errors or flaws.
- Total productive maintenance (TPM) is a technique that operates on the premise that everyone in a facility, rather than just the maintenance personnel, should engage in maintenance.

Further Readings

1. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
2. Production and Operations Management; Dr. Sarojrani Pattnaik, Dr. Swagatika Mishra Assistant Professor Department of Mechanical Engineering VSSUT Burla

Check Your Understanding – Answers

- 1) a)
- 2) b)

Notes

- 3) c)
- 4) a)
- 5) c)
- 6) b)
- 7) c)
- 8) c)
- 9) a)
- 10) b)
- 11) c)
- 12) b)
- 13) a)
- 14) b)
- 15) a)

Unit - 5.5: Issues in Operation Management

Notes

Objectives

At the end of this unit, you will be able to:

- Learn about operations management
- Issues of operations management

Introduction

Operations management (OM) is the administration of business procedures inside an organisation to achieve the best level of efficiency achievable. It is concerned with transforming materials and labour as effectively as feasible into goods and services in order to maximise an organization's profitability. Operations management teams strive to produce the maximum net operational profit feasible by balancing expenses and income.

Operations management include making use of resources such as people, materials, equipment, and technology. Operations managers buy, create, and supply items to clients depending on client requirements and the company's capabilities.

Operations management is in charge of a variety of strategic concerns, such as selecting the size of manufacturing plants, project management procedures, and the layout of information technology networks. Other operational concerns include inventory management, including work-in-process levels and raw material acquisition, quality assurance, handling of materials, and maintenance policies.

Operations have an impact on every area of an organisation. Operations management is at work when a firm wants to run smoothly, develop their business, please their consumers, and generate money.

In the context of production and manufacturing, operations management include planning, coordinating, and overseeing. A well-managed operations department allows a firm to be efficient and productive by utilising the fewest resources while satisfying client needs. Essentially, operations management emphasises the relationship between process and value.

5.5.1 Issues in Operation Management

With rising consumer demands and dwindling supply of certain resources, operations management has become increasingly vital for corporate performance and profitability. More accountability and accuracy are brought about by effective and appropriate operations management, allowing for the successful supply of products and services.

While operations management has greatly assisted manufacturing facilities, there are still several issues that operations managers encounter on a regular basis. Operations management is a difficult area that necessitates a thorough grasp of company processes in order for operations managers to provide value to their customers. These are as follows:

Notes

- Managing overheads
- Monitoring performance
- Listening to feedback
- Responding to competition
- Regulation and compliance
- Uncertainty
- Cyber risks
- Finding the right staff
- Retaining customers
- Cash flow

1. Managing Overheads

Overheads may quickly add up, whether they're in the shape of administrative costs, insurance, rent, or electricity bills. Excessive overheads may be particularly destructive if they are not controlled in a timely manner, thus they are a significant issue for small firms to deal with.

Example

Spiralling overheads can take many forms, including rent arrears, missing utility payments, office equipment, and corporate overspending.

The Solution: Examine your expenditures to identify where you might reduce back. Any service that is overkill for your team or your clients' needs should be reduced. Quick remedies like as cloud computing, free video call services, telecommuting when possible, and renting rather than buying office equipment can all be beneficial in this situation.

2. Monitoring Performance

What metrics do you use to assess performance? And, how can prospective indicators show that your team is working to its maximum potential? If specific people of your team aren't doing as effectively as they should, your company may suffer as a result.

Example

The bad performance of one of your employees is beginning to damage the mood of the rest of the team, affecting their own productivity as a result.

The Solution: Describe a set of performance metrics that demonstrate how well your team and business are operating. Rather of relying on simple financial metrics, it may be extremely advantageous to define, explain, and agree on the expectations of separate teams. Determine how outcomes will be measured, track progress, and assess success at the conclusion of the management cycle.

3. Listening to Feedback

Negative internet reviews may quickly harm your image and reputation in today's

business environment. Customers, too, demand prompt and effective customer service. It is important to businesses to listen to the feedback they receive so that they may provide good service to their main audience.

Example

There has been a complaint concerning the usability of your website, which has resulted in a recent decrease in traffic and business. You must now act...

The Solution: Consider hiring a consultant or a third party to track how you're viewed online and identify where you're going wrong with customer service. They will assist with everything from selecting the best customer-facing workers to optimising the handoff process.

4. Responding to Competition

Starting a business nowadays is more reasonable and achievable than ever before. However, with so much freely accessible information and assistance available, there may be greater competition in the same industry. And, whether your rivalry is focused on a specific service or a larger market, how can you differentiate yourself?

Example

A rival offers the same product/service at a cheaper price than you, stealing a significant portion of your consumers.

The Solution: Monitoring rival behaviour can provide you with information on how to stay ahead of the competition. Monitoring their procedures and practises can assist you in anticipating their future actions – and adapting your approach accordingly.

As previously said, optimise your customer service such that it exceeds expectations. Identify the components of your business or product that your consumers appreciate the most and give them the best service possible.

Be a pioneer in the adoption of new technology, processes, products, and solutions. Make your website mobile-friendly, use social media successfully, and leave a lasting impact in all you do. Showing your consumers that you're a forward-thinking, progressive firm with a matching product or service is a powerful method to stand out in a competitive market.

5. Regulation and Compliance

Regulation and compliance may be a huge problem if not handled appropriately. As markets and technology develop, so do the rules and laws that govern them. Unfortunately, companies are frequently unaware of, or misunderstand, what is expected of them. At this point, fines and penalties enter the picture, and something that should have been simply settled becomes a problem.

Example

The recent implementation of GDPR, which went into effect in May 2018.

The Solution: Trying to grasp the intricate terminology of any new legislation, depending on your industry, might be a bit of a challenge, especially with everything

Notes

else going on in your organisation. If necessary, consider engaging a consultant to assist you with the compliance process.

6. Uncertainty

You can never predict what will happen in the future, but you may plan for consumer and market trends with some foresight. Most organisations will have some form of long-term aim in mind, but it's astonishing how many will fail to plan for moments of uncertainty.

Example

In a shifting economic climate, the volatility of market trends might drive enterprises to restrain themselves, causing dissatisfaction among their staff.

The Solution: Short-term goals take primacy when there is uncertainty at the leadership. Long-term ambitions, on the other hand, should never be underestimated. Maintain control of your firm by developing a five- or ten-year strategy and balancing it with more ad hoc short-term initiatives.

7. Cyber Risks

Cyber security is quickly becoming a danger for all firms, rather than just those in the IT industry. If cyber security is not adequately handled, firms expose themselves to an expanding number of dangers, which may result in data theft, hacking, and other assaults.

Example

Data theft has resulted in the exposure of sensitive consumer information, generating distrust within your customer base.

The Solution: It is advised that a strategy that integrates cyber security into all component of the businesses, from the IT department to staff training, be used. Businesses can use cyber security frameworks, and many of them are utilitarian in nature, including corporate context, policy and standards, security capabilities and controls, and the environment.

8. Finding the Right Staff

As the firm expands and your accomplishments rise, you should have a better sense of the type of individual who is a good match for your organisation. In a small firm, competence and personality are more important than other attributes, yet not everyone is the appropriate fit for a specific organisation.

Example

You've invested in a lengthy hiring procedure just to discover that you've recruited someone who isn't a good fit for the company.

The Solution: It is critical to customise your management style to the many personalities in the workplace before, during, and after the recruiting process. Consider what works best for each individual; some people prefer to work in groups, while others

prefer to do projects on their own. If feasible, introduce prospects to your team to see how the dynamics will play out. This might include inviting the candidate's potential line manager to the interview.

Create an atmosphere in which your employees may thrive. Be open about their expectations, problems, and triumphs, and establish a work/life balance that assures comfort and satisfaction for everyone on the team.

9. Retaining Customers

You'll have discovered a consumer base that gravitates toward your service or product at this point, but how can you maintain them and keep them coming back?

Example

Business has been sporadic, with clients not returning as frequently as you'd planned, and things are going to dry up even more...

The Solution: Consider the qualities of your present top consumers, individuals that use your service on a regular basis and rave about your brand. What characteristics do they all share? Don't be scared to ask your greatest consumers what they want — feedback is critical to attracting and maintaining loyal customers.

10. Cash Flow

Money troubles may be a huge concern, whether they are due to unpaid bills, unexpected outgoings, or awaiting obligations. With so much to consider, keeping a strong cash flow is critical to a company's survival.

Example

The business completes a job and gets paid after a month. However, you've got infrastructure costs, employee salary and your personal expenses to consider too...

The Solution: Plan and budget as much as possible. There are plenty of money management tools out there to keep track of your cash flow. Look into apps that can create budgets, calculate VAT, automate bill payments and alert you when unusual outgoings occur.

It could be advisable to look into fixing costs wherever possible. Essential overheads such as energy costs can be fixed, our Shield Gas and Shield Electricity products being perfect examples. You'll also want to ensure that you're paying the correct business rates on your premises.

Check Your Understanding

- 1) _____ include planning, coordinating, and overseeing.
 - a) Money Management
 - b) Operations Management
 - c) Inventory Management
 - d) Production Management

Notes

- e) None of the above
- 2) _____ is quickly becoming a danger for all firms, rather than just those in the IT industry.
- Cyber security
 - Monitoring Behaviour
 - Cyber Risks
 - Uncertainty
 - None of the above
- 3) _____ has resulted in the exposure of sensitive consumer information, generating distrust within your customer base.
- Managing Overheads
 - Operations Management
 - Monitoring Performance
 - Data Theft
 - None of the above
- 4) _____ is the administration of business practices to create the highest level of efficiency possible within an organization.
- Operations management
 - Cash Flow
 - Total Productive Management
 - Productive Management
 - None of the above

Summary

- A well-managed operations department allows a firm to be efficient and productive by utilising the fewest resources while satisfying client needs. Essentially, operations management emphasises the relationship between process and value.
- Overheads may quickly add up, whether they're in the shape of administrative costs, insurance, rent, or electricity bills.
- Regulation and compliance may be a huge problem if not handled appropriately. As markets and technology develop, so do the rules and laws that govern them.
- Operations Management is concerned with converting materials and labour into goods and services as efficiently as possible to maximize the profit of an organization. Operations management teams attempt to balance costs with revenue to achieve the highest net operating profit possible.

Activity

- Visit an organization and understand their operation process and issue faced in managing the operations.

Questions & Exercises

1. Elaborate the issues faced by an organization in operations management.

Glossary

- In a small firm, competence and personality are more important than other attributes, yet not everyone is the appropriate fit for a specific organisation.
- Cyber security is quickly becoming a danger for all firms, rather than just those in the IT industry. If cyber security is not adequately handled, firms expose themselves to an expanding number of dangers, which may result in data theft, hacking, and other assaults.
- Data theft has resulted in the exposure of sensitive consumer information, generating distrust within your customer base.
- Operations management (OM) is the administration of business practices to create the highest level of efficiency possible within an organization.

Further Readings

1. Production and Operations Management; Prepared and Compiled By: Gowtham Kumar C. K CMA (Inter), M. Com, K.SET, B. ED Assistant professor
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Check Your Understanding - Answers

- 1) b)
- 2) a)
- 3) d)

Notes